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COMPOSITIONS AND METHODS FOR MODULATING THE INTERACTION BETWEEN SS18-SSX FUSION ONCOPROTEIN AND NUCLEOSOMES

Abstract

The present invention is based, in part, on the identification of a minimal region of the SS18-SSX fusion oncoprotein that mediates a direct, high-affinity interaction between the mSWI/SNF complex and the nucleosome acidic patch, and methods and agents of modulating the interaction between the SS18-SSX fusion protein and H2A K119Ub-marked nucleosomes to treat synovial sarcoma.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is the U.S. national phase of International Patent Application No. PCT/US2021/014367, filed on 21 Jan. 2021, which claims the benefit of priority to U.S. Provisional Application Ser. No. 62/989,238, filed on 13 Mar. 2020; the entire contents of each of said applications are incorporated herein in its their entirety by this reference.

SEQUENCE LISTING

[0003] The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Feb. 26, 2021, is named DFS-24825_SL.txt and is 1,165,513 bytes in size.

BACKGROUND OF THE INVENTION

[0004] A synchronous combination of histone reader domains, chromatin complex conformations, DNA-binding transcription factors (TFs), and other features are required to orchestrate the appropriate targeting of chromatin regulatory machinery in eukaryotic cells. Chromatin reader proteins play critical roles in mediating the engagement of regulatory proteins and protein complexes to specific features of nucleosomal architecture, often to facilitate site-specific catalytic activities. These include bromodomains which recognize acetylated lysines (Fujisawa and Filippakopoulos (2017) *Nat. Rev. Mol. Cell Biol.* 18:246-262), PHD domains which recognize methylation and crotonylatation of histone tails (Hyun et al. (2017) *Exp. Mol. Med.* 49:e324; Xiong et al. (2016) *Nat. Chem. Biol.* 12:1111-1118), and increasingly appreciated, nucleosome acidic patch interacting domains of SNF2 helicase-based chromatin remodeling complexes (Dann et al. (2017) *Nature* 548:607-611; Levendosky and Bowman (2019) *eLife* 8, doi:10.7554/eLife.45472; Dao et al. (2019) *Nat. Chem. Biol.* doi:10.1038/s41589-019-0413-4). In parallel, TFs recognize their cognate DNA motifs genome-wide, and, when tethered to other proteins or protein complexes, such as chromatin remodeling complexes, can direct their global positioning on chromatin to achieve cell-, tissue- and cancer-specific gene expression programs. For example, TFs have been shown to tether transiently to the surfaces of mammalian SWI/SNF (BAF) ATP-dependent chromatin remodeling complexes to globally reposition them to sites enriched for specific TF DNA-binding motifs (Sandoval et al. (2018) *Mol. Cell* 71:554-566; Boulay et al. (2017) *Cell* 171:163-178). Importantly, results of recent large-scale human genetic sequencing studies indicate that perturbations across each of the above classes of chromatin-bound factors represent frequent and recurrent events in human cancer (Kadoch et al. (2013) *Nat. Genet.* 45:592-601; Valencia and Kadoch (2019) *Nat. Cell Biol.* 21:152-161; Kadoch and Crabtree (2015) *Sci. Advances* 1:e1500447), intellectual disability (Iwase et al. (2017) *J. Neurosci.* 37:10773-10782), and other disorders, with mutations ranging from point mutations and deletions to fusion proteins which alter target engagement and activity of chromatin regulatory complexes on the genome (Wan et al. (2017) *Nature* 543:265-269; McBride et al. (2018) *Cancer Cell* 33:1128-1141; Kadoch and Crabtree (2013) *Cell* 153:71-85).

[0005] It has remained elusive, however, how nuclear fusion oncoproteins that lack canonical TF DNA-binding or recognizable chromatin reader domains yield altered, region-specific targeting of chromatin regulatory proteins and protein complexes. For example, the SS18-SSX fusion oncoprotein involving the BAF complex subunit, SS18, and 78 amino acids of one of the SSX proteins normally expressed only in testes (Clark et al. (1994) *Nat. Genet.* 7:502-508; Crew et al. (1995) *EMBO J.* 14:2333-2340; De Leeuw et al. (1995) *Hum. Mol. Genet.* 4:1097-1099; Smith and McNeel (2010) *Clinic. & Dev. Immunol.* 2010:150591), is hallmark to 100% of cases of synovial sarcoma. Incorporation of SS18-SSX in to BAF complexes causes biochemical changes, such as the destabilization of the SMARCB1 (BAF47) subunit, and results in de novo BAF complex targeting to a highly cancer-specific set of sites, particularly, broad, polycomb-repressed regions at which polycomb complex occupancy is reduced and gene expression is activated (McBride et al. (2018) *Cancer Cell* 33:1128-1141). Although some studies have indicated SSX interactions with chromatin-associated factors (Banito et al. (2018) *Cancer cell* 33:527-541), the mechanism by which the site-specific binding and unique biochemical properties are achieved remains largely unknown.

[0006] Accordingly, there remains a great need in the art to identify therapeutic agents and methods that target SS18-SSX fusion oncoprotein to treat synovial sarcoma.

SUMMARY OF THE INVENTION

[0007] The present invention is based, at least in part, on the identification of a minimal region of the SS18-SSX fusion oncoprotein, the hallmark oncogenic driver of synovial sarcoma (SS), that mediates a direct, high-affinity interaction between the mSWI/SNF complex and the nucleosome acidic patch. This engagement results in altered mSWI/SNF composition and orientation on nucleosomes, driving cancer-specific mSWI/SNF complex targeting and gene expression. Furthermore, an acidic C-terminal region of SSX confers preferential affinity to repressed, H2AK119Ub-marked nucleosomes, underlying the selective targeting to polycomb-marked genomic regions and SS-specific dependency on PRC1 function. Together, these results describe a functional interplay between a key nucleosome binding hub and a histone modification that underlies the disease-specific chromatin recruitment of a major chromatin remodeling complex.

[0008] Accordingly, in one aspect, a method of treating a subject afflicted with synovial sarcoma comprising administering to the subject a therapeutically effective amount of an agent that inhibits binding of a SS18-SSX fusion protein to a nucleosome, optionally wherein the nucleosome is an H2A K119Ub-marked nucleosome, is provided.

[0009] Numerous embodiments are further provided that can be applied to any aspect of the present invention and/or combined with any other embodiment described herein. For example, in one embodiment, the SS18-SSX fusion protein comprises a C-terminal region containing a basic region, and an acidic region of a SSX protein, optionally wherein the basic region comprises a minimal 34-amino acid region. In another embodiment, the SS18-SSX fusion protein is selected from Table 2. In still another embodiment, the agent inhibits binding of the basic region of the SS18-SSX fusion protein to an acidic patch of the nucleosome, optionally wherein the nucleosome is an H2A K119Ub-marked nucleosome. In yet another embodiment, the agent is a small molecule inhibitor, a small molecule degrader, CRISPR guide RNA (gRNA), RNA interfering agent, oligonucleotide, peptide or peptidomimetic inhibitor, aptamer, antibody, or intrabody. In another embodiment, the RNA interfering agent is a small interfering RNA (siRNA), CRISPR RNA (crRNA), CRISPR guide RNA (gRNA), a small hairpin RNA (shRNA), a microRNA (miRNA), or a piwi-interacting RNA (piRNA). In still another embodiment, the agent comprises an antibody and/or intrabody, or an antigen binding fragment thereof, which specifically binds to the SS18-SSX fusion protein, the SSX tail, and/or the H2AK119Ub-marked nucleosome, optionally wherein the SSX tail is SSX tail (34 amino acid) and/or SSX tail (78 amino acid). In yet another embodiment, the agent comprises an antibody and/or intrabody, or an antigen binding fragment thereof, which specifically binds to at least one of the following regions: (1) the basic region of the SS18-SSX fusion protein; (2) the acidic region of the SS18-SSX fusion protein; (3) the acidic patch of the H2AK119Ub-marked nucleosome; and/or (4) the H2AK119Ub mark. In another embodiment, the antibody and/or intrabody, or antigen binding fragment thereof, is chimeric, humanized, composite, or human. In still another embodiment, the antibody and/or intrabody, or antigen binding fragment thereof, comprises an effector domain, comprises an Fc domain, and/or is selected from the group consisting of Fv, Fav, F(ab′)2, Fab′, dsFv, scFv, sc(Fv)2, and diabodies fragments. In yet another embodiment, the agent induces deletion or mutation of the basic region of the SS18-SSX fusion protein, the acidic region of the SS18-SSX fusion protein, the acidic patch of the H2AK119Ub-marked nucleosome, and/or a region within the SSX tail (34 amino acid). In another embodiment, the agent inhibits H2A ubiquitination. In still another embodiment, the agent inhibits ubiquitin ligase activity of a PRC1 complex. In yet another embodiment, the agent reduces expression, copy number, and/or ubiquitin ligase activity of RING1A and/or RING1B. In another embodiment, the agent inhibits recruitment of a SS18-SSX fusion protein-bound BAF complex to an H2AK119Ub-marked nucleosome. In still another embodiment, the agent inhibits activation of at least one oncogenic target gene of the SS18-SSX fusion protein. In yet another embodiment, the oncogenic target gene of the SS18-SSX fusion protein is selected from the group consisting of WNT16 and oncogenic target genes listed in McBride et al. (2018) *Cancer Cell* 33:1128-1141. In another embodiment, the agent reduces the number of viable or proliferating cells in the cancer, and/or reduces the volume or size of a tumor comprising the cancer cells. In still another embodiment, the method further comprises administering to the subject an immunotherapy and/or cancer therapy, optionally wherein the immunotherapy and/or cancer therapy is administered before, after, or concurrently with the agent. In yet another embodiment, the immunotherapy is cell-based. In another embodiment, the immunotherapy comprises a cancer vaccine and/or virus. In still another embodiment, the immunotherapy inhibits an immune checkpoint, such as an immune checkpoint selected from the group consisting of CTLA-4, PD-1, VISTA, B7-H2, B7-H3, PD-L1, B7-H4, B7-H6, ICOS, HVEM, PD-L2, CD160, gp49B, PIR-B, KIR family receptors, TIM-1, TIM-3, TIM-4, LAG-3, GITR, 4-IBB, OX-40, BTLA, SIRPalpha (CD47), CD48, 2B4 (CD244), B7.1, B7.2, ILT-2, ILT-4, TIGIT, HHLA2, butyrophilins, and A2aR. In yet another embodiment, the cancer therapy is selected from the group consisting of radiation, a radiosensitizer, and a chemotherapy. In another embodiment, the method further comprises administering to the subject at least one additional therapeutic agent or regimen for treating the cancer.

[0010] In another aspect, a method of reducing viability or proliferation of synovial sarcoma cells comprising contacting the synovial sarcoma cells with an agent that inhibits binding of a SS18-SSX fusion protein to a nucleosome, optionally wherein the nucleosome is an H2AK119Ub-marked nucleosome, is provided.

[0011] As described above, numerous embodiments are further provided that can be applied to any aspect of the present invention and/or combined with any other embodiment described herein. For example, in one embodiment, the SS18-SSX fusion protein comprises a C-terminal region containing a basic region, and an acidic region of a SSX protein, optionally wherein the basic region comprises a minimal 34-amino acid region. In another embodiment, the SS18-SSX fusion protein is selected from Table 2. In still another embodiment, the agent inhibits binding of the basic region of the SS18-SSX fusion protein to an acidic patch of the H2AK119Ub-marked nucleosome. In yet another embodiment, the agent is a small molecule inhibitor, a small molecule degrader, CRISPR guide RNA (gRNA), RNA interfering agent, oligonucleotide, peptide or peptidomimetic inhibitor, aptamer, antibody, or intrabody. In another embodiment, the RNA interfering agent is a small interfering RNA (siRNA), CRISPR RNA (crRNA), CRISPR guide RNA (gRNA), a small hairpin RNA (shRNA), a microRNA (miRNA), or a piwi-interacting RNA (piRNA). In still another embodiment, the agent comprises an antibody and/or intrabody, or an antigen binding fragment thereof, which specifically binds to the SS18-SSX fusion protein, or the H2AK119Ub-marked nucleosome. In yet another embodiment, the agent comprises an antibody and/or intrabody, or an antigen binding fragment thereof, which specifically binds to at least one of the following regions: (1) the basic region of the SS18-SSX fusion protein; (2) the acidic region of the SS18-SSX fusion protein; (3) the acidic patch of the H2AK119Ub-marked nucleosome; and/or (4) the H2AK119Ub mark. In another embodiment, the antibody and/or intrabody, or antigen binding fragment thereof, is chimeric, humanized, composite, or human. In still another embodiment, the antibody and/or intrabody, or antigen binding fragment thereof, comprises an effector domain, comprises an Fc domain, and/or is selected from the group consisting of Fv, Fav, F(ab′)2, Fab′, dsFv, scFv, sc(Fv)2, and diabodies fragments. In yet another embodiment, the agent induces deletion or mutation of the basic region of the SS18-SSX fusion protein, the acidic region of the SS18-SSX fusion protein, and/or the acidic patch of the H2AK119Ub-marked nucleosome. In another embodiment, the agent inhibits H2A ubiquitination. In still another embodiment, the agent inhibits ubiquitin ligase activity of a PRC1 complex. In yet another embodiment, the agent reduces expression, copy number, and/or ubiquitin ligase activity of RING1A and/or RING1B. In another embodiment, the agent inhibits recruitment of a SS18-SSX fusion protein-bound BAF complex to an H2AK119Ub-marked nucleosome. In still another embodiment, the agent inhibits activation of at least one oncogenic target gene of the SS18-SSX fusion protein. In yet another embodiment, the oncogenic target gene of the SS18-SSX fusion protein is selected from the group consisting of WNT16 and oncogenic target genes listed in McBride et al. (2018) *Cancer Cell* 33:1128-1141. In another embodiment, the method further comprises contacting the cancer cells with an immunotherapy and/or cancer therapy, optionally wherein the immunotherapy and/or cancer therapy is administered before, after, or concurrently with the agent. In still another embodiment, the immunotherapy is cell-based. In yet another embodiment, the immunotherapy comprises a cancer vaccine and/or virus. In another embodiment, the immunotherapy inhibits an immune checkpoint, such as an immune checkpoint selected from the group consisting of CTLA-4, PD-1, VISTA, B7-H2, B7-H3, PD-L1, B7-H4, B7-H6, ICOS, HVEM, PD-L2, CD160, gp49B, PIR-B, KIR family receptors, TIM-1, TIM-3, TIM-4, LAG-3, GITR, 4-IBB, OX-40, BTLA, SIRPalpha (CD47), CD48, 2B4 (CD244), B7.1, B7.2, ILT-2, ILT-4, TIGIT, HHLA2, butyrophilins, and A2aR. In still another embodiment, the cancer therapy is selected from the group consisting of radiation, a radiosensitizer, and a chemotherapy.

[0012] In still another aspect, a method of assessing the efficacy of an agent for treating synovial sarcoma in a subject, comprising: a) detecting in a subject sample at a first point in time the number of viable and/or proliferating cancer cells; b) repeating step a) during at least one subsequent point in time after administration of the agent; and c) comparing number of viable and/or proliferating cancer cells detected in steps a) and b), wherein the absence of, or a significant decrease in number of viable and/or proliferating cancer cells in the subsequent sample as compared to the amount in the sample at the first point in time, indicates that the agent treats synovial sarcoma in the subject, is provided.

[0013] As described above, numerous embodiments are further provided that can be applied to any aspect of the present invention and/or combined with any other embodiment described herein. For example, in one embodiment, the subject has undergone treatment, completed treatment, and/or is in remission for synovial sarcoma between the first point in time and the subsequent point in time. In another embodiment, the first and/or at least one subsequent sample is selected from the group consisting of ex vivo and in vivo samples. In still another embodiment, the first and/or at least one subsequent sample is obtained from an animal model of synovial sarcoma. In yet another embodiment, the first and/or at least one subsequent sample is a portion of a single sample or pooled samples obtained from the subject. In another embodiment, the sample comprises cells, serum, peritumoral tissue, and/or intratumoral tissue obtained from the subject. In still another embodiment, the method further comprises determining responsiveness to the agent by measuring at least one criteria selected from the group consisting of clinical benefit rate, survival until mortality, pathological complete response, semi-quantitative measures of pathologic response, clinical complete remission, clinical partial remission, clinical stable disease, recurrence-free survival, metastasis free survival, disease free survival, circulating tumor cell decrease, circulating marker response, and RECIST criteria. In yet another embodiment, the agent is administered in a pharmaceutically acceptable formulation. In another embodiment, the step of administering or contacting occurs in vivo, ex vivo, or in vitro.

[0014] In yet another aspect, a cell-based assay for screening for agents that reduce viability or proliferation of a synovial sarcoma cell comprising: a) contacting the synovial sarcoma cell with a test agent; and b) determining the ability of the test agent to inhibit binding of a SS18-SSX fusion protein, a SSX (78 amino acid) region, and/or a SSX (34 amino acid) minimal region to a nucleosome, optionally wherein the nucleosome is a H2AK119Ub-marked nucleosome, is provided.

[0015] As described above, numerous embodiments are further provided that can be applied to any aspect of the present invention and/or combined with any other embodiment described herein. For example, in one embodiment, the SS18-SSX fusion protein comprises a C-terminal region containing a basic region, and an acidic region of a SSX protein, optionally wherein the basic region comprises a minimal 34-amino acid regbion. In another embodiment, the SS18-SSX fusion protein is selected from Table 2. In still another embodiment, the step of contacting occurs in vivo, ex vivo, or in vitro. In yet another embodiment, the assay further comprising determining the ability of the test agent to inhibit recruitment of a SS18-SSX fusion protein-bound BAF complex to an H2AK119Ub-marked nucleosome and/or H2AK 119Ub-marked region of chromatin in cells, optionally wherein the cellular chromatin comprises a PRC1/H2A Ub domain. In another embodiment, the assay further comprises determining the ability of the test agent to inhibit activation of at least one oncogenic target gene of the SS18-SSX fusion protein. In still another embodiment, the oncogenic target gene of the SS18-SSX fusion protein is selected from the group consisting of WNT16 and oncogenic target genes listed in McBride et al. (2018) *Cancer Cell* 33:1128-1141. In yet another embodiment, the assay further comprises determining a reduction in the viability or proliferation of the cancer cells.

[0016] In another aspect, an in vitro assay for screening for agents that reduce viability or proliferation of a synovial sarcoma cell comprising: a) mixing a protein comprising a c-terminal basic region and a c-terminal acidic region of a SSX protein and a nucleosome together, optionally wherein the nucleosome is a H2AK119Ub-marked nucleosome; b) adding a test agent to the mixture; and c) determining the ability of the test agent to decrease binding of the protein to the nucleosome, is provided.

[0017] As described above, numerous embodiments are further provided that can be applied to any aspect of the present invention and/or combined with any other embodiment described herein. For example, in one embodiment, the protein comprises c-terminal 34 amino acids (aa155-188) of a SSX protein. In another embodiment, the protein comprises c-terminal 78 amino acids (aa 111-188) of a SSX protein. In still another embodiment, the protein is a SS18-SSX fusion protein. In yet another embodiment, the SS18-SSX fusion protein is selected from Table 2. In another embodiment, the SS18-SSX fusion protein comprises SS18 protein fused with a c-terminal portion of a SSX protein. In still another embodiment, the SS18-SSX fusion protein comprises c-terminal 34 amino acids (aa155-188) of a SSX protein. In yet another embodiment, the SS18-SSX fusion protein comprises c-terminal 78 amino acids (aa 111-188) of a SSX protein. In another embodiment, the SSX protein is selected form the group consisting of human SSX1, SSX2, SSX3, SSX4, SSX6, SSX7, SSX8, and SSX9. In still another embodiment, the SS18-SSX fusion protein comprises W164, R167, L168, R169 and/or R171 of SEQ ID: 3, 7, 13, 17, 21, 25, or 31, or orthologs thereof. In yet another embodiment, the SS18-SSX fusion protein is a part of a BAF complex. In another embodiment, the nucleosome comprises H2A protein comprising E56, E64, D90, E91, E92 and/or E113 of human, mouse, rat, or *Xenopus* H2A, or orthologs thereof; and/or H2B protein comprising E105 and/or E113 of human, mouse, rat, or *Xenopus* H2B, or orthologs thereof. In still another embodiment, the subject is an animal model of the cancer, optionally wherein the animal model is a mouse model. In yet another embodiment, the subject is a mammal. In another embodiment, the mammal is a mouse or human. In still another embodiment, the mammal is a human.

[0018] In still another aspect, an isolated modified protein complex selected from the group consisting of protein complexes listed in Table 3, wherein the isolated modified protein complex comprises at least one subunit that is modified, is provided.

[0019] As described above, numerous embodiments are further provided that can be applied to any aspect of the present invention and/or combined with any other embodiment described herein. For example, in one embodiment, the at least one modified subunit is a fragment of the subunit. In another embodiment, the fragment of the subunit binds to at least one binding partner of the subunit to form the isolated modified protein complex. In still another embodiment, the fragment of the subunit comprises the basic region and/or the acidic region of a SSX protein. In yet another embodiment, the fragment of the subunit comprises c-terminal 34 amino acids (aa155-188) of a SSX protein. In another embodiment, the fragment of the subunit comprises c-terminal 78 amino acids (aa 111-188) of a SSX protein. In still another embodiment, the SSX protein is selected form the group consisting of human SSX1, SSX2, SSX3, SSX4, SSX6, SSX7, SSX8, and SSX9. In yet another embodiment, the fragment of the subunit comprises the acidic patch of a nucleosome and/or the H2A K119 Ub mark. In another embodiment, at least one subunit is linked to at least another subunit. In still another embodiment, at least one subunit is linked to at least another subunit through covalent cross-links. In yet another embodiment, at least one subunit is linked to at least another subunit through a peptide linker. In another embodiment, the at least one subunit comprises a heterologous amino acid sequence. In still another embodiment, the heterologous amino acid sequence comprises an affinity tag or a label. In yet another embodiment, the affinity tag is selected from the group consisting of Glutathione-S-Transferase (GST), calmodulin binding protein (CBP), protein C tag, Myc tag, HaloTag, HA tag, Flag tag, His tag, biotin tag, and V5 tag. In yet another embodiment, the label is a fluorescent protein. In another embodiment, the at least one subunit is selected from the group consisting of HA-SS18-SSX1, V5-SS18-SSX1, V5-SS18-SSX1 34aa tail, V5-SS18-SSX1 78aa tail, H2A, and H2B.

[0020] In yet another aspect, a pharmaceutical composition comprising an isolated modified protein complex described herein, and a carrier, is provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] FIG. **1**A-FIG. **1**E show that SS18-SSX-containing BAF complexes exhibit significantly increased affinity for chromatin. FIG. **1**A shows colloidal blue staining performed on purifications of wild-type BAF complexes (from HA-SS18 WT-expressing 293T cells) and SS18-SSX-containing BAF complexes (from HA-SS18-SSX1-expressing cells), from soluble nuclear extract (NE) and chromatin-bound (CHR) fractions. Equal amounts (by volume) of nuclei in each condition were isolated and subsequently purified in to NE and CHR fractions. FIG. **1**B shows MS spectral counts for BAF complex subunits (green) and histone proteins (orange) from HA-SS18 WT and HA-SS18-SSX purifications from NE and CHR fractions in (FIG. **1**A). Peptides counts are log 2 normalized to bait (SS18 peptides). FIG. **1**C shows density sedimentation gradients using 10-30% glycerol performed on HA-SS18 WT and HA-SS18-SSX1 purifications from HEK-293T cells. BAF complex subunits and histone proteins are indicated.

SYPRO® Ruby staining was used for visualization. FIG. **1**D shows immunoblot for SMARCA4 and SMARCC1 performed on Aska SS cells in shCtrl (control, non-targeting harpin shRNA) and shSSX (shRNA targeted to SSX) conditions following differential salt extraction (0-1000 mM NaCl). FIG. **1**E shows FRAP studies performed on HEK293T cells expressing either GFP-SS18 WT or GFP-SS18-SSX1. Recovery kinetics were recorded and the recovery half-times were calculated to be 10.1 and 33.9 seconds for GFP-SS18 WT and GFP-SS18-SSX1, respectively (values represent mean of n=30 cells per condition, with error bars indicating standard deviation at each time point).

[0022] FIG. **2**A-FIG. **2**H show that SS18-SSX-containing BAF complexes exhibit high-affinity interactions with histones and longer residency times on chromatin. FIG. **2**A shows MS spectral counts for BAF complex subunits and histone proteins from HA-SS18 WT and HA-SS18-SSX purifications from soluble nuclear extract NE and CHR fractions from FIG. **1**A. Total number of peptides (number of peptides normalized to bait, SS18) are shown. FIG. **2**B shows ranked peptides captured in HA-SS18-SSX purification (chromatin-bound fraction). Red indicates mSWI/SNF complex subunits. Green indicates histones. Orange indicates members of PRC1 and PRC2 complexes, shown for comparison. See also Tables 5A-5E. FIG. **2**C has two panels. The top panel shows immunoblot for GFP and H2A performed on HEK-293T cells infected with either GFP-SS18 WT or GFP-SS18-SSX following differential salt extraction (0-1000 mM NaCl). The bottom panel shows immunoblot for SS18 and H2A K119Ub performed on HEK-293T cells (naive) and Aska-SS cells following differential salt extraction (0-1000 mM NaCl) experiments. FIG. **2**D shows immunoblot for SMARCA4 and SS18 performed HEK-293T cells or Aska SS cells (SS18-SSX+) following differential salt extraction (0-1000 mM NaCl). FIG. **2**E shows FRAP experiments performed in the Aska SS cell line modified to express BRG1 (SMARCA4)-Halo. Aska-SS cells were treated with either shControl shRNA hairpin or shSSX (targeting the SS18-SSX fusion). Recovery t½ times (seconds) with 95% CIs are shown; n=20 cells. FIG. **2**F shows SYPRO® Ruby staining indicating identified proteins from FIG. **1***c* in Fraction 13 (HA-SS18 WT) and Fraction 18 (HA-SS18-SSX). FIG. **2**G shows SMARCB1 peptide abundance (normalized to SMARCA4) and relative to SS18 WT-bound complexes (soluble NE fraction). NE indicates nuclear extract; CHR indicates chromatin-bound fraction. FIG. **2**H has two panesl. The left panel shows cyber-gold staining of complexes purified from untreated (no benzonase) nuclear extracts isolated via ammonium sulfate extraction. The right panel shows that H3 immunoblot reveals prominent histone binding in HA-SS18-SSX-bound complexes but not in HA-SS18 WT-bound complexes.

[0023] FIG. **3**A-FIG. **3**I show that conserved basic and acidic regions within a minimal SSX domain are necessary and sufficient to bind nucleosomes and promote specialized BAF complex chromatin recruitment and activity. FIG. **3**A shows GST (control) and GST-SSX1 (78aa) purified recombinant proteins incubated with mammalian mononucleosomes (purified by MNase digestion), captured using glutathione resin, visualized using colloidal blue. FIG. **3**B shows quantitative targeted MS analysis of MBP pull down experiments using the MBP-SSX 78aa protein and endogenous mammalian nucleosomes purified using MNase digestion from 293T cells. Log 2 (FC) calculated relative to input sample. Red indicates enriched; blue indicates depleted. FIG. **3**C shows immunofluorescence analysis of V5-tagged SS18 and SS18-SSX relative to RING1B and SUZ12 in 293T cells. Arrows indicate positions of the Barr bodies (inactive X). Scale bar indicates 5 μm. FIG. **3**D shows alignment of SSX1 protein across species and relative to related PRDM7/9 proteins. Highly conserved basic and acidic regions are indicated in blue and red, respectively. FIG. **3**D discloses SEQ ID NOS 231-233, 233-236, 232, 237, 232 and 238-243, respectively, in order of appearance. FIG. **3**E shows pull-down experiments of N-terminally biotinylated SSX peptides (scrambled (aa155-188), SSX 34aa (aa155-188), SSX 24aa (aa164-188) and SSX 23aa (aa165-188) incubated with mammalian mononucleosomes and visualized with colloidal blue. FIG. **3**F shows pull-down experiments of N-terminally biotinylated SSX peptides including scrambled control, wildtype (WT) and mutant variants (single alanine substitutions as well as regional substitutions (i.e., Basic/A, basic region RLRERK-->AAAAAA (SEQ ID NOS 219-220, respectively, in order of appearance); Acidic/A, acidic region DPEEDDE-->AAAAAAA (SEQ ID NOS 221-222, respectively, in order of appearance)) incubated with mammalian mononucleosomes and visualized with colloidal blue. FIG. **3**F discloses SEQ ID NO: 232. FIG. **3**G shows ChIP-seq density heatmaps reflecting chromatin occupancy of V5-SS18-SSX1, V5-SS18, V5-SS18-SSX (24aa) and V5-SS18-SSX (34aa) over all V5 Peaks (38,014 total peaks). FIG. **3**H shows heatmap reflecting top 5% upregulated and downregulated genes (Z-score) by RNA-seq for each condition. FIG. **3**I shows proliferation experiments performed on SYO-1 SS cells infected with either control hairpin (shCt) or shSSX (knockdown of endogenous SS18-SSX) with overexpression of empty vector control, SS18-SSX 78aa or SS18-SSX 34aa variants. n=3 independent experimental replicates; error bars represent standard deviation; ** indicate $p<0.01$.

[0024] FIG. **4**A-FIG. **4**H show the SSX 78aa protein binds mononucleosomes, with preference for nucleosomes decorated with repressive histone modifications. FIG. **4**A shows coomassie-stained gel of recombinantly purified GST, GST-SSX (78aa) proteins, run next to BSA protein as control. FIG. **4**B shows purification of mammalian mononucleosomes from HEK-293T cells using MNase digestion. FIG. **4**C shows incubation of GST or GST-SSX (78aa) with either recombinant or mammalian mononucleosomes, resolved by immunoblot for GST and histone H3 or Coomassie and histone H3. Two representative experiments are shown. FIG. **4**D shows purification of MBP and

MBP-SSX (78aa) proteins for targeted, quantitative histone mass-spectrometry. Quantitative histone mass spectrometry performed on MBP-SSX1 (versus MBP control) incubated with pooled mononucleosomes isolated from HEK-293T cells via MNase digestion. Experiment performed in n=2 replicates. See also Tables 3A-3C. FIG. **4**E shows a schematic diagram for targeted MS experiments. FIG. **4**F shows enrichment of SSX-bound histone peptides, over input. Enriched and depleted proteins are shown in red and blue, respectively. FIG. **4**G shows quantitative densitometry normalized to input reflecting GST-SSX 78aa preferential binding to mammalian mononucleosomes (prepared via MNase digestion in HEK-293T cells) versus recombinant, unmodified nucleosomes. Bars represent averages of n=3 independent experiments, error bars represent standard deviation; p-value=0.0164. FIG. **4**H shows heatmap reflecting enrichment or depletion of selected histone marks, including H2AZ and H3K4 methylation states. Scale=log 2FC.

[0025] FIG. **5**A-FIG. **5**G show nucleosome binding and nuclear localization properties of SS18-SSX and SSX variants. FIG. **5**A shows immunofluorescence imaging performed on IMR90 fibroblasts and HEK293T cells infected with either V5-SS18-SSX or V5-SS18. Visualized in red for H3K9me3, SMARCA4, PBRM1, SMARCC1, H3K9Ac across experiments. DAPI is shown as nuclear stain and merged images are provided with scale bars; Scale bar indicates 5 μm. FIG. **5**B shows IF-based localization of SS18 FL (1-188aa) in fibroblasts. H2AUb119, DAPI counterstain, and merged images are shown. Scale bar indicates 5 μm. FIG. **5**C shows peptide competition experiment using Biotinylated SSX peptide (aa 155-188) and unlabeled SSX (aa 155-188). Visualization for Histone H3 uses immunoblot. FIG. **5**D shows SSX peptide hybridization experiments performed on methanol-fixed cells. Streptavidin (SA) used for biotinylated SSX peptide visualization, H2AUb119 for Barr bodies. DAPI counterstain and merged images shown. Scale bar indicates 5 μm. FIG. **5**E has two panels. The top panel shows conservation analysis among SSX and PRDM 7/9 human protein regions. The bottom panel shows peptide pull down experiments with recombinant nucleosomes performed with Scrambled control SSX1, SSX1, PRDM7, PRDM9. Visualization is by colloidal blue staining. FIG. **5**E discloses SEQ ID NOS 232-233, 236-237, 244 and 224, respectively, in order of appearance. FIG. **5**F has two panels. The left panel shows alignment of SSX proteins (SSX 1-9). The right panel shows peptide pull down experiments with recombinant nucleosomes performed with aa 155-188 of SSX family members. Visualization is by colloidal blue staining. FIG. **5**F discloses SEQ ID NOS 232, 232, 232-233 and 233-237, respectively, in order of appearance. FIG. **5**G shows peptide competition experiment using Biotinylated SSX peptide (aa 155-188) and Scrambled control SSX peptide (aa 155-188). Visualization for Histone H3 is by immunoblot.

[0026] FIG. **6**A-FIG. **6**E show defining a minimal 34-aa SSX region responsible for chromatin engagement and oncogenic gene expression. FIG. **6**A shows additional representative V5 ChIP-seq and RNA-seq tracks, here shown at the SOX2 and GALNT9 loci. FIG. **6**B shows differential salt experiments ([0-1000 mM NaCl]) performed on HEK-293T cells infected with either SS18-SSX 34aa versus SS18-SSX 24aa. Immunoblots for V5 as well as GAPDH and H3 (controls) are shown. FIG. **6**C shows immunofluoroscence imaging of IMR90 fibroblasts infected with SS18 and SS18-SSX variants, as indicated, and stained for V5 (SS18-SSX or SSX variant) and DAPI; merged images are shown. Localization to H2AUb119-high sites (Barr bodies) is highlighted. Scale bar indicates 5 μm. FIG. **6**D shows beta-gal senescence assay performed on IMR90 cells infected with WT SS18, SS18-SSX and SSX FL and 78aa variants, as indicated. FIG. **6**E shows that SYO-1 synovial sarcoma cells were treated with either shCtrl (control hairpin) or shSSX (shRNA targeting SSX) to reduce levels of endogenous fusion, followed by rescue of SS18-SSX WT and mutant variants or empty vector control. Proliferation was evaluated over 16 days (see also FIG. **3**I).

[0027] FIG. **7**A-FIG. **7**J show that the SSX basic region outcompetes the SMARCB1 C-terminal alpha-helical domain for nucleosome acidic patch binding. FIG. **7**A shows incubation of biotinylated SSX peptides (aa 155-188) in either WT or RLR motif-mutant forms (R167A, R169A, R171A) with nucleosomes. FIG. **7**B shows photocrosslinking experiments performed with reactive diazarine probes localized throughout the nucleosome acidic patch region indicate strongest binding to H2A E56 and 12B E113 residues. FIG. **7**C shows SSX binding sites mapped on nucleosome PDB: 1KX5. Acidic patch crosslinked sites are labeled. FIG. **7**D shows incubation of GST-SSX 78aa tail with either WT or acidic patch mutant nucleosomes (D90N, E92K, and E113K). Visualization of binding is by histone H3 immunoblot. FIG. **7**E shows LANA peptide competition experiment with SSX 34aa biotinylated peptide bound to nucleosomes. FIG. **7**F shows TALOS secondary structure prediction of the SSX 78aa region. An alpha helical probability (aa HAWTHRLRERK (SEQ ID NO: 223)) is indicated in red. The protein is largely disordered with a short helical--like segment (aa164-171) and a beta-strand like segment (aa174-179). FIG. **7**F discloses SEQ ID NO: 232. FIG. **7**G shows V5 ChIP-seg heat map reflecting genome-wide localization of V5-tagged SS18-SSX, SS18 WT and SS18-SSX RLR-->RLA (RI69A) mutant in CRL7250 fibroblasts. FIG. **7**H shows reciprocal competition experiments performed with either SMARCB1 C-terminal alpha helical domain bound to nucleosomes or SSX 34aa bound to nucleosomes and competed with indicated peptide. FIG. **7**I shows REAA nucleosome remodeling assay performed with BAF complexes containing either WT SS18 or SS18-SSX. Experiment performed at 37 degrees C., 0-40 min time course, BAF complex capture performed using ARDD1A IP. FIG. **7**J shows ATAC-seq DNA accessibility (log 2FC(RPKM+1) performed in CRL7250 fibroblasts over SS18-SSX-specific sites and SS18 WT/SS18-SSX shared sites, defined in FIG. **7**G.

[0028] FIG. **8**A-FIG. **8**G show that the SSX basic region and SMARCB1 C-terminal alpha helical domain compete for nucleosome acidic patch binding. FIG. **8**A shows strategy for nucleosome-peptide photocrosslinking. FIG. **8**B shows additional (replicate) photocrosslinking experiments performed with reactive diazarine probes localized throughout the nucleosome acidic patch region indicate strongest binding to H2A E56 and H2B E113 residues, weaker binding to H2A E91, and no binding to E61, E92, and D90 residues. Experimental conditions are as follows: 0.3 μM mononucleosomes, 3 μM SSX, 150 mM KCl. FIG. **8**C shows pulldown experiments performed with either Scrambled or SSX 34aa peptides (biotinylated) incubated with mammalian mononucleosomes prepared from cells infected with WT H2A, or H2AD90N, H2A E92K mutant variants. FIG. **8**D shows 15N-HSQC spectrum of SSX1 mutant having 7 C-terminal residue deletion, with assignments marked in red. The data were collected using 330 IM protein in pH 6.5 buffer at 15° C. on a 700 MHz spectrometer. FIG. **8**E shows a model indicating docking of solved LANA peptide-nucleosome binding region and SSX peptide crosslinking in the nucleosome acidic patch. Interacting residues from photocrosslinking are highlighted. FIG. **8**F shows modeling of SSX C-term (34aa) alpha helical peptide on nucleosome structure (PDB: 1KX5) using ZDOCK, in full nucleosome and zoomed-in view of acidic patch region. FIG. **8**G shows photocrosslinking experiments performed with SSX 34aa peptide incubated with nucleosomes modified at the H2A E56 residue, with and without LANA peptide competition.

[0029] FIG. **9**A-FIG. **9**G show that mutations in the basic region of SSX affect the targeting and function of SS18-SSX-containing BAF complexes. FIG. **9**A shows gene expression changes across each SS18 WT and SS18-SSX variant conditions from FIG. **7**G. FIG. **9**B shows proliferative rescue experiments performed in SYO-1 SS cell line treated with shSSX and rescued with either vector control, SS18-SSX or SS18-SSX (R169A or W164A) variants. n=3 independent experiments performed; error bars represent standard deviation; * indicates $p<0.05$. FIG. **9**C shows peptide hybridization of IMR90 cells using SSX and mutant basic region mutant peptides. Arrows indicate positions of the Barr bodies. Scale bar indicates 5 μm. FIG. **9**C discloses SEQ ID NOS 221-222, respectively, in order of appearance. FIG. **9**D shows immunoblot performed on whole-cell extracts (RIPA extraction) from SYO1 cells treated with either shCtrl or shSSX and infected with either empty vector or SS18-SSX variants, used in proliferation experiments in FIG. **9**B. FIG. **9**E shows peptide hybridization of IMR90 cells using SSX and mutant basic region mutant (W164A and R169A) peptides. Arrows indicate positions of the Barr bodies. Scale bar indicates 5 μm. FIG. **9**F shows ChIP-seq studies (anti-V5) performed in CRL7250 cells infected with either SS18-SSX or SS18-SSX W164A mutant, mapped as summary plot over SS18-SSX target sites. FIG. **9**G shows RNA-seq (gene expression) data, box and whisker plots indicating average expression in SS18-SSX versus SS18-SSX W164A mutant conditions.

[0030] FIG. **10**A-FIG. **10**G show subunit composition, chromatin binding, and functional properties of SS18-SSX-bound BAF complexes. FIG. **10**A shows SMARCB1 peptide abundance calculated from MS experiments (anti-SMARCA4 (BRG1) IPs) performed in Aska-SS synovial sarcoma cells, human Fibroblasts, and HEK-293T cells. Peptide abundance normalized to SMARCA4 abundance. FIG. **10**B shows input and GFP IPs performed in Aska-SS cells infected with either GFP-SS18 or GFP-SS18-SSX. SMARCC1, SS18, GFP, SMARCB1, and TBP levels are shown. FIG. **10**C shows SS18-SMARCA4 crosslinks detected in CX-MS experiments of intact, fully-formed BAF complexes in (Mashtalir et al. (2018) *Cell* 175:1272-1288). FIG. **10**D shows immunoblot studies performed on CRL7250 cells infected with SS18-SSX variants indicated. FIG. **10**E shows the immunoblot performed for ARID1A and SS18 on complexes captured via ARID1A, used for nucleosome remodeling and ATPase assays. FIG. **10**F shows ATAC-seq experiments performed in SYO-1 SS cells in shCtrl and shSSX conditions, mapped over SS18 ChIP-seq. FIG. **10**G shows ATPase activity calculated by ADP-Glo for SS18 WT- and SS18-SSX-containing BAF complexes. T indicates 0-40 min timecourse; n=3 experimental replicates at each time point; error bars represent standard deviation.

[0031] FIG. **11**A-FIG. **11**K show that SSX preferentially binds H2A K119Ub-marked nucleosomes to promote BAF complex targeting to polycomb-repressed loci. FIG. **11**A shows CERES dependency scores (fitness dropout) derived from genome-scale fitness screens performed using CRISPR-Cas9-based methods (Achilles, Broad Institute; available on the World Wide Web at depmap.org/portal/achilles/). Difference is the score calculated between SYO1 (SS18-SSX+) cells and SW982 cells (negative for fusion, histologic mimic). mSWI/SNF, PRC1, PRC2 members are shown. FIG. **11**B shows SS18 localization (by ChIP-seq) in SYO-1 cells treated with either scrambled KD or shSS18-SSX, aligned with H2AUb119 ChIP-seq in the scrambled KD condition. FIG. **11**C shows example tracks at the SLIT3 locus reflecting co-localization of SS18-SSX BAF complexes, H2AUb, and RING1B (PRC1). FIG. **11**D shows GST-SSX pull down experiments performed using either WT nucleosomes or H2A K119Ub nucleosomes. H3 immunoblot is used for assessment of nucleosome binding to GST-SSX. FIG. **11**E shows alphalisa experiment performed with GST-SSX and 10 nM biotinylated nucleosomes of either WT, H2AUb or H2BUb nucleosomes. EC50 measurements are shown. n=3 experiments. FIG. **11**F shows pull down experiments using endogenous, fully-assembled HA-SS18- or HA-SS18-SSX-bound BAF complexes incubated with either WT nucleosomes (unmodified) or H2A K119Ub-modified nucleosomes. SMARCA4 and H3 immunoblots are shown. FIG. **11**G has two panels. The left panel shows the representation of PRC1 complex-nucleosome structure (McGinty et al. 2018; PDB: 4R8P), indicating regions mutagenized. The right panel shows the immunoblot of representative mutations which inhibit

H2A K119Ub deposition absent changes to PRC1 structural integrity. FIG. **11**H shows immunofluorescence imaging for RING1B (red), V5 SS18-SSX (green), with DAPI nuclear stain, and merged images in WT and RING1A/B dKO 293T cells with rescued conditions as indicated. FIG. **11**I shows quantification of Barr body (inactive X Chr) localization for each condition, meanAU is plotted. Peptides were incubated −/+presence of USP2 treatment. Error bars=st.dev. FIG. **11**J shows pull down experiments performed using either GST-SSX or GST-SSXdel7aa (acidic C-term DPEEDDE-->AAAAAAA (SEQ ID NOS 221-222, respectively, in order of appearance)) with WT nucleosomes or H2A K119Ub nucleosomes. H3 immunoblot is used for assessment of nucleosome binding to GST-SSX. FIG. **11**K shows an alphalisa experiment performed with GST-SSX or GST-SSXdel7aa (acidic C-term) and 10 nM biotinylated nucleosomes of either WT or H2AUb nucleosomes. EC50 measurements are shown. n=3 experiments; error bars=st.dev. Data for GST-SSX with WT nucleosomes and H2A K119Ub nucleosomes are shared between FIG. **11**E.

[0032] FIG. **12**A-FIG. **12**L show that SS18-SSX-bound BAF complexes preferentially bind H2A K119Ub-marked nucleosomes. FIG. **12**A shows waterfall dependency plots for RING1B, PCGF3, PCGF5 and KDM2B genes across n=387 cell lines (Project DRIVE Dataset; available on the World Wide Web at oncologynibr.shinyapps.io/drive/; Novartis). SS cell lines containing the SS18-SSX fusion oncoprotein are highlighted in red. FIG. **12**B shows H2A K119Ub and RING1B ChIP-seq tracks over selected loci, aligned with SS18 (BAF) localization in SYO-1 cells treated with shScramble or shSS18-SSX. FIG. **12**C shows MBP-SSX1 (78aa) pull down experiments which indicate capture of histones, and specifically, H2AUb species. FIG. **12**D shows CERES dependency scores (fitness dropout) derived from genome-scale fitness screens performed using CRISPR-Cas9-based methods (Achilles, Broad Institute; available on the World Wide Web at depmap.org/portal/achilles/). Difference is the score calculated between SYO1, Yamato-SS, SCS241 (SS18-SSX+) cells and SW982 cells (negative for fusion, histologic mimic). Blue indicates enriched for dependency. mSWI/SNF, PRC1, PRC2 members are shown. FIG. **12**E shows CERES and DEMETER Dependency scores for SSX1 and SS18 genes for CRISPR-Cas9 and RNAi datasets, respectively. Synovial sarcoma cell lines are indicated in pink; all other cell lines are represented in gray. FIG. **12**F shows CERES and DEMETER Dependency scores for SSX1 and SS18 genes for CRISPR-Cas9 and RNAi datasets, respectively. Synovial sarcoma and soft tissue (SS cell lines) exhibit preferential dependency. (Project DRIVE; available on the World Wide Web at oncologynibr.shinyapps.io/drive/). SS cell lines containing the SS18-SSX fusion oncoprotein are highlighted in red. FIG. **12** G shows GST-SSX pull down experiments performed using either WT nucleosomes or H2A K119Ub nucleosomes. H3 immunoblot is used for assessment of nucleosome binding to GST-SSX. FIG. **12**H shows streptavidin-based pull-down experiments using endogenous, fully-assembled HA-SS18- or HA-SS18-SSX-bound BAF complexes incubated with biotinylated WT nucleosomes (unmodified) or H2A K119Ub-modified nucleosomes. SMARCA4 and H3 immunoblots are shown. FIG. **12**I shows that silver stain of the WT SS18 complexes and SS18-SSX fusion complexes isolated usin ammonium sulfate nuclear extraction protocol. Identified proteins labeled (Left). WB of the samples on the right indicating presence of histone H3 (Right). FIG. **12**J shows pull down experiments performed using GST-SSX incubated with unmodified or a series of modified recombinant mononucleosomes, or endogenous mononucleosomes (mammalian, purified via MNase digestion from HEK-239T cells). FIG. **12**K shows quantitative densitometry performed on experiment in FIG. **6**D. FIG. **12**L shows fluorescence polarization assays performed with fluorescently-labeled SSX1 (78aa) and either unmodified nucleosomes (blue curve) or H2A K119Ub-modified nucleosomes (red curve).

[0033] FIG. **13**A-FIG. **13**G show that SSX targeting requires PRC1 complex-mediated H2A K119Ub placement. FIG. **13**A shows immunoblots performed on V5 IP and input protein levels in WT and RING1A/B double KO (dKO) HEK-293T cells. FIG. **13**B shows an immunoblot of representative, structurally-guided RING1B mutations which inhibit H2AK119Ub deposition partially, fully, or not at all. FIG. **13**C shows immunofluorescence imaging for RING1B (red), V5 SS18-SSX (green), with DAPI nuclear stain, and merged images. FIG. **13**D shows peptide hybridization experiments. Representative images of SSX labeling of Barr bodies (inactive X) identified for each condition using H3K27me3 staining. Peptides (SSX or Scrambled) were incubated methanol-fixed cells, untreated or treated with USP2 deubiquitinating enzyme. FIG. **13**E shows incubation of GST-SSX WT, SSX mutant variants, or UBQLN1-TUBE2 or hHR23A-TUBE1 (pos controls) with Ub-coated beads. FIG. **13**F shows V5-SS18-SSX, H2A K119Ub, and H3K27me3 IF studies performed in WT and RING1A/B dKO 293T cells. FIG. **13**G shows DMSO control or EZH2 inhibitor treatment (to inhibit H3K27me3 placement) indicates no change to SS18-SSX foci localized to Barr bodies.

[0034] FIG. **14**A-FIG. **14**B show a model for SS18-SSX-bound BAF complex nucleosome engagement. FIG. **14**A shows a schematic of SS18 WT and the SS18-SSX fusion oncoprotein. FIG. **14**B shows a model for BAF complex engagement on nucleosomes in WT and SS18-SSX fusion oncoprotein states. In WT complexes, the core module of BAF complexes engages the nucleosome acidic patch via the SMARCB1 C-terminal alpha helical domain (aa 351-385). Upon expression of SS18-SSX, the SSX alpha helical basic region (RLRERK (SEQ ID NO: 219)) dominantly engages the acidic patch, displacing SMARCB1, leading to its degradation, and changing the orientation of the BAF core module (Mashtalir et al. (2018) *Cell* 175:1272-1288) on the nucleosome. This SS18-SSX-specific conformation of BAF complexes exhibits strong preference for H2AUbK119-decorated nucleosomes, underpinning their preference for polycomb chromatin regions.

[0035] For any figure showing a bar histogram, curve, or other data associated with a legend, the bars, curve, or other data presented from left to right for each indication correspond directly and in order to the boxes from top to bottom of the legend.

DETAILED DESCRIPTION OF THE INVENTION

[0036] The present invention is based, at least in part, on the discovery of the mechanism by which the SS18-SSX oncogenic fusion protein engages with chromatin and directs BAF chromatin remodeling complexes to specialized target sites. Specifically, it was found that SSX contains a basic region that directly binds the nucleosome acidic patch, altering BAF complex subunit configuration and activity. Further, SSX-nucleosome binding is augmented by the presence of ubiquitylated H2A (H2A K119Ub) on nucleosomes, preferential recognition of which requires a second, conserved region of SSX. These dual reader-like features of SSX underlie the highly disease-specific, hallmark chromatin remodeling complex targeting, gene expression, and functional dependencies in synovial sarcoma. Collectively, these studies reveal a novel mechanism of chromatin localization with important biological and disease implications.

[0037] There is current no direct way to treat human synovial sarcoma, driven by the SS18-SSX fusion oncoprotein. A major reason behind this is that, until this invention, little is known about how SS18-SSX specifically engages with chromatin to “hijack” BAF chromatin remodeling complexes to new sites genome-wide to activate cancer-promoting gene expression. The present disclosure unveils an unexpected, direct interaction between SSX and the nucleosome, specifically, the acidic patch region of the nucleosome, and the preference for repressed heterochromatin marked by the H2A K119Ub mark. Thus, the present disclosure provides an accurate and biologically meaningful screening strategy to identify agents that break SS18-SSX or SS18-SSX-containing BAF complex- H2A K119Ub nucleosome contacts. Chemical matter revealed from such a screening is capable of treating and potentially curing this disease in a highly specific manner.

[0038] Accordingly, the present invention relates, in part, to methods and agents for treating synovial sarcoma by modulating the interaction between SS18-SSX oncogenic fusion protein and H2A K119Ub nucleosomes.

I. Definitions

[0039] The articles “a” and “an” are used herein to refer to one or to more than one (i.e. to at least one) of the grammatical object of the article. By way of example, “an element” means one element or more than one element.

[0040] The term “administering” is intended to include routes of administration which allow an agent to perform its intended function. Examples of routes of administration for treatment of a body which can be used include injection (subcutaneous, intravenous, parenterally, intraperitoneally, intrathecal, etc.), oral, inhalation, and transdermal routes. The injection can be bolus injections or can be continuous infusion. Depending on the route of administration, the agent can be coated with or disposed in a selected material to protect it from natural conditions which may detrimentally affect its ability to perform its intended function. The agent may be administered alone, or in conjunction with a pharmaceutically acceptable carrier. The agent also may be administered as a prodrug, which is converted to its active form in vivo.

[0041] The term “altered amount” or “altered level” refers to increased or decreased copy number (e.g., germline and/or somatic) of a biomarker nucleic acid, e.g., increased or decreased expression level in a cancer sample, as compared to the expression level or copy number of the biomarker nucleic acid in a control sample. The term “altered amount” of a biomarker also includes an increased or decreased protein level of a biomarker protein in a sample, e.g., a cancer sample, as compared to the corresponding protein level in a normal, control sample. Furthermore, an altered amount of a biomarker protein may be determined by detecting posttranslational modification such as methylation status of the marker, which may affect the expression or activity of the biomarker protein.

[0042] The amount of a biomarker in a subject is “significantly” higher or lower than the normal amount of the biomarker, if the amount of the biomarker is greater or less, respectively, than the normal level by an amount greater than the standard error of the assay employed to assess amount, and preferably at least 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 150%, 200%, 300%, 350%, 400%, 500%, 600%, 700%, 800%, 900%, 1000% or than that amount. Alternately, the amount of the biomarker in the subject can be considered “significantly” higher or lower than the normal amount if the amount is at least about two, and preferably at least about three, four, or five times, higher or lower, respectively, than the normal amount of the biomarker. Such “significance” can also be applied to any other measured parameter described herein, such as for expression, inhibition, cytotoxicity, cell growth, and the like.

[0043] The term “altered level of expression” of a biomarker refers to an expression level or copy number of the biomarker in a test sample, e.g., a sample derived from a patient suffering from cancer, that is greater or less than the standard error of the assay employed to assess expression or copy number, and is preferably at least twice, and more preferably three, four, five or ten or more times the expression level or copy number of the biomarker in a control sample (e.g., sample from a healthy subjects not having the associated disease) and preferably, the average expression level or copy number of the biomarker in several control samples. The altered level of expression is greater or less than the standard error of the assay employed to assess expression or copy number, and is preferably at least 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 150%, 200%, 300%, 350%, 400%, 500%, 600%, 700%, 800%, 900%, 1000% or more times the expression level or copy number of the biomarker in a control sample (e.g., sample from a healthy subjects not having the associated disease) and preferably, the average expression level or copy number of the biomarker in several control samples. In some embodiments, the level of the biomarker refers to the level of the biomarker itself, the level of a modified biomarker (e.g., phosphorylated biomarker), or to the level of a biomarker relative to another measured variable, such as a control (e.g., phosphorylated biomarker relative to an unphosphorylated biomarker).

[0044] The term “altered activity” of a biomarker refers to an activity of the biomarker which is increased or decreased in a disease state, e.g., in a cancer sample, as compared to the activity of the biomarker in a normal, control sample. Altered activity of the biomarker may be the result of, for example, altered expression of the biomarker, altered protein level of the biomarker, altered structure of the biomarker, or, e.g., an altered interaction with other proteins involved in the same or different pathway as the biomarker or altered interaction with transcriptional activators or inhibitors.

[0045] The term “altered structure” of a biomarker refers to the presence of mutations or allelic variants within a biomarker nucleic acid or protein, e.g., mutations which affect expression or activity of the biomarker nucleic acid or protein, as compared to the normal or wild-type gene or protein. For example, mutations include, but are not limited to substitutions, deletions, or addition mutations. Mutations may be present in the coding or non-coding region of the biomarker nucleic acid.

[0046] The term “SWI/SNF complex” refers to SWItch/Sucrose Non-Fermentable, a nucleosome remodeling complex found in both eukaryotes and prokaryotes (Neigeborn Carlson (1984) *Genetics* 108:845-858; Stern et al. (1984) *J. Mol. Biol.* 178:853-868). The SWI/SNF complex was first discovered in the yeast, *Saccharomyces cerevisiae*, named after yeast mating types switching (SWI) and sucrose nonfermenting (SNF) pathways (Workman and Kingston (1998) *Annu Rev Biochem.* 67:545-579; Sudarsanam and Winston (2000) *Trends Genet.* 16:345-351). It is a group of proteins comprising, at least, SWI1, SWI2/SNF2, SWI3, SWI5, and SWI6, as well as other polypeptides (Pazin and Kadonaga (1997) *Cell* 88:737-740). A genetic screening for suppressive mutations of the SWI/SNF phenotypes identified different histones and chromatin components, indicating that these proteins were possibly involved in histone binding and chromatin organization (Winston and Carlson (1992) *Trends Genet.* 8:387-391). Biochemical purification of the SWI/SNF2p in *S. cerevisiae* demonstrated that this protein was part of a complex containing an additional 11 polypeptides, with a combined molecular weight over 1.5 MDa. The SWI/SNF complex contains the ATPase Swi2/Snf2p, two actin-related proteins (Arp7p and Arp9) and other subunits involved in DNA and protein-protein interactions. The purified SWI/SNF complex was able to alter the nucleosome structure in an ATP-dependent manner (Workman and Kingston (1998), supra; Vignali et al. (2000) *Mol Cell Biol.* 20:1899-1910). The structures of the SWI/SNF and RSC complexes are highly conserved but not identical, reflecting an increasing complexity of chromatin (e.g., an increased genome size, the presence of DNA methylation, and more complex genetic organization) through evolution. For this reason, the SWI/SNF complex in higher eukaryotes maintains core components, but also substitute or add on other components with more specialized or tissue-specific domains. Yeast contains two distinct and similar remodeling complexes, SWI/SNF and RSC (Remodeling the Structure of Chromatin). In *Drosophila*, the two complexes are called BAP (Brahma Associated Protein) and PBAP (Polybromo-associated BAP) complexes. The human analogs are BAF (Brgl Associated Factors, or SWI/SNF-A) and PBAF (Polybromo-associated BAF, or SWI/SNF-B). The BAF complex comprises, at least, BAF250A (ARID1A), BAF250B (ARID1B), BAF57 (SMARCE1), BAF190/BRM (SMARCA2), BAF47 (SMARCB1), BAF53A (ACTL6A), BRG1/BAF190 (SMARCA4), BAF155 (SMARCC1), and BAF170 (SMARCC2). The PBAF complex comprises, at last, BAF200 (ARID2), BAF180 (PBRM1), BRD7, BAF45A (PHF10), BRG1/BAF190 (SMARCA4), BAF155 (SMARCC1), and BAF170 (SMARCC2). As in *Drosophila*, human BAF and PBAF share the different core components BAF47, BAF57, BAF60, BAF155, BAF170, BAF45 and the two actins b-Actin and BAF53 (Mohrmann and Verrijzer (2005) *Biochim Biophys Acta.* 1681:59-73). The central core of the BAF and PBAF is the ATPase catalytic subunit BRG1/hBRM, which contains multiple domains to bind to other protein subunits and acetylated histones. For a summary of different complex subunits and their domain structure, see Tang et al. (2010) *Prog Biophys Mol Biol.* 102:122-128 (e.g., FIG. **3**), Hohmann and Vakoc (2014) *Trends Genet.* 30:356-363 (e.g., FIG. **1**), and Kadoch and Crabtree (2015) *Sci. Adv.* 1:e1500447. For chromatin remodeling, the SWI/SNF complex use the energy of ATP hydrolysis to slide the DNA around the nucleosome. The first step consists in the binding between the remodeler and the nucleosome. This binding occurs with nanomolar affinity and reduces the digestion of nucleosomal DNA by nucleases. The 3-D structure of the yeast RSC complex was first solved and imaged using negative stain electron microscopy (Asturias et al. (2002) *Proc Natl Acad Sci USA* 99:13477-13480). The first Cryo-EM structure of the yeast SWI/SNF complex was published in 2008 (Dechassa et al. 2008). DNA footprinting data showed that the SWI/SNF complex makes close contacts with only one gyre of nucleosomal DNA. Protein crosslinking showed that the ATPase SWI2/SNF2p and Swi5p (the homologue of Ini1p in human), Snf6, Swi29, Snf11 and Sw82p (not conserved in human) make close contact with the histones. Several individual SWI/SNF subunits are encoded by gene families, whose protein products are mutually exclusive in the complex (Wu et al. (2009) *Cell* 136:200-206). Thus, only one paralog is incorporated in a given SWI/SNF assembly. The only exceptions are BAF155 and BAF170, which are always present in the complex as homo- or hetero-dimers.

[0047] Combinatorial association of SWI/SNF subunits could in principle give rise to hundreds of distinct complexes, although the exact number has yet to be determined (Wu et al. (2009), supra). Genetic evidence indicates that distinct subunit configurations of SWI/SNF are equipped to perform specialized functions. As an example, SWI/SNF contains one of two ATPase subunits, BRG1 or BRM/SMARCA2, which share 75% amino acid sequence identity (Khavari et al. (1993) *Nature* 366:170-174). While in certain cell types BRG1 and BRM can compensate for loss of the other subunit, in other contexts these two ATPases perform divergent functions (Strobeck et al. (2002) *J Biol Chem.* 277:4782-4789; Hoffman et al. (2014) *Proc Natl Acad Sci USA.* 111:3128-3133). In some cell types, BRG1 and BRM can even functionally oppose one another to regulate differentiation (Flowers et al. (2009) *J Biol Chem.* 284:10067-10075). The functional specificity of BRG1 and BRM has been linked to sequence variations near their N-terminus, which have different interaction specificities for transcription factors (Kadam and Emerson (2003) *Mol Cell.* 11:377-389). Another example of paralogous subunits that form mutually exclusive SWI/SNF complexes are ARID1A/BAF250A, ARID1B/BAF250B, and ARID2/BAF200. ARID1A and ARID1B share 60% sequence identity, but yet can perform opposing functions in regulating the cell cycle, with MYC being an important downstream target of each paralog (Nagl et al. (2007) *EMBO J.* 26:752-763). ARID2 has diverged considerably from ARID1A/ARID1B and exists in a unique SWI/SNF assembly known as PBAF (or SWI/SNF-B), which contains several unique subunits not found in ARID1A/B-containing complexes. The composition of SWI/SNF can also be dynamically reconfigured during cell fate transitions through cell type-specific expression patterns of certain subunits. For example, BAF53A/ACTL6A is repressed and replaced by BAF53B/ACTL6B during neuronal differentiation, a switch that is essential for proper neuronal functions in vivo (Lessard et al. (2007) *Neuron* 55:201-215). These studies stress that SWI/SNF in fact represents a collection of multi-subunit complexes whose integrated functions control diverse cellular processes, which is also incorporated in the scope of definitions of the instant disclosure. Two recently published meta-analyses of cancer genome sequencing data estimate that nearly 20% of human cancers harbor mutations in one (or more) of the genes encoding SWI/SNF (Kadoch et al. (2013) *Nat Genet.* 45:592-601; Shain and Pollack (2013) *PLoS One.* 8:e55119). Such mutations are generally loss-of-function, implicating SWI/SNF as a major tumor suppressor in diverse cancers. Specific SWI/SNF gene mutations are generally linked to a specific subset of cancer lineages: SNF5 is mutated in malignant rhabdoid tumors (MRT), PBRM1/BAF180 is frequently inactivated in renal carcinoma, and BRG1 is mutated in non-small cell lung cancer (NSCLC) and several other cancers. In the instant disclosure, the scope of “SWI/SNF complex” may cover at least one fraction or the whole complex (e.g., some or all subunit proteins/other components), either in the human BAF/PBAF forms or their homologs/orthologs in other species (e.g., the yeast and *drosophila* forms described herein). Preferably, a “SWI/SNF complex” described herein contains at least part of the full complex bio-functionality, such as binding to other subunits/components, binding to DNA/histone, catalyzing ATP, promoting chromatin remodeling, etc.

[0048] The term “BAF complex” refers to at least one type of mammalian SWI/SNF complexes. Its nucleosome remodeling activity can be reconstituted with a set of four core subunits (BRG1/SMARCA4, SNF5/SMARCB1, BAF155/SMARCC1, and BAF170/SMARCC2), which have orthologs in the yeast complex (Phelan et al. (1999) *Mol Cell.* 3:247-253). However, mammalian SWI/SNF contains several subunits not found in the yeast counterpart, which can provide interaction surfaces for chromatin (e.g. acetyl-lysine recognition by bromodomains) or transcription factors and thus contribute to the genomic targeting of the complex (Wang et al. (1996) *EMBO J.* 15:5370-5382; Wang et al. (1996) *Genes Dev.* 10:2117-2130; Nie et al. (2000)). A key attribute of mammalian SWI/SNF is the heterogeneity of subunit configurations that can exist in different tissues and even in a single cell type (e.g., as BAF, PBAF, neural progenitor BAF (npBAF), neuron BAF (nBAF), embryonic stem cell BAF (esBAF), etc.). In some embodiments, the BAF complex described herein refers to one type of mammalian SWI/SNF complexes, which is different from PBAF complexes.

[0049] The term “PBAF complex” refers to one type of mammalian SWI/SNF complexes originally known as SWI/SNF-B. It is highly related to the BAF complex and can be separated with conventional chromatographic approaches. For example, human BAF and PBAF complexes share multiple identical subunits (such as BRG, BAF170, BAF155, BAF60, BAF57, BAF53, BAF45, actin, SS18, and hSNF5/INI1). However, while BAF contains BAF250 subunit, PBAF contains BAF180 and BAF200, instead (Lemon et al. (2001) *Nature* 414:924-998; Yan et al. (2005) *Genes Dev.* 19:1662-1667). Moreover, they do have selectivity in regulating interferon-responsive genes (Yan et al. (2005), supra, showing that BAF200, but not BAF180, is required for PBAF to mediate expression of IFITMI gene induced by IFN-α, while the IFITM3 gene expression is dependent on BAF but not PBAF). Due to these differences, PBAF, but not BAF, was able to activate vitamin D receptor-dependent transcription on a chromatinized template in vitro (Lemon et al. (2001), supra). The 3-D structure of human PBAF complex preserved in negative stain was found to be similar to yeast RSC but dramatically different from yeast SWI/SNF (Leschziner et al. (2005) *Structure* 13:267-275).

[0050] The term “BRG” or “BRG1/BAF190 (SMARCA4)” refers to a subunit of the SWI/SNF complex, which can be find in either BAF or PBAF complex. It is an ATP-depedendent helicase and a transcription activator, encoded by the SMARCA4 gene. BRG1 can also bind BRCA1, as well as regulate the expression of the tumorigenic protein CD44.

[0051] BRG1 is important for development past the pre-implantation stage. Without having a functional BRG1, exhibited with knockout research, the embryo will not hatch out of the zona pellucida, which will inhibit implantation from occurring on the endometrium (uterine wall). BRG1 is also crucial to the development of sperm. During the first stages of meiosis in spermatogenesis there are high levels of BRG1. When BRG1 is genetically damaged, meiosis is stopped in prophase 1, hindering the development of sperm and would result in infertility. More knockout research has concluded BRG1's aid in the development of smooth muscle. In a BRG1 knockout, smooth muscle in the gastrointestinal tract lacks contractility, and intestines are incomplete in some cases. Another defect occurring in knocking out BRG1 in smooth muscle development is heart complications such as an open ductus arteriosus after birth (Kim et al. (2012) *Development* 139:1133-1140; Zhang et al. (2011) *Mol. Cell. Biol.* 31:2618-2631). Mutations in SMARCA4 were first recognized in human lung cancer cell lines (Medina et al. (2008) *Hum. Mut.* 29:617-622). Later it was recognized that mutations exist in a significant frequency of medulloblastoma and pancreatic cancers among other tumor subtypes (Jones et al. (2012) *Nature* 488:100-105; Shain et al. (2012) *Proc Natl Acad Sci USA* 109:E252-E259; Shain and Pollack (2013), supra). Mutations in BRG1 (or SMARCA4) appear to be mutually exclusive with the presence of activation at any of the MYC-genes, which indicates that the BRG1 and MYC proteins are functionally related. Another recent study demonstrated a causal role of BRG1 in the control of retinoic acid and glucocorticoid-induced cell differentiation in lung cancer and in other tumor types. This enables the cancer cell to sustain undifferentiated gene expression programs that affect the control of key cellular processes. Furthermore, it explains why lung cancer and other solid tumors are completely refractory to treatments based on these compounds that are effective therapies for some types of leukemia (Romero et al. (2012) *EMBO Mol. Med.* 4:603-616). The role of BRG1 in sensitivity or resistance to anti-cancer drugs had been recently highlighted by the elucidation of the mechanisms of action of darinaparsin, an arsenic-based anti-cancer drugs. Darinaparsin has been shown to induce phosphorylation of BRG1, which leads to its exclusion from the chromatin. When excluded from the chromatin, BRG1 can no longer act as a transcriptional co-regulator.

[0052] This leads to the inability of cells to express HO-1, a cytoprotective enzyme. BRG1 has been shown to interact with proteins such as ACTL6A, ARID1A, ARID1B, BRCA1, CTNNB1, CBX5, CREBBP, CCNE1, ESR1, FANCA, HSP90B1, ING1, Myc, NR3C1, P53, POLR2A, PHB, SIN3A, SMARCB1, SMARCC1, SMARCC2, SMARCE1, STAT2, STK11, etc.

[0053] The term “BRG” or “BRG1/BAF190 (SMARCA4)” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human BRG1(SMARCA4) cDNA and human BRG1 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, seven different human BRG1 isoforms are known. Human BRG1 isoform A (NP_001122321.1) is encodable by the transcript variant 1 (NM_001128849.1), which is the longest transcript. Human BRG1 isoform B (NP_001122316.1 or NP_003063.2) is encodable by the transcript variant 2 (NM_001128844.1), which differs in the 5′ UTR and lacks an alternate exon in the 3′ coding region, compared to the variant 1, and also by the transcript variant 3 (NM_003072.3), which lacks an alternate exon in the 3′ coding region compared to variant 1. Human BRG1 isoform C (NP_001122317.1) is encodable by the transcript variant 4 (NM_001128845.1), which lacks two alternate in-frame exons and uses an alternate splice site in the 3′ coding region, compared to variant 1. Human BRG1 isoform D (NP_001122318.1) is encodable by the transcript variant 5 (NM_001128846.1), which lacks two alternate in-frame exons and uses two alternate splice sites in the 3′ coding region, compared to variant 1. Human BRG1 isoform E (NP_001122319.1) is encodable by the transcript variant 6 (NM_001128847.1), which lacks two alternate in-frame exons in the 3′ coding region, compared to variant 1. Human BRG1 isoform F (NP_001122320.1) is encodable by the transcript variant 7 (NM_001128848.1), which lacks two alternate in-frame exons and uses an alternate splice site in the 3′ coding region, compared to variant 1. Nucleic acid and polypeptide sequences of BRG1 orthologs in organisms other than humans are well known and include, for example, chimpanzee BRG1 (XM_016935029.1 and XP_016790518.1, XM_016935038.1 and XP_016790527.1, XM_016935039.1 and XP_016790528.1, XM_016935036.1 and XP_016790525.1, XM_016935037.1 and XP_016790526.1, XM_016935041.1 and XP_016790530.1, XM_016935040.1 and XP_016790529.1, XM_016935042.1 and XP_016790531.1, XM_016935043.1 and XP_016790532.1, XM_016935035.1 and XP_016790524.1, XM_016935032.1 and XP_016790521.1, XM_016935033.1 and XP_016790522.1, XM_016935030.1 and XP_016790519.1, XM_016935031.1 and XP_016790520.1, and XM_016935034.1 and XP_016790523.1), Rhesus monkey BRG1 (XM_015122901.1 and XP_014978387.1, XM_015122902.1 and XP_014978388.1, XM_015122903.1 and XP_014978389.1, XM_015122906.1 and XP_014978392.1, XM_015122905.1 and XP_014978391.1, XM_015122904.1 and XP_014978390.1, XM_015122907.1 and XP_014978393.1, XM_015122909.1 and XP_014978395.1, and XM_015122910.1 and XP_014978396.1), dog BRG1 (XM_014122046.1 and XP_013977521.1, XM_014122043.1 and XP_013977518.1, XM_014122042.1 and

XP_013977517.1, XM_014122041.1 and XP_013977516.1, XM_014122045.1 and XP_013977520.1, and XM_014122044.1 and XP_013977519.1), cattle BRG1 (NM_001105614.1 and NP_001099084.1), rat BRG1 (NM_134368.1 and NP_599195.1).

[0054] Anti-BRG1 antibodies suitable for detecting BRG1 protein are well-known in the art and include, for example, MABE1118, MABE121, MABE60, and 07-478 (poly- and mono-clonal antibodies from EMD Millipore, Billerica, MA), AM26021PU-N, AP23972PU-N, TA322909, TA322910, TA327280, TA347049, TA347050, TA347851, and TA349038 (antibodies from OriGene Technologies, Rockville, MD), NB100-2594, AF5738, NBP2-22234, NBP2-41270, NBP1-51230, and NBP1-40379 (antibodes from Novus Biologicals, Littleton, CO), ab110641, ab4081, ab215998, ab108318, ab70558, ab118558, ab133257, ab92496, ab196535, and ab196315 (antibodies from AbCam, Cambridge, MA), Cat #: 720129, 730011, 730051, MA1-10062, PA5-17003, and PA5-17008 (antibodies from ThermoFisher Scientific, Waltham, MA), GTX633391, GTX32478, GTX31917, GTX16472, and GTX50842 (antibodies from GeneTex, Irvine, CA), antibody 7749 (ProSci, Poway, CA), Brg-1 (N-15), Brg-1 (N-15) X, Brg-1 (H-88), Brg-1 (H-88) X, Brg-1 (P-18), Brg-1 (P-18) X, Brg-1 (G-7), Brg-1 (G-7) X, Brg-1 (H-10), and Brg-1 (H-10) X (antibodies from Santa Cruz Biotechnology, Dallas, TX), antibody of Cat. AF5738 (R&D Systmes, Minneapolis, MN), etc. In addition, reagents are well-known for detecting BRG1 expression. Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing BRG1 Expression can be found in the commercial product lists of the above-referenced companies. PFI 3 is a known small molecule inhibitor of polybromo 1 and BRG1 (e.g., Cat. B7744 from APExBIO, Houston, TX). It is to be noted that the term can further be used to refer to any combination of features described herein regarding BRG1 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe an BRG1 molecule of the present invention.

[0055] The term “BRM” or “BRM/BAF190 (SMARCA2)” refers to a subunit of the SWI/SNF complex, which can be found in either BAF or PBAF complexes. It is an ATP-dependent helicase and a transcription activator, encoded by the SMARCA2 gene. The catalytic core of the SWI/SNF complex can be either of two closely related ATPases, BRM or BRG1, with the potential that the choice of alternative subunits is a key determinant of specificity. Instead of impeding differentiation as was seen with BRG1 depletion, depletion of BRM caused accelerated progression to the differentiation phenotype. BRM was found to regulate genes different from those as BRG1 targets and be capable of overriding BRG1-dependent activation of the osteocalcin promoter, due to its interaction with different ARID family members (Flowers et al. (2009), supra). The known binding partners for BRM include, for example, ACTL6A, ARID1B, CEBPB, POLR2A, Prohibitin, SIN3A, SMARCB1, and SMARCC1.

[0056] The term “BRM” or “BRM/BAF190 (SMARCA2)” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human BRM (SMARCA2) cDNA and human BRM protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, seven different human BRM isoforms are known. Human BRM isoform A (NP_003061.3 or NP_001276325.1) is encodable by the transcript variant 1 (NM_003070.4), which is the longest transcript, or the transcript variant 3 (NM_001289396.1), which differs in the 5′ UTR, compared to variant 1. Human BRM isoform B (NP_620614.2) is encodable by the transcript variant 2 (NM_139045.3), which lacks an alternate in-frame exon in the coding region, compared to variant 1. Human BRM isoform C (NP_001276326.1) is encodable by the transcript variant 4 (NM_001289397.1), which uses an alternate in-frame splice site and lacks an alternate in-frame exon in the 3′ coding region, compared to variant 1. Human BRM isoform D (NP_001276327.1) is encodable by the transcript variant 5 (NM_001289398.1), which differs in the 5′ UTR, lacks a portion of the 5′ coding region, and initiates translation at an alternate downstream start codon, compared to variant 1. Human BRM isoform E (NP_001276328.1) is encodable by the transcript variant 6 (NM_001289399.1), which differs in the 5′ UTR, lacks a portion of the 5′ coding region, and initiates translation at an alternate downstream start codon, compared to variant 1. Human BRM isoform F (NP_001276329.1) is encodable by the transcript variant 7 (NM_001289400.1), which differs in the 5′ UTR, lacks a portion of the 5′ coding region, and initiates translation at an alternate downstream start codon, compared to variant 1. Nucleic acid and polypeptide sequences of BRM orthologs in organisms other than humans are well known and include, for example, chimpanzee BRM (XM_016960529.1 and XP_016816018.1), dog BRG1 (XM_005615906.2 and XP_005615963.1, XM_845066.4 and XP_850159.1, XM_005615905.2 and XP_005615962.1, XM_005615904.2 and XP_005615961.1, XM_005615903.2 and XP_005615960.1, and XM_005615902.2 and XP_005615959.1), cattle BRM (NM_001099115.2 and NP_001092585.1), rat BRM (NM_001004446.1 and NP_001004446.1).

[0057] Anti-BRM antibodies suitable for detecting BRM protein are well-known in the art and include, for example, antibody MABE89 (EMD Millipore, Billerica, MA), antibody TA351725 (OriGene Technologies, Rockville, MD), NBP1-90015, NBP1-80042, NB100-55308, NB100-55309, NB100-55307, and H00006595-M06 (antibodes from Novus Biologicals, Littleton, CO), ab15597, ab12165, ab58188, and ab200480 (antibodies from AbCam, Cambridge, MA), Cat #: 11966 and 6889 (antibodies from Cell Signaling, Danvers, MA), etc. In addition, reagents are well-known for detecting BRM expression. Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing BRM Expression can be found in the commercial product lists of the above-referenced companies. For example, BRM

RNAi product H00006595-R02 (Novus Biologicals), CRISPER gRNA products from GenScript, Piscataway, NJ, and other inhibitory RNA products from Origene, ViGene Biosciences (Rockville, MD), and Santa Cruz. It is to be noted that the term can further be used to refer to any combination of features described herein regarding BRM molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe an BRM molecule of the present invention.

[0058] The term "BAF250A" or "ARID1A" refers to AT-rich interactive domain-containing protein 1A, a subunit of the SWI/SNF complex, which can be find in BAF but not PBAF complex. In humans there are two BAF250 isoforms, BAF250A/ARID1A and BAF250B/ARID1B. They are thought to be E3 ubiquitin ligases that target histone H2B (Li et al. (2010) *Mol. Cell. Biol.* 30:1673-1688). ARID1A is highly expressed in the spleen, thymus, prostate, testes, ovaries, small intestine, colon and peripheral leukocytes. ARID1A is involved in transcriptional activation and repression of select genes by chromatin remodeling. It is also involved in vitamin D-coupled transcription regulation by associating with the WINAC complex, a chromatin-remodeling complex recruited by vitamin D receptor. ARID1A belongs to the neural progenitors-specific chromatin remodeling (npBAF) and the neuron-specific chromatin remodeling (nBAF) complexes, which are involved in switching developing neurons from stem/progenitors to post-mitotic chromatin remodeling as they exit the cell cycle and become committed to their adult state. ARID1A also plays key roles in maintaining embryonic stem cell pluripotency and in cardiac development and function (Lei et al. (2012) *J. Biol. Chem.* 287:24255-24262; Gao et al. (2008) *Proc. Natl. Acad. Sci. U.S.A.* 105:6656-6661). Loss of BAF250a expression was seen in 42% of the ovarian clear cell carcinoma samples and 21% of the endometrioid carcinoma samples, compared with just 1% of the high-grade serous carcinoma samples. ARID1A deficiency also impairs the DNA damage checkpoint and sensitizes cells to PARP inhibitors (Shen et al. (2015) *Cancer Discov.* 5:752-767). Human ARID1A protein has 2285 amino acids and a molecular mass of 242045 Da, with at least a DNA-binding domain that can specifically bind an AT-rich DNA sequence, recognized by a SWI/SNF complex at the beta-globin locus, and a C-terminus domain for glucocorticoid receptor-dependent transcriptional activation. ARID1A has been shown to interact with proteins such as SMARCB1/BAF47 (Kato et al. (2002) *J. Biol. Chem.* 277:5498-505; Wang et al. (1996) *EMBO J.* 15:5370-5382) and SMARCA4/BRG1 (Wang et al. (1996), supra; Zhao et al. (1998) *Cell* 95:625-636), etc.

[0059] The term "BAF250A" or "ARID1A" is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human BAF250A (ARID1A) cDNA and human BAF250A (ARID1A) protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human ARID1A isoforms are known. Human ARID1A isoform A (NP_006006.3) is encodable by the transcript variant 1 (NM_006015.4), which is the longer transcript. Human ARID1A isoform B (NP_624361.1) is encodable by the transcript variant 2 (NM_139135.2), which lacks a segment in the coding region compared to variant 1. Isoform B thus lacks an internal segment, compared to isoform A. Nucleic acid and polypeptide sequences of ARID1A orthologs in organisms other than humans are well known and include, for example, chimpanzee ARID1A (XM_016956953.1 and XP_016812442.1, XM_016956958.1 and XP_016812447.1, and XM_009451423.2 and XP_009449698.2), Rhesus monkey ARID1A (XM_015132119.1 and XP_014987605.1, and XM_015132127.1 and XP_014987613.1), dog ARID1A (XM_847453.5 and XP_852546.3, XM_005617743.2 and XP_005617800.1, XM_005617742.2 and XP_005617799.1, XM_005617744.2 and XP_005617801.1, XM_005617746.2 and XP_005617803.1, and XM_005617745.2 and XP_005617802.1), cattle ARID1A (NM_001205785.1 and NP_001192714.1), rat ARID1A (NM_001106635.1 and NP_001100105.1).

[0060] Anti-ARID1A antibodies suitable for detecting ARID1A protein are well-known in the art and include, for example, antibody Cat #04-080 (EMD Millipore, Billerica, MA), antibodies TA349170, TA350870, and TA350871 (OriGene Technologies, Rockville, MD), antibodies NBP1-88932, NB100-55334, NBP2-43566, NB100-55333, and H00008289-QO1 (Novus Biologicals, Littleton, CO), antibodies ab182560, ab182561, ab176395, and ab97995 (AbCam, Cambridge, MA), antibodies Cat #: 12354 and 12854 (Cell Signaling Technology, Danvers, MA), antibodies GTX129433, GTX129432, GTX632013, GTX12388, and GTX31619 (GeneTex, Irvine, CA), etc. In addition, reagents are well-known for detecting ARID1A expression. For example, multiple clinical tests for ARID1A are available at NIH Genetic Testing Registry (GTR®) (e.g., GTR Test ID: GTR000520952.1 for mental retardation, offered by Centogene AG, Germany). Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing ARID1A Expression can be found in the commercial product lists of the above-referenced companies, such as RNAi products H00008289-RO1, H00008289-R02, and H00008289-R03 (Novus Biologicals) and CRISPR products KN301547G1 and KN301547G2 (Origene). Other CRISPR products include sc-400469 (Santa Cruz Biotechnology) and those from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding ARID1A molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe an ARID1A molecule of the present invention.

[0061] The term "loss-of-function mutation" for BAF250A/ARID1A refers to any mutation in an ARID1A-related nucleic acid or protein that results in reduced or eliminated ARID1A protein amounts and/or function. For example, nucleic acid mutations include single-base substitutions, multi-base substitutions, insertion mutations, deletion mutations, frameshift mutations, missesnse mutations, nonsense mutations, splice-site mutations, epigenetic modifications (e.g., methylation, phosphorylation, acetylation, ubiquitylation, sumoylation, histone acetylation, histone deacetylation, and the like), and combinations thereof. In some embodiments, the mutation is a "nonsynonymous mutation," meaning that the mutation alters the amino acid sequence of ARID1A. Such mutations reduce or eliminate ARID1A protein amounts and/or function by eliminating proper coding sequences required for proper ARID1A protein translation and/or coding for ARID1A proteins that are non-functional or have reduced function (e.g., deletion of enzymatic and/or structural domains, reduction in protein stability, alteration of sub-cellular localization, and the like). Such mutations are well-known in the art. In addition, a representative list describing a wide variety of structural mutations correlated with the functional result of reduced or eliminated ARID1A protein amounts and/or function is described in the Tables and the Examples.

[0062] The term "BAF250B" or "ARID1B" refers to AT-rich interactive domain-containing protein 1B, a subunit of the SWI/SNF complex, which can be find in BAF but not PBAF complex. ARID1B and ARID1A are alternative and mutually exclusive ARID-subunits of the SWI/SNF complex. Germline mutations in ARID1B are associated with Coffin-Siris syndrome (Tsurusaki et al. (2012) *Nat. Genet.* 44:376-378; Santen et al. (2012) *Nat. Genet.* 44:379-380). Somatic mutations in ARID1B are associated with several cancer subtypes, indicating that it is a tumor suppressor gene (Shai and Pollack (2013) *PLoS ONE* 8:e55119; Sausen et al. (2013) *Nat. Genet.* 45:12-17; Shain et al. (2012) *Proc. Natl. Acad. Sci. U.S.A.* 109:E252-E259; Fujimoto et al. (2012) *Nat. Genet.* 44:760-764). Human ARID1A protein has 2236 amino acids and a molecular mass of 236123 Da, with at least a DNA-binding domain that can specifically bind an AT-rich DNA sequence, recognized by a SWI/SNF complex at the beta-globin locus, and a C-terminus domain for glucocorticoid receptor-dependent transcriptional activation. ARID1B has been shown to interact with SMARCA4/BRG1 (Hurlstone et al. (2002) *Biochem. J.* 364:255-264; Inoue et al. (2002) *J. Biol. Chem.* 277:41674-41685 and SMARCA2/BRM (Inoue et al. (2002), supra).

[0063] The term "BAF250B" or "ARID1B" is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human BAF250B (ARID1B) cDNA and human BAF250B (ARID1B) protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, three different human ARID1B isoforms are known. Human ARID1B isoform A (NP_059989.2) is encodable by the transcript variant 1 (NM_017519.2). Human ARID1B isoform B (NP_065783.3) is encodable by the transcript variant 2 (NM_020732.3). Human ARID1B isoform C (NP_001333742.1) is encodable by the transcript variant 3 (NM_001346813.1). Nucleic acid and polypeptide sequences of ARID1B orthologs in organisms other than humans are well known and include, for example, Rhesus monkey ARID1B (XM_015137088.1 and XP_014992574.1), dog ARID1B (XM_014112912.1 and XP_013968387.1), cattle ARID1B (XM_010808714.2 and XP_010807016.1, and XM_015464874.1 and XP_015320360.1), rat ARID1B (XM_017604567.1 and XP_017460056.1).

[0064] Anti-ARID1B antibodies suitable for detecting ARID1B protein are well-known in the art and include, for example, antibody Cat #ABE316 (EMD Millipore, Billerica, MA), antibody TA315663 (OriGene Technologies, Rockville, MD), antibodies H00057492-M02, H00057492-MO1, NB100-57485, NBP1-89358, and NB100-57484 (Novus Biologicals, Littleton, CO), antibodies ab57461, ab69571, ab84461, and ab163568 (AbCam, Cambridge, MA), antibodies Cat #: PA5-38739, PA5-49852, and PA5-50918 (ThermoFisher Scientific, Danvers, MA), antibodies GTX130708, GTX60275, and GTX56037 (GeneTex, Irvine, CA), ARID1B (KMN1) Antibody and other antibodies (Santa Cruz Biotechnology), etc. In addition, reagents are well-known for detecting ARID1B expression. For example, multiple clinical tests for ARID1B are available at NIH Genetic Testing Registry (GTR®) (e.g., GTR Test ID: GTR000520953.1 for mental retardation, offered by Centogene AG, Germany). Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing ARID1B Expression can be found in the commercial product lists of the above-referenced companies, such as RNAi products H00057492-R03, H00057492-RO1, and H00057492-R02 (Novus Biologicals) and CRISPR products KN301548 and KN214830 (Origene). Other CRISPR products include sc-402365 (Santa Cruz Biotechnology) and those from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding ARID1B molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe an ARID1B molecule of the present invention.

[0065] The term "loss-of-function mutation" for BAF250B/ARID1B refers to any mutation in an ARID1B-related nucleic acid or protein that results in reduced or eliminated ARID1B protein amounts and/or function. For example, nucleic acid mutations include single-base substitutions, multi-base substitutions, insertion mutations, deletion mutations, frameshift mutations, missesnse mutations, nonsense mutations, splice-site mutations, epigenetic modifications (e.g., methylation, phosphorylation, acetylation, ubiquitylation, sumoylation, histone acetylation, histone deacetylation, and the like), and combinations thereof. In some embodiments, the mutation is a "nonsynonymous mutation," meaning that the mutation alters the amino acid sequence of ARID1B. Such mutations reduce or eliminate ARID1B protein amounts and/or function by eliminating proper coding sequences required for proper ARID1B protein translation and/or coding for ARID1B proteins that are non-functional or have reduced function (e.g., deletion of enzymatic and/or structural domains, reduction in protein stability, alteration of sub-cellular localization, and the like). Such mutations are well-known in the art. In addition, a representative list describing a wide variety of structural mutations correlated with the functional result of reduced or eliminated ARID1B protein amounts and/or function is described in the Tables and the Examples.

[0066] The term “SMARCC1” refers to SWI/SNF related, matrix associated, actin dependent regulator of chromatin subfamily c member 1. SMARCC1 is a member of the SWI/SNF family of proteins, whose members display helicase and ATPase activities and which are thought to regulate transcription of certain genes by altering the chromatin structure around those genes. The encoded protein is part of the large ATP-dependent chromatin remodeling complex SNF/SWI and contains a predicted leucine zipper motif typical of many transcription factors. SMARCC1 is a component of SWI/SNF chromatin remodeling complexes that carry out key enzymatic activities, changing chromatin structure by altering DNA-histone contacts within a nucleosome in an ATP-dependent manner. SMARCC1 stimulates the ATPase activity of the catalytic subunit of the complex (Phelan et al. (1999) *Mol Cell* 3:247-253). SMARCC1 belongs to the neural progenitors-specific chromatin remodeling complex (npBAF complex) and the neuron-specific chromatin remodeling complex (nBAF complex). During neural development a switch from a stem/progenitor to a postmitotic chromatin remodeling mechanism occurs as neurons exit the cell cycle and become committed to their adult state. The transition from proliferating neural stem/progenitor cells to postmitotic neurons requires a switch in subunit composition of the npBAF and nBAF complexes. As neural progenitors exit mitosis and differentiate into neurons, npBAF complexes which contain ACTL6A/BAF53A and PHF10/BAF45A, are exchanged for homologous alternative ACTL6B/BAF53B and DPF1/BAF45B or DPF3/BAF45C subunits in neuron-specific complexes (nBAF). The npBAF complex is essential for the self-renewal/proliferative capacity of the multipotent neural stem cells. The nBAF complex along with CREST plays a role regulating the activity of genes essential for dendrite growth. Human SMARCC1 protein has 1105 amino acids and a molecular mass of 122867 Da. Binding partners of SMARCC1 include, e.g., NR3C1, SMARD1, TRIP12, CEBPB, KDM6B, and MKKS.

[0067] The term “SMARCC1” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human SMARCC1 cDNA and human SMARCC1 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, human SMARCC1 protein (NP_003065.3) is encodable by the transcript (NM_003074.3). Nucleic acid and polypeptide sequences of SMARCC1 orthologs in organisms other than humans are well known and include, for example, chimpanzee SMARCC1 (XM_016940956.2 and XP_016796445.1, XM_001154676.6 and XP_001154676.1, XM_016940957.1 and XP_016796446.1, and XM_009445383.3 and XP_009443658.1), Rhesus monkey SMARCC1 (XM_015126104.1 and XP_014981590.1, XM_015126103.1 and XP_014981589.1, XM_001083389.3 and XP_001083389.2, and XM_015126105.1 and XP_014981591.1), dog SMARCC1 (XM_533845.6 and XP_533845.2, XM_014122183.2 and XP_013977658.1, and XM_014122184.2 and XP_013977659.1), cattle SMARCC1 (XM_024983285.1 and XP_024839053.1), mouse SMARCC1 (NM_009211.2 and NP_033237.2), rat SMARCC1 (NM_001106861.1 and NP_001100331.1), chicken SMARCC1 (XM_025147375.1 and XP_025003143.1, and XM_015281170.2 and XP_015136656.2), tropical clawed frog SMARCC1 (XM_002942718.4 and XP_002942764.2), and zebrafish SMARCC1 (XM_003200246.5 and XP_003200294.1, and XM_005158282.4 and XP_005158339.1). Representative sequences of SMARCC1 orthologs are presented below in Table 1.

[0068] Anti-SMARCC1 antibodies suitable for detecting SMARCC1 protein are well-known in the art and include, for example, antibody TA334040 (Origene), antibodies NBP1-88720, NBP2-20415, NBP1-88721, and NB100-55312 (Novus Biologicals, Littleton, CO), antibodies ab172638, ab126180, and ab22355 (AbCam, Cambridge, MA), antibody Cat #PA5-30174 (ThermoFisher Scientific), antibody Cat #27-825 (ProSci, Poway, CA), etc. In addition, reagents are well-known for detecting SMARCC1. A clinical test of SMARCC1 for hereditary disese is available with the test ID no. GTR000558444.1 in NIH Genetic Testing Registry (GTR®), offered by Tempus Labs, Inc., (Chicago, IL). Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing SMARCC1 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-29780 and sc-29781 and CRISPR product #sc-400838 from Santa Cruz Biotechnology, RNAi products SR304474 and TL309245V, and CRISPR product KN208534 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding SMARCC1 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a SMARCC1 molecule encompassed by the present invention.

[0069] The term “SMARCC2” refers to SWI/SNF related, matrix associated, actin dependent regulator of chromatin subfamily c member 2. SMARCC2 is an important paralog of gene SMARCC1. SMARCC2 is a member of the SWI/SNF family of proteins, whose members display helicase and ATPase activities and which are thought to regulate transcription of certain genes by altering the chromatin structure around those genes. The encoded protein is part of the large ATP-dependent chromatin remodeling complex SNF/SWI and contains a predicted leucine zipper motif typical of many transcription factors. SMARCC2 is a component of SWI/SNF chromatin remodeling complexes that carry out key enzymatic activities, changing chromatin structure by altering DNA-histone contacts within a nucleosome in an ATP-dependent manner (Kadam et al. (2000) *Genes Dev* 14:2441-2451). SMARCC2 can stimulate the ATPase activity of the catalytic subunit of the complex (Phelan et al. (1999) *Mol Cell* 3:247-253). SMARCC2 is required for CoREST dependent repression of neuronal specific gene promoters in non-neuronal cells (Battaglioli et al. (2002) *J Biol Chem* 277:41038-41045). SMARCC2 belongs to the neural progenitors-specific chromatin remodeling complex (npBAF complex) and the neuron-specific chromatin remodeling complex (nBAF complex). SMARCC2 is a critical regulator of myeloid differentiation, controlling granulocytopoiesis and the expression of genes involved in neutrophil granule formation. Human SMARCC2 protein has 1214 amino acids and a molecular mass of 132879 Da. Binding partners of SMARCC2 include, e.g., SIN3A, SMARD1, KDM6B, and RCOR1.

[0070] The term “SMARCC2” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human SMARCC2 cDNA (NM_003074.3) and human SMARCC2 protein sequences (NP_003065.3) are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, four different human SMARCC2 isoforms are known. Human SMARCC2 isoform a (NP_003066.2) is encodable by the transcript variant 1 (NM_003075.4). Human SMARCC2 isoform b (NP_620706.1) is encodable by the transcript variant 2 (NM_139067.3), which contains an alternate in-frame exon in the central coding region and uses an alternate in-frame splice site in the 3′ coding region, compared to variant 1. The encoded isoform (b), contains a novel internal segment, lacks a segment near the C-terminus, and is shorter than isoform a. Human SMARCC2 isoform c (NP_001123892.1) is encodable by the transcript variant 3 (NM_001130420.2), which contains an alternate in-frame exon in the central coding region and contains alternate in-frame segment in the 3′ coding region, compared to variant 1. The encoded isoform (c), contains a novel internal segment, lacks a segment near the C-terminus, and is shorter than isoform a. Human SMARCC2 isoform d (NP_001317217.1) is encodable by the transcript variant 4 (NM_001330288.1), which contains an alternate in-frame exon in the central coding region compared to variant 1. The encoded isoform (d), contains the same N- and C-termini, but is longer than isoform a. Nucleic acid and polypeptide sequences of SMARCC2 orthologs in organisms other than humans are well known and include, for example, chimpanzee SMARCC2 (XM_016923208.2 and XP_016778697.1, XM_016923212.2 and XP_016778701.1, XM_016923214.2 and XP_016778703.1, XM_016923210.2 and XP_016778699.1, XM_016923209.2 and XP_016778698.1, XM_016923213.2 and XP_016778702.1, XM_016923211.2 and XP_016778700.1, and XM_016923216.2 and XP_016778705.1), Rhesus monkey SMARCC2 (XM_015151975.1 and XP_015007461.1, XM_015151976.1 and XP_015007462.1, XM_015151974.1 and XP_015007460.1, XM_015151969.1 and XP_015007455.1, XM_015151972.1 and XP_015007458.1, XM_015151973.1 and XP_015007459.1, and XM_015151970.1 and XP_015007456.1), dog SMARCC2 (XM_022424046.1 and XP_022279754.1, XM_014117150.2 and XP_013972625.1, XM_014117149.2 and XP_013972624.1, XM_005625493.3 and XP_005625550.1, XM_014117151.2 and XP_013972626.1, XM_005625492.3 and XP_005625549.1, XM_005625495.3 and XP_005625552.1, XM_005625494.3 and XP_005625551.1, and XM_022424047.1 and XP_022279755.1), cattle SMARCC2 (NM_001172224.1 and NP_001165695.1), mouse SMARCC1 (NM_001114097.1 and NP_001107569.1, NM_001114096.1 and NP_001107568.1, and NM_198160.2 and NP_937803.1), rat SMARCC2 (XM_002729767.5 and XP_002729813.2, XM_006240805.3 and XP_006240867.1, XM_006240806.3 and XP_006240868.1, XM_001055795.6 and XP_001055795.1, XM_006240807.3 and XP_006240869.1, XM_008765050.2 and XP_008763272.1, XM_017595139.1 and XP_017450628.1, XM_001055673.6 and XP_001055673.1, and XM_001055738.6 and XP_001055738.1), and zebrafish SMARCC2 (XM_021474611.1 and XP_021330286.1). Representative sequences of SMARCC2 orthologs are presented below in Table 1.

[0071] Anti-SMARCC2 antibodies suitable for detecting SMARCC2 protein are well-known in the art and include, for example, antibody TA314552 (Origene), antibodies NBP1-90017 and NBP2-57277 (Novus Biologicals, Littleton, CO), antibodies ab71907, ab84453, and ab64853 (AbCam, Cambridge, MA), antibody Cat #PA5-54351 (ThermoFisher Scientific), etc. In addition, reagents are well-known for detecting SMARCC2. A clinical test of SMARCC2 for hereditary disese is available with the test ID no. GTR000546600.2 in NIH Genetic Testing Registry (GTR®), offered by Fulgent Clinical Diagnostics Lab (Temple City, CA). Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing SMARCC2 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-29782 and sc-29783 and CRISPR product #sc-402023 from Santa Cruz Biotechnology, RNAi products SR304475 and TL301505V, and CRISPR product KN203744 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding SMARCC2 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a SMARCC2 molecule encompassed by the present invention.

[0072] The term “SMARCD1” refers to SWI/SNF related, matrix associated, actin dependent regulator of chromatin subfamily D member 1. SMARCD1 is a member of the SWI/SNF family of proteins, whose members display helicase and ATPase activities and which are thought to regulate transcription of certain genes by altering the chromatin structure around those genes. The encoded protein is part of the large ATP-dependent chromatin remodeling complex SNF/SWI and has sequence similarity to the yeast Swp73 protein. SMARCD1 is a component of SWI/SNF chromatin remodeling complexes that carry out key enzymatic activities, changing chromatin structure by altering DNA-histone contacts within a nucleosome in an ATP-dependent manner (Wang et al. (1996) *Genes Dev* 10:2117-2130). SMARCD1 belongs to the neural progenitors-specific chromatin remodeling complex (npBAF complex) and the neuron-specific chromatin remodeling complex (nBAF complex). SMARCD1 has a strong influence on vitamin D-mediated transcriptional activity from an enhancer vitamin D receptor element (VDRE). SMARCD1 a link between mammalian SWI-SNF-like chromatin remodeling complexes and the vitamin D receptor (VDR) heterodimer (Koszewski et al. (2003) *J Steroid Biochem Mol Biol* 87:223-231). SMARCD1 mediates critical interactions between nuclear receptors and the BRG1/SMARCA4 chromatin-remodeling complex for transactivation (Hsiao et al. (2003) *Mol Cell Biol* 23:6210-6220). Human SMARCD1 protein has 515 amino acids and a molecular mass of 58233 Da. Binding partners of SMARCD1 include, e.g., ESR1, NR3C1, NR1H4, PGR, SMARCA4, SMARCC1 and SMARCC2.

[0073] The term “SMARCD1” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human SMARCD1 cDNA and human SMARCD1 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human SMARCD1 isoforms are known. Human SMARCD1 isoform a (NP_003067.3) is encodable by the transcript variant 1 (NM_003076.4), which is the longer transcript.

[0074] Human SMARCD1 isoform b (NP_620710.2) is encodable by the transcript variant 2 (NM_139071.2), which lacks an alternate in-frame exon, compared to variant 1, resulting in a shorter protein (isoform b), compared to isoform a. Nucleic acid and polypeptide sequences of SMARCD1 orthologs in organisms other than humans are well known and include, for example, chimpanzee SMARCD1 (XM_016923432.2 and XP_016778921.1, XM_016923431.2 and XP_016778920.1, and XM_016923433.2 and XP_016778922.1), Rhesus monkey SMARCD1 (XM_001111275.3 and XP_001111275.3, XM_001111166.3 and XP_001111166.3, and XM_001111207.3 and XP_001111207.3), dog SMARCD1 (XM_543674.6 and XP_543674.4), cattle SMARCD1 (NM_001038559.2 and NP_001033648.1), mouse SMARCD1 (NM_031842.2 and NP_114030.2), rat SMARCD1 (NM_001108752.1 and NP_001102222.1), chicken SMARCD1 (XM_424488.6 and XP_424488.3), tropical clawed frog SMARCD1 (NM_001004862.1 and NP_001004862.1), and zebrafish SMARCD1 (NM_198358.1 and NP_938172.1). Representative sequences of SMARCD1 orthologs are presented below in Table 1.

[0075] Anti-SMARCD1 antibodies suitable for detecting SMARCD1 protein are well-known in the art and include, for example, antibody TA344378 (Origene), antibodies NBP1-88719 and NBP2-20417 (Novus Biologicals, Littleton, CO), antibodies ab224229, ab83208, and ab86029 (AbCam, Cambridge, MA), antibody Cat #PA5-52049 (ThermoFisher Scientific), etc. In addition, reagents are well-known for detecting SMARCD1. A clinical test of SMARCD1 for hereditary disese is available with the test ID no. GTR000558444.1 in NIH Genetic Testing Registry (GTR®), offered by Tempus Labs, Inc., (Chicago, IL). Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing SMARCD1 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-72597 and sc-725983 and CRISPR product #sc-402641 from Santa Cruz Biotechnology, RNAi products SR304476 and TL301504V, and CRISPR product KN203474 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding SMARCD1 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a SMARCD1 molecule encompassed by the present invention.

[0076] The term “SMARCD2” refers to SWI/SNF related, matrix associated, actin dependent regulator of chromatin subfamily D member 2. SMARCD2 is a member of the SWI/SNF family of proteins, whose members display helicase and ATPase activities and which are thought to regulate transcription of certain genes by altering the chromatin structure around those genes. The encoded protein is part of the large ATP-dependent chromatin remodeling complex SNF/SWI and has sequence similarity to the yeast Swp73 protein. SMARCD2 is a component of SWI/SNF chromatin remodeling complexes that carry out key enzymatic activities, changing chromatin structure by altering DNA-histone contacts within a nucleosome in an ATP-dependent manner (Euskirchen et al. (2012) *J Biol Chem* 287:30897-30905; Kadoch et al. (2015) *Sci Adv* 1(5):e1500447). SMARCD2 is a critical regulator of myeloid differentiation, controlling granulocytopoiesis and the expression of genes involved in neutrophil granule formation (Witzel et al. (2017) *Nat Genet* 49:742-752). Human SMARCD2 protein has 531 amino acids and a molecular mass of 589213 Da. Binding partners of SMARCD2 include, e.g., UNKL and CEBPE.

[0077] The term “SMARCD2” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human SMARCD2 cDNA and human SMARCD2 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, three different human SMARCD2 isoforms are known. Human SMARCD2 isoform 1 (NP_001091896.1) is encodable by the transcript variant 1 (NM_001098426.1). Human SMARCD2 isoform 2 (NP_001317368.1) is encodable by the transcript variant 2 (NM_001330439.1). Human SMARCD2 isoform 3 (NP_001317369.1) is encodable by the transcript variant 3 (NM_001330440.1). Nucleic acid and polypeptide sequences of SMARCD2 orthologs in organisms other than humans are well known and include, for example, chimpanzee SMARCD2 (XM_009433047.3

and XP_009431322.1, XM_001148723.6 and XP_001148723.1, XM_009433048.3 and XP_009431323.1, XM_009433049.3 and XP_009431324.1, XM_024350546.1 and XP_024206314.1, and XM_024350547.1 and XP_024206315.1), Rhesus monkey SMARCD2 (XM_015120093.1 and XP_014975579.1), dog SMARCD2 (XM_022422831.1 and XP_022278539.1, XM_005624251.3 and XP_005624308.1, XM_845276.5 and XP_850369.1, and XM_005624252.3 and XP_005624309.1), cattle SMARCD2 (NM_001205462.3 and NP_001192391.1), mouse SMARCC1 (NM_001130187.1 and NP_001123659.1, and NM_031878.2 and NP_114084.2), rat SMARCD2 (NM_031983.2 and NP_114189.1), chicken SMARCD2 (XM_015299406.2 and XP_015154892.1), tropical clawed frog SMARCD2 (NM_001045802.1 and NP_001039267.1), and zebrafish SMARCD2 (XM_687657.6 and XP_692749.2, and XM_021480266.1 and XP_021335941.1). Representative sequences of SMARCD2 orthologs are presented below in Table 1.

[0078] Anti-SMARCD2 antibodies suitable for detecting SMARCD2 protein are well-known in the art and include, for example, antibody TA335791 (Origene), antibodies H00006603-M02 and H00006603-MO1 (Novus Biologicals, Littleton, CO), antibodies ab81622, ab56241, and ab221084 (AbCam, Cambridge, MA), antibody Cat #51-805 (ProSci, Poway, CA), etc. In addition, reagents are well-known for detecting SMARCD2. A clinical test of SMARCD2 for hereditary disese is available with the test ID no. GTR000558444.1 in NIH Genetic Testing Registry (GTR®), offered by Tempus Labs, Inc., (Chicago, IL). Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing SMARCD2 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-93762 and sc-153618 and CRISPR product #sc-403091 from Santa Cruz Biotechnology, RNAi products SR304477 and TL309244V, and CRISPR product KN214286 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding SMARCD2 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a SMARCD2 molecule encompassed by the present invention.

[0079] The term “SMARCD3” refers to SWI/SNF related, matrix associated, actin dependent regulator of chromatin subfamily D member 3. SMARCD3 is a member of the SWI/SNF family of proteins, whose members display helicase and ATPase activities and which are thought to regulate transcription of certain genes by altering the chromatin structure around those genes. The encoded protein is part of the large ATP-dependent chromatin remodeling complex SNF/SWI and has sequence similarity to the yeast Swp73 protein. SMARCD3 is a component of SWI/SNF chromatin remodeling complexes that carry out key enzymatic activities, changing chromatin structure by altering DNA-histone contacts within a nucleosome in an ATP-dependent manner. SMARCD3 stimulates nuclear receptor mediated transcription. SMARCD3 belongs to the neural progenitors-specific chromatin remodeling complex (npBAF complex) and the neuron-specific chromatin remodeling complex (nBAF complex). Human SMARCD3 protein has 483 amino acids and a molecular mass of 55016 Da. Binding partners of SMARCD3 include, e.g., PPARG/NR1C3, RXRA/NR1F1, ESR1, NR5A1, NR5A2/LRH1 and other transcriptional activators including the HLH protein SREBF1/SREBP1 and the homeobox protein PBX1.

[0080] The term “SMARCD3” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human SMARCD3 cDNA and human SMARCD3 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human SMARCD3 isoforms are known. Human SMARCD3 isoform 1 (NP_001003802.1 and NP_003069.2) is encodable by the transcript variant 1 (NM_001003802.1) and the transcript variant 2 (NM_003078.3). Human SMARCD2 isoform 2 (NP_001003801.1) is encodable by the transcript variant 3 (NM_001003801.1). Nucleic acid and polypeptide sequences of SMARCD3 orthologs in organisms other than humans are well known and include, for example, chimpanzee SMARCD3 (XM_016945944.2 and XP_016801433.1, XM_016945946.2 and XP_016801435.1, XM_016945945.2 and XP_016801434.1, and XM_016945943.2 and XP_016801432.1), Rhesus monkey SMARCD3 (NM_001260684.1 and NP_001247613.1), cattle SMARCD3 (NM_001078154.1 and NP_001071622.1), mouse SMARCC3 (NM_025891.3 and NP_080167.3), rat SMARCD3 (NM_001011966.1 and NP_001011966.1). Representative sequences of SMARCD3 orthologs are presented below in Table 1.

[0081] Anti-SMARCD3 antibodies suitable for detecting SMARCD3 protein are well-known in the art and include, for example, antibody TA811107 (Origene), antibodies H00006604-MO1 and NBP2-39013 (Novus Biologicals, Littleton, CO), antibodies ab171075, ab131326, and ab50556 (AbCam, Cambridge, MA), antibody Cat #720131 (ThermoFisher Scientific), antibody Cat #28-327 (ProSci, Poway, CA), etc. In addition, reagents are well-known for detecting SMARCD3. A clinical test of SMARCD3 for hereditary disese is available with the test ID no. GTR000558444.1 in NIH Genetic Testing Registry (GTR®), offered by Tempus Labs, Inc., (Chicago, IL). Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing SMARCD3 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-89355 and sc-108054 and CRISPR product #sc-402705 from Santa Cruz Biotechnology, RNAi products SR304478 and TL309243V, and CRISPR product KN201135 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding SMARCD3 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a SMARCD3 molecule encompassed by the present invention.

[0082] The term “SMARCE1” refers to SWI/SNF related, matrix associated, actin dependent regulator of chromatin subfamily E member 1. The protein encoded by this gene is part of the large ATP-dependent chromatin remodeling complex SWI/SNF, which is required for transcriptional activation of genes normally repressed by chromatin. The encoded protein, either alone or when in the SWI/SNF complex, can bind to 4-way junction DNA, which is thought to mimic the topology of DNA as it enters or exits the nucleosome. The protein contains a DNA-binding HMG domain, but disruption of this domain does not abolish the DNA-binding or nucleosome-displacement activities of the SWI/SNF complex. Unlike most of the SWI/SNF complex proteins, this protein has no yeast counterpart. SMARCE1 is a component of SWI/SNF chromatin remodeling complexes that carry out key enzymatic activities, changing chromatin structure by altering DNA-histone contacts within a nucleosome in an ATP-dependent manner. SMARCE1 belongs to the neural progenitors-specific chromatin remodeling complex (npBAF complex) and the neuron-specific chromatin remodeling complex (nBAF complex). SMARCE1 is required for the coactivation of estrogen responsive promoters by SWI/SNF complexes and the SRC/p160 family of histone acetyltransferases (HATs). SMARCE1 also specifically interacts with the CoREST corepressor resulting in repression of neuronal specific gene promoters in non-neuronal cells. Human SMARCE1 protein has 411 amino acids and a molecular mass of 46649 Da. SMARCE1 interacts with BRDT, and also binds to the SRC/p160 family of histone acetyltransferases (HATs) composed of NCOA1, NCOA2, and NCOA3. SMARCE1 interacts with RCOR1/CoREST, NR3C1 and ZMIM2/ZIMP7.

[0083] The term “SMARCE1” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human SMARCE1 cDNA and human SMARCE1 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, human SMARCE1 protein (NP_003070.3) is encodable by transcript (NM_003079.4). Nucleic acid and polypeptide sequences of SMARCE1 orthologs in organisms other than humans are well known and include, for example, chimpanzee SMARCE1 (XM_009432223.3 and XP_009430498.1, XM_511478.7 and XP_511478.2, XM_009432222.3 and XP_009430497.1, and XM_001169953.6 and XP_001169953.1), Rhesus monkey SMARCE1 (NM_001261306.1 and NP_001248235.1), cattle SMARCE1 (NM_001099116.2 and NP_001092586.1), mouse SMARCE1 (NM_020618.4 and NP_065643.1), rat SMARCE1 (NM_001024993.1 and NP_001020164.1), chicken SMARCE1 (NM_001006335.2 and NP_001006335.2), tropical clawed frog SMARCE1 (NM_001005436.1 and NP_001005436.1), and zebrafish SMARCE1 (NM_201298.1 and NP_958455.2). Representative sequences of SMARCE1 orthologs are presented below in Table 1.

[0084] Anti-SMARCE1 antibodies suitable for detecting SMARCE1 protein are well-known in the art and include, for example, antibody TA335790 (Origene), antibodies NBP1-90012 and NB100-2591 (Novus Biologicals, Littleton, CO), antibodies ab131328, ab228750, and ab137081 (AbCam, Cambridge, MA), antibody Cat #PA5-18185 (ThermoFisher Scientific), antibody Cat #57-670 (ProSci, Poway, CA), etc. In addition, reagents are well-known for detecting SMARCE1. A clinical test of SMARCE1 for hereditary disese is available with the test ID no. GTR000558444.1 in NIH Genetic Testing Registry (GTR®), offered by Tempus Labs, Inc., (Chicago, IL). Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing SMARCE1 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-45940 and sc-45941 and CRISPR product #sc-404713 from Santa Cruz Biotechnology, RNAi products SR304479 and TL309242, and CRISPR product KN217885 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding SMARCE1 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a SMARCE1 molecule encompassed by the present invention.

[0085] The term “DPF1” refers to Double PHD Fingers 1. DPF1 has an important role in developing neurons by participating in regulation of cell survival, possibly as a neurospecific transcription factor. DPF1 belongs to the neuron-specific chromatin remodeling complex (nBAF complex). During neural development a switch from a stem/progenitor to a post-mitotic chromatin remodeling mechanism occurs as neurons exit the cell cycle and become committed to their adult state. The transition from proliferating neural stem/progenitor cells to post-mitotic neurons requires a switch in subunit composition of the npBAF and nBAF complexes. As neural progenitors exit mitosis and differentiate into neurons, npBAF complexes which contain ACTL6A/BAF53A and PHF10/BAF45A, are exchanged for homologous alternative ACTL6B/BAF53B and DPF1/BAF45B or DPF3/BAF45C subunits in neuron-specific complexes (nBAF). The npBAF complex is essential for the self-renewal/proliferative capacity of the multipotent neural stem cells. The nBAF complex along with CREST plays a role regulating the activity of genes essential for dendrite growth. Human DPF1 protein has 380 amino acids and a molecular mass of 425029 Da. DPF1 is a component of neuron-specific chromatin remodeling complex (nBAF complex) composed of at least, ARID1A/BAF250A or ARID1B/BAF250B, SMARCD1/BAF60A, SMARCD3/BAF60C,

SMARCA2/BRM/BAF190B, SMARCA4/BRG1/BAF190A, SMARCB1/BAF47, SMARCC1/BAF155, SMARCE1/BAF57, SMARCC2/BAF170, DPF1/BAF45B, DPF3/BAF45C, ACTL6B/BAF53B and actin.

[0086] The term “DPF1” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human DPF1 cDNA and human DPF1 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, five different human DPF1 isoforms are known. Human DPF1 isoform a (NP_001128627.1) is encodable by the transcript variant 1 (NM_001135155.2). Human DPF1 isoform b (NP_004638.2) is encodable by the transcript variant 2 (NM_004647.3). Human DPF1 isoform c (NP_001128628.1) is encodable by the transcript variant 3 (NM_001135156.2). Human DPF1 isoform d (NP_001276907.1) is encodable by the transcript variant 4 (NM_001289978.1). Human DPF1 isoform e (NP_001350508.1) is encodable by the transcript variant 5 (NM_001363579.1). Nucleic acid and polypeptide sequences of DPF1 orthologs in organisms other than humans are well known and include, for example, Rhesus monkey DPF1 (XM_015123830.1 and XP_014979316.1, XM_015123829.1 and XP_014979315.1, XM_015123835.1 and XP_014979321.1, XM_015123831.1 and XP_014979317.1, XM_015123833.1 and XP_014979319.1, and XM_015123832.1 and XP_014979318.1), cattle DPF1 (NM_001076855.1 and NP_001070323.1), mouse DPF1 (NM_013874.2 and NP_038902.1), rat DPF1 (NM_001105729.3 and NP_001099199.2), and tropical clawed frog DPF1 (NM_001097276.1 and NP_001090745.1). Representative sequences of DPF1 orthologs are presented below in Table 1.

[0087] Anti-DPF1 antibodies suitable for detecting DPF1 protein are well-known in the art and include, for example, antibody TA311193 (Origene), antibodies NBP2-13932 and NBP2-19518 (Novus Biologicals, Littleton, CO), antibodies ab199299, ab173160, and ab3940 (AbCam, Cambridge, MA), antibody Cat #PA5-61895 (ThermoFisher Scientific), antibody Cat #28-079 (ProSci, Poway, CA), etc. In addition, reagents are well-known for detecting DPF1. Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing DPF1 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-97084 and sc-143155 and CRISPR product #sc-409539 from Santa Cruz Biotechnology, RNAi products SR305389 and TL313388V, and CRISPR product KN213721 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding DPF1 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a DPF1 molecule encompassed by the present invention.

[0088] The term “DPF2” refers to Double PHD Fingers 2. DPF2 protein is a member of the d4 domain family, characterized by a zinc finger-like structural motif. It functions as a transcription factor which is necessary for the apoptotic response following deprivation of survival factors. It likely serves a regulatory role in rapid hematopoietic cell growth and turnover. This gene is considered a candidate gene for multiple endocrine neoplasia type I, an inherited cancer syndrome involving multiple parathyroid, enteropancreatic, and pituitary tumors. DPF2 is a transcription factor required for the apoptosis response following survival factor withdrawal from myeloid cells. DPF2 also has a role in the development and maturation of lymphoid cells. Human DPF2 protein has 391 amino acids and a molecular mass of 44155 Da.

[0089] The term “DPF2” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human DPF2 cDNA and human DPF2 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human DPF2 isoforms are known. Human DPF2 isoform 1 (NP_006259.1) is encodable by the transcript variant 1 (NM_006268.4). Human DPF2 isoform 2 (NP_001317237.1) is encodable by the transcript variant 2 (NM_001330308.1). Nucleic acid and polypeptide sequences of DPF2 orthologs in organisms other than humans are well known and include, for example, chimpanzee DPF2 (NM_001246651.1 and NP_001233580.1), Rhesus monkey DPF2 (XM_002808062.2 and XP_002808108.2, and XM_015113800.1 and XP_014969286.1), dog DPF2 (XM_861495.5 and XP_866588.1, and XM_005631484.3 and XP_005631541.1), cattle DPF2 (NM_001100356.1 and NP_001093826.1), mouse DPF2 (NM_001291078.1 and NP_001278007.1, and NM_011262.5 and NP_035392.1), rat DPF2 (NM_001108516.1 and NP_001101986.1), chicken DPF2 (NM_204331.1 and NP_989662.1), tropical clawed frog DPF2 (NM_001197172.2 and NP_001184101.1), and zebrafish DPF2 (NM_001007152.1 and NP_001007153.1). Representative sequences of DPF2 orthologs are presented below in Table 1.

[0090] Anti-DPF2 antibodies suitable for detecting DPF2 protein are well-known in the art and include, for example, antibody TA312307 (Origene), antibodies NBP1-76512 and NBP1-87138 (Novus Biologicals, Littleton, CO), antibodies ab134942, ab232327, and ab227095 (AbCam, Cambridge, MA), etc. In addition, reagents are well-known for detecting DPF2. A clinical test of DPF2 for hereditary disese is available with the test ID no. GTR000536833.2 in NIH Genetic Testing Registry (GTR®), offered by Fulgent Genetics Clinical Diagnostics Lab (Temple City, CA). Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing DPF2 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-97031 and sc-143156 and CRISPR product #sc-404801-KO-2 from Santa Cruz Biotechnology, RNAi products SR304035 and TL313387V, and CRISPR product KN202364 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding DPF2 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a DPF2 molecule encompassed by the present invention.

[0091] The term “DPF3” refers to Double PHD Fingers 3, a member of the D4 protein family. The encoded protein is a transcription regulator that binds acetylated histones and is a component of the BAF chromatin remodeling complex. DPF3 belongs to the neuron-specific chromatin remodeling complex (nBAF complex). During neural development a switch from a stem/progenitor to a post-mitotic chromatin remodeling mechanism occurs as neurons exit the cell cycle and become committed to their adult state. The transition from proliferating neural stem/progenitor cells to post-mitotic neurons requires a switch in subunit composition of the npBAF and nBAF complexes. As neural progenitors exit mitosis and differentiate into neurons, npBAF complexes which contain ACTL6A/BAF53A and PHF10/BAF45A, are exchanged for homologous alternative ACTL6B/BAF53B and DPF1/BAF45B or DPF3/BAF45C subunits in neuron-specific complexes (nBAF). The npBAF complex is essential for the self-renewal/proliferative capacity of the multipotent neural stem cells. The nBAF complex along with CREST plays a role regulating the activity of genes essential for dendrite growth (By similarity). DPF3 is a muscle-specific component of the BAF complex, a multiprotein complex involved in transcriptional activation and repression of select genes by chromatin remodeling (alteration of DNA-nucleosome topology). DPF3 specifically binds acetylated lysines on histone 3 and 4 (H3K14ac, H3K9ac, H4K5ac, H4K8ac, H4K12ac, H4K16ac). In the complex, DPF3 acts as a tissue-specific anchor between histone acetylations and methylations and chromatin remodeling. DPF3 plays an essential role in heart and skeletal muscle development. Human DPF3 protein has 378 amino acids and a molecular mass of 43084 Da. The PHD-type zinc fingers of DPF3 mediate its binding to acetylated histones. DPF3 belongs to the requiem/DPF family.

[0092] The term “DPF3” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human DPF3 cDNA and human DPF3 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, four different human DPF3 isoforms are known. Human DPF3 isoform 1 (NP_036206.3) is encodable by the transcript variant 1 (NM_012074.4). Human DPF3 isoform 2 (NP_001267471.1) is encodable by the transcript variant 2 (NM_001280542.1). Human DPF3 isoform 3 (NP_001267472.1) is encodable by the transcript variant 3 (NM_001280543.1). Human DPF3 isoform 4 (NP_001267473.1) is encodable by the transcript variant 4 (NM_001280544.1). Nucleic acid and polypeptide sequences of DPF3 orthologs in organisms other than humans are well known and include, for example, chimpanzee DPF3 (XM_016926314.2 and XP_016781803.1, XM_016926316.2 and XP_016781805.1, and XM_016926315.2 and XP_016781804.1), dog DPF3 (XM_014116039.1 and XP_013971514.1), mouse DPF3 (NM_001267625.1 and NP_001254554.1, NM_001267626.1 and NP_001254555.1, and NM_058212.2 and NP_478119.1), chicken DPF3 (NM_204639.2 and NP_989970.1), tropical clawed frog DPF3 (NM_001278413.1 and NP_001265342.1), and zebrafish DPF3 (NM_001111169.1 and NP_001104639.1). Representative sequences of DPF3 orthologs are presented below in Table 1.

[0093] Anti-DPF3 antibodies suitable for detecting DPF3 protein are well-known in the art and include, for example, antibody TA335655 (Origene), antibodies NBP2-49494 and NBP2-14910 (Novus Biologicals, Littleton, CO), antibodies ab180914, ab127703, and ab85360 (AbCam, Cambridge, MA), antibody PA5-38011 (ThermoFisher Scientific), antibody Cat #7559 (ProSci, Poway, CA), etc. In addition, reagents are well-known for detecting DPF3. Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing DPF3 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-97031 and sc-92150 and CRISPR product #sc-143157 from Santa Cruz Biotechnology, RNAi products SR305368 and TL313386V, and CRISPR product KN218937 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding DPF3 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a DPF3 molecule encompassed by the present invention.

[0094] The term “ACTL6A” refers to Actin Like 6A, a family member of actin-related proteins (ARPs), which share significant amino acid sequence identity to conventional actins. Both actins and ARPs have an actin fold, which is an ATP-binding cleft, as a common feature. The ARPs are involved in diverse cellular processes, including vesicular transport, spindle orientation, nuclear migration and chromatin remodeling. This gene encodes a 53 kDa subunit protein of the BAF (BRG1/brm-associated factor) complex in mammals, which is functionally related to SWI/SNF complex in *S. cerevisiae* and *Drosophila*; the latter is thought to facilitate transcriptional activation of specific genes by antagonizing chromatin-mediated transcriptional repression. Together with beta-actin, it is required for maximal ATPase activity of BRG1, and for the association of the BAF complex with chromatin/matrix. ACTL6A is a component of SWI/SNF chromatin remodeling complexes that carry out key enzymatic activities, changing chromatin structure by altering DNA-histone contacts within a nucleosome in an ATP-dependent manner. ACTL6A is required for maximal ATPase activity of SMARCA4/BRG1/BAF190A and for association of the SMARCA4/BRG1/BAF190A containing remodeling complex BAF with chromatin/nuclear matrix. ACTL6A belongs to the neural progenitors-specific chromatin remodeling complex (npBAF complex) and is required for the proliferation of neural progenitors. During neural development a switch from a stem/progenitor to a post-mitotic chromatin remodeling mechanism occurs as neurons exit the cell cycle and become committed to their adult state. The transition from proliferating neural stem/progenitor cells to post-mitotic neurons requires a switch in subunit composition of the npBAF and nBAF complexes. As neural progenitors exit mitosis and differentiate into neurons, npBAF complexes which contain ACTL6A/BAF53A and PHF10/BAF45A, are exchanged for homologous alternative ACTL6B/BAF53B and DPF1/BAF45B or DPF3/BAF45C subunits in neuron-specific complexes (nBAF). The npBAF complex is essential for the self-renewal/proliferative capacity of the multipotent neural stem cells. The nBAF complex along with CREST plays a role regulating the activity of genes essential for dendrite growth. ACTL6A is a component of the NuA4 histone acetyltransferase (HAT) complex which is involved in transcriptional activation of select genes principally by acetylation of nucleosomal histones H4 and H2A. This modification may both alter nucleosome—DNA interactions and promote interaction of the modified histones with other proteins which positively regulate transcription. This complex may be required for the activation of transcriptional programs associated with oncogene and proto-oncogene mediated growth induction, tumor suppressor mediated growth arrest and replicative senescence, apoptosis, and DNA repair. NuA4 may also play a direct role in DNA repair when recruited to sites of DNA damage. Putative core component of the chromatin remodeling IN080 complex which is involved in transcriptional regulation, DNA replication and probably DNA repair. Human ACTL6A protein has 429 amino acids and a molecular mass of 47461 Da.

[0095] The term “ACTL6A” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human ACTL6A cDNA and human ACTL6A protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human ACTL6A isoforms are known. Human ACTL6A isoform 1 (NP_004292.1) is encodable by the transcript variant 1 (NM_004301.4). Human ACTL6A isoform 2 (NP_817126.1 and NP_829888.1) is encodable by the transcript variant 2 (NM_177989.3) and transcript variant 3 (NM_178042.3). Nucleic acid and polypeptide sequences of ACTL6A orthologs in organisms other than humans are well known and include, for example, chimpanzee ACTL6A (NM_001271671.1 and NP_001258600.1), Rhesus monkey ACTL6A (NM_001104559.1 and NP_001098029.1), cattle ACTL6A (NM_001105035.1 and NP_001098505.1), mouse ACTL6A (NM_019673.2 and NP_062647.2), rat ACTL6A (NM_001039033.1 and NP_001034122.1), chicken ACTL6A (XM_422784.6 and XP_422784.3), tropical clawed frog ACTL6A (NM_204006.1 and NP_989337.1), and zebrafish ACTL6A (NM_173240.1 and NP_775347.1). Representative sequences of ACTL6A orthologs are presented below in Table 1.

[0096] Anti-ACTL6A antibodies suitable for detecting ACTL6A protein are well-known in the art and include, for example, antibody TA345058 (Origene), antibodies NB100-61628 and NBP2-55376 (Novus Biologicals, Littleton, CO), antibodies ab131272 and ab189315 (AbCam, Cambridge, MA), antibody 702414 (ThermoFisher Scientific), antibody Cat #45-314 (ProSci, Poway, CA), etc. In addition, reagents are well-known for detecting ACTL6A. Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing ACTL6A expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-60239 and sc-60240 and CRISPR product #sc-403200-KO-2 from Santa Cruz Biotechnology, RNAi products SR300052 and TL306860V, and CRISPR product KN201689 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding ACTL6A molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe an ACTL6A molecule encompassed by the present invention.

[0097] The term “β-Actin” refers to Actin Beta. This gene encodes one of six different actin proteins. Actins are highly conserved proteins that are involved in cell motility, structure, integrity, and intercellular signaling. The encoded protein is a major constituent of the contractile apparatus and one of the two nonmuscle cytoskeletal actins that are ubiquitously expressed. Mutations in this gene cause Baraitser-Winter syndrome 1, which is characterized by intellectual disability with a distinctive facial appearance in human patients. Numerous pseudogenes of this gene have been identified throughout the human genome. Actins are highly conserved proteins that are involved in various types of cell motility and are ubiquitously expressed in all eukaryotic cells. Actin is found in two main states: G-actin is the globular monomeric form, whereas F-actin forms helical polymers. Both G- and F-actin are intrinsically flexible structures. Human β-Actin protein has 375 amino acids and a molecular mass of 41737 Da. The binding partners of β-Actin include, e.g., CPNE1, CPNE4, DHX9, GCSAM, ERBB2, XPO6, and EMD.

[0098] The term “β-Actin” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human β-Actin cDNA and human β-Actin protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, human β-Actin (NP_001092.1) is encodable by the transcript (NM_001101.4). Nucleic acid and polypeptide sequences of $3-Actin orthologs in organisms other than humans are well known and include, for example, chimpanzee β-Actin (NM_001009945.1 and NP_001009945.1), Rhesus monkey β-Actin (NM_001033084.1 and NP_001028256.1), dog β-Actin (NM_001195845.2 and NP_001182774.2), cattle β-Actin (NM_173979.3 and NP_776404.2), mouse β-Actin

(NM_007393.5 and NP_031419.1), rat β-Actin (NM_031144.3 and NP_112406.1), chicken β-Actin (NM_205518.1 and NP_990849.1), and tropical clawed frog β-Actin (NM_213719.1 and NP_998884.1). Representative sequences of β-Actin orthologs are presented below in Table 1.

[0099] Anti-β-Actin antibodies suitable for detecting β-Actin protein are well-known in the art and include, for example, antibody TA353557 (Origene), antibodies NB600-501 and NB600-503 (Novus Biologicals, Littleton, CO), antibodies ab8226 and ab8227 (AbCam, Cambridge, MA), antibody AM4302 (ThermoFisher Scientific), antibody Cat #PM-7669-biotin (ProSci, Poway, CA), etc. In addition, reagents are well-known for detecting $-Actin. Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing β-Actin expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-108069 and sc-108070 and CRISPR product #sc-400000-KO-2 from Santa Cruz Biotechnology, RNAi products SR300047 and TL314976V, and CRISPR product KN203643 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding β-Actin molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a β-Actin molecule encompassed by the present invention.

[0100] The term “BCL7A” refers to BCL Tumor Suppressor 7A. This gene is directly involved, with Myc and IgH, in a three-way gene translocation in a Burkitt lymphoma cell line. As a result of the gene translocation, the N-terminal region of the gene product is disrupted, which is thought to be related to the pathogenesis of a subset of high-grade B cell non-Hodgkin lymphoma. The N-terminal segment involved in the translocation includes the region that shares a strong sequence similarity with those of BCL7B and BCL7C. Diseases associated with BCL7A include Lymphoma and Burkitt Lymphoma. An important paralog of this gene is BCL7C. Human BCL7A protein has 210 amino acids and a molecular mass of 22810 Da.

[0101] The term “BCL7A” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human BCL7A cDNA and human BCL7A protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human BCL7A isoforms are known. Human BCL7A isoform a (NP_066273.1) is encodable by the transcript variant 1 (NM_020993.4). Human BCL7A isoform b (NP_001019979.1) is encodable by the transcript variant 2 (NM_001024808.2). Nucleic acid and polypeptide sequences of BCL7A orthologs in organisms other than humans are well known and include, for example, chimpanzee BCL7A (XM_009426452.3 and XP_009424727.2, and XM_016924434.2 and XP_016779923.1), Rhesus monkey BCL7A (XM_015153012.1 and XP_015008498.1, and XM_015153013.1 and XP_015008499.1), dog BCL7A (XM_543381.6 and XP_543381.2, and XM_854760.5 and XP_859853.1), cattle BCL7A (XM_024977701.1 and XP_024833469.1, and XM_024977700.1 and XP_024833468.1), mouse BCL7A (NM_029850.3 and NP_084126.1), rat BCL7A (XM_017598515.1 and XP_017454004.1), chicken BCL7A (XM_004945565.3 and XP_004945622.1, and XM_415148.6 and XP_415148.2), tropical clawed frog BCL7A (NM_001006871.1 and NP_001006872.1), and zebrafish BCL7A (NM_212560.1 and NP_997725.1). Representative sequences of BCL7A orthologs are presented below in Table 1.

[0102] Anti-BCL7A antibodies suitable for detecting BCL7A protein are well-known in the art and include, for example, antibody TA344744 (Origene), antibodies NBP1-30941 and NBP1-91696 (Novus Biologicals, Littleton, CO), antibodies ab137362 and ab1075 (AbCam, Cambridge, MA), antibody PA5-27123 (ThermoFisher Scientific), antibody Cat #45-325 (ProSci, Poway, CA), etc. In addition, reagents are well-known for detecting BCL7A. Multiple clinical tests of BCL7A are available in NIH Genetic Testing Registry (GTR®) (e.g., GTR Test ID: GTR000541481.2, offered by Fulgent Clinical Diagnostics Lab (Temple City, CA)). Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing BCL7A expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-96136 and sc-141671 and CRISPR product #sc-410702 from Santa Cruz Biotechnology, RNAi products SR300417 and TL314490V, and CRISPR product KN210489 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding BCL7A molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a BCL7A molecule encompassed by the present invention.

[0103] The term “BCL7B” refers to BCL Tumor Suppressor 7B, a member of the BCL7 family including BCL7A, BCL7B and BCL7C proteins. This member is BCL7B, which contains a region that is highly similar to the N-terminal segment of BCL7A or BCL7C proteins. The BCL7A protein is encoded by the gene known to be directly involved in a three-way gene translocation in a Burkitt lymphoma cell line. This gene is located at a chromosomal region commonly deleted in Williams syndrome. This gene is highly conserved from *C. elegans* to human. BCL7B is a positive regulator of apoptosis. BCL7B plays a role in the Wnt signaling pathway, negatively regulating the expression of Wnt signaling components CTNNB1 and HMGA1 (Uehara et al. (2015) *PLoS Genet* 11(1):e1004921). BCL7B is involved in cell cycle progression, maintenance of the nuclear structure and stem cell differentiation (Uehara et al. (2015) *PLoS Genet* 11(1):e1004921). It plays a role in lung tumor development or progression. Human BCL7B protein has 202 amino acids and a molecular mass of 22195 Da.

[0104] The term “BCL7B” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof.

Representative human BCL7B cDNA and human BCL7B protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, three different human BCL7B isoforms are known. Human BCL7B isoform 1 (NP_001698.2) is encodable by the transcript variant 1 (NM_001707.3). Human BCL7B isoform 2 (NP_001184173.1) is encodable by the transcript variant 2 (NM_001197244.1). Human BCL7B isoform 3 (NP_001287990.1) is encodable by the transcript variant 3 (NM_001301061.1). Nucleic acid and polypeptide sequences of BCL7B orthologs in organisms other than humans are well known and include, for example, chimpanzee BCL7B (XM_003318671.3 and XP_003318719.1, and XM_003318672.3 and XP_003318720.1), Rhesus monkey BCL7B (NM_001194509.1 and NP_001181438.1), dog BCL7B (XM_546926.6 and XP_546926.1, and XM_005620975.2 and XP_005621032.1), cattle BCL7B (NM_001034775.2 and NP_001029947.1), mouse BCL7B (NM_009745.2 and NP_033875.2), chicken BCL7B (XM_003643231.4 and XP_003643279.1, XM_004949975.3 and XP_004950032.1, and XM_025142155.1 and XP_024997923.1), tropical clawed frog BCL7B (NM_001103072.1 and NP_001096542.1), and zebrafish BCL7B (NM_001006018.1 and NP_001006018.1, and NM_213165.1 and NP_998330.1). Representative sequences of BCL7B orthologs are presented below in Table 1.

[0105] Anti-BCL7B antibodies suitable for detecting BCL7B protein are well-known in the art and include, for example, antibody TA809485 (Origene), antibodies H00009275-M01 and NBP2-34097 (Novus Biologicals, Littleton, CO), antibodies ab130538 and ab172358 (AbCam, Cambridge, MA), antibody MA527163 (ThermoFisher Scientific), antibody Cat #58-996 (ProSci, Poway, CA), etc. In addition, reagents are well-known for detecting BCL7B. Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing BCL7B expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-89728 and sc-141672 and CRISPR product #sc-411262 from Santa Cruz Biotechnology, RNAi products SR306141 and TL306418V, and CRISPR product KN201696 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding BCL7B molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a BCL7B molecule encompassed by the present invention.

[0106] The term “BCL7C” refers to BCL Tumor Suppressor 7C, a member of the BCL7 family including BCL7A, BCL7B and BCL7C proteins. This gene is identified by the similarity of its product to the N-terminal region of BCL7A protein. BCL7C may play an anti-apoptotic role. Diseases associated with BCL7C include Lymphoma. Human BCL7C protein has 217 amino acids and a molecular mass of 23468 Da.

[0107] The term “BCL7C” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human BCL7C cDNA and human BCL7C protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human BCL7C isoforms are known. Human BCL7C isoform 1 (NP_001273455.1) is encodable by the transcript variant 1 (NM_001286526.1). Human BCL7C isoform 2 (NP_004756.2) is encodable by the transcript variant 2 (NM_004765.3). Nucleic acid and polypeptide sequences of BCL7C orthologs in organisms other than humans are well known and include, for example, chimpanzee BCL7C (XM_016929717.2 and XP_016785206.1, XM_016929716.2 and XP_016785205.1, and XM_016929718.2 and XP_016785207.1), Rhesus monkey BCL7C (NM_001265776.2 and NP_001252705.1), cattle BCL7C (NM_001099722.1 and NP_001093192.1), mouse BCL7C (NM_001347652.1 and NP_001334581.1, and NM_009746.2 and NP_033876.1), and rat BCL7C (NM_001106298.1 and NP_001099768.1). Representative sequences of BCL7C orthologs are presented below in Table 1.

[0108] Anti-BCL7C antibodies suitable for detecting BCL7C protein are well-known in the art and include, for example, antibody TA347083 (Origene), antibodies NBP2-15559 and NBP1-86441 (Novus Biologicals, Littleton, CO), antibodies ab126944 and ab231278 (AbCam, Cambridge, MA), antibody PA5-30308 (ThermoFisher Scientific), etc. In addition, reagents are well-known for detecting BCL7C. Multiple clinical tests of BCL7C are available in NIH Genetic Testing Registry (GTR®) (e.g., GTR Test ID: GTR000540637.2, offered by Fulgent Clinical Diagnostics Lab (Temple City, CA)). Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing BCL7C expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-93022 and sc-141673 and CRISPR product #sc-411261 from Santa Cruz Biotechnology, RNAi products SR306140 and TL315552V, and CRISPR product KN205720 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding BCL7C molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a BCL7C molecule encompassed by the present invention.

[0109] The term “SMARCA4” refers to SWI/SNF related, matrix associated, actin dependent regulator of chromatin, subfamily a, member 4, a member of the SWI/SNF family of proteins and is highly similar to the brahma protein of *Drosophila*. Members of this family have helicase and ATPase activities and are thought to regulate transcription of certain genes by altering the chromatin structure around those genes. The encoded protein is part of the large ATP-dependent chromatin remodeling complex SNF/SWI, which is required for transcriptional activation of genes normally repressed by chromatin. In addition, this protein can bind BRCA1, as well as regulate the expression of the tumorigenic protein CD44. Mutations in this gene cause rhabdoid tumor predisposition syndrome type 2. SMARCA4 is a component of SWI/SNF chromatin remodeling complexes that carry out key enzymatic activities, changing chromatin structure by altering DNA-histone contacts within a nucleosome in an ATP-dependent manner. SMARCA4 is a component of the CREST-BRG1 complex, a multiprotein complex that regulates promoter activation by orchestrating a calcium-dependent release of a repressor complex and a recruitment of an activator complex. In resting neurons, transcription of the c-FOS promoter is inhibited by BRG1-dependent recruitment of a phospho-RB1-HDAC repressor complex. Upon calcium influx, RB1 is dephosphorylated by calcineurin, which leads to release of the repressor complex. At the same time, there is increased recruitment of CREBBP to the promoter by a CREST-dependent mechanism, which leads to transcriptional activation. The CREST-BRG1 complex also binds to the NR2B promoter, and activity-dependent induction of NR2B expression involves a release of HDAC1 and recruitment of CREBBP. SMARCA4 belongs to the neural progenitors-specific chromatin remodeling complex (npBAF complex) and the neuron-specific chromatin remodeling complex (nBAF complex). During neural development a switch from a stem/progenitor to a postmitotic chromatin remodeling mechanism occurs as neurons exit the cell cycle and become committed to their adult state. The transition from proliferating neural stem/progenitor cells to postmitotic neurons requires a switch in subunit composition of the npBAF and nBAF complexes. As neural progenitors exit mitosis and differentiate into neurons, npBAF complexes which contain ACTL6A/BAF53A and PHF10/BAF45A, are exchanged for homologous alternative ACTL6B/BAF53B and DPF1/BAF45B or DPF3/BAF45C subunits in neuron-specific complexes (nBAF). The npBAF complex is essential for the self-renewal/proliferative capacity of the multipotent neural stem cells. The nBAF complex along with CREST plays a role regulating the activity of genes essential for dendrite growth. SMARCA4/BAF190A promote neural stem cell self-renewal/proliferation by enhancing Notch-dependent proliferative signals, while concurrently making the neural stem cell insensitive to SHH-dependent differentiating cues. SMARCA4 acts as a corepressor of ZEB1 to regulate E-cadherin transcription and is required for induction of epithelial-mesenchymal transition (EMT) by ZEB1. Human SMARCA4 protein has 1647 amino acids and a molecular mass of 184646 Da. The known binding partners of SMARCA4 include, e.g., PHF10/BAF45A, MYOG, IKFZ1, ZEB1, NR3C1, PGR, SMARD1, TOPBP1 and ZMIM2/ZIMP7.

[0110] The term “SMARCA4” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human SMARCA4 cDNA and human SMARCA4 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, six different human SMARCA4 isoforms are known. Human SMARCA4 isoform A (NP_001122321.1) is encodable by the transcript variant 1 (NM_001128849.1). Human SMARCA4 isoform B (NP_001122316.1 and NP_003063.2) is encodable by the transcript variant 2 (NM_001128844.1) and the transcript variant 3 (NM_003072.3). Human SMARCA4 isoform C (NP_001122317.1) is encodable by the transcript variant 4 (NM_001128845.1). Human SMARCA4 isoform D (NP_001122318.1) is encodable by the transcript variant 5 (NM_001128846.1). Human SMARCA4 isoform E (NP_001122319.1) is encodable by the transcript variant 6 (NM_001128847.1). Human SMARCA4 isoform F (NP_001122320.1) is encodable by the transcript variant 7 (NM_001128848.1). Nucleic acid and polypeptide sequences of SMARCA4 orthologs in organisms other than humans are well known and include, for example, Rhesus monkey SMARCA4 (XM_015122901.1 and XP_014978387.1, XM_015122902.1 and XP_014978388.1, XM_015122903.1 and XP_014978389.1, XM_015122906.1 and XP_014978392.1, XM_015122905.1 and XP_014978391.1, XM_015122904.1 and XP_014978390.1, XM_015122907.1 and XP_014978393.1, XM_015122909.1 and XP_014978395.1, and XM_015122910.1 and XP_014978396.1), cattle SMARCA4 (NM_001105614.1 and NP_001099084.1), mouse SMARCA4 (NM_001174078.1 and NP_001167549.1, NM_011417.3 and NP 035547.2, NM_001174079.1 and NP_001167550.1, NM_001357764.1 and NP_001344693.1), rat SMARCA4 (NM_134368.1 and NP_599195.1), chicken SMARCA4 (NM_205059.1 and NP_990390.1), and zebrafish SMARCA4 (NM_181603.1 and NP_853634.1). Representative sequences of SMARCA4 orthologs are presented below in Table 1.

[0111] Anti-SMARCA4 antibodies suitable for detecting SMARCA4 protein are well-known in the art and include, for example, antibody AM26021PU-N(Origene), antibodies NB100-2594 and AF5738 (Novus Biologicals, Littleton, CO), antibodies ab110641 and ab4081 (AbCam, Cambridge, MA), antibody 720129 (ThermoFisher Scientific), antibody 7749 (ProSci), etc. In addition, reagents are well-known for detecting SMARCA4. Multiple clinical tests of SMARCA4 are available in NIH Genetic Testing Registry (GTR®) (e.g., GTR Test ID: GTR000517106.2, offered by Fulgent Clinical Diagnostics Lab (Temple City, CA)). Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing SMARCA4 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-29827 and sc-44287 and CRISPR product #sc-400168 from Santa Cruz Biotechnology, RNAi products SR321835 and TL309249V, and CRISPR product KN219258 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding SMARCA4 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a SMARCA4 molecule encompassed by the present invention.

[0112] The term "SMARCE1" refers to SWI/SNF related, matrix associated, actin dependent regulator of chromatin subfamily E member 1. The protein encoded by this gene is part of the large ATP-dependent chromatin remodeling complex SWI/SNF, which is required for transcriptional activation of genes normally repressed by chromatin. The encoded protein, either alone or when in the SWI/SNF complex, can bind to 4-way junction DNA, which is thought to mimic the topology of DNA as it enters or exits the nucleosome. The protein contains a DNA-binding HMG domain, but disruption of this domain does not abolish the DNA-binding or nucleosome-displacement activities of the SWI/SNF complex. Unlike most of the SWI/SNF complex proteins, this protein has no yeast counterpart. SMARCE1 is a component of SWI/SNF chromatin remodeling complexes that carry out key enzymatic activities, changing chromatin structure by altering DNA-histone contacts within a nucleosome in an ATP-dependent manner. SMARCE1 belongs to the neural progenitors-specific chromatin remodeling complex (npBAF complex) and the neuron-specific chromatin remodeling complex (nBAF complex). SMARCE1 is required for the coactivation of estrogen responsive promoters by SWI/SNF complexes and the SRC/p160 family of histone acetyltransferases (HATs). SMARCE1 also specifically interacts with the CoREST corepressor resulting in repression of neuronal specific gene promoters in non-neuronal cells. Human SMARCE1 protein has 411 amino acids and a molecular mass of 46649 Da. SMARCE1 interacts with BRDT, and also binds to the SRC/p160 family of histone acetyltransferases (HATs) composed of NCOA1, NCOA2, and NCOA3. SMARCE1 interacts with RCOR1/CoREST, NR3C1 and ZMIM2/ZIMP7.

[0113] The term "SMARCE1" is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human SMARCE1 cDNA and human SMARCE1 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, human SMARCE1 protein (NP_003070.3) is encodable by transcript (NM_003079.4). Nucleic acid and polypeptide sequences of SMARCE1 orthologs in organisms other than humans are well known and include, for example, chimpanzee SMARCE1 (XM_009432223.3 and XP_009430498.1, XM_511478.7 and XP_511478.2, XM_009432222.3 and XP_009430497.1, and XM_001169953.6 and XP_001169953.1), Rhesus monkey SMARCE1 (NM_001261306.1 and NP_001248235.1), cattle SMARCE1 (NM_001099116.2 and NP_001092586.1), mouse SMARCE1 (NM_020618.4 and NP_065643.1), rat SMARCE1 (NM_001024993.1 and NP_001020164.1), chicken SMARCE1 (NM_001006335.2 and NP_001006335.2), tropical clawed frog SMARCE1 (NM_001005436.1 and NP_001005436.1), and zebrafish SMARCE1 (NM_201298.1 and NP_958455.2). Representative sequences of SMARCE1 orthologs are presented below in Table 1.

[0114] Anti-SMARCE1 antibodies suitable for detecting SMARCE1 protein are well-known in the art and include, for example, antibody TA335790 (Origene), antibodies NBP1-90012 and NB100-2591 (Novus Biologicals, Littleton, CO), antibodies ab131328, ab228750, and ab137081 (AbCam, Cambridge, MA), antibody Cat #PA5-18185 (ThermoFisher Scientific), antibody Cat #57-670 (ProSci, Poway, CA), etc. In addition, reagents are well-known for detecting SMARCE1. A clinical test of SMARCE1 for hereditary disese is available with the test ID no. GTR000558444.1 in NIH Genetic Testing Registry (GTR®), offered by Tempus Labs, Inc., (Chicago, IL). Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing SMARCE1 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-45940 and sc-45941 and CRISPR product #sc-404713 from Santa Cruz Biotechnology, RNAi products SR304479 and TL309242, and CRISPR product KN217885 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding SMARCE1 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a SMARCE1 molecule encompassed by the present invention.

[0115] The term "DPF1" refers to Double PHD Fingers 1. DPF1 has an important role in developing neurons by participating in regulation of cell survival, possibly as a neurospecific transcription factor. DPF1 belongs to the neuron-specific chromatin remodeling complex (nBAF complex). During neural development a switch from a stem/progenitor to a post-mitotic chromatin remodeling mechanism occurs as neurons exit the cell cycle and become committed to their adult state. The transition from proliferating neural stem/progenitor cells to post-mitotic neurons requires a switch in subunit composition of the npBAF and nBAF complexes. As neural progenitors exit mitosis and differentiate into neurons, npBAF complexes which contain ACTL6A/BAF53A and PHF10/BAF45A, are exchanged for homologous alternative ACTL6B/BAF53B and DPF1/BAF45B or DPF3/BAF45C subunits in neuron-specific complexes (nBAF). The npBAF complex is essential for the self-renewal/proliferative capacity of the multipotent neural stem cells. The nBAF complex along with CREST plays a role regulating the activity of genes essential for dendrite growth. Human DPF1 protein has 380 amino acids and a molecular mass of 425029 Da. DPF1 is a component of neuron-specific chromatin remodeling complex (nBAF complex) composed of at least, ARID1A/BAF250A or ARID1B/BAF250B, SMARCD1/BAF60A, SMARCD3/BAF60C, SMARCA2/BRM/BAF190B, SMARCA4/BRG1/BAF190A, SMARCB1/BAF47, SMARCC1/BAF155, SMARCE1/BAF57, SMARCC2/BAF170, DPF1/BAF45B, DPF3/BAF45C, ACTL6B/BAF53B and actin.

[0116] The term "DPF1" is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof.

Representative human DPF1 cDNA and human DPF1 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, five different human DPF1 isoforms are known. Human DPF1 isoform a (NP_001128627.1) is encodable by the transcript variant 1 (NM_001135155.2). Human DPF1 isoform b (NP_004638.2) is encodable by the transcript variant 2 (NM_004647.3). Human DPF1 isoform c (NP_001128628.1) is encodable by the transcript variant 3 (NM_001135156.2). Human DPF1 isoform d (NP_001276907.1) is encodable by the transcript variant 4 (NM_001289978.1). Human DPF1 isoform e (NP_001350508.1) is encodable by the transcript variant 5 (NM_001363579.1). Nucleic acid and polypeptide sequences of DPF1 orthologs in organisms other than humans are well known and include, for example, Rhesus monkey DPF1 (XM_015123830.1 and XP_014979316.1, XM_015123829.1 and XP_014979315.1, XM_015123835.1 and XP_014979321.1, XM_015123831.1 and XP_014979317.1, XM_015123833.1 and XP_014979319.1, and XM_015123832.1 and XP_014979318.1), cattle DPF1 (NM_001076855.1 and NP_001070323.1), mouse DPF1 (NM_013874.2 and NP_038902.1), rat DPF1 (NM_001105729.3 and NP_001099199.2), and tropical clawed frog DPF1 (NM_001097276.1 and NP_001090745.1). Representative sequences of DPF1 orthologs are presented below in Table 1.

[0117] Anti-DPF1 antibodies suitable for detecting DPF1 protein are well-known in the art and include, for example, antibody TA311193 (Origene), antibodies NBP2-13932 and NBP2-19518 (Novus Biologicals, Littleton, CO), antibodies ab199299, ab173160, and ab3940 (AbCam, Cambridge, MA), antibody Cat #PA5-61895 (ThermoFisher Scientific), antibody Cat #28-079 (ProSci, Poway, CA), etc. In addition, reagents are well-known for detecting DPF1. Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing DPF1 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-97084 and sc-143155 and CRISPR product #sc-409539 from Santa Cruz Biotechnology, RNAi products SR305389 and TL313388V, and CRISPR product KN213721 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding DPF1 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a DPF1 molecule encompassed by the present invention.

[0118] The term “DPF2” refers to Double PHD Fingers 2. DPF2 protein is a member of the d4 domain family, characterized by a zinc finger-like structural motif. It functions as a transcription factor which is necessary for the apoptotic response following deprivation of survival factors. It likely serves a regulatory role in rapid hematopoietic cell growth and turnover. This gene is considered a candidate gene for multiple endocrine neoplasia type I, an inherited cancer syndrome involving multiple parathyroid, enteropancreatic, and pituitary tumors. DPF2 is a transcription factor required for the apoptosis response following survival factor withdrawal from myeloid cells. DPF2 also has a role in the development and maturation of lymphoid cells. Human DPF2 protein has 391 amino acids and a molecular mass of 44155 Da.

[0119] The term “DPF2” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human DPF2 cDNA and human DPF2 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human DPF2 isoforms are known. Human DPF2 isoform 1 (NP_006259.1) is encodable by the transcript variant 1 (NM_006268.4). Human DPF2 isoform 2 (NP_001317237.1) is encodable by the transcript variant 2 (NM_001330308.1). Nucleic acid and polypeptide sequences of DPF2 orthologs in organisms other than humans are well known and include, for example, chimpanzee DPF2 (NM_001246651.1 and NP_001233580.1), Rhesus monkey DPF2 (XM_002808062.2 and XP_002808108.2, and XM_015113800.1 and XP_014969286.1), dog DPF2 (XM_861495.5 and XP_866588.1, and XM_005631484.3 and XP_005631541.1), cattle DPF2 (NM_001100356.1 and NP_001093826.1), mouse DPF2 (NM_001291078.1 and NP_001278007.1, and NM_011262.5 and NP_035392.1), rat DPF2 (NM_001108516.1 and NP_001101986.1), chicken DPF2 (NM_204331.1 and NP_989662.1), tropical clawed frog DPF2 (NM_001197172.2 and NP_001184101.1), and zebrafish DPF2 (NM_001007152.1 and NP_001007153.1). Representative sequences of DPF2 orthologs are presented below in Table 1.

[0120] Anti-DPF2 antibodies suitable for detecting DPF2 protein are well-known in the art and include, for example, antibody TA312307 (Origene), antibodies NBP1-76512 and NBP1-87138 (Novus Biologicals, Littleton, CO), antibodies ab134942, ab232327, and ab227095 (AbCam, Cambridge, MA), etc. In addition, reagents are well-known for detecting DPF2. A clinical test of DPF2 for hereditary disese is available with the test ID no. GTR000536833.2 in NIH Genetic Testing Registry (GTR®), offered by Fulgent Genetics Clinical Diagnostics Lab (Temple City, CA). Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing DPF2 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-97031 and sc-143156 and CRISPR product #sc-404801-KO-2 from Santa Cruz Biotechnology, RNAi products SR304035 and TL313387V, and CRISPR product KN202364 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding DPF2 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a DPF2 molecule encompassed by the present invention.

[0121] The term "DPF3" refers to Double PHD Fingers 3, a member of the D4 protein family. The encoded protein is a transcription regulator that binds acetylated histones and is a component of the BAF chromatin remodeling complex. DPF3 belongs to the neuron-specific chromatin remodeling complex (nBAF complex). During neural development a switch from a stem/progenitor to a post-mitotic chromatin remodeling mechanism occurs as neurons exit the cell cycle and become committed to their adult state. The transition from proliferating neural stem/progenitor cells to post-mitotic neurons requires a switch in subunit composition of the npBAF and nBAF complexes. As neural progenitors exit mitosis and differentiate into neurons, npBAF complexes which contain ACTL6A/BAF53A and PHF10/BAF45A, are exchanged for homologous alternative ACTL6B/BAF53B and DPF1/BAF45B or DPF3/BAF45C subunits in neuron-specific complexes (nBAF). The npBAF complex is essential for the self-renewal/proliferative capacity of the multipotent neural stem cells. The nBAF complex along with CREST plays a role regulating the activity of genes essential for dendrite growth (By similarity). DPF3 is a muscle-specific component of the BAF complex, a multiprotein complex involved in transcriptional activation and repression of select genes by chromatin remodeling (alteration of DNA-nucleosome topology). DPF3 specifically binds acetylated lysines on histone 3 and 4 (H3K14ac, H3K9ac, H4K5ac, H4K8ac, H4K12ac, H4K16ac). In the complex, DPF3 acts as a tissue-specific anchor between histone acetylations and methylations and chromatin remodeling. DPF3 plays an essential role in heart and skeletal muscle development. Human DPF3 protein has 378 amino acids and a molecular mass of 43084 Da. The PHD-type zinc fingers of DPF3 mediate its binding to acetylated histones. DPF3 belongs to the requiem/DPF family.
[0122] The term "DPF3" is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human DPF3 cDNA and human DPF3 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, four different human DPF3 isoforms are known. Human DPF3 isoform 1 (NP_036206.3) is encodable by the transcript variant 1 (NM_012074.4). Human DPF3 isoform 2 (NP_001267471.1) is encodable by the transcript variant 2 (NM_001280542.1). Human DPF3 isoform 3 (NP_001267472.1) is encodable by the transcript variant 3 (NM_001280543.1). Human DPF3 isoform 4 (NP_001267473.1) is encodable by the transcript variant 4 (NM_001280544.1). Nucleic acid and polypeptide sequences of DPF3 orthologs in organisms other than humans are well known and include, for example, chimpanzee DPF3 (XM_016926314.2 and XP_016781803.1, XM_016926316.2 and XP_016781805.1, and XM_016926315.2 and XP_016781804.1), dog DPF3 (XM_014116039.1 and XP_013971514.1), mouse DPF3 (NM_001267625.1 and NP_001254554.1, NM_001267626.1 and NP_001254555.1, and NM_058212.2 and NP_478119.1), chicken DPF3 (NM_204639.2 and NP_989970.1), tropical clawed frog DPF3 (NM_001278413.1 and NP_001265342.1), and zebrafish DPF3 (NM_001111169.1 and NP_001104639.1). Representative sequences of DPF3 orthologs are presented below in Table 1.
[0123] Anti-DPF3 antibodies suitable for detecting DPF3 protein are well-known in the art and include, for example, antibody TA335655 (Origene), antibodies NBP2-49494 and NBP2-14910 (Novus Biologicals, Littleton, CO), antibodies ab180914, ab127703, and ab85360 (AbCam, Cambridge, MA), antibody PA5-38011 (ThermoFisher Scientific), antibody Cat #7559 (ProSci, Poway, CA), etc. In addition, reagents are well-known for detecting DPF3. Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing DPF3 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-97031 and sc-92150 and CRISPR product #sc-143157 from Santa Cruz Biotechnology, RNAi products SR305368 and TL313386V, and CRISPR product KN218937 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding DPF3 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a DPF3 molecule encompassed by the present invention.
[0124] The term "ACTL6A" refers to Actin Like 6A, a family member of actin-related proteins (ARPs), which share significant amino acid sequence identity to conventional actins. Both actins and ARPs have an actin fold, which is an ATP-binding cleft, as a common feature. The ARPs are involved in diverse cellular processes, including vesicular transport, spindle orientation, nuclear migration and chromatin remodeling. This gene encodes a 53 kDa subunit protein of the BAF (BRG1/brm-associated factor) complex in mammals, which is functionally related to SWI/SNF complex in *S. cerevisiae* and *Drosophila*; the latter is thought to facilitate transcriptional activation of specific genes by antagonizing chromatin-mediated transcriptional repression. Together with beta-actin, it is required for maximal ATPase activity of BRG1, and for the association of the BAF complex with chromatin/matrix. ACTL6A is a component of SWI/SNF chromatin remodeling complexes that carry out key enzymatic activities, changing chromatin structure by altering DNA-histone contacts within a nucleosome in an ATP-dependent manner. ACTL6A is required for maximal ATPase activity of SMARCA4/BRG1/BAF190A and for association of the SMARCA4/BRG1/BAF190A containing remodeling complex BAF with chromatin/nuclear matrix. ACTL6A belongs to the neural progenitors-specific chromatin remodeling complex (npBAF complex) and is required for the proliferation of neural progenitors. During neural development a switch from a stem/progenitor to a post-mitotic chromatin remodeling mechanism occurs as neurons exit the cell cycle and become committed to their adult state.

The transition from proliferating neural stem/progenitor cells to post-mitotic neurons requires a switch in subunit composition of the npBAF and nBAF complexes. As neural progenitors exit mitosis and differentiate into neurons, npBAF complexes which contain ACTL6A/BAF53A and PHF10/BAF45A, are exchanged for homologous alternative ACTL6B/BAF53B and DPF1/BAF45B or DPF3/BAF45C subunits in neuron-specific complexes (nBAF). The npBAF complex is essential for the self-renewal/proliferative capacity of the multipotent neural stem cells. The nBAF complex along with CREST plays a role regulating the activity of genes essential for dendrite growth. ACTL6A is a component of the NuA4 histone acetyltransferase (HAT) complex which is involved in transcriptional activation of select genes principally by acetylation of nucleosomal histones H4 and H2A. This modification may both alter nucleosome—DNA interactions and promote interaction of the modified histones with other proteins which positively regulate transcription. This complex may be required for the activation of transcriptional programs associated with oncogene and proto-oncogene mediated growth induction, tumor suppressor mediated growth arrest and replicative senescence, apoptosis, and DNA repair. NuA4 may also play a direct role in DNA repair when recruited to sites of DNA damage. Putative core component of the chromatin remodeling IN080 complex which is involved in transcriptional regulation, DNA replication and probably DNA repair. Human ACTL6A protein has 429 amino acids and a molecular mass of 47461 Da.

[0125] The term “ACTL6A” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human ACTL6A cDNA and human ACTL6A protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human ACTL6A isoforms are known. Human ACTL6A isoform 1 (NP_004292.1) is encodable by the transcript variant 1 (NM_004301.4). Human ACTL6A isoform 2 (NP_817126.1 and NP_829888.1) is encodable by the transcript variant 2 (NM_177989.3) and transcript variant 3 (NM_178042.3). Nucleic acid and polypeptide sequences of ACTL6A orthologs in organisms other than humans are well known and include, for example, chimpanzee ACTL6A (NM_001271671.1 and NP_001258600.1), Rhesus monkey ACTL6A (NM_001104559.1 and NP_001098029.1), cattle ACTL6A (NM_001105035.1 and NP_001098505.1), mouse ACTL6A (NM_019673.2 and NP_062647.2), rat ACTL6A (NM_001039033.1 and NP_001034122.1), chicken ACTL6A (XM_422784.6 and XP_422784.3), tropical clawed frog ACTL6A (NM_204006.1 and NP_989337.1), and zebrafish ACTL6A (NM_173240.1 and NP_775347.1). Representative sequences of ACTL6A orthologs are presented below in Table 1.

[0126] Anti-ACTL6A antibodies suitable for detecting ACTL6A protein are well-known in the art and include, for example, antibody TA345058 (Origene), antibodies NB100-61628 and NBP2-55376 (Novus Biologicals, Littleton, CO), antibodies ab131272 and ab189315 (AbCam, Cambridge, MA), antibody 702414 (ThermoFisher Scientific), antibody Cat #45-314 (ProSci, Poway, CA), etc. In addition, reagents are well-known for detecting ACTL6A. Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing ACTL6A expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-60239 and sc-60240 and CRISPR product #sc-403200-KO-2 from Santa Cruz Biotechnology, RNAi products SR300052 and TL306860V, and CRISPR product KN201689 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding ACTL6A molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe an ACTL6A molecule encompassed by the present invention.

[0127] The term “β-Actin” refers to Actin Beta. This gene encodes one of six different actin proteins. Actins are highly conserved proteins that are involved in cell motility, structure, integrity, and intercellular signaling. The encoded protein is a major constituent of the contractile apparatus and one of the two nonmuscle cytoskeletal actins that are ubiquitously expressed. Mutations in this gene cause Baraitser-Winter syndrome 1, which is characterized by intellectual disability with a distinctive facial appearance in human patients. Numerous pseudogenes of this gene have been identified throughout the human genome. Actins are highly conserved proteins that are involved in various types of cell motility and are ubiquitously expressed in all eukaryotic cells. Actin is found in two main states: G-actin is the globular monomeric form, whereas F-actin forms helical polymers. Both G- and F-actin are intrinsically flexible structures. Human β-Actin protein has 375 amino acids and a molecular mass of 41737 Da. The binding partners of β-Actin include, e.g., CPNE1, CPNE4, DHX9, GCSAM, ERBB2, XPO6, and EMD.

[0128] The term “β-Actin” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human β-Actin cDNA and human β-Actin protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, human β-Actin (NP_001092.1) is encodable by the transcript (NM_001101.4). Nucleic acid and polypeptide sequences of $3-Actin orthologs in organisms other than humans are well known and include, for example, chimpanzee β-Actin (NM_001009945.1 and NP_001009945.1), Rhesus monkey β-Actin (NM_001033084.1 and NP_001028256.1), dog β-Actin (NM_001195845.2 and NP_001182774.2), cattle β-Actin (NM_173979.3 and NP_776404.2), mouse β-Actin (NM_007393.5 and NP_031419.1), rat β-Actin (NM_031144.3 and NP_112406.1), chicken β-Actin (NM_205518.1 and NP_990849.1), and tropical clawed frog β-Actin (NM_213719.1 and NP_998884.1). Representative sequences of β-Actin orthologs are presented below in Table 1.

[0129] Anti-β-Actin antibodies suitable for detecting β-Actin protein are well-known in the art and include, for example, antibody TA353557 (Origene), antibodies NB600-501 and NB600-503 (Novus Biologicals, Littleton, CO), antibodies ab8226 and ab8227 (AbCam, Cambridge, MA), antibody AM4302 (ThermoFisher Scientific), antibody Cat #PM-7669-biotin (ProSci, Poway, CA), etc. In addition, reagents are well-known for detecting $-Actin. Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing β-Actin expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-108069 and sc-108070 and CRISPR product #sc-400000-KO-2 from Santa Cruz Biotechnology, RNAi products SR300047 and TL314976V, and CRISPR product KN203643 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding β-Actin molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a β-Actin molecule encompassed by the present invention.
[0130] The term "BCL7A" refers to BCL Tumor Suppressor 7A. This gene is directly involved, with Myc and IgH, in a three-way gene translocation in a Burkitt lymphoma cell line. As a result of the gene translocation, the N-terminal region of the gene product is disrupted, which is thought to be related to the pathogenesis of a subset of high-grade B cell non-Hodgkin lymphoma. The N-terminal segment involved in the translocation includes the region that shares a strong sequence similarity with those of BCL7B and BCL7C. Diseases associated with BCL7A include Lymphoma and Burkitt Lymphoma. An important paralog of this gene is BCL7C. Human BCL7A protein has 210 amino acids and a molecular mass of 22810 Da.
[0131] The term "BCL7A" is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human BCL7A cDNA and human BCL7A protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human BCL7A isoforms are known. Human BCL7A isoform a (NP_066273.1) is encodable by the transcript variant 1 (NM_020993.4). Human BCL7A isoform b (NP_001019979.1) is encodable by the transcript variant 2 (NM_001024808.2). Nucleic acid and polypeptide sequences of BCL7A orthologs in organisms other than humans are well known and include, for example, chimpanzee BCL7A (XM_009426452.3 and XP_009424727.2, and XM_016924434.2 and XP_016779923.1), Rhesus monkey BCL7A (XM_015153012.1 and XP_015008498.1, and XM_015153013.1 and XP_015008499.1), dog BCL7A (XM_543381.6 and XP_543381.2, and XM_854760.5 and XP_859853.1), cattle BCL7A (XM_024977701.1 and XP_024833469.1, and XM_024977700.1 and XP_024833468.1), mouse BCL7A (NM_029850.3 and NP_084126.1), rat BCL7A (XM_017598515.1 and XP_017454004.1), chicken BCL7A (XM_004945565.3 and XP_004945622.1, and XM_415148.6 and XP_415148.2), tropical clawed frog BCL7A (NM_001006871.1 and NP_001006872.1), and zebrafish BCL7A (NM_212560.1 and NP_997725.1). Representative sequences of BCL7A orthologs are presented below in Table 1.
[0132] Anti-BCL7A antibodies suitable for detecting BCL7A protein are well-known in the art and include, for example, antibody TA344744 (Origene), antibodies NBP1-30941 and NBP1-91696 (Novus Biologicals, Littleton, CO), antibodies ab137362 and ab1075 (AbCam, Cambridge, MA), antibody PA5-27123 (ThermoFisher Scientific), antibody Cat #45-325 (ProSci, Poway, CA), etc. In addition, reagents are well-known for detecting BCL7A. Multiple clinical tests of BCL7A are available in NIH Genetic Testing Registry (GTR®) (e.g., GTR Test ID: GTR000541481.2, offered by Fulgent Clinical Diagnostics Lab (Temple City, CA)). Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing BCL7A expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-96136 and sc-141671 and CRISPR product #sc-410702 from Santa Cruz Biotechnology, RNAi products SR300417 and TL314490V, and CRISPR product KN210489 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding BCL7A molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a BCL7A molecule encompassed by the present invention.
[0133] The term "BCL7B" refers to BCL Tumor Suppressor 7B, a member of the BCL7 family including BCL7A, BCL7B and BCL7C proteins. This member is BCL7B, which contains a region that is highly similar to the N-terminal segment of BCL7A or BCL7C proteins. The BCL7A protein is encoded by the gene known to be directly involved in a three-way gene translocation in a Burkitt lymphoma cell line. This gene is located at a chromosomal region commonly deleted in Williams syndrome. This gene is highly conserved from *C. elegans* to human. BCL7B is a positive regulator of apoptosis. BCL7B plays a role in the Wnt signaling pathway, negatively regulating the expression of Wnt signaling components CTNNB1 and HMGA1 (Uehara et al. (2015) *PLoS Genet* 11(1):e1004921). BCL7B is involved in cell cycle progression, maintenance of the nuclear structure and stem cell differentiation (Uehara et al. (2015) *PLoS Genet* 11(1):e1004921). It plays a role in lung tumor development or progression. Human BCL7B protein has 202 amino acids and a molecular mass of 22195 Da.
[0134] The term "BCL7B" is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human BCL7B cDNA and human BCL7B protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, three different human BCL7B isoforms are known. Human BCL7B isoform 1 (NP_001698.2) is encodable by the transcript variant

1 (NM_001707.3). Human BCL7B isoform 2 (NP_001184173.1) is encodable by the transcript variant 2 (NM_001197244.1). Human BCL7B isoform 3 (NP_001287990.1) is encodable by the transcript variant 3 (NM_001301061.1). Nucleic acid and polypeptide sequences of BCL7B orthologs in organisms other than humans are well known and include, for example, chimpanzee BCL7B (XM_003318671.3 and XP_003318719.1, and XM_003318672.3 and XP_003318720.1), Rhesus monkey BCL7B (NM_001194509.1 and NP_001181438.1), dog BCL7B (XM_546926.6 and XP_546926.1, and XM_005620975.2 and XP_005621032.1), cattle BCL7B (NM_001034775.2 and NP_001029947.1), mouse BCL7B (NM_009745.2 and NP_033875.2), chicken BCL7B (XM_003643231.4 and XP_003643279.1, XM_004949975.3 and XP_004950032.1, and XM_025142155.1 and XP_024997923.1), tropical clawed frog BCL7B (NM_001103072.1 and NP_001096542.1), and zebrafish BCL7B (NM_001006018.1 and NP_001006018.1, and NM_213165.1 and NP_998330.1). Representative sequences of BCL7B orthologs are presented below in Table 1.

[0135] Anti-BCL7B antibodies suitable for detecting BCL7B protein are well-known in the art and include, for example, antibody TA809485 (Origene), antibodies H00009275-MO1 and NBP2-34097 (Novus Biologicals, Littleton, CO), antibodies ab130538 and ab172358 (AbCam, Cambridge, MA), antibody MA527163 (ThermoFisher Scientific), antibody Cat #58-996 (ProSci, Poway, CA), etc. In addition, reagents are well-known for detecting BCL7B. Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing BCL7B expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-89728 and sc-141672 and CRISPR product #sc-411262 from Santa Cruz Biotechnology, RNAi products SR306141 and TL306418V, and CRISPR product KN201696 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding BCL7B molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a BCL7B molecule encompassed by the present invention.

[0136] The term “BCL7C” refers to BCL Tumor Suppressor 7C, a member of the BCL7 family including BCL7A, BCL7B and BCL7C proteins. This gene is identified by the similarity of its product to the N-terminal region of BCL7A protein. BCL7C may play an anti-apoptotic role. Diseases associated with BCL7C include Lymphoma. Human BCL7C protein has 217 amino acids and a molecular mass of 23468 Da.

[0137] The term “BCL7C” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human BCL7C cDNA and human BCL7C protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human BCL7C isoforms are known. Human BCL7C isoform 1 (NP_001273455.1) is encodable by the transcript variant 1 (NM_001286526.1). Human BCL7C isoform 2 (NP_004756.2) is encodable by the transcript variant 2 (NM_004765.3). Nucleic acid and polypeptide sequences of BCL7C orthologs in organisms other than humans are well known and include, for example, chimpanzee BCL7C (XM_016929717.2 and XP_016785206.1, XM_016929716.2 and XP_016785205.1, and XM_016929718.2 and XP_016785207.1), Rhesus monkey BCL7C (NM_001265776.2 and NP_001252705.1), cattle BCL7C (NM_001099722.1 and NP_001093192.1), mouse BCL7C (NM_001347652.1 and NP_001334581.1, and NM_009746.2 and NP_033876.1), and rat BCL7C (NM_001106298.1 and NP_001099768.1). Representative sequences of BCL7C orthologs are presented below in Table 1.

[0138] Anti-BCL7C antibodies suitable for detecting BCL7C protein are well-known in the art and include, for example, antibody TA347083 (Origene), antibodies NBP2-15559 and NBP1-86441 (Novus Biologicals, Littleton, CO), antibodies ab126944 and ab231278 (AbCam, Cambridge, MA), antibody PA5-30308 (ThermoFisher Scientific), etc. In addition, reagents are well-known for detecting BCL7C. Multiple clinical tests of BCL7C are available in NIH Genetic Testing Registry (GTR®) (e.g., GTR Test ID: GTR000540637.2, offered by Fulgent Clinical Diagnostics Lab (Temple City, CA)). Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing BCL7C expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-93022 and sc-141673 and CRISPR product #sc-411261 from Santa Cruz Biotechnology, RNAi products SR306140 and TL315552V, and CRISPR product KN205720 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding BCL7C molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a BCL7C molecule encompassed by the present invention.

[0139] The term “SMARCA4” refers to SWI/SNF related, matrix associated, actin dependent regulator of chromatin, subfamily a, member 4, a member of the SWI/SNF family of proteins and is highly similar to the brahma protein of *Drosophila*. Members of this family have helicase and ATPase activities and are thought to regulate transcription of certain genes by altering the chromatin structure around those genes. The encoded protein is part of the large ATP-dependent chromatin remodeling complex SNF/SWI, which is required for transcriptional activation of genes normally repressed by chromatin. In addition, this protein can bind BRCA1, as well as regulate the expression of the tumorigenic protein CD44. Mutations in this gene cause rhabdoid tumor predisposition syndrome type 2. SMARCA4 is a component of SWI/SNF chromatin remodeling complexes that carry out key enzymatic activities, changing chromatin structure by altering DNA-histone contacts within a nucleosome in an ATP-dependent manner. SMARCA4 is a component of the CREST-BRG1 complex, a multiprotein complex that regulates promoter activation by orchestrating a calcium-dependent release of a repressor complex and a recruitment of an activator complex. In resting neurons, transcription of the c-FOS promoter is inhibited by BRG1-dependent recruitment of a phospho-RB1-HDAC repressor complex. Upon calcium influx, RB1 is dephosphorylated by calcineurin, which leads to release of the repressor complex. At the same time, there is increased recruitment of CREBBP to the promoter by a CREST-dependent mechanism, which leads to transcriptional activation. The CREST-BRG1 complex also binds to the NR2B promoter, and activity-dependent induction of NR2B expression involves a release of HDAC1 and recruitment of CREBBP. SMARCA4 belongs to the neural progenitors-specific chromatin remodeling complex (npBAF complex) and the neuron-specific chromatin remodeling complex (nBAF complex). During neural development a switch from a stem/progenitor to a postmitotic chromatin remodeling mechanism occurs as neurons exit the cell cycle and become committed to their adult state. The transition from proliferating neural stem/progenitor cells to postmitotic neurons requires a switch in subunit composition of the npBAF and nBAF complexes. As neural progenitors exit mitosis and differentiate into neurons, npBAF complexes which contain ACTL6A/BAF53A and PHF10/BAF45A, are exchanged for homologous alternative ACTL6B/BAF53B and DPF1/BAF45B or DPF3/BAF45C subunits in neuron-specific complexes (nBAF). The npBAF complex is essential for the self-renewal/proliferative capacity of the multipotent neural stem cells. The nBAF complex along with CREST plays a role regulating the activity of genes essential for dendrite growth. SMARCA4/BAF190A promote neural stem cell self-renewal/proliferation by enhancing Notch-dependent proliferative signals, while concurrently making the neural stem cell insensitive to SHH-dependent differentiating cues. SMARCA4 acts as a corepressor of ZEB1 to regulate E-cadherin transcription and is required for induction of epithelial-mesenchymal transition (EMT) by ZEB1. Human SMARCA4 protein has 1647 amino acids and a molecular mass of 184646 Da. The known binding partners of SMARCA4 include, e.g., PHF10/BAF45A, MYOG, IKFZ1, ZEB1, NR3C1, PGR, SMARD1, TOPBP1 and ZMIM2/ZIMP7.

[0140] The term “SMARCA4” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human SMARCA4 cDNA and human SMARCA4 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, six different human SMARCA4 isoforms are known. Human SMARCA4 isoform A (NP_001122321.1) is encodable by the transcript variant 1 (NM_001128849.1). Human SMARCA4 isoform B (NP_001122316.1 and NP_003063.2) is encodable by the transcript variant 2 (NM_001128844.1) and the transcript variant 3 (NM_003072.3). Human SMARCA4 isoform C (NP_001122317.1) is encodable by the transcript variant 4 (NM_001128845.1). Human SMARCA4 isoform D (NP_001122318.1) is encodable by the transcript variant 5 (NM_001128846.1). Human SMARCA4 isoform E (NP_001122319.1) is encodable by the transcript variant 6 (NM_001128847.1). Human SMARCA4 isoform F (NP_001122320.1) is encodable by the transcript variant 7 (NM_001128848.1). Nucleic acid and polypeptide sequences of SMARCA4 orthologs in organisms other than humans are well known and include, for example, Rhesus monkey SMARCA4 (XM_015122901.1 and XP_014978387.1, XM_015122902.1 and XP_014978388.1, XM_015122903.1 and XP_014978389.1, XM_015122906.1 and XP_014978392.1, XM_015122905.1 and XP_014978391.1, XM_015122904.1 and XP_014978390.1, XM_015122907.1 and XP_014978393.1, XM_015122909.1 and XP_014978395.1, and XM_015122910.1 and XP_014978396.1), cattle SMARCA4 (NM_001105614.1 and NP_001099084.1), mouse SMARCA4 (NM_001174078.1 and NP_001167549.1, NM 011417.3 and NP_035547.2, NM_001174079.1 and NP_001167550.1, NM_001357764.1 and NP_001344693.1), rat SMARCA4 (NM_134368.1 and NP_599195.1), chicken SMARCA4 (NM_205059.1 and NP_990390.1), and zebrafish SMARCA4 (NM_181603.1 and NP_853634.1). Representative sequences of SMARCA4 orthologs are presented below in Table 1.

[0141] Anti-SMARCA4 antibodies suitable for detecting SMARCA4 protein are well-known in the art and include, for example, antibody AM26021PU-N(Origene), antibodies NB100-2594 and AF5738 (Novus Biologicals, Littleton, CO), antibodies ab110641 and ab4081 (AbCam, Cambridge, MA), antibody 720129 (ThermoFisher Scientific), antibody 7749 (ProSci), etc. In addition, reagents are well-known for detecting SMARCA4. Multiple clinical tests of SMARCA4 are available in NIH Genetic Testing Registry (GTR®) (e.g., GTR Test ID: GTR000517106.2, offered by Fulgent Clinical Diagnostics Lab (Temple City, CA)). Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing SMARCA4 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-29827 and sc-44287 and CRISPR product #sc-400168 from Santa Cruz Biotechnology, RNAi products SR321835 and TL309249V, and CRISPR product KN219258 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding SMARCA4 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a SMARCA4 molecule encompassed by the present invention.

[0142] The term “SS18” refers to SS18, NBAF Chromatin Remodeling Complex Subunit. SS18 functions synergistically with RBM14 as a transcriptional coactivator. Isoform 1 and isoform 2 of SS18 function in nuclear receptor coactivation. Isoform 1 and isoform 2 of SS18 function in general transcriptional coactivation. Diseases associated with SS18 include Sarcoma, Synovial Cell Sarcoma. Among its related pathways are transcriptional misregulation in cancer and chromatin regulation/acetylation. Human SS18 protein has 418 amino acids and a molecular mass of 45929 Da. The known binding partners of SS18 include, e.g., MLLT10 and RBM14 isoform 1.

[0143] The term "SS18" is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human SS18 cDNA and human SS18 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, three different human SS18 isoforms are known. Human SS18 isoform 1 (NP_001007560.1) is encodable by the transcript variant 1 (NM_001007559.2). Human SS18 isoform 2 (NP_005628.2) is encodable by the transcript variant 2 (NM_005637.3). Human SS18 isoform 3 (NP_001295130.1) is encodable by the transcript variant 3 (NM_001308201.1). Nucleic acid and polypeptide sequences of SS18 orthologs in organisms other than humans are well known and include, for example, dog SS18 (XM_005622940.3 and XP_005622997.1, XM_537295.6 and XP_537295.3, XM_003434925.4 and XP_003434973.1, and XM_005622941.3 and XP_005622998.1), mouse SS18 (NM_009280.2 and NP_033306.2, NM_001161369.1 and NP_001154841.1, NM_001161370.1 and NP_001154842.1, and NM_001161371.1 and NP_001154843.1), rat SS18 (NM_001100900.1 and NP_001094370.1), chicken SS18 (XM_015277943.2 and XP_015133429.1, and XM_015277944.2 and XP_015133430.1), tropical clawed frog SS18 (XM_012964966.1 and XP_012820420.1, XM_018094711.1 and XP_017950200.1, XM_012964964.2 and XP_012820418.1, and XM_012964965.2 and XP_012820419.1), and zebrafish SS18 (NM_001291325.1 and NP_001278254.1, and NM_199744.2 and NP_956038.1). Representative sequences of BRD7 orthologs are presented below in Table 1.

[0144] Anti-SS18 antibodies suitable for detecting SS18 protein are well-known in the art and include, for example, antibody TA314572 (Origene), antibodies NBP2-31777 and NBP2-31612 (Novus Biologicals, Littleton, CO), antibodies ab179927 and ab89086 (AbCam, Cambridge, MA), antibody PA5-63745 (ThermoFisher Scientific), etc. In addition, reagents are well-known for detecting SS18. Multiple clinical tests of SS18 are available in NIH Genetic Testing Registry (GTR®) (e.g., GTR Test ID: GTR000546059.2, offered by Fulgent Clinical Diagnostics Lab (Temple City, CA)). Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing SS18 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-38449 and sc-38450 and CRISPR product #sc-401575 from Santa Cruz Biotechnology, RNAi products SR304614 and TL309102V, and CRISPR product KN215192 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding SS18 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a SS18 molecule encompassed by the present invention.

[0145] The term "SSX" refers to a family of highly homologous synovial sarcoma X (SSX) breakpoint proteins. The mammalian SSX family proteins include, e.g., human SSX1-9. These proteins can function as transcriptional repressors. They are also capable of eliciting spontaneous humoral and cellular immune responses in cancer patients, and are useful targets in cancer vaccine-based immunotherapy. SSX1, SSX2 and SSX4 family members, have been involved in t(X;18)(p11.2;q11.2) translocations that are characteristically found in all synovial sarcomas. This translocation results in the fusion of the synovial sarcoma translocation gene on chromosome 18 to one of the SSX genes on chromosome X. The encoded hybrid proteins are responsible for transforming activity. While some of the related SSX genes are involved in t(X;18)(p11.2;q11.2) translocations that are characteristically found in all synovial sarcomas, SSX3, SSX5, and SSX7 do not appear to be involved in such translocations. SSX6, or SSX6P is classified as a pseudogene because a splice donor in the 3′ UTR has changed compared to other family numbers, rendering the transcript a candidate for nonsense-mediated mRNA decay (NMD). SSX8, or SSX8P (SSX Family Member 8, Pseudogene) is a Pseudogene. SSX9, or SSX9P (SSX Family Member 9, Pseudogene) is a Pseudogene. SSX C-terminus comprises a 6-amino acid basic region and a 7-amino adic acidic region. The. representative basic regions and acidic regions for SSX1 to SSX9 are shown in FIG. **3**D.

[0146] The term "SSX1" is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human SSX1 cDNA and human SSX1 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human SSX1 transcript variants are known. Human transcript variant 1 (NM_001278691.2) and human transcript variant 2 (NM_005635.4) encode the same human SSX1 protein (NP_001265620.1 and NP_005626.1). Transcript variant 1 represents the longer transcript. Transcript variant 2 differs in the 5′ UTR compared to variant 1. Nucleic acid and polypeptide sequences of SSX1 orthologs in organisms other than humans are well known and include, for example, monkey SS18 (XM_017854812.1 and XP_017710301.1), and chimpanzee SS18 (XM_016944028.1 and XP_016799517.1, XM_016944029.1 and XP_016799518.1, XM_016944031.1 and XP_016799520.1, and XM_016944030.1 and XP_016799519.1). A representative SSX1 has 188 amino acids with a molecular mass of 21931 Da. Representative sequences of SSX1 orthologs are presented below in Table 1.

[0147] Anti-SSX1 antibodies suitable for detecting SSX1 protein are well-known in the art and include, for example, antibodies CF502523 and CF502693 (Origene), antibodies NBP2-00614 and H00006756-MO1 (Novus Biologicals, Littleton, CO), antibodies ab206839 and ab234815 (AbCam, Cambridge, MA), antibody MA5-25511 (ThermoFisher Scientific), etc. In addition, reagents are well-known for detecting SSX1. Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing SSX1 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-44120 and sc-44120-SH and CRISPR product #sc-403551 from Santa Cruz Biotechnology, RNAi products SR304610 and TL309084, and CRISPR product KN401600 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding SSX1 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a SSX1 molecule encompassed by the present invention.

[0148] The term "SSX2" is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human SSX2 cDNA and human SSX2 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, three different human SSX2 transcript variants are known. Human SSX2 isoform 1 (NP_003138.3) is encodable by the transcript variant 1 (NM_003147.5). Human SSX2 isoform 2 (NP_783629.1) is encodable by the transcript variant 2 (NM_175698.2). Human SSX2 isoform 3 (NP_001265626.1) is encodable by the transcript variant 3 (NM_001278697.1). SSX2 has an identical duplicate, SSX2B (GeneID: 727837), located about 45 kb downstream in the opposite orientation on chromosome X. Three different human SSX2B transcript variants are known. Human SSX2B isoform 1 (NP_001265630.1) is encodable by the transcript variant 1 (NM_001278701.2). Human SSX2B isoform 2 (NP_001157889.1) is encodable by the transcript variant 2 (NM_001164417.3). Human SSX2B isoform 3 (NP_001265631.1) is encodable by the transcript variant 3 (NM_001278702.2). Nucleic acid and polypeptide sequences of SSX2 orthologs in organisms other than humans are well known. Representative sequences of SSX2 orthologs are presented below in Table 1.

[0149] Anti-SSX2 antibodies suitable for detecting SSX2 protein are well-known in the art and include, for example, antibodies CF500618 and CF500620 (Origene), antibodies NBP1-48008 and H00006757-MO1 (Novus Biologicals, Littleton, CO), antibodies ab236415 and ab48571 (AbCam, Cambridge, MA), antibody MA5-24971 (ThermoFisher Scientific), etc. In addition, reagents are well-known for detecting SSX2. Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing SSX2 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA product #sc-38446 and CRISPR product #sc-417124 from Santa Cruz Biotechnology, RNAi products SR304611 and TL309083, and CRISPR product KN401214 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding SSX2 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a SSX2 molecule encompassed by the present invention.

[0150] The term "SSX4" is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human SSX4 cDNA and human SSX4 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human SSX4 transcript variants are known. Human SSX4 isoform 1 (NP_005627.1) is encodable by the transcript variant 1 (NM_005636.4). Human SSX4 isoform 2 (NP_783856.1) is encodable by the transcript variant 2 (NM_175729.1). Chromosome Xp11 contains a segmental duplication resulting in two identical copies of synovial sarcoma, X breakpoint 4, SSX4 and SSX4B, in tail-to-tail orientation. Two different human SSX4B transcript variants are known. Human SSX4B isoform a (NP_001030004.1) is encodable by the transcript variant 1 (NM_001034832.3). Human SSX4B isoform 2 (NP_001035702.1) is encodable by the transcript variant 2 (NM_001040612.2). Nucleic acid and polypeptide sequences of SSX4 orthologs in organisms other than humans are well known, for example, dog putative protein SSX6-like (XM_005641306.2 and XP_005641363.1 and XM_022416309.1 and XP_022272017.1), cattle protein SSX1-like (XM_024988534.1 and XP_024844302.1), cattle synovial sarcoma, X breakpoint 5 (XM_024988283.1 and XP_024844051.1, and XM_024988284.1 and XP_024844052.1), and mouse synovial sarcoma, X member B, breakpoint 2 (NM_001001450.4 and NP_001001450.1, and NM_001134226.1 and NP_001127698.1). Representative sequences of SSX4 orthologs are presented below in Table 1.

[0151] Anti-SSX4 antibodies suitable for detecting SSX4 protein are well-known in the art and include, for example, antibodies TA339114 and TA339115 (Origene), antibodies H00006759-M02 and H00006759-B01P (Novus Biologicals, Littleton, CO), antibody ab172215 (AbCam, Cambridge, MA), antibody PA5-41117 (ThermoFisher Scientific), etc. In addition, reagents are well-known for detecting SSX4. Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing SSX4 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-106732 and sc-106800 and CRISPR product #sc-416410 from Santa Cruz Biotechnology, RNAi products SR304613 and TL309081, and CRISPR product KN422659 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding SSX4 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a SSX4 molecule encompassed by the present invention.

[0152] The term “SSX3” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human SSX3 cDNA and human SSX3 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, Human SSX3 (NP_066294.1) is encodable by the transcript (NM_021014.4). Nucleic acid and polypeptide sequences of SSX3 orthologs in organisms other than humans are well known, for example, monkey SSX3 (XM_002806224.3 and XP_002806270.1). Representative sequences of SSX3 orthologs are presented below in Table 1.
[0153] Anti-SSX3 antibodies suitable for detecting SSX3 protein are well-known in the art and include, for example, antibody TA345316 (Origene), antibodies H00010214-M03 and H00010214-B01P (Novus Biologicals, Littleton, CO), antibody ab160884 (AbCam, Cambridge, MA), antibodies MA5-24431 and PA5-69016 (ThermoFisher Scientific), etc. In addition, reagents are well-known for detecting SSX3. Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing SSX3 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-38447 and sc-38447-SH and CRISPR product #sc-417585 from Santa Cruz Biotechnology, RNAi products SR306902 and TL301375, and CRISPR product KN403244 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding SSX3 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a SSX3 molecule encompassed by the present invention.
[0154] The term “SSX5” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human SSX5 cDNA and human SSX5 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human SSX5 transcript variants are known. Human SSX5 isoform 1 (NP_066295.3) is encodable by the transcript variant 1 (NM_021015.4). Human SSX5 isoform 2 (NP_783729.1) is encodable by the transcript variant 2 (NM_175723.1). Nucleic acid and polypeptide sequences of SSX5 orthologs in organisms other than humans are well known. Representative sequences of SSX5 orthologs are presented below in Table 1.
[0155] Anti-SSX5 antibodies suitable for detecting SSX5 protein are well-known in the art and include, for example, antibodies CF504221 and CF504223 (Origene), antibodies NBP2-01842 and H00006758-B01P (Novus Biologicals, Littleton, CO), antibodies PA5-92141 and MA5-25901 (ThermoFisher Scientific), etc. In addition, reagents are well-known for detecting SSX5. Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing SSX5 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-38448 and sc-38448-SH and CRISPR product #sc-403552 from Santa Cruz Biotechnology, RNAi products SR304612 and TL301374, and CRISPR product KN402208 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding SSX5 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a SSX5 molecule encompassed by the present invention.
[0156] The term “SSX7” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human SSX7 cDNA and human SSX7 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, Human SSX7 (NP_775494.1) is encodable by the transcript (NM_173358.2). Nucleic acid and polypeptide sequences of SSX7 orthologs in organisms other than humans are well known. Representative sequences of SSX7 orthologs are presented below in Table 1.
[0157] Anti-SSX7 antibodies suitable for detecting SSX7 protein are well-known in the art and include, for example, antibody TA339916 (Origene), antibody NBP1-79468 (Novus Biologicals, Littleton, CO), antibody PA5-49262 (ThermoFisher Scientific), etc. In addition, reagents are well-known for detecting SSX7. Moreover, mutilple siRNA, shRNA, CRISPR constructs for reducing SSX7 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA products #sc-106568 and sc-106568-SH and CRISPR product #sc-403553 from Santa Cruz Biotechnology, RNAi products SR316959 and TL301372, and CRISPR product KN413920 (Origene), and multiple CRISPR products from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding SSX7 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe a SSX7 molecule encompassed by the present invention.
[0158] The SS18-SSX fusion protein is formed by chromosomal translocation, which results in a fusion of SS18 protein with the C-terminal of the SSX family member (e.g., SSX1, SSX2, and SSX4). Many of these function as oncoproteins which play important roles in tumorgenesis. For example, the molecular hallmark of synovial sarcoma is a pathognomonic reciprocal translocation t(X;18)(p11; q11), leading to the fusion of SS18 (SYT) to one of the homologs SSX genes (most frequently SSX1 or SSX2, in rare cases SSX4), generating oncogenic SS18-SSX chimeric proteins. Representative sequences of SS18-SSX fusion proteins are presented below in Table 2.
[0159] Unless otherwise specified here within, the terms “antibody” and “antibodies” broadly encompass naturally-occurring forms of antibodies (e.g. IgG, IgA, IgM, IgE) and recombinant antibodies, such as single-chain antibodies,

chimeric and humanized antibodies and multi-specific antibodies, as well as fragments and derivatives of all of the foregoing, which fragments and derivatives have at least an antigenic binding site. Antibody derivatives may comprise a protein or chemical moiety conjugated to an antibody.

[0160] In addition, intrabodies are well-known antigen-binding molecules having the characteristic of antibodies, but that are capable of being expressed within cells in order to bind and/or inhibit intracellular targets of interest (Chen et al. (1994) *Human Gene Ther.* 5:595-601). Methods are well-known in the art for adapting antibodies to target (e.g., inhibit) intracellular moieties, such as the use of single-chain antibodies (scFvs), modification of immunoglobulin VL domains for hyperstability, modification of antibodies to resist the reducing intracellular environment, generating fusion proteins that increase intracellular stability and/or modulate intracellular localization, and the like. Intracellular antibodies can also be introduced and expressed in one or more cells, tissues or organs of a multicellular organism, for example for prophylactic and/or therapeutic purposes (e.g., as a gene therapy) (see, at least PCT Publs. WO 08/020079, WO 94/02610, WO 95/22618, and WO 03/014960; U.S. Pat. No. 7,004,940; Cattaneo and Biocca (1997) *Intracellular Antibodies: Development and Applications* (*Landes and* Springer-Verlag publs.); *Kontermann* (2004) *Methods* 34:163-170; Cohen et al. (1998) *Oncogene* 17:2445-2456; Auf der Maur et al. (2001) *FEBS Lett.* 508:407-412; Shaki-Loewenstein et al. (2005) *J. Immunol*. Meth. 303:19-39).

[0161] The term “antibody” as used herein also includes an “antigen-binding portion” of an antibody (or simply “antibody portion”). The term “antigen-binding portion”, as used herein, refers to one or more fragments of an antibody that retain the ability to specifically bind to an antigen (e.g., a biomarker polypeptide or fragment thereof). It has been shown that the antigen-binding function of an antibody can be performed by fragments of a full-length antibody. Examples of binding fragments encompassed within the term “antigen-binding portion” of an antibody include (i) a Fab fragment, a monovalent fragment consisting of the VL, VH, CL and CHi domains; (ii) a F(ab′).sub.2 fragment, a bivalent fragment comprising two Fab fragments linked by a disulfide bridge at the hinge region; (iii) a Fd fragment consisting of the VH and CHi domains; (iv) a Fv fragment consisting of the VL and VH domains of a single arm of an antibody, (v) a dAb fragment (Ward et al., (1989) *Nature* 341:544-546), which consists of a VH domain; and (vi) an isolated complementarity determining region (CDR). Furthermore, although the two domains of the Fv fragment, VL and VH, are coded for by separate genes, they can be joined, using recombinant methods, by a synthetic linker that enables them to be made as a single protein chain in which the VL and VH regions pair to form monovalent polypeptides (known as single chain Fv (scFv); see e.g., Bird et al. (1988) *Science* 242:423-426; and Huston et al. (1988) *Proc. Natl. Acad. Sci. USA* 85:5879-5883; and Osbourn et al. 1998, Nature Biotechnology 16: 778). Such single chain antibodies are also intended to be encompassed within the term “antigen-binding portion” of an antibody. Any VH and VL sequences of specific scFv can be linked to human immunoglobulin constant region cDNA or genomic sequences, in order to generate expression vectors encoding complete IgG polypeptides or other isotypes. VH and VL can also be used in the generation of Fab, Fv or other fragments of immunoglobulins using either protein chemistry or recombinant DNA technology. Other forms of single chain antibodies, such as diabodies are also encompassed. Diabodies are bivalent, bispecific antibodies in which VH and VL domains are expressed on a single polypeptide chain, but using a linker that is too short to allow for pairing between the two domains on the same chain, thereby forcing the domains to pair with complementary domains of another chain and creating two antigen binding sites (see e.g., Holliger et al. (1993) *Proc. Natl. Acad. Sci. U.S.A.* 90:6444-6448; Poljak et al. (1994) *Structure* 2:1121-1123).

[0162] Still further, an antibody or antigen-binding portion thereof may be part of larger immunoadhesion polypeptides, formed by covalent or noncovalent association of the antibody or antibody portion with one or more other proteins or peptides. Examples of such immunoadhesion polypeptides include use of the streptavidin core region to make a tetrameric scFv polypeptide (Kipriyanov et al. (1995) *Human Antibodies and Hybridomas* 6:93-101) and use of a cysteine residue, biomarker peptide and a C-terminal polyhistidine tag to make bivalent and biotinylated scFv polypeptides (Kipriyanov et al. (1994) *Mol. Immunol.* 31:1047-1058). Antibody portions, such as Fab and F(ab′).sub.2 fragments, can be prepared from whole antibodies using conventional techniques, such as papain or pepsin digestion, respectively, of whole antibodies. Moreover, antibodies, antibody portions and immunoadhesion polypeptides can be obtained using standard recombinant DNA techniques, as described herein.

[0163] Antibodies may be polyclonal or monoclonal; xenogeneic, allogeneic, or syngeneic; or modified forms thereof (e.g. humanized, chimeric, etc.). Antibodies may also be fully human. Preferably, antibodies encompassed by the present invention bind specifically or substantially specifically to a biomarker polypeptide or fragment thereof. The terms “monoclonal antibodies” and “monoclonal antibody composition”, as used herein, refer to a population of antibody polypeptides that contain only one species of an antigen binding site capable of immunoreacting with a particular epitope of an antigen, whereas the term “polyclonal antibodies” and “polyclonal antibody composition” refer to a population of antibody polypeptides that contain multiple species of antigen binding sites capable of interacting with a particular antigen. A monoclonal antibody composition typically displays a single binding affinity for a particular antigen with which it immunoreacts.

[0164] Antibodies may also be “humanized,” which is intended to include antibodies made by a non-human cell having variable and constant regions which have been altered to more closely resemble antibodies that would be

made by a human cell. For example, by altering the non-human antibody amino acid sequence to incorporate amino acids found in human germline immunoglobulin sequences. The humanized antibodies encompassed by the present invention may include amino acid residues not encoded by human germline immunoglobulin sequences (e.g., mutations introduced by random or site-specific mutagenesis in vitro or by somatic mutation in vivo), for example in the CDRs. The term “humanized antibody”, as used herein, also includes antibodies in which CDR sequences derived from the germline of another mammalian species, have been grafted onto human framework sequences.
[0165] The term “biomarker” refers to a measurable entity of the present invention that has been determined to be predictive of cancer therapy effects (e.g., SS18-SSX target genes described described herein, such as those in the tables, figures, examples, and otherwise described in the specification). Biomarkers can include, without limitation, nucleic acids (e.g., genomic nucleic acids and/or transcribed nucleic acids) and proteins. Many biomarkers are also useful as therapeutic targets.
[0166] A “blocking” antibody or an antibody “antagonist” is one which inhibits or reduces at least one biological activity of the antigen(s) it binds. In certain embodiments, the blocking antibodies or antagonist antibodies or fragments thereof described herein substantially or completely inhibit a given biological activity of the antigen(s).
[0167] The term “body fluid” refers to fluids that are excreted or secreted from the body as well as fluids that are normally not (e.g. amniotic fluid, aqueous humor, bile, blood and blood plasma, cerebrospinal fluid, cerumen and earwax, cowper's fluid or pre-ejaculatory fluid, chyle, chyme, stool, female ejaculate, interstitial fluid, intracellular fluid, lymph, menses, breast milk, mucus, pleural fluid, pus, saliva, sebum, semen, serum, sweat, synovial fluid, tears, urine, vaginal lubrication, vitreous humor, vomit).
[0168] The terms “cancer” or “tumor” or “hyperproliferative” refer to the presence of cells possessing characteristics typical of cancer-causing cells, such as uncontrolled proliferation, immortality, metastatic potential, rapid growth and proliferation rate, and certain characteristic morphological features. In some embodiments, such cells exhibit such characteristics in part or in full due to the expression and activity of SS18-SSX oncogenic fusion protein target genes.
[0169] Cancer cells are often in the form of a tumor, but such cells may exist alone within an animal, or may be a non-tumorigenic cancer cell, such as a leukemia cell. As used herein, the term “cancer” includes premalignant as well as malignant cancers. Cancers include, but are not limited to, B cell cancer, e.g., multiple myeloma, Waldenström's macroglobulinemia, the heavy chain diseases, such as, for example, alpha chain disease, gamma chain disease, and mu chain disease, benign monoclonal gammopathy, and immunocytic amyloidosis, melanomas, breast cancer, lung cancer, bronchus cancer, colorectal cancer, prostate cancer, pancreatic cancer, stomach cancer, ovarian cancer, urinary bladder cancer, brain or central nervous system cancer, peripheral nervous system cancer, esophageal cancer, cervical cancer, uterine or endometrial cancer, cancer of the oral cavity or pharynx, liver cancer, kidney cancer, testicular cancer, biliary tract cancer, small bowel or appendix cancer, salivary gland cancer, thyroid gland cancer, adrenal gland cancer, osteosarcoma, chondrosarcoma, cancer of hematologic tissues, and the like. Other non-limiting examples of types of cancers applicable to the methods encompassed by the present invention include human sarcomas and carcinomas, e.g., fibrosarcoma, myxosarcoma, liposarcoma, chondrosarcoma, osteogenic sarcoma, chordoma, angiosarcoma, endotheliosarcoma, lymphangiosarcoma, lymphangioendotheliosarcoma, synovioma, mesothelioma, Ewing's tumor, leiomyosarcoma, rhabdomyosarcoma, colon carcinoma, colorectal cancer, pancreatic cancer, breast cancer, ovarian cancer, prostate cancer, squamous cell carcinoma, basal cell carcinoma, adenocarcinoma, sweat gland carcinoma, sebaceous gland carcinoma, papillary carcinoma, papillary adenocarcinomas, cystadenocarcinoma, medullary carcinoma, bronchogenic carcinoma, renal cell carcinoma, hepatoma, bile duct carcinoma, liver cancer, choriocarcinoma, seminoma, embryonal carcinoma, Wilms' tumor, cervical cancer, bone cancer, brain tumor, testicular cancer, lung carcinoma, small cell lung carcinoma, bladder carcinoma, epithelial carcinoma, glioma, astrocytoma, medulloblastoma, craniopharyngioma, ependymoma, pinealoma, hemangioblastoma, acoustic neuroma, oligodendroglioma, meningioma, melanoma, neuroblastoma, retinoblastoma; leukemias, e.g., acute lymphocytic leukemia and acute myelocytic leukemia (myeloblastic, promyelocytic, myelomonocytic, monocytic and erythroleukemia); chronic leukemia (chronic myelocytic (granulocytic) leukemia and chronic lymphocytic leukemia); and polycythemia vera, lymphoma (Hodgkin's disease and non-Hodgkin's disease), multiple myeloma, Waldenstrom's macroglobulinemia, and heavy chain disease. In some embodiments, cancers are epithlelial in nature and include but are not limited to, bladder cancer, breast cancer, cervical cancer, colon cancer, gynecologic cancers, renal cancer, laryngeal cancer, lung cancer, oral cancer, head and neck cancer, ovarian cancer, pancreatic cancer, prostate cancer, or skin cancer. In other embodiments, the cancer is breast cancer, prostate cancer, lung cancer, or colon cancer. In still other embodiments, the epithelial cancer is non-small-cell lung cancer, nonpapillary renal cell carcinoma, cervical carcinoma, ovarian carcinoma (e.g., serous ovarian carcinoma), or breast carcinoma. The epithelial cancers may be characterized in various other ways including, but not limited to, serous, endometrioid, mucinous, clear cell, Brenner, or undifferentiated.
[0170] In certain embodiments, the cancer encompasses synovial sarcoma. Synovial sarcoma is an aggressive malignancy comprising 7-10% of all soft tissue tumors with a predominance in adolescents and young adults. The molecular hallmark of synovial sarcoma is a pathognomonic reciprocal translocation t(X;18)(p11; q11), leading to

the fusion of SS18 (SYT) to one of the homologs SSX genes (most frequently SSX1 or SSX2, in rare cases SSX4), generating oncogenic SS18-SSX chimeric proteins.

[0171] Synovial sarcoma is a rare cancer. Only about 1 to 3 individuals in a million people are diagnosed with this disease each year. The diagnosis starts with imaging studies. X-ray, sonogram, CT scan, and MRI may be used in the course of evaluating a suspicious mass. After imaging studies, the next step in diagnosis is a biopsy to remove a sample of the tumor for further analysis. Among the different types of biopsies, open biopsy (a surgical incision is made to remove the sample) or core needle biopsy (a large needle is used to take the sample) are preferred. Normally, the sample tissue obtained from the biopsy is sent directly from the procedure room to a pathology laboratory to be sliced and fixed on small glass plates (slides). The pathologist commonly uses a technique called immunohistochemistry to learn about the tumor cells. Another technique called cytogenetics is often used to detect the chromosomal translocation specific to synovial sarcoma, which helps to confirm the diagnosis. Once a tumor has been deemed malignant, further imaging studies such as a PET scan of the whole body and/or CT scan of the chest, abdomen or pelvis may be used to look for possible metastases.

[0172] The primary treatment for synovial sarcoma is surgery to remove the entire tumor with clear margins when possible. “Clear margins” are achieved when healthy tissue surrounding the tumor is removed along with the tumor, making it more likely that all cancer cells have been removed from the area. Depending on the location and size of the mass, it may be difficult for a surgeon to remove adequate margins around the tumor while preserving function. Radiotherapy may also be used, either before or after surgery, to reduce the risk of leaving cells behind. Chemotherapy (typically Doxorubicin and/or Ifosfamide) may be recommended in the treatment of synovial sarcoma, especially in advanced or metastatic disease.

[0173] Prognosis in synovial sarcoma patients is influenced by the quality of surgery patients receive and the characteristics of the disease (including tumor size, local invasiveness, histological subtype, presence of metastases, and lymph node involvement). Patients with small tumors that can be completely removed with adequate margins at diagnosis have an excellent prognosis. The risk of developing distant metastases is higher for patients with tumors that are larger than 5 cm. Patients with the poorly differentiated subtype are considered to have a worse prognosis than those with other subtypes, and patients with metastases that cannot be removed have a poor prognosis.

[0174] The term “coding region” refers to regions of a nucleotide sequence comprising codons which are translated into amino acid residues, whereas the term “noncoding region” refers to regions of a nucleotide sequence that are not translated into amino acids (e.g., 5′ and 3′ untranslated regions).

[0175] The term “complementary” refers to the broad concept of sequence complementarity between regions of two nucleic acid strands or between two regions of the same nucleic acid strand. It is known that an adenine residue of a first nucleic acid region is capable of forming specific hydrogen bonds (“base pairing”) with a residue of a second nucleic acid region which is antiparallel to the first region if the residue is thymine or uracil. Similarly, it is known that a cytosine residue of a first nucleic acid strand is capable of base pairing with a residue of a second nucleic acid strand which is antiparallel to the first strand if the residue is guanine. A first region of a nucleic acid is complementary to a second region of the same or a different nucleic acid if, when the two regions are arranged in an antiparallel fashion, at least one nucleotide residue of the first region is capable of base pairing with a residue of the second region. Preferably, the first region comprises a first portion and the second region comprises a second portion, whereby, when the first and second portions are arranged in an antiparallel fashion, at least about 50%, and preferably at least about 75%, at least about 90%, or at least about 95% of the nucleotide residues of the first portion are capable of base pairing with nucleotide residues in the second portion. More preferably, all nucleotide residues of the first portion are capable of base pairing with nucleotide residues in the second portion.

[0176] The terms “conjoint therapy” and “combination therapy,” as used herein, refer to the administration of two or more therapeutic substances. The different agents comprising the combination therapy may be administered concomitant with, prior to, or following the administration of one or more therapeutic agents.

[0177] The term “control” refers to any reference standard suitable to provide a comparison to the expression products in the test sample. In one embodiment, the control comprises obtaining a “control sample” from which expression product levels are detected and compared to the expression product levels from the test sample. Such a control sample may comprise any suitable sample, including but not limited to a sample from a control cancer patient (can be stored sample or previous sample measurement) with a known outcome; normal tissue or cells isolated from a subject, such as a normal patient or the cancer patient, cultured primary cells/tissues isolated from a subject such as a normal subject or the cancer patient, adjacent normal cells/tissues obtained from the same organ or body location of the cancer patient, a tissue or cell sample isolated from a normal subject, or a primary cells/tissues obtained from a depository. In another preferred embodiment, the control may comprise a reference standard expression product level from any suitable source, including but not limited to housekeeping genes, an expression product level range from normal tissue (or other previously analyzed control sample), a previously determined expression product level range within a test sample from a group of patients, or a set of patients with a certain outcome (for example, survival for one, two, three, four years, etc.) or receiving a certain treatment (for example, standard of care cancer therapy). It will be understood by those of skill in the art that such control samples and reference standard expression product levels can be used in combination as controls in the methods of the present invention. In one embodiment, the control may comprise normal or non-cancerous cell/tissue sample. In another preferred embodiment, the control may comprise an expression level for a set of patients, such as a set of cancer patients, or for a set of cancer patients receiving a certain treatment, or for a set of patients with one outcome versus another outcome. In the former case, the specific expression product level of each patient can be assigned to a percentile level of expression, or expressed as either higher or lower than the mean or average of the reference standard expression level. In another preferred embodiment, the control may comprise normal cells, cells from patients treated with combination chemotherapy, and cells from patients having benign cancer. In another embodiment, the control may also comprise a measured value for example, average level of expression of a particular gene in a population compared to the level of expression of a housekeeping gene in the same population. Such a population may comprise normal subjects, cancer patients who have not undergone any treatment (i.e., treatment naive), cancer patients undergoing standard of care therapy, or patients having benign cancer. In another preferred embodiment, the control comprises a ratio transformation of expression product levels, including but not limited to determining a ratio of expression product levels of two genes in the test sample and comparing it to any suitable ratio of the same two genes in a reference standard; determining expression product levels of the two or more genes in the test sample and determining a difference in expression product levels in any suitable control; and determining expression product levels of the two or more genes in the test sample, normalizing their expression to expression of housekeeping genes in the test sample, and comparing to any suitable control. In particularly preferred embodiments, the control comprises a control sample which is of the same lineage and/or type as the test sample. In another embodiment, the control may comprise expression product levels grouped as percentiles within or based on a set of patient samples, such as all patients with cancer. In one embodiment a control expression product level is established wherein higher or lower levels of expression product relative to, for instance, a particular percentile, are used as the basis for predicting outcome. In another preferred embodiment, a control expression product level is established using expression product levels from cancer control patients with a known outcome, and the expression product levels from the test sample are compared to the control expression product level as the basis for predicting outcome. As demonstrated by the data below, the methods encompassed by the present invention are not limited to use of a specific cut-point in comparing the level of expression product in the test sample to the control.

[0178] The “copy number” of a biomarker nucleic acid refers to the number of DNA sequences in a cell (e.g., germline and/or somatic) encoding a particular gene product. Generally, for a given gene, a mammal has two copies of each gene. The copy number can be increased, however, by gene amplification or duplication, or reduced by deletion. For example, germline copy number changes include changes at one or more genomic loci, wherein said one or more genomic loci are not accounted for by the number of copies in the normal complement of germline copies in a control (e.g., the normal copy number in germline DNA for the same species as that from which the specific germline DNA and corresponding copy number were determined). Somatic copy number changes include changes at one or more genomic loci, wherein said one or more genomic loci are not accounted for by the number of copies in germline DNA of a control (e.g., copy number in germline DNA for the same subject as that from which the somatic DNA and corresponding copy number were determined).

[0179] The term “immune cell” refers to cells that play a role in the immune response. Immune cells are of hematopoietic origin, and include lymphocytes, such as B cells and T cells; natural killer cells; myeloid cells, such as monocytes, macrophages, eosinophils, mast cells, basophils, and granulocytes.

[0180] Conventional T cells, also known as Tconv or Teffs, have effector functions (e.g., cytokine secretion, cytotoxic activity, anti-self-recognization, and the like) to increase immune responses by virtue of their expression of one or more T cell receptors. Tcons or Teffs are generally defined as any T cell population that is not a Treg and include, for example, naive T cells, activated T cells, memory T cells, resting Tcons, or Tcons that have differentiated toward, for example, the Th1 or Th2 lineages. In some embodiments, Teffs are a subset of non-Treg T cells. In some embodiments, Teffs are CD4+ Teffs or CD8+ Teffs, such as CD4+ helper T lymphocytes (e.g., Th0, Th1, Tfh, or Th17) and CD8+ cytotoxic T lymphocytes. As described further herein, cytotoxic T cells are CD8+T lymphocytes. “Naive Tcons” are CD4.sup.+ T cells that have differentiated in bone marrow, and successfully underwent a positive and negative processes of central selection in a thymus, but have not yet been activated by exposure to an antigen. Naive Tcons are commonly characterized by surface expression of L-selectin (CD62L), absence of activation markers such as CD25, CD44 or CD69, and absence of memory markers such as CD45RO. Naive Tcons are therefore believed to be quiescent and non-dividing, requiring interleukin-7 (IL-7) and interleukin-15 (IL- 15) for homeostatic survival (see, at least WO 2010/101870). The presence and activity of such cells are undesired in the context of suppressing immune responses. Unlike Tregs, Tcons are not anergic and can proliferate in response to antigen-based T cell receptor activation (Lechler et al. (2001) *Philos. Trans. R. Soc. Lond. Biol. Sci.* 356:625-637). In tumors, exhausted cells can present hallmarks of anergy.

[0181] The term “immunotherapy” or “immunotherapies” refer to any treatment that uses certain parts of a subject's immune system to fight diseases such as cancer. The subject's own immune system is stimulated (or suppressed), with or without administration of one or more agent for that purpose. Immunotherapies that are designed to elicit or amplify an immune response are referred to as "activation immunotherapies." Immunotherapies that are designed to reduce or suppress an immune response are referred to as "suppression immunotherapies." Any agent believed to have an immune system effect on the genetically modified transplanted cancer cells can be assayed to determine whether the agent is an immunotherapy and the effect that a given genetic modification has on the modulation of immune response. In some embodiments, the immunotherapy is cancer cell-specific. In some embodiments, immunotherapy can be "untargeted," which refers to administration of agents that do not selectively interact with immune system cells, yet modulates immune system function. Representative examples of untargeted therapies include, without limitation, chemotherapy, gene therapy, and radiation therapy.

[0182] Immunotherapy is one form of targeted therapy that may comprise, for example, the use of cancer vaccines and/or sensitized antigen presenting cells. For example, an oncolytic virus is a virus that is able to infect and lyse cancer cells, while leaving normal cells unharmed, making them potentially useful in cancer therapy. Replication of oncolytic viruses both facilitates tumor cell destruction and also produces dose amplification at the tumor site. They may also act as vectors for anticancer genes, allowing them to be specifically delivered to the tumor site. The immunotherapy can involve passive immunity for short-term protection of a host, achieved by the administration of pre-formed antibody directed against a cancer antigen or disease antigen (e.g., administration of a monoclonal antibody, optionally linked to a chemotherapeutic agent or toxin, to a tumor antigen). For example, anti-VEGF and mTOR inhibitors are known to be effective in treating renal cell carcinoma. Immunotherapy can also focus on using the cytotoxic lymphocyte-recognized epitopes of cancer cell lines. Alternatively, antisense polynucleotides, ribozymes, RNA interference molecules, triple helix polynucleotides and the like, can be used to selectively modulate biomolecules that are linked to the initiation, progression, and/or pathology of a tumor or cancer.

[0183] Immunotherapy can involve passive immunity for short-term protection of a host, achieved by the administration of pre-formed antibody directed against a cancer antigen or disease antigen (e.g., administration of a monoclonal antibody, optionally linked to a chemotherapeutic agent or toxin, to a tumor antigen). Immunotherapy can also focus on using the cytotoxic lymphocyte-recognized epitopes of cancer cell lines. Alternatively, antisense polynucleotides, ribozymes, RNA interference molecules, triple helix polynucleotides and the like, can be used to selectively modulate biomolecules that are linked to the initiation, progression, and/or pathology of a tumor or cancer.

[0184] In some embodiments, immunotherapy comprises inhibitors of one or more immune checkpoints. The term "immune checkpoint" refers to a group of molecules on the cell surface of CD4+ and/or CD8+T cells that fine-tune immune responses by down-modulating or inhibiting an anti-tumor immune response. Immune checkpoint proteins are well-known in the art and include, without limitation, CTLA-4, PD-1, VISTA, B7-H2, B7-H3, PD-L1, B7-H4, B7-H6, ICOS, HVEM, PD-L2, CD160, gp49B, PIR-B, KIR family receptors, TIM-1, TIM-3, TIM-4, LAG-3, GITR, 4-IBB, OX-40, BTLA, SIRPalpha (CD47), CD48, 2B4 (CD244), B7.1, B7.2, ILT-2, ILT-4, TIGIT, HHLA2, butyrophilins, and A2aR (see, for example, WO 2012/177624). The term further encompasses biologically active protein fragment, as well as nucleic acids encoding full-length immune checkpoint proteins and biologically active protein fragments thereof. In some embodiment, the term further encompasses any fragment according to homology descriptions provided herein. In one embodiment, the immune checkpoint is PD-1.

[0185] Immune checkpoints and their sequences are well-known in the art and representative embodiments are described below. For example, the term "PD-1" refers to a member of the immunoglobulin gene superfamily that functions as a coinhibitory receptor having PD-L1 and PD-L2 as known ligands. PD-1 was previously identified using a subtraction cloning based approach to select for genes upregulated during TCR-induced activated T cell death. PD-1 is a member of the CD28/CTLA-4 family of molecules based on its ability to bind to PD-L1. Like CTLA-4, PD-1 is rapidly induced on the surface of T-cells in response to anti-CD3 (Agata et al. 25 (1996) *Int. Immunol.* 8:765). In contrast to CTLA-4, however, PD-1 is also induced on the surface of B-cells (in response to anti-IgM). PD-1 is also expressed on a subset of thymocytes and myeloid cells (Agata et al. (1996) supra; Nishimura et al. (1996) *Int. Immunol.* 8:773).

[0186] The nucleic acid and amino acid sequences of a representative human PD-1 biomarker is available to the public at the GenBank database under NM_005018.2 and NP_005009.2 (see also Ishida et al. (1992) 20 *EMBO J* 11:3887; Shinohara et al. (1994) *Genomics* 23:704; U.S. Pat. No. 5,698,520). PD-1 has an extracellular region containing immunoglobulin superfamily domain, a transmembrane domain, and an intracellular region including an immunoreceptor tyrosine-based inhibitory motif (ITIM) (Ishida et al. (1992) *EMBO J.* 11:3887; Shinohara et al. (1994) *Genomics* 23:704; and U.S. Pat. No. 5,698,520) and an immunoreceptor tyrosine-based switch motif (ITSM). These features also define a larger family of polypeptides, called the immunoinhibitory receptors, which also includes gp49B, PIR-B, and the killer inhibitory receptors (KIRs) (Vivier and Daeron (1997) *Immunol. Today* 18:286). It is often assumed that the tyrosyl phosphorylated ITIM and ITSM motif of these receptors interacts with SH2-domain containing phosphatases, which leads to inhibitory signals. A subset of these immunoinhibitory receptors bind to MHC polypeptides, for example the KIRs, and CTLA4 binds to B7-1 and B7-2. It has been proposed that there is a phylogenetic relationship between the MHC and B7 genes (Henry et al. (1999) *Immunol. Today* 20(6):285-8). Nucleic acid and polypeptide sequences of PD-1 orthologs in organisms other than humans are well-known and include, for example, rat PD-1 (NM_001106927.1 and NP_001100397.1), dog PD-1 (XM_543338.3 and XP_543338.3), cow PD-1 (NM_001083506.1 and NP_001076975.1), and chicken PD-1 (XM_422723.3 and XP_422723.2).

[0187] PD-1 polypeptides are inhibitory receptors capable of transmitting an inhibitory signal to an immune cell to thereby inhibit immune cell effector function, or are capable of promoting costimulation (e.g., by competitive inhibition) of immune cells, e.g., when present in soluble, monomeric form. Preferred PD-1 family members share sequence identity with PD-1 and bind to one or more B7 family members, e.g., B7-1, B7-2, PD-1 ligand, and/or other polypeptides on antigen presenting cells.

[0188] The term “PD-1 activity,” includes the ability of a PD-1 polypeptide to modulate an inhibitory signal in an activated immune cell, e.g., by engaging a natural PD-1 ligand on an antigen presenting cell. Modulation of an inhibitory signal in an immune cell results in modulation of proliferation of, and/or cytokine secretion by, an immune cell. Thus, the term “PD-1 activity” includes the ability of a PD-1 polypeptide to bind its natural ligand(s), the ability to modulate immune cell costimulatory or inhibitory signals, and the ability to modulate the immune response.

[0189] The term “PD-1 ligand” refers to binding partners of the PD-1 receptor and includes both PD-L1 (Freeman et al. (2000) *J. Exp. Med.* 192:1027-1034) and PD-L2 (Latchman et al. (2001) *Nat. Immunol.* 2:261). At least two types of human PD-1 ligand polypeptides exist. PD-1 ligand proteins comprise a signal sequence, and an IgV domain, an IgC domain, a transmembrane domain, and a short cytoplasmic tail. Both PD-L1 (See Freeman et al. (2000) for sequence data) and PD-L2 (See Latchman et al. (2001) Nat. *Immunol.* 2:261 for sequence data) are members of the B7 family of polypeptides. Both PD-L1 and PD-L2 are expressed in placenta, spleen, lymph nodes, thymus, and heart. Only PD-L2 is expressed in pancreas, lung and liver, while only PD-L1 is expressed in fetal liver. Both PD-1 ligands are upregulated on activated monocytes and dendritic cells, although PD-L1 expression is broader. For example, PD-L1 is known to be constitutively expressed and upregulated to higher levels on murine hematopoietic cells (e.g., T cells, B cells, macrophages, dendritic cells (DCs), and bone marrow-derived mast cells) and non-hematopoietic cells (e.g., endothelial, epithelial, and muscle cells), whereas PD-L2 is inducibly expressed on DCs, macrophages, and bone marrow-derived mast cells (see Butte et al. (2007) *Immunity* 27:111).

[0190] PD-1 ligands comprise a family of polypeptides having certain conserved structural and functional features. The term “family” when used to refer to proteins or nucleic acid molecules, is intended to mean two or more proteins or nucleic acid molecules having a common structural domain or motif and having sufficient amino acid or nucleotide sequence homology, as defined herein. Such family members can be naturally or non-naturally occurring and can be from either the same or different species. For example, a family can contain a first protein of human origin, as well as other, distinct proteins of human origin or alternatively, can contain homologues of non-human origin. Members of a family may also have common functional characteristics. PD-1 ligands are members of the B7 family of polypeptides. The term “B7 family” or “B7 polypeptides” as used herein includes costimulatory polypeptides that share sequence homology with B7 polypeptides, e.g., with B7-1, B7-2, B7h (Swallow et al. (1999) *Immunity* 11:423), and/or PD-1 ligands (e.g., PD-L1 or PD-L2). For example, human B7-1 and B7-2 share approximately 26% amino acid sequence identity when compared using the BLAST program at NCBI with the default parameters (Blosum62 matrix with gap penalties set at existence 11 and extension 1 (See the NCBI website). The term B7 family also includes variants of these polypeptides which are capable of modulating immune cell function. The B7 family of molecules share a number of conserved regions, including signal domains, IgV domains and the IgC domains. IgV domains and the IgC domains are art-recognized Ig superfamily member domains. These domains correspond to structural units that have distinct folding patterns called Ig folds. Ig folds are comprised of a sandwich of two R sheets, each consisting of anti-parallel β strands of 5-10 amino acids with a conserved disulfide bond between the two sheets in most, but not all, IgC domains of Ig, TCR, and MHC molecules share the same types of sequence patterns and are called the C1-set within the Ig superfamily. Other IgC domains fall within other sets. IgV domains also share sequence patterns and are called V set domains. IgV domains are longer than IgC domains and contain an additional pair of R strands.

[0191] Preferred B7 polypeptides are capable of providing costimulatory or inhibitory signals to immune cells to thereby promote or inhibit immune cell responses. For example, B7 family members that bind to costimulatory receptors increase T cell activation and proliferation, while B7 family members that bind to inhibitory receptors reduce costimulation. Moreover, the same B7 family member may increase or decrease T cell costimulation. For example, when bound to a costimulatory receptor, PD-1 ligand can induce costimulation of immune cells or can inhibit immune cell costimulation, e.g., when present in soluble form. When bound to an inhibitory receptor, PD-1 ligand polypeptides can transmit an inhibitory signal to an immune cell. Preferred B7 family members include B7-1, B7-2, B7h, PD-L1 or PD-L2 and soluble fragments or derivatives thereof. In one embodiment, B7 family members bind to one or more receptors on an immune cell, e.g., CTLA4, CD28, ICOS, PD-1 and/or other receptors, and, depending on the receptor, have the ability to transmit an inhibitory signal or a costimulatory signal to an immune cell, preferably a T cell.

[0192] Modulation of a costimulatory signal results in modulation of effector function of an immune cell. Thus, the term “PD-1 ligand activity” includes the ability of a PD-1 ligand polypeptide to bind its natural receptor(s) (e.g. PD-

1 or B7-1), the ability to modulate immune cell costimulatory or inhibitory signals, and the ability to modulate the immune response.

[0193] The term "PD-L1" refers to a specific PD-1 ligand. Two forms of human PD-L1 molecules have been identified. One form is a naturally occurring PD-L1 soluble polypeptide, i.e., having a short hydrophilic domain and no transmembrane domain, and is referred to herein as PD-L1S. The second form is a cell-associated polypeptide, i.e., having a transmembrane and cytoplasmic domain, referred to herein as PD-L1M. The nucleic acid and amino acid sequences of representative human PD-L1 biomarkers regarding PD-L1M are also available to the public at the GenBank database under NM_014143.3 and NP_054862.I. PD-L1 proteins comprise a signal sequence, and an IgV domain and an IgC domain. The signal sequence of PD-L1S is from about amino acid 1 to about amino acid 18. The signal sequence of PD-L1M is from about amino acid 1 to about amino acid 18. The IgV domain of PD-L1S is from about amino acid 19 to about amino acid 134 and the IgV domain of PD-L1M is from about amino acid 19 to about amino acid 134. The IgC domain of PD-L1S is from about amino acid 135 to about amino acid 227 and the IgC domain of PD-L1M is from about amino acid 135 to about amino acid 227. The hydrophilic tail of the PD-L1 exemplified in PD-L1S comprises a hydrophilic tail shown from about amino acid 228 to about amino acid 245. The PD-L1 polypeptide of PD-L1M comprises a transmembrane domain from about amino acids 239 to about amino acid 259 of PD-L1M and a cytoplasmic domain shown from about amino acid 260 to about amino acid 290 of PD-L1M. In addition, nucleic acid and polypeptide sequences of PD-L1 orthologs in organisms other than humans are well-known and include, for example, rat PD-L1 (NM_001191954.1 and NP_001178883.1), dog PD-L1 (XM_541302.3 and XP_541302.3), cow PD-L1 (NM_001163412.1 and NP_001156884.1), and chicken PD-L1 (XM_424811.3 and XP_424811.3).

[0194] The term "PD-L2" refers to another specific PD-1 ligand. PD-L2 is a B7 family member expressed on various APCs, including dendritic cells, macrophages and bone-marrow derived mast cells (Zhong et al. (2007) Eur. J. Immunol. 37:2405). APC-expressed PD-L2 is able to both inhibit T cell activation through ligation of PD-1 and costimulate T cell activation, through a PD-1 independent mechanism (Shin et al. (2005) J. Exp. Med. 201:1531). In addition, ligation of dendritic cell-expressed PD-L2 results in enhanced dendritic cell cytokine expression and survival (Radhakrishnan et al. (2003) J. Immunol. 37:1827; Nguyen et al. (2002) J. Exp. Med. 196:1393). The nucleic acid and amino acid sequences of representative human PD-L2 biomarkers are well-known in the art and are also available to the public at the GenBank database under NM_025239.3 and NP_079515.2. PD-L2 proteins are characterized by common structural elements. In some embodiments, PD-L2 proteins include at least one or more of the following domains: a signal peptide domain, a transmembrane domain, an IgV domain, an IgC domain, an extracellular domain, a transmembrane domain, and a cytoplasmic domain. For example, amino acids 1-19 of PD-L2 comprises a signal sequence. As used herein, a "signal sequence" or "signal peptide" serves to direct a polypeptide containing such a sequence to a lipid bilayer, and is cleaved in secreted and membrane bound polypeptides and includes a peptide containing about 15 or more amino acids which occurs at the N-terminus of secretory and membrane bound polypeptides and which contains a large number of hydrophobic amino acid residues. For example, a signal sequence contains at least about 10-30 amino acid residues, preferably about 15-25 amino acid residues, more preferably about 18-20 amino acid residues, and even more preferably about 19 amino acid residues, and has at least about 35-65%, preferably about 38-50%, and more preferably about 40-45% hydrophobic amino acid residues (e.g., valine, leucine, isoleucine or phenylalanine). In another embodiment, amino acid residues 220-243 of the native human PD-L2 polypeptide and amino acid residues 201-243 of the mature polypeptide comprise a transmembrane domain. As used herein, the term "transmembrane domain" includes an amino acid sequence of about 15 amino acid residues in length which spans the plasma membrane. More preferably, a transmembrane domain includes about at least 20, 25, 30, 35, 40, or 45 amino acid residues and spans the plasma membrane. Transmembrane domains are rich in hydrophobic residues, and typically have an alpha-helical structure. In a preferred embodiment, at least 50%, 60%, 70%, 80%, 90%, 95% or more of the amino acids of a transmembrane domain are hydrophobic, e.g., leucines, isoleucines, tyrosines, or tryptophans. Transmembrane domains are described in, for example, Zagotta, W. N. et al. (1996) Annu. Rev. Neurosci. 19: 235-263. In still another embodiment, amino acid residues 20-120 of the native human PD-L2 polypeptide and amino acid residues 1-101 of the mature polypeptide comprise an IgV domain. Amino acid residues 121-219 of the native human PD-L2 polypeptide and amino acid residues 102-200 of the mature polypeptide comprise an IgC domain. As used herein, IgV and IgC domains are recognized in the art as Ig superfamily member domains. These domains correspond to structural units that have distinct folding patterns called Ig folds. Ig folds are comprised of a sandwich of two B sheets, each consisting of antiparallel (3 strands of 5-10 amino acids with a conserved disulfide bond between the two sheets in most, but not all, domains. IgC domains of Ig, TCR, and MHC molecules share the same types of sequence patterns and are called the Cl set within the Ig superfamily. Other IgC domains fall within other sets. IgV domains also share sequence patterns and are called V set domains. IgV domains are longer than C-domains and form an additional pair of strands. In yet another embodiment, amino acid residues 1-219 of the native human PD-L2 polypeptide and amino acid residues 1-200 of the mature polypeptide comprise an extracellular domain. As used herein, the term "extracellular domain" represents the N-terminal amino acids which extend as a tail from the surface of a cell. An extracellular domain of the present invention includes an IgV domain and an IgC domain, and may include a signal peptide domain. In still another embodiment, amino acid residues 244-273 of the native human PD-L2 polypeptide and amino acid residues 225-273 of the mature polypeptide comprise a cytoplasmic domain. As used herein, the term "cytoplasmic domain" represents the C-terminal amino acids which extend as a tail into the cytoplasm of a cell. In addition, nucleic acid and polypeptide sequences of PD-L2 orthologs in organisms other than humans are well-known and include, for example, rat PD-L2 (NM_001107582.2 and NP_001101052.2), dog PD-L2 (XM_847012.2 and XP_852105.2), cow PD-L2 (XM_586846.5 and XP_586846.3), and chimpanzee PD-L2 (XM_001140776.2 and XP_001140776.1).

[0195] The term "PD-L2 activity," "biological activity of PD-L2," or "functional activity of PD-L2," refers to an activity exerted by a PD-L2 protein, polypeptide or nucleic acid molecule on a PD-L2-responsive cell or tissue, or on a PD-L2 polypeptide binding partner, as determined in vivo, or in vitro, according to standard techniques. In one embodiment, a PD-L2 activity is a direct activity, such as an association with a PD-L2 binding partner. As used herein, a "target molecule" or "binding partner" is a molecule with which a PD-L2 polypeptide binds or interacts in nature, such that PD-L2-mediated function is achieved. In an exemplary embodiment, a PD-L2 target molecule is the receptor RGMb. Alternatively, a PD-L2 activity is an indirect activity, such as a cellular signaling activity mediated by interaction of the PD-L2 polypeptide with its natural binding partner (i.e., physiologically relevant interacting macromolecule involved in an immune function or other biologically relevant function), e.g., RGMb. The biological activities of PD-L2 are described herein. For example, the PD-L2 polypeptides of the present invention can have one or more of the following activities: 1) bind to and/or modulate the activity of the receptor RGMb, PD-1, or other PD-L2 natural binding partners, 2) modulate intra- or intercellular signaling, 3) modulate activation of immune cells, e.g., T lymphocytes, and 4) modulate the immune response of an organism, e.g., a human organism.

[0196] "Anti-immune checkpoint therapy" refers to the use of agents that inhibit immune checkpoint nucleic acids and/or proteins. Inhibition of one or more immune checkpoints can block or otherwise neutralize inhibitory signaling to thereby upregulate an immune response in order to more efficaciously treat cancer. Exemplary agents useful for inhibiting immune checkpoints include antibodies, small molecules, peptides, peptidomimetics, natural ligands, and derivatives of natural ligands, that can either bind and/or inactivate or inhibit immune checkpoint proteins, or fragments thereof; as well as RNA interference, antisense, nucleic acid aptamers, etc. that can downregulate the expression and/or activity of immune checkpoint nucleic acids, or fragments thereof. Exemplary agents for upregulating an immune response include antibodies against one or more immune checkpoint proteins block the interaction between the proteins and its natural receptor(s); a non-activating form of one or more immune checkpoint proteins (e.g., a dominant negative polypeptide); small molecules or peptides that block the interaction between one or more immune checkpoint proteins and its natural receptor(s); fusion proteins (e.g. the extracellular portion of an immune checkpoint inhibition protein fused to the Fc portion of an antibody or immunoglobulin) that bind to its natural receptor(s); nucleic acid molecules that block immune checkpoint nucleic acid transcription or translation; and the like. Such agents can directly block the interaction between the one or more immune checkpoints and its natural receptor(s) (e.g., antibodies) to prevent inhibitory signaling and upregulate an immune response. Alternatively, agents can indirectly block the interaction between one or more immune checkpoint proteins and its natural receptor(s) to prevent inhibitory signaling and upregulate an immune response. For example, a soluble version of an immune checkpoint protein ligand such as a stabilized extracellular domain can binding to its receptor to indirectly reduce the effective concentration of the receptor to bind to an appropriate ligand. In one embodiment, anti-PD-1 antibodies, anti-PD-L1 antibodies, and/or anti-PD-L2 antibodies, either alone or in combination, are used to inhibit immune checkpoints. These embodiments are also applicable to specific therapy against particular immune checkpoints, such as the PD-1 pathway (e.g., anti-PD-1 pathway therapy, otherwise known as PD-1 pathway inhibitor therapy).

[0197] The term "immune response" includes T cell mediated and/or B cell mediated immune responses. Exemplary immune responses include T cell responses, e.g., cytokine production and cellular cytotoxicity. In addition, the term immune response includes immune responses that are indirectly effected by T cell activation, e.g., antibody production (humoral responses) and activation of cytokine responsive cells, e.g., macrophages.

[0198] The term "immunotherapeutic agent" can include any molecule, peptide, antibody or other agent which can stimulate a host immune system to generate an immune response to a tumor or cancer in the subject. Various immunotherapeutic agents are useful in the compositions and methods described herein.

[0199] The term "inhibit" includes decreasing, reducing, limiting, and/or blocking, of, for example a particular action, function, and/or interaction. In some embodiments, the interation between two molecules is "inhibited" if the interaction is reduced, blocked, disrupted or destabilized.

[0200] In some embodiments, cancer is "inhibited" if at least one symptom of the cancer is alleviated, terminated, slowed, or prevented. As used herein, cancer is also "inhibited" if recurrence or metastasis of the cancer is reduced, slowed, delayed, or prevented.

[0201] The term "interaction", when referring to an interaction between two molecules, refers to the physical contact (e.g., binding) of the molecules with one another. Generally, such an interaction results in an activity (which

produces a biological effect) of one or both of said molecules.
[0202] An "isolated protein" refers to a protein that is substantially free of other proteins, cellular material, separation medium, and culture medium when isolated from cells or produced by recombinant DNA techniques, or chemical precursors or other chemicals when chemically synthesized. An "isolated" or "purified" protein or biologically active portion thereof is substantially free of cellular material or other contaminating proteins from the cell or tissue source from which the antibody, polypeptide, peptide or fusion protein is derived, or substantially free from chemical precursors or other chemicals when chemically synthesized. The language "substantially free of cellular material" includes preparations of a biomarker polypeptide or fragment thereof, in which the protein is separated from cellular components of the cells from which it is isolated or recombinantly produced. In one embodiment, the language "substantially free of cellular material" includes preparations of a biomarker protein or fragment thereof, having less than about 30% (by dry weight) of non-biomarker protein (also referred to herein as a "contaminating protein"), more preferably less than about 20% of non-biomarker protein, still more preferably less than about 10% of non-biomarker protein, and most preferably less than about 5% non-biomarker protein. When antibody, polypeptide, peptide or fusion protein or fragment thereof, e.g., a biologically active fragment thereof, is recombinantly produced, it is also preferably substantially free of culture medium, i.e., culture medium represents less than about 20%, more preferably less than about 10%, and most preferably less than about 5% of the volume of the protein preparation.
[0203] As used herein, the term "isotype" refers to the antibody class (e.g., IgM, IgG1, IgG2C, and the like) that is encoded by heavy chain constant region genes.
[0204] The "normal" level of expression of a biomarker is the level of expression of the biomarker in cells of a subject, e.g., a human patient, not afflicted with a cancer. An "over-expression" or "significantly higher level of expression" of a biomarker refers to an expression level in a test sample that is greater than the standard error of the assay employed to assess expression, and is preferably at least 10%, and more preferably 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0, 2.1, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10, 10.5, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 times or more higher than the expression activity or level of the biomarker in a control sample (e.g., sample from a healthy subject not having the biomarker associated disease) and preferably, the average expression level of the biomarker in several control samples. A "significantly lower level of expression" of a biomarker refers to an expression level in a test sample that is at least 10%, and more preferably 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0, 2.1, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10, 10.5, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 times or more lower than the expression level of the biomarker in a control sample (e.g., sample from a healthy subject not having the biomarker associated disease) and preferably, the average expression level of the biomarker in several control samples.
[0205] An "over-expression" or "significantly higher level of expression" of a biomarker refers to an expression level in a test sample that is greater than the standard error of the assay employed to assess expression, and is preferably at least 10%, and more preferably 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0, 2.1, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10, 10.5, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 times or more higher than the expression activity or level of the biomarker in a control sample (e.g., sample from a healthy subject not having the biomarker associated disease) and preferably, the average expression level of the biomarker in several control samples. A "significantly lower level of expression" of a biomarker refers to an expression level in a test sample that is at least 10%, and more preferably 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0, 2.1, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10, 10.5, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 times or more lower than the expression level of the biomarker in a control sample (e.g., sample from a healthy subject not having the biomarker associated disease) and preferably, the average expression level of the biomarker in several control samples.
[0206] The term "predictive" includes the use of a biomarker nucleic acid and/or protein status, e.g., over- or under-activity, emergence, expression, growth, remission, recurrence or resistance of tumors before, during or after therapy, for determining the likelihood of response of a cancer to an agent that inhibits binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome alone or in combination with an immunotherapy and/or cancer therapy. Such predictive use of the biomarker may be confirmed by, e.g., (1) increased or decreased copy number (e.g., by FISH, FISH plus SKY, single-molecule sequencing, e.g., as described in the art at least at J. Biotechnol., 86:289-301, or qPCR), overexpression or underexpression of a biomarker nucleic acid (e.g., by ISH, Northern Blot, or qPCR), increased or decreased biomarker protein (e.g., by IHC), or increased or decreased activity, e.g., in more than about 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 20%, 25%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 100%, or more of assayed human cancers types or cancer samples; (2) its absolute or relatively modulated presence or absence in a biological sample, e.g., a sample containing tissue, whole blood, serum, plasma, buccal scrape, saliva, cerebrospinal fluid, urine, stool, or bone marrow, from a subject, e.g. a human, afflicted with cancer; (3) its absolute or relatively modulated presence or absence in clinical subset of patients with cancer (e.g., those responding to an agent that inhibits binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome alone or in combination with an immunotherapy and/or cancer therapy, or those developing resistance thereto).
[0207] The terms "prevent," "preventing," "prevention," "prophylactic treatment," and the like refer to reducing the

probability of developing a disease, disorder, or condition in a subject, who does not have, but is at risk of or susceptible to developing a disease, disorder, or condition.

[0208] The term “cancer response,” “response to immunotherapy,” or “response to modulators of T-cell mediated cytotoxicity/immunotherapy combination therapy” relates to any response of the hyperproliferative disorder (e.g., cancer) to a cancer agent, such as a modulator of T-cell mediated cytotoxicity, and an immunotherapy, preferably to a change in tumor mass and/or volume after initiation of neoadjuvant or adjuvant therapy. Hyperproliferative disorder response may be assessed, for example for efficacy or in a neoadjuvant or adjuvant situation, where the size of a tumor after systemic intervention can be compared to the initial size and dimensions as measured by CT, PET, mammogram, ultrasound or palpation. Responses may also be assessed by caliper measurement or pathological examination of the tumor after biopsy or surgical resection. Response may be recorded in a quantitative fashion like percentage change in tumor volume or in a qualitative fashion like “pathological complete response” (pCR), “clinical complete remission” (cCR), “clinical partial remission” (cPR), “clinical stable disease” (cSD), “clinical progressive disease” (cPD) or other qualitative criteria. Assessment of hyperproliferative disorder response may be done early after the onset of neoadjuvant or adjuvant therapy, e.g., after a few hours, days, weeks or preferably after a few months. A typical endpoint for response assessment is upon termination of neoadjuvant chemotherapy or upon surgical removal of residual tumor cells and/or the tumor bed. This is typically three months after initiation of neoadjuvant therapy. In some embodiments, clinical efficacy of the therapeutic treatments described herein may be determined by measuring the clinical benefit rate (CBR). The clinical benefit rate is measured by determining the sum of the percentage of patients who are in complete remission (CR), the number of patients who are in partial remission (PR) and the number of patients having stable disease (SD) at a time point at least 6 months out from the end of therapy. The shorthand for this formula is CBR=CR+PR+SD over 6 months. In some embodiments, the CBR for a particular cancer therapeutic regimen is at least 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, or more. Additional criteria for evaluating the response to cancer therapies are related to “survival,” which includes all of the following: survival until mortality, also known as overall survival (wherein said mortality may be either irrespective of cause or tumor related); “recurrence-free survival” (wherein the term recurrence shall include both localized and distant recurrence); metastasis free survival; disease free survival (wherein the term disease shall include cancer and diseases associated therewith). The length of said survival may be calculated by reference to a defined start point (e.g., time of diagnosis or start of treatment) and end point (e.g., death, recurrence or metastasis). In addition, criteria for efficacy of treatment can be expanded to include response to chemotherapy, probability of survival, probability of metastasis within a given time period, and probability of tumor recurrence. For example, in order to determine appropriate threshold values, a particular cancer therapeutic regimen can be administered to a population of subjects and the outcome can be correlated to biomarker measurements that were determined prior to administration of any cancer therapy. The outcome measurement may be pathologic response to therapy given in the neoadjuvant setting. Alternatively, outcome measures, such as overall survival and disease-free survival can be monitored over a period of time for subjects following cancer therapy for which biomarker measurement values are known. In certain embodiments, the doses administered are standard doses known in the art for cancer therapeutic agents. The period of time for which subjects are monitored can vary. For example, subjects may be monitored for at least 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 25, 30, 35, 40, 45, 50, 55, or 60 months. Biomarker measurement threshold values that correlate to outcome of a cancer therapy can be determined using well-known methods in the art, such as those described in the Examples section.

[0209] The term “resistance” refers to an acquired or natural resistance of a cancer sample or a mammal to a cancer therapy (i.e., being nonresponsive to or having reduced or limited response to the therapeutic treatment), such as having a reduced response to a therapeutic treatment by 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more, such 2-fold, 3-fold, 4-fold, 5-fold, 10-fold, 15-fold, 20-fold or more, or any range in between, inclusive. The reduction in response can be measured by comparing with the same cancer sample or mammal before the resistance is acquired, or by comparing with a different cancer sample or a mammal that is known to have no resistance to the therapeutic treatment. A typical acquired resistance to chemotherapy is called “multidrug resistance.” The multidrug resistance can be mediated by P-glycoprotein or can be mediated by other mechanisms, or it can occur when a mammal is infected with a multi-drug-resistant microorganism or a combination of microorganisms. The determination of resistance to a therapeutic treatment is routine in the art and within the skill of an ordinarily skilled clinician, for example, can be measured by cell proliferative assays and cell death assays as described herein as “sensitizing.” In some embodiments, the term “reverses resistance” means that the use of a second agent in combination with a primary cancer therapy (e.g., chemotherapeutic or radiation therapy) is able to produce a significant decrease in tumor volume at a level of statistical significance (e.g., $p<0.05$) when compared to tumor volume of untreated tumor in the circumstance where the primary cancer therapy (e.g., chemotherapeutic or radiation therapy) alone is unable to produce a statistically significant decrease in tumor volume compared to tumor volume of untreated tumor. This generally applies to tumor volume measurements made at a time when the untreated tumor is growing log rhythmically.

[0210] The terms “response” or “responsiveness” refers to an cancer response, e.g. in the sense of reduction of tumor size or inhibiting tumor growth. The terms can also refer to an improved prognosis, for example, as reflected by an increased time to recurrence, which is the period to first recurrence censoring for second primary cancer as a first event or death without evidence of recurrence, or an increased overall survival, which is the period from treatment to death from any cause. To respond or to have a response means there is a beneficial endpoint attained when exposed to a stimulus. Alternatively, a negative or detrimental symptom is minimized, mitigated or attenuated on exposure to a stimulus. It will be appreciated that evaluating the likelihood that a tumor or subject will exhibit a favorable response is equivalent to evaluating the likelihood that the tumor or subject will not exhibit favorable response (i.e., will exhibit a lack of response or be non-responsive).

[0211] An "RNA interfering agent" as used herein, is defined as any agent which interferes with or inhibits expression of a target biomarker gene by RNA interference (RNAi). Such RNA interfering agents include, but are not limited to, nucleic acid molecules including RNA molecules which are homologous to the target biomarker gene of the present invention, or a fragment thereof, short interfering RNA (siRNA), and small molecules which interfere with or inhibit expression of a target biomarker nucleic acid by RNA interference (RNAi). "RNA interference (RNAi)" is an evolutionally conserved process whereby the expression or introduction of RNA of a sequence that is identical or highly similar to a target biomarker nucleic acid results in the sequence specific degradation or specific post-transcriptional gene silencing (PTGS) of messenger RNA (mRNA) transcribed from that targeted gene (see Coburn and Cullen (2002) *J. Virol.* 76:9225), thereby inhibiting expression of the target biomarker nucleic acid. In one embodiment, the RNA is double stranded RNA (dsRNA). This process has been described in plants, invertebrates, and mammalian cells. In nature, RNAi is initiated by the dsRNA-specific endonuclease Dicer, which promotes processive cleavage of long dsRNA into double-stranded fragments termed siRNAs. siRNAs are incorporated into a protein complex that recognizes and cleaves target mRNAs. RNAi can also be initiated by introducing nucleic acid molecules, e.g., synthetic siRNAs or RNA interfering agents, to inhibit or silence the expression of target biomarker nucleic acids. As used herein, "inhibition of target biomarker nucleic acid expression" or "inhibition of marker gene expression" includes any decrease in expression or protein activity or level of the target biomarker nucleic acid or protein encoded by the target biomarker nucleic acid. The decrease may be of at least 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95% or 99% or more as compared to the expression of a target biomarker nucleic acid or the activity or level of the protein encoded by a target biomarker nucleic acid which has not been targeted by an RNA interfering agent.

[0212] In addition to RNAi, genome editing can be used to modulate the copy number or genetic sequence of a biomarker of interest, such as constitutive or induced knockout or mutation of a biomarker of interest. For example, the CRISPR-Cas system can be used for precise editing of genomic nucleic acids (e.g., for creating non-functional or null mutations). In such embodiments, the CRISPR guide RNA and/or the Cas enzyme may be expressed. For example, a vector containing only the guide RNA can be administered to an animal or cells transgenic for the Cas9 enzyme. Similar strategies may be used (e.g., designer zinc finger, transcription activator-like effectors (TALEs) or homing meganucleases). Such systems are well-known in the art (see, for example, U.S. Pat. No. 8,697,359; Sander and Joung (2014) *Nat. Biotech.* 32:347-355; Hale et al. (2009) *Cell* 139:945-956; Karginov and Hannon (2010) *Mol. Cell* 37:7; U.S. Pat. Publ. 2014/0087426 and 2012/0178169; Boch et al. (2011) *Nat. Biotech.* 29:135-136; Boch et al. (2009) *Science* 326:1509-1512; Moscou and Bogdanove (2009) *Science* 326:1501; Weber et al. (2011) *PLoS One* 6:e19722; Li et al. (2011) *Nucl. Acids Res.* 39:6315-6325; Zhang et al. (2011) *Nat. Biotech.* 29:149-153; Miller et al. (2011) *Nat. Biotech.* 29:143-148; Lin et al. (2014) *Nucl. Acids Res.* 42:e47). Such genetic strategies can use constitutive expression systems or inducible expression systems according to well-known methods in the art.

[0213] The term "sample" used for detecting or determining the presence or level of at least one biomarker is typically whole blood, plasma, serum, saliva, urine, stool (e.g., feces), tears, and any other bodily fluid (e.g., as described above under the definition of "body fluids"), or a tissue sample (e.g., biopsy) such as bone marrow and bone sample, or surgical resection tissue. In certain instances, the method of the present invention further comprises obtaining the sample from the individual prior to detecting or determining the presence or level of at least one marker in the sample.

[0214] The term "sensitize" means to alter cancer cells or tumor cells in a way that allows for more effective treatment of the associated cancer with a cancer therapy (e.g., anti-immune checkpoint, chemotherapeutic, and/or radiation therapy). In some embodiments, normal cells are not affected to an extent that causes the normal cells to be unduly injured by the therapies. An increased sensitivity or a reduced sensitivity to a therapeutic treatment is measured according to a known method in the art for the particular treatment and methods described herein below, including, but not limited to, cell proliferative assays (Tanigawa N, Kern D H, Kikasa Y, Morton D L, Cancer Res 1982; 42: 2159-2164), cell death assays (Weisenthal L M, Shoemaker R H, Marsden J A, Dill P L, Baker J A, Moran E M, Cancer Res 1984; 94: 161-173; Weisenthal L M, Lippman M E, Cancer Treat Rep 1985; 69: 615-632; Weisenthal L M, In: Kaspers G J L, Pieters R, Twentyman P R, Weisenthal L M, Veerman A J P, eds. Drug Resistance in Leukemia and Lymphoma. Langhorne, P A: Harwood Academic Publishers, 1993: 415-432; Weisenthal L M, Contrib Gynecol Obstet 1994; 19: 82-90). The sensitivity or resistance may also be measured in animal by measuring the tumor size reduction over a period of time, for example, 6 month for human. A composition or a method

sensitizes response to a therapeutic treatment if the increase in treatment sensitivity or the reduction in resistance is 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 100%, or more, such 2-fold, 3-fold, 4-fold, 5-fold, 10-fold, 15-fold, 20-fold or more, or any range in between, inclusive, compared to treatment sensitivity or resistance in the absence of such composition or method. The determination of sensitivity or resistance to a therapeutic treatment is routine in the art and within the skill of an ordinarily skilled clinician. It is to be understood that any method described herein for enhancing the efficacy of a cancer therapy can be equally applied to methods for sensitizing hyperproliferative or otherwise cancerous cells (e.g., resistant cells) to the cancer therapy.
[0215] “Short interfering RNA” (siRNA), also referred to herein as “small interfering RNA” is defined as an agent which functions to inhibit expression of a target biomarker nucleic acid, e.g., by RNAi. An siRNA may be chemically synthesized, may be produced by in vitro transcription, or may be produced within a host cell. In one embodiment, siRNA is a double stranded RNA (dsRNA) molecule of about 15 to about 40 nucleotides in length, preferably about 15 to about 28 nucleotides, more preferably about 19 to about 25 nucleotides in length, and more preferably about 19, 20, 21, or 22 nucleotides in length, and may contain a 3′ and/or 5′ overhang on each strand having a length of about 0, 1, 2, 3, 4, or 5 nucleotides. The length of the overhang is independent between the two strands, i.e., the length of the overhang on one strand is not dependent on the length of the overhang on the second strand. Preferably the siRNA is capable of promoting RNA interference through degradation or specific post-transcriptional gene silencing (PTGS) of the target messenger RNA (mRNA).
[0216] In another embodiment, an siRNA is a small hairpin (also called stem loop) RNA (shRNA). In one embodiment, these shRNAs are composed of a short (e.g., 19-25 nucleotide) antisense strand, followed by a 5-9 nucleotide loop, and the analogous sense strand. Alternatively, the sense strand may precede the nucleotide loop structure and the antisense strand may follow. These shRNAs may be contained in plasmids, retroviruses, and lentiviruses and expressed from, for example, the pol III U6 promoter, or another promoter (see, e.g., Stewart, et al. (2003) RNA April; 9(4):493-501 incorporated by reference herein).
[0217] RNA interfering agents, e.g., siRNA molecules, may be administered to a patient having or at risk for having cancer, to inhibit expression of a biomarker gene which is overexpressed in cancer and thereby treat, prevent, or inhibit cancer in the subject.
[0218] The term “small molecule” is a term of the art and includes molecules that are less than about 1000 molecular weight or less than about 500 molecular weight. In one embodiment, small molecules do not exclusively comprise peptide bonds. In another embodiment, small molecules are not oligomeric. Exemplary small molecule compounds which can be screened for activity include, but are not limited to, peptides, peptidomimetics, nucleic acids, carbohydrates, small organic molecules (e.g., polyketides) (Cane et al. (1998) *Science* 282:63), and natural product extract libraries. In another embodiment, the compounds are small, organic non-peptidic compounds. In a further embodiment, a small molecule is not biosynthetic.
[0219] The term “specific binding” refers to antibody binding to a predetermined antigen. Typically, the antibody binds with an affinity (K.sub.D) of approximately less than 10-7 M, such as approximately less than 10.sup.−8 M, 10.sup.−9 μM or 10.sup.−10 μM or even lower when determined by surface plasmon resonance (SPR) technology in a BIACORE® assay instrument using an antigen of interest as the analyte and the antibody as the ligand, and binds to the predetermined antigen with an affinity that is at least 1.1-, 1.2-, 1.3-, 1.4-, 1.5-, 1.6-, 1.7-, 1.8-, 1.9-, 2.0-, 2.5-, 3.0-, 3.5-, 4.0-, 4.5-, 5.0-, 6.0-, 7.0-, 8.0-, 9.0-, or 10.0-fold or greater than its affinity for binding to a non-specific antigen (e.g., BSA, casein) other than the predetermined antigen or a closely-related antigen. The phrases “an antibody recognizing an antigen” and “an antibody specific for an antigen” are used interchangeably herein with the term “an antibody which binds specifically to an antigen.” Selective binding is a relative term referring to the ability of an antibody to discriminate the binding of one antigen over another.
[0220] The term “subject” refers to any healthy animal, mammal or human, or any animal, mammal or human afflicted with a cancer, e.g., brain, lung, ovarian, pancreatic, liver, breast, prostate, and/or colorectal cancers, melanoma, multiple myeloma, and the like. The term “subject” is interchangeable with “patient.”
[0221] The term “survival” includes all of the following: survival until mortality, also known as overall survival (wherein said mortality may be either irrespective of cause or tumor related); “recurrence-free survival” (wherein the term recurrence shall include both localized and distant recurrence); metastasis free survival; disease free survival (wherein the term disease shall include cancer and diseases associated therewith). The length of said survival may be calculated by reference to a defined start point (e.g. time of diagnosis or start of treatment) and end point (e.g. death, recurrence or metastasis). In addition, criteria for efficacy of treatment can be expanded to include response to chemotherapy, probability of survival, probability of metastasis within a given time period, and probability of tumor recurrence.
[0222] The term “synergistic effect” refers to the combined effect of two or more cancer agents (e.g., an agent that inhibits binding of a SS18-SSX fusion protein with a H2AK119Ub-marked nucleosome in combination with immunotherapy) can be greater than the sum of the separate effects of the cancer agents/therapies alone.
[0223] The term “T cell” includes CD4.sup.+ T cells and CD8.sup.+ T cells. The term T cell also includes both T

helper 1 type T cells and T helper 2 type T cells. The term "antigen presenting cell" includes professional antigen presenting cells (e.g., B lymphocytes, monocytes, dendritic cells, Langerhans cells), as well as other antigen presenting cells (e.g., keratinocytes, endothelial cells, astrocytes, fibroblasts, and oligodendrocytes).

[0224] The term "therapeutic effect" refers to a local or systemic effect in animals, particularly mammals, and more particularly humans, caused by a pharmacologically active substance. The term thus means any substance intended for use in the diagnosis, cure, mitigation, treatment or prevention of disease or in the enhancement of desirable physical or mental development and conditions in an animal or human. The phrase "therapeutically-effective amount" means that amount of such a substance that produces some desired local or systemic effect at a reasonable benefit/risk ratio applicable to any treatment. In certain embodiments, a therapeutically effective amount of a compound will depend on its therapeutic index, solubility, and the like. For example, certain compounds discovered by the methods of the present invention may be administered in a sufficient amount to produce a reasonable benefit/risk ratio applicable to such treatment.

[0225] The terms "therapeutically-effective amount" and "effective amount" as used herein means that amount of a compound, material, or composition comprising a compound of the present invention which is effective for producing some desired therapeutic effect in at least a sub-population of cells in an animal at a reasonable benefit/risk ratio applicable to any medical treatment. Toxicity and therapeutic efficacy of subject compounds may be determined by standard pharmaceutical procedures in cell cultures or experimental animals, e.g., for determining the LD.sub.50 and the ED.sub.50. Compositions that exhibit large therapeutic indices are preferred. In some embodiments, the LD.sub.50 (lethal dosage) can be measured and can be, for example, at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 200%, 300%, 400%, 500%, 600%, 700%, 800%, 900%, 1000% or more reduced for the agent relative to no administration of the agent. Similarly, the ED.sub.50 (i.e., the concentration which achieves a half-maximal inhibition of symptoms) can be measured and can be, for example, at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 200%, 300%, 400%, 500%, 600%, 700%, 800%, 900%, 1000% or more increased for the agent relative to no administration of the agent. Also, Similarly, the IC.sub.50 (i.e., the concentration which achieves half-maximal cytotoxic or cytostatic effect on cancer cells) can be measured and can be, for example, at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 200%, 300%, 400%, 500%, 600%, 700%, 800%, 900%, 1000% or more increased for the agent relative to no administration of the agent. In some embodiments, cancer cell growth in an assay can be inhibited by at least about 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or even 100%. In another embodiment, at least about a 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or even 100% decrease in a solid malignancy can be achieved.

[0226] A "transcribed polynucleotide" or "nucleotide transcript" is a polynucleotide (e.g. an mRNA, hnRNA, a cDNA, or an analog of such RNA or cDNA) which is complementary to or homologous with all or a portion of a mature mRNA made by transcription of a biomarker nucleic acid and normal post-transcriptional processing (e.g. splicing), if any, of the RNA transcript, and reverse transcription of the RNA transcript.

[0227] As used herein, the term "unresponsiveness" includes refractivity of cancer cells to therapy or refractivity of therapeutic cells, such as immune cells, to stimulation, e.g., stimulation via an activating receptor or a cytokine. Unresponsiveness can occur, e.g., because of exposure to immunosuppressants or exposure to high doses of antigen. As used herein, the term "anergy" or "tolerance" includes refractivity to activating receptor-mediated stimulation. Such refractivity is generally antigen-specific and persists after exposure to the tolerizing antigen has ceased. For example, anergy in T cells (as opposed to unresponsiveness) is characterized by lack of cytokine production, e.g., IL-2. T cell anergy occurs when T cells are exposed to antigen and receive a first signal (a T cell receptor or CD-3 mediated signal) in the absence of a second signal (a costimulatory signal). Under these conditions, reexposure of the cells to the same antigen (even if reexposure occurs in the presence of a costimulatory polypeptide) results in failure to produce cytokines and, thus, failure to proliferate. Anergic T cells can, however, proliferate if cultured with cytokines (e.g., IL-2). For example, T cell anergy can also be observed by the lack of IL-2 production by T lymphocytes as measured by ELISA or by a proliferation assay using an indicator cell line. Alternatively, a reporter gene construct can be used. For example, anergic T cells fail to initiate IL-2 gene transcription induced by a heterologous promoter under the control of the 5′ IL-2 gene enhancer or by a multimer of the API sequence that can be found within the enhancer (Kang et al. (1992) *Science* 257:1134).

[0228] As used herein, the term "protein complex" means a composite unit that is a combination of two or more proteins formed by interaction between the proteins. Typically, but not necessarily, a "protein complex" is formed by the binding of two or more proteins together through specific non-covalent binding interactions. However, covalent bonds may also be present between the interacting partners. For instance, the two interacting partners can be covalently crosslinked so that the protein complex becomes more stable. The protein complex may or may not include and/or be associated with other molecules such as nucleic acid, such as RNA or DNA, or lipids or further cofactors or moieties selected from a metal ions, hormones, second messengers, phosphate, sugars. A "protein complex" encompassed by the present invention may also be part of or a unit of a larger physiological protein assembly.

[0229] The term “isolated protein complex” means a protein complex present in a composition or environment that is different from that found in nature, in its native or original cellular or body environment. Preferably, an “isolated protein complex” is separated from at least 50%, more preferably at least 75%, most preferably at least 90% of other naturally co-existing cellular or tissue components. Thus, an “isolated protein complex” may also be a naturally existing protein complex in an artificial preparation or a non-native host cell. An “isolated protein complex” may also be a “purified protein complex”, that is, a substantially purified form in a substantially homogenous preparation substantially free of other cellular components, other polypeptides, viral materials, or culture medium, or, when the protein components in the protein complex are chemically synthesized, free of chemical precursors or by-products associated with the chemical synthesis. A “purified protein complex” typically means a preparation containing preferably at least 75%, more preferably at least 85%, and most preferably at least 95% of a particular protein complex. A “purified protein complex” may be obtained from natural or recombinant host cells or other body samples by standard purification techniques, or by chemical synthesis.
[0230] The term “modified protein complex” refers to a protein complex present in a composition that is different from that found in nature, in its native or original cellular or body environment. The term “modification” as used herein refers to all modifications of a protein or protein complex encompassed by the present invention including cleavage and addition or removal of a group. In some embodiments, the “modified protein complex” comprises at least one subunit that is modified, i.e., different from that found in nature, in its native or original cellular or body environment. The “modified subunit” may be, e.g., a derivative or fragment of the native subunit from which it derives from.
[0231] As used herein, the term “domain” means a functional portion, segment or region of a protein, or polypeptide. “Interaction domain” refers specifically to a portion, segment or region of a protein, polypeptide or protein fragment that is responsible for the physical affinity of that protein, protein fragment or isolated domain for another protein, protein fragment or isolated domain.
[0232] The terms “polypeptide fragment” or “fragment”, when used in reference to a reference polypeptide, refers to a polypeptide in which amino acid residues are deleted as compared to the reference polypeptide itself, but where the remaining amino acid sequence is usually identical to the corresponding positions in the reference polypeptide. Such deletions may occur at the amino-terminus, internally, or at the carboxyl-terminus of the reference polypeptide, or alternatively both. Fragments typically are at least 5, 6, 8 or 10 amino acids long, at least 14 amino acids long, at least 20, 30, 40 or 50 amino acids long, at least 75 amino acids long, or at least 100, 150, 200, 300, 500 or more amino acids long. They can be, for example, at least and/or including 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 120, 140, 160, 180, 200, 220, 240, 260, 280, 300, 320, 340, 360, 380, 400, 420, 440, 460, 480, 500, 520, 540, 560, 580, 600, 620, 640, 660, 680, 700, 720, 740, 760, 780, 800, 820, 840, 860, 880, 900, 920, 940, 960, 980, 1000, 1020, 1040, 1060, 1080, 1100, 1120, 1140, 1160, 1180, 1200, 1220, 1240, 1260, 1280, 1300, 1320, 1340 or more long so long as they are less than the length of the full-length polypeptide. Alternatively, they can be no longer than and/or excluding such a range so long as they are less than the length of the full-length polypeptide.
[0233] The term “tag” as used herein is meant to be understood in its broadest sense and to include, but is not limited to any suitable enzymatic, fluorescent, or radioactive labels and suitable epitopes, including but not limited to HA-tag, Myc-tag, T7, His-tag, FLAG-tag, Calmodulin binding proteins, glutathione-S-transferase, strep-tag, KT3-epitope, EEF-epitopes, green-fluorescent protein and variants thereof.
[0234] The term “nucleosome” refers to the fundamental unit of chromatin. The term “chromatin” refer to the larger-scale nucleoprotein structure comprising the cellular genome. Cellular chromatin comprises nucleic acid, primarily DNA, and protein, including histones and non-histone chromosomal proteins. The majority of eukaryotic cellular chromatin exists in the form of nucleosomes, wherein a “nucleosome” core comprises approximately 150 base pairs of DNA associated with an octamer comprising two each of histones H2A, H2B, H3 and H4; and linker DNA (of variable length depending on the organism) extends between nucleosome cores. A molecule of histone H1 is generally associated with the linker DNA. For the purposes of the present disclosure, the term “chromatin” is meant to encompass all types of cellular nucleoprotein, both prokaryotic and eukaryotic. Cellular chromatin includes both chromosomal and episomal chromatin.
[0235] The term “histone” refers to highly alkaline proteins found in eukaryotic cell nuclei that package and order DNA into structural units called nucleosomes. They are the chief protein components of chromatin, acting as spools around which DNA winds, and play a role in gene regulation. In certain embodiments, the histone is histone H2A (e.g., human, mouse, rat, and/or *Xenopus*, optionally canonical Histone H2A). In certain embodiments, the histone is histone H2B (e.g., human, mouse, rat, and/or *Xenopus*, optionally canonical Histone H2B). As described below, H2A and H2B sequences, variation, and structure-function relationships are well-known in the art and are functionally similar, such that, for example, working examples described herein use *Xenopus* H2A and H2B sequences because they are structurally and functionally similar to Human H2A and H2B sequences.
[0236] An “accessible region” is a site in cellular chromatin in which a target site present in the nucleic acid can be bound by an exogenous molecule which recognizes the target site. Without wishing to be bound by any particular

theory, it is believed that an accessible region is one that is not packaged into a nucleosomal structure. The distinct structure of an accessible region can often be detected by its sensitivity to chemical and enzymatic probes, for example, nucleases The accessibility of chromatin is mediated in part by interactions with SWI/SNF (BAF) complexes via interactions with the nucleosome “acidic patch.” The “acidic patch” of a nucleosome is formed from six H2A and two H2B residues, which together create a highly contoured and negatively charged binding interface on the nucleosome surface. This canonical structural region of nucleosomes is well-known in the art (see, for example, Dann et al. (2017) *Nature* 548:607-611 and Luger et al. (1997) *J. Mol. Biol.* 272:301-311) and is described further herein. In certain assays useful according to the present invention, nucleosomal interactions with DNA and/or proteins (e.g., SS18-SSX-containing BAF complexes), can be analyzed. Certain such assays measure changes to DNA lengths. The preferential protection against degradation may be due to the DNA being wrapped around one or more histone proteins, preferably an octomer of histone proteins. The threshold size may be the size of a complete turn of the DNA about a histone core+/−22 bases. The threshold size may be between 100 and 160 bases, preferably between 110 and b 140 bases, more preferably between 120 and 130 bases and ideally 125 bases+/−1 base. The threshold size may be a size equal to or greater than 100 bases, more preferably equal to or greater than 110 bases still more preferably equal to or greater than 120 bases and ideally 125 bases or more.

[0237] Eukaryotes have chromatin arranged around proteins in the form of nucleosomes, which are the smallest subunits of chromatin and include approximately 146-147 base pairs of DNA wrapped around an octamer of core histone proteins (two each of H2A, H2B, H3, and H4). As used herein, the term “Histone H3” refers to the H3 member of the Histone family, which comprises proteins used to form the structure of nucleosomes in eukaryotic cells. Mammalian cells have three known sequence variants of Histone H3 proteins, denoted H3.1, H3.2 and H3.3, that are highly conserved differing in sequence by only a few amino acids.

[0238] As used herein, the term “Histone H3” can refer to H3.1, H3.2, or H3.3 individually or collectively. These amino acid sequences include a methionine as residue number 1 that is cleaved off when the protein is processed. Thus, for example, serine 11 in the Histone H3 amino acid sequences shown in Table 1 below corresponds to serine (Ser) 10 of the present invention. These three protein variants are encoded by at least fifteen different genes/transcripts. Sequences encoding the Histone H3.1 variant are publicly available as HIST1H3A (NM 003529.2; NP_003520.1), HIST1H3B (NM_003537.3; NP 003528.1), HIST1H3C (NM 003531.2; NP_003522.1), HIST1H3D (NM_003530.3; NP 003521.2), HIST1H3E (NM 003532.2; NP_003523.1), HIST1H3F (NM_021018.2; NP 066298.1), HIST1H3G (NM 003534.2; NP_003525.1), HIST1H3H (NM_003536.2; NP_003527.1), HIST1H3I (NM 003533.2; NP_003524.1), and HIST1H3J (NM_003535.2; NP_003526.1). Sequences encoding the Histone H3.2 variant are publicly available as HIST2H3A (NM_001005464.2; NP 001005464.1), HIST2H3C (NM_021059.2; NP_066403.2), and HIST2H3D (NM_001123375.1; NP 001116847.1). Sequences encoding the Histone H3.3 variant are publicly available as H3F3A (NM_002107.3; NP_002098.1) and H3F3B (NM 005324.3; NP_005315.1). See U.S. Pat. Publ. 2012/0202843 for additional details. Moreover, polypeptide sequences for Histone H3 orthologs, as well as nucleic acid sequences that encode such polypeptides, are well-known in many species, and include, for example, Histone H3.1 orthologs in mice (NM_013550.4; NP_038578.2), chimpanzee (XM_527253.4; XP_527253.2), monkey (XM_001088298.2; XP 001088298.1), dog (XM_003434195.1; XP 003434243.1), cow (XM_002697460.1; XP_002697506.1), rat (XM_001055231.2; XP 001055231.1), and zebrafish (NM_001100173.1; NP_001093643.1). Histone H3.2 orthologs in mice (NM_178215.1; NP_835587.1), chimpanzee (XM_524859.4; XP_524859.2), monkey (XM_001084245.2; XP_001084245.1), dog (XM_003640147.1; XP 003640195.1), cow (XM_002685500.1; XP_002685546.1), rat (NM_001107698.1; NP 001101168.1), chicken (XM_001233027.2; XP_001233028.1), and zebrafish (XM_002662732.1; XP_002662778.1). Similarly, Histone H3.3 orthologs in mice (XM_892026.4; XP_897119.3), monkey (XM_001085836.2; XP_001085836.1), cow (NM_001099370.1; NP 001092840.1), rat (NM_053985.2; NP_446437.1), chicken (NM_205296.1; NP_990627.1), and zebrafish (NM_200003.1; NP_956297.1), are well-known. Representative Histone H3 orthologs are provided in Table 1.

[0239] As used herein, the term “Histone H2” can refer to H2A or H2B individually or collectively. The structure of H2A consists of histone fold domain extended by a short alphaC-helix and has both N- and C-terminal tails. The alphaC-helix and C-terminal tail form a docking domain that locks the H2A-H2B dimer onto the surface of H3-H4 tetramer. H2A protein sequences, and nucleic acids encoding same, are well-known in the art and include many useful variants, including canonical H2A, H2A.1, H2A.B, H2A.L, H2A.P, H2A.W, H2A.X, H2A.Z, and macroH2A (see Draizen et al. (2016) *Database* PMID: 26989147 and HistoneDB 2.0 available on the World Wide Web). The structure of H2B consists of histone fold with a long flexible N-terminal tail which protrudes between the DNA gyres. H2B interats with H4 in the nucleosome vore via four helix bundle motif and alphaC-helix of H2B decorates the nucleosome surface. H2B protein sequences, and nucleic acids ecndogin same, are well-known in the art and include many useful variants, including canonical H2B, H2B.1, H2B.W, H2B.Z, sperm H2B, and subH2B (see Draizen et al. (2016) *Database* PMID: 26989147 and HistoneDB 2.0 available on the World Wide Web).

[0240] There is a known and definite correspondence between the amino acid sequence of a particular protein and the nucleotide sequences that can code for the protein, as defined by the genetic code (shown below). Likewise, there is a known and definite correspondence between the nucleotide sequence of a particular nucleic acid and the amino

acid sequence encoded by that nucleic acid, as defined by the genetic code.
TABLE-US-00001 GENETIC CODE Alanine (Ala, A) GCA, GCC, GCG, GCT Arginine (Arg, R) AGA, ACG, CGA, CGC, CGG, CGT Asparagine (Asn, N) AAC, AAT Aspartic acid (Asp, D) GAC, GAT Cysteine (Cys, C) TGC, TGT Glutamic acid (Glu, E) GAA, GAG Glutamine (Gln, Q) CAA, CAG Glycine (Gly, G) GGA, GGC, GGG, GGT Histidine (His, H) CAC, CAT Isoleucine (Ile, I) ATA, ATC, ATT Leucine (Leu, L) CTA, CTC, CTG, CTT, TTA, TTG Lysine (Lys, K) AAA, AAG Methionine (Met, M) ATG Phenylalanine (Phe, F) TTC, TTT Proline (Pro, P) CCA, CCC, CCG, CCT Serine (Ser, S) AGC, AGT, TCA, TCC, TCG, TCT Threonine (Thr, T) ACA, ACC, ACG, ACT Tryptophan (Trp, W) TGG Tyrosine (Tyr, Y) TAC, TAT Valine (Val, V) GTA, GTC, GTG, GTT Termination signal (end) TAA, TAG, TGA
[0241] An important and well-known feature of the genetic code is its redundancy, whereby, for most of the amino acids used to make proteins, more than one coding nucleotide triplet may be employed (illustrated above). Therefore, a number of different nucleotide sequences may code for a given amino acid sequence. Such nucleotide sequences are considered functionally equivalent since they result in the production of the same amino acid sequence in all organisms (although certain organisms may translate some sequences more efficiently than they do others). Moreover, occasionally, a methylated variant of a purine or pyrimidine may be found in a given nucleotide sequence. Such methylations do not affect the coding relationship between the trinucleotide codon and the corresponding amino acid.
[0242] In view of the foregoing, the nucleotide sequence of a DNA or RNA encoding a biomarker nucleic acid (or any portion thereof) can be used to derive the polypeptide amino acid sequence, using the genetic code to translate the DNA or RNA into an amino acid sequence. Likewise, for polypeptide amino acid sequence, corresponding nucleotide sequences that can encode the polypeptide can be deduced from the genetic code (which, because of its redundancy, will produce multiple nucleic acid sequences for any given amino acid sequence). Thus, description and/or disclosure herein of a nucleotide sequence which encodes a polypeptide should be considered to also include description and/or disclosure of the amino acid sequence encoded by the nucleotide sequence. Similarly, description and/or disclosure of a polypeptide amino acid sequence herein should be considered to also include description and/or disclosure of all possible nucleotide sequences that can encode the amino acid sequence.
[0243] Finally, nucleic acid and amino acid sequence information for the loci and biomarkers of the present invention are well-known in the art and readily available on publicly available databases, such as the National Center for Biotechnology Information (NCBI). In addition, nucleic acid and amino acid sequence information for the SS18, SSX, SS18-SSX fusion proteins of the present invention are provided below.
TABLE-US-00002 TABLE 1 SEQ ID NO: 1 Human SSX1 cDNA Sequence variant 1 (NM_001278691.2; CDS: 148-714) 1 accactgctg ccgacctcgc aaccactgct ttgtctctga atagagacag ggtttcctta 61 tgttggccga actgggcttg acctcctcgg ctcaagtgat cctcccacct cggcctcgga 121 actacaggtg agactgctcc tggtgccatg aacggagacg acacctttgc aaagagaccc 181 agggatgatg ctaaagcatc agagaagaga agcaaggcct ttgatgatat tgccacatac 241 ttctctaaga aagagtggaa aaagatgaaa tactcggaga aaatcagcta tgtgtatatg 301 aagagaaact ataaggccat gactaaacta ggtttcaaag tcaccctccc acctttcatg 361 tgtaataaac aggccacaga cttccagggg aatgattttg ataatgacca taaccgcagg 421 attcaggttg aacatcctca gatgactttc ggcaggctcc acagaatcat cccgaagatc 481 atgcccaaga agccagcaga ggacgaaaat gattcgaagg gagtgtcaga agcatctggc 541 ccacaaaacg atgggaaaca actgcacccc ccaggaaaag caaatatttc tgagaagatt 601 aataagagat ctggacccaa aagggggaaa catgcctgga cccacagact gcgtgagaga 661 aagcagctgg tgatttatga agagatcagt gaccctgagg aagatgacga gtaactcccc 721 tgggggatac gacacatgcc cttgatgaga agcagaacgt ggtgaccttt cacgaacatg 781 ggcatggctg cggctccctc gtcatcaggt gcatagcaag tgaaagcaag tgttcacaac 841 ggtgaaactt gagcgtcatt tttcttagtg tgccaagagt tcgatgttag tgtttccatt 901 gtattttctt acagtgtgcc attctgttag atactatcct tataattgat gagcaagaca 961 tactgaatgc atatttcggt ttgtgtatcc atgcacctac gtcagaaaac aagtattgtc 1021 aggtattctc tccatagaac agcactatcc tcatctctcc ccagatgtga ctactgaggg 1081 cagttctgag tgtttaattt cagacttttt cctctgcatt tacacacaca cacacacaca 1141 cacgcacaca cacacaccaa gtaccagtat aagcatctcc catctgcttt tcccattgcc 1201 atgcgtcctg gtcaagcccc cctcactctg tttcctgttc agcatgtact cccctcatcc 1261 gattcccctg tatcagtcac tgacagttaa taaacctttg caaacgttc SEQ ID NO: 2 Human SSX1 cDNA Sequence variant 2 (NM_005635.4; CDS: 62- 628) 1 accactgctg ccgacctcgc aaccactgct ttgtctctga agtgagactg ctcctggtgc 61 catgaacgga gacgacacct ttgcaaagag acccagggat gatgctaaag catcagagaa 121 gagaagcaag gcctttgatg atattgccac atacttctct aagaaagagt ggaaaaagat 181 gaaatactcg gagaaaatca gctatgtgta tatgaagaga aactataagg ccatgactaa 241 actaggtttc aaagtcaccc tcccaccttt catgtgtaat aaacaggcca cagacttcca 301 ggggaatgat tttgataatg accataaccg caggattcag gttgaacatc ctcagatgac 361 tttcggcagg ctccacagaa tcatcccgaa gatcatgccc aagaagccag cagaggacga 421 aaatgattcg aagggagtgt cagaagcatc tggcccacaa aacgatggga aacaactgca 481 ccccccagga aaagcaaata tttctgagaa gattaataag agatctggac ccaaaagggg 541 gaaacatgcc tggacccaca gactgcgtga gagaaagcag ctggtgattt atgaagagat 601 cagtgaccct gaggaagatg acgagtaact cccctggggg atacgacaca tgcccttgat 661 gagaagcaga acgtggtgac ctttcacgaa catgggcatg gctgcggctc cctcgtcatc 721 aggtgcatag caagtgaaag caagtgttca caacggtgaa acttgagcgt catttttctt 781 agtgtgccaa gagttcgatg ttagtgtttc cattgtattt tcttacagtg tgccattctg 841 ttagatacta tccttataat tgatgagcaa gacatactga atgcatattt cggtttgtgt 901

atccatgcac ctacgtcaga aaacaagtat tgtcaggtat tctctccata gaacagcact 961 atcctcatct ctccccagat gtgactactg agggcagttc tgagtgttta atttcagact 1021 ttttcctctg catttacaca cacacacaca cacacacgca cacacacaca ccaagtacca 1081 gtataagcat ctcccatctg cttttcccat tgccatgcgt cctggtcaag cccccctcac 1141 tctgtttcct gttcagcatg tactcccctc atccgattcc cctgtatcag tcactgacag 1201 ttaataaacc tttgcaaacg ttc SEQ ID NO: 3 Human SSX1 Amino Acid Sequence isoform 1 (NP_001265620.1 and NP_005626.1) 1 mngddtfakr prddakasek rskafddiat yfskkewkkm kysekisyvy mkrnykamtk 61 lgfkvtlppf mcnkqatdfq gndfdndhnr riqvehpqmt fgrlhriipk impkkpaede 121 ndskgvseas gpqndgkqlh ppgkanisek inkrsgpkrg khawthrlre rkqlviyeei 181 sdpeedde SEQ ID NO: 4 Human SSX2 cDNA Sequence variant 1 (NM_003147.5; CDS: 137- 808) 1 gggattggct actttaagtt cagagtacgc atgctctgac tttctctctc tttcgattct 61 tccatactca gagtacgcac ggtctgattt tctctttgga ttcttccaaa atcagagtca 121 gactgctccc ggtgccatga acggagacga cgcctttgca aggagaccca cggttggtgc 181 tcaaatacca gagaagatcc aaaaggcctt cgatgatatt gccaaatact tctctaagga 241 agagtgggaa aagatgaaag cctcggagaa aatcttctat gtgtatatga agagaaagta 301 tgaggctatg actaaactag gtttcaaggc caccctccca cctttcatgt gtaataaacg 361 ggccgaagac ttccaggggga atgatttgga taatgaccct aaccgtggga atcaggttga 421 acgtcctcag atgactttcg gcaggctcca gggaatctcc ccgaagatca tgcccaagaa 481 gccagcagag gaaggaaatg attcggagga agtgccagaa gcatctggcc cacaaaatga 541 tgggaaagag ctgtgccccc cgggaaaacc aactacctct gagaagattc acgagagatc 601 tggaaatagg gaggcccaag aaaaggaaga gagacgcgga acagctcatc ggtggagcag 661 tcagaacaca cacaacattg gtcgattcag tttgtcaact tctatgggtg cagttcatgg 721 tacccccaaa acaattacac acaacaggga cccaaaaggg gggaacatgc ctggacccac 781 agactgcgtg agagaaaaca gctggtgatt tatgaagaga tcagcgaccc tgaggaagat 841 gacgagtaac tcccctcagg gatacgacac atgcccatga tgagaagcag aacgtggtga 901 cctttcacga acatgggcat ggctgcggac ccctcgtcat caggtgcata gcaagtgaaa 961 gcaagtgttc acaacagtga aaagttgagc gtcatttttc ttagtgtgcc aagagttcga 1021 tgttagcgtt tacgttgtat tttcttacac tgtgtcattc tgttagatac taacattttc 1081 attgatgagc aagacatact taatgcatat tttggtttgt gtatccatgc acctacctta 1141 gaaaacaagt attgtcggtt acctctgcat ggaacagcat taccctcctc tctccccaga 1201 tgtgactact gagggcagtt ctgagtgttt aatttcagat tttttcctct gcatttacac 1261 acacacgcac acaaaccaca ccacacacac acacacacac acacacacac acacacacac 1321 acaccaagta ccagtataag catctgccat ctgcttttcc cattgccatg cgtcctggtc 1381 aagctcccct cactctgttt cctggtcagc atgtactccc ctcatccgat tcccctgtag 1441 cagtcactga cagttaataa acctttgcaa acgttcaaaa aaaaaaaaaa aaaa SEQ ID NO: 5 Human SSX2 Amino Acid Sequence isoform 1 (NP_003138.3) 1 mngddafarr ptvgaqipek iqkafddiak yfskeewekm kasekifyvy mkrkyeamtk 61 lgfkatlppf mcnkraedfq gndldndpnr gnqverpqmt fgrlqgispk impkkpaeeg 121 ndseevpeas gpqndgkelc ppgkpttsek ihersgnrea qekeerrgta hrwssqnthn 181 igrfslstsm gavhgtpkti thnrdpkggn mpgptdcvre nsw SEQ ID NO: 6 Human SSX2 cDNA Sequence variant 2 (NM_175698.2; CDS: 137- 703) 1 gggattggct actttaagtt cagagtacgc atgctctgac tttctctctc tttcgattct 61 tccatactca gagtacgcac ggtctgattt ttctttgga ttcttccaaa atcagagtca 121 gactgctccc ggtgccatga acggagacga cgcctttgca aggagaccca cggttggtgc 181 tcaaatacca gagaagatcc aaaaggcctt cgatgatatt gccaaatact tctctaagga 241 agagtgggaa aagatgaaag cctcggagaa aatcttctat gtgtatatga agagaaagta 301 tgaggctatg actaaactag gtttcaaggc caccctccca cctttcatgt gtaataaacg 361 ggccgaagac ttccaggggga atgatttgga taatgaccct aaccgtggga atcaggttga 421 acgtcctcag atgactttcg gcaggctcca gggaatctcc ccgaagatca tgcccaagaa 481 gccagcagag gaaggaaatg attcggagga agtgccagaa gcatctggcc cacaaaatga 541 tgggaaagag ctgtgccccc cgggaaaacc aactacctct gagaagattc acgagagatc 601 tggacccaaa agggggggaac atgcctggac ccacagactg cgtgagagaa aacagctggt 661 gatttatgaa gagatcagcg accctgagga agatgacgag taactcccct cagggatacg 721 acacatgccc atgatgagaa gcagaacgtg gtgacctttc acgaacatgg gcatggctgc 781 ggacccctcg tcatcaggtg catagcaagt gaaagcaagt gttcacaaca gtgaaaagtt 841 gagcgtcatt tttcttagtg tgccaagagt tcgatgttag cgtttacgtt gtattttctt 901 acactgtgtc attctgttag atactaacat tttcattgat gagcaagaca tacttaatgc 961 atattttggt ttgtgtatcc atgcacctac cttagaaaac aagtattgtc ggttacctct 1021 gcatggaaca gcattaccct cctctctccc cagatgtgac tactgagggc agttctgagt 1081 gtttaatttc agattttttc ctctgcattt acacacacac gcacacaaac cacaccacac 1141 acacacacac acacacacac acacacacac acacacacca agtaccagta taagcatctg 1201 ccatctgctt ttcccattgc catgcgtcct ggtcaagctc ccctcactct gtttcctggt 1261 cagcatgtac tcccctcatc cgattcccct gtagcagtca ctgacagtta ataaaccttt 1321 gcaaacgttc aaaaaaaaaa aaaaaaaa SEQ ID NO: 7 Human SSX2 Amino Acid Sequence isoform 2 (NP_783629.1) 1 mngddafarr ptvgaqipek iqkafddiak yfskeewekm kasekifyvy mkrkyeamtk 61 lgfkatlppf mcnkraedfq gndldndpnr gnqverpqmt fgrlqgispk impkkpaeeg 121 ndseevpeas gpqndgkelc ppgkpttsek ihersgpkrg ehawthrlre rkqlviyeei 181 sdpeedde SEQ ID NO: 8 Human SSX2 cDNA Sequence variant 3 (NM_001278697.1; CDS: 137-781) 1 gggattggct actttaagtt cagagtacgc atgctctgac tttctctctc tttcgattct 61 tccatactca gagtacgcac ggtctgattt tctctttgga ttcttccaaa atcagagtca 121 gactgctccc ggtgccatga acggagacga cgcctttgca aggagaccca cggttggtgc 181 tcaaatacca gagaagatcc aaaaggcctt cgatgatatt gccaaatact tctctaagga 241 agagtgggaa aagatgaaag cctcggagaa aatcttctat gtgtatatga agagaaagta 301 tgaggctatg actaaactag gtttcaaggc caccctccca cctttcatgt gtaataaacg 361 ggccgaagac ttccaggggga atgatttgga taatgaccct aaccgtggga atcaggttga 421 acgtcctcag atgactttcg gcaggctcca gggaatctcc ccgaagatca tgcccaagaa 481 gccagcagag gaaggaaatg attcggagga agtgccagaa gcatctggcc cacaaaatga 541 tgggaaagag ctgtgccccc cgggaaaacc aactacctct gagaagattc acgagagatc 601 tggaaatagg gaggcccaag aaaaggaaga gagacgcgga acagctcatc ggtggagcag 661 tcagaacaca cacaacattg gacccaaaag gggggaacat gcctggaccc acagactgcg 721 tgagagaaaa cagctggtga tttatgaaga gatcagcgac cctgaggaag atgacgagta 781 actcccctca gggatacgac acatgcccat gatgagaagc agaacgtggt gacctttcac 841 gaacatgggc atggctgcgg acccctcgtc atcaggtgca tagcaagtga aagcaagtgt 901 tcacaacagt gaaaagttga gcgtcatttt tcttagtgtg ccaagagttc gatgttagcg 961 tttacgttgt attttcttac actgtgtcat tctgttagat actaacattt tcattgatga 1021 gcaagacata cttaatgcat attttggttt gtgtatccat gcacctacct tagaaaacaa 1081 gtattgtcgg ttacctctgc atggaacagc attaccctcc tctctcccca gatgtgacta 1141 ctgagggcag ttctgagtgt ttaatttcag attttttcct ctgcatttac acacacacgc 1201 acacaaacca caccacacac acacacacac acacacacac acacacacac acacaccaag 1261 taccagtata agcatctgcc atctgctttt cccattgcca tgcgtcctgg tcaagctccc 1321 ctcactctgt ttcctggtca gcatgtactc ccctcatccg attcccctgt agcagtcact 1381 gacagttaat aaacctttgc aaacgttcaa aaaaaaaaaa aaaaaa SEQ ID NO: 9 Human SSX2 Amino Acid Sequence isoform 3 (NP_001265626.1) 1 mngddafarr ptvgaqipek iqkafddiak yfskeewekm kasekifyvy mkrkyeamtk 61 lgfkatlppf mcnkraedfq gndldndpnr gnqverpqmt fgrlqgispk impkkpaeeg 121 ndseevpeas gpqndgkelc ppgkpttsek ihersgnrea qekeerrgta hrwssqnthn 181 igpkrgehaw thrlrerkql viyeeisdpe edde SEQ ID NO: 10 Human SSX2B cDNA Sequence variant 1 (NM_001278701.2; CDS: 88-759) 1 ctttcgattc ttccatactc agagtacgca cggtctgatt ttctctttgg attcttccaa 61 aatcagagtc agactgctcc cggtgccatg aacggagacg acgcctttgc aaggagaccc 121 acggttggtg ctcaaatacc agagaagatc caaaaggcct tcgatgatat tgccaaatac 181 ttctctaagg aagagtggga aaagatgaaa gcctcggaga aaatcttcta tgtgtatatg 241 aagagaaagt atgaggctat gactaaacta ggtttcaagg ccaccctccc acctttcatg 301 tgtaataaac gggccgaaga cttccagggg aatgatttgg ataatgaccc taaccgtggg 361 aatcaggttg aacgtcctca gatgactttc ggcaggctcc agggaatctc cccgaagatc 421 atgcccaaga agccagcaga ggaaggaaat gattcggagg aagtgccaga agcatctggc 481 ccacaaaatg atgggaaaga gctgtgcccc ccgggaaaac caactacctc tgagaagatt 541 cacgagagat ctggaaatag ggaggcccaa gaaaaggaag agagacgcgg aacagctcat 601 cggtggagca gtcagaacac acacaacatt ggtcgattca gtttgtcaac ttctatgggt 661 gcagttcatg gtaccccctaa aacaattaca cacaacaggg acccaaaagg ggggaacatg 721 cctggaccca cagactgcgt gagagaaaac agctggtgat ttatgaagag atcagcgacc 781 ctgaggaaga tgacgagtaa ctcccctcag ggatacgaca catgcccatg atgagaagca 841 gaacgtggtg acctttcacg aacatgggca tggctgcgga cccctcgtca tcaggtgcat 901 agcaagtgaa agcaagtgtt cacaacagtg aaaagttgag cgtcattttt cttagtgtgc 961 caagagttcg atgttagcgt ttacgttgta ttttcttaca ctgtgtcatt ctgttagata 1021 ctaacatttt cattgatgag caagacatac ttaatgcata ttttggtttg tgtatccatg 1081 cacctacctt agaaaacaag tattgtcggt tacctctgca tggaacagca ttaccctect 1141 ctctccccag atgtgactac tgagggcagt tctgagtgtt taatttcaga tttttcctc 1201 tgcatttaca cacacacgca cacaaaccac accacacaca cacacacaca cacacacaca 1261 cacacacaca cacaccaagt accagtataa gcatctgcca tctgcttttc ccattgccat 1321 gcgtcctggt caagctcccc tcactctgtt tcctggtcag catgtactcc cctcatccga 1381 ttcccctgta gcagtcactg acagttaata aacctttgca aacgttc SEQ ID NO: 11 Human SSX2B Amino Acid Sequence isoform 1 (NP_001265630.1) 1 mngddafarr ptvgaqipek iqkafddiak yfskeewekm kasekifyvy mkrkyeamtk 61 lgfkatlppf mcnkraedfq gndldndpnr gnqverpqmt fgrlqgispk impkkpaeeg 121 ndseevpeas gpqndgkelc ppgkpttsek ihersgnrea qekeerrgta hrwssqnthn 181 igrfslstsm gavhgtpkti thnrdpkggn mpgptdcvre nsw SEQ ID NO: 12 Human SSX2B cDNA Sequence variant 2 (NM_001164417.3; CDS: 88-654) 1 ctttcgattc ttccatactc agagtacgca cggtctgatt ttctctttgg attcttccaa 61 aatcagagtc agactgctcc cggtgccatg aacggagacg acgcctttgc aaggagaccc 121 acggttggtg ctcaaatacc agagaagatc caaaaggcct tcgatgatat tgccaaatac 181 ttctctaagg aagagtggga aaagatgaaa gcctcggaga aaatcttcta tgtgtatatg 241 aagagaaagt atgaggctat gactaaacta ggtttcaagg ccaccctccc acctttcatg 301 tgtaataaac gggccgaaga cttccagggg aatgatttgg ataatgaccc taaccgtggg 361 aatcaggttg aacgtcctca gatgactttc ggcaggctcc agggaatctc cccgaagatc 421 atgcccaaga agccagcaga ggaaggaaat gattcggagg aagtgccaga agcatctggc 481 ccacaaaatg atgggaaaga gctgtgcccc ccgggaaaac caactacctc tgagaagatt 541 cacgagagat ctggacccaa aagggggggaa catgcctgga cccacagact gcgtgagaga 601 aaacagctgg tgatttatga agagatcagc gaccctgagg aagatgacga gtaactcccc 661 tcagggatac gacacatgcc catgatgaga agcagaacgt ggtgaccttt cacgaacatg 721 ggcatggctg cggacccctc gtcatcaggt gcatagcaag tgaaagcaag tgttcacaac 781 agtgaaaagt tgagcgtcat ttttcttagt gtgccaagag ttcgatgtta gcgtttacgt 841 tgtattttct tacactgtgt cattctgtta gatactaaca ttttcattga tgagcaagac 901 atacttaatg catattttgg tttgtgtatc catgcaccta ccttagaaaa caagtattgt 961 cggttacctc tgcatggaac agcattaccc tectctctcc ccagatgtga ctactgaggg 1021 cagttctgag tgtttaattt cagatttttt cctctgcatt tacacacaca cgcacacaaa 1081 ccacaccaca cacacacaca cacacacaca cacacacacc aagtaccagt 1141 ataagcatct gccatctgct tttcccattg ccatgcgtcc tggtcaagct cccctcactc 1201 tgtttcctgg tcagcatgta ctcccctcat ccgattcccc tgtagcagtc actgacagtt 1261 aataaacctt tgcaaacgtt c SEQ ID NO: 13 Human SSX2B Amino Acid Sequence isoform 2 (NP_001157889.1) 1 mngddafarr ptvgaqipek iqkafddiak yfskeewekm kasekifyvy mkrkyeamtk 61 lgfkatlppf mcnkraedfq gndldndpnr gnqverpqmt fgrlqgispk impkkpaeeg 121 ndseevpeas gpqndgkelc ppgkpttsek ihersgpkrg ehawthrlre rkqlviyeei 181 sdpeedde SEQ ID NO: 14 Human SSX2B cDNA Sequence variant 3 (NM_001278702.2; CDS: 88-732) 1 ctttcgattc ttccatactc agagtacgca cggtctgatt ttctctttgg attcttccaa 61 aatcagagtc agactgctcc cggtgccatg aacggagacg acgcctttgc aaggagaccc 121 acggttggtg ctcaaatacc agagaagatc caaaaggcct tcgatgatat tgccaaatac 181 ttctctaagg aagagtggga aaagatgaaa gcctcggaga aaatcttcta tgtgtatatg 241 aagagaaagt atgaggctat gactaaacta ggtttcaagg ccaccctccc acctttcatg 301 tgtaataaac gggccgaaga cttccagggg aatgatttgg ataatgaccc taaccgtggg 361 aatcaggttg aacgtcctca gatgactttc ggcaggctcc agggaatctc cccgaagatc 421 atgcccaaga agccagcaga ggaaggaaat gattcggagg aagtgccaga agcatctggc 481 ccacaaaatg atgggaaaga gctgtgcccc ccgggaaaac caactacctc tgagaagatt 541 cacgagagat ctggaaatag ggaggcccaa gaaaaggaag agagacgcgg aacagctcat 601 cggtggagca gtcagaacac acacaacatt ggacccaaaa ggggggaaca tgcctggacc 661 cacagactgc gtgagagaaa acagctggtg atttatgaag agatcagcga ccctgaggaa 721 gatgacgagt aactcccctc agggatacga cacatgccca tgatgagaag cagaacgtgg 781 tgacctttca cgaacatggg catggctgcg gacccctcgt catcaggtgc atagcaagtg 841 aaagcaagtg ttcacaacag tgaaaagttg agcgtcattt ttcttagtgt gccaagagtt 901 cgatgttagc gtttacgttg tattttctta cactgtgtca ttctgttaga tactaacatt 961 ttcattgatg agcaagacat acttaatgca tattttggtt tgtgtatcca tgcacctacc 1021 ttagaaaaca agtattgtcg gttacctctg catggaacag cattaccctc ctctctcccc 1081 agatgtgact actgagggca gttctgagtg tttaatttca gattttttcc tctgcattta 1141 cacacacacg cacacaaacc acaccacaca cacacacaca cacacacaca cacacacaca 1201 cacacaccaa gtaccagtat aagcatctgc catctgcttt tcccattgcc atgcgtcctg 1261 gtcaagctcc cctcactctg tttcctggtc agcatgtact cccctcatcc gattccctg 1321 tagcagtcac tgacagttaa taaacctttg caaacgttc SEQ ID NO: 15 Human SSX2B Amino Acid Sequence isoform 3 (NP_001265631.1) 1 mngddafarr ptvgaqipek iqkafddiak yfskeewekm kasekifyvy mkrkyeamtk 61 lgfkatlppf mcnkraedfq gndldndpnr gnqverpqmt fgrlqgispk impkkpaeeg 121 ndseevpeas gpqndgkelc ppgkpttsek ihersgnrea qekeerrgta hrwssqnthn 181 igpkrgehaw thrlrerkql viyeeisdpe edde SEQ ID NO: 16 Human SSX4 cDNA Sequence variant 1 (NM_005636.4; CDS: 47- 613) 1 gcccttttga ttcttccaca atcagggtga gactgctccc agtgccatga acggagacga 61 cgcctttgca aggagaccca gggatgatgc tcaaatatca gagaagttac gaaaggcctt 121 cgatgatatt gccaaatact tctctaagaa agagtgggaa aagatgaaat cctcggagaa 181 aatcgtctat gtgtatatga agctaaacta tgaggtcatg actaaactag gtttcaaggt 241 caccctccca cctttcatgc gtagtaaacg ggctgcagac ttccacggga atgattttgg 301 taacgatcga aaccacagga atcaggttga acgtcctcag atgactttcg gcagcctcca 361 gagaatcttc ccgaagatca tgcccaagaa gccagcagag gaagaaaatg gtttgaagga 421 agtgccagag gcatctggcc cacaaaatga tgggaaacag ctgtgccccc cgggaaatcc 481 aagtaccttg gagaagatta acaagacatc tggacccaaa agggggaaac atgcctggac 541 ccacagactg cgtgagagaa agcagctggt ggtttatgaa gagatcagcg accctgagga 601 agatgacgag taactcccct cggggatatg acacatgccc atgatgagaa gcagaacgtg 661 gtgacctttc acgaacatgg gcatggctgc ggacccctcg tcatcaggtg catagcaagt 721 gaaagcaagt gttcacaaca gtgaaaagtt gagcgtcatt tttcttagtg tgccaagagt 781 tcgatgttgg cgtttccgct gtattttctt gcagtgtgcc attctgttag acattagcgt 841 tttcgttgat gagcaagaca tgcttaatgc atatttcggc ttgtgtatcc atgcacctac 901 ctcagaaaac aagtattgtc aggtattctc tccatagaac agcactaccc tectctctcc 961 ccagatgtga ctactgaggg gaggtctgag tgtttaattt ccgatttttt cctctgcatt 1021 tacacacaca ccacacacgc acacacacac accaagtacc agtataagca tctcccatct 1081 gcttttctcc attgccatgc gacctggtca agcccccctc actctgtttc ctgttcagca 1141 tgtactcccc tcatccgatt ccgttgtatc agtcactgac agttaataaa cctttgcaaa 1201 cgttcccca SEQ ID NO: 17 Human SSX4 Amino Acid Sequence isoform 1 (NP_005627.1) 1 mngddafarr prddaqisek lrkafddiak yfskkewekm kssekivyvy mklnyevmtk 61 lgfkvtlppf mrskraadfh gndfgndrnh rnqverpqmt fgslqrifpk impkkpaeee 121 nglkevpeas gpqndgkqlc ppgnpstlek inktsgpkrg khawthrlre rkqlvvyeei 181 sdpeedde SEQ ID NO: 18 Human SSX4 cDNA Sequence variant 2 (NM_175729.1; CDS: 59- 520) 1 acacgccgat ttgccctttt gattcttcca caatcagggt gagactgctc ccagtgccat 61 gaacggagac gacgcctttg caaggagacc cagggatgat gctcaaatat cagagaagtt 121 acgaaaggcc ttcgatgata ttgccaaata cttctctaag aaagagtggg aaaagatgaa 181 atcctcggag aaaatcgtct atgtgtatat gaagctaaac tatgaggtca tgactaaact 241 aggtttcaag gtcaccctcc cacctttcat gcgtagtaaa cgggctgcag acttccacgg 301 gaatgatttt ggtaacgatc gaaaccacag gaatcaggtt gaacgtcctc agatgacttt 361 cggcagcctc cagagaatct tcccgaagga cccaaaaggg ggaaacatgc ctggacccac 421 agactgcgtg agagaaagca gctggtggtt tatgaagaga tcagcgaccc tgaggaagat 481 gacgagtaac tcccctcggg gatatgacac atgcccatga tgagaagcag aacgtggtga 541 cctttcacga acatgggcat ggctgcggac ccctcgtcat caggtgcata gcaagtgaaa 601 gcaagtgttc acaacagtga aaagttgagc gtcatttttc ttagtgtgcc aagagttcga 661 tgttggcgtt tccgctgtat tttcttgcag tgtgccattc tgttagacat tagcgttttc 721 gttgatgagc aagacatgct taatgcatat ttcggcttgt gtatccatgc acctacctca 781 gaaaacaagt attgtcaggt attctctcca tagaacagca ctaccctect ctctccccag 841 atgtgactac tgaggggagg tctgagtgtt taatttccga tttttttcctc tgcatttaca 901 cacacaccac acgcacacac acacaccacc agtaccagta taagcatctc ccatctgctt 961 ttctccattg ccatgcgacc tggtcaagcc cccctcactc tgtttcctgt tcagcatgta 1021 ctcccctcat ccgattccgt tgtatcagtc actgacagtt aataaacctt tgcaaacgtt 1081 caaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaa SEQ ID NO: 19 Human SSX4 Amino Acid Sequence isoform 2 (NP_783856.1) 1 mngddafarr prddaqisek lrkafddiak yfskkewekm kssekivyvy mklnyevmtk 61 lgfkvtlppf mrskraadfh gndfgndrnh rnqverpqmt fgslqrifpk dpkggnmpgp 121 tdcvresww fmkrsatlrk mtsnsprgyd tcp SEQ ID NO: 20 Human SSX4B cDNA Sequence variant 1 (NM_001034832.3; CDS: 70-636) 1 tcagagtacg cacacgccga tttgcccttt tgattcttcc acaatcaggg tgagactgct 61 cccagtgcca tgaacggaga cgacgccttt gcaaggagac ccagggatga tgctcaaata 121 tcagagaagt tacgaaaggc cttcgatgat attgccaaat acttctctaa gaaagagtgg 181 gaaaagatga aatcctcgga gaaaatcgtc tatgtgtata tgaagctaaa ctatgaggtc 241 atgactaaac taggtttcaa ggtcaccctc ccacctttca tgcgtagtaa acgggctgca 301 gacttccacg ggaatgattt tggtaacgat cgaaaccaca ggaatcaggt tgaacgtcct 361 cagatgactt tcggcagcct ccagagaatc ttcccgaaga tcatgcccaa gaagccagca 421 gaggaagaaa atggtttgaa ggaagtgcca gaggcatctg gcccacaaaa tgatgggaaa 481 cagctgtgcc ccccgggaaa tccaagtacc ttggagaaga ttaacaagac atctggaccc 541 aaaaggggga aacatgcctg gacccacaga ctgcgtgaga gaaagcagct ggtggtttat 601 gaagagatca gcgaccctga ggaagatgac gagtaactcc cctcggggat atgacacatg 661 cccatgatga gaagcagaac gtggtgacct ttcacgaaca tgggcatggc tgcggacccc 721 tcgtcatcag gtgcatagca agtgaaagca agtgttcaca acagtgaaaa gttgagcgtc 781 atttttctta gtgtgccaag agttcgatgt tggcgtttcc gctgtatttt cttgcagtgt 841 gccattctgt tagacattag cgttttogtt gatgagcaag acatgcttaa tgcatatttc 901 ggcttgtgta tccatgcacc tacctcagaa aacaagtatt gtcaggtatt ctctccatag 961 aacagcacta ccctcctctc tccccagatg tgactactga ggggaggtct gagtgtttaa 1021 tttccgattt tttcctctgc atttacacac acaccacaca cgcacacaca cacaccaagt 1081 accagtataa gcatctccca tctgcttttc tccattgcca tgcgacctgg tcaagccccc 1141 ctcactctgt ttcctgttca gcatgtactc ccctcatccg attccgttgt atcagtcact 1201 gacagttaat aaacctttgc aaacgttcaa aaaaaaaaaa aaaa SEQ ID NO: 21 Human SSX4B Amino Acid Sequence isoform 1 (NP_001030004.1) 1 mngddafarr prddaqisek lrkafddiak yfskkewekm kssekivyvy mklnyevmtk 61 lgfkvtlppf mrskraadfh gndfgndrnh rnqverpqmt fgslqrifpk impkkpaeee 121 nglkevpeas gpqndgkqlc ppgnpstlek inktsgpkrg khawthrlre rkqlvvyeei 181 sdpeedde SEQ ID NO: 22 Human SSX4B cDNA Sequence variant 2 (NM_001040612.2; CDS: 70-531) 1 tcagagtacg cacacgccga tttgcccttt tgattcttcc acaatcaggg tgagactgct 61 cccagtgcca tgaacggaga cgacgccttt gcaaggagac ccagggatga tgctcaaata 121 tcagagaagt tacgaaaggc cttcgatgat attgccaaat acttctctaa gaaagagtgg 181 gaaaagatga aatcctcgga gaaaatcgtc tatgtgtata tgaagctaaa ctatgaggtc 241 atgactaaac taggtttcaa ggtcaccctc ccacctttca tgcgtagtaa acgggctgca 301 gacttccacg ggaatgattt tggtaacgat cgaaaccaca ggaatcaggt tgaacgtcct 361 cagatgactt tcggcagcct ccagagaatc ttcccgaagg acccaaaagg gggaaacatg 421 cctggaccca cagactgcgt gagagaaagc agctggtggt ttatgaagag atcagcgacc 481 ctgaggaaga tgacgagtaa ctcccctcgg ggatatgaca catgcccatg atgagaagca 541 gaacgtggtg acctttcacg aacatgggca tggctgcgga cccctcgtca tcaggtgcat 601 agcaagtgaa agcaagtgtt cacaacagtg aaaagttgag cgtcattttt cttagtgtgc 661 caagagttcg atgttggcgt ttccgctgta ttttcttgca gtgtgccatt ctgttagaca 721 ttagcgtttt cgttgatgag caagacatgc ttaatgcata tttcggcttg tgtatccatg 781 cacctacctc agaaaacaag tattgtcagg tattctctcc atagaacage actaccctcc 841 tctctcccca gatgtgacta ctgaggggag gtctgagtgt ttaatttccg atttttcct 901 ctgcatttac acacacacca cacacgcaca cacacacacc aagtaccagt ataagcatct 961 cccatctgct tttctccatt gccatgcgac ctggtcaagc ccccctcact ctgtttcctg 1021 ttcagcatgt actcccctca tccgattccg ttgtatcagt cactgacagt taataaacct 1081 ttgcaaacgt tcaaaaaaaa aaaaaaaa SEQ ID NO: 23 Human SSX4B Amino Acid Sequence isoform 2 (NP_001035702.1) 1 mngddafarr prddaqisek lrkafddiak yfskkewekm kssekivyvy mklnyevmtk 61 lgfkvtlppf mrskraadfh gndfgndrnh rnqverpqmt fgslqrifpk dpkggnmpgp 121 tdcvresww fmkrsatlrk mtsnsprgyd tcp SEQ ID NO: 24 Human SSX3 cDNA Sequence (NM_021014.4; 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TABLE-US-00003 SEQ ID NO: 26 Human SSX5 cDNA Sequence variant 1 (NM_021015.4; CDS: 86 . . . 775) 1 ctctctctct cgatttttcc acagagtacg cacgctctga ttgtttcgat tcttccaaaa 61 tcagagacag agtgctcccg gtgccatgaa cggagacgat gcctttgtac ggagacctag 121 ggttggttct caaataccag agaagatgca aaagcatccc tggagacaag tctgtgaccg 181 tggaatacat ttggtgaatc tcagtccgtt ctggaaggtg ggaagagagc cagccagcag 241 cattaaagct ctactgtgtg gcaggggaga agctagggcc ttcgatgata ttgccaaata 301 cttctctgag aaagagtggg aaaagatgaa agcctcggag aaaatcatct atgtgtatat 361 gaagagaaag tatgaggcca tgactaaact aggtttcaag gccaccctcc cacctttcat 421 gcgtaataaa cgggtcgcag acttccaggg gaatgatttt gataatgacc ctaaccgtgg 481 gaatcaggtt gaacatcctc agatgacttt cggcaggctc cagggaatct tcccgaagat 541 cacgcccgag aagccagcag aggaaggaaa tgattcgaag ggagtgccag aagcatctgg 601 cccacagaac aatgggaaac agctgcgccc ctcaggaaaa ctaaatacct ctgagaaggt 661 taacaagaca tctggaccca aaagggggaa acatgcctgg acccacagag tgcgtgagag 721 aaagcaactg gtgatttatg aagagatcag cgaccctcag gaagatgacg agtaactccc 781 ctcggggata tgacacatgc ccatgatgag aagcagaacg tggtgacctt tcacgaacat 841 gggcatggct gcggatccct cgtcatcagg tgtatagcaa gtgaaagcaa gtgttcacaa 901 cagtgaaaag ttgagcgtca tttttcttag tgtgccaaga gttcgatgtt agtgtttctg 961 ttgtattttg ttacagtgtg ccattctgtt agatattagc gttttcactg atgagcaaga 1021 catacttaat gcatatttca gtttgtgtat ccatgcacct acctcagaaa acaagtatcg 1081 tcaggtattc tctgcataga acaacactac cctcctctct tcccagatgt gaccactgag 1141 ggcagttctg agtgtttaat ttcagatttt ttcctctgca tttacacaaa cacacacaca 1201 tgccacacag acacacatgc gcgcgcgcgc gcacacacac acacacacac acacacacac 1261 acacacacac caagtaccag tataggcatc tcccaactgc ttttccccat gtgtcctggt 1321 caagcccccc tcactctgtt tcctgttcag catgtactcc cctcatccga ttcccctcta 1381 tcagtcactg ccagttaata aacctttgca aacgtt SEQ ID NO: 27 Human SSX5 Amino Acid Sequence isoform 1 (NP_066295.3) 1 mngddafvrr prvgsqipek mqkhpwrqvc drgihlvnls pfwkvgrepa ssikallcgr 61 gearafddia kyfsekewek mkasekiiyv ymkrkyeamt klgfkatlpp fmrnkrvadf 121 qgndfdndpn rgnqvehpqm tfgrlqgifp kitpekpaee gndskgvpea sgpqnngkql 181 rpsgklntse kvnktsgpkr gkhawthrvr erkqlviyee isdpqedde SEQ ID NO: 28 Human SSX5 cDNA Sequence variant 2 (NM_175723.1; CDS: 54 . . . 620) 1 cgctctgatt gtttcgattc ttccaaaatc agagacagag tgctcccggt gccatgaacg 61 gagacgatgc ctttgtacgg agacctaggg ttggttctca aataccagag aagatgcaaa 121 aggccttcga tgatattgcc aaatacttct ctgagaaaga gtgggaaaag atgaaagcct 181 cggagaaaat catctatgtg tatatgaaga gaaagtatga ggccatgact aaactaggtt 241 tcaaggccac cctcccacct ttcatgcgta ataaacgggt cgcagacttc caggggaatg 301 attttgataa tgaccctaac cgtgggaatc aggttgaaca tcctcagatg actttcggca 361 ggctccaggg aatcttcccg aagatcacgc ccgagaagcc agcagaggaa ggaaatgatt 421 cgaagggagt gccagaagca tctggcccac agaacaatgg gaaacagctg cgcccctcag 481 gaaaactaaa tacctctgag aaggttaaca agacatctgg acccaaaagg gggaaacatg 541 cctggaccca cagagtgcgt gagagaaagc aactggtgat ttatgaagag atcagcgacc 601 ctcaggaaga tgacgagtaa ctcccctcgg ggatatgaca catgcccatg atgagaagca 661 gaacgtggtg acctttcacg aacatgggca tggctgcgga tccctcgtca tcaggtgtat 721 agcaagtgaa agcaagtgtt cacaacagtg aaaagttgag cgtcattttt cttagtgtgc 781 caagagttcg atgttagtgt ttctgttgta ttttgttaca gtgtgccatt ctgttagata 841 ttagcgtttt cactgatgag caagacatac ttaatgcata tttcagtttg tgtatccatg 901 cacctacctc agaaaacaag tatcgtcagg tattctctgc atagaacaac actaccctcc 961 tctcttccca gatgtgacca ctgagggcag ttctgagtgt ttaatttcag atttttcct 1021 ctgcatttac acaaacacac acacatgcca cacagacaca catgcgcgcg cgcgcgcaca 1081 cacacacaca cacacacaca cacacacaca cacaccaagt accagtatag gcatctccca 1141 actgcttttc cccatgtgtc ctggtcaagc ccccctcact ctgtttcctg ttcagcatgt 1201 actcccctca tccgattccc ctctatcagt cactgccagt taataaacct ttgcaaacgt 1261 taaaaaaaaa aaaaaa SEQ ID NO: 29 Human SSX5 Amino Acid Sequence isoform 2 (NP_783729.1) 1 mngddafvrr prvgsqipek mqkafddiak yfsekewekm kasekiiyvy mkrkyeamtk 61 lgfkatlppf mrnkrvadfq gndfdndpnr gnqvehpqmt fgrlqgifpk itpekpaeeg 121 ndskgvpeas gpqnngkqlr psgklntsek vnktsgpkrg khawthrvre rkqlviyeei 181 sdpqedde SEQ ID NO: 30 Human SSX7 cDNA Sequence (NM_173358.2. CDS: 160-726) 1 ccaggctcca gggacagaac cttctcaaag tgggggtgga gactctgatt ttcccgccta 61 aagcatcccc tgggattggc tactttaagt tcagagtatg catgctctga ctttctctct 121 cgattcttcc atactcagag tcagactgct cctggtgcca tgaacggaga cgacgccttt 181 gcaaggagac ctagggctgg tgctcaaata ccagagaaga tccaaaagtc cttcgatgat 241 attgccaaat acttctctaa gaaagagtgg gaaaagatga aatccttgga gaaaatcagc 301 tatgtgtata tgaagagaaa gtatgaggcc atgactaaac taggcttcaa ggccaccctc 361 ccacctttca tgcataatac aggggccaca gacctccagg ggaatgattt tgataatgac 421 cgtaaccaag ggaatcaggt tgaacgtcct cagatgactt tttgcaggct ccagagaatc 481 ttcccgaaga tcatgcccaa gaagccagca gaggaaggaa atgattcgaa gggagtgcca 541 gaagcatctg gctcacagaa cgatgggaaa cacctgtgcc ctccaggaaa accaagtacc 601 tctgagaaga ttaacaagac atccggaccc aaaaggggga aacatgcctg gacccacaga 661 ctgcgtgaga gaaagcagct ggtgatttat gaagagatca gcgaccctga agaagacgac 721 gagtaactcc cctcggggat acgacatatg cccatgatga gaagcagaac gtggtgacct 781 ttcacgaaca tgggcatggc tgcggacccc tcgtcatcag gtgcatagca agtgaaagca 841 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TABLE-US-00005 SEQ ID NO: 76 Mouse SMARCC2 cDNA Sequence Variant 3 (NM_198160.2, CDS: 92-3391) 1 gtggcggcgg gaggcggcgg gaggcgggcg gaggaggagg cgggagctga gctgagcggg 61 gcgggcggcg gcggggcccg agcccgagaa gatggcggtg cggaagaagg acggcggccc 121 caacgtgaag tactacgagg ccgcggacac cgtgacccag ttcgacaacg tgcggctctg 181 gctcggcaag aactacaaga agtacataca agcagaaccg ccaaccaaca agtctctgtc 241 cagcctggtg gtgcagttgc tccagtttca ggaagaggtt tttggcaaac atgtcagcaa 301 cgcaccgctt actaaactgc cgatcaaatg tttcctagat ttcaaagcag gaggatccct 361 ctgccatatt cttgcagctg cctacaaatt caagagtgac cagggatggc ggcgttacga 421 tttccagaat ccatcacgca tggaccgcaa tgtggaaatg ttcatgacca ttgagaagtc 481 cttggtacag aataattgcc tgtcacgacc taacattttc ctctgcccag aaattgagcc 541 caaactgcta gggaaattaa aagacattgt taagagacac cagggaacca tctctgagga 601 taagagcaat gcctcccatg ttgtgtatcc tgtcccaggg aacctagaag aagaggaatg 661 ggtacggcca gtcatgaaga gggataaaca ggttcttctg cactggggct actatcctga 721 cagctacgac acgtggatcc cagcgagtga aattgaagca tctgtggagg 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glhtqsrsai vqalwqyvkt nrlqdshdke yingdkyfqq 301 ifdcprlkfs eipqrltall lppdpivinh visvdpsdqk ktacydidve veeplkgqms 361 sfllstanqq eisaldskih etiesinqlk iqrdfmlsfs rdpkgyvqdl lrsqsrdlkv 421 mtdvagnpee erraaefyhq wsqeavsryf yckiqqrrqe leqslvvrnt SEQ ID NO: 96 Human SMARCD3 cDNA Sequence Variant 2 (NM_003078.3, CDS: 169-1581) 1 gccgggccga gccgagcgcc gagcagggag cgggcggccg cgctccgggc cggggtcccg 61 ggggagcaga tcctcagaat ggcccttggt gctgcaggcg cggtgggctc cgggcccagg 121 caccgagggg gcactggatg actctccagg tgcaggaccc tgccatctat gactccaggt 181 cttcagcacc cacccaccgt ggtacagcgc cccgggatgc cgtctggagc ccggatgccc 241 caccaggggg cgcccatggg ccccccgggc tccccgtaca tgggcagccc cgccgtgcga 301 cccggcctgg cccccgcggg catggagccc gcccgcaagc gagcagcgcc cccgcccggg 361 cagagccagg cacagagcca gggccagccg gtgcccaccg cccccgcgcg gagccgcagt 421 gccaagagga ggaagatggc tgacaaaatc ctccctcaaa ggattcggga gctggtcccc 481 gagtcccagg cttacatgga cctcttggca tttgagagga aactggatca aaccatcatg 541 cggaagcggg tggacatcca ggaggctctg aagaggccca 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gacagatgta 1441 gccggcaacc ctgaagagga gcgccgggct gagttctacc accagccctg gtcccaggag 1501 gccgtcagtc gctacttcta ctgcaagatc cagcagcgca ggcaggagct ggagcagtcg 1561 ctggttgtgc gcaacaccta ggagcccaaa aataagcagc acgacggaac tttcagccgt 1621 gtcccgggcc ccagcatttt gccccgggct ccagcatcac tcctctgcca ccttggggtg 1681 tggggctgga ttaaaagtca ttcatctgac aaaaaaaaaa aaaaaaaa SEQ ID NO: 97 Human SMARCD3 Amino Acid Sequence Isoform 2 (NP_001317368.1) 1 mspgnrmpma glqvgppags pfgaaaplrp gmpptmmdpf rkrllvpqaq ppmpaqrrgl 61 krrkmadkvl pqrirelvpe sqaymdllaf erkldqtiar krmeiqeaik kpltqkrklr 121 iyisntfsps kaegdsagta gtpggtpagd kvaswelrve gkllddpskq krkfssffks 181 lvieldkely gpdnhlvewh rmpttqetdg fqvkrpgdln vkctlllmld hqppqykldp 241 rlarllgvht qtraaimqal wlyikhnqlq dghereyinc nryfrqifsc grlrfseipm 301 klagllqhpd pivinhvisv dpndqkktac ydidvevddp lkaqmsnfla sttnqqeias 361 ldvkihetie sinqlktqrd fmlsfstdpq dfiqewlrsq rrdlkiitdv ignpeeerra 421 afyhqpwaqe avgrhifakv qqrrqeleqv lgirlt SEQ ID NO: 98 Human SMARCD3 cDNA Sequence Variant 3 (NM_001003801.1, CDS: 102-1553) 1 agcaggactc agaggggaga gttggaggaa aaaaaaaggc agaaaaggga aagaaagagg 61 aagagagaga gagagtgaga ggagccgctg agcccacccc gatggccgcg gacgaagttg 121 ccggaggggc gcgcaaagcc acgaaaagca aactttttga gtttctggtc catggggtgc 181 gccccgggat gccgtctgga gcccggatgc cccaccaggg ggcgcccatg ggcccccccgg 241 gctccccgta catgggcagc cccgccgtgc gacccggcct ggcccccgcg ggcatggagc 301 ccgcccgcaa gcgagcagcg cccccgcccg ggcagagcca ggcacagagc cagggccagc 361 cggtgcccac cgcccccgcg cggagccgca gtgccaagag gaggaagatg gctgacaaaa 421 tcctccctca aaggattcgg gagctggtcc ccgagtccca ggcttacatg gacctcttgg 481 catttgagag gaaactggat caaaccatca tgcggaagcg ggtggacatc caggaggctc 541 tgaagaggcc catgaagcaa aagcggaagc tgcgactcta tatctccaac acttttaacc 601 ctgcgaagcc tgatgctgag gattccgacg gcagcattgc ctcctgggag ctacgggtgg 661 aggggaagct cctggatgat cccagcaaac agaagcggaa gttctcttct ttcttcaaga 721 gtttggtcat cgagctggac aaagatcttt atggccctga caaccacctc gttgagtggc 781 atcggacacc cacgacccag gagacggacg gcttccaggt 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cattcatctg 1681 acaaaaaaaa aaaaaaaaaa SEQ ID NO: 99 Mouse SMARCD3 cDNA Sequence (NM_025891.3, CDS: 145- 1596) 1 gggccccctc cccactccgc tcgagtagaa gtgtgagaga gcccagcagg actcagaggg 61 gagagttgga ggaaaaaaaa ggcagaaaag ggaaagaaag aggaagagag agagagagtg 121 agaggagccg ctgagcccac cccgatggcc gcggacgaag ttgccggagg ggcgcgcaaa 181 gccacgaaaa gcaaactttt tgagtttctg gtccatgggg tgcgccccgg gatgccgtct 241 ggagcccgaa tgccccacca gggggcgccc atgggccccc cgggctcccc gtacatgggc 301 agccccgcgg tacgacccgg cctggccccc gcgggcatgg agcccgcccg caagcgagca 361 gcgccccccgc ccgggcagag ccaggcacag ggccagggcc agcccgtgcc caccgcccca 421 gcgcggagcc gcagtgccaa gaggaggaag atggctgaca aaatcctccc tcaaaggatt 481 cgggagctgg tacccgagtc ccaggcttac atggacctcc tagcatttga gaggaaactg 541 gatcaaacca tcatgcggaa gcgggtggac atccaggagg ccctgaagag gcccatgaag 601 caaaagcgaa agctgcgcct ttatatctcc aatactttta accctgcgaa gcctgatgcg 661 gaagactctg atggcagcat tgcctcctgg gagctgcggg tggaggggaa gctcttggat 721 gatcctagta agcagaagag gaagttttct tccttcttca 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TABLE-US-00007 SEQ ID NO: 126 Human DPF3 Amino Acid Sequence Isoform 1 (NP_036206.3) 1 matvihnplk algdqfykea iehcrsynsr lcaersvrlp fldsqtgvaq nncyiwmekr 61 hrgpglapgq lytyparcwr kkrrlhpped pklrlleikp evelplkkdg ftsesttlea 121 llrgegvekk vdareeesiq eiqrvlende nveegneeed leedipkrkn rtrgrargsa 181 ggrrrhdaas qedhdkpyvc dicgkryknr pglsyhyaht hlaseegdea qdqetrsppn 241 hrnenhrpqk gpdgtvipnn ycdfclggsn mnkksgrpee lvscadcgrs ahlggegrke 301 keaaaaartt edlfgstses dtstfhgfde ddleeprscr grrsgrgspt adkkgsc SEQ ID NO: 127 Human DPF3 cDNA Sequence Variant 2 (NM_001280542.1, CDS: 29-1165) 1 agacaatatt ctgttacatt gtagcaaaat ggcgactgtc attcacaacc ccctgaaagc 61 gctcggggac cagttctaca aggaagccat tgagcactgc cggagttaca actcacggct 121 gtgtgcagag cgcagcgtgc gtcttccctt cctggactca cagactgggg tggcccagaa 181 caactgctac atctggatgg agaagaggca ccgaggccca ggccttgccc cgggccagct 241 gtatacatac cctgcccgct gctggcgcaa gaagagacga ttgcacccac ctgaagatcc 301 aaaactgcgg ctgctggaga taaaacctga agtggagctt cccctgaaga aggatgggtt 361 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aslgtfqqmw iskqeyeegg 421 kqcverkcp SEQ ID NO: 141 Human ACTL6A cDNA Sequence variant 2 (NM_177989.3; CDS:

196-1359) 1 agacttaggc ctggacccta gtgattggct gataggagga gccagcaagt gtggctgagc 61 tccggggtgt gtggacgccg ctttgttgcc tgagatgaag ttggagccct tgtttttgac 121 attggatcct atactgtgag agctggttat gctggtgagg actgccccaa ggtggatttt 181 cctacagcta ttggtatggt ggtagaaaga gatgacggaa gcacattaat ggaaatagat 241 ggcgataaag gcaaacaagg cggtcccacc tactacatag atactaatgc tctgcgtgtt 301 ccgagggaga atatggaggc catttcacct ctaaaaaatg ggatggttga agactgggat 361 agtttccaag ctattttgga tcatacctac aaaatgcatg tcaaatcaga agccagtctc 421 catcctgttc tcatgtcaga ggcaccgtgg aatactagag caaagagaga gaaactgaca 481 gagttaatgt ttgaacacta caacatccct gccttcttcc tttgcaaaac tgcagttttg 541 acagcatttg ctaatggtcg ttctactggg ctgattttgg acagtggagc cactcatacc 601 actgcaattc cagtccacga tggctatgtc cttcaacaag gcattgtgaa atccccctctt 661 gctggagact ttattactat gcagtgcaga gaactcttcc aagaaatgaa tattgaattg 721 gttcctccat atatgattgc atcaaaagaa gctgttcgtg aaggatctcc agcaaactgg 781 aaaagaaaag agaagttgcc tcaggttacg aggtcttggc acaattatat gtgtaattgt 841 gttatccagg attttcaagc ttcggtactt caagtgtcag attcaactta tgatgaacaa 901 gtggctgcac agatgccaac tgttcattat gaattcccca atggctacaa ttgtgatttt 961 ggtgcagagc ggctaaagat tccagaagga ttatttgacc cttccaatgt aaaggggtta 1021 tcaggaaaca caatgttagg agtcagtcat gttgtcacca caagtgttgg gatgtgtgat 1081 attgacatca gaccaggtct ctatggcagt gtaatagtgg caggaggaaa cacactaata 1141 cagagtttta ctgacaggtt gaatagagag ctgtctcaga aaactcctcc aagtatgcgg 1201 ttgaaattga ttgcaaataa tacaacagtg gaacggaggt ttagctcatg gattggcggc 1261 tccattctag cctctttggg tacctttcaa cagatgtgga tttccaagca agaatatgaa 1321 gaaggaggga agcagtgtgt agaaagaaaa tgcccttgag aaagagttcc caagcttcta 1381 ccttcctttt gtcaccttac gtttcatagc tttagtatac tcaggaaaag aatgaccatc 1441 ttttgtagaa tgtttataca tttttgcata tttcaatttc cacttaaatt ttttaaagct 1501 ttaactggct ctataaatta agtttgtgct ttccttgaaa tgcacttatt cttattacaa 1561 gcattttata attttgtata aatgtctatt ttctctaaat attttgcttt cagtaaaatg 1621 ctttccaact ctgtttagtg tattaattac cagtggattg gtagaactgc tttttattga 1681 ctagtaaaag ttactgccta tgctttttac cttaggctta cagaattaaa taaaaattag 1741 ccattccaga aataaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaa SEQ ID NO: 142 Human ACTL6A cDNA Sequence variant 3 (NM_178042.3; CDS: 388-1551) 1 agacttaggc ctggacccta gtgattggct gataggagga gccagcaagt gtggctgagc 61 tccggggtgt gtggacgccg ctttgttgcc tgaggtgggt ggcggtggaa gttaagggag 121 tcaggggcta tcgctcctcg agactcgcag tcgcggccac tgcagtcact tcgccagtta 181 gcccttaggg taggagtcgc gccggcagca gccatgagcg gcggcgtgta cgggggaggt 241 gagtgagtgc ggccggacga gagagcgcgc cttttcggcg tgtgggatga agttggagcc 301 cttgtttttg acattggatc ctatactgtg agagctggtt atgctggtga ggactgcccc 361 aaggtggatt ttcctacagc tattggtatg gtggtagaaa gagatgacgg aagcacatta 421 atggaaatag atggcgataa aggcaaacaa ggcggtccca cctactacat agatactaat 481 gctctgcgtg ttccgaggga gaatatggag gccatttcac ctctaaaaaa tgggatggtt 541 gaagactggg atagtttcca agctattttg gatcatacct acaaaatgca tgtcaaatca 601 gaagccagtc tccatcctgt tctcatgtca gaggcaccgt ggaatactag agcaaagaga 661 gagaaactga cagagttaat gtttgaacac tacaacatcc ctgccttctt cctttgcaaa 721 actgcagttt tgacagcatt tgctaatggt cgttctactg ggctgatttt ggacagtgga 781 gccactcata ccactgcaat tccagtccac gatggctatg tccttcaaca aggcattgtg 841 aaatcccctc ttgctggaga ctttattact atgcagtgca gagaactctt ccaagaaatg 901 aatattgaat tggttcctcc atatatgatt gcatcaaaag aagctgttcg tgaaggatct 961 ccagcaaact ggaaaagaaa agagaagttg cctcaggtta cgaggtcttg gcacaattat 1021 atgtgtaatt gtgttatcca ggattttcaa gcttcggtac ttcaagtgtc agattcaact 1081 tatgatgaac aagtggctgc acagatgcca actgttcatt atgaattccc caatggctac 1141 aattgtgatt ttggtgcaga gcggctaaag attccagaag gattatttga cccttccaat 1201 gtaaaggggt tatcaggaaa cacaatgtta ggagtcagtc atgttgtcac cacaagtgtt 1261 gggatgtgtg atattgacat cagaccaggt ctctatggca gtgtaatagt ggcaggagga 1321 aacacactaa tacagagttt tactgacagg ttgaatagag agctgtctca gaaaactcct 1381 ccaagtatgc ggttgaaatt gattgcaaat aatacaacag tggaacggag gtttagctca 1441 tggattggcg gctccattct agcctctttg ggtacctttc aacagatgtg gatttccaag 1501 caagaatatg aagaaggagg gaagcagtgt gtagaaagaa aatgcccttg agaaagagtt 1561 cccaagcttc taccttcctt ttgtcacctt acgtttcata gctttagtat actcaggaaa 1621 agaatgacca tcttttgtag aatgtttata catttttgca tatttcaatt tccacttaaa 1681 ttttttaaag ctttaactgg ctctataaat taagtttgtg ctttccttga aatgcactta 1741 ttcttattac aagcatttta taattttgta taaatgtcta ttttctctaa atattttgct 1801 ttcagtaaaa tgctttccaa ctctgtttag tgtattaatt accagtggat tggtagaact 1861 gctttttatt gactagtaaa agttactgcc tatgcttttt accttaggct tacagaatta 1921 aataaaaatt agccattcca gaaataaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa SEQ ID NO: 143 Human ACTL6A Amino Acid Sequence isoform 2 (NP_817126.1 and NP_829888.1) 1 mvverddgst lmeidgdkgk qggptyyidt nalrvprenm eaisplkngm vedwdsfqai 61 ldhtykmhvk seaslhpvlm seapwntrak rekltelmfe hynipafflc ktavltafan 121 grstglilds gathttaipv hdgyvlqqgi vksplagdfi tmqcrelfqe mnielvppym 181 iaskeavreg spanwkrkek lpqvtrswhn ymcncviqdf qasvlqvsds tydeqvaaqm 241 ptvhyefpng yncdfgaerl kipeglfdps nvkglsgntm lgvshvvtts vgmcdidirp 301 glygsvivag gntliqsftd rlnrelsqkt ppsmrlklia nnttverrfs swiggsilas 361 lgtfqqmwis kqeyeeggkq cverkcp SEQ ID NO: 144 Mouse ACTL6A cDNA Sequence (NM_019673.2; CDS: 311-1600) 1 cttcttctgt cgcttctccc tctccctgcc cctacggatg ccttccattg gctaagacgg 61 ctaaaccgcg cggggatgca gcagcgccac actctgattg gctaatgact aagccggacc 121 ctttgtcatt ggttgatacg agaaaccagc aagagtggct gtgcagcggg cgtgcggccg 181 ctgctttgtt gccggagggg gcggcgttgg aagttgcagg cttgcggggc cggcgttctc 241 agggagagga gtcacgccgc tgttatcttt cgtccggtag tcttcggcca gtccccgcca 301 gacagtagcc atgagcggcg gcgtgtacgg cggagatgaa gttggcgctc ttgtttttga 361 cattggatcg tacacagtga gggctggcta tgctggcgag gactgcccta aggttgattt 421 ccccacggct atcggtgtgg tgctggagag agatgacgga agtacaatga tggagattga 481 tggtgacaaa ggcaagcagg gcgggcccac ctactacata gacaccaatg ccctccgcgt 541 gcccagggag aacatggagg ccatctcacc actcaagaat ggcatggttg aagactggga 601 tagtttccag gccattttgg atcatacata caagatgcat gtcaaatccg aagccagcct 661 gcatcctgtt ctcatgtcgg aagcaccgtg gaacaccagg gcgaagagag agaaactgac 721 agagttgatg tttgagcact acagcatccc tgcattcttc ctttgcaaaa ctgcagtttt 781 gacggcattt gctaatggtc gttctactgg gctgattttg gacagtggag ctacccacac 841 cactgcgatt ccagtccacg atggctatgt tcttcaacaa ggcattgtga aatcccctct 901 ggctggagac ttcattacca tgcagtgcag agaactcttc caggaaatga acatagaact 961 cattcctcct tacatgattg catcaaaaga ggctgttcga gaaggttctc cagccaactg 1021 gaaaagaaaa gagaaactgc cccaggttac aaggtcttgg cacaattaca tgtgcaactg 1081 cgtcatccag gattttcaag cttccgttct tcaggtgtca gactccacct acgacgaaca 1141 agtggctgca cagatgccaa ccgtccacta cgaattcccc aatggctaca actgtgattt 1201 tggggcagag cggctgaaaa ttcctgaagg gttatttgac ccttccaacg taaagggact 1261 gtctgggaac acgatgctgg gagtcagtca cgttgtcaca accagcgtcg gaatgtgtga 1321 catcgacatc agaccaggtc tctacggcag tgtgatcgta gcaggaggaa acacgctaat 1381 acagagtttc actgacaggt taaatagaga gctttctcag aaaactccac caagtatgcg 1441 gttgaaactg attgcaaaca acacgacggt ggagcggagg ttcagctcat ggattggtgg 1501 ctctatccta gcatctttgg gtacctttca acagatgtgg atttctaaac aggaatatga 1561 agaaggaggg aagcagtgtg tagaaagaaa atgcccttga gggctccacc ctgcctgccc 1621 gtcacctcaa cgtctgtagc tttagtacac tcaggaaaag atgaccatct tttgtagaat 1681 gtttatacat gtttgcatat ttcaatttcc acttaaattt tttaaggctt taactggctc 1741 tataaattaa atgagtttgt gctttccttg aaatgcactt attcttatta caggcatttt 1801 ataattttgt atgaatgtct attttctcta aatattttgc tttcagtaag tactctccag 1861 ctctcctggg ggttggttgg tggaattact ctgtattgac aagtacaagt tactgcctat 1921 gctttgtacc ttaggctaca aaactaaata aaaatcacta ctgtcctag SEQ ID NO: 145 Mouse ACTL6A Amino Acid Sequence (NP_062647.2) 1 msggvyggde vgalvfdigs ytvragyage dcpkvdfpta igvvlerddg stmmeidgdk 61 gkqggptyyi dtnalrvpre nmeaisplkn gmvedwdsfq aildhtykmh vkseaslhpv 121 lmseapwntr akrekltelm fehysipaff lcktavltaf angrstglil dsgathttai 181 pvhdgyvlqq givksplagd fitmqcrelf qemnielipp ymiaskeavr egspanwkrk 241 eklpqvtrsw hnymcncviq dfqasvlqvs dstydeqvaa qmptvhyefp ngyncdfgae 301 rlkipeglfd psnvkglsgn tmlgvshvvt tsvgmcdidi rpglygsviv aggntliqsf 361 tdrinrelsq ktppsmrlkl iannttverr fsswiggsil aslgtfqqmw iskqeyeegg 421 kqcverkcp SEQ ID NO: 146 Human β-Actin cDNA Sequence (NM_001101.4; CDS: 193-1320) 1 gagtgagcgg cgcggggcca atcagcgtgc gccgttccga aagttgcctt ttatggctcg 61 agcggccgcg gcggcgccct ataaaaccca gcggcgcgac gcgccaccac cgccgagacc 121 gcgtccgccc cgcgagcaca gagcctcgcc tttgccgatc cgccgcccgt ccacacccgc 181 cgccagctca ccatggatga tgatatcgcc gcgctcgtcg tcgacaacgg ctccggcatg 241 tgcaaggccg gcttcgcggg cgacgatgcc ccccgggccg tcttcccctc catcgtgggg 301 cgccccaggc accagggcgt gatggtgggc atgggtcaga aggattccta tgtgggcgac 361 gaggcccaga gcaagagagg catcctcacc ctgaagtacc ccatcgagca cggcatcgtc 421 accaactggg acgacatgga gaaaatctgg caccacacct tctacaatga gctgcgtgtg 481 gctcccgagg agcaccccgt gctgctgacc gaggcccccc tgaaccccaa ggccaaccgc 541 gagaagatga cccagatcat gtttgagacc ttcaacaccc cagccatgta cgttgctatc 601 caggctgtgc tatccctgta cgcctctggc cgtaccactg gcatcgtgat ggactccggt 661 gacggggtca cccacactgt gcccatctac gaggggtatg ccctccccca tgccatcctg 721 cgtctggacc tggctggccg ggacctgact gactacctca tgaagatcct caccgagcgc 781 ggctacagct tcaccaccac ggccgagcgg gaaatcgtgc gtgacattaa ggagaagctg 841 tgctacgtcg ccctggactt cgagcaagag atggccacgg ctgcttccag ctcctccctg 901 gagaagagct acgagctgcc tgacggccag gtcatcacca ttggcaatga gcggttccgc 961 tgccctgagg cactcttcca gccttccttc ctgggcatgg agtcctgtgg catccacgaa 1021 actaccttca actccatcat gaagtgtgac gtggacatcc gcaaagacct gtacgccaac 1081 acagtgctgt ctggcggcac caccatgtac cctggcattg ccgacaggat gcagaaggag 1141 atcactgccc tggcacccag cacaatgaag atcaagatca ttgctcctcc tgagcgcaag 1201 tactccgtgt ggatcggcgg ctccatcctg gcctcgctgt ccaccttcca gcagatgtgg 1261 atcagcaagc aggagtatga cgagtccggc ccctccatcg tccaccgcaa atgcttctag 1321 gcggactatg acttagttgc gttacaccct ttcttgacaa aacctaactt gcgcagaaaa 1381 caagatgaga ttggcatggc tttatttgtt ttttttgttt tgttttggtt ttttttttt 1441 ttttggcttg actcaggatt taaaaactgg aacggtgaag gtgacagcag tcggttggag 1501 cgagcatccc ccaaagttca caatgtggcc gaggactttg attgcacatt gttgtttttt 1561 taatagtcat tccaaatatg agatgcgttg ttacaggaag tcccttgcca tcctaaaagc 1621 caccccactt ctctctaagg agaatggccc agtcctctcc caagtccaca caggggaggt 1681 gatagcattg ctttcgtgta aattatgtaa tgcaaaattt ttttaatctt cgccttaata 1741 ctttttttatt ttgttttatt ttgaatgatg agccttcgtg cccccccttc cccctttttt 1801 gtcccccaac ttgagatgta tgaaggcttt tggtctccct gggagtgggt ggaggcagcc 1861 agggcttacc tgtacactga cttgagacca gttgaataaa agtgcacacc ttaaaaatga 1921 ggaaaaaaaa aaaaaaaaaa SEQ ID NO: 147 Human β-Actin Amino Acid Sequence (NP_001092.1) 1 mdddiaalvv dngsgmckag fagddaprav fpsivgrprh qgvmvgmgqk dsyvgdeaqs 61 krgiltlkyp iehgivtnwd dmekiwhhtf ynelrvapee hpvllteapl npkanrekmt 121 qimfetfntp amyvaiqavl slyasgrttg ivmdsgdgvt htvpiyegya lphailrldl 181 agrdltdylm kiltergysf tttaereivr dikeklcyva ldfeqemata asssleksy 241 elpdgqviti gnerfrcpea lfqpsflgme scgihettfn simkcdvdir kdlyantvls 301 ggttmypgia drmqkeital apstmkikii apperkysvw iggsilasls tfqqmwiskq 361 eydesgpsiv hrkcf SEQ ID NO: 148 Mouse β-Actin cDNA Sequence (NM_007393.5; CDS: 110-1237) 1 tataaaaccc ggcggcgcaa cgcgcagcca ctgtcgagtc gcgtccaccc gcgagcacag 61 cttctttgca gctccttcgt tgccggtcca cacccgccac cagttcgcca tggatgacga 121 tatcgctgcg ctggtcgtcg acaacggctc cggcatgtgc aaagccggct tcgcgggcga 181 cgatgctccc cgggctgtat tcccctccat cgtgggccgc cctaggcacc agggtgtgat 241 ggtgggaatg ggtcagaagg actcctatgt gggtgacgag gcccagagca agagaggtat 301 cctgaccctg aagtacccca ttgaacatgg cattgttacc aactgggacg acatggagaa 361 gatctggcac cacaccttct acaatgagct gcgtgtggcc cctgaggagc accctgtgct 421 gctcaccgag gcccccctga accctaaggc caaccgtgaa aagatgaccc agatcatgtt 481 tgagaccttc aacaccccag ccatgtacgt agccatccag gctgtgctgt ccctgtatgc 541 ctctggtcgt accacaggca ttgtgatgga ctccggagac ggggtcaccc acactgtgcc 601 catctacgag ggctatgctc tccctcacgc catcctgcgt ctggacctgg ctggccggga 661 cctgacagac tacctcatga agatcctgac cgagcgtggc tacagcttca ccaccacagc 721 tgagagggaa atcgtgcgtg acatcaaaga gaagctgtgc tatgttgctc tagacttcga 781 gcaggagatg gccactgccg catcctcttc ctccctggag aagagctatg agctgcctga 841 cggccaggtc atcactattg gcaacgagcg gttccgatgc cctgaggctc ttttccagcc 901 ttccttcttg ggtatggaat cctgtggcat ccatgaaact acattcaatt ccatcatgaa 961 gtgtgacgtt gacatccgta aagacctcta tgccaacaca gtgctgtctg gtggtaccac 1021 catgtaccca ggcattgctg acaggatgca gaaggagatt actgctctgg ctcctagcac 1081 catgaagatc aagatcattg ctcctcctga gcgcaagtac tctgtgtgga tcggtggctc 1141 catcctggcc tcactgtcca ccttccagca gatgtggatc agcaagcagg agtacgatga 1201 gtccggcccc tccatcgtgc accgcaagtg cttctaggcg gactgttact gagctgcgtt 1261 ttacaccctt tctttgacaa aacctaactt gcgcagaaaa aaaaaaaata agagacaaca 1321 ttggcatggc tttgtttttt taaatttttt ttaaagtttt tttttttttt tttttttttt 1381 tttttaagtt tttttgtttt gttttggcgc ttttgactca ggatttaaaa actggaacgg 1441 tgaaggcgac agcagttggt tggagcaaac atcccccaaa gttctacaaa tgtggctgag 1501 gactttgtac attgttttgt tttttttttt ttttggtttt gtcttttttt aatagtcatt 1561 ccaagtatcc atgaaataag tggttacagg aagtccctca ccctcccaaa agccaccccc 1621 actcctaaga ggaggatggt cgcgtccatg ccctgagtcc accccggggа aggtgacagc 1681 attgcttctg tgtaaattat gtactgcaaa aattttttta aatcttccgc cttaatactt 1741 catttttgtt tttaatttct gaatggccca ggtctgaggc ctccctttttt tttgtccccc 1801 caacttgatg tatgaaggct ttggtctccc tgggaggggg ttgaggtgtt gaggcagcca 1861 gggctggcct gtacactgac ttgagaccaa taaaagtgca caccttacct tacacaaaca 1921 aaaaaaaaaa aaaaa SEQ ID NO: 149 Mouse β-Actin Amino Acid Sequence (NP_031419.1) 1 mdddiaalvv dngsgmckag fagddaprav fpsivgrprh qgvmvgmgqk dsyvgdeaqs 61 krgiltlkyp iehgivtnwd dmekiwhhtf ynelrvapee hpvllteapl npkanrekmt 121 qimfetfntp amyvaiqavl slyasgrttg ivmdsgdgvt htvpiyegya lphailrldl 181 agrdltdylm kiltergysf tttaereivr dikeklcyva ldfeqemata asssleksy 241 elpdgqviti gnerfrcpea lfqpsflgme scgihettfn simkcdvdir kdlyantvls 301 ggttmypgia drmqkeital apstmkikii apperkysvw iggsilasls tfqqmwiskq 361 eydesgpsiv hrkcf SEQ ID NO: 150 Human BCL7A cDNA Sequence variant 1 (NM_020993.4; CDS: 207-902) 1 actgggccag gcgcgcggcg gccccgggct ttgtgtgtgt gtgtatgtgt gtgtgtgtgt 61 gtgtgtgtgt gtgagtgtgt gcgtgtgaga gtgcgagtgt ctgtgcgcga gtgagtgagc 121 ggcgggcggg cgcgagtgtg gccgccgcgg agcgcgagca ggacccggcg ggcgcgctcc 181 ccagcctccg tctccccgcc ggaaccatgt cgggcaggtc ggttcgagcc gagacgagga 241 gccgggccaa agatgatatc aagagggtca tggcggcgat cgagaaagtg cgcaaatggg 301 agaagaaatg ggtgaccgtt ggtgacacat ccctacgaat ctacaaatgg gtccctgtga 361 cggagcccaa ggttgatgac aaaaacaaga ataagaaaaa aggcaaggac gagaagtgtg 421 gctcagaggt gaccactccg gagaacagtt cctccccagg gatgatggac atgcatgacg 481 ataacagcaa ccagagctcc atcgcagatg cctcccccat caaacaggag aacagcagca 541 actccagccc cgctccagag cccaactcgg ctgtgcccag cgacggcacc gaggccaagg 601 tggatgaggc ccaggctgat gggaaggagc acccaggagc tgaagatgct tctgatgagc 661 agaattcaca gtcctcgatg gaacattcga tgaacagctc agagaaagta gatcggcagc 721 cgtctggaga ctcgggtctg gccgcagaga cgtctgcaat ctctcaggta cctcgctcga 781 ggtctcagag gggcagccag atcggccggg agcccattgg gttgtcgggg gatttggaag 841 gagtgccacc ctctaaaaag atgaaactgg aggcctctca acaaaactcc gaagagatgt 901 agacgatgct ttaaagcctc cgatccatgt tccatggaag gtacatcagc aattaattct 961 agagcaactt tgccccagcg attcctcttg ggtgcgaaca gaactactaa cgtttcaagt 1021 ttaccaagtg caaatccaag aagacccaga acggcgtcac ttctcagaca ctgaagaact 1081 ctgctgtgaa gcaaaacact caaaccttta agggactgtc cttggggagg caggcggggc 1141 tgacagctca ggagtgtctg cacactgtct cggaagccag gattccattt gtgttgctgc 1201 tgtattttcc ccccacttct ctatgtaacg atataagcta tcggagggtg gtaccgatca 1261 ggaacgcttt ttggcggggc tttccactgt tcaaccgatt ccttccgctt tcttttttg 1321 tgccttgtgc ccttgaggtg acctctggca tgtatcctgg tggttcttac atccccctct 1381 gcaaagtgcc ctcttggttt ggttcgggcg gcggctgcca ccctactcac cgctctcctc 1441 cctgccccag gacttcatcg gagcaggcag ggtggagcga aggagctcct tagcccacct 1501 ggtttgcagg tgcaggggga ccttaggcac gccccaagca ccaggcacca gggcccaagg 1561 acgcgcaggt gttggggcac agtccccaag ggctcggccc cttggatcag gctgggcact 1621 cgctgtgctc tcccctcctt ggggcgttta ggactgggcg tctccaagcc caccatggcc 1681 cagatggacg tgcaaagccc ttggaatttt ctggcacttc ctctctattg cccccaccac 1741 caccaccccc atcactgctt tctcccagac ctccgaatac gaaatggctt ctctggctga 1801 ctgcaaggct gtctccttaa ggcactgagt gggccgggga ggctgggagc cggcggcagg 1861 attagctggt gctgaacttt ctctcatagg acgtcgcttg gatttcaaat ccacggtcac 1921 ctgctgccct ttgcctcccc cgacgcccca gcctgtgccc cggagaggca ggatcgcagt 1981 ggtcagaatc cacgtgcttt cctattctca ggctgttctg actctgagcc aacagctgga 2041 ccgtgtctca tccccagaac atgccgtctg tccccaccgg ggagtgggcc ttgatggccg 2101 ggcctcgaag gccacaaaca aggcgtcgag gaattggaaa gatttgcaca ccctccagaa 2161 aggagagacg caatctcccc tccctcccat cccccacctt cgctggaaca gcttcctctc 2221 actgaacgga gacgcccccct tggacgaact gcctaatcgt ttggttctga ggcctggttt 2281 gctcttaatt aatatatgaa ctcctcagac cttaaacctt ttcctaagct ttctttactg 2341 cactggagtt ctgactccct ttgagttgtg tgttactggg ggtggggtgg ggtcatgggt 2401 tttgttgttt ttgggggcta attggtgcat attcaggtac cacctttgac gtgtggctct 2461 ttctcctgac catcatggga agtgtctgct ggattccatt ttctaagagt ttctgaggt 2521 gaggctctta tttttttttt taagggatcc tgtctatttc ctgcacttcg agaagaatca 2581 aaatgttcct gaatttcaaa tacctcatgc aaaatgtctc ctgaaataag ggaaaaaaaa 2641 aaaaccacaa ctttgaaaat cttaatgttg aagttagcaa tgccgaaagg tttctgtctt 2701 aaaaaaaaaa atccttgtac ttatcaattt tgccccttag gcagtcagtt ttgttgagaa 2761 ctgtgtcctg catcctggcg cagaacctac ctgatgcggt tcctctccac gcatctcgag 2821 gcggcgttac ctccagattc cgtagagtta gagtcacatt tttctttgca gcgaaactcc 2881 atcttggtga gagatgaatt tggatattta tttccttctc tgtttttggg aaacgagagg 2941 ctacaaccaa gacagctgaa ggagaatgaa acacacacat ccacagaaac agagaggcgt 3001 aggtggccct gccgttgacc gcagcctctc tggacaggca aggggagttg gcgcaggtga 3061 ggactcagac gacgtccacc gtcccaaggc tgtcactagt atttctctga agtgcctgaa 3121 ggtaggaatg ggccggcgat tgggaccagc tgggccccac cacggccacg ccaggcaaag 3181 cgccagcagc cctgcactcc acgctggcca agaaggcctt ccacgcagaa tgacaagact 3241 gcaaaaatcc gatgtgcttc cttccctggc gcagtcgctc ctcgagccgc tgccccccac 3301 ccaccctgca cccctcgccc tccccccacc acagaatcta agacctttca gcttcgagcc 3361 agggggcggg ggatcccgag caaaagcctt ccgtggacat caggccccgt ggcctcaagg 3421 gctcccaggg caaacctaat tccccccaaa acgtgaagtc ggggaagctg cggctacaca 3481 ttccacaaag tgctggcact tacacccaca acccggaagg ctgtggaccg attcctctag 3541 ggtggtgacc tcccattagc aaacggtgtc atggtttgga atgttcatta tcgccaagaa 3601 cctggttaga ggcataaaga cctttttca ccgttaccta attttttccc ctttcaagaa 3661 ttttttttt ttttggtgtg ttgtacagca gtataatttt tcacttattt attccatcag 3721 tagatatggt ttgtacaatg tacaattgtt tcatttcaga aaataaaaat ttcaaatcat 3781 gaa

TABLE-US-00008 SEQ ID NO: 151 Human BCL7A Amino Acid Sequence isoform A (NP_066273.1) 1 msgrsvraet rsrakddikr vmaaiekvrk wekkwvtvgd tslriykwvp vtepkvddkn 61 knkkkgkdek cgsevttpen ssspgmmdmh ddnsnqssia daspikqens snsspapepn 121 savpsdgtea kvdeaqadgk ehpgaedasd eqnsqssmeh smnssekvdr qpsgdsglaa 181 etsaisqvpr srsqrgsqig repiglsgdl egvppskkmk leasqqnsee m

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SEQ ID NO: 212 Mouse SMARCA4 cDNA Sequence variant 1 (NM_001174078.1; 261-5114) 1 ggcaagtgga gcgggtagac agggaggcgg gggcgcgcgg cgggcgcgtg cggtgggggg 61 gggtggcctg gcgaagccca gcgggcgcgc gcgcgaggct ttcccactcg cttggcagcg 121 gcggagacgg cttctttgtt tcctgaggag aagcgagacg cccactctgt ccccgacccc 181 tcgtggaggg ttgggggcgg cgccaggaag gttacggcgc cgttacctcc aggagaccag 241 tgcctgtagc tccagtaaag atgtctactc cagacccacc cttgggtggg actcctcggc 301 ctggtccttc cccaggccct ggtccttcac ctggtgcaat gctgggtcct agccctggcc 361 cctcaccagg ttctgcccac agcatgatgg ggccaagccc aggacctcct tcagcaggac 421 atcccatgcc cacccagggg cctggagggt acccccagga caacatgcat cagatgcaca 481 agcctatgga gtccatgcac gagaagggca tgcctgatga cccacgatac aaccagatga 541 aagggatggg catgcggtca ggggcccaca caggcatggc acctccacct agtcccatgg 601 accagcattc tcaaggttac ccctcacccc tcggcggctc tgaacatgcc tccagtcctg 661 tcccagccag tggcccatct tcaggccccc agatgtcctc tgggccagga ggggccccac 721 tagatggttc tgatccccag gccttgggac agcaaaacag aggcccaacc ccatttaacc 781 agaaccagct gcatcaactc agagctcaga taatggccta caagatgttg gccaggggcc 841 agccattgcc cgaccacctg cagatggccg tgcaaggcaa gcggccgatg cctggaatgc 901 agcaacagat gccaacacta cctccaccct cagtgtccgc cacaggaccc ggacctggac 961 ccggccctgg ccctggccct ggcccaggac cagcccctcc aaattacagt agaccccatg 1021 gtatgggagg gcccaacatg cctcccccag gaccctcagg tgtgcccccc gggatgcctg 1081 gtcagccgcc tggagggcct cccaagccat ggcctgaagg acccatggcc aatgctgctg 1141 cccccacaag caccccacag aagctgattc ctccgcaacc aacaggccgt ccttcacctg 1201 cacctcctgc tgtcccgcct gctgcctcac ctgtaatgcc accacaaaca cagtccccag 1261 ggcagccagc ccagcctgct ccattggtgc cactgcacca gaagcagagc cgaatcaccc 1321 ccatccagaa gccccgaggc cttgaccctg tggagatcct acaagagcgg gagtacaggc 1381 ttcaggctcg aatcgcacac agaattcagg aacttgaaaa cctccctggg tccctggctg 1441 gggaccttcg aaccaaagca accatcgaac tcaaggccct taggttgctg aacttccaga 1501 ggcagctgcg ccaggaggtg gtggtgtgca tgcgaagaga cacagccctg gagacagccc 1561 tcaatgccaa ggcctacaag cgcagcaaac gtcagtcact acgggaggcc cgcatcactg 1621

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6361 gtaggacttg ccttgg SEQ ID NO: 213 HumanHistone H3.1 Amino Acid Sequence (NP_003520.1) 1 martkqtark stggkaprkq latkaarksa patggvkkph ryrpgtvalr eirryqkste 61 llirklpfqr lvreiaqdfk tdlrfqssav malqeaceay lvglfedtnl caihakrvti 121 mpkdiqlarr irgera SEQ ID NO: 214 Mouse Histone H3.1 Amino Acid Sequence (NP_038578.2): 1 martkqtark stggkaprkq latkaarksa patggvkkph ryrpgtvalr eirryqkste 61 llirklpfqr lvreiaqdfk tdlrfqssav malqeaceay lvglfedtnl caihakrvti 121 mpkdiqlarr irgera SEQ ID NO: 215 Human Histone H3.2 Amino Acid Sequence (NP_001005464.1): 1 martkqtark stggkaprkq latkaarksa patggvkkph ryrpgtvalr eirryqkste 61 llirklpfqr lvreiaqdfk tdlrfqssav malqeaseay lvglfedtnl caihakrvti 121 mpkdiqlarr irgera SEQ ID NO: 216 Mouse Histone H3.2 Amino Acid Sequence (NP_835587.1): 1 martkqtark stggkaprkq latkaarksa patggvkkph ryrpgtvalr eirryqkste 61 llirklpfqr lvreiaqdfk tdlrfqssav malqeaseay lvglfedtnl caihakrvti 121 mpkdiqlarr irgera SEQ ID NO: 217 Human Histone H3.3 Amino Acid Sequence (NP_002098.1): 1 martkqtark stggkaprkq latkaarksa pstggvkkph ryrpgtvalr eirryqkste 61 llirklpfqr lvreiaqdfk tdlrfqsaai galqeaseay lvglfedtnl caihakrvti 121 mpkdiqlarr irgera SEQ ID NO: 218 Mouse Histone H3.3 Amino Acid Sequence (NP_032237.1); 1 martkqtark stggkaprkq latkaarksa pstggvkkph ryrpgtvalr eirryqkste 61 llirklpfqr lvreiaqdfk tdlrfqsaai galqeaseay lvglfedtnl caihakrvti 121 mpkdiqlarr irgera [0244] H2A protein and cDNA sequences described herein, including human, mouse, rat, and *Xenopus* H2A sequences [0245] H2B protein and cDNA sequences described herein, including human, mouse, rat, and *Xenopus* H2B sequences

TABLE-US-00011 TABLE 2 SS18-SSX fusion protein sequences
MSVAFAAPRQRGKGEITPAAIQKMLDDNNHLIQCIMDSQNKGKTSECSQYQQMLHTNLVYLATIADSNQN
MQSLLPAPPTQNMPMGPGGMNQSGPPPPPPSHNMPSDGMVGGGPPAPHMQNQMNGQMPGPNHMPMQGPGP
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PSQQYNMPQGGGQHYQGQQPPMGMMGQVNQGNHMMGQPQIPPYRPPQQGPPQQYSGQEDYYGDQYSHGGQ
GPPEGMNQQYYPDGNSQYGQQQDAYQGPPPQQGYPPQQQQYPGQQGYPGQQQGYGPSQGGPGPQYPNYPQ
GQGQQYGGYRPTQPGPPQPPQQRPYGYDQIMPKKPAEDENDSKGVSEASGPQNDGKQLH
PPGKANISEKINKRSGPKPGKHAWTHRLRERKQLVIYEETSDPEEDDE (SEQ ID NO: 225) SS18-SSX fusion protein cDNA sequences SS18 AA 1-379 aa + SSX1 (C-terminal 78 AA)
ATGTCTGTGGCTTTCGCGGCCCCGAGGCAGCGAGGCAAGGGGGAGATCACTCCCGCTGCGATTCAGAAGA
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CTTATGGATATGACCAGATCATGCCCAAGAAGCCAGCAGAGGACGAAAATGATTCGAAGGGAGTGTCAGAAGC
ATCTGGCCCACAAAACGATGGGAAACAACTGCACCCCCCAGGAAAAGCAAATATTTCTGAGAAGATTAATAAG
AGATCTGGACCCAAAAGGGGGAAACATGCCTGGACCCACAGACTGCGTGAGAGAAAGCAGCTGGTGATTTATC
AAGAGATCAGTGACCCTGAGGAAGATGACGAGTAA (SEQ ID NO: 226) * Included in Tables 1 and 2 are RNA nucleic acid molecules (e.g., thymines replaced with uredines), nucleic acid molecules encoding orthologs of the encoded proteins, as well as DNA or RNA nucleic acid sequences comprising a nucleic acid sequence having at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or more identity across their full length with the nucleic acid sequence of any sequence listed in Table 1, or a portion thereof. Such nucleic acid molecules can have a function of the full-length nucleic acid as described further herein. * Included in Tables 1 and 2 are orthologs of the proteins, as well as polypeptide molecules comprising an amino acid sequence having at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or more identity across their full length with an amino acid sequence of any sequence listed in Table 1, or a portion thereof. Such polypeptides can have a function of the full-length polypeptide as described further herein.

II. Subjects

[0246] In one embodiment, the subject for whom an agent that inhibits binding of a SS18-SSX fusion protein to an

H2A K119Ub-marked nucleosome is administered, or whose predicted likelihood of efficacy of the agent for treating a cancer is determined, is a mammal (e.g., rat, primate, non-human mammal, domestic animal, such as a dog, cat, cow, horse, and the like), and is preferably a human. In another embodiment, the subject is an animal model of cancer. For example, the animal model can be an orthotopic xenograft animal model of a human-derived cancer.

[0247] In another embodiment of the methods of the present invention, the subject has not undergone treatment, such as chemotherapy, radiation therapy, targeted therapy, and/or immunotherapies. In still another embodiment, the subject has undergone treatment, such as chemotherapy, radiation therapy, targeted therapy, and/or immunotherapies.

[0248] In certain embodiments, the subject has had surgery to remove cancerous or precancerous tissue. In other embodiments, the cancerous tissue has not been removed, e.g., the cancerous tissue may be located in an inoperable region of the body, such as in a tissue that is essential for life, or in a region where a surgical procedure would cause considerable risk of harm to the patient.

[0249] The methods of the present invention can be used to determine the responsiveness to the agent for treating a cancer. In one embodiment, the cancer is synovial sarcoma.

III. Sample Collection, Preparation and Separation

[0250] In some embodiments, biomarker amount and/or activity measurement(s) in a sample from a subject is compared to a predetermined control (standard) sample. The sample from the subject is typically from a diseased tissue, such as cancer cells or tissues. The control sample can be from the same subject or from a different subject. The control sample is typically a normal, non-diseased sample. However, in some embodiments, such as for staging of disease or for evaluating the efficacy of treatment, the control sample can be from a diseased tissue. The control sample can be a combination of samples from several different subjects. In some embodiments, the biomarker amount and/or activity measurement(s) from a subject is compared to a pre-determined level. This pre-determined level is typically obtained from normal samples. As described herein, a “pre-determined” biomarker amount and/or activity measurement(s) may be a biomarker amount and/or activity measurement(s) used to, by way of example only, evaluate a subject that may be selected for treatment, evaluate a response to cancer therapy (e.g., an agent that inhibits binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome), and/or evaluate a response to a combination cancer therapy (e.g., an agent that inhibits binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome in combination of at least one immunotherapy). A pre-determined biomarker amount and/or activity measurement(s) may be determined in populations of patients with or without cancer. The pre-determined biomarker amount and/or activity measurement(s) can be a single number, equally applicable to every patient, or the pre-determined biomarker amount and/or activity measurement(s) can vary according to specific subpopulations of patients. Age, weight, height, and other factors of a subject may affect the pre-determined biomarker amount and/or activity measurement(s) of the individual. Furthermore, the pre-determined biomarker amount and/or activity can be determined for each subject individually. In one embodiment, the amounts determined and/or compared in a method described herein are based on absolute measurements.

[0251] In another embodiment, the amounts determined and/or compared in a method described herein are based on relative measurements, such as ratios (e.g., biomarker copy numbers, level, and/or activity before a treatment vs. after a treatment, such biomarker measurements relative to a spiked or man-made control, such biomarker measurements relative to the expression of a housekeeping gene, and the like). For example, the relative analysis can be based on the ratio of pre-treatment biomarker measurement as compared to post-treatment biomarker measurement. Pre-treatment biomarker measurement can be made at any time prior to initiation of cancer therapy. Post-treatment biomarker measurement can be made at any time after initiation of cancer therapy. In some embodiments, post-treatment biomarker measurements are made 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 weeks or more after initiation of cancer therapy, and even longer toward indefinitely for continued monitoring. Treatment can comprise cancer therapy, such as a therapeutic regimen comprising an agent that inhibits binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome, or in combination with other cancer agents, such as with immune checkpoint inhibitors.

[0252] The pre-determined biomarker amount and/or activity measurement(s) can be any suitable standard. For example, the pre-determined biomarker amount and/or activity measurement(s) can be obtained from the same or a different human for whom a patient selection is being assessed. In one embodiment, the pre-determined biomarker amount and/or activity measurement(s) can be obtained from a previous assessment of the same patient. In such a manner, the progress of the selection of the patient can be monitored over time. In addition, the control can be obtained from an assessment of another human or multiple humans, e.g., selected groups of humans, if the subject is a human. In such a manner, the extent of the selection of the human for whom selection is being assessed can be compared to suitable other humans, e.g., other humans who are in a similar situation to the human of interest, such as those suffering from similar or the same condition(s) and/or of the same ethnic group.

[0253] In some embodiments of the present invention the change of biomarker amount and/or activity measurement(s) from the pre-determined level is about 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, or 5.0 fold or greater, or any range in between, inclusive. Such cutoff values apply equally when the measurement is based on relative changes, such as based on the ratio of pre-treatment biomarker measurement as compared to post-treatment biomarker measurement.

[0254] Biological samples can be collected from a variety of sources from a patient including a body fluid sample, cell sample, or a tissue sample comprising nucleic acids and/or proteins. “Body fluids” refer to fluids that are excreted or secreted from the body as well as fluids that are normally not (e.g., amniotic fluid, aqueous humor, bile, blood and blood plasma, cerebrospinal fluid, cerumen and earwax, cowper's fluid or pre-ejaculatory fluid, chyle, chyme, stool, female ejaculate, interstitial fluid, intracellular fluid, lymph, menses, breast milk, mucus, pleural fluid, pus, saliva, sebum, semen, serum, sweat, synovial fluid, tears, urine, vaginal lubrication, vitreous humor, vomit). In a preferred embodiment, the subject and/or control sample is selected from the group consisting of cells, cell lines, histological slides, paraffin embedded tissues, biopsies, whole blood, nipple aspirate, serum, plasma, buccal scrape, saliva, cerebrospinal fluid, urine, stool, and bone marrow. In one embodiment, the sample is serum, plasma, or urine. In another embodiment, the sample is serum.

[0255] The samples can be collected from individuals repeatedly over a longitudinal period of time (e.g., once or more on the order of days, weeks, months, annually, biannually, etc.). Obtaining numerous samples from an individual over a period of time can be used to verify results from earlier detections and/or to identify an alteration in biological pattern as a result of, for example, disease progression, drug treatment, etc. For example, subject samples can be taken and monitored every month, every two months, or combinations of one, two, or three month intervals according to the present invention. In addition, the biomarker amount and/or activity measurements of the subject obtained over time can be conveniently compared with each other, as well as with those of normal controls during the monitoring period, thereby providing the subject's own values, as an internal, or personal, control for long-term monitoring.

[0256] Sample preparation and separation can involve any of the procedures, depending on the type of sample collected and/or analysis of biomarker measurement(s). Such procedures include, by way of example only, concentration, dilution, adjustment of pH, removal of high abundance polypeptides (e.g., albumin, gamma globulin, and transferrin, etc.), addition of preservatives and calibrants, addition of protease inhibitors, addition of denaturants, desalting of samples, concentration of sample proteins, extraction and purification of lipids.

[0257] The sample preparation can also isolate molecules that are bound in non-covalent complexes to other protein (e.g., carrier proteins). This process may isolate those molecules bound to a specific carrier protein (e.g., albumin), or use a more general process, such as the release of bound molecules from all carrier proteins via protein denaturation, for example using an acid, followed by removal of the carrier proteins.

[0258] Removal of undesired proteins (e.g., high abundance, uninformative, or undetectable proteins) from a sample can be achieved using high affinity reagents, high molecular weight filters, ultracentrifugation and/or electrodialysis. High affinity reagents include antibodies or other reagents (e.g., aptamers) that selectively bind to high abundance proteins. Sample preparation could also include ion exchange chromatography, metal ion affinity chromatography, gel filtration, hydrophobic chromatography, chromatofocusing, adsorption chromatography, isoelectric focusing and related techniques. Molecular weight filters include membranes that separate molecules on the basis of size and molecular weight. Such filters may further employ reverse osmosis, nanofiltration, ultrafiltration and microfiltration.

[0259] Ultracentrifugation is a method for removing undesired polypeptides from a sample. Ultracentrifugation is the centrifugation of a sample at about 15,000-60,000 rpm while monitoring with an optical system the sedimentation (or lack thereof) of particles. Electrodialysis is a procedure which uses an electromembrane or semipermable membrane in a process in which ions are transported through semi-permeable membranes from one solution to another under the influence of a potential gradient. Since the membranes used in electrodialysis may have the ability to selectively transport ions having positive or negative charge, reject ions of the opposite charge, or to allow species to migrate through a semipermable membrane based on size and charge, it renders electrodialysis useful for concentration, removal, or separation of electrolytes.

[0260] Separation and purification in the present invention may include any procedure known in the art, such as capillary electrophoresis (e.g., in capillary or on-chip) or chromatography (e.g., in capillary, column or on a chip). Electrophoresis is a method which can be used to separate ionic molecules under the influence of an electric field. Electrophoresis can be conducted in a gel, capillary, or in a microchannel on a chip. Examples of gels used for electrophoresis include starch, acrylamide, polyethylene oxides, agarose, or combinations thereof. A gel can be modified by its cross-linking, addition of detergents, or denaturants, immobilization of enzymes or antibodies (affinity electrophoresis) or substrates (zymography) and incorporation of a pH gradient. Examples of capillaries used for electrophoresis include capillaries that interface with an electrospray.

[0261] Capillary electrophoresis (CE) is preferred for separating complex hydrophilic molecules and highly charged solutes. CE technology can also be implemented on microfluidic chips. Depending on the types of capillary and buffers used, CE can be further segmented into separation techniques such as capillary zone electrophoresis (CZE), capillary isoelectric focusing (CIEF), capillary isotachophoresis (cITP) and capillary electrochromatography (CEC). An embodiment to couple CE techniques to electrospray ionization involves the use of volatile solutions, for example, aqueous mixtures containing a volatile acid and/or base and an organic such as an alcohol or acetonitrile.

[0262] Capillary isotachophoresis (cITP) is a technique in which the analytes move through the capillary at a constant speed but are nevertheless separated by their respective mobilities. Capillary zone electrophoresis (CZE), also known as free-solution CE (FSCE), is based on differences in the electrophoretic mobility of the species, determined by the charge on the molecule, and the frictional resistance the molecule encounters during migration which is often directly proportional to the size of the molecule. Capillary isoelectric focusing (CIEF) allows weakly-ionizable amphoteric molecules, to be separated by electrophoresis in a pH gradient. CEC is a hybrid technique between traditional high performance liquid chromatography (HPLC) and CE.

[0263] Separation and purification techniques used in the present invention include any chromatography procedures known in the art. Chromatography can be based on the differential adsorption and elution of certain analytes or partitioning of analytes between mobile and stationary phases. Different examples of chromatography include, but not limited to, liquid chromatography (LC), gas chromatography (GC), high performance liquid chromatography (HPLC), etc.

IV. Biomarker Nucleic Acids and Polypeptides

[0264] One aspect of the present invention pertains to the use of isolated nucleic acid molecules that correspond to biomarker nucleic acids that encode a biomarker polypeptide or a portion of such a polypeptide. As used herein, the term “nucleic acid molecule” is intended to include DNA molecules (e.g., cDNA or genomic DNA) and RNA molecules (e.g., mRNA) and analogs of the DNA or RNA generated using nucleotide analogs. The nucleic acid molecule can be single-stranded or double-stranded, but preferably is double-stranded DNA.

[0265] An “isolated” nucleic acid molecule is one which is separated from other nucleic acid molecules which are present in the natural source of the nucleic acid molecule. Preferably, an “isolated” nucleic acid molecule is free of sequences (preferably protein-encoding sequences) which naturally flank the nucleic acid (i.e., sequences located at the 5′ and 3′ ends of the nucleic acid) in the genomic DNA of the organism from which the nucleic acid is derived. For example, in various embodiments, the isolated nucleic acid molecule can contain less than about 5 kB, 4 kB, 3 kB, 2 kB, 1 kB, 0.5 kB or 0.1 kB of nucleotide sequences which naturally flank the nucleic acid molecule in genomic DNA of the cell from which the nucleic acid is derived. Moreover, an “isolated” nucleic acid molecule, such as a cDNA molecule, can be substantially free of other cellular material or culture medium when produced by recombinant techniques, or substantially free of chemical precursors or other chemicals when chemically synthesized.

[0266] A biomarker nucleic acid molecule of the present invention can be isolated using standard molecular biology techniques and the sequence information in the database records described herein. Using all or a portion of such nucleic acid sequences, nucleic acid molecules of the present invention can be isolated using standard hybridization and cloning techniques (e.g., as described in Sambrook et al., ed., *Molecular Cloning: A Laboratory Manual*, 2nd ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, NY, 1989).

[0267] A nucleic acid molecule of the present invention can be amplified using cDNA, mRNA, or genomic DNA as a template and appropriate oligonucleotide primers according to standard PCR amplification techniques. The nucleic acid molecules so amplified can be cloned into an appropriate vector and characterized by DNA sequence analysis. Furthermore, oligonucleotides corresponding to all or a portion of a nucleic acid molecule of the present invention can be prepared by standard synthetic techniques, e.g., using an automated DNA synthesizer.

[0268] Moreover, a nucleic acid molecule of the present invention can comprise only a portion of a nucleic acid sequence, wherein the full length nucleic acid sequence comprises a marker of the present invention or which encodes a polypeptide corresponding to a marker of the present invention. Such nucleic acid molecules can be used, for example, as a probe or primer. The probe/primer typically is used as one or more substantially purified oligonucleotides. The oligonucleotide typically comprises a region of nucleotide sequence that hybridizes under stringent conditions to at least about 7, preferably about 15, more preferably about 25, 50, 75, 100, 125, 150, 175, 200, 250, 300, 350, or 400 or more consecutive nucleotides of a biomarker nucleic acid sequence. Probes based on the sequence of a biomarker nucleic acid molecule can be used to detect transcripts or genomic sequences corresponding to one or more markers of the present invention. The probe comprises a label group attached thereto, e.g., a radioisotope, a fluorescent compound, an enzyme, or an enzyme co-factor.

[0269] A biomarker nucleic acid molecules that differ, due to degeneracy of the genetic code, from the nucleotide sequence of nucleic acid molecules encoding a protein which corresponds to the biomarker, and thus encode the same protein, are also contemplated.

[0270] In addition, it will be appreciated by those skilled in the art that DNA sequence polymorphisms that lead to changes in the amino acid sequence can exist within a population (e.g., the human population). Such genetic polymorphisms can exist among individuals within a population due to natural allelic variation. An allele is one of a group of genes which occur alternatively at a given genetic locus. In addition, it will be appreciated that DNA polymorphisms that affect RNA expression levels can also exist that may affect the overall expression level of that gene (e.g., by affecting regulation or degradation).

[0271] The term “allele,” which is used interchangeably herein with “allelic variant,” refers to alternative forms of a gene or portions thereof. Alleles occupy the same locus or position on homologous chromosomes. When a subject has two identical alleles of a gene, the subject is said to be homozygous for the gene or allele. When a subject has two different alleles of a gene, the subject is said to be heterozygous for the gene or allele. For example, biomarker alleles can differ from each other in a single nucleotide, or several nucleotides, and can include substitutions, deletions, and insertions of nucleotides. An allele of a gene can also be a form of a gene containing one or more mutations.

[0272] The term "allelic variant of a polymorphic region of gene" or "allelic variant", used interchangeably herein, refers to an alternative form of a gene having one of several possible nucleotide sequences found in that region of the gene in the population. As used herein, allelic variant is meant to encompass functional allelic variants, non-functional allelic variants, SNPs, mutations and polymorphisms.

[0273] The term "single nucleotide polymorphism" (SNP) refers to a polymorphic site occupied by a single nucleotide, which is the site of variation between allelic sequences. The site is usually preceded by and followed by highly conserved sequences of the allele (e.g., sequences that vary in less than 1/100 or 1/1000 members of a population). A SNP usually arises due to substitution of one nucleotide for another at the polymorphic site. SNPs can also arise from a deletion of a nucleotide or an insertion of a nucleotide relative to a reference allele. Typically the polymorphic site is occupied by a base other than the reference base. For example, where the reference allele contains the base "T" (thymidine) at the polymorphic site, the altered allele can contain a "C" (cytidine), "G" (guanine), or "A" (adenine) at the polymorphic site. SNP's may occur in protein-coding nucleic acid sequences, in which case they may give rise to a defective or otherwise variant protein, or genetic disease. Such a SNP may alter the coding sequence of the gene and therefore specify another amino acid (a "missense" SNP) or a SNP may introduce a stop codon (a "nonsense" SNP). When a SNP does not alter the amino acid sequence of a protein, the SNP is called "silent." SNP's may also occur in noncoding regions of the nucleotide sequence. This may result in defective protein expression, e.g., as a result of alternative spicing, or it may have no effect on the function of the protein.

[0274] As used herein, the terms "gene" and "recombinant gene" refer to nucleic acid molecules comprising an open reading frame encoding a polypeptide corresponding to a marker of the present invention. Such natural allelic variations can typically result in 1-5% variance in the nucleotide sequence of a given gene. Alternative alleles can be identified by sequencing the gene of interest in a number of different individuals. This can be readily carried out by using hybridization probes to identify the same genetic locus in a variety of individuals. Any and all such nucleotide variations and resulting amino acid polymorphisms or variations that are the result of natural allelic variation and that do not alter the functional activity are intended to be within the scope of the present invention.

[0275] In another embodiment, a biomarker nucleic acid molecule is at least 7, 15, 20, 25, 30, 40, 60, 80, 100, 150, 200, 250, 300, 350, 400, 450, 550, 650, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1500, 1600, 1700, 1800, 1900, 2000, 2200, 2400, 2600, 2800, 3000, 3500, 4000, 4500, or more nucleotides in length and hybridizes under stringent conditions to a nucleic acid molecule corresponding to a marker of the present invention or to a nucleic acid molecule encoding a protein corresponding to a marker of the present invention. As used herein, the term "hybridizes under stringent conditions" is intended to describe conditions for hybridization and washing under which nucleotide sequences at least 60% (65%, 70%, 75%, 80%, preferably 85%) identical to each other typically remain hybridized to each other. Such stringent conditions are known to those skilled in the art and can be found in sections 6.3.1-6.3.6 of *Current Protocols in Molecular Biology*, John Wiley & Sons, N.Y. (1989). A preferred, non-limiting example of stringent hybridization conditions are hybridization in 6× sodium chloride/sodium citrate (SSC) at about 45° C., followed by one or more washes in 0.2×SSC, 0.1% SDS at 50-65° C.

[0276] In addition to naturally-occurring allelic variants of a nucleic acid molecule of the present invention that can exist in the population, the skilled artisan will further appreciate that sequence changes can be introduced by mutation thereby leading to changes in the amino acid sequence of the encoded protein, without altering the biological activity of the protein encoded thereby. For example, one can make nucleotide substitutions leading to amino acid substitutions at "non-essential" amino acid residues. A "non-essential" amino acid residue is a residue that can be altered from the wild-type sequence without altering the biological activity, whereas an "essential" amino acid residue is required for biological activity. For example, amino acid residues that are not conserved or only semi-conserved among homologs of various species may be non-essential for activity and thus would be likely targets for alteration. Alternatively, amino acid residues that are conserved among the homologs of various species (e.g., murine and human) may be essential for activity and thus would not be likely targets for alteration.

[0277] Accordingly, another aspect of the present invention pertains to nucleic acid molecules encoding a polypeptide of the present invention that contain changes in amino acid residues that are not essential for activity. Such polypeptides differ in amino acid sequence from the naturally-occurring proteins which correspond to the markers of the present invention, yet retain biological activity. In one embodiment, a biomarker protein has an amino acid sequence that is at least about 40% identical, 50%, 60%, 70%, 75%, 80%, 83%, 85%, 87.5%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or identical to the amino acid sequence of a biomarker protein described herein.

[0278] An isolated nucleic acid molecule encoding a variant protein can be created by introducing one or more nucleotide substitutions, additions or deletions into the nucleotide sequence of nucleic acids of the present invention, such that one or more amino acid residue substitutions, additions, or deletions are introduced into the encoded protein. Mutations can be introduced by standard techniques, such as site-directed mutagenesis and PCR-mediated mutagenesis. Preferably, conservative amino acid substitutions are made at one or more predicted non-essential amino acid residues. A “conservative amino acid substitution” is one in which the amino acid residue is replaced with an amino acid residue having a similar side chain. Families of amino acid residues having similar side chains have been defined in the art. These families include amino acids with basic side chains (e.g., lysine, arginine, histidine), acidic side chains (e.g., aspartic acid, glutamic acid), uncharged polar side chains (e.g., glycine, asparagine, glutamine, serine, threonine, tyrosine, cysteine), non-polar side chains (e.g., alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine, tryptophan), beta-branched side chains (e.g., threonine, valine, isoleucine) and aromatic side chains (e.g., tyrosine, phenylalanine, tryptophan, histidine). Alternatively, mutations can be introduced randomly along all or part of the coding sequence, such as by saturation mutagenesis, and the resultant mutants can be screened for biological activity to identify mutants that retain activity. Following mutagenesis, the encoded protein can be expressed recombinantly and the activity of the protein can be determined.

[0279] In some embodiments, the present invention further contemplates the use of anti-biomarker antisense nucleic acid molecules, i.e., molecules which are complementary to a sense nucleic acid of the present invention, e.g., complementary to the coding strand of a double-stranded cDNA molecule corresponding to a marker of the present invention or complementary to an mRNA sequence corresponding to a marker of the present invention. Accordingly, an antisense nucleic acid molecule of the present invention can hydrogen bond to (i.e. anneal with) a sense nucleic acid of the present invention. The antisense nucleic acid can be complementary to an entire coding strand, or to only a portion thereof, e.g., all or part of the protein coding region (or open reading frame). An antisense nucleic acid molecule can also be antisense to all or part of a non-coding region of the coding strand of a nucleotide sequence encoding a polypeptide of the present invention. The non-coding regions (“5′ and 3′ untranslated regions”) are the 5′ and 3′ sequences which flank the coding region and are not translated into amino acids.

[0280] An antisense oligonucleotide can be, for example, about 5, 10, 15, 20, 25, 30, 35, 40, 45, or 50 or more nucleotides in length. An antisense nucleic acid can be constructed using chemical synthesis and enzymatic ligation reactions using procedures known in the art. For example, an antisense nucleic acid (e.g., an antisense oligonucleotide) can be chemically synthesized using naturally occurring nucleotides or variously modified nucleotides designed to increase the biological stability of the molecules or to increase the physical stability of the duplex formed between the antisense and sense nucleic acids, e.g., phosphorothioate derivatives and acridine substituted nucleotides can be used. Examples of modified nucleotides which can be used to generate the antisense nucleic acid include 5-fluorouracil, 5-bromouracil, 5-chlorouracil, 5-iodouracil, hypoxanthine, xanthine, 4-acetylcytosine, 5-(carboxyhydroxylmethyl) uracil, 5-carboxymethylaminomethyl-2-thiouridine, 5-carboxymethylaminomethyluracil, dihydrouracil, beta-D-galactosylqueosine, inosine, N6-isopentenyladenine, 1-methylguanine, 1-methylinosine, 2,2-dimethylguanine, 2-methyladenine, 2-methylguanine, 3-methylcytosine, 5-methylcytosine, N6-adenine, 7-methylguanine, 5-methylaminomethyluracil, 5-methoxyaminomethyl-2-thiouracil, beta-D-mannosylqueosine, 5′-methoxycarboxymethyluracil, 5-methoxyuracil, 2-methylthio-N6-isopentenyladenine, uracil-5-oxyacetic acid (v), wybutoxosine, pseudouracil, queosine, 2-thiocytosine, 5-methyl-2-thiouracil, 2-thiouracil, 4-thiouracil, 5-methyluracil, uracil-5-oxyacetic acid methylester, uracil-5-oxyacetic acid (v), 5-methyl-2-thiouracil, 3-(3-amino-3-N-2-carboxypropyl) uracil, (acp3)w, and 2,6-diaminopurine. Alternatively, the antisense nucleic acid can be produced biologically using an expression vector into which a nucleic acid has been sub-cloned in an antisense orientation (i.e., RNA transcribed from the inserted nucleic acid will be of an antisense orientation to a target nucleic acid of interest, described further in the following subsection).

[0281] The antisense nucleic acid molecules of the present invention are typically administered to a subject or generated in situ such that they hybridize with or bind to cellular mRNA and/or genomic DNA encoding a polypeptide corresponding to a selected marker of the present invention to thereby inhibit expression of the marker, e.g., by inhibiting transcription and/or translation. The hybridization can be by conventional nucleotide complementarity to form a stable duplex, or, for example, in the case of an antisense nucleic acid molecule which binds to DNA duplexes, through specific interactions in the major groove of the double helix. Examples of a route of administration of antisense nucleic acid molecules of the present invention includes direct injection at a tissue site or infusion of the antisense nucleic acid into a blood- or bone marrow-associated body fluid.

[0282] Alternatively, antisense nucleic acid molecules can be modified to target selected cells and then administered systemically. For example, for systemic administration, antisense molecules can be modified such that they specifically bind to receptors or antigens expressed on a selected cell surface, e.g., by linking the antisense nucleic acid molecules to peptides or antibodies which bind to cell surface receptors or antigens. The antisense nucleic acid molecules can also be delivered to cells using the vectors described herein. To achieve sufficient intracellular concentrations of the antisense molecules, vector constructs in which the antisense nucleic acid molecule is placed under the control of a strong pol II or pol III promoter are preferred.

[0283] An antisense nucleic acid molecule of the present invention can be an a-anomeric nucleic acid molecule. An a-anomeric nucleic acid molecule forms specific double-stranded hybrids with complementary RNA in which, contrary to the usual α-units, the strands run parallel to each other (Gaultier et al. (1987) *Nucleic Acids Res.* 15:6625-6641). The antisense nucleic acid molecule can also comprise a 2′-o-methylribonucleotide (Inoue et al. (1987) *Nucleic Acids Res.* 15:6131-6148) or a chimeric RNA-DNA analogue (Inoue et al. (1987) *FEBS Lett.* 215:327-330).

[0284] The present invention also encompasses ribozymes. Ribozymes are catalytic RNA molecules with ribonuclease activity which are capable of cleaving a single-stranded nucleic acid, such as an mRNA, to which they have a complementary region. Thus, ribozymes (e.g., hammerhead ribozymes as described in Haselhoff and Gerlach (1988) *Nature* 334:585-591) can be used to catalytically cleave mRNA transcripts to thereby inhibit translation of the protein encoded by the mRNA. A ribozyme having specificity for a nucleic acid molecule encoding a polypeptide corresponding to a marker of the present invention can be designed based upon the nucleotide sequence of a cDNA corresponding to the marker. For example, a derivative of a Tetrahymena L-19 IVS RNA can be constructed in which the nucleotide sequence of the active site is complementary to the nucleotide sequence to be cleaved (see Cech et al. U.S. Pat. No. 4,987,071; and Cech et al. U.S. Pat. No. 5,116,742). Alternatively, an mRNA encoding a polypeptide of the present invention can be used to select a catalytic RNA having a specific ribonuclease activity from a pool of RNA molecules (see, e.g., Bartel and Szostak (1993) *Science* 261:1411-1418).

[0285] The present invention also encompasses nucleic acid molecules which form triple helical structures. For example, expression of a biomarker protein can be inhibited by targeting nucleotide sequences complementary to the regulatory region of the gene encoding the polypeptide (e.g., the promoter and/or enhancer) to form triple helical structures that prevent transcription of the gene in target cells. See generally Helene (1991) *Anticancer Drug Des.* 6(6):569-84; Helene (1992) *Ann. N. Y. Acad. Sci.* 660:27-36; and Maher (1992) *Bioassays* 14(12):807-15.

[0286] In various embodiments, the nucleic acid molecules of the present invention can be modified at the base moiety, sugar moiety or phosphate backbone to improve, e.g., the stability, hybridization, or solubility of the molecule. For example, the deoxyribose phosphate backbone of the nucleic acid molecules can be modified to generate peptide nucleic acid molecules (see Hyrup et al. (1996) *Bioorganic* & *Medicinal Chemistry* 4(1): 5-23). As used herein, the terms “peptide nucleic acids” or “PNAs” refer to nucleic acid mimics, e.g., DNA mimics, in which the deoxyribose phosphate backbone is replaced by a pseudopeptide backbone and only the four natural nucleobases are retained. The neutral backbone of PNAs has been shown to allow for specific hybridization to DNA and RNA under conditions of low ionic strength. The synthesis of PNA oligomers can be performed using standard solid phase peptide synthesis protocols as described in Hyrup et al. (1996), supra; Perry-O'Keefe et al. (1996) *Proc. Natl. Acad. Sci. USA* 93:14670-675.

[0287] PNAs can be used in therapeutic and diagnostic applications. For example, PNAs can be used as antisense or antigene agents for sequence-specific modulation of gene expression by, e.g., inducing transcription or translation arrest or inhibiting replication. PNAs can also be used, e.g., in the analysis of single base pair mutations in a gene by, e.g., PNA directed PCR clamping; as artificial restriction enzymes when used in combination with other enzymes, e.g., S1 nucleases (Hyrup (1996), supra; or as probes or primers for DNA sequence and hybridization (Hyrup (1996), supra; Perry-O'Keefe et al. (1996) *Proc. Natl. Acad. Sci. USA* 93:14670-14675).

[0288] In another embodiment, PNAs can be modified, e.g., to enhance their stability or cellular uptake, by attaching lipophilic or other helper groups to PNA, by the formation of PNA-DNA chimeras, or by the use of liposomes or other techniques of drug delivery known in the art. For example, PNA-DNA chimeras can be generated which can combine the advantageous properties of PNA and DNA. Such chimeras allow DNA recognition enzymes, e.g., RNASE H and DNA polymerases, to interact with the DNA portion while the PNA portion would provide high binding affinity and specificity. PNA-DNA chimeras can be linked using linkers of appropriate lengths selected in terms of base stacking, number of bonds between the nucleobases, and orientation (Hyrup (1996), supra). The synthesis of PNA-DNA chimeras can be performed as described in Hyrup (1996), supra, and Finn et al. (1996) *Nucleic Acids Res.* 24(17):3357-3363. For example, a DNA chain can be synthesized on a solid support using standard phosphoramidite coupling chemistry and modified nucleoside analogs. Compounds such as 5′-(4-methoxytrityl)amino-5′-deoxy-thymidine phosphoramidite can be used as a link between the PNA and the 5′ end of DNA (Mag et al. (1989) *Nucleic Acids Res.* 17:5973-5988). PNA monomers are then coupled in a step-wise manner to produce a chimeric molecule with a 5′ PNA segment and a 3′ DNA segment (Finn et al. (1996) *Nucleic Acids Res.* 24:3357-3363). Alternatively, chimeric molecules can be synthesized with a 5′ DNA segment and a 3′ PNA segment (Peterser et al. (1975) *Bioorganic Med. Chem. Lett.* 5:1119-11124).

[0289] In other embodiments, the oligonucleotide can include other appended groups such as peptides (e.g., for targeting host cell receptors in vivo), or agents facilitating transport across the cell membrane (see, e.g., Letsinger et al. (1989) *Proc. Natl. Acad. Sci. USA* 86:6553-6556; Lemaitre et al. (1987) *Proc. Natl. Acad. Sci. USA* 84:648-652; PCT Publication No. WO 88/09810) or the blood-brain barrier (see, e.g., PCT Publication No. WO 89/10134). In addition, oligonucleotides can be modified with hybridization-triggered cleavage agents (see, e.g., Krol et al. (1988) *Bio/Techniques* 6:958-976) or intercalating agents (see, e.g., Zon (1988) *Pharm. Res.* 5:539-549). To this end, the oligonucleotide can be conjugated to another molecule, e.g., a peptide, hybridization triggered cross-linking agent, transport agent, hybridization-triggered cleavage agent, etc.

[0290] Another aspect of the present invention pertains to the use of biomarker proteins and biologically active portions thereof. In one embodiment, the native polypeptide corresponding to a marker can be isolated from cells or tissue sources by an appropriate purification scheme using standard protein purification techniques. In another embodiment, polypeptides corresponding to a marker of the present invention are produced by recombinant DNA techniques. Alternative to recombinant expression, a polypeptide corresponding to a marker of the present invention can be synthesized chemically using standard peptide synthesis techniques.

[0291] An “isolated” or “purified” protein or biologically active portion thereof is substantially free of cellular material or other contaminating proteins from the cell or tissue source from which the protein is derived, or substantially free of chemical precursors or other chemicals when chemically synthesized. The language “substantially free of cellular material” includes preparations of protein in which the protein is separated from cellular components of the cells from which it is isolated or recombinantly produced. Thus, protein that is substantially free of cellular material includes preparations of protein having less than about 30%, 20%, 10%, or 5% (by dry weight) of heterologous protein (also referred to herein as a “contaminating protein”). When the protein or biologically active portion thereof is recombinantly produced, it is also preferably substantially free of culture medium, i.e., culture medium represents less than about 20%, 10%, or 5% of the volume of the protein preparation. When the protein is produced by chemical synthesis, it is preferably substantially free of chemical precursors or other chemicals, i.e., it is separated from chemical precursors or other chemicals which are involved in the synthesis of the protein. Accordingly such preparations of the protein have less than about 30%, 20%, 10%, 5% (by dry weight) of chemical precursors or compounds other than the polypeptide of interest.

[0292] Biologically active portions of a biomarker polypeptide include polypeptides comprising amino acid sequences sufficiently identical to or derived from a biomarker protein amino acid sequence described herein, but which includes fewer amino acids than the full length protein, and exhibit at least one activity of the corresponding full-length protein. Typically, biologically active portions comprise a domain or motif with at least one activity of the corresponding protein. A biologically active portion of a protein of the present invention can be a polypeptide which is, for example, 10, 25, 50, 100 or more amino acids in length. Moreover, other biologically active portions, in which other regions of the protein are deleted, can be prepared by recombinant techniques and evaluated for one or more of the functional activities of the native form of a polypeptide of the present invention.

[0293] Preferred polypeptides have an amino acid sequence of a biomarker protein encoded by a nucleic acid molecule described herein. Other useful proteins are substantially identical (e.g., at least about 40%, preferably 50%, 60%, 70%, 75%, 80%, 83%, 85%, 88%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%) to one of these sequences and retain the functional activity of the protein of the corresponding naturally-occurring protein yet differ in amino acid sequence due to natural allelic variation or mutagenesis.

[0294] To determine the percent identity of two amino acid sequences or of two nucleic acids, the sequences are aligned for optimal comparison purposes (e.g., gaps can be introduced in the sequence of a first amino acid or nucleic acid sequence for optimal alignment with a second amino or nucleic acid sequence). The amino acid residues or nucleotides at corresponding amino acid positions or nucleotide positions are then compared. When a position in the first sequence is occupied by the same amino acid residue or nucleotide as the corresponding position in the second sequence, then the molecules are identical at that position. The percent identity between the two sequences is a function of the number of identical positions shared by the sequences (i.e., % identity=# of identical positions/total # of positions (e.g., overlapping positions)×100). In one embodiment the two sequences are the same length.

[0295] The determination of percent identity between two sequences can be accomplished using a mathematical algorithm. A preferred, non-limiting example of a mathematical algorithm utilized for the comparison of two sequences is the algorithm of Karlin and Altschul (1990) *Proc. Natl. Acad. Sci. USA* 87:2264-2268, modified as in Karlin and Altschul (1993) *Proc. Natl. Acad. Sci. USA* 90:5873-5877. Such an algorithm is incorporated into the NBLAST and XBLAST programs of Altschul, et al. (1990) *J. Mol. Biol.* 215:403-410. BLAST nucleotide searches can be performed with the NBLAST program, score=100, wordlength=12 to obtain nucleotide sequences homologous to a nucleic acid molecules of the present invention. BLAST protein searches can be performed with the XBLAST program, score=50, wordlength=3 to obtain amino acid sequences homologous to a protein molecules of the present invention. To obtain gapped alignments for comparison purposes, Gapped BLAST can be utilized as described in Altschul et al. (1997) *Nucleic Acids Res.* 25:3389-3402. Alternatively, PSI-Blast can be used to perform an iterated search which detects distant relationships between molecules. When utilizing BLAST, Gapped BLAST, and PSI-Blast programs, the default parameters of the respective programs (e.g., XBLAST and NBLAST) can be used. See world wide web ncbi.nlm.nih.gov. Another preferred, non-limiting example of a mathematical algorithm utilized for the comparison of sequences is the algorithm of Myers and Miller, (1988) *Comput Appl Biosci,* 4:11-7. Such an algorithm is incorporated into the ALIGN program (version 2.0) which is part of the GCG sequence alignment software package. When utilizing the ALIGN program for comparing amino acid sequences, a PAM120 weight residue table, a gap length penalty of 12, and a gap penalty of 4 can be used. Yet another useful algorithm for identifying regions of local sequence similarity and alignment is the FASTA algorithm as described in Pearson and Lipman (1988) *Proc. Natl. Acad. Sci. USA* 85:2444-2448. When using the FASTA algorithm for comparing nucleotide or amino acid sequences, a PAM120 weight residue table can, for example, be used with a k-tuple value of 2.

[0296] The percent identity between two sequences can be determined using techniques similar to those described above, with or without allowing gaps. In calculating percent identity, only exact matches are counted.

[0297] The present invention also provides chimeric or fusion proteins corresponding to a biomarker protein. As used herein, a “chimeric protein” or “fusion protein” comprises all or part (preferably a biologically active part) of a polypeptide corresponding to a marker of the present invention operably linked to a heterologous polypeptide (i.e., a polypeptide other than the polypeptide corresponding to the marker). Within the fusion protein, the term “operably linked” is intended to indicate that the polypeptide of the present invention and the heterologous polypeptide are fused in-frame to each other. The heterologous polypeptide can be fused to the amino-terminus or the carboxyl-terminus of the polypeptide of the present invention.

[0298] One useful fusion protein is a GST fusion protein in which a polypeptide corresponding to a marker of the present invention is fused to the carboxyl terminus of GST sequences. Such fusion proteins can facilitate the purification of a recombinant polypeptide of the present invention.

[0299] In another embodiment, the fusion protein contains a heterologous signal sequence, immunoglobulin fusion protein, toxin, or other useful protein sequence. Chimeric and fusion proteins of the present invention can be produced by standard recombinant DNA techniques. In another embodiment, the fusion gene can be synthesized by conventional techniques including automated DNA synthesizers. Alternatively, PCR amplification of gene fragments can be carried out using anchor primers which give rise to complementary overhangs between two consecutive gene fragments which can subsequently be annealed and re-amplified to generate a chimeric gene sequence (see, e.g., Ausubel et al., supra). Moreover, many expression vectors are commercially available that already encode a fusion moiety (e.g., a GST polypeptide). A nucleic acid encoding a polypeptide of the present invention can be cloned into such an expression vector such that the fusion moiety is linked in-frame to the polypeptide of the present invention.

[0300] A signal sequence can be used to facilitate secretion and isolation of the secreted protein or other proteins of interest. Signal sequences are typically characterized by a core of hydrophobic amino acids which are generally cleaved from the mature protein during secretion in one or more cleavage events. Such signal peptides contain processing sites that allow cleavage of the signal sequence from the mature proteins as they pass through the secretory pathway. Thus, the present invention pertains to the described polypeptides having a signal sequence, as well as to polypeptides from which the signal sequence has been proteolytically cleaved (i.e., the cleavage products). In one embodiment, a nucleic acid sequence encoding a signal sequence can be operably linked in an expression vector to a protein of interest, such as a protein which is ordinarily not secreted or is otherwise difficult to isolate. The signal sequence directs secretion of the protein, such as from a eukaryotic host into which the expression vector is transformed, and the signal sequence is subsequently or concurrently cleaved. The protein can then be readily purified from the extracellular medium by art recognized methods. Alternatively, the signal sequence can be linked to the protein of interest using a sequence which facilitates purification, such as with a GST domain.

[0301] The present invention also pertains to variants of the biomarker polypeptides described herein. Such variants have an altered amino acid sequence which can function as either agonists (mimetics) or as antagonists. Variants can be generated by mutagenesis, e.g., discrete point mutation or truncation. An agonist can retain substantially the same, or a subset, of the biological activities of the naturally occurring form of the protein. An antagonist of a protein can inhibit one or more of the activities of the naturally occurring form of the protein by, for example, competitively binding to a downstream or upstream member of a cellular signaling cascade which includes the protein of interest. Thus, specific biological effects can be elicited by treatment with a variant of limited function. Treatment of a subject with a variant having a subset of the biological activities of the naturally occurring form of the protein can have fewer side effects in a subject relative to treatment with the naturally occurring form of the protein.

[0302] Variants of a biomarker protein which function as either agonists (mimetics) or as antagonists can be identified by screening combinatorial libraries of mutants, e.g., truncation mutants, of the protein of the present invention for agonist or antagonist activity. In one embodiment, a variegated library of variants is generated by combinatorial mutagenesis at the nucleic acid level and is encoded by a variegated gene library. A variegated library of variants can be produced by, for example, enzymatically ligating a mixture of synthetic oligonucleotides into gene sequences such that a degenerate set of potential protein sequences is expressible as individual polypeptides, or alternatively, as a set of larger fusion proteins (e.g., for phage display). There are a variety of methods which can be used to produce libraries of potential variants of the polypeptides of the present invention from a degenerate oligonucleotide sequence. Methods for synthesizing degenerate oligonucleotides are known in the art (see, e.g., Narang (1983) *Tetrahedron* 39:3; Itakura et al. (1984) *Annu. Rev. Biochem.* 53:323; Itakura et al. (1984) *Science* 198:1056; Ike et al. (1983) *Nucleic Acid Res.* 11:477).

[0303] In addition, libraries of fragments of the coding sequence of a polypeptide corresponding to a marker of the present invention can be used to generate a variegated population of polypeptides for screening and subsequent selection of variants. For example, a library of coding sequence fragments can be generated by treating a double stranded PCR fragment of the coding sequence of interest with a nuclease under conditions wherein nicking occurs only about once per molecule, denaturing the double stranded DNA, renaturing the DNA to form double stranded

DNA which can include sense/antisense pairs from different nicked products, removing single stranded portions from reformed duplexes by treatment with S1 nuclease, and ligating the resulting fragment library into an expression vector. By this method, an expression library can be derived which encodes amino terminal and internal fragments of various sizes of the protein of interest.

[0304] Several techniques are known in the art for screening gene products of combinatorial libraries made by point mutations or truncation, and for screening cDNA libraries for gene products having a selected property. The most widely used techniques, which are amenable to high throughput analysis, for screening large gene libraries typically include cloning the gene library into replicable expression vectors, transforming appropriate cells with the resulting library of vectors, and expressing the combinatorial genes under conditions in which detection of a desired activity facilitates isolation of the vector encoding the gene whose product was detected. Recursive ensemble mutagenesis (REM), a technique which enhances the frequency of functional mutants in the libraries, can be used in combination with the screening assays to identify variants of a protein of the present invention (Arkin and Yourvan (1992) *Proc. Natl. Acad. Sci. USA* 89:7811-7815; Delgrave et al. 91993) *Protein Engineering* 6(3):327-331).

[0305] The production and use of biomarker nucleic acid and/or biomarker polypeptide molecules described herein can be facilitated by using standard recombinant techniques. In some embodiments, such techniques use vectors, preferably expression vectors, containing a nucleic acid encoding a biomarker polypeptide or a portion of such a polypeptide. As used herein, the term “vector” refers to a nucleic acid molecule capable of transporting another nucleic acid to which it has been linked. One type of vector is a “plasmid”, which refers to a circular double stranded DNA loop into which additional DNA segments can be ligated. Another type of vector is a viral vector, wherein additional DNA segments can be ligated into the viral genome. Certain vectors are capable of autonomous replication in a host cell into which they are introduced (e.g., bacterial vectors having a bacterial origin of replication and episomal mammalian vectors). Other vectors (e.g., non-episomal mammalian vectors) are integrated into the genome of a host cell upon introduction into the host cell, and thereby are replicated along with the host genome. Moreover, certain vectors, namely expression vectors, are capable of directing the expression of genes to which they are operably linked. In general, expression vectors of utility in recombinant DNA techniques are often in the form of plasmids (vectors). However, the present invention is intended to include such other forms of expression vectors, such as viral vectors (e.g., replication defective retroviruses, adenoviruses and adeno-associated viruses), which serve equivalent functions.

[0306] The recombinant expression vectors of the present invention comprise a nucleic acid of the present invention in a form suitable for expression of the nucleic acid in a host cell. This means that the recombinant expression vectors include one or more regulatory sequences, selected on the basis of the host cells to be used for expression, which is operably linked to the nucleic acid sequence to be expressed. Within a recombinant expression vector, “operably linked” is intended to mean that the nucleotide sequence of interest is linked to the regulatory sequence(s) in a manner which allows for expression of the nucleotide sequence (e.g., in an in vitro transcription/translation system or in a host cell when the vector is introduced into the host cell). The term “regulatory sequence” is intended to include promoters, enhancers and other expression control elements (e.g., polyadenylation signals). Such regulatory sequences are described, for example, in Goeddel, *Methods in Enzymology: Gene Expression Technology* vol. 185, Academic Press, San Diego, CA (1991). Regulatory sequences include those which direct constitutive expression of a nucleotide sequence in many types of host cell and those which direct expression of the nucleotide sequence only in certain host cells (e.g., tissue-specific regulatory sequences). It will be appreciated by those skilled in the art that the design of the expression vector can depend on such factors as the choice of the host cell to be transformed, the level of expression of protein desired, and the like. The expression vectors of the present invention can be introduced into host cells to thereby produce proteins or peptides, including fusion proteins or peptides, encoded by nucleic acids as described herein.

[0307] The recombinant expression vectors for use in the present invention can be designed for expression of a polypeptide corresponding to a marker of the present invention in prokaryotic (e.g., *E. coli*) or eukaryotic cells (e.g., insect cells {using baculovirus expression vectors}, yeast cells or mammalian cells). Suitable host cells are discussed further in Goeddel, supra. Alternatively, the recombinant expression vector can be transcribed and translated in vitro, for example using T7 promoter regulatory sequences and T7 polymerase.

[0308] Expression of proteins in prokaryotes is most often carried out in *E. coli* with vectors containing constitutive or inducible promoters directing the expression of either fusion or non-fusion proteins. Fusion vectors add a number of amino acids to a protein encoded therein, usually to the amino terminus of the recombinant protein. Such fusion vectors typically serve three purposes: 1) to increase expression of recombinant protein; 2) to increase the solubility of the recombinant protein; and 3) to aid in the purification of the recombinant protein by acting as a ligand in affinity purification. Often, in fusion expression vectors, a proteolytic cleavage site is introduced at the junction of the fusion moiety and the recombinant protein to enable separation of the recombinant protein from the fusion moiety subsequent to purification of the fusion protein. Such enzymes, and their cognate recognition sequences, include Factor Xa, thrombin and enterokinase. Typical fusion expression vectors include pGEX (Pharmacia Biotech Inc; Smith and Johnson, 1988, *Gene* 67:31-40), pMAL (New England Biolabs, Beverly, MA) and pRIT5 (Pharmacia,

Piscataway, NJ) which fuse glutathione S-transferase (GST), maltose E binding protein, or protein A, respectively, to the target recombinant protein.
[0309] Examples of suitable inducible non-fusion *E. coli* expression vectors include pTrc (Amann et al. (1988) *Gene* 69:301-315) and pET 11d (Studier et al., p. 60-89, In *Gene Expression Technology: Methods in Enzymology* vol. 185, Academic Press, San Diego, CA, 1991). Target biomarker nucleic acid expression from the pTrc vector relies on host RNA polymerase transcription from a hybrid trp-lac fusion promoter. Target biomarker nucleic acid expression from the pET 11d vector relies on transcription from a T7 gn10-lac fusion promoter mediated by a co-expressed viral RNA polymerase (T7 gn1). This viral polymerase is supplied by host strains BL21 (DE3) or HMS174(DE3) from a resident prophage harboring a T7 gn1 gene under the transcriptional control of the lacUV 5 promoter.
[0310] One strategy to maximize recombinant protein expression in *E. coli* is to express the protein in a host bacterium with an impaired capacity to proteolytically cleave the recombinant protein (Gottesman, p. 119-128, In *Gene Expression Technology: Methods in Enzymology* vol. 185, Academic Press, San Diego, CA, 1990. Another strategy is to alter the nucleic acid sequence of the nucleic acid to be inserted into an expression vector so that the individual codons for each amino acid are those preferentially utilized in *E. coli* (Wada et al., (1992) *Nucleic Acids Res.* 20:2111-2118). Such alteration of nucleic acid sequences of the present invention can be carried out by standard DNA synthesis techniques.
[0311] In another embodiment, the expression vector is a yeast expression vector. Examples of vectors for expression in yeast *S. cerevisiae* include pYepSec1 (Baldari et al. (1987) *EMBO J.* 6:229-234), pMFa (Kurjan and Herskowitz (1982) *Cell* 30:933-943), pJRY88 (Schultz et al. (1987) *Gene* 54:113-123), pYES2 (Invitrogen Corporation, San Diego, CA), and pPicZ (Invitrogen Corp, San Diego, CA).
[0312] Alternatively, the expression vector is a baculovirus expression vector. Baculovirus vectors available for expression of proteins in cultured insect cells (e.g., Sf 9 cells) include the pAc series (Smith et al. (1983) *Mol. Cell Biol.* 3:2156-2165) and the pVL series (Lucklow and Summers (1989) *Virology* 170:31-39).
[0313] In yet another embodiment, a nucleic acid of the present invention is expressed in mammalian cells using a mammalian expression vector. Examples of mammalian expression vectors include pCDM8 (Seed (1987) *Nature* 329:840) and pMT2PC (Kaufman et al. (1987) *EMBO J.* 6:187-195). When used in mammalian cells, the expression vector's control functions are often provided by viral regulatory elements. For example, commonly used promoters are derived from polyoma, Adenovirus 2, cytomegalovirus and Simian Virus 40. For other suitable expression systems for both prokaryotic and eukaryotic cells see chapters 16 and 17 of Sambrook et al., supra.
[0314] In another embodiment, the recombinant mammalian expression vector is capable of directing expression of the nucleic acid preferentially in a particular cell type (e.g., tissue-specific regulatory elements are used to express the nucleic acid). Tissue-specific regulatory elements are known in the art. Non-limiting examples of suitable tissue-specific promoters include the albumin promoter (liver-specific; Pinkert et al. (1987) *Genes Dev.* 1:268-277), lymphoid-specific promoters (Calame and Eaton (1988) *Adv. Immunol.* 43:235-275), in particular promoters of T cell receptors (Winoto and Baltimore (1989) *EMBO J.* 8:729-733) and immunoglobulins (Banerji et al. (1983) *Cell* 33:729-740; Queen and Baltimore (1983) *Cell* 33:741-748), neuron-specific promoters (e.g., the neurofilament promoter; Byrne and Ruddle (1989) *Proc. Natl. Acad. Sci. USA* 86:5473-5477), pancreas-specific promoters (Edlund et al. (1985) *Science* 230:912-916), and mammary gland-specific promoters (e.g., milk whey promoter; U.S. Pat. No. 4,873,316 and European Application Publication No. 264,166). Developmentally-regulated promoters are also encompassed, for example the murine hox promoters (Kessel and Gruss (1990) *Science* 249:374-379) and the a-fetoprotein promoter (Camper and Tilghman (1989) *Genes Dev.* 3:537-546).
[0315] The present invention further provides a recombinant expression vector comprising a DNA molecule cloned into the expression vector in an antisense orientation. That is, the DNA molecule is operably linked to a regulatory sequence in a manner which allows for expression (by transcription of the DNA molecule) of an RNA molecule which is antisense to the mRNA encoding a polypeptide of the present invention. Regulatory sequences operably linked to a nucleic acid cloned in the antisense orientation can be chosen which direct the continuous expression of the antisense RNA molecule in a variety of cell types, for instance viral promoters and/or enhancers, or regulatory sequences can be chosen which direct constitutive, tissue-specific or cell type specific expression of antisense RNA. The antisense expression vector can be in the form of a recombinant plasmid, phagemid, or attenuated virus in which antisense nucleic acids are produced under the control of a high efficiency regulatory region, the activity of which can be determined by the cell type into which the vector is introduced. For a discussion of the regulation of gene expression using antisense genes (see Weintraub et al. (1986) *Trends in Genetics*, Vol. 1(1)).
[0316] Another aspect of the present invention pertains to host cells into which a recombinant expression vector of the present invention has been introduced. The terms “host cell” and “recombinant host cell” are used interchangeably herein. It is understood that such terms refer not only to the particular subject cell but to the progeny or potential progeny of such a cell. Because certain modifications may occur in succeeding generations due to either mutation or environmental influences, such progeny may not, in fact, be identical to the parent cell, but are still included within the scope of the term as used herein.
[0317] A host cell can be any prokaryotic (e.g., *E. coli*) or eukaryotic cell (e.g., insect cells, yeast or mammalian

cells).

[0318] Vector DNA can be introduced into prokaryotic or eukaryotic cells via conventional transformation or transfection techniques. As used herein, the terms "transformation" and "transfection" are intended to refer to a variety of art-recognized techniques for introducing foreign nucleic acid into a host cell, including calcium phosphate or calcium chloride co-precipitation, DEAE-dextran-mediated transfection, lipofection, or electroporation. Suitable methods for transforming or transfecting host cells can be found in Sambrook, et al. (supra), and other laboratory manuals.

[0319] For stable transfection of mammalian cells, it is known that, depending upon the expression vector and transfection technique used, only a small fraction of cells may integrate the foreign DNA into their genome. In order to identify and select these integrants, a gene that encodes a selectable marker (e.g., for resistance to antibiotics) is generally introduced into the host cells along with the gene of interest. Preferred selectable markers include those which confer resistance to drugs, such as G418, hygromycin and methotrexate. Cells stably transfected with the introduced nucleic acid can be identified by drug selection (e.g., cells that have incorporated the selectable marker gene will survive, while the other cells die).

V. Analyzing Biomarker Nucleic Acids and Polypeptides

[0320] Biomarker nucleic acids and/or biomarker polypeptides can be analyzed according to the methods described herein and techniques known to the skilled artisan to identify such genetic or expression alterations useful for the present invention including, but not limited to, 1) an alteration in the level of a biomarker transcript or polypeptide, 2) a deletion or addition of one or more nucleotides from a biomarker gene, 4) a substitution of one or more nucleotides of a biomarker gene, 5) aberrant modification of a biomarker gene, such as an expression regulatory region, and the like.

a. Methods for Detection of Copy Number and/or Genomic Nucleic Acid Mutations

[0321] Methods of evaluating the copy number and/or genomic nucleic acid status (e.g., mutations) of a biomarker nucleic acid are well-known to those of skill in the art. The presence or absence of chromosomal gain or loss can be evaluated simply by a determination of copy number of the regions or markers identified herein.

[0322] In one embodiment, a biological sample is tested for the presence of copy number changes in genomic loci containing the genomic marker. A copy number of at least 3, 4, 5, 6, 7, 8, 9, or 10 of a biomarker is predictive of poorer outcome of treatment with the agent inhibiting binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome.

[0323] Methods of evaluating the copy number of a biomarker locus include, but are not limited to, hybridization-based assays. Hybridization-based assays include, but are not limited to, traditional "direct probe" methods, such as Southern blots, in situ hybridization (e.g., FISH and FISH plus SKY) methods, and "comparative probe" methods, such as comparative genomic hybridization (CGH), e.g., cDNA-based or oligonucleotide-based CGH. The methods can be used in a wide variety of formats including, but not limited to, substrate (e.g. membrane or glass) bound methods or array-based approaches.

[0324] In one embodiment, evaluating the biomarker gene copy number in a sample involves a Southern Blot. In a Southern Blot, the genomic DNA (typically fragmented and separated on an electrophoretic gel) is hybridized to a probe specific for the target region. Comparison of the intensity of the hybridization signal from the probe for the target region with control probe signal from analysis of normal genomic DNA (e.g., a non-amplified portion of the same or related cell, tissue, organ, etc.) provides an estimate of the relative copy number of the target nucleic acid. Alternatively, a Northern blot may be utilized for evaluating the copy number of encoding nucleic acid in a sample. In a Northern blot, mRNA is hybridized to a probe specific for the target region. Comparison of the intensity of the hybridization signal from the probe for the target region with control probe signal from analysis of normal RNA (e.g., a non-amplified portion of the same or related cell, tissue, organ, etc.) provides an estimate of the relative copy number of the target nucleic acid. Alternatively, other methods well-known in the art to detect RNA can be used, such that higher or lower expression relative to an appropriate control (e.g., a non-amplified portion of the same or related cell tissue, organ, etc.) provides an estimate of the relative copy number of the target nucleic acid.

[0325] An alternative means for determining genomic copy number is in situ hybridization (e.g., Angerer (1987) *Meth. Enzymol* 152: 649). Generally, in situ hybridization comprises the following steps: (1) fixation of tissue or biological structure to be analyzed; (2) prehybridization treatment of the biological structure to increase accessibility of target DNA, and to reduce nonspecific binding; (3) hybridization of the mixture of nucleic acids to the nucleic acid in the biological structure or tissue; (4) post-hybridization washes to remove nucleic acid fragments not bound in the hybridization and (5) detection of the hybridized nucleic acid fragments. The reagent used in each of these steps and the conditions for use vary depending on the particular application. In a typical in situ hybridization assay, cells are fixed to a solid support, typically a glass slide. If a nucleic acid is to be probed, the cells are typically denatured with heat or alkali. The cells are then contacted with a hybridization solution at a moderate temperature to permit annealing of labeled probes specific to the nucleic acid sequence encoding the protein. The targets (e.g., cells) are then typically washed at a predetermined stringency or at an increasing stringency until an appropriate signal to noise ratio is obtained. The probes are typically labeled, e.g., with radioisotopes or fluorescent reporters. In one embodiment, probes are sufficiently long so as to specifically hybridize with the target nucleic acid(s) under stringent conditions. Probes generally range in length from about 200 bases to about 1000 bases. In some applications it is necessary to block the hybridization capacity of repetitive sequences. Thus, in some embodiments, tRNA, human genomic DNA, or Cot-I DNA is used to block non-specific hybridization.

[0326] An alternative means for determining genomic copy number is comparative genomic hybridization. In general, genomic DNA is isolated from normal reference cells, as well as from test cells (e.g., tumor cells) and amplified, if necessary. The two nucleic acids are differentially labeled and then hybridized in situ to metaphase chromosomes of a reference cell. The repetitive sequences in both the reference and test DNAs are either removed or their hybridization capacity is reduced by some means, for example by prehybridization with appropriate blocking nucleic acids and/or including such blocking nucleic acid sequences for said repetitive sequences during said hybridization. The bound, labeled DNA sequences are then rendered in a visualizable form, if necessary. Chromosomal regions in the test cells which are at increased or decreased copy number can be identified by detecting regions where the ratio of signal from the two DNAs is altered. For example, those regions that have decreased in copy number in the test cells will show relatively lower signal from the test DNA than the reference compared to other regions of the genome. Regions that have been increased in copy number in the test cells will show relatively higher signal from the test DNA. Where there are chromosomal deletions or multiplications, differences in the ratio of the signals from the two labels will be detected and the ratio will provide a measure of the copy number. In another embodiment of CGH, array CGH (aCGH), the immobilized chromosome element is replaced with a collection of solid support bound target nucleic acids on an array, allowing for a large or complete percentage of the genome to be represented in the collection of solid support bound targets. Target nucleic acids may comprise cDNAs, genomic DNAs, oligonucleotides (e.g., to detect single nucleotide polymorphisms) and the like. Array-based CGH may also be performed with single-color labeling (as opposed to labeling the control and the possible tumor sample with two different dyes and mixing them prior to hybridization, which will yield a ratio due to competitive hybridization of probes on the arrays). In single color CGH, the control is labeled and hybridized to one array and absolute signals are read, and the possible tumor sample is labeled and hybridized to a second array (with identical content) and absolute signals are read. Copy number difference is calculated based on absolute signals from the two arrays. Methods of preparing immobilized chromosomes or arrays and performing comparative genomic hybridization are well-known in the art (see, e.g., U.S. Pat. Nos. 6,335,167; 6,197,501; 5,830,645; and 5,665,549 and Albertson (1984) *EMBO J.* 3: 1227-1234; Pinkel (1988) *Proc. Natl. Acad. Sci. USA* 85: 9138-9142; EPO Pub. No. 430,402*; Methods in Molecular Biology*, Vol. 33: In situ Hybridization Protocols, Choo, ed., Humana Press, Totowa, N.J. (1994), etc.) In another embodiment, the hybridization protocol of Pinkel, et al. (1998) *Nature Genetics* 20: 207-211, or of Kallioniemi (1992) *Proc. Natl Acad Sci USA* 89:5321-5325 (1992) is used.

[0327] In still another embodiment, amplification-based assays can be used to measure copy number. In such amplification-based assays, the nucleic acid sequences act as a template in an amplification reaction (e.g., Polymerase Chain Reaction (PCR). In a quantitative amplification, the amount of amplification product will be proportional to the amount of template in the original sample. Comparison to appropriate controls, e.g. healthy tissue, provides a measure of the copy number.

[0328] Methods of “quantitative” amplification are well-known to those of skill in the art. For example, quantitative PCR involves simultaneously co-amplifying a known quantity of a control sequence using the same primers. This provides an internal standard that may be used to calibrate the PCR reaction. Detailed protocols for quantitative PCR are provided in Innis, et al. (1990) PCR Protocols, *A Guide to Methods and Applications*, Academic Press, Inc. N.Y.). Measurement of DNA copy number at microsatellite loci using quantitative PCR analysis is described in Ginzonger, et al. (2000) *Cancer Research* 60:5405-5409. The known nucleic acid sequence for the genes is sufficient to enable one of skill in the art to routinely select primers to amplify any portion of the gene. Fluorogenic quantitative PCR may also be used in the methods of the present invention. In fluorogenic quantitative PCR, quantitation is based on amount of fluorescence signals, e.g., TaqMan and SYBR green.

[0329] Other suitable amplification methods include, but are not limited to, ligase chain reaction (LCR) (see Wu and Wallace (1989) *Genomics* 4: 560, Landegren, et al. (1988) *Science* 241:1077, and Barringer et al. (1990) *Gene* 89: 117), transcription amplification (Kwoh, et al. (1989) *Proc. Natl. Acad. Sci. USA* 86: 1173), self-sustained sequence replication (Guatelli, et al. (1990) *Proc. Nat. Acad. Sci. USA* 87: 1874), dot PCR, and linker adapter PCR, etc.

[0330] Loss of heterozygosity (LOH) and major copy proportion (MCP) mapping (Wang, Z. C., et al. (2004) *Cancer Res* 64(1):64-71; Seymour, A. B., et al. (1994) *Cancer Res* 54, 2761-4; Hahn, S. A., et al. (1995) *Cancer Res* 55, 4670-5; Kimura, M., et al. (1996) *Genes Chromosomes Cancer* 17, 88-93; Li et al., (2008) *MBC Bioinform.* 9, 204-219) may also be used to identify regions of amplification or deletion.

b. Methods for Detection of Biomarker Nucleic Acid Expression

[0331] Biomarker expression may be assessed by any of a wide variety of well-known methods for detecting expression of a transcribed molecule or protein. Non-limiting examples of such methods include immunological methods for detection of secreted, cell-surface, cytoplasmic, or nuclear proteins, protein purification methods, protein function or activity assays, nucleic acid hybridization methods, nucleic acid reverse transcription methods,

and nucleic acid amplification methods.
[0332] In preferred embodiments, activity of a particular gene is characterized by a measure of gene transcript (e.g. mRNA), by a measure of the quantity of translated protein, or by a measure of gene product activity. Marker expression can be monitored in a variety of ways, including by detecting mRNA levels, protein levels, or protein activity, any of which can be measured using standard techniques. Detection can involve quantification of the level of gene expression (e.g., genomic DNA, cDNA, mRNA, protein, or enzyme activity), or, alternatively, can be a qualitative assessment of the level of gene expression, in particular in comparison with a control level. The type of level being detected will be clear from the context.
[0333] In another embodiment, detecting or determining expression levels of a biomarker and functionally similar homologs thereof, including a fragment or genetic alteration thereof (e.g., in regulatory or promoter regions thereof) comprises detecting or determining RNA levels for the marker of interest. In one embodiment, one or more cells from the subject to be tested are obtained and RNA is isolated from the cells. In a preferred embodiment, a sample of breast tissue cells is obtained from the subject.
[0334] In one embodiment, RNA is obtained from a single cell. For example, a cell can be isolated from a tissue sample by laser capture microdissection (LCM). Using this technique, a cell can be isolated from a tissue section, including a stained tissue section, thereby assuring that the desired cell is isolated (see, e.g., Bonner et al. (1997) Science 278: 1481; Emmert-Buck et al. (1996) Science 274:998; Fend et al. (1999) Am. J. Path. 154: 61 and Murakami et al. (2000) Kidney Int. 58:1346). For example, Murakami et al., supra, describe isolation of a cell from a previously immunostained tissue section.
[0335] It is also be possible to obtain cells from a subject and culture the cells in vitro, such as to obtain a larger population of cells from which RNA can be extracted. Methods for establishing cultures of non-transformed cells, i.e., primary cell cultures, are known in the art.
[0336] When isolating RNA from tissue samples or cells from individuals, it may be important to prevent any further changes in gene expression after the tissue or cells has been removed from the subject. Changes in expression levels are known to change rapidly following perturbations, e.g., heat shock or activation with lipopolysaccharide (LPS) or other reagents. In addition, the RNA in the tissue and cells may quickly become degraded. Accordingly, in a preferred embodiment, the tissue or cells obtained from a subject is snap frozen as soon as possible.
[0337] RNA can be extracted from the tissue sample by a variety of methods, e.g., the guanidium thiocyanate lysis followed by CsCl centrifugation (Chirgwin et al., 1979, Biochemistry 18:5294-5299). RNA from single cells can be obtained as described in methods for preparing cDNA libraries from single cells, such as those described in Dulac, C. (1998) Curr. Top. Dev. Biol. 36, 245 and Jena et al. (1996) J. Immunol. Methods 190:199. Care to avoid RNA degradation must be taken, e.g., by inclusion of RNAsin.
[0338] The RNA sample can then be enriched in particular species. In one embodiment, poly(A)+ RNA is isolated from the RNA sample. In general, such purification takes advantage of the poly-A tails on mRNA. In particular and as noted above, poly-T oligonucleotides may be immobilized within on a solid support to serve as affinity ligands for mRNA. Kits for this purpose are commercially available, e.g., the MessageMaker kit (Life Technologies, Grand Island, NY).
[0339] In a preferred embodiment, the RNA population is enriched in marker sequences. Enrichment can be undertaken, e.g., by primer-specific cDNA synthesis, or multiple rounds of linear amplification based on cDNA synthesis and template-directed in vitro transcription (see, e.g., Wang et al. (1989) *Proc. Natl. Acad. Sci. U.S.A.* 86: 9717; Dulac et al., supra, and Jena et al., supra).
[0340] The population of RNA, enriched or not in particular species or sequences, can further be amplified. As defined herein, an “amplification process” is designed to strengthen, increase, or augment a molecule within the RNA. For example, where RNA is mRNA, an amplification process such as RT-PCR can be utilized to amplify the mRNA, such that a signal is detectable or detection is enhanced. Such an amplification process is beneficial particularly when the biological, tissue, or tumor sample is of a small size or volume.
[0341] Various amplification and detection methods can be used. For example, it is within the scope of the present invention to reverse transcribe mRNA into cDNA followed by polymerase chain reaction (RT-PCR); or, to use a single enzyme for both steps as described in U.S. Pat. No. 5,322,770, or reverse transcribe mRNA into cDNA followed by symmetric gap ligase chain reaction (RT-AGLCR) as described by R. L. Marshall, et al., PCR Methods and Applications 4: 80-84 (1994). Real time PCR may also be used.
[0342] Other known amplification methods which can be utilized herein include but are not limited to the so-called “NASBA” or “3SR” technique described in PNAS USA 87: 1874-1878 (1990) and also described in Nature 350 (No. 6313): 91-92 (1991); Q-beta amplification as described in published European Patent Application (EPA) No. 4544610; strand displacement amplification (as described in G. T. Walker et al., Clin. Chem. 42: 9-13 (1996) and European Patent Application No. 684315; target mediated amplification, as described by PCT Publication WO9322461; PCR; ligase chain reaction (LCR) (see, e.g., Wu and Wallace, Genomics 4, 560 (1989), Landegren et al., Science 241, 1077 (1988)); self-sustained sequence replication (SSR) (see, e.g., Guatelli et al., Proc. Nat. Acad. Sci. USA, 87, 1874 (1990)); and transcription amplification (see, e.g., Kwoh et al., Proc. Natl. Acad. Sci. USA 86,

1173 (1989)).
[0343] Many techniques are known in the state of the art for determining absolute and relative levels of gene expression, commonly used techniques suitable for use in the present invention include Northern analysis, RNase protection assays (RPA), microarrays and PCR-based techniques, such as quantitative PCR and differential display PCR. For example, Northern blotting involves running a preparation of RNA on a denaturing agarose gel, and transferring it to a suitable support, such as activated cellulose, nitrocellulose or glass or nylon membranes. Radiolabeled cDNA or RNA is then hybridized to the preparation, washed and analyzed by autoradiography.
[0344] In situ hybridization visualization may also be employed, wherein a radioactively labeled antisense RNA probe is hybridized with a thin section of a biopsy sample, washed, cleaved with RNase and exposed to a sensitive emulsion for autoradiography. The samples may be stained with hematoxylin to demonstrate the histological composition of the sample, and dark field imaging with a suitable light filter shows the developed emulsion.
[0345] Non-radioactive labels such as digoxigenin may also be used.
[0346] Alternatively, mRNA expression can be detected on a DNA array, chip or a microarray. Labeled nucleic acids of a test sample obtained from a subject may be hybridized to a solid surface comprising biomarker DNA. Positive hybridization signal is obtained with the sample containing biomarker transcripts. Methods of preparing DNA arrays and their use are well-known in the art (see, e.g., U.S. Pat. Nos: 6,618,6796; 6,379,897; 6,664,377; 6,451,536; 548,257; U.S. 20030157485 and Schena et al. (1995) *Science* 20, 467-470; Gerhold et al. (1999) *Trends In Biochem. Sci.* 24, 168-173; and Lennon et al. (2000) *Drug Discovery Today* 5, 59-65, which are herein incorporated by reference in their entirety). Serial Analysis of Gene Expression (SAGE) can also be performed (See for example U.S. Patent Application 20030215858).
[0347] To monitor mRNA levels, for example, mRNA is extracted from the biological sample to be tested, reverse transcribed, and fluorescently-labeled cDNA probes are generated. The microarrays capable of hybridizing to marker cDNA are then probed with the labeled cDNA probes, the slides scanned and fluorescence intensity measured. This intensity correlates with the hybridization intensity and expression levels.
[0348] Types of probes that can be used in the methods described herein include cDNA, riboprobes, synthetic oligonucleotides and genomic probes. The type of probe used will generally be dictated by the particular situation, such as riboprobes for in situ hybridization, and cDNA for Northern blotting, for example. In one embodiment, the probe is directed to nucleotide regions unique to the RNA. The probes may be as short as is required to differentially recognize marker mRNA transcripts, and may be as short as, for example, 15 bases; however, probes of at least 17, 18, 19 or 20 or more bases can be used. In one embodiment, the primers and probes hybridize specifically under stringent conditions to a DNA fragment having the nucleotide sequence corresponding to the marker. As herein used, the term “stringent conditions” means hybridization will occur only if there is at least 95% identity in nucleotide sequences. In another embodiment, hybridization under “stringent conditions” occurs when there is at least 97% identity between the sequences.
[0349] The form of labeling of the probes may be any that is appropriate, such as the use of radioisotopes, for example, .sup.32P and .sup.35S. Labeling with radioisotopes may be achieved, whether the probe is synthesized chemically or biologically, by the use of suitably labeled bases.
[0350] In one embodiment, the biological sample contains polypeptide molecules from the test subject. Alternatively, the biological sample can contain mRNA molecules from the test subject or genomic DNA molecules from the test subject.
[0351] In another embodiment, the methods further involve obtaining a control biological sample from a control subject, contacting the control sample with a compound or agent capable of detecting marker polypeptide, mRNA, genomic DNA, or fragments thereof, such that the presence of the marker polypeptide, mRNA, genomic DNA, or fragments thereof, is detected in the biological sample, and comparing the presence of the marker polypeptide, mRNA, genomic DNA, or fragments thereof, in the control sample with the presence of the marker polypeptide, mRNA, genomic DNA, or fragments thereof in the test sample.
c. Methods for Detection of Biomarker Protein Expression
[0352] The activity or level of a biomarker protein can be detected and/or quantified by detecting or quantifying the expressed polypeptide. The polypeptide can be detected and quantified by any of a number of means well-known to those of skill in the art. Aberrant levels of polypeptide expression of the polypeptides encoded by a biomarker nucleic acid and functionally similar homologs thereof, including a fragment or genetic alteration thereof (e.g., in regulatory or promoter regions thereof) are associated with the likelihood of response of a cancer to a modulator of T cell mediated cytotoxicity alone or in combination with an immunotherapy treatment. Any method known in the art for detecting polypeptides can be used. Such methods include, but are not limited to, immunodiffusion, immunoelectrophoresis, radioimmunoassay (RIA), enzyme-linked immunosorbent assays (ELISAs), immunofluorescent assays, Western blotting, binder-ligand assays, immunohistochemical techniques, agglutination, complement assays, high performance liquid chromatography (HPLC), thin layer chromatography (TLC), hyperdiffusion chromatography, and the like (e.g., Basic and Clinical Immunology, Sites and Terr, eds., Appleton and Lange, Norwalk, Conn. pp 217-262, 1991 which is incorporated by reference). Preferred are binder-ligand

immunoassay methods including reacting antibodies with an epitope or epitopes and competitively displacing a labeled polypeptide or derivative thereof.

[0353] For example, ELISA and RIA procedures may be conducted such that a desired biomarker protein standard is labeled (with a radioisotope such as .sup.125I or .sup.35S, or an assayable enzyme, such as horseradish peroxidase or alkaline phosphatase), and, together with the unlabeled sample, brought into contact with the corresponding antibody, whereon a second antibody is used to bind the first, and radioactivity or the immobilized enzyme assayed (competitive assay). Alternatively, the biomarker protein in the sample is allowed to react with the corresponding immobilized antibody, radioisotope- or enzyme-labeled anti-biomarker protein antibody is allowed to react with the system, and radioactivity or the enzyme assayed (ELISA-sandwich assay). Other conventional methods may also be employed as suitable.

[0354] The above techniques may be conducted essentially as a “one-step” or “two-step” assay. A “one-step” assay involves contacting antigen with immobilized antibody and, without washing, contacting the mixture with labeled antibody. A “two-step” assay involves washing before contacting, the mixture with labeled antibody. Other conventional methods may also be employed as suitable.

[0355] In one embodiment, a method for measuring biomarker protein levels comprises the steps of: contacting a biological specimen with an antibody or variant (e.g., fragment) thereof which selectively binds the biomarker protein, and detecting whether said antibody or variant thereof is bound to said sample and thereby measuring the levels of the biomarker protein.

[0356] Enzymatic and radiolabeling of biomarker protein and/or the antibodies may be effected by conventional means. Such means will generally include covalent linking of the enzyme to the antigen or the antibody in question, such as by glutaraldehyde, specifically so as not to adversely affect the activity of the enzyme, by which is meant that the enzyme must still be capable of interacting with its substrate, although it is not necessary for all of the enzyme to be active, provided that enough remains active to permit the assay to be effected. Indeed, some techniques for binding enzyme are non-specific (such as using formaldehyde), and will only yield a proportion of active enzyme.

[0357] It is usually desirable to immobilize one component of the assay system on a support, thereby allowing other components of the system to be brought into contact with the component and readily removed without laborious and time-consuming labor. It is possible for a second phase to be immobilized away from the first, but one phase is usually sufficient.

[0358] It is possible to immobilize the enzyme itself on a support, but if solid-phase enzyme is required, then this is generally best achieved by binding to antibody and affixing the antibody to a support, models and systems for which are well-known in the art. Simple polyethylene may provide a suitable support.

[0359] Enzymes employable for labeling are not particularly limited, but may be selected from the members of the oxidase group, for example. These catalyze production of hydrogen peroxide by reaction with their substrates, and glucose oxidase is often used for its good stability, ease of availability and cheapness, as well as the ready availability of its substrate (glucose). Activity of the oxidase may be assayed by measuring the concentration of hydrogen peroxide formed after reaction of the enzyme-labeled antibody with the substrate under controlled conditions well-known in the art.

[0360] Other techniques may be used to detect biomarker protein according to a practitioner's preference based upon the present disclosure. One such technique is Western blotting (Towbin et at., Proc. Nat. Acad. Sci. 76:4350 (1979)), wherein a suitably treated sample is run on an SDS-PAGE gel before being transferred to a solid support, such as a nitrocellulose filter. Anti-biomarker protein antibodies (unlabeled) are then brought into contact with the support and assayed by a secondary immunological reagent, such as labeled protein A or anti-immunoglobulin (suitable labels including .sup.125I, horseradish peroxidase and alkaline phosphatase). Chromatographic detection may also be used.

[0361] Immunohistochemistry may be used to detect expression of biomarker protein, e.g., in a biopsy sample. A suitable antibody is brought into contact with, for example, a thin layer of cells, washed, and then contacted with a second, labeled antibody. Labeling may be by fluorescent markers, enzymes, such as peroxidase, avidin, or radiolabeling. The assay is scored visually, using microscopy.

[0362] Anti-biomarker protein antibodies, such as intrabodies, may also be used for imaging purposes, for example, to detect the presence of biomarker protein in cells and tissues of a subject. Suitable labels include radioisotopes, iodine (.sup.125I, .sup.121I), carbon (.sup.14C), sulphur (.sup.35S), tritium (.sup.3H), indium (.sup.I12In), and technetium (.sup.99mTc), fluorescent labels, such as fluorescein and rhodamine, and biotin.

[0363] For in vivo imaging purposes, antibodies are not detectable, as such, from outside the body, and so must be labeled, or otherwise modified, to permit detection. Markers for this purpose may be any that do not substantially interfere with the antibody binding, but which allow external detection. Suitable markers may include those that may be detected by X-radiography, NMR or MRI. For X-radiographic techniques, suitable markers include any radioisotope that emits detectable radiation but that is not overtly harmful to the subject, such as barium or cesium, for example. Suitable markers for NMR and MRI generally include those with a detectable characteristic spin, such as deuterium, which may be incorporated into the antibody by suitable labeling of nutrients for the relevant hybridoma, for example.

[0364] The size of the subject, and the imaging system used, will determine the quantity of imaging moiety needed to produce diagnostic images. In the case of a radioisotope moiety, for a human subject, the quantity of radioactivity injected will normally range from about 5 to 20 millicuries of technetium-99. The labeled antibody or antibody fragment will then preferentially accumulate at the location of cells which contain biomarker protein. The labeled antibody or antibody fragment can then be detected using known techniques.
[0365] Antibodies that may be used to detect biomarker protein include any antibody, whether natural or synthetic, full length or a fragment thereof, monoclonal or polyclonal, that binds sufficiently strongly and specifically to the biomarker protein to be detected. An antibody may have a Ka of at most about 10.sup.−6M, 10.sup.−7M, 10.sup.−8M, 10.sup.−9M, 10.sup.−10M, 10.sup.−11 M, 10.sup.−12M. The phrase “specifically binds” refers to binding of, for example, an antibody to an epitope or antigen or antigenic determinant in such a manner that binding can be displaced or competed with a second preparation of identical or similar epitope, antigen or antigenic determinant. An antibody may bind preferentially to the biomarker protein relative to other proteins, such as related proteins.
[0366] Antibodies are commercially available or may be prepared according to methods known in the art.
[0367] Antibodies and derivatives thereof that may be used encompass polyclonal or monoclonal antibodies, chimeric, human, humanized, primatized (CDR-grafted), veneered or single-chain antibodies as well as functional fragments, i.e., biomarker protein binding fragments, of antibodies. For example, antibody fragments capable of binding to a biomarker protein or portions thereof, including, but not limited to, Fv, Fab, Fab′ and F(ab′) 2 fragments can be used. Such fragments can be produced by enzymatic cleavage or by recombinant techniques. For example, papain or pepsin cleavage can generate Fab or F(ab′) 2 fragments, respectively. Other proteases with the requisite substrate specificity can also be used to generate Fab or F(ab′) 2 fragments. Antibodies can also be produced in a variety of truncated forms using antibody genes in which one or more stop codons have been introduced upstream of the natural stop site. For example, a chimeric gene encoding a F(ab′) 2 heavy chain portion can be designed to include DNA sequences encoding the CH, domain and hinge region of the heavy chain.
[0368] Synthetic and engineered antibodies are described in, e.g., Cabilly et al., U.S. Pat. No. 4,816,567 Cabilly et al., European Patent No. 0,125,023 B1; Boss et al., U.S. Pat. No. 4,816,397; Boss et al., European Patent No. 0,120,694 B1; Neuberger, M. S. et al., WO 86/01533; Neuberger, M. S. et al., European Patent No. 0,194,276 B1; Winter, U.S. Pat. No. 5,225,539; Winter, European Patent No. 0,239,400 B1; Queen et al., European Patent No. 0451216 B1; and Padlan, E. A. et al., EP 0519596 A1. See also, Newman, R. et al., BioTechnology, 10: 1455-1460 (1992), regarding primatized antibody, and Ladner et al., U.S. Pat. No. 4,946,778 and Bird, R. E. et al., Science, 242: 423-426 (1988)) regarding single-chain antibodies. Antibodies produced from a library, e.g., phage display library, may also be used.
[0369] In some embodiments, agents that specifically bind to a biomarker protein other than antibodies are used, such as peptides. Peptides that specifically bind to a biomarker protein can be identified by any means known in the art. For example, specific peptide binders of a biomarker protein can be screened for using peptide phage display libraries.
d. Methods for Detection of Biomarker Structural Alterations
[0370] The following illustrative methods can be used to identify the presence of a structural alteration in a biomarker nucleic acid and/or biomarker polypeptide molecule in order to, for example, identify the SS18-SSX fusion protein/H2A K119Ub nucleosomes pathway proteins that are overexpressed, overfunctional, and the like.
[0371] In certain embodiments, detection of the alteration involves the use of a probe/primer in a polymerase chain reaction (PCR) (see, e.g., U.S. Pat. Nos. 4,683,195 and 4,683,202), such as anchor PCR or RACE PCR, or, alternatively, in a ligation chain reaction (LCR) (see, e.g., Landegran et al. (1988) Science 241:1077-1080; and Nakazawa et al. (1994) Proc. Natl. Acad. Sci. USA 91:360-364), the latter of which can be particularly useful for detecting point mutations in a biomarker nucleic acid such as a biomarker gene (see Abravaya et al. (1995) Nucleic Acids Res. 23:675-682). This method can include the steps of collecting a sample of cells from a subject, isolating nucleic acid (e.g., genomic, mRNA or both) from the cells of the sample, contacting the nucleic acid sample with one or more primers which specifically hybridize to a biomarker gene under conditions such that hybridization and amplification of the biomarker gene (if present) occurs, and detecting the presence or absence of an amplification product, or detecting the size of the amplification product and comparing the length to a control sample. It is anticipated that PCR and/or LCR may be desirable to use as a preliminary amplification step in conjunction with any of the techniques used for detecting mutations described herein.
[0372] Alternative amplification methods include: self-sustained sequence replication (Guatelli, J. C. et al. (1990) Proc. Natl. Acad. Sci. USA 87:1874-1878), transcriptional amplification system (Kwoh, D. Y. et al. (1989) Proc. Natl. Acad. Sci. USA 86:1173-1177), Q-Beta Replicase (Lizardi, P. M. et al. (1988) Bio-Technology 6:1197), or any other nucleic acid amplification method, followed by the detection of the amplified molecules using techniques well-known to those of skill in the art. These detection schemes are especially useful for the detection of nucleic acid molecules if such molecules are present in very low numbers.
[0373] In an alternative embodiment, mutations in a biomarker nucleic acid from a sample cell can be identified by alterations in restriction enzyme cleavage patterns. For example, sample and control DNA is isolated, amplified

(optionally), digested with one or more restriction endonucleases, and fragment length sizes are determined by gel electrophoresis and compared. Differences in fragment length sizes between sample and control DNA indicates mutations in the sample DNA. Moreover, the use of sequence specific ribozymes (see, for example, U.S. Pat. No. 5,498,531) can be used to score for the presence of specific mutations by development or loss of a ribozyme cleavage site.

[0374] In other embodiments, genetic mutations in biomarker nucleic acid can be identified by hybridizing a sample and control nucleic acids, e.g., DNA or RNA, to high density arrays containing hundreds or thousands of oligonucleotide probes (Cronin, M. T. et al. (1996) Hum. Mutat. 7:244-255; Kozal, M. J. et al. (1996) Nat. Med. 2:753-759). For example, biomarker genetic mutations can be identified in two dimensional arrays containing light-generated DNA probes as described in Cronin et al. (1996) supra. Briefly, a first hybridization array of probes can be used to scan through long stretches of DNA in a sample and control to identify base changes between the sequences by making linear arrays of sequential, overlapping probes. This step allows the identification of point mutations. This step is followed by a second hybridization array that allows the characterization of specific mutations by using smaller, specialized probe arrays complementary to all variants or mutations detected. Each mutation array is composed of parallel probe sets, one complementary to the wild-type gene and the other complementary to the mutant gene.

[0375] Such biomarker genetic mutations can be identified in a variety of contexts, including, for example, germline and somatic mutations.

[0376] In yet another embodiment, any of a variety of sequencing reactions known in the art can be used to directly sequence a biomarker gene and detect mutations by comparing the sequence of the sample biomarker with the corresponding wild-type (control) sequence. Examples of sequencing reactions include those based on techniques developed by Maxam and Gilbert (1977) *Proc. Natl. Acad. Sci. USA* 74:560 or Sanger (1977) *Proc. Natl. Acad Sci*. USA 74:5463. It is also contemplated that any of a variety of automated sequencing procedures can be utilized when performing the diagnostic assays (Naeve (1995) *Biotechniques* 19:448-53), including sequencing by mass spectrometry (see, e.g., PCT International Publication No. WO 94/16101; Cohen et al. (1996) *Adv. Chromatogr.* 36:127-162; and Griffin et al. (1993) *Appl. Biochem. Biotechnol.* 38:147-159).

[0377] Other methods for detecting mutations in a biomarker gene include methods in which protection from cleavage agents is used to detect mismatched bases in RNA/RNA or RNA/DNA heteroduplexes (Myers et al. (1985) Science 230:1242). In general, the art technique of “mismatch cleavage” starts by providing heteroduplexes formed by hybridizing (labeled) RNA or DNA containing the wild-type biomarker sequence with potentially mutant RNA or DNA obtained from a tissue sample. The double-stranded duplexes are treated with an agent which cleaves single-stranded regions of the duplex such as which will exist due to base pair mismatches between the control and sample strands. For instance, RNA/DNA duplexes can be treated with RNase and DNA/DNA hybrids treated with SI nuclease to enzymatically digest the mismatched regions. In other embodiments, either DNA/DNA or RNA/DNA duplexes can be treated with hydroxylamine or osmium tetroxide and with piperidine in order to digest mismatched regions. After digestion of the mismatched regions, the resulting material is then separated by size on denaturing polyacrylamide gels to determine the site of mutation. See, for example, Cotton et al. (1988) Proc. Natl. Acad. Sci. USA 85:4397 and Saleeba et al. (1992) Methods Enzymol. 217:286-295. In a preferred embodiment, the control DNA or RNA can be labeled for detection.

[0378] In still another embodiment, the mismatch cleavage reaction employs one or more proteins that recognize mismatched base pairs in double-stranded DNA (so called “DNA mismatch repair” enzymes) in defined systems for detecting and mapping point mutations in biomarker cDNAs obtained from samples of cells. For example, the mutY enzyme of *E. coli* cleaves A at G/A mismatches and the thymidine DNA glycosylase from HeLa cells cleaves T at G/T mismatches (Hsu et al. (1994) *Carcinogenesis* 15:1657-1662). According to an exemplary embodiment, a probe based on a biomarker sequence, e.g., a wild-type biomarker treated with a DNA mismatch repair enzyme, and the cleavage products, if any, can be detected from electrophoresis protocols or the like (e.g., U.S. Pat. No. 5,459,039.)

[0379] In other embodiments, alterations in electrophoretic mobility can be used to identify mutations in biomarker genes. For example, single strand conformation polymorphism (SSCP) may be used to detect differences in electrophoretic mobility between mutant and wild type nucleic acids (Orita et al. (1989) *Proc Natl. Acad. Sci USA* 86:2766; see also Cotton (1993) *Mutat. Res.* 285:125-144 and Hayashi (1992) *Genet. Anal. Tech. Appl.* 9:73-79). Single-stranded DNA fragments of sample and control biomarker nucleic acids will be denatured and allowed to renature. The secondary structure of single-stranded nucleic acids varies according to sequence, the resulting alteration in electrophoretic mobility enables the detection of even a single base change. The DNA fragments may be labeled or detected with labeled probes. The sensitivity of the assay may be enhanced by using RNA (rather than DNA), in which the secondary structure is more sensitive to a change in sequence. In a preferred embodiment, the subject method utilizes heteroduplex analysis to separate double stranded heteroduplex molecules on the basis of changes in electrophoretic mobility (Keen et al. (1991) *Trends Genet.* 7:5).

[0380] In yet another embodiment the movement of mutant or wild-type fragments in polyacrylamide gels containing a gradient of denaturant is assayed using denaturing gradient gel electrophoresis (DGGE) (Myers et al. (1985) *Nature*

313:495). When DGGE is used as the method of analysis, DNA will be modified to ensure that it does not completely denature, for example by adding a GC clamp of approximately 40 bp of high-melting GC-rich DNA by PCR. In a further embodiment, a temperature gradient is used in place of a denaturing gradient to identify differences in the mobility of control and sample DNA (Rosenbaum and Reissner (1987) *Biophys. Chem.* 265:12753).

[0381] Examples of other techniques for detecting point mutations include, but are not limited to, selective oligonucleotide hybridization, selective amplification, or selective primer extension. For example, oligonucleotide primers may be prepared in which the known mutation is placed centrally and then hybridized to target DNA under conditions which permit hybridization only if a perfect match is found (Saiki et al. (1986) *Nature* 324:163; Saiki et al. (1989) *Proc. Natl. Acad. Sci. USA* 86:6230). Such allele specific oligonucleotides are hybridized to PCR amplified target DNA or a number of different mutations when the oligonucleotides are attached to the hybridizing membrane and hybridized with labeled target DNA.

[0382] Alternatively, allele specific amplification technology which depends on selective PCR amplification may be used in conjunction with the instant invention. Oligonucleotides used as primers for specific amplification may carry the mutation of interest in the center of the molecule (so that amplification depends on differential hybridization) (Gibbs et al. (1989) *Nucleic Acids Res.* 17:2437-2448) or at the extreme 3′ end of one primer where, under appropriate conditions, mismatch can prevent, or reduce polymerase extension (Prossner (1993) *Tibtech* 11:238). In addition it may be desirable to introduce a novel restriction site in the region of the mutation to create cleavage-based detection (Gasparini et al. (1992) *Mol. Cell Probes* 6:1). It is anticipated that in certain embodiments amplification may also be performed using Taq ligase for amplification (Barany (1991) *Proc. Natl. Acad. Sci USA* 88:189). In such cases, ligation will occur only if there is a perfect match at the 3′ end of the 5′ sequence making it possible to detect the presence of a known mutation at a specific site by looking for the presence or absence of amplification.

VI. Cancer Therapies

[0383] The efficacy of a cancer therapy with an agent that inhibits binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome is predicted according to biomarker presence, absence, amount and/or activity associated with a cancer in a subject according to the methods described herein. In one embodiment, such cancer therapy (e.g., an agent inhibiting binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome) or combinations of therapies (e.g., an agent inhibiting binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome, in combination with at least one immunotherapy) can be administered to a desired subject or once a subject is indicated as being a likely responder to cancer therapy (e.g., an agent inhibiting binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome). In another embodiment, such cancer therapy (e.g., an agent inhibiting binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome) can be avoided once a subject is indicated as not being a likely responder to the cancer therapy (e.g., an agent inhibiting binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome) and an alternative treatment regimen, such as targeted and/or untargeted cancer therapies can be administered. Combination therapies are also contemplated and can comprise, for example, one or more chemotherapeutic agents and radiation, one or more chemotherapeutic agents and immunotherapy, or one or more chemotherapeutic agents, radiation and chemotherapy, each combination of which can be with or without the agent inhibiting binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome.

[0384] The term “targeted therapy” refers to administration of agents that selectively interact with a chosen biomolecule to thereby treat cancer. One example includes administration of an agent that inhibits binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome. These agents block or otherwise reduce the interaction between a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome such that the activation of the SS18-SSX fusion protein target genes otherwise induced by the interaction is blocked or otherwise reduced. These agents may inhibit binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome in a direct or indirect way.

[0385] Targeted therapy regarding the inhibition of immune checkpoint inhibitor is useful in combination with the methods of the present invention. The term “immune checkpoint inhibitor” means a group of molecules on the cell surface of CD4+ and/or CD8+ T cells that fine-tune immune responses by down-modulating or inhibiting an anti-tumor immune response. Immune checkpoint proteins are well-known in the art and include, without limitation, CTLA-4, PD-1, VISTA, B7-H2, B7-H3, PD-L1, B7-H4, B7-H6, 2B4, ICOS, HVEM, PD-L2, CD160, gp49B, PIR-B, KIR family receptors, TIM-1, TIM-3, TIM-4, LAG-3, BTLA, SIRPalpha (CD47), CD48, 2B4 (CD244), B7.1, B7.2, ILT-2, ILT-4, TIGIT, and A2aR (see, for example, WO 2012/177624). Inhibition of one or more immune checkpoint inhibitors can block or otherwise neutralize inhibitory signaling to thereby upregulate an immune response in order to more efficaciously treat cancer.

[0386] Immunotherapy is one form of targeted therapy that may comprise, for example, the use of cancer vaccines and/or sensitized antigen presenting cells. For example, an oncolytic virus is a virus that is able to infect and lyse cancer cells, while leaving normal cells unharmed, making them potentially useful in cancer therapy. Replication of oncolytic viruses both facilitates tumor cell destruction and also produces dose amplification at the tumor site. They may also act as vectors for anticancer genes, allowing them to be specifically delivered to the tumor site. The

immunotherapy can involve passive immunity for short-term protection of a host, achieved by the administration of pre-formed antibody directed against a cancer antigen or disease antigen (e.g., administration of a monoclonal antibody, optionally linked to a chemotherapeutic agent or toxin, to a tumor antigen). For example, anti-VEGF and mTOR inhibitors are known to be effective in treating renal cell carcinoma. Immunotherapy can also focus on using the cytotoxic lymphocyte-recognized epitopes of cancer cell lines. Alternatively, antisense polynucleotides, ribozymes, RNA interference molecules, triple helix polynucleotides and the like, can be used to selectively modulate biomolecules that are linked to the initiation, progression, and/or pathology of a tumor or cancer.
[0387] Similarly, agents and therapies other than immunotherapy or in combination thereof can be used with in combination with agents inhibiting binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome to treat a cancer that would benefit therefrom. For example, chemotherapy, radiation, epigenetic modifiers (e.g., histone deacetylase (HDAC) modifiers, methylation modifiers, phosphorylation modifiers, and the like), targeted therapy, and the like are well-known in the art.
[0388] The term “untargeted therapy” refers to administration of agents that do not selectively interact with a chosen biomolecule yet treat cancer. Representative examples of untargeted therapies include, without limitation, chemotherapy, gene therapy, and radiation therapy.
[0389] In one embodiment, chemotherapy is used. Chemotherapy includes the administration of a chemotherapeutic agent. Such a chemotherapeutic agent may be, but is not limited to, those selected from among the following groups of compounds: platinum compounds, cytotoxic antibiotics, antimetabolities, anti-mitotic agents, alkylating agents, arsenic compounds, DNA topoisomerase inhibitors, taxanes, nucleoside analogues, plant alkaloids, and toxins; and synthetic derivatives thereof. Exemplary compounds include, but are not limited to, alkylating agents: cisplatin, treosulfan, and trofosfamide; plant alkaloids: vinblastine, paclitaxel, docetaxol; DNA topoisomerase inhibitors: teniposide, crisnatol, and mitomycin; anti-folates: methotrexate, mycophenolic acid, and hydroxyurea; pyrimidine analogs: 5-fluorouracil, doxifluridine, and cytosine arabinoside; purine analogs: mercaptopurine and thioguanine; DNA antimetabolites: 2′-deoxy-5-fluorouridine, aphidicolin glycinate, and pyrazoloimidazole; and antimitotic agents: halichondrin, colchicine, and rhizoxin. Compositions comprising one or more chemotherapeutic agents (e.g., FLAG, CHOP) may also be used. FLAG comprises fludarabine, cytosine arabinoside (Ara-C) and G-CSF. CHOP comprises cyclophosphamide, vincristine, doxorubicin, and prednisone. In another embodiment, PARP (e.g., PARP-1 and/or PARP-2) inhibitors are used and such inhibitors are well-known in the art (e.g., Olaparib, ABT-888, BSI-201, BGP-15 (N-Gene Research Laboratories, Inc.); INO-1001 (Inotek Pharmaceuticals Inc.); PJ34 (Soriano et al., 2001; Pacher et al., 2002b); 3-aminobenzamide (Trevigen); 4-amino-1,8-naphthalimide; (Trevigen); 6(5H)-phenanthridinone (Trevigen); benzamide (U.S. Pat. No. Re. 36,397); and NU1025 (Bowman et al.). The mechanism of action is generally related to the ability of PARP inhibitors to bind PARP and decrease its activity. PARP catalyzes the conversion of .beta.-nicotinamide adenine dinucleotide (NAD+) into nicotinamide and poly-ADP-ribose (PAR). Both poly (ADP-ribose) and PARP have been linked to regulation of transcription, cell proliferation, genomic stability, and carcinogenesis (Bouchard V. J. et. al. Experimental Hematology, Volume 31, Number 6, June 2003, pp. 446-454(9); Herceg Z.; Wang Z.-Q. Mutation Research/Fundamental and Molecular Mechanisms of Mutagenesis, Volume 477, Number 1, 2 Jun. 2001, pp. 97-110(14)). Poly(ADP-ribose) polymerase 1 (PARP1) is a key molecule in the repair of DNA single-strand breaks (SSBs) (de Murcia J. et al. 1997. Proc Natl Acad Sci USA 94:7303-7307; Schreiber V, Dantzer F, Ame J C, de Murcia G (2006) Nat Rev Mol Cell Biol 7:517-528; Wang Z Q, et al. (1997) Genes Dev 11:2347-2358). Knockout of SSB repair by inhibition of PARP1 function induces DNA double-strand breaks (DSBs) that can trigger synthetic lethality in cancer cells with defective homology-directed DSB repair (Bryant H E, et al. (2005) Nature 434:913-917; Farmer H, et al. (2005) Nature 434:917-921). The foregoing examples of chemotherapeutic agents are illustrative, and are not intended to be limiting.
[0390] In another embodiment, radiation therapy is used. The radiation used in radiation therapy can be ionizing radiation. Radiation therapy can also be gamma rays, X-rays, or proton beams. Examples of radiation therapy include, but are not limited to, external-beam radiation therapy, interstitial implantation of radioisotopes (I-125, palladium, iridium), radioisotopes such as strontium-89, thoracic radiation therapy, intraperitoneal P-32 radiation therapy, and/or total abdominal and pelvic radiation therapy. For a general overview of radiation therapy, see Hellman, Chapter 16: Principles of Cancer Management: Radiation Therapy, 6th edition, 2001, DeVita et al., eds., J. B. Lippencott Company, Philadelphia. The radiation therapy can be administered as external beam radiation or teletherapy wherein the radiation is directed from a remote source. The radiation treatment can also be administered as internal therapy or brachytherapy wherein a radioactive source is placed inside the body close to cancer cells or a tumor mass. Also encompassed is the use of photodynamic therapy comprising the administration of photosensitizers, such as hematoporphyrin and its derivatives, Vertoporfin (BPD-MA), phthalocyanine, photosensitizer Pc4, demethoxy-hypocrellin A; and 2BA-2-DMHA.
[0391] In another embodiment, surgical intervention can occur to physically remove cancerous cells and/or tissues.
[0392] In still another embodiment, hormone therapy is used. Hormonal therapeutic treatments can comprise, for example, hormonal agonists, hormonal antagonists (e.g., flutamide, bicalutamide, tamoxifen, raloxifene, leuprolide acetate (LUPRON), LH-RH antagonists), inhibitors of hormone biosynthesis and processing, and steroids (e.g.,

dexamethasone, retinoids, deltoids, betamethasone, cortisol, cortisone, prednisone, dehydrotestosterone, glucocorticoids, mineralocorticoids, estrogen, testosterone, progestins), vitamin A derivatives (e.g., all-trans retinoic acid (ATRA)); vitamin D3 analogs; antigestagens (e.g., mifepristone, onapristone), or antiandrogens (e.g., cyproterone acetate).

[0393] In yet another embodiment, hyperthermia, a procedure in which body tissue is exposed to high temperatures (up to 106° F.) is used. Heat may help shrink tumors by damaging cells or depriving them of substances they need to live. Hyperthermia therapy can be local, regional, and whole-body hyperthermia, using external and internal heating devices. Hyperthermia is almost always used with other forms of therapy (e.g., radiation therapy, chemotherapy, and biological therapy) to try to increase their effectiveness. Local hyperthermia refers to heat that is applied to a very small area, such as a tumor. The area may be heated externally with high-frequency waves aimed at a tumor from a device outside the body. To achieve internal heating, one of several types of sterile probes may be used, including thin, heated wires or hollow tubes filled with warm water; implanted microwave antennae; and radiofrequency electrodes. In regional hyperthermia, an organ or a limb is heated. Magnets and devices that produce high energy are placed over the region to be heated. In another approach, called perfusion, some of the patient's blood is removed, heated, and then pumped (perfused) into the region that is to be heated internally. Whole-body heating is used to treat metastatic cancer that has spread throughout the body. It can be accomplished using warm-water blankets, hot wax, inductive coils (like those in electric blankets), or thermal chambers (similar to large incubators). Hyperthermia does not cause any marked increase in radiation side effects or complications. Heat applied directly to the skin, however, can cause discomfort or even significant local pain in about half the patients treated. It can also cause blisters, which generally heal rapidly.

[0394] In still another embodiment, photodynamic therapy (also called PDT, photoradiation therapy, phototherapy, or photochemotherapy) is used for the treatment of some types of cancer. It is based on the discovery that certain chemicals known as photosensitizing agents can kill one-celled organisms when the organisms are exposed to a particular type of light. PDT destroys cancer cells through the use of a fixed-frequency laser light in combination with a photosensitizing agent. In PDT, the photosensitizing agent is injected into the bloodstream and absorbed by cells all over the body. The agent remains in cancer cells for a longer time than it does in normal cells. When the treated cancer cells are exposed to laser light, the photosensitizing agent absorbs the light and produces an active form of oxygen that destroys the treated cancer cells. Light exposure must be timed carefully so that it occurs when most of the photosensitizing agent has left healthy cells but is still present in the cancer cells. The laser light used in PDT can be directed through a fiber-optic (a very thin glass strand). The fiber-optic is placed close to the cancer to deliver the proper amount of light. The fiber-optic can be directed through a bronchoscope into the lungs for the treatment of lung cancer or through an endoscope into the esophagus for the treatment of esophageal cancer. An advantage of PDT is that it causes minimal damage to healthy tissue. However, because the laser light currently in use cannot pass through more than about 3 centimeters of tissue (a little more than one and an eighth inch), PDT is mainly used to treat tumors on or just under the skin or on the lining of internal organs. Photodynamic therapy makes the skin and eyes sensitive to light for 6 weeks or more after treatment. Patients are advised to avoid direct sunlight and bright indoor light for at least 6 weeks. If patients must go outdoors, they need to wear protective clothing, including sunglasses. Other temporary side effects of PDT are related to the treatment of specific areas and can include coughing, trouble swallowing, abdominal pain, and painful breathing or shortness of breath. In December 1995, the U.S. Food and Drug Administration (FDA) approved a photosensitizing agent called porfimer sodium, or Photofrin®, to relieve symptoms of esophageal cancer that is causing an obstruction and for esophageal cancer that cannot be satisfactorily treated with lasers alone. In January 1998, the FDA approved porfimer sodium for the treatment of early nonsmall cell lung cancer in patients for whom the usual treatments for lung cancer are not appropriate. The National Cancer Institute and other institutions are supporting clinical trials (research studies) to evaluate the use of photodynamic therapy for several types of cancer, including cancers of the bladder, brain, larynx, and oral cavity.

[0395] In yet another embodiment, laser therapy is used to harness high-intensity light to destroy cancer cells. This technique is often used to relieve symptoms of cancer such as bleeding or obstruction, especially when the cancer cannot be cured by other treatments. It may also be used to treat cancer by shrinking or destroying tumors. The term “laser” stands for light amplification by stimulated emission of radiation. Ordinary light, such as that from a light bulb, has many wavelengths and spreads in all directions. Laser light, on the other hand, has a specific wavelength and is focused in a narrow beam. This type of high-intensity light contains a lot of energy. Lasers are very powerful and may be used to cut through steel or to shape diamonds. Lasers also can be used for very precise surgical work, such as repairing a damaged retina in the eye or cutting through tissue (in place of a scalpel). Although there are several different kinds of lasers, only three kinds have gained wide use in medicine: Carbon dioxide (CO.sub.2) laser—This type of laser can remove thin layers from the skin's surface without penetrating the deeper layers. This technique is particularly useful in treating tumors that have not spread deep into the skin and certain precancerous conditions. As an alternative to traditional scalpel surgery, the CO.sub.2 laser is also able to cut the skin. The laser is used in this way to remove skin cancers. Neodymium:yttrium-aluminum-garnet (Nd:YAG) laser—Light from this laser can penetrate deeper into tissue than light from the other types of lasers, and it can cause blood to clot quickly. It can be carried through optical fibers to less accessible parts of the body. This type of laser is sometimes used to treat throat cancers. Argon laser—This laser can pass through only superficial layers of tissue and is therefore useful in dermatology and in eye surgery. It also is used with light-sensitive dyes to treat tumors in a procedure known as photodynamic therapy (PDT). Lasers have several advantages over standard surgical tools, including: Lasers are more precise than scalpels. Tissue near an incision is protected, since there is little contact with surrounding skin or other tissue. The heat produced by lasers sterilizes the surgery site, thus reducing the risk of infection. Less operating time may be needed because the precision of the laser allows for a smaller incision. Healing time is often shortened; since laser heat seals blood vessels, there is less bleeding, swelling, or scarring. Laser surgery may be less complicated. For example, with fiber optics, laser light can be directed to parts of the body without making a large incision. More procedures may be done on an outpatient basis. Lasers can be used in two ways to treat cancer: by shrinking or destroying a tumor with heat, or by activating a chemical--known as a photosensitizing agent--that destroys cancer cells. In PDT, a photosensitizing agent is retained in cancer cells and can be stimulated by light to cause a reaction that kills cancer cells. CO.sub.2 and Nd:YAG lasers are used to shrink or destroy tumors. They may be used with endoscopes, tubes that allow physicians to see into certain areas of the body, such as the bladder. The light from some lasers can be transmitted through a flexible endoscope fitted with fiber optics. This allows physicians to see and work in parts of the body that could not otherwise be reached except by surgery and therefore allows very precise aiming of the laser beam. Lasers also may be used with low-power microscopes, giving the doctor a clear view of the site being treated. Used with other instruments, laser systems can produce a cutting area as small as 200 microns in diameter--less than the width of a very fine thread. Lasers are used to treat many types of cancer. Laser surgery is a standard treatment for certain stages of glottis (vocal cord), cervical, skin, lung, vaginal, vulvar, and penile cancers. In addition to its use to destroy the cancer, laser surgery is also used to help relieve symptoms caused by cancer (palliative care). For example, lasers may be used to shrink or destroy a tumor that is blocking a patient's trachea (windpipe), making it easier to breathe. It is also sometimes used for palliation in colorectal and anal cancer. Laser-induced interstitial thermotherapy (LITT) is one of the most recent developments in laser therapy. LITT uses the same idea as a cancer treatment called hyperthermia; that heat may help shrink tumors by damaging cells or depriving them of substances they need to live. In this treatment, lasers are directed to interstitial areas (areas between organs) in the body. The laser light then raises the temperature of the tumor, which damages or destroys cancer cells.

[0396] The duration and/or dose of treatment with therapies may vary according to the particular therapeutic agent or combination thereof. An appropriate treatment time for a particular cancer therapeutic agent will be appreciated by the skilled artisan. The present invention contemplates the continued assessment of optimal treatment schedules for each cancer therapeutic agent, where the phenotype of the cancer of the subject as determined by the methods of the present invention is a factor in determining optimal treatment doses and schedules.

[0397] Any means for the introduction of a polynucleotide into mammals, human or non-human, or cells thereof may be adapted to the practice of this invention for the delivery of the various constructs encompassed by the present invention into the intended recipient. In one embodiment encompassed by the present invention, the DNA constructs are delivered to cells by transfection, i.e., by delivery of “naked” DNA or in a complex with a colloidal dispersion system. A colloidal system includes macromolecule complexes, nanocapsules, microspheres, beads, and lipid-based systems including oil-in-water emulsions, micelles, mixed micelles, and liposomes. The preferred colloidal system of this invention is a lipid-complexed or liposome-formulated DNA. In the former approach, prior to formulation of DNA, e.g., with lipid, a plasmid containing a transgene bearing the desired DNA constructs may first be experimentally optimized for expression (e.g., inclusion of an intron in the 5′ untranslated region and elimination of unnecessary sequences (Felgner, et al., Ann NY Acad Sci 126-139, 1995). Formulation of DNA, e.g. with various lipid or liposome materials, may then be effected using known methods and materials and delivered to the recipient mammal. See, e.g., Canonico et al, Am J Respir Cell Mol Biol 10:24-29, 1994; Tsan et al, Am J Physiol 268; Alton et al., Nat Genet. 5:135-142, 1993 and U.S. Pat. No. 5,679,647 by Carson et al.

[0398] The targeting of liposomes can be classified based on anatomical and mechanistic factors. Anatomical classification is based on the level of selectivity, for example, organ-specific, cell-specific, and organelle-specific. Mechanistic targeting can be distinguished based upon whether it is passive or active. Passive targeting utilizes the natural tendency of liposomes to distribute to cells of the reticulo-endothelial system (RES) in organs, which contain sinusoidal capillaries. Active targeting, on the other hand, involves alteration of the liposome by coupling the liposome to a specific ligand such as a monoclonal antibody, sugar, glycolipid, or protein, or by changing the composition or size of the liposome in order to achieve targeting to organs and cell types other than the naturally occurring sites of localization.

[0399] The surface of the targeted delivery system may be modified in a variety of ways. In the case of a liposomal targeted delivery system, lipid groups can be incorporated into the lipid bilayer of the liposome in order to maintain the targeting ligand in stable association with the liposomal bilayer. Various linking groups can be used for joining the lipid chains to the targeting ligand. Naked DNA or DNA associated with a delivery vehicle, e.g., liposomes, can be administered to several sites in a subject (see below).

[0400] Nucleic acids can be delivered in any desired vector. These include viral or non-viral vectors, including adenovirus vectors, adeno-associated virus vectors, retrovirus vectors, lentivirus vectors, and plasmid vectors. Exemplary types of viruses include HSV (herpes simplex virus), AAV (adeno associated virus), HIV (human immunodeficiency virus), BIV (bovine immunodeficiency virus), and MLV (murine leukemia virus). Nucleic acids can be administered in any desired format that provides sufficiently efficient delivery levels, including in virus particles, in liposomes, in nanoparticles, and complexed to polymers.

[0401] The nucleic acids encoding a protein or nucleic acid of interest may be in a plasmid or viral vector, or other vector as is known in the art. Such vectors are well-known and any can be selected for a particular application. In one embodiment encompassed by the present invention, the gene delivery vehicle comprises a promoter and a demethylase coding sequence. Preferred promoters are tissue-specific promoters and promoters which are activated by cellular proliferation, such as the thymidine kinase and thymidylate synthase promoters. Other preferred promoters include promoters which are activatable by infection with a virus, such as the a- and 0-interferon promoters, and promoters which are activatable by a hormone, such as estrogen. Other promoters which can be used include the Moloney virus LTR, the CMV promoter, and the mouse albumin promoter. A promoter may be constitutive or inducible.

[0402] In another embodiment, naked polynucleotide molecules are used as gene delivery vehicles, as described in WO 90/11092 and U.S. Pat. No. 5,580,859. Such gene delivery vehicles can be either growth factor DNA or RNA and, in certain embodiments, are linked to killed adenovirus. Curiel et al., Hum. Gene. Ther. 3:147-154, 1992. Other vehicles which can optionally be used include DNA-ligand (Wu et al., J. Biol. Chem. 264:16985-16987, 1989), lipid-DNA combinations (Felgner et al., Proc. Natl. Acad. Sci. USA 84:7413 7417, 1989), liposomes (Wang et al., Proc. Natl. Acad. Sci. 84:7851-7855, 1987) and microprojectiles (Williams et al., Proc. Natl. Acad. Sci. 88:2726-2730, 1991).

[0403] A gene delivery vehicle can optionally comprise viral sequences such as a viral origin of replication or packaging signal. These viral sequences can be selected from viruses such as astrovirus, coronavirus, orthomyxovirus, papovavirus, paramyxovirus, parvovirus, picornavirus, poxvirus, retrovirus, togavirus or adenovirus. In a preferred embodiment, the growth factor gene delivery vehicle is a recombinant retroviral vector. Recombinant retroviruses and various uses thereof have been described in numerous references including, for example, Mann et al., Cell 33:153, 1983, Cane and Mulligan, Proc. Nat'l. Acad. Sci. USA 81:6349, 1984, Miller et al., Human Gene Therapy 1:5-14, 1990, U.S. Pat. Nos. 4,405,712, 4,861,719, and 4,980,289, and PCT Application Nos. WO 89/02,468, WO 89/05,349, and WO 90/02,806. Numerous retroviral gene delivery vehicles can be utilized in the present invention, including for example those described in EP 0,415,731; WO 90/07936; WO 94/03622; WO 93/25698; WO 93/25234; U.S. Pat. No. 5,219,740; WO 9311230; WO 9310218; Vile and Hart, Cancer Res. 53:3860-3864, 1993; Vile and Hart, Cancer Res. 53:962-967, 1993; Ram et al., Cancer Res. 53:83-88, 1993; Takamiya et al., J. Neurosci. Res. 33:493-503, 1992; Baba et al., J. Neurosurg. 79:729-735, 1993 (U.S. Pat. No. 4,777,127, GB 2,200,651, EP 0,345,242 and WO91/02805).

[0404] Other viral vector systems that can be used to deliver a polynucleotide encompassed by the present invention have been derived from herpes virus, e.g., Herpes Simplex Virus (U.S. Pat. No. 5,631,236 by Woo et al., issued May 20, 1997 and WO 00/08191 by Neurovex), vaccinia virus (Ridgeway (1988) Ridgeway, “Mammalian expression vectors,” In: Rodriguez R L, Denhardt D T, ed. Vectors: A survey of molecular cloning vectors and their uses. Stoneham: Butterworth; Baichwal and Sugden (1986) “Vectors for gene transfer derived from animal DNA viruses: Transient and stable expression of transferred genes,” In: Kucherlapati R, ed. Gene transfer. New York: Plenum Press; Coupar et al. (1988) Gene, 68:1-10), and several RNA viruses. Preferred viruses include an alphavirus, a poxivirus, an arena virus, a vaccinia virus, a polio virus, and the like. They offer several attractive features for various mammalian cells (Friedmann (1989) Science, 244:1275-1281; Ridgeway, 1988, supra; Baichwal and Sugden, 1986, supra; Coupar et al., 1988; Horwich et al. (1990) J. Virol., 64:642-650).

[0405] In other embodiments, target DNA in the genome can be manipulated using well-known methods in the art. For example, the target DNA in the genome can be manipulated by deletion, insertion, and/or mutation are retroviral insertion, artificial chromosome techniques, gene insertion, random insertion with tissue specific promoters, gene targeting, transposable elements and/or any other method for introducing foreign DNA or producing modified DNA/modified nuclear DNA. Other modification techniques include deleting DNA sequences from a genome and/or altering nuclear DNA sequences. Nuclear DNA sequences, for example, may be altered by site-directed mutagenesis.

[0406] In other embodiments, recombinant biomarker polypeptides, and fragments thereof, can be administered to subjects. In some embodiments, fusion proteins can be constructed and administered which have enhanced biological properties. In addition, the biomarker polypeptides, and fragment thereof, can be modified according to well-known pharmacological methods in the art (e.g., pegylation, glycosylation, oligomerization, etc.) in order to further enhance desirable biological activities, such as increased bioavailability and decreased proteolytic degradation.

VII. Clinical Efficacy

[0407] Clinical efficacy can be measured by any method known in the art. For example, the response to a cancer therapy (e.g., an agent inhibiting binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome), relates to any response of the cancer, e.g., a tumor, to the therapy, preferably to a change in tumor mass and/or volume after initiation of neoadjuvant or adjuvant chemotherapy. Tumor response may be assessed in a neoadjuvant or adjuvant situation where the size of a tumor after systemic intervention can be compared to the initial size and dimensions as measured by CT, PET, mammogram, ultrasound or palpation and the cellularity of a tumor can be estimated histologically and compared to the cellularity of a tumor biopsy taken before initiation of treatment. Response may also be assessed by caliper measurement or pathological examination of the tumor after biopsy or surgical resection. Response may be recorded in a quantitative fashion like percentage change in tumor volume or cellularity or using a semi-quantitative scoring system such as residual cancer burden (Symmans et al., *J. Clin. Oncol*. (2007) 25:4414-4422) or Miller-Payne score (Ogston et al., (2003) *Breast* (Edinburgh, Scotland) 12:320-327) in a qualitative fashion like “pathological complete response” (pCR), “clinical complete remission” (cCR), “clinical partial remission” (cPR), “clinical stable disease” (cSD), “clinical progressive disease” (cPD) or other qualitative criteria. Assessment of tumor response may be performed early after the onset of neoadjuvant or adjuvant therapy, e.g., after a few hours, days, weeks or preferably after a few months. A typical endpoint for response assessment is upon termination of neoadjuvant chemotherapy or upon surgical removal of residual tumor cells and/or the tumor bed.

[0408] In some embodiments, clinical efficacy of the therapeutic treatments described herein may be determined by measuring the clinical benefit rate (CBR). The clinical benefit rate is measured by determining the sum of the percentage of patients who are in complete remission (CR), the number of patients who are in partial remission (PR) and the number of patients having stable disease (SD) at a time point at least 6 months out from the end of therapy. The shorthand for this formula is CBR=CR+PR+SD over 6 months. In some embodiments, the CBR for a particular agent that inhibits binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome therapeutic regimen is at least 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, or more.

[0409] Additional criteria for evaluating the response to cancer therapy (e.g., an agent that inhibits binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome) are related to “survival,” which includes all of the following: survival until mortality, also known as overall survival (wherein said mortality may be either irrespective of cause or tumor related); “recurrence-free survival” (wherein the term recurrence shall include both localized and distant recurrence); metastasis free survival; disease free survival (wherein the term disease shall include cancer and diseases associated therewith). The length of said survival may be calculated by reference to a defined start point (e.g., time of diagnosis or start of treatment) and end point (e.g., death, recurrence or metastasis). In addition, criteria for efficacy of treatment can be expanded to include response to chemotherapy, probability of survival, probability of metastasis within a given time period, and probability of tumor recurrence.

[0410] For example, in order to determine appropriate threshold values, a particular agent inhibiting binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome can be administered to a population of subjects and the outcome can be correlated to biomarker measurements that were determined prior to administration of any cancer therapy (e.g., an agent inhibiting binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome). The outcome measurement may be pathologic response to therapy given in the neoadjuvant setting. Alternatively, outcome measures, such as overall survival and disease-free survival can be monitored over a period of time for subjects following cancer therapy (e.g., an agent inhibiting binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome) for whom biomarker measurement values are known. In certain embodiments, the same doses of the agent inhibiting binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome are administered to each subject. In related embodiments, the doses administered are standard doses known in the art for the agent inhibiting binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome. The period of time for which subjects are monitored can vary. For example, subjects may be monitored for at least 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 25, 30, 35, 40, 45, 50, 55, or 60 months. Biomarker measurement threshold values that correlate to outcome of a cancer therapy (e.g., inhibiting binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome) can be determined using methods such as those described in the Examples section.

VIII. Further Uses and Methods of the Present Invention

[0411] The compositions described herein can be used in a variety of diagnostic, prognostic, and therapeutic applications. In any method described herein, such as a diagnostic method, prognostic method, therapeutic method, or combination thereof, all steps of the method can be performed by a single actor or, alternatively, by more than one actor. For example, diagnosis can be performed directly by the actor providing therapeutic treatment. Alternatively, a person providing a therapeutic agent can request that a diagnostic assay be performed. The diagnostician and/or the therapeutic interventionist can interpret the diagnostic assay results to determine a therapeutic strategy. Similarly, such alternative processes can apply to other assays, such as prognostic assays.

a. Screening Methods

[0412] One aspect of the present invention relates to screening assays, including non-cell based assays and xenograft animal model assays. In one embodiment, the assays provide a method for identifying whether a cancer is likely to respond to cancer therapy (e.g., an agent inhibiting binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome), such as in a human by using a xenograft animal model assay, and/or whether an agent can inhibit the growth of or kill a cancer cell that is unlikely to respond to cancer therapy (e.g., an agent inhibiting binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome).

[0413] In one embodiment, an assay is a cell-based assay, comprising contacting a synovial sarcoma cancer cell with a test agent, and determining the ability of the test agent to decrease (1) binding of a SS18-SSX fusion protein to a H2A K119Ub nucleosome; (2) recruitment of a SS18-SSX fusion protein-bound BAF complex to a H2A K119Ub nucleosome; and/or (3) expression of at least one a SS18-SSX fusion protein target gene.

[0414] In another embodiment, an assay is a cell-free assay, comprising a) mixing a protein comprising a c-terminal basic region and a c-terminal acidic region of a SSX protein, and a H2A K119Ub nucleosome together; b) adding a test agent to the mixture; and c) determining the ability of the test agent to decrease binding of the protein to the H2A K119Ub nucleosome, and/or recruitment of the BAF complex to the H2A K119Ub nucleosome.

[0415] For example, in a direct binding assay, one protein (or their respective target polypeptides or molecules) can be coupled with a radioisotope or enzymatic label such that binding can be determined by detecting the labeled protein or molecule in a complex. For example, the targets can be labeled with .sup.125I, .sup.35S, .sup.14C, or .sup.3H, either directly or indirectly, and the radioisotope detected by direct counting of radioemmission or by scintillation counting. Alternatively, the targets can be enzymatically labeled with, for example, horseradish peroxidase, alkaline phosphatase, or luciferase, and the enzymatic label detected by determination of conversion of an appropriate substrate to product.

[0416] Determining the interaction between two molecules (e.g., a nucleosome and a SS18-SSX fusion protein) can be accomplished using standard binding or enzymatic analysis assays. These assays may include thermal shift assays (measure of variation of the melting temperature of the protein alone and in the presence of a molecule) (R. Zhang, F. Monsma, Curr. Opin. Drug Discov. Devel., 13 (4) (2010), pp. 389-402), SPR (surface plasmon resonance) (T. Neumann, et al. Curr. Top Med. Chem., 7 (16) (2007), pp. 1630-1642), FRET/BRET (Fluorescence or Bioluminescence Resonance Excitation Transfer) (A. L. Mattheyses, A. I. Marcus, Methods Mol. Biol., 1278 (2015), pp. 329-339; J. Bacart, et al. Biotechnol. J., 3 (3) (2008), pp. 311-324), Elisa (Enzyme-linked immunosorbent assay) (Z. Weng, Q. Zhao, Methods Mol. Biol., 1278 (2015), pp. 341-352), fluorescence polarization (Y. Du, Methods Mol. Biol., 1278 (2015), pp. 529-544), and Far western (U. Mahlknecht, O. G. Ottmann, D. Hoelzer J. Biotechnol., 88 (2) (2001), pp. 89-94) or other techniques. More sophisticated (and lower throughput) biophysical methods that provide structural or thermodynamic details of the molecule binding mode (using isothermal calorimetry (ITC), Nuclear Magnetic Resonance (NMR), and X-ray crystallography) may also be needed for further validation and characterization of potential hits.

[0417] Alternatively, high throughput cellular screens measuring the loss of interaction using reverse two hybrid or BRET may be used and offer the advantage of selecting only cell penetrable molecules (A. R. Horswill, S. N. Savinov, S. J. Benkovic Proc. Natl. Acad. Sci. USA, 101 (44) (2004), pp. 15591-15596; A. Hamdi, P. Colas Trends Pharmacol. Sci., 33 (2) (2012), pp. 109-118). The latter approaches require further validation to assess the “on target” effect. In one or more embodiments of the above described assay methods, it may be desirable to immobilize polypeptides or molecules to facilitate separation of complexed from uncomplexed forms of one or both of the proteins or molecules, as well as to accommodate automation of the assay.

[0418] Binding of a test agent to a target can be accomplished in any vessel suitable for containing the reactants. Non-limiting examples of such vessels include microtiter plates, test tubes, and micro-centrifuge tubes. Immobilized forms of the antibodies of the present invention can also include antibodies bound to a solid phase like a porous, microporous (with an average pore diameter less than about one micron) or macroporous (with an average pore diameter of more than about 10 microns) material, such as a membrane, cellulose, nitrocellulose, or glass fibers; a bead, such as that made of agarose or polyacrylamide or latex; or a surface of a dish, plate, or well, such as one made of polystyrene.

[0419] In an alternative embodiment, determining the ability of the agent to inhibit binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosomecan be accomplished by determining the ability of the test agent to modulate the activity of a polypeptide or other product that functions downstream or upstream of its position within the pathway. For example, it can be accomplished by measuring the activity of the downstream target genes of SS18-SSX fusion protein.

[0420] The present invention further pertains to novel agents identified by the above-described screening assays. Accordingly, it is within the scope of this invention to further use an agent identified as described herein in an appropriate animal model. For example, an agent identified as described herein can be used in an animal model to determine the efficacy, toxicity, or side effects of treatment with such an agent. Alternatively, an antibody identified as described herein can be used in an animal model to determine the mechanism of action of such an agent.

b. Predictive Medicine

[0421] The present invention also pertains to the field of predictive medicine in which diagnostic assays, prognostic assays, and monitoring clinical trials are used for prognostic (predictive) purposes to thereby treat an individual prophylactically. Accordingly, one aspect of the present invention relates to diagnostic assays for determining the amount and/or activity level of a biomarker described herein in the context of a biological sample (e.g., blood, serum, cells, or tissue) to thereby determine whether an individual afflicted with a cancer is likely to respond to an agent inhibiting binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome, such as in a cancer. Such assays can be used for prognostic or predictive purpose to thereby prophylactically treat an individual prior to the onset or after recurrence of a disorder characterized by or associated with biomarker polypeptide, nucleic acid expression or activity. The skilled artisan will appreciate that any method can use one or more (e.g., combinations) of biomarkers described herein, such as those in the tables, figures, examples, and otherwise described in the specification.

[0422] Another aspect of the present invention pertains to monitoring the influence of agents (e.g., drugs, compounds, and small nucleic acid-based molecules) on the expression or activity of a biomarker described herein. These and other agents are described in further detail in the following sections.

[0423] The skilled artisan will also appreciate that, in certain embodiments, the methods of the present invention implement a computer program and computer system. For example, a computer program can be used to perform the algorithms described herein. A computer system can also store and manipulate data generated by the methods of the present invention which comprises a plurality of biomarker signal changes/profiles which can be used by a computer system in implementing the methods of this invention. In certain embodiments, a computer system receives biomarker expression data; (ii) stores the data; and (iii) compares the data in any number of ways described herein (e.g., analysis relative to appropriate controls) to determine the state of informative biomarkers from cancerous or pre-cancerous tissue. In other embodiments, a computer system (i) compares the determined expression biomarker level to a threshold value; and (ii) outputs an indication of whether said biomarker level is significantly modulated (e.g., above or below) the threshold value, or a phenotype based on said indication.

[0424] In certain embodiments, such computer systems are also considered part of the present invention. Numerous types of computer systems can be used to implement the analytic methods of this invention according to knowledge possessed by a skilled artisan in the bioinformatics and/or computer arts. Several software components can be loaded into memory during operation of such a computer system. The software components can comprise both software components that are standard in the art and components that are special to the present invention (e.g., dCHIP software described in Lin et al. (2004) Bioinformatics 20, 1233-1240; radial basis machine learning algorithms (RBM) known in the art).

[0425] The methods encompassed by the present invention can also be programmed or modeled in mathematical software packages that allow symbolic entry of equations and high-level specification of processing, including specific algorithms to be used, thereby freeing a user of the need to procedurally program individual equations and algorithms. Such packages include, e.g., Matlab from Mathworks (Natick, Mass.), Mathematica from Wolfram Research (Champaign, Ill.) or S-Plus from MathSoft (Seattle, Wash.).

[0426] In certain embodiments, the computer comprises a database for storage of biomarker data. Such stored profiles can be accessed and used to perform comparisons of interest at a later point in time. For example, biomarker expression profiles of a sample derived from the non-cancerous tissue of a subject and/or profiles generated from population-based distributions of informative loci of interest in relevant populations of the same species can be stored and later compared to that of a sample derived from the cancerous tissue of the subject or tissue suspected of being cancerous of the subject.

[0427] In addition to the exemplary program structures and computer systems described herein, other, alternative program structures and computer systems will be readily apparent to the skilled artisan. Such alternative systems, which do not depart from the above described computer system and programs structures either in spirit or in scope, are therefore intended to be comprehended within the accompanying claims.

c. Diagnostic Assays

[0428] The present invention provides, in part, methods, systems, and code for accurately classifying whether a biological sample is associated with a cancer that is likely to respond to cancer therapy (e.g., an agent that inhibits binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome). In some embodiments, the present invention is useful for classifying a sample (e.g., from a subject) as associated with or at risk for responding to or not responding to cancer therapy (e.g., an agent that inhibits binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome) using a statistical algorithm and/or empirical data (e.g., the amount or activity of a biomarker described herein, such as in the tables, figures, examples, and otherwise described in the specification).

[0429] An exemplary method for detecting the amount or activity of a biomarker described herein, and thus useful for classifying whether a sample is likely or unlikely to respond to cancer therapy (e.g., an agent that inhibits binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome) involves obtaining a biological sample from a test subject and contacting the biological sample with an agent, such as a protein-binding agent like an antibody or antigen-binding fragment thereof, or a nucleic acid-binding agent like an oligonucleotide, capable of detecting the amount or activity of the biomarker in the biological sample. In some embodiments, at least one antibody or antigen-binding fragment thereof is used, wherein two, three, four, five, six, seven, eight, nine, ten, or more such antibodies or antibody fragments can be used in combination (e.g., in sandwich ELISAs) or in serial. In certain instances, the statistical algorithm is a single learning statistical classifier system. For example, a single learning statistical classifier system can be used to classify a sample as a based upon a prediction or probability value and the presence or level of the biomarker. The use of a single learning statistical classifier system typically classifies the sample as, for example, a likely cancer therapy (e.g., an agent that inhibits binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome) responder or progressor sample with a sensitivity, specificity, positive predictive value, negative predictive value, and/or overall accuracy of at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%.

[0430] Other suitable statistical algorithms are well-known to those of skill in the art. For example, learning statistical classifier systems include a machine learning algorithmic technique capable of adapting to complex data sets (e.g., panel of markers of interest) and making decisions based upon such data sets. In some embodiments, a single learning statistical classifier system such as a classification tree (e.g., random forest) is used. In other embodiments, a combination of 2, 3, 4, 5, 6, 7, 8, 9, 10, or more learning statistical classifier systems are used, preferably in tandem. Examples of learning statistical classifier systems include, but are not limited to, those using inductive learning (e.g., decision/classification trees such as random forests, classification and regression trees (C&RT), boosted trees, etc.), Probably Approximately Correct (PAC) learning, connectionist learning (e.g., neural networks (NN), artificial neural networks (ANN), neuro fuzzy networks (NFN), network structures, perceptrons such as multi-layer perceptrons, multi-layer feed-forward networks, applications of neural networks, Bayesian learning in belief networks, etc.), reinforcement learning (e.g., passive learning in a known environment such as naive learning, adaptive dynamic learning, and temporal difference learning, passive learning in an unknown environment, active learning in an unknown environment, learning action-value functions, applications of reinforcement learning, etc.), and genetic algorithms and evolutionary programming. Other learning statistical classifier systems include support vector machines (e.g., Kernel methods), multivariate adaptive regression splines (MARS), Levenberg-Marquardt algorithms, Gauss-Newton algorithms, mixtures of Gaussians, gradient descent algorithms, and learning vector quantization (LVQ). In certain embodiments, the method of the present invention further comprises sending the sample classification results to a clinician, e.g., an oncologist.

[0431] In another embodiment, the diagnosis of a subject is followed by administering to the individual a therapeutically effective amount of a defined treatment based upon the diagnosis.

[0432] In one embodiment, the methods further involve obtaining a control biological sample (e.g., biological sample from a subject who does not have a cancer or whose cancer is susceptible to cancer therapy (e.g., an agent that inhibits binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome), a biological sample from the subject during remission, or a biological sample from the subject during treatment for developing a cancer progressing despite cancer therapy (e.g., an agent that inhibits binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome).

d. Prognostic Assays

[0433] The diagnostic methods described herein can furthermore be utilized to identify subjects having or at risk of developing a cancer that is likely or unlikely to be responsive to cancer therapy (e.g., an agent that inhibits binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome). The assays described herein, such as the preceding diagnostic assays or the following assays, can be utilized to identify a subject having or at risk of developing a disorder associated with a misregulation of the amount or activity of at least one biomarker described in, such as in cancer. Alternatively, the prognostic assays can be utilized to identify a subject having or at risk for developing a disorder associated with a misregulation of the at least one biomarker described herein, such as in cancer. Furthermore, the prognostic assays described herein can be used to determine whether a subject can be administered an agent (e.g., an agonist, antagonist, peptidomimetic, polypeptide, peptide, nucleic acid, small molecule, or other drug candidate) to treat a disease or disorder associated with the aberrant biomarker expression or activity.

e. Treatment Methods

[0434] The therapeutic compositions described herein, such as the agent that inhibits binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome, can be used in a variety of in vitro and in vivo therapeutic applications using the formulations and/or combinations described herein. In one embodiment, the therapeutic agents can be used to treat cancers determined to be responsive thereto. For example, single or multiple agents that inhibit binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome can be used to treat cancers in subjects identified as likely responders thereto.

[0435] Treatment methods of the present invention involve contacting a cell, such as a cancer cell with an agent that inhibits binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome. An agent that inhibits binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome can be an agent as described herein, such as a small molecule, a nucleic acid, a polypeptide, an antibody, or a peptidomimetic. In one embodiment, the agent binds to H2A K119Ub-marked nucleosomes or the SS18-SSX fusion protein at the interaction interface between the H2A K119Ub-marked nucleosomes and the SS18-SSX fusion protein, thereby blocking or competing with the H2A K119Ub-marked nucleosomes and the SS18-SSX fusion protein interaction formation. For example, the agent may bind to the basic region (e.g., the RLR motif) and/or the acidic region of the SS18-SSX fusion protein. The agent may bind to the acidic patch or the H2A K119Ub mark of the H2A K119Ub-marked neucleosomes. In another embodiment, the agent binds to another site of the H2A K119Ub-marked nucleosomes or the the SS18-SSX fusion protein and capable of inducing a conformational change leading to a loss of interaction with the targeted partner. In yet another embodiment, the agent inhibits the function or activity of a domain or a site of the H2A K119Ub-marked nucleosomes or the SS18-SSX fusion protein that is necessary for the H2A K119Ub-marked nucleosomes and the SS18-SSX fusion protein interaction formation. In still another embodiment, the agent inhibits the H2A ubiquitination of neucleosomes, induces deletion or mutation of the acidic patch of the H2A K119Ub-marked nucleosomes, and/or induces deletion or mutation of the basic region (e.g., RLR motif) of the SS18-SSX fusion protein itself, thus breaking the H2A K119Ub-marked nucleosomes and the SS18-SSX fusion protein interaction. In one embodiment, the agent inhibits ubiquitin ligase activity of a PRC1 complex. For example, the agent may reduces expression, copy number, and/or ubiquitin ligase activity of RING1A and/or RING1B. In another embodiment, the agent is a CRISPR/Cas9 reagent that targets the critical residues on the SS18-SSX fusion protein or the H2A K119Ub-marked nucleosomes important for the SS18-SSX fusion protein and the H2A K119Ub-marked nucleosomes interaction, which include but are no tlimited to the critical residues identified in the examples herein.

[0436] These treatment methods can be performed in vitro (e.g., by contacting the cell with the agent) or, alternatively, by contacting an agent with cells in vivo (e.g., by administering the agent to a subject). As such, the present invention provides methods useful for treating an individual afflicted with a condition that would benefit from a decreased activity of SS18-SSX target genes by inhibiting binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome, such as a cancer like synovial sarcoma. In one embodiment, the method involves administering an agent (e.g., an agent identified by a screening assay described herein), or combination of agents that inhibit SS18-SSX target genes expression or activity.

[0437] In addition, these inhibitory agents can also be administered in combination therapy with, e.g., chemotherapeutic agents, hormones, antiangiogens, radiolabelled, compounds, or with surgery, cryotherapy, and/or radiotherapy. The preceding treatment methods can be administered in conjunction with other forms of conventional therapy (e.g., standard-of-care treatments for cancer well-known to the skilled artisan), either consecutively with, pre- or post-conventional therapy. For example, these modulatory agents can be administered with a therapeutically effective dose of chemotherapeutic agent. In another embodiment, these modulatory agents are administered in conjunction with chemotherapy to enhance the activity and efficacy of the chemotherapeutic agent. The Physicians' Desk Reference (PDR) discloses dosages of chemotherapeutic agents that have been used in the treatment of various cancers. The dosing regimen and dosages of these aforementioned chemotherapeutic drugs that are therapeutically effective will depend on the particular melanoma, being treated, the extent of the disease and other factors familiar to the physician of skill in the art and can be determined by the physician.

IX. Isolated Modified Protein Complexes

[0438] The present invention relates, in part, to an isolated modified protein complex selected from the group consisting of protein complexes listed in Table 3, wherein the isolated modified protein complex comprises at least one subunit that is modified.

[0439] In certain embodiments, at least one subunit of a complex encompassed by the present invention is a homolog, a derivative, e.g., a functionally active derivative, a fragment, e.g., a functionally active fragment, of a protein subunit of a complex encompassed by the present invention. In certain embodiments encompassed by the present invention, a homolog, derivative or fragment of a protein subunit of a complex encompassed by the present invention is still capable of forming a complex with the other subunit(s). Complex-formation can be tested by any method known to the skilled artisan. Such methods include, but are not limited to, non-denaturing PAGE, FRET, and Fluorescence Polarization Assay.

[0440] Homologs (e.g., nucleic acids encoding subunit proteins from other species) or other related sequences (e.g., paralogs) which are members of a native cellular protein complex can be identified and obtained by low, moderate or high stringency hybridization with all or a portion of the particular nucleic acid sequence as a probe, using methods well known in the art for nucleic acid hybridization and cloning.

[0441] Exemplary moderately stringent hybridization conditions are as follows: prehybridization of filters containing DNA is carried out for 8 hours to overnight at 65° C. in buffer composed of 6×SSC, 50 mM Tris-HCl (pH 7.5), 1 mM EDTA, 0.02% PVP, 0.02% Ficoll, 0.02% BSA, and 500 μg/ml denatured salmon sperm DNA. Filters are hybridized for 48 hours at 65° C. in prehybridization mixture containing 100 μg/ml denatured salmon sperm DNA and 5-20×10.sup.6 cpm of .sup.32P-labeled probe. Washing of filters is done at 37° C. for 1 hour in a solution containing 2×SSC, 0.01% PVP, 0.01% Ficoll, and 0.01% BSA. This is followed by a wash in 0.1×SSC at 50° C. for 45 min before autoradiography. Alternatively, exemplary conditions of high stringency are as follows: e.g., hybridization to filter-bound DNA in 0.5 μM NaHPO.sub.4, 7% sodium dodecyl sulfate (SDS), 1 mM EDTA at 65° C., and washing in 0.1×SSC/0.1% SDS at 68° C. (Ausubel et al., eds., 1989, *Current Protocols in Molecular Biology*, Vol. I, Green Publishing Associates, Inc., and John Wiley & sons, Inc., New York, at p.2.10.3). Other conditions of high stringency which may be used are well known in the art. Exemplary low stringency hybridization conditions comprise hybridization in a buffer comprising 35% formamide, 5×SSC, 50 mM Tris-HCl (pH 7.5), 5 mM EDTA, 0.02% PVP, 0.02% Ficoll, 0.2% BSA, 100 μg/ml denatured salmon sperm DNA, and 1 0% (wt/vol) dextran sulfate for 18-20 hours at 40° C., washing in a buffer consisting of 2×SSC, 25 mM Tris-HCl (pH 7.4), 5 mM EDTA, and 0.1% SDS for 1.5 hours at 55° C., and washing in a buffer consisting of 2×SSC, 25 mM Tris-HCl (pH 7.4), 5 mM EDTA, and 0.1% SDS for 1.5 hours at 60° C.

[0442] In certain embodiments, a homolog of a subunit binds to the same proteins to which the subunit binds. In certain, more specific embodiments, a homolog of a subunit binds to the same proteins to which the subunit binds wherein the binding affinity between the homolog and the binding partner of the subunit is at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95% or at least 98% of the binding affinity between the subunit and the binding partner. Binding affinities between proteins can be determined by any method known to the skilled artisan.

[0443] In certain embodiments, a fragment of a protein subunit of the complex consists of at least 6 (continuous) amino acids, of at least 10, at least 20 amino acids, at least 30 amino acids, at least 40 amino acids, at least 50 amino acids, at least 75 amino acids, at least 100 amino acids, at least 150 amino acids, at least 200 amino acids, at least 250 amino acids, at least 300 amino acids, at least 400 amino acids, or at least 500 amino acids of the protein subunit of the naturally occurring protein complex. In specific embodiments. Such fragments are not larger than 40 amino acids, 50 amino acids, 75 amino acids, 100 amino acids, 150 amino acids, 200 amino acids, 250 amino acids, 300 amino acids, 400 amino acids, or than 500 amino acids. In more specific embodiments, the functional fragment is capable of forming a complex encompassed by the present invention, i.e., the fragment can still bind to at least one other protein subunit to form a complex encompassed by the present invention. In one embodiment, the fragment of the subunit comprises the basic region and/or the acidic region of a SSX protein. In another embodiment, the fragment of the subunit comprises c-terminal 34 amino acids (aa155-188) of a SSX protein. In still another embodiment, the fragment of the subunit comprises c-terminal 78 amino acids (aa 111-188) of a SSX protein. The SSX protein may be selected form the group comsisting of human SSX1, SSX2, SSX3, SSX4, SSX6, SSX7, SSX8, and SSX9. In yet another embodiment, the fragment of the subunit comprises the acidic patch of a nucleosome and/or the H2A K119 Ub mark.

[0444] Derivatives or analogs of subunit proteins include, but are not limited, to molecules comprising regions that are substantially homologous to the subunit proteins, in various embodiments, by at least 30%, 40%, 50%, 60%, 70%, 80%, 90% or 95% identity over an amino acid sequence of identical size or when compared to an aligned sequence in which the alignment is done by a computer homology program known in the art, or whose encoding nucleic acid is capable of hybridizing to a sequence encoding the subunit protein under stringent, moderately stringent, or nonstringent conditions.

[0445] Derivatives of a protein subunit include, but are not limited to, fusion proteins of a protein subunit of a complex encompassed by the present invention to a heterologous amino acid sequence, mutant forms of a protein subunit of a complex encompassed by the present invention, and chemically modified forms of a protein subunit of a complex encompassed by the present invention. In a specific embodiment, the functional derivative of a protein subunit of a complex encompassed by the present invention is capable of forming a complex encompassed by the present invention, i.e., the derivative can still bind to at least one other protein subunit to form a complex encompassed by the present invention.

[0446] In certain embodiments encompassed by the present invention, at least two subunits of a complex encompassed by the present invention are linked to each other via at least one covalent bond. A covalent bond between subunits of a complex encompassed by the present invention increases the stability of the complex encompassed by the present invention because it prevents the dissociation of the subunits. Any method known to the skilled artisan can be used to achieve a covalent bond between at least two subunits encompassed by the present invention.

[0447] In specific embodiments, covalent cross-links are introduced between adjacent subunits. Such cross-links can be between the side chains of amino acids at opposing sides of the dimer interface. Any functional groups of amino acid residues at the dimer interface in combination with suitable cross-linking agents can be used to create covalent bonds between the protein subunits at the dimer interface. Existing amino acids at the dimer interface can be used or, alternatively, suitable amino acids can be introduced by site-directed mutagenesis.

[0448] In exemplary embodiments, cysteine residues at opposing sides of the dimer interface are oxidized to form disulfide bonds. See, e.g., Reznik et al., (1996) Nat Bio Technol 14:1007-1011, at page 1008. 1,3-dibromoacetone can also be used to create an irreversible covalent bond between two sulfhydryl groups at the dimer interface. In certain other embodiments, lysine residues at the dimer inter face are used to create a covalent bond between the protein subunits of the complex. Crosslinkers that can be used to create covalent bonds between the epsilon amino groups of lysine residues are, e.g., but are not limited to, bis(sulfosuccinimidyl)suberate; dimethyladipimidate-2HD1; disuccinimidyl glutarate; N-hydroxysuccinimidyl 2,3-dibromoproprionate.

[0449] In other specific embodiments, two or more interacting subunits, or homologues, derivatives or fragments thereof, are directly fused together, or covalently linked together through a peptide linker, forming a hybrid protein

having a single unbranched polypeptide chain. Thus, the protein complex may be formed by "intramolecular interactions between two portions of the hybrid protein. In still another embodiment, at least one of the fused or linked interacting subunit in this protein complex is a homologue, derivative or fragment of a native protein.
[0450] In specific embodiments, at least one subunit, or a homologue, derivative or fragment thereof, may be expressed as fusion or chimeric protein comprising the subunit, homologue, derivative or fragment, joined via a peptide bond to a heterologous amino acid sequence.
[0451] As used herein, a "chimeric protein" or "fusion protein" comprises all or part (preferably a biologically active part) of a polypeptide corresponding to a subunit or a fragment, homologue or derivative thereof, operably linked to a heterologous polypeptide (i.e., a polypeptide other than the polypeptide corresponding to the subunit or a fragment, homologue or derivative thereof). Within the fusion protein, the term "operably linked" is intended to indicate that the polypeptide encompassed by the present invention and the heterologous polypeptide are fused in-frame to each other. The heterologous polypeptide can be fused to the amino-terminus or the carboxyl-terminus of the polypeptide encompassed by the present invention.
[0452] In one embodiment, the heterologous amino acid sequence comprises an affinity tag that can be used for affinity purification. In another embodiment, the heterologous amino acid sequence includes a fluorescent label. In still another embodiment, the fusion protein contains a heterologous signal sequence, immunoglobulin fusion protein, toxin, or other useful protein sequences.
[0453] A variety of peptide tags known in the art may be used to generate fusion proteins of the protein subunits of a complex encompassed by the present invention, such as but not limited to the immunoglobulin constant regions, polyhistidine sequence (Petty, 1996, Metal-chelate affinity chromatography, in Current Protocols in Molecular Biology, Vol. 2, Ed. Ausubel et al., Greene Publish. Assoc. & Wiley Interscience), glutathione S-transferase (GST: Smith, 1993, *Methods Mol. Cell Bio.* 4:220-229), the E. coli maltose binding protein (Guanetal., 1987, *Gene* 67:21-30), and various cellulose binding domains (U. S. Pat. Nos. 5,496,934: 5,202,247; 5,137,819; Tomme et al., 1994, *Protein Eng.* 7:117-123), etc.
[0454] One possible peptide tags are short amino acid sequences to which monoclonal antibodies are available, such as but not limited to the following well known examples, the FLAG epitope, the myc epitope at amino acids 408-439, the influenza virus hemaglutinin (HA) epitope. Other peptide tags are recognized by specific binding partners and thus facilitate isolation by affinity binding to the binding partner, which is preferably immobilized and/or on a solid support. As will be appreciated by those skilled in the art, many methods can be used to obtain the coding region of the above-mentioned peptide tags, including but not limited to, DNA cloning, DNA amplification, and synthetic methods. Some of the peptide tags and reagents for their detection and isolation are available commercially.
[0455] In certain embodiments, a combination of different peptide tags is used for the purification of the protein subunits of a complex encompassed by the present invention or for the purification of a complex. In certain embodiments, at least one subunit has at least two peptide tags, e.g., a FLAG tag and a His tag. The different tags can be fused together or can be fused in different positions to the protein subunit. In the purification procedure, the different peptide tags are used subsequently or concurrently for purification. In certain embodiments, at least two different subunits are fused to a peptide tag, wherein the peptide tags of the two subunits can be identical or different. Using different tagged subunits for the purification of the complex ensures that only complex will be purified and minimizes the amount of uncomplexed protein subunits, such as monomers or homodimers.
[0456] Various leader sequences known in the art can be used for the efficient secretion of a protein subunit of a complex encompassed by the present invention from bacterial and mammalian cells (von Heijne, 1985, J. Mol. Biol. 184:99-105). Leader peptides are selected based on the intended host cell, and may include bacterial, yeast, viral, animal, and mammalian sequences. For example, the herpes virus glycoprotein D leader peptide is suitable for use in a variety of mammalian cells. A preferred leader peptide for use in mammalian cells can be obtained from the V-J2-C region of the mouse immunoglobulin kappa chain (Bernard et al., 1981. *Proc. Natl. Acad. Sci.* 78:5812-5816).
[0457] DNA sequences encoding desired peptide tag or leader peptide which are known or readily available from libraries or commercial suppliers are suitable in the practice of this invention.
[0458] In certain embodiments, the protein subunits of a complex encompassed by the present invention are derived from the same species. In more specific embodiments, the protein subunits are all derived from human. In another specific embodiment, the protein subunits are all derived from a mammal.
[0459] In certain other embodiments, the protein subunits of a complex encompassed by the present invention are derived from a non-human species, such as, but not limited to, cow, pig, horse, cat, dog, rat, mouse, a primate (e.g., a chimpanzee, a monkey Such as a cynomolgous monkey). In certain embodiments, one or more subunits are derived from human and the other subunits are derived from a mammal other than a human to give rise to chimeric complexes.
[0460] Included within the scope encompassed by the present invention is an isolated modified protein complex in which the subunits, or homologs, derivatives, or fragments thereof, are differentially modified during or after translation, e.g., by glycosylation, acetylation, phosphorylation, amidation, derivatization by known protecting/blocking groups, proteolytic cleavage, linkage to an antibody molecule or other cellular ligand, etc. Any of

numerous chemical modifications may be carried out by known techniques, including but not limited to specific chemical cleavage by cyanogen bromide, trypsin, chymotrypsin, papain, V8 protease, NaBH4, acetylation, formylation, oxidation, reduction, metabolic synthesis in the presence of tunicamycin, etc. In still another embodiment, the protein sequences are modified to have a heterofunctional reagent; such heterofunctional reagents can be used to crosslink the members of the complex.

[0461] The protein complexes encompassed by the present invention can also be in a modified form. For example, an antibody selectively immunoreactive with the protein complex can be bound to the protein complex. In another example, a non-antibody modulator capable of enhancing the interaction between the interacting partners in the protein complex may be included.

[0462] The above-described protein complexes may further include any additional components, e.g., other proteins, nucleic acids, lipid molecules, monosaccharides or polysaccharides, ions, etc.

TABLE-US-00012 TABLE 3 Protein complex Subunits of the protein complex SS18-SSX Subunit_1: SMARCC1 or SMARCC2 BAF-NCP complex Subunit_2: SMARCC1 or SMARCC2 Subunit_3: SMARCD1, SMARCD2, or SMARCD3 Subunit_4: SS18-SSX fusion protein Subunit_5: SMARCE1 Subunit_6: ARID1A or ARID1B Subunit_7: DPF1, DPF2, or DPF3 Subunit_8: ACTL6A Subunit_9: β-Actin Subunit_10: BCL7A, BCL7B, or BCL7C Subunit_11: SMARCA2 or SMARCA4 Subunit_12: H2A (with K119 Ub) Subunit_13: H2B Subunit_14: H3 Subunit_15: H4

X. Pharmaceutical Compositions

[0463] In another aspect, the present invention provides pharmaceutically acceptable compositions which comprise a therapeutically-effective amount of an agent that inhibits binding of a SS18-SSX fusion protein to an H2A K119Ub-marked nucleosome, formulated together with one or more pharmaceutically acceptable carriers (additives) and/or diluents.

[0464] As described in detail below, the pharmaceutical compositions of the present invention may be specially formulated for administration in solid or liquid form, including those adapted for the following: (1) oral administration, for example, drenches (aqueous or non-aqueous solutions or suspensions), tablets, boluses, powders, granules, pastes; (2) parenteral administration, for example, by subcutaneous, intramuscular or intravenous injection as, for example, a sterile solution or suspension; (3) topical application, for example, as a cream, ointment or spray applied to the skin; (4) intravaginally or intrarectally, for example, as a pessary, cream or foam; or (5) aerosol, for example, as an aqueous aerosol, liposomal preparation or solid particles containing the compound.

[0465] The phrase “therapeutically-effective amount” as used herein means that amount of an agent that modulates (e.g., inhibits) biomarker expression and/or activity which is effective for producing some desired therapeutic effect, e.g., cancer treatment, at a reasonable benefit/risk ratio.

[0466] The phrase “pharmaceutically acceptable” is employed herein to refer to those agents, materials, compositions, and/or dosage forms which are, within the scope of sound medical judgment, suitable for use in contact with the tissues of human beings and animals without excessive toxicity, irritation, allergic response, or other problem or complication, commensurate with a reasonable benefit/risk ratio.

[0467] The phrase “pharmaceutically-acceptable carrier” as used herein means a pharmaceutically-acceptable material, composition or vehicle, such as a liquid or solid filler, diluent, excipient, solvent or encapsulating material, involved in carrying or transporting the subject chemical from one organ, or portion of the body, to another organ, or portion of the body. Each carrier must be “acceptable” in the sense of being compatible with the other ingredients of the formulation and not injurious to the subject. Some examples of materials which can serve as pharmaceutically-acceptable carriers include: (1) sugars, such as lactose, glucose and sucrose; (2) starches, such as corn starch and potato starch; (3) cellulose, and its derivatives, such as sodium carboxymethyl cellulose, ethyl cellulose and cellulose acetate; (4) powdered tragacanth; (5) malt; (6) gelatin; (7) talc; (8) excipients, such as cocoa butter and suppository waxes; (9) oils, such as peanut oil, cottonseed oil, safflower oil, sesame oil, olive oil, corn oil and soybean oil; (10) glycols, such as propylene glycol; (11) polyols, such as glycerin, sorbitol, mannitol and polyethylene glycol; (12) esters, such as ethyl oleate and ethyl laurate; (13) agar; (14) buffering agents, such as magnesium hydroxide and aluminum hydroxide; (15) alginic acid; (16) pyrogen-free water; (17) isotonic saline; (18) Ringer's solution; (19) ethyl alcohol; (20) phosphate buffer solutions; and (21) other non-toxic compatible substances employed in pharmaceutical formulations.

[0468] The term “pharmaceutically-acceptable salts” refers to the relatively non-toxic, inorganic and organic acid addition salts of the agents that modulates (e.g., inhibits) biomarker expression and/or activity. These salts can be prepared in situ during the final isolation and purification of the respiration uncoupling agents, or by separately reacting a purified respiration uncoupling agent in its free base form with a suitable organic or inorganic acid, and isolating the salt thus formed. Representative salts include the hydrobromide, hydrochloride, sulfate, bisulfate, phosphate, nitrate, acetate, valerate, oleate, palmitate, stearate, laurate, benzoate, lactate, phosphate, tosylate, citrate, maleate, fumarate, succinate, tartrate, napthylate, mesylate, glucoheptonate, lactobionate, and laurylsulphonate salts and the like (See, for example, Berge et al. (1977) “Pharmaceutical Salts”, *J. Pharm. Sci.* 66:1-19).

[0469] In other cases, the agents useful in the methods of the present invention may contain one or more acidic functional groups and, thus, are capable of forming pharmaceutically-acceptable salts with pharmaceutically-acceptable bases. The term "pharmaceutically-acceptable salts" in these instances refers to the relatively non-toxic, inorganic and organic base addition salts of agents that modulates (e.g., inhibits) biomarker expression. These salts can likewise be prepared in situ during the final isolation and purification of the respiration uncoupling agents, or by separately reacting the purified respiration uncoupling agent in its free acid form with a suitable base, such as the hydroxide, carbonate or bicarbonate of a pharmaceutically-acceptable metal cation, with ammonia, or with a pharmaceutically-acceptable organic primary, secondary or tertiary amine. Representative alkali or alkaline earth salts include the lithium, sodium, potassium, calcium, magnesium, and aluminum salts and the like. Representative organic amines useful for the formation of base addition salts include ethylamine, diethylamine, ethylenediamine, ethanolamine, diethanolamine, piperazine and the like (see, for example, Berge et al., supra).

[0470] Wetting agents, emulsifiers and lubricants, such as sodium lauryl sulfate and magnesium stearate, as well as coloring agents, release agents, coating agents, sweetening, flavoring and perfuming agents, preservatives and antioxidants can also be present in the compositions.

[0471] Examples of pharmaceutically-acceptable antioxidants include: (1) water soluble antioxidants, such as ascorbic acid, cysteine hydrochloride, sodium bisulfate, sodium metabisulfite, sodium sulfite and the like; (2) oil-soluble antioxidants, such as ascorbyl palmitate, butylated hydroxyanisole (BHA), butylated hydroxytoluene (BHT), lecithin, propyl gallate, alpha-tocopherol, and the like; and (3) metal chelating agents, such as citric acid, ethylenediamine tetraacetic acid (EDTA), sorbitol, tartaric acid, phosphoric acid, and the like.

[0472] Formulations useful in the methods of the present invention include those suitable for oral, nasal, topical (including buccal and sublingual), rectal, vaginal, aerosol and/or parenteral administration. The formulations may conveniently be presented in unit dosage form and may be prepared by any methods well-known in the art of pharmacy. The amount of active ingredient which can be combined with a carrier material to produce a single dosage form will vary depending upon the host being treated, the particular mode of administration. The amount of active ingredient, which can be combined with a carrier material to produce a single dosage form will generally be that amount of the compound which produces a therapeutic effect. Generally, out of one hundred percent, this amount will range from about 1 percent to about ninety-nine percent of active ingredient, preferably from about 5 percent to about 70 percent, most preferably from about 10 percent to about 30 percent.

[0473] Methods of preparing these formulations or compositions include the step of bringing into association an agent that modulates (e.g., inhibits) biomarker expression and/or activity, with the carrier and, optionally, one or more accessory ingredients. In general, the formulations are prepared by uniformly and intimately bringing into association a respiration uncoupling agent with liquid carriers, or finely divided solid carriers, or both, and then, if necessary, shaping the product.

[0474] Formulations suitable for oral administration may be in the form of capsules, cachets, pills, tablets, lozenges (using a flavored basis, usually sucrose and acacia or tragacanth), powders, granules, or as a solution or a suspension in an aqueous or non-aqueous liquid, or as an oil-in-water or water-in-oil liquid emulsion, or as an elixir or syrup, or as pastilles (using an inert base, such as gelatin and glycerin, or sucrose and acacia) and/or as mouth washes and the like, each containing a predetermined amount of a respiration uncoupling agent as an active ingredient. A compound may also be administered as a bolus, electuary or paste.

[0475] In solid dosage forms for oral administration (capsules, tablets, pills, dragees, powders, granules and the like), the active ingredient is mixed with one or more pharmaceutically-acceptable carriers, such as sodium citrate or dicalcium phosphate, and/or any of the following: (1) fillers or extenders, such as starches, lactose, sucrose, glucose, mannitol, and/or silicic acid; (2) binders, such as, for example, carboxymethylcellulose, alginates, gelatin, polyvinyl pyrrolidone, sucrose and/or acacia; (3) humectants, such as glycerol; (4) disintegrating agents, such as agar-agar, calcium carbonate, potato or tapioca starch, alginic acid, certain silicates, and sodium carbonate; (5) solution retarding agents, such as paraffin; (6) absorption accelerators, such as quaternary ammonium compounds; (7) wetting agents, such as, for example, acetyl alcohol and glycerol monostearate; (8) absorbents, such as kaolin and bentonite clay; (9) lubricants, such a talc, calcium stearate, magnesium stearate, solid polyethylene glycols, sodium lauryl sulfate, and mixtures thereof; and (10) coloring agents. In the case of capsules, tablets and pills, the pharmaceutical compositions may also comprise buffering agents. Solid compositions of a similar type may also be employed as fillers in soft and hard-filled gelatin capsules using such excipients as lactose or milk sugars, as well as high molecular weight polyethylene glycols and the like.

[0476] A tablet may be made by compression or molding, optionally with one or more accessory ingredients. Compressed tablets may be prepared using binder (for example, gelatin or hydroxypropylmethyl cellulose), lubricant, inert diluent, preservative, disintegrant (for example, sodium starch glycolate or cross-linked sodium carboxymethyl cellulose), surface-active or dispersing agent. Molded tablets may be made by molding in a suitable machine a mixture of the powdered peptide or peptidomimetic moistened with an inert liquid diluent.

[0477] Tablets, and other solid dosage forms, such as dragees, capsules, pills and granules, may optionally be scored or prepared with coatings and shells, such as enteric coatings and other coatings well-known in the pharmaceutical-formulating art. They may also be formulated so as to provide slow or controlled release of the active ingredient therein using, for example, hydroxypropylmethyl cellulose in varying proportions to provide the desired release profile, other polymer matrices, liposomes and/or microspheres. They may be sterilized by, for example, filtration through a bacteria-retaining filter, or by incorporating sterilizing agents in the form of sterile solid compositions, which can be dissolved in sterile water, or some other sterile injectable medium immediately before use. These compositions may also optionally contain opacifying agents and may be of a composition that they release the active ingredient(s) only, or preferentially, in a certain portion of the gastrointestinal tract, optionally, in a delayed manner. Examples of embedding compositions, which can be used include polymeric substances and waxes. The active ingredient can also be in micro-encapsulated form, if appropriate, with one or more of the above-described excipients.

[0478] Liquid dosage forms for oral administration include pharmaceutically acceptable emulsions, microemulsions, solutions, suspensions, syrups and elixirs. In addition to the active ingredient, the liquid dosage forms may contain inert diluents commonly used in the art, such as, for example, water or other solvents, solubilizing agents and emulsifiers, such as ethyl alcohol, isopropyl alcohol, ethyl carbonate, ethyl acetate, benzyl alcohol, benzyl benzoate, propylene glycol, 1,3-butylene glycol, oils (in particular, cottonseed, groundnut, corn, germ, olive, castor and sesame oils), glycerol, tetrahydrofuryl alcohol, polyethylene glycols and fatty acid esters of sorbitan, and mixtures thereof.

[0479] Besides inert diluents, the oral compositions can also include adjuvants such as wetting agents, emulsifying and suspending agents, sweetening, flavoring, coloring, perfuming and preservative agents.

[0480] Suspensions, in addition to the active agent may contain suspending agents as, for example, ethoxylated isostearyl alcohols, polyoxyethylene sorbitol and sorbitan esters, microcrystalline cellulose, aluminum metahydroxide, bentonite, agar-agar and tragacanth, and mixtures thereof.

[0481] Formulations for rectal or vaginal administration may be presented as a suppository, which may be prepared by mixing one or more respiration uncoupling agents with one or more suitable nonirritating excipients or carriers comprising, for example, cocoa butter, polyethylene glycol, a suppository wax or a salicylate, and which is solid at room temperature, but liquid at body temperature and, therefore, will melt in the rectum or vaginal cavity and release the active agent.

[0482] Formulations which are suitable for vaginal administration also include pessaries, tampons, creams, gels, pastes, foams or spray formulations containing such carriers as are known in the art to be appropriate.

[0483] Dosage forms for the topical or transdermal administration of an agent that modulates (e.g., inhibits) biomarker expression and/or activity include powders, sprays, ointments, pastes, creams, lotions, gels, solutions, patches and inhalants. The active component may be mixed under sterile conditions with a pharmaceutically-acceptable carrier, and with any preservatives, buffers, or propellants which may be required.

[0484] The ointments, pastes, creams and gels may contain, in addition to a respiration uncoupling agent, excipients, such as animal and vegetable fats, oils, waxes, paraffins, starch, tragacanth, cellulose derivatives, polyethylene glycols, silicones, bentonites, silicic acid, talc and zinc oxide, or mixtures thereof.

[0485] Powders and sprays can contain, in addition to an agent that modulates (e.g., inhibits) biomarker expression and/or activity, excipients such as lactose, talc, silicic acid, aluminum hydroxide, calcium silicates and polyamide powder, or mixtures of these substances. Sprays can additionally contain customary propellants, such as chlorofluorohydrocarbons and volatile unsubstituted hydrocarbons, such as butane and propane.

[0486] The agent that modulates (e.g., inhibits) biomarker expression and/or activity, can be alternatively administered by aerosol. This is accomplished by preparing an aqueous aerosol, liposomal preparation or solid particles containing the compound. A nonaqueous (e.g., fluorocarbon propellant) suspension could be used. Sonic nebulizers are preferred because they minimize exposing the agent to shear, which can result in degradation of the compound.

[0487] Ordinarily, an aqueous aerosol is made by formulating an aqueous solution or suspension of the agent together with conventional pharmaceutically acceptable carriers and stabilizers. The carriers and stabilizers vary with the requirements of the particular compound, but typically include nonionic surfactants (Tweens, Pluronics, or polyethylene glycol), innocuous proteins like serum albumin, sorbitan esters, oleic acid, lecithin, amino acids such as glycine, buffers, salts, sugars or sugar alcohols. Aerosols generally are prepared from isotonic solutions.

[0488] Transdermal patches have the added advantage of providing controlled delivery of a respiration uncoupling agent to the body. Such dosage forms can be made by dissolving or dispersing the agent in the proper medium. Absorption enhancers can also be used to increase the flux of the peptidomimetic across the skin. The rate of such flux can be controlled by either providing a rate controlling membrane or dispersing the peptidomimetic in a polymer matrix or gel.

[0489] Ophthalmic formulations, eye ointments, powders, solutions and the like, are also contemplated as being within the scope of this invention.

[0490] Pharmaceutical compositions of this invention suitable for parenteral administration comprise one or more respiration uncoupling agents in combination with one or more pharmaceutically-acceptable sterile isotonic aqueous or nonaqueous solutions, dispersions, suspensions or emulsions, or sterile powders which may be reconstituted into sterile injectable solutions or dispersions just prior to use, which may contain antioxidants, buffers, bacteriostats, solutes which render the formulation isotonic with the blood of the intended recipient or suspending or thickening agents.

[0491] Examples of suitable aqueous and nonaqueous carriers which may be employed in the pharmaceutical compositions encompassed by the present invention include water, ethanol, polyols (such as glycerol, propylene glycol, polyethylene glycol, and the like), and suitable mixtures thereof, vegetable oils, such as olive oil, and injectable organic esters, such as ethyl oleate. Proper fluidity can be maintained, for example, by the use of coating materials, such as lecithin, by the maintenance of the required particle size in the case of dispersions, and by the use of surfactants.

[0492] These compositions may also contain adjuvants such as preservatives, wetting agents, emulsifying agents and dispersing agents. Prevention of the action of microorganisms may be ensured by the inclusion of various antibacterial and antifungal agents, for example, paraben, chlorobutanol, phenol sorbic acid, and the like. It may also be desirable to include isotonic agents, such as sugars, sodium chloride, and the like into the compositions. In addition, prolonged absorption of the injectable pharmaceutical form may be brought about by the inclusion of agents which delay absorption such as aluminum monostearate and gelatin.

[0493] In some cases, in order to prolong the effect of a drug, it is desirable to slow the absorption of the drug from subcutaneous or intramuscular injection. This may be accomplished by the use of a liquid suspension of crystalline or amorphous material having poor water solubility. The rate of absorption of the drug then depends upon its rate of dissolution, which, in turn, may depend upon crystal size and crystalline form. Alternatively, delayed absorption of a parenterally-administered drug form is accomplished by dissolving or suspending the drug in an oil vehicle.

[0494] Injectable depot forms are made by forming microencapsule matrices of an agent that modulates (e.g., inhibits) biomarker expression and/or activity, in biodegradable polymers such as polylactide-polyglycolide. Depending on the ratio of drug to polymer, and the nature of the particular polymer employed, the rate of drug release can be controlled. Examples of other biodegradable polymers include poly(orthoesters) and poly(anhydrides). Depot injectable formulations are also prepared by entrapping the drug in liposomes or microemulsions, which are compatible with body tissue.

[0495] When the respiration uncoupling agents of the present invention are administered as pharmaceuticals, to humans and animals, they can be given per se or as a pharmaceutical composition containing, for example, 0.1 to 99.5% (more preferably, 0.5 to 90%) of active ingredient in combination with a pharmaceutically acceptable carrier.

[0496] Actual dosage levels of the active ingredients in the pharmaceutical compositions of this invention may be determined by the methods of the present invention so as to obtain an amount of the active ingredient, which is effective to achieve the desired therapeutic response for a particular subject, composition, and mode of administration, without being toxic to the subject.

[0497] The nucleic acid molecules encompassed by the present invention can be inserted into vectors and used as gene therapy vectors. Gene therapy vectors can be delivered to a subject by, for example, intravenous injection, local administration (see U.S. Pat. No. 5,328,470) or by stereotactic injection (see e.g., Chen et al. (1994) *Proc. Natl. Acad. Sci. USA* 91:3054 3057). The pharmaceutical preparation of the gene therapy vector can include the gene therapy vector in an acceptable diluent, or can comprise a slow release matrix in which the gene delivery vehicle is imbedded. Alternatively, where the complete gene delivery vector can be produced intact from recombinant cells, e.g., retroviral vectors, the pharmaceutical preparation can include one or more cells which produce the gene delivery system.

[0498] The present invention also encompasses kits for detecting and/or modulating biomarkers described herein. A kit of the present invention may also include instructional materials disclosing or describing the use of the kit or an antibody of the disclosed invention in a method of the disclosed invention as provided herein. A kit may also include additional components to facilitate the particular application for which the kit is designed. For example, a kit may additionally contain means of detecting the label (e.g., enzyme substrates for enzymatic labels, filter sets to detect fluorescent labels, appropriate secondary labels such as a sheep anti-mouse-HRP, etc.) and reagents necessary for controls (e.g., control biological samples or standards). A kit may additionally include buffers and other reagents recognized for use in a method of the disclosed invention. Non-limiting examples include agents to reduce non-specific binding, such as a carrier protein or a detergent.

[0499] Other embodiments of the present invention are described in the following Examples. The present invention is further illustrated by the following examples which should not be construed as further limiting.

EXAMPLES

Example 1: Materials and Methods for Examples 2-5

a. Cell Lines and Cell Culture

[0500] The two synovial sarcoma cell lines, Aska and SYO1, were generous gifts from Kazuyuki Itoh, Norifumi Naka, and Satoshi Takenaka (Osaka University, Japan) and Akira Kawai (National Cancer Center Hospital, Japan), respectively. The CRL7250 human fibroblast cell line was obtained from Drs. Berkeley Gryder and Javed Khan (National Cancer Institute, Bethesda, MD). The HEK293T cell line was purchased ATCC (CRL-3216). Each cell line was cultured using standard protocols in DMEM medium (Gibco) supplemented with 10-20% fetal bovine serum, 1% Glutamax (Gibco), 1% Sodium Pyruvate (Gibco) and 1% Penicillin-Streptomycin (Gibco) and grown in a humidified incubator at 37° C. with 5% CO.sub.2.

b. Stable Gene Expression and shRNA Knockdown Constructs

[0501] Constitutive expression of SS18 wild-type (SS18), SS18-SSX1 (SS18-SSX1) and SS18-SSX1 mutations with HA or V5 N-terminus tag was obtained using an EF1alpha-driven expression vector (modified from Clonetech, dual Promoter EF-1a-MCS-PGK-Puro or EF-la-MCS-PGK-Blast) expressed in cells by lentiviral infection and selected with puromycin (2 μg/mL) or blasticidin (10 μg/mL). Constitutive expression of shRNA hairpins targeting the 3′UTR region of SSX of the SS18-SSX fusion (5′-CAGTCACTGACAGTTAATAAA-3′ (SEQ ID NO: 227)) or a scramble non-targeting control (5′-CCTAAGGTTAAGTCGCCCTCGCTCGAGCGAGGGCGACTTAACCTTAGG-3′ (SEQ ID NO: 228)) was obtained using lentiviral infection of the pLKO.1 vector with puromycin (2 μg/mL) selection.

c. Lentivirus Generation and Harvesting

[0502] Lentivirus production was obtained from PEI (Polysciences) transfection of HEK293T LentiX™ cells (Clontech) with co-transfection of the packaging vectors pspax2 and pMD2.G along with the gene delivery vector. Viral supernatants were collected 72 hours after transfection, underwent ultracentrifugation at 20,000 rpm for 2.5 hr at 4° C. to concentrate, and then virus pellets were resuspended in PBS. For infection, the viral pellets were added to cells in a drop wise manner in the presence of polybrene (10 μg/mL). After 48 hours, the media containing the lentivirus was replaced and infected cells were selected by addition of puromycin (2 μg/mL) or blasticidin (10 μg/mL).

d. Western Blot Analysis

[0503] Detection of proteins by western blot (WB) analysis was achieved using standard protocols with primary antibodies (Table 4). Samples were separated on 4-12% Bis-Tris SDS PAGE gel (Invitrogen) and transferred to PVDF membrane. The membranes were then blocked in 5% milk and incubated with primary antibody in PBST over night at 4° C. Following incubation with the primary antibody, membranes were washed 3× in PBST, incubated with IRDye® (LI-COR Biosciences) secondary antibodies for 3 hours, washed 3× in PBST with a final PBS wash, and then visualized by the LI-COR Odyssey® Imaging System (LI-COR Biosciences).

TABLE-US-00013 TABLE 4 Description and characterization of antibodies used herein. Protein Clone Lot # Catelog ID Supplier Application BRG1 G-7 G0115 sc-17796 Santa Cruz Immunoblotting BRG1 D1Q7F 1 49360S Cell Signaling Technology Immunoprecipitation SMARCB1/BAF47 A-5 K0515 sc-166165 Santa Cruz Immunoblotting SMARCC1/BAF155 H-76 sc-10756 Cell Signaling Technology Immunoblotting SS18 D6I4Z 1 21792S Cell Signaling Technology Immunoblotting & ChIP RING1B D22F2 5694 Cell Signaling Technology Immunoblotting & ChIP H2AK119Ub D27C4 8240 Cell Signaling Technology Immunoblotting & ChIP H3 GR135489-1 ab1791 Abcam Immunoblotting GAPDH G-9 K2316 sc-365062 Santa Cruz Immunoblotting V5 tag 1805125 P/N-46-0705 Thermo Fisher Scientific Immunoblotting V5 tag D3H8Q 4 13202S Cell Signaling Technology ChIP GFP A-11120 Thermo Fisher Scientific Immunoblotting GST G7781 Sigma-Aldrich Immunoblotting HA Ab9110 Abcam Immunoblotting MBP E8032 New England Biolabs Immunoblotting

e. Cell Lysate Collection

[0504] Whole cell extractions (WCE) were obtained by washing harvested cell pellets with PBS pH 7.4, resuspending in whole cell lysis buffer (PBS pH 7.4 and 1% SDS) and then heating for three minutes at 95° C. Lysates were sonicated until fully liquid. Nuclear extractions (NE) were obtained by suspending the harvested cells in Buffer 0 (50 mM Tris pH 7.5, 0.1% NP-40, 1 mM EDTA, 1 mM MgCl2 with protease inhibitor (Roche, C756U27), 1 mM DTT and 1 mM phenylmethylsulfonyl fluoride (PMSF)), centrifuging at 5,000 rpm for 5 minutes at 4° C., and discarding the supernatant. The pellet (nuclei) were resuspended in EB300 (50 mM Tris pH 7.5, 0.1% NP-40, 1 mM EDTA, 1 mM MgCl2, 300 mM NaCl with protease inhibitor cocktail (Roche, C756U27), 1 mM DTT and 1 mM phenylmethylsulfonyl fluoride (PMSF)), vortexed, incubated on ice, centrifuged at 15,000 rpm for 10 minutes at 4° C. and supernatant containing the nuclear extract collected.

f. Co-Immunoprecipitations

[0505] Nuclear extracts were quantified by Bradford assay and 150-200 μg of protein was incubated with 2 μg of antibody in Buffer EB300 (50 mM Tris pH 7.5, 0.1% NP-40, 1 mM EDTA, 1 mM MgCl2, 150 mM NaCl with protease inhibitor (Roche, C756U27), 1 mM DTT and 1 mM phenylmethylsulfonyl fluoride (PMSF)) overnight at 4° C. Each sample was then incubated with Protein G Dynabeads® (Thermo Scientific) for 2-3 hours. Beads were washed three times with Buffer EB300 followed by elution with 20 μL of elution buffer (NuPage™ LDS buffer (2×) (Life Technologies) containing 100 mM DTT and water).

g. Cell Proliferation Assay

[0506] To measure cell proliferation following lentiviral infection, 2.5×10.sup.4 cells per well were seeded in 12-well plates following 48-hour exposure to lentivirus and 5-day selection with puromycin or blasticidin, with Day 7 denoting the day cells were plated after infection and selection. The cell viability in three wells was then measured using a Vi-CELL™ Cell Counter (Beckman, Brea, CA) every 72 hours.

h. Differential Salt Extraction

[0507] Following collection of 5.0×10.sup.7 cells, cells were resuspended in elution 0 buffer (50 mM Tris-HCl pH

7.5, 1 mM EDTA, 0.1% NP40 with protease inhibitor mixture (Roche, C756U27) and 1 mM PMSF), incubated on ice for 5 minutes, and pelleted by centrifugation. The supernatant was collected (0 mM fraction), and the cell pellet was resuspended in elution 150 buffer (50 mM Tris-HCl pH 7.5, 150 mM NaCl) 1 mM EDTA, 0.1% NP40 with protease inhibitor mixture (Roche, C756U27) and 1 mM PMSF) and vortexed. This process was repeated sequentially with elution 300 buffer, elution 500 buffer, and elution 1000 buffer that contained increasing concentrations of NaCl in order to obtain 0, 150, 300, 500, and 1,000 mM NaCl soluble fractions. Each of these soluble fractions, along with a total sample (5×10.sup.6 cells in elution buffer) and the chromatin pellet (non-soluble material remaining following extraction with 1000 mM NaCl) fractions, was denatured in SDS to a final concentration of 1%, protein quantified by Pierce™ BCA Protein Assay Kit (Thermo Fisher Scientific), and analyzed (1.5 μg of protein) by immunoblot.

i. Purification of mSWI/SNF (BAF) Complexes

[0508] Stable HEK293T cell lines expressing by lentiviral infection HA-SS18 WT or HA-SS18-SSX1 were grown in 150 mm dishes. Complexes were purified using methods previously described with a few modifications (Mashtalir et al. (2014) *Mol. Cell* 54:392-406). Confluent plates were scraped to remove cells and cells were washed with PBS. Cell suspension was spun down by centrifugation at 3000 rpm for 5 minutes at 4° C. and pellets were resuspended in hypotonic buffer (10 mM Tris HCl pH 7.5, 10 mM KCl, 1.5 mM MgCL2, 1 mM DTT, 1 mM PMSF) and incubated on ice for 5 minutes. Following incubation, cell suspension was spun down by centrifugation at 5000 rpm for 5 minutes at 4° C., and pellets were resuspended in 5× volume of fresh hypotonic buffer (with protease inhibitor cocktail, Roche C756U27) and then cells were homogenized using a Dounce homogenizer (glass). Cell suspension was layered onto hypotonic buffer sucrose cushion made with 30% sucrose w/v, spun down by centrifugation at 5000 rpm for 1 hour at 4° C. followed by removal of the cytosol-containing layer. The nuclei containing pellets were resuspended in high salt buffer (50 mM Tris HCl pH 7.5, 300 mM KCl, 1 mM MgCL2, 1 mM EDTA, 1 mM, 1% NP40, 1 mM DTT, 1 mM PMSF and protease inhibitor cocktail) and then the homogenate rotated for 1 hour at 4° C. Homogenates were then spun down by centrifugation at 20,000 rpm for 1 hour at 4° C. in a SW32Ti rotor (Beckman Coulter). The soluble proteins, consisting of the nuclear extract (NE) fraction, was separated from the insoluble chromatin pellet, consisting of the chromatin (CHR) fraction. The chromatin pellet was further solubilized by treatment with Benzonase® (Sigma Aldrich) for 30 minutes and subsequently additional KCl was added to final concentration of 700 mM (50 mM Tris HCl pH 7.5, 700 mM KCl, 1 mM MgCL2, 1 mM EDTA, 1 mM, 1% NP40, 1 mM DTT, 1 mM PMSF and protease inhibitor cocktail), and sonicated 3 times for 30 seconds with 5-minute intervals. The solubilized chromatin fraction was then spun down by centrifugation at 20,000 rpm for 1 hour at 4° C. in a SW32Ti rotor (Beckman Coulter) and supernatant was collected. The collected nuclear extract and chromatin fractions were filtered with a 0.45 μm filter and rotated overnight at 4° C. with HA magnetic resin. HA beads were washed in high salt buffer and eluted with 1 mg/mL of HA peptide for 4 times at durations of 1.5 hour each. Eluted proteins were then subjected to density gradient centrifugation or dialysis.

J. Colloidal Blue and Silver Stain

[0509] HA-SS18 WT and HA-SS18-SSX1 mSWI/SNF complexes were purified via HA-epitope-dependent complex purification. Importantly, for FIG. **1**A, the same number of cells were used for both HA-SS18 WT and HA-SS18-SSX expressing cells, and nuclear material from both cell lines was split into NE and CHR fractions, representing an equal total amount of complexes in the nucleus. Hence, equal input/output loading by volume was achieved. Samples were run on a 4-12% Bis-Tris SDS PAGE gel, stained using Colloidal blue kit or SilverQuest™ Silver Staining Kit (Invitrogen), and imaged using LI-COR Odyssey® Imaging System (LI-COR Biosciences) or Epson-Perfection V600 Photo scanner, respectibly.

k. Density Sedimentation Gradients

[0510] Purified protein complexes were added to the top of a linear, 11 ml 10%-30% glycerol gradients containing 25 mM HEPES pH 7.9, 0.1 mM EDTA, 12.5 mM MgCl2, 100 mM KCl with 1 mM DTT and protease inhibitors (Roche, C756U27). Gradient tubes were placed into SW41 rotor (Beckman Coulter) and spun by centrifugation at 40000 rpm for 16 hours at 4° C. Fractions of 550 μL volume were collected sequentially from the top of the gradient. 100 μL of each fraction was concentrated with 10 μL of Strataclean beads (Agilent Technologies, 400714), loaded and run on a SDS-PAGE gel, and then analyzed by SYPRO® Ruby Protein Gel Stain (Thermo Fisher Scientific) and scanned using Typhoon™ FLA 9500 scanner.

l. Mass Spectrometry Proteomics Analysis of Purified Complexes

[0511] Equal amounts of purified HA-SS18 WT and HA-SS18-SSX1 complexes were loaded onto SDS-PAGE gels from both the nuclear extract (NE) and chromatin (CHR) fractions. Samples were migrated into the gel for a length of 2 cm, gels were stained with colloidal blue stain and protein bands were excised for protein detection by mass spectrometry. The samples were then prepared and data were analyzed by the Taplin Biological Mass Spectrometry Facility directed by Dr. Steven Gygi (Harvard Medical School).

m. Protein and Peptide Pull Downs

[0512] Recombinant purified proteins with affinity tags (MBP or GST) or biotinylated peptides were purified using magnetic beads (Maltose, glutathione or streptavidin respectably) by incubation in EB150 buffer (50 mM Tris-HCl pH 7.5, 150 mM NaCl) 1 mM EDTA, 0.1% NP40 with protease inhibitor mixture (Roche, C756U27) and 1 mM PMSF) at 4° C. overnight. The flow through was removed, the immobilized bait was incubated with 1-2 μg of purified mammalian mono-nucleosomes from HEK293T cells, recombinant mono-nucleosomes (EpiCypher, 16-0006), recombinant H2AK119Ub mono-nucleosomes (EpiCypher, 16-0020) or recombinant protein for 3 hours at 4° C., and the beads were washed 3× with EB150 buffer and then eluted in 2×LDS with 200 mM DTT with heating at 95° C. for 5 minutes. The pull downs were then visualized by immunoblot analysis or colloidal blue staining.

n. Peptide Competition Experiments

[0513] The peptide competition experiments were set up in a similar manner as the peptide pull down experiments with the following exceptions: SSX1 (aa 55-78) or SMARCB1-CC (aa 351-385) biotin-labeled peptides at 10 μM in EB150 were bound to Streptavidin Dynabeads® (Pierce Streptavidin Magnetic Beads, Thermo Scientific) in parallel to 1-2 μg of mononucleosomes incubated with LANA, SSX (aa 155-188) or SMARCB1-CC (aa 351-385) peptide (KE Biochem) at varying concentrations ranging from 0-30 μM overnight at 4° C. Beads were washed 3 times in EB150, and resuspended with the mononucleosome/LANA peptide solutions. The suspension was rotated for 3-5 hours at 4° C. The beads were washed 5 times in EB150, and eluted in Sample Buffer (2×LDS with 200 mM DTT) to load onto 10-20% Tricine gels.

o. Quantitative Targeted Mass Spectrometry

[0514] Mammalian mono-nucleosomes purified from MBP-SSX1 78aa pull downs along with representative input samples were prepared and analyzed by the targeted mass spectrometry pipeline described previously (Creech et al. (2015) *Methods* 72:57-64). Briefly, samples were prepared by histone extraction by acid precipitation followed by protein digestion from incubation with trypsin. To these prepared samples, synthesized isotopically labeled peptides of histone tails with numerous modifications were added at a known quantity. Each sample was then separated using a Proxeon EASY-nLC™ 1000 UHPLC system (Thermo Scientific) and detected with a Q Exactive™ mass spectrometer (Thermo Scientific). The fold change in abundance of each histone peptide from the input sample compared to the pull down was calculated from the light:heavy ratio in detected peak size.

p. Detection of Nucleosome Acidic Patch Interactions by Photocrosslinking

[0515] Details of the design and preparation of diazirine containing nucleosomes for photo-crosslinking studies were described elsewhere (Dao et al. (2019) *Nat. Chem. Biol.* doi:10.1038/s41589-019-0413-4). Briefly, diazirine-containing recombinant nucleosomes (0.5 uM) were incubated with biotinylated SSX peptides (12.5 uM) in binding buffer (20 mM HEPES, pH 7.9, 4 mM Tris, pH 7.5, 150 mM KCl, 10 mM MgCl2, 10% glycerol, and 0.02% (v/v) IGEPAL CA-630) at 30° C. for 30 mins, and cooled on ice for 5 mins. The reaction mixtures were then irradiated at 365 nm for 10 minutes. Reactions were then analyzed by western blotting employing IRDye® 800CW streptavidin on a LI-COR Odyssey® Infrared Imager. Additional details are found in Dao et al. (2019) *Nat. Chem. Biol.* doi:10.1038/s41589-019-0413-4.

q. Immunofluorescence

[0516] Immunofluorescent images were obtained as previously described (Daou et al. (2011) *Pro. Nat. Acad. Sci. U.S.A* 108:2747-2752). Following lentiviral infection and/or drug treatment, cells were prepared by fixation in 3% PFA-PBS and then were permeabilized with PBS 0.1% NP40. Following incubation with primary antibodies, the Anti-rabbit Alexa Fluor® 594 and Anti-mouse Alexa Fluor® 488 (Life Technologies) secondary antibodies were used for visualization. Staining with 4′,6-diamidino-2-phenylindole (DAPI) was used to visualize nuclei. Images were acquired using Zeiss Axio Imager Z2 microscope and images were processed using ImageJ program (NIH).

r. Fluorescent Recovery after Photobleaching (FRAP)

[0517] The FRAP experiments were carried out in the same manner as previously described (Carvalho et al. (2004) *Dev. Cell* 6:815-829). Briefly, HEK293T cells expressing GFP-SS18 WT or GFP-SS18-SSX1 by lentiviral infection or Aska cells co-expressing BRG1-Halo fusion with pLKO.1 shScramble control or shSSX were imaged to measure the mean fluorescence intensity of a defined nuclear region pre and post-photobleaching at 5 second intervals. The relative fluorescence intensity (RFI) for each image was calculated by normalizing the maximal difference in fluorescence intensity post-bleaching to 1. The ti/2 values and mobile fractions were determined using the software Prism (GraphPad Software) from >n=27-30 cells in each condition over two biological replicates.

s. Chromatin Immunoprecipitation (ChIP)

[0518] For chromatin immunoprecipitation (ChIP) experiments, prepared cells were harvested following 48 hours of lentiviral infection and 5 day selection (unless otherwise indicate) with puromycin or blasticidin. Capture of chromatin bound proteins was performed using standard protocols (Millipore, Billerica, MA). Briefly, cells were cross-linked with 1% formaldehyde for 10 minutes at 37° C., reaction was quenched by addition of 125 mM glycine for 5 min and then 5 (for synovial sarcoma cell lines) or 10 (for fibroblast cell lines) million fixed cells were used per experiment. Chromatin was fragmented by sonication with a Covaris E220 and the solubilized chromatin was incubated with a primary antibody overnight at 4° C. to form antibody-chromatin complexes. These complexes were incubated with Protein G-Dynabeads® (Thermo Scientific) for 3 hours at 4° C. Beads were then washed 3× and eluted. The samples then underwent crosslink reversal, treatment with RNase A (Roche), and treatment with proteinase K (Thermo Scientific) followed by DNA capture with AMPure beads (Beckman Coulter).

t. RNA Isolation from Cell Lines

[0519] Cells (1×10^6) were collected following 48 hours of lentiviral infection and 5 days (7 days post-infection) of selection with puromycin or blasticidin for extraction of RNA for RNA-seq experiments. Samples for RNA-seq were prepared in biological duplicates (collected using independent production of lentivirus, infection, selection, and cell culture). Total RNA was collected using the RNeasy® Mini Kit (Qiagen) following homogenization of cell lysates using the QIAshredder (Qiagen).

u. Library Preparation and Sequencing for RNA and ChIP Samples

[0520] Library preparations for next-generation sequencing of RNA-seq samples were performed using the NEBNext® Poly(A) mRNA Magnetic Isolation Module (New England BioLabs) to purify mRNA from 1 μg of total RNA isolated from cells. Next, the isolated mRNA was used with the NEBNext® Ultra™ II Directional RNA Library Prep Kit for Illumina (New England BioLabs) to generate DNA. The DNA from these prepared RNA samples as well as the ChIP-seq samples were then prepared for sequencing using the NEBNext® Ultra™ II (New England BioLabs) to amplify and barcode each sample. The fragments sizes were determined using a D1000 ScreenTape system (Agilent) and the DNA quantified by Kapa Library Quantification Kit Illumina® Platforms (Kapa Biosystems). The samples were then diluted and loaded on a buffer cartridge for 75 bp single end sequencing on the NextSeq™ 500 system (Illumina).

v. Data Processing and Visualization for ChIP Samples

[0521] Alignment of ChIP-seq data was done using Bowtie2, version 2.1.0 (Langmead and Salzberg (2012) *Nat. Meth.* 9:357-359) and reads were mapped to the hg19 human reference genome, using the parameter -k 1.

[0522] To process the aligned data, peaks were called using MACS2 (Zhang et al. (2008) *Genome Biol.* 9:R137) version 2.1.0 against an input sample with a q=0.001 cutoff and broad peaks were called for each antibody in each cell line and condition. Those peaks that were mapped to unmappable chromosomes (any that were not chr1-22, X or Y) or were located in blacklisted regions of ENCODE were excluded. For downstream analysis of data, bam files were generated with duplicates removed using the samtools rmdup command and the -b option. All ChIP-seq tracks were obtained from the bedGraphToBigWig script (UCSC) using bedgraph files generated with MACS2 using the -B-SPMR options. ChIP-seq tracks were visualized using IGV version 2.4.16 (Broad Institute).

[0523] To identify peaks of BAF complex localization, the merged peak set for V5 in V5-SS18 WT and V5-SS18-SSX1 conditions was used with bedtools merge -d 2000 to cause neighboring broad peaks to be called as a single peak. Read counts across peak sets were determined by calling the Rsubread v1.26.1 bioconductor package function feature Counts( ) on bam files. Subsequently, these values were divided by the total number of mapped reads divided by one million to give a normalized value of RPM for each interval contained within the input bed.

[0524] HTSeq was used to calculate metagene read densities with fragment lengths of 200 bp to account for fragment size selection that occurs during sonication. Total read counts for each region was normalized by the number of mapped reads to calculate reads per million mapped reads. The metagene plots were created using mean read densities over all sites for each condition around the center of the peak. All ChIP-seq heatmaps were created using these same HTSeq read densities with sites were then ranked by mean ChIP-seq signal for the indicated antibody and condition. Heatmap visualization was obtained from Python matplotlib using a midpoint of 0.5 reads per million to set the threshold of visualization for the heatmap color scale.

w. Data Processing and Visualization for RNA Samples

[0525] STAR was used to determine RPM values for each sample. Significance was determined with the DESeq2 R package with input raw read counts obtained from Rsubread featureCounts against the hg19 refFlat annotation. Small RNA genes (MIR & SNO) were filtered out from the gene lists for all analyses. Genes with a significant change in expression were determined with a Bonferri-corrected p-value of less than 1e-5, a two-fold change in gene expression ($|\log 2FC|>1$), and inclusion of expressed genes (RPKM≥1 in a minimum of one sample) to identify significantly changing genes. For visualization of RNA-seq data, heatmaps were generated by plotting the z-scores of RPKM values across each sample of the comparison conditions.

X. CRISPR-Cas9 and shRNA Synthetic Lethal Screening Data Analyses

[0526] CRISPR-Cas9 datasets (Avana-19Q3) were obtained from the Project Achilles Data Portal (available on the World Wide Web at depmap.org/portal/achilles/). Fitness (CERES) scores were extracted for each cell line and hierarchical clustering was performed using complete linkage and correlation as a distance measure. Heatmaps were generated using pheatmap in RStudio. DRIVE data is publicly available and can be downloaded from the Novartis DRIVE Data Portal (available on the World Wide Web at oncologynibr.shinyapps.io/drive/). Waterfall plots were generated using ggplot2 in RStudio.

y. Purification of Mammalian Mononucleosomes

[0527] Mammalian mononucleosomes were purified from HEK293T cells similar to as previously described (Mashtalir et al. (2014) *Mol. Cell* 54:392-406). Cells were scraped from plates, washed with cold PBS, and centrifuged at 5,000 rpm for 5 min at 4° C. Pellets were resuspended in hypotonic buffer (EBO: 50 mM Tris HCl, pH 7.5, 1 mM EDTA, 1 mM $MgCl_2$, 0.1% NP40 supplemented with 1 mM DTT, 1 mM PMSF, and protease inhibitor cocktail (Roche, C756U27) and incubated for 5 min on ice. The suspension was centrifuged at 5,000 rpm for 5 min at 4° C., and pellets were resuspended in 5 volumes of EB420 (EBO: 50 mM Tris HCl, pH 7.5, 420 mM NaCl, 1 mM MgCl2, 0.1% NP40 with supplemented with 1 mM DTT and 1 mM PMSF containing protease inhibitor cocktail (Roche, C756U27). Homogenate incubated on rotator for 1 hour at 4° C. The supernatant was then centrifuged at 20,000 rpm (30,000×g) for 1 hour at 4° C. using a SW32Ti rotor. Supernatant was then discarded and chromatin pellet was washed in MNAse buffer (20 mM Tris-HCl pH 7.5, 100 mM KCl, 2 mM MgCl.sub.2, 1 mM CaCl.sub.2, 0.3 μM sucrose, 0.1% NP-40, and protease inhibitor cocktail) three times. Following MNase treatment (3 U/mL for 30 min at room temperature, Sigma-Aldrich), the reaction was quenched with 5 mM of EGTA and 5 mM of EDTA. The samples were then centrifuged at 20,000×g for 1 hour at 4° C. to obtain the soluble chromatin fraction. Soluble chromatin fraction was loaded onto 10-30% glycerol gradient (Mashtalir et al. (2014) *Mol. Cell* 54:392-406) and fractions containing mononucleosomes were isolated and concentrated using centrifugal filter (Amicon, EMD Millipore).

z. Restriction Enzyme Accessibility Assay (REAA) Nucleosome Remodeling Assay

[0528] SMARCA4 (BRG1) levels of the ammonium sulfate nuclear extracts were normalized via BCA protein quantification and Silver Stain analyses for HA-SS18 and HA-SS18-SSX conditions. Protein was diluted for final reaction concentration of 150 μg/mL in REAA buffer (20 mM HEPES, pH 8.0, 50 mM KCl, 5 mM MgCl2) containing 0.1 mg/mL BSA, 1 mM DTT, 20 nM nucleosomes (EpiDyne Nucleosome Remodeling Assay Substrate ST601-GATC1, EpiCypher). The REAA mixture was incubated at 37° C. for 10 min, and reaction was initiated using 1-2 mM ATP (Ultrapure ATP, Promega) and 0.005 U/mL DpnJJ Restriction Enzyme (New England Biolabs). The REAA reaction mixture was quenched with 20-24 mM EDTA and placed on ice. Proteinase K (Ambion) was added at 100 mg/mL for 30-60 min, followed by either AMPure bead DNA purification and D1000 HS DNA ScreenTape Analysis (Agilent) or mixing with GelPilot® Loading Dye (QIAGEN) and loading onto 8% TBE gel (Novex 8% TBE Gels, Thermo Fisher). TBE gels were stained with either SYBR®-Safe (Invitrogen) or Syto®-60 Red Fluorescent Nucleic Acid Stain (Invitrogen), followed by imaging with UV light on an Alpha Innotech AlphaImager™ 2200 and/or with 652 nm light excitation on a Li-Cor Odyssey® CLx imaging system (LI-COR).

aa. Preparation of Peptides

[0529] Custom peptide sequences were prepared using standard synthesis techniques from KE Biochem. The peptides were confirmed to have >95% purity by HPLC and obtained as a white to off-white lyophilized powder. The powder was re-suspended in DMSO (Sigma) for use in experiments.

ab. Expression and Purification of Recombinant Proteins

[0530] DNA constructs of human SSX1 aa111-188 and related mutates in pGEX-6P2 expression vector were transformed in E. coli BL21 (DE3) cells and overexpressed in TB medium in the presence of 100 μg/ml of ampicillin. Cells were grown at 37° C. to an OD600 of 0.6, cooled to 17° C., induced with 500 μM isopropyl-1-thio-D-galactopyranoside (IPTG), incubated overnight at 17° C., collected by centrifugation, and stored at −80° C. For .sup.13C- and .sup.15N-labeled protein expression for NMR analysis, minimal media containing .sup.13C-labeled glucose and .sup.15N-labeled ammonium chloride was used for E. coli growth and protein expression, following an established protocol (Marley et al. (2001) *J. Biomol. NMR* 20:71-75). Cell pellets were resuspended in buffer A (25 mM HEPES, pH 7.5, 200 mM NaCl, 5% glycerol, and 0.5 mM TCEP) supplemented with 1 mM PMSF, lysed in a Microfluidizer (Microfluidics) and centrifuged at 16,000×g for 45 min. Glutathione sepharose beads (GE healthcare) were incubated with lysate supernatant for 90 min to captured GST-tagged proteins and washed with buffer A. Beads with bound protein were transferred to an FPLC-compatible column and the bound protein was washed with high salt buffer (buffer A containing 1M NaCl) followed by elution with buffer A supplemented with 15 mM glutathione (Sigma). Eluted protein fractions were collected, concentrated and purified by size exclusion chromatography using a Superdex® 75 10/300 column (GE healthcare) equilibrated with buffer A. Eluted protein was incubated with GST-3C protease at 4° C. overnight. Cleaved samples were incubated with a second round of glutathione beads to remove GST-3C and free GST, and desired protein product contained within the flow-through fractions was further purified by ion-exchange chromatography using mono-Q column (GE healthcare). Fractions containing the cleaved protein product were pooled, concentrated and stored at −80° C.

Ac. Peptide Hybridization Assay

[0531] IMR90 fibroblasts were grown on coverslips, washed with PBS and fixed using 100% ice-cold methanol for 3 minutes. Coverslips were then washed with IF wash buffer (PBS 0.1% NP40 1 mM Sodium azide) 3 times. Selected groups were treated with 200 ng/ml of recombinant USP2 catalytic domain (Boston biochem) for 1 hour. Coverslips were then washed 3 times with IF wash buffer and incubated with 2 μM of biotinylated peptides. Coverslips were subsequently washed 3 times with IF wash buffer and fixed in 3% PFA-PBS for 20 minutes. The rest of the procedure followed accordingly to standard IF protocol. In brief, following incubation with primary antibodies, the Anti-rabit Alexa Fluor® 594 and Streptavidin Alexa Fluor® 488 (Life Technologies) secondary antibodies/reagent were used for visualization of primary antibodies or biotinylated peptides. Staining with 4′,6-diamidino-2-phenylindole (DAPI) was used to visualize nuclei. Images were acquired using Zeiss Axio Imager Z2 microscope and images were processed using ImageJ program (NIH).

ad. NMR Structure Prediction

[0532] .sup.15N and .sup.13C doubly-labeled C-terminal deletion mutant SSX1-7aa (aal11-181) protein were expressed from E. coli in M9 minimal medium containing .sup.15NH.sub.4Cl and .sup.13C-glucose as the sole nitrogen and carbon sources. Non-uniformly-sampled (NUS) triple resonance experiments, HNCA, HN(CO)CA, HNCO, HN(CA)CO, HN(CA)CB, HN(COCA)CB, and C(CO)NH, using 0.33 mM .sub.15N/.sub.13C-SSX1-7aa(aa 111-181) protein in PBS buffer, pH 6.5 with 10% D20, were performed at 15° C. on a 700 MHz Agilent DD2 spectrometer equipped with a cryogenic probe. The data were processed using NMRPipe (Delaglio et al. (1995) *J. Biomol. NMR* 6:277-293) and Iterative Soft Thresholding reconstruction approach (istHMS) (Hyberts et al. (2012) *J. Biomol. NMR* 52:315-327) and analyzed by CARA (Keller (2005) ETH). Backbone dihedral angle restraints and secondary structure predications based on assigned chemical shifts were obtained using the TALOS+ software (Shen et al. (2009) *J. Biomol. NMR* 44:213-223).

ae. Nuclear Extraction

[0533] Nuclear extracts for 293T V5-SS18WT and V5-SS18-SSX1 cells were prepared as described in Mashtalir et al. (2018) *Cell* 175:1272-1288. Specifically, cells were scraped from plates, washed with cold PBS, pelleted at 3,000 rpm for 5 min at 4° C., and resuspended in Buffer A hypotonic buffer (50 mM Hepes, pH 7.6, 25 mM KCl, 10% Glycerol, 0.1% NP-40, 0.05 mM EDTA, 5 mM MgCl2 supplemented with protease inhibitor (Roche), and 1 mM phenylmethylsulfonyl fluoride (PMSF)). Lysates were pelleted at 3,000 rpm for 5 min at 4 C. Supernatants were discarded, and nuclei were resuspended in Buffer C high salt buffer (10 mM Hepes, pH 7.6, 100 mM KCl, 10% Glycerol, 0.5 mM EDTA, 3 mM MgCl2 supplemented with protease inhibitor and 1 mM PMSF). Lysates were incubated at 4° C. at constant rotation. Lysates were then pelleted at 40,000×rpm for 1 hour at 4° C. Supernatants were collected, and mixed with (NH4)2SO4 at 300 mg/ml for 30 min. Samples were pelleted at 15,000 rpm for 30 minutes and supernatant was discarded. Protein concentrations were quantified via bicinchonic acid (BCA) assay (Pierce). Finally, samples were supplemented with 1 mM DTT.

af. ATPase Assays

[0534] ATPase consumption assays were performed using the ADP-Glo Kinase Assay kit (Promega). The same conditions as the REAA nucleosome remodeling assay described above were used. Following incubation with desired substrates for 40 min at 37° C., 1× volume of ADP-Glo Reagent was used to quench the reaction and incubated at RT for 40 min. 2× volume of the Kinase Detection Reagent was then added and incubated at RT for 1 h. Luminescence readout was recorded. Substrates used for this assay were purified recombinant mononucleosome (EpiDyne Nucleosome Remodeling Assay Substrate ST601-GATC1, EpiCypher, Cat #16-4101). Nuclear extract material was used at 150ug for each ARID1A-IP using ARID1A antibody (Cell Signaling, Cat #12354S).

Example 2: SS18-SSX-Bound BAF Complexes Bind Chromatin with Uniquely High Affinity Via Stoichiometric Histone Binding

[0535] Interactions between chromatin-associated proteins and the histone landscape play major roles in dictating genome topology and gene expression. Cancer-specific fusion oncoproteins display unique chromatin localization patterns, yet often lack classical transcription factor-like DNA-binding domains, presenting challenges in identifying mechanisms governing their site-specific chromatin targeting and function. Recent studies indicate that SS18-SSX-bound BAF complexes have specialized biochemical and chromatin localization properties (McBride et al. (2018) *Cancer Cell* 33:1128-1141; Kadoch and Crabtree (2013) *Cell* 153:71-85). To explore the underlying molecular recognition mechanisms driving these associations and activities, HA-tagged versions of either wild-type (WT) SS18 or SS18-SSX were expressed in HEK-293T cells and BAF complex purifications were performed from soluble nuclear extract (NE) and nuclease-treated solubilized chromatin (CHR) (FIG. **1**A). Strikingly, fusion oncoprotein SS18-SSX-bound BAF complexes preferentially eluted in the CHR material, in contrast to WT complexes, which eluted nearly completely in the soluble NE material, as expected from previous studies examining WT (and other loss-of-function mutant variants of) BAF complexes (Kadoch et al. (2013) *Nat. Genetics* 45:592-601; Mashtalir et al. (2018) *Cell* 175:1272-1288). Importantly, SS18-SSX-bound complexes captured near-stoichiometric amounts of core histone proteins H2A, H2B, H3, and H4 (FIG. **1**A). The complexes were next subjected to mass-spectrometric (MS) analyses and selective co-enrichment of histone peptides with HA-SS18-SSX, but not with HA-WT SS18 was found in the chromatin-bound fractions (FIG. **1**B, FIGS. **2**A-**2**B). Notably, peptides corresponding to the H2A K119Ub mark were captured only in the purifications of SS18-SSX-bound complexes but not in SS18 WT complexes, in agreement with the visualization of this mark upon colloidal blue staining (FIG. **1**A, Tables 5A-5E). In addition, it was found that SS18-SSX purifications most substantially enriched for ATPase subunits SMARCA4 and SMARCA2, BCL7A, and ACTL6A, consistent with the fact that SS18 is part of the ATPase module of mSWI/SNF complexes (Mashtalir et al. (2018) *Cell* 175:1272-1288), while core module components, particularly SMARCB1 were less enriched compared to WT SS18 purifications (FIG. **1**B, FIGS. **2**F and **2**G, Tables 5A-5E). Binding to PRC1 components was not detected, as has been previously indicated (Banito et al. (2018) *Cancer cell* 33:527-541) (FIG. **2**B).

TABLE-US-00014 TABLE 5A HA-SS18SSX1_CHR_peptides Unique Total reference Gene Symbol MWT(kDa) AVG 55 584 sp|P51532|SMCA4_HUMAN SMARCA4 184.53 3.196 84 490 sp|P51531|SMCA2_HUMAN SMARCA2 181.17 2.9458 12 366 sp|P33778|H2B1B_HUMAN HIST1H2BB 13.94 2.4497 9 214 sp|Q96KK5|H2A1H_HUMAN HIST1H2AH 13.9 2.5116 69 167 sp|O14497|ARI1A_HUMAN ARID1A 241.89 3.2645 51 165 sp|Q8TAQ2|SMRC2_HUMAN SMARCC2 132.8 2.9885 29 160 sp|O96019|ACL6A_HUMAN ACTL6A 47.43 3.1707 39 133 sp|Q92922|SMRC1_HUMAN SMARCC1 122.79 3.0345 64 125 sp|Q8NFD5|ARI1B_HUMAN ARID1B 235.97 3.0308 12 125 sp|P62805|H4_HUMAN HIST1H4A 11.36 2.7795 12 119 sp|P62736|ACTA_HUMAN ACTA2 41.98 2.5744 34 78 sp|Q9NYF8|BCLF1_HUMAN BCLAF1 106.06 3.1403 57 73 sp|Q14839|CHD4_HUMAN CHD4 217.87 3.3514 9 73 sp|P63261|ACTG_HUMAN ACTG1 41.77 3.283 12 70 sp|Q4VC05|BCL7A_HUMAN BCL7A 22.8 3.4877 62 68 sp|O75691|UTP20_HUMAN UTP20 318.18 3.3196 25 67 sp|Q9H307|PININ_HUMAN PNN 81.56 3.0645 27 66 sp|Q6STE5|SMRD3_HUMAN SMARCD3 54.98 3.1596 25 61 sp|Q969G3|SMCE1_HUMAN SMARCE1 46.62 3.3347 25 61 sp|Q96GM5|SMRD1_HUMAN SMARCD1 58.2 3.3292 4 60 sp|Q16384|SSX1_HUMAN SSX1 21.92 3.3725 20 57 sp|Q12824|SNF5_HUMAN SMARCB1 44.11 3.1529 21 55 sp|Q00839|HNRPU_HUMAN HNRNPU 90.53 3.0794 27 54 sp|Q9UKV3|ACINU_HUMAN ACIN1 151.77 2.983 33 51 sp|Q9H0A0|NAT10_HUMAN NAT10 115.66 3.4608 17 49 sp|P22087|FBRL_HUMAN FBL 33.76 2.9946 23 47 sp|Q92925|SMRD2_HUMAN SMARCD2 58.88 3.3416 42 45 sp|Q8WYP5|ELYS_HUMAN AHCTF1 252.34 3.4462 22 45 sp|Q9Y2W1|TR150_HUMAN THRAP3 108.6 3.1033 42 44 sp|Q14980|NUMA1_HUMAN NUMA1 238.12 3.8461 42 44 sp|O75643|U520_HUMAN SNRNP200 244.35 3.2527 38 42 sp|Q6P2Q9|PRP8_HUMAN PRPF8 273.43 3.3056 38 42 sp|P11388|TOP2A_HUMAN TOP2A 174.28 3.2238 31 42 sp|O00567|NOP56_HUMAN NOP56 66.01 3.3301 24 38 sp|P52272|HNRPM_HUMAN HNRNPM 77.46 3.1869 20 38 sp|P08670|VIME_HUMAN VIM 53.62 3.2563 31 37 sp|Q08211|DHX9_HUMAN DHX9 140.87 3.568 13 36 sp|Q14978|NOLC1_HUMAN NOLC1 73.56 2.6167 21 33 sp|P20700|LMNB1_HUMAN LMNB1 66.37 3.162 24 32 sp|O60264|SMCA5_HUMAN SMARCA5 121.83 3.1653 5 32 sp|P62987|RL40_HUMAN UBA52 14.72 2.7242 31 31 sp|Q9H583|HEAT1_HUMAN HEATR1 242.22 3.443 4 31 sp|Q71DI3|H32_HUMAN HIST2H3A 15.38 1.8646 11 29 sp|Q8WUZ0|BCL7C_HUMAN BCL7C 23.45 3.4264 9 29 sp|P45973|CBX5_HUMAN CBX5 22.21 2.805 27 28 sp|O75533|SF3B1_HUMAN SF3B1 145.74 3.3041 26 28 sp|Q9UIG0|BAZ1B_HUMAN BAZ1B 170.8 3.4813 26 28 sp|Q9Y5B9|SP16H_HUMAN SUPT16H 119.84 3.4761 20 27 sp|P17480|UBF1_HUMAN UBTF 89.35 3.043 26 26 sp|Q14690|RRP5_HUMAN PDCD11 208.57 3.5217 24 26 sp|Q96T58|MINT_HUMAN SPEN 402 3.2629 19 26 sp|Q9Y2X3|NOP58_HUMAN NOP58 59.54 4.0996 19 26 sp|Q08945|SSRP1_HUMAN SSRP1 81.02 3.2852 23 25 sp|Q9NZM4|BICRA_HUMAN BICRA 158.39 3.5192 11 25 sp|Q07955|SRSF1_HUMAN SRSF1 27.73 2.7723 23 24 sp|Q6PL18|ATAD2_HUMAN ATAD2 158.46 3.5607 20 23 sp|P46100|ATRX_HUMAN ATRX 282.41 3.3981 17 23 sp|P25440|BRD2_HUMAN BRD2 88.01 3.4653 8 23 sp|Q15287|RNPS1_HUMAN RNPS1 34.19 2.513 20 22 sp|Q14676|MDC1_HUMAN MDC1 226.53 3.4061 19 20 sp|Q14692|BMS1_HUMAN BMS1 145.72 3.6681 19 20 sp|P46013|KI67_HUMAN MKI67 358.47 3.0296 18 20 sp|Q9H8M2|BRD9_HUMAN BRD9 66.96 3.7468 18 20 sp|Q96T23|RSF1_HUMAN RSF1 163.72 3.644 18 20 sp|Q15393|SF3B3_HUMAN SF3B3 135.49 3.5038 18 20 sp|Q02880|TOP2B_HUMAN TOP2B 183.15 3.4782 14 20 sp|Q92785|REQU_HUMAN DPF2 44.13 4.0099 14 20 sp|O75367|H2AY_HUMAN H2AFY 39.59 3.3191 13 20 sp|P43243|MATR3_HUMAN MATR3 94.56 3.2566 12 20 sp|Q1KMD3|HNRL2_HUMAN HNRNPUL2 85.05 3.3094 3 20 sp|Q01130|SRSF2_HUMAN SRSF2 25.46 4.6781 18 19 sp|Q6KC79|NIPBL_HUMAN NIPBL 315.85 3.4965 18 19 sp|O14646|CHD1_HUMAN CHD1 196.57 3.3084 17 19 sp|Q9BQG0|MBB1A_HUMAN MYBBP1A 148.76 3.1516 15 19 sp|P46087|NOP2_HUMAN NOP2 89.25 3.5025 17 18 sp|Q8WWQ0|PHIP_HUMAN PHIP 206.56 3.1097 15 18 sp|Q12905|ILF2_HUMAN ILF2 43.04 3.9059 15 18 sp|Q8IXT5|RB12B_HUMAN RBM12B 118.03 2.9192 2 18 sp|Q9BRL6|SRSF8_HUMAN SRSF8 32.27 1.8838 17 17 sp|P78527|PRKDC_HUMAN PRKDC 468.79 3.4649 17 17 sp|Q15029|U5S1_HUMAN EFTUD2 109.37 3.3215 15 17 sp|Q13435|SF3B2_HUMAN SF3B2 100.16 3.6324 9 17 sp|P50402|EMD_HUMAN EMD 28.98 2.4119 7 17 sp|Q5BKZ1|ZN326_HUMAN ZNF326 65.61 2.8312 15 16 sp|Q14683|SMC1A_HUMAN SMC1A 143.14 3.0529 14 16 sp|Q86U86|PB1_HUMAN PBRM1 192.83 3.4056 13 16 sp|Q12906|ILF3_HUMAN ILF3 95.28 3.2276 11 16 sp|Q9BVJ6|UT14A_HUMAN UTP14A 87.92 3.5193 14 15 sp|Q9H2P0|ADNP_HUMAN ADNP 123.49 3.5487 14 15 sp|Q7Z3K3|POGZ_HUMAN POGZ 155.24 3.486 14 15 sp|Q9Y3T9|NOC2L_HUMAN NOC2L 84.87 3.4078 13 15 sp|Q9P0M6|H2AW_HUMAN H2AFY2 40.03 3.4783 12 15 sp|O76021|RL1D1_HUMAN RSL1D1 54.94 3.0238 14 14 sp|Q03164|KMT2A_HUMAN KMT2A 431.5 4.0265 14 14 sp|Q9UIF9|BAZ2A_HUMAN BAZ2A 211.07 3.4659 13 14 sp|Q13620|CUL4B_HUMAN CUL4B 103.92 3.6501 12 14 sp|O60306|AQR_HUMAN AQR 171.19 3.1564 10 14 sp|Q6AI39|BICRL_HUMAN BICRAL 115.01 3.6798 10 14 sp|Q9NY61|AATF_HUMAN AATF 63.09 3.3347 3 14 tr|B9EGQ8|B9EGQ8_HUMAN SMARCA4 189.33 3.1403 2 14 sp|Q15532|SSXT_HUMAN SS18 45.9 2.6438 13 13 sp|Q9UK61|TASOR_HUMAN FAM208A 188.91 3.4606 13 13 sp|Q9NTI5|PDS5B_HUMAN PDS5B 164.56 2.9322 12 13 sp|P23396|RS3_HUMAN RPS3 26.67 2.7633 11 13 sp|Q49A26|GLYR1_HUMAN GLYR1 60.52 3.509 9 13 sp|P55795|HNRH2_HUMAN HNRNPH2 49.23 3.2887 9 13 sp|Q86VM9|ZCH18_HUMAN ZC3H18 106.32 2.9032 9 13 sp|Q13247|SRSF6_HUMAN SRSF6 39.56 2.4674 3 13 sp|Q71UI9|H2AV_HUMAN H2AFV 13.5 2.7796 12 12 sp|P24928|RPB1_HUMAN POLR2A 217.04 3.3463 12 12 sp|O43143|DHX15_HUMAN DHX15 90.88 3.2449 12 12 sp|Q96GQ7|DDX27_HUMAN DDX27 89.78 3.0281 6 12 sp|P31943|HNRH1_HUMAN HNRNPH1 49.2 2.9519 11 11 sp|O60216|RAD21_HUMAN RAD21 71.64 3.9259 11 11 sp|P09874|PARP1_HUMAN PARP1 113.01 3.4647 11 11 sp|Q12788|TBL3_HUMAN TBL3 88.98 3.4469 11 11 sp|P33993|MCM7_HUMAN MCM7 81.26 3.3527 11 11 sp|Q9NYH9|UTP6_HUMAN UTP6 70.15 3.3108 11 11 sp|Q8TDD1|DDX54_HUMAN DDX54 98.53 3.2741 11 11 sp|Q15397|PUM3_HUMAN PUM3 73.54 3.2129 11 11 sp|P28370|SMCA1_HUMAN SMARCA1 122.53 2.7975 11 11 sp|Q13619|CUL4A_HUMAN CUL4A 87.62 2.6132 10 11 sp|Q12873|CHD3_HUMAN CHD3 226.45 3.7322 10 11 sp|Q8IZL8|PELP1_HUMAN PELP1 119.62 3.6196 10 11 sp|P18583|SON_HUMAN SON 263.66 3.395 10 11 sp|Q99549|MPP8_HUMAN MPHOSPH8 97.12 3.3621 10 11 sp|Q9UKM9|RALY_HUMAN RALY 32.44 3.1183 9 11 sp|O00541|PESC_HUMAN PES1 67.96 2.4036 7 11 sp|Q13243|SRSF5_HUMAN SRSF5 31.25 3.5127 6 11 sp|Q16629|SRSF7_HUMAN SRSF7 27.35 2.6912 4 11 sp|P16403|H12_HUMAN HIST1H1C 21.35 2.8282 2 11 sp|Q02539|H11_HUMAN HIST1H1A 21.83 2.9123 2 11 sp|Q93079|H2B1H_HUMAN HIST1H2BH 13.88 2.2042 10 10 sp|Q99459|CDC5L_HUMAN CDC5L 92.19 3.4411 10 10 sp|P38919|IF4A3_HUMAN EIF4A3 46.84 3.4146 10 10 sp|Q8N7H5|PAF1_HUMAN PAF1 59.94 3.3744 10 10 sp|Q6PD62|CTR9_HUMAN CTR9 133.42 3.368 10 10 sp|Q92794|KAT6A_HUMAN KAT6A 224.89 3.2929 10 10 sp|Q8IWA0|WDR75_HUMAN WDR75 94.44 3.1652 10 10 sp|Q9ULI0|ATD2B_HUMAN ATAD2B 164.81 3.1043 10 10 sp|Q03188|CENPC_HUMAN CENPC 106.77 3.0478 10 10 sp|Q9NVP1|DDX18_HUMAN DDX18 75.36 2.7626 10 10 sp|Q9UQ35|SRRM2_HUMAN SRRM2 299.44 2.7014 9 10 sp|Q9Y4W2|LAS1L_HUMAN LAS1L 83.01 3.8251 9 10 sp|Q9Y2R4|DDX52_HUMAN DDX52 67.46 3.626 9 10 sp|Q14202|ZMYM3_HUMAN ZMYM3 152.28 3.4516 9 10 sp|Q9NR30|DDX21_HUMAN DDX21 87.29 3.4294 9 10 sp|P62701|RS4X_HUMAN RPS4X 29.58 3.093 9 10 sp|Q13601|KRR1_HUMAN KRR1 43.64 3.0329 9 10 sp|Q92841|DDX17_HUMAN DDX17 80.22 2.8111 8 10 sp|Q8NC56|LEMD2_HUMAN LEMD2 56.94 3.249 8 10 sp|Q7Z7K6|CENPV_HUMAN CENPV 29.93 3.2113 9 9 sp|P57740|NU107_HUMAN NUP107 106.31 3.6499 9 9 sp|Q03701|CEBPZ_HUMAN CEBPZ 120.9 3.5391 9 9 sp|O60281|ZN292_HUMAN ZNF292 304.62 3.2897 9 9 sp|Q13330|MTA1_HUMAN MTA1 80.74 3.2111 9 9 sp|O43390|HNRPR_HUMAN HNRNPR 70.9 3.1736 9 9 sp|P49750|YLPM1_HUMAN YLPM1 219.85 3.1531 9 9 sp|Q9BSC4|NOL10_HUMAN NOL10 80.25 3.0247 9 9 sp|Q8N1F7|NUP93_HUMAN NUP93 93.43 2.9512 9 9 sp|Q7Z5J4|RAI1_HUMAN RAI1 203.23 2.939 9 9 sp|Q92621|NU205_HUMAN NUP205 227.78 2.9163 8 9 sp|Q9Y5J1|UTP18_HUMAN UTP18 61.96 3.6186 8 9 sp|Q9Y5B6|PAXB1_HUMAN PAXBP1 104.74 3.4559 8 9 sp|P12236|ADT3_HUMAN SLC25A6 32.85 2.8579 8 9 sp|P22626|ROA2_HUMAN HNRNPA2B1 37.41 2.8007 8 9 sp|P02545|LMNA_HUMAN LMNA 74.09 2.7943 4 9 sp|P07305|H10_HUMAN H1F0 20.85 3.1846 8 8 sp|Q9UH99|SUN2_HUMAN SUN2 80.26 4.2804 8 8 sp|Q9NXF1|TEX10_HUMAN TEX10 105.61 3.9844 8 8 sp|O15213|WDR46_HUMAN WDR46 68.03 3.9455 8 8 sp|O00159|MYO1C_HUMAN MYO1C 121.61 3.5705 8 8 sp|Q8N3U4|STAG2_HUMAN STAG2 141.24 3.4872 8 8 sp|Q9BVP2|GNL3_HUMAN GNL3 61.95 3.4513 8 8 sp|Q5QJE6|TDIF2_HUMAN DNTTIP2 84.42 3.3455 8 8 sp|Q16531|DDB1_HUMAN DDB1 126.89 3.3262 8 8 sp|Q6ZRS2|SRCAP_HUMAN SRCAP 343.34 3.1877 7 8 sp|Q9UMS4|PRP19_HUMAN PRPF19 55.15 3.6093 7 8 sp|Q9H6F5|CCD86_HUMAN CCDC86 40.21 3.5931 7 8 sp|Q9BZE4|NOG1_HUMAN GTPBP4 73.92 3.1582 7 8 sp|O60832|DKC1_HUMAN DKC1 57.64 3.1341 7 8 sp|P39019|RS19_HUMAN RPS19 16.05 2.5209 5 8 sp|Q9BQE9|BCL7B_HUMAN BCL7B 22.18 2.9657 4 8 sp|Q13595|TRA2A_HUMAN TRA2A 32.67 2.8133 7 7 sp|O00566|MPP10_HUMAN MPHOSPH10 78.82 4.0635 7 7 sp|Q8IY81|SPB1_HUMAN FTSJ3 96.5 3.8802 7 7 sp|Q8WXH0|SYNE2_HUMAN SYNE2 795.94 3.7401 7 7 sp|O95251|KAT7_HUMAN KAT7 70.6 3.5459 7 7 sp|P42285|SK2L2_HUMAN SKIV2L2 117.73 3.4754 7 7 sp|P42167|LAP2B_HUMAN TMPO 50.64 3.3395 7 7 sp|O14647|CHD2_HUMAN CHD2 211.21 3.2097 7 7 sp|Q9NRL2|BAZ1A_HUMAN BAZ1A 178.59 3.2083 7 7 sp|Q13129|RLF_HUMAN RLF 217.81 3.1787 7 7 sp|Q13111|CAF1A_HUMAN CHAF1A 106.86 3.1007 7 7 sp|P68371|TBB4B_HUMAN TUBB4B 49.8 2.9823 7 7 sp|P78316|NOP14_HUMAN NOP14 97.61 2.902 7 7 sp|O15042|SR140_HUMAN U2SURP 118.22 2.8143 7 7 sp|Q5SSJ5|HP1B3_HUMAN HP1BP3 61.17 2.7313 7 7 sp|Q9NQS7|INCE_HUMAN INCENP 105.36 2.6037 6 7 sp|Q13185|CBX3_HUMAN CBX3 20.8 3.6562 6 7 sp|P21796|VDAC1_HUMAN VDAC1 30.75 3.204 6 7 sp|P36578|RL4_HUMAN RPL4 47.67 2.6157 5 7 sp|Q92784|DPF3_HUMAN DPF3 43.06 3.5672 5 7 sp|O00422|SAP18_HUMAN SAP18 17.55 3.3178 5 7 sp|Q9H9B1|EHMT1_HUMAN EHMT1 141.38 3.3147 5 7 sp|P62995|TRA2B_HUMAN TRA2B 33.65 3.2146 5 7 IGH1M_MOUSE Ighg1 43.36 2.9952 5 7 sp|Q8TDI0|CHD5_HUMAN CHD5 222.91 2.86 4 7 sp|P07910|HNRPC_HUMAN HNRNPC 33.65 3.3974 6 6 sp|Q15061|WDR43_HUMAN WDR43 74.84 4.277 6 6 sp|Q9NQZ2|SAS10_HUMAN UTP3 54.53 4.2012 6 6 sp|Q15269|PWP2_HUMAN PWP2 102.39 4.0268 6 6 sp|P61978|HNRPK_HUMAN HNRNPK 50.94 3.9336 6 6 sp|P19338|NUCL_HUMAN NCL 76.57 3.9301 6 6 sp|Q7KZ85|SPT6H_HUMAN SUPT6H 198.95 3.8795 6 6 sp|Q8WTT2|NOC3L_HUMAN NOC3L 92.49 3.7112 6 6 sp|P08865|RSSA_HUMAN RPSA 32.83 3.5636 6 6 sp|Q8TED0|UTP15_HUMAN UTP15 58.38 3.547 6 6 sp|Q96KQ7|EHMT2_HUMAN EHMT2 132.29 3.4908 6 6 sp|P51398|RT29_HUMAN DAP3 45.54 3.4773 6 6 sp|Q9BVI4|NOC4L_HUMAN NOC4L 58.43 3.4602 6 6 sp|P07199|CENPB_HUMAN CENPB 65.13 3.337 6 6 sp|Q9BQE3|TBA1C_HUMAN TUBA1C 49.86 3.3028 6 6 sp|Q03252|LMNB2_HUMAN LMNB2 69.91 3.2923 6 6 sp|Q5VWN6|F208B_HUMAN FAM208B 268.68 3.2658 6 6 sp|Q9HC52|CBX8_HUMAN CBX8 43.37 3.2482 6 6 sp|Q8IWI9|MGAP_HUMAN MGA 331.63 3.2248 6 6 sp|Q8WXI9|P66B_HUMAN GATAD2B 65.22 3.223 6 6 sp|Q13573|SNW1_HUMAN SNW1 61.46 3.1062 6 6 sp|Q9Y2P8|RCL1_HUMAN RCL1 40.82 3.0802 6 6 sp|B2RXH8|HNRC2_HUMAN HNRNPCL2 32.05 3.0374 6 6 sp|P52701|MSH6_HUMAN MSH6 152.69 2.9401 6 6 sp|Q16891|MIC60_HUMAN IMMT 83.63 2.895 5 6 sp|P52597|HNRPF_HUMAN HNRNPF 45.64 3.7652 5 6

sp|Q8NEJ9|NGDN_HUMAN NGDN 35.87 3.6102 5 6 sp|Q8ND82|Z280C_HUMAN ZNF280C 83.04 3.3478 5 6
sp|P38159|RBMX_HUMAN RBMX 42.31 3.2052 5 6 sp|P15880|RS2_HUMAN RPS2 31.3 3.1733 5 6
sp|Q5VZL5|ZMYM4_HUMAN ZMYM4 172.68 2.9892 5 6 sp|P62753|RS6_HUMAN RPS6 28.66 2.8134 5 6
sp|P29375|KDM5A_HUMAN KDM5A 191.97 2.7763 4 6 sp|Q5TAP6|UT14C_HUMAN UTP14C 87.13 3.2114 3 6
sp|P16104|H2AX_HUMAN H2AFX 15.14 3.72 5 5 sp|P83916|CBX1_HUMAN CBX1 21.4 4.365 5 5
sp|P52292|IMA1_HUMAN KPNA2 57.83 4.1255 5 5 sp|Q9HCS7|SYF1_HUMAN XAB2 99.95 4.0454 5 5
sp|Q9UNX4|WDR3_HUMAN WDR3 106.03 3.9842 5 5 sp|P55265|DSRAD_HUMAN ADAR 135.98 3.8215 5 5
sp|Q9NU22|MDN1_HUMAN MDN1 632.42 3.7989 5 5 sp|Q12830|BPTF_HUMAN BPTF 338.05 3.7277 5 5
sp|O75400|PR40A_HUMAN PRPF40A 108.74 3.6993 5 5 sp|P49792|RBP2_HUMAN RANBP2 357.97 3.6351 5 5
sp|O94901|SUN1_HUMAN SUN1 90.01 3.6143 5 5 tr|A0A1P0AZG4|A0A1P0AZG4_HUMAN LCOR 137.14
3.5821 5 5 sp|Q9UQE7|SMC3_HUMAN SMC3 141.45 3.5623 5 5 sp|Q14137|BOP1_HUMAN BOP1 83.58 3.5549
5 5 sp|P26368|U2AF2_HUMAN U2AF2 53.47 3.477 5 5 sp|P68104|EF1A1_HUMAN EEF1A1 50.11 3.3958 5 5
sp|Q8WVM7|STAG1_HUMAN STAG1 144.34 3.3576 5 5 sp|Q969X6|UTP4_HUMAN UTP4 76.84 3.3295 5 5
sp|Q12769|NU160_HUMAN NUP160 162.02 3.3228 5 5 sp|Q8WUM0|NU133_HUMAN NUP133 128.9 3.2892 5 5
sp|Q53HL2|BOREA_HUMAN CDCA8 31.3 3.2744 5 5 sp|Q14684|RRP1B_HUMAN RRP1B 84.38 3.2428 5 5
sp|P49411|EFTU_HUMAN TUFM 49.51 3.1661 5 5 sp|Q9H582|ZN644_HUMAN ZNF644 149.47 3.1597 5 5
sp|P62241|RS8_HUMAN RPS8 24.19 3.0978 5 5 sp|Q14103|HNRPD_HUMAN HNRNPD 38.41 2.9787 5 5
sp|Q12931|TRAP1_HUMAN TRAP1 80.06 2.9565 5 5 sp|Q09028|RBBP4_HUMAN RBBP4 47.63 2.9205 5 5
sp|Q9BWN1|PRR14_HUMAN PRR14 64.29 2.8386 5 5 sp|Q96GD4|AURKB_HUMAN AURKB 39.29 2.8373 5 5
sp|Q14739|LBR_HUMAN LBR 70.66 2.8151 5 5 sp|Q13895|BYST_HUMAN BYSL 49.57 2.5969 5 5
sp|Q9NY12|GAR1_HUMAN GAR1 22.33 2.4622 4 5 sp|Q96EU6|RRP36_HUMAN RRP36 29.8 3.6465 4 5
sp|P14866|HNRPL_HUMAN HNRNPL 64.09 3.6299 4 5 sp|P14678|RSMB_HUMAN SNRPB 24.59 3.375 4 5
sp|Q96KR1|ZFR_HUMAN ZFR 116.94 2.776 4 5 sp|P63244|RACK1_HUMAN RACK1 35.05 2.595 4 5
sp|Q8WYB5|KAT6B_HUMAN KAT6B 231.23 2.3396 4 5 sp|P62269|RS18_HUMAN RPS18 17.71 2.3307 3 5
sp|Q7Z4V5|HDGR2_HUMAN HDGFL2 74.27 3.5225 3 5 sp|Q00325|MPCP_HUMAN SLC25A3 40.07 3.4595 3 5
sp|Q8TF01|PNISR_HUMAN PNISR 92.52 3.272 2 5 sp|P84103|SRSF3_HUMAN SRSF3 19.32 2.3291 2 5
sp|Q96PV6|LENG8_HUMAN LENG8 86.07 1.9933 4 4 sp|P32969|RL9_HUMAN RPL9 21.85 4.8881 4 4
sp|Q8N201|INT1_HUMAN INTS1 244.14 4.2509 4 4 sp|Q14498|RBM39_HUMAN RBM39 59.34 4.0875 4 4
sp|P12956|XRCC6_HUMAN XRCC6 69.8 3.9984 4 4 sp|Q13263|TIF1B_HUMAN TRIM28 88.49 3.7834 4 4
sp|Q96QD9|UIF_HUMAN FYTTD1 35.8 3.783 4 4 sp|Q9NPI1|BRD7_HUMAN BRD7 74.09 3.7737 4 4
sp|O60287|NPA1P_HUMAN URB1 254.23 3.7233 4 4 sp|P45880|VDAC2_HUMAN VDAC2 31.55 3.6583 4 4
sp|P62424|RL7A_HUMAN RPL7A 29.98 3.6514 4 4 sp|Q9Y6K1|DNM3A_HUMAN DNMT3A 101.79 3.5442 4 4
sp|Q9BYG3|MK671_HUMAN NIFK 34.2 3.5075 4 4 sp|P35580|MYH10_HUMAN MYH10 228.86 3.5027 4 4
sp|P11142|HSP7C_HUMAN HSPA8 70.85 3.4819 4 4 sp|O60508|PRP17_HUMAN CDC40 65.48 3.4393 4 4
sp|O96028|NSD2_HUMAN NSD2 152.16 3.359 4 4 sp|O15164|TIF1A_HUMAN TRIM24 116.76 3.3318 4 4
sp|Q8IX01|SUGP2_HUMAN SUGP2 120.13 3.3282 4 4 sp|Q14966|ZN638_HUMAN ZNF638 220.49 3.3182 4 4
sp|Q9NVI7|ATD3A_HUMAN ATAD3A 71.32 3.3045 4 4 sp|Q9Y6X3|SCC4_HUMAN MAU2 69.04 3.2931 4 4
sp|O94776|MTA2_HUMAN MTA2 74.98 3.2496 4 4 sp|Q69YN4|VIR_HUMAN KIAA1429 201.9 3.2469 4 4
sp|Q96L73|NSD1_HUMAN NSD1 296.46 3.2347 4 4 sp|Q9BTV4|TMM43_HUMAN TMEM43 44.85 3.22 4 4
sp|Q9H9B4|SFXN1_HUMAN SFXN1 35.6 3.2098 4 4 sp|O14979|HNRDL_HUMAN HNRNPDL 46.41 3.2084 4 4
sp|Q5JTV8|TOIP1_HUMAN TOR1AIP1 66.21 3.1934 4 4 sp|Q08170|SRSF4_HUMAN SRSF4 56.65 3.1185 4 4
sp|Q7L2E3|DHX30_HUMAN DHX30 133.85 3.0745 4 4 sp|O94906|PRP6_HUMAN PRPF6 106.86 3.0464 4 4
sp|P18124|RL7_HUMAN RPL7 29.21 3.0421 4 4 sp|P33991|MCM4_HUMAN MCM4 96.5 3.0269 4 4
sp|Q8IY37|DHX37_HUMAN DHX37 129.46 3.0184 4 4 sp|Q53GS7|GLE1_HUMAN GLE1 79.79 3.016 4 4
sp|P17844|DDX5_HUMAN DDX5 69.1 2.9945 4 4 sp|Q9H0D6|XRN2_HUMAN XRN2 108.51 2.9774 4 4
sp|Q15059|BRD3_HUMAN BRD3 79.49 2.9604 4 4 sp|Q96PK6|RBM14_HUMAN RBM14 69.45 2.9572 4 4
sp|Q96G21|IMP4_HUMAN IMP4 33.74 2.9255 4 4 sp|P35658|NU214_HUMAN NUP214 213.49 2.9007 4 4
sp|Q86YP4|P66A_HUMAN GATAD2A 68.02 2.8966 4 4 sp|Q07021|C1QBP_HUMAN C1QBP 31.34 2.8754 4 4
sp|Q6DKI1|RL7L_HUMAN RPL7L1 28.64 2.8526 4 4 sp|O43795|MYO1B_HUMAN MYO1B 131.9 2.8423 4 4
sp|Q86U38|NOP9_HUMAN NOP9 69.39 2.814 4 4 sp|Q96ME7|ZN512_HUMAN ZNF512 64.64 2.8133 4 4
sp|Q13242|SRSF9_HUMAN SRSF9 25.53 2.7693 4 4 sp|P62277|RS13_HUMAN RPS13 17.21 2.7392 4 4
sp|Q8N8A6|DDX51_HUMAN DDX51 72.41 2.7334 4 4 sp|P49756|RBM25_HUMAN RBM25 100.12 2.7244 4 4
sp|O75152|ZC11A_HUMAN ZC3H11A 89.08 2.6256 4 4 sp|P62750|RL23A_HUMAN RPL23A 17.68 2.5769 4 4
sp|Q96HS1|PGAM5_HUMAN PGAM5 31.98 2.4728 4 4 sp|P26373|RL13_HUMAN RPL13 24.25 2.3533 3 4
sp|O95218|ZRAB2_HUMAN ZRANB2 37.38 3.9985 3 4 sp|Q9NS69|TOM22_HUMAN TOMM22 15.51 3.9897 3 4
sp|Q9Y6A4|CFA20_HUMAN CFAP20 22.76 3.5149 3 4 sp|P78364|PHC1_HUMAN PHC1 105.47 3.4121 3 4
sp|Q96DI7|SNR40_HUMAN SNRNP40 39.29 3.1631 3 4 sp|Q9Y277|VDAC3_HUMAN VDAC3 30.64 2.7054 2 4
sp|P46783|RS10_HUMAN RPS10 18.89 2.7067 3 3 sp|P35453|HXD13_HUMAN HOXD13 36.08 4.379 3 3
sp|Q15459|SF3A1_HUMAN SF3A1 88.83 4.3736 3 3 sp|P56182|RRP1_HUMAN RRP1 52.81 4.3342 3 3

sp|P25705|ATPA_HUMAN ATP5A1 59.71 4.272 3 3 sp|P08708|RS17_HUMAN RPS17 15.54 4.2613 3 3
sp|P11021|GRP78_HUMAN HSPA5 72.29 4.2177 3 3 sp|Q92769|HDAC2_HUMAN HDAC2 55.33 4.101 3 3
sp|Q15424|SAFB1_HUMAN SAFB 102.58 4.0441 3 3 sp|Q8WVC0|LEO1_HUMAN LEO1 75.36 3.8904 3 3
sp|O15523|DDX3Y_HUMAN DDX3Y 73.11 3.8037 3 3 sp|P55201|BRPF1_HUMAN BRPF1 137.41 3.7974 3 3
sp|Q15050|RRS1_HUMAN RRS1 41.17 3.7946 3 3 sp|P62316|SMD2_HUMAN SNRPD2 13.52 3.6706 3 3
sp|P38432|COIL_HUMAN COIL 62.57 3.6496 3 3 sp|Q9P035|HACD3_HUMAN HACD3 43.13 3.5854 3 3
sp|Q9H8H0|NOL11_HUMAN NOL11 81.07 3.5795 3 3 sp|Q9HAF1|EAF6_HUMAN MEAF6 21.62 3.578 3 3
sp|Q8IWX8|CHERP_HUMAN CHERP 103.64 3.5545 3 3 sp|Q9BUJ2|HNRL1_HUMAN HNRNPUL1 95.68 3.5307
3 3 sp|Q9H8H2|DDX31_HUMAN DDX31 94.03 3.4706 3 3 sp|O94880|PHF14_HUMAN PHF14 99.99 3.4646 3 3
sp|Q5JTH9|RRP12_HUMAN RRP12 143.61 3.4367 3 3 sp|P56537|IF6_HUMAN EIF6 26.58 3.4285 3 3
sp|Q9UKJ3|GPTC8_HUMAN GPATCH8 164.1 3.3578 3 3 sp|Q07020|RL18_HUMAN RPL18 21.62 3.3553 3 3
sp|P06748|NPM_HUMAN NPM1 32.55 3.329 3 3 sp|Q3ZCQ8|TIM50_HUMAN TIMM50 39.62 3.3211 3 3
sp|Q14781|CBX2_HUMAN CBX2 56.05 3.3022 3 3 sp|Q9H7B2|RPF2_HUMAN RPF2 35.56 3.2995 3 3
sp|Q9UBB9|TFP11_HUMAN TFIP11 96.76 3.2993 3 3 sp|P61247|RS3A_HUMAN RPS3A 29.93 3.2546 3 3
sp|P35251|RFC1_HUMAN RFC1 128.18 3.2541 3 3 sp|Q99848|EBP2_HUMAN EBNA1BP2 34.83 3.2005 3 3
sp|P34931|HS71L_HUMAN HSPA1L 70.33 3.1654 3 3 sp|O95831|AIFM1_HUMAN AIFM1 66.86 3.1432 3 3
tr|F8VXC8|F8VXC8_HUMAN SMARCC2 136.1 3.1344 3 3 sp|Q5VT52|RPRD2_HUMAN RPRD2 155.92 3.1314
3 3 sp|Q13206|DDX10_HUMAN DDX10 100.83 3.0819 3 3 sp|P62081|RS7_HUMAN RPS7 22.11 3.0752 3 3
sp|Q01831|XPC_HUMAN XPC 105.89 3.0057 3 3 sp|O15226|NKRF_HUMAN NKRF 77.62 3.0054 3 3
sp|Q92522|H1X_HUMAN H1FX 22.47 2.9991 3 3 sp|Q8WXF0|SRS12_HUMAN SRSF12 30.49 2.9422 3 3
sp|Q9NRZ9|HELLS_HUMAN HELLS 97.01 2.8286 3 3 sp|P62851|RS25_HUMAN RPS25 13.73 2.8129 3 3
sp|P46781|RS9_HUMAN RPS9 22.58 2.7809 3 3 sp|Q13428|TCOF_HUMAN TCOF1 152.02 2.6949 3 3
sp|Q9H501|ESF1_HUMAN ESF1 98.73 2.6558 3 3 sp|Q9BZJ0|CRNL1_HUMAN CRNKL1 100.39 2.5946 3 3
sp|Q99496|RING2_HUMAN RNF2 37.63 2.5349 3 3 sp|Q8NI36|WDR36_HUMAN WDR36 105.26 2.4894 3 3
sp|Q9Y2R9|RT07_HUMAN MRPS7 28.12 2.4687 3 3 sp|P62249|RS16_HUMAN RPS16 16.44 2.4589 3 3
sp|P53999|TCP4_HUMAN SUB1 14.39 2.4304 3 3 sp|Q13123|RED_HUMAN IK 65.56 2.3999 3 3
sp|P30876|RPB2_HUMAN POLR2B 133.81 2.367 3 3 sp|P62906|RL10A_HUMAN RPL10A 24.82 2.2409 3 3
sp|Q96H22|CENPN_HUMAN CENPN 39.53 2.2018 3 3 sp|Q96T37|RBM15_HUMAN RBM15 107.12 2.0773 2 3
tr|B2R5W2|B2R5W2_HUMAN HNRNPC 31.93 5.509 2 3 sp|P67809|YBOX1_HUMAN YBX1 35.9 4.8382 2 3
sp|Q9Y5S9|RBM8A_HUMAN RBM8A 19.88 3.4783 2 3 sp|Q9UIS9|MBD1_HUMAN MBD1 66.56 3.2337 2 3
sp|Q66PJ3|AR6P4_HUMAN ARL6IP4 44.89 3.2302 2 3 sp|P17661|DESM_HUMAN DES 53.5 2.986 2 3
sp|P82650|RT22_HUMAN MRPS22 41.25 2.8833 2 3 sp|P04406|G3P_HUMAN GAPDH 36.03 2.5682 2 3
sp|P22090|RS4Y1_HUMAN RPS4Y1 29.44 2.4726 1 3 tr|Q6PJV4|Q6PJV4_HUMAN THRAP3 41.83 4.322 2 2
sp|P49711|CTCF_HUMAN CTCF 82.73 5.3155 2 2 sp|Q68CP9|ARID2_HUMAN ARID2 197.27 4.6815 2 2
sp|Q9GZL7|WDR12_HUMAN WDR12 47.68 4.5006 2 2 sp|Q96IZ7|RSRC1_HUMAN RSRC1 38.65 4.4356 2 2
sp|Q9UGU0|TCF20_HUMAN TCF20 211.64 4.4306 2 2 sp|P0DMV9|HS71B_HUMAN HSPA1B 70.01 4.4149 2 2
sp|Q14669|TRIPC_HUMAN TRIP12 220.3 4.3086 2 2 sp|Q5UIP0|RIF1_HUMAN RIF1 274.29 4.3075 2 2
sp|O00267|SPT5H_HUMAN SUPT5H 120.92 4.2498 2 2 sp|P05388|RLA0_HUMAN RPLP0 34.25 4.2162 2 2
sp|Q9BYN8|RT26_HUMAN MRPS26 24.2 4.1072 2 2 sp|Q5SRE5|NU188_HUMAN NUP188 195.92 4.0991 2 2
sp|Q15637|SF01_HUMAN SF1 68.29 4.0782 2 2 sp|Q96SB8|SMC6_HUMAN SMC6 126.25 3.9457 2 2
sp|P20226|TBP_HUMAN TBP 37.67 3.9353 2 2 sp|Q96MU7|YTDC1_HUMAN YTHDC1 84.65 3.9179 2 2
sp|Q92665|RT31_HUMAN MRPS31 45.29 3.8434 2 2 sp|Q92552|RT27_HUMAN MRPS27 47.58 3.8003 2 2
sp|Q9NV31|IMP3_HUMAN IMP3 21.84 3.7393 2 2 sp|P04844|RPN2_HUMAN RPN2 69.24 3.7317 2 2
sp|Q32P51|RA1L2_HUMAN HNRNPA1L2 34.2 3.6858 2 2 sp|P05141|ADT2_HUMAN SLC25A5 32.83 3.6825 2 2
sp|Q8N1GO|ZN687_HUMAN ZNF687 129.45 3.6672 2 2 sp|Q96EY7|PTCD3_HUMAN PTCD3 78.5 3.6382 2 2
sp|O75494|SRS10_HUMAN SRSF10 31.28 3.634 2 2 sp|O75531|BAF_HUMAN BANF1 10.05 3.6281 2 2
sp|Q13427|PPIG_HUMAN PPIG 88.56 3.6088 2 2 tr|A0A0A0MQS2|A0A0A0MQS2_HUMAN CLASRP 77.12
3.5848 2 2 sp|Q9UNQ2|DIM1_HUMAN DIMT1 35.21 3.5846 2 2 sp|P39023|RL3_HUMAN RPL3 46.08 3.569 2 2
sp|P41219|PERI_HUMAN PRPH 53.62 3.5672 2 2 sp|Q8WYH8|ING5_HUMAN ING5 27.73 3.5567 2 2
sp|Q15022|SUZ12_HUMAN SUZ12 83 3.543 2 2 sp|P26599|PTBP1_HUMAN PTBP1 57.19 3.5251 2 2
sp|P07196|NFL_HUMAN NEFL 61.48 3.5125 2 2 sp|Q69YH5|CDCA2_HUMAN CDCA2 112.61 3.5117 2 2
sp|O95478|NSA2_HUMAN NSA2 30.05 3.5067 2 2 sp|O75151|PHF2_HUMAN PHF2 120.7 3.5008 2 2
sp|P49736|MCM2_HUMAN MCM2 101.83 3.4994 2 2 sp|P35637|FUS_HUMAN FUS 53.39 3.4823 2 2
sp|P07437|TBB5_HUMAN TUBB 49.64 3.4431 2 2 sp|Q9Y3A2|UTP11_HUMAN UTP11 30.43 3.4209 2 2
sp|Q8TDN6|BRX1_HUMAN BRIX1 41.37 3.3781 2 2 sp|Q9HCG8|CWC22_HUMAN CWC22 105.4 3.3528 2 2
sp|Q9NQ39|RS10L_HUMAN RPS10P5 20.11 3.3483 2 2 sp|Q9UJS0|CMC2_HUMAN SLC25A13 74.13 3.345 2 2
sp|Q6PK04|CC137_HUMAN CCDC137 33.21 3.344 2 2 sp|Q5T280|CI114_HUMAN SPOUT1 41.98 3.3243 2 2
sp|Q9UJV9|DDX41_HUMAN DDX41 69.79 3.3239 2 2 sp|Q9GZS3|WDR61_HUMAN WDR61 33.56 3.3203 2 2
sp|Q13084|RM28_HUMAN MRPL28 30.14 3.314 2 2 sp|Q9NPF5|DMAP1_HUMAN DMAP1 52.96 3.3061 2 2

sp|P82933|RT09_HUMAN MRPS9 45.81 3.2952 2 2 sp|Q9UGL1|KDM5B_HUMAN KDM5B 175.54 3.2822 2 2 sp|Q5T3J3|LRIF1_HUMAN LRIF1 84.52 3.2788 2 2 sp|Q9GZR7|DDX24_HUMAN DDX24 96.27 3.2775 2 2 sp|P62304|RUXE_HUMAN SNRPE 10.8 3.2748 2 2 sp|Q8N0S6|CENPL_HUMAN CENPL 38.97 3.245 2 2 sp|Q14146|URB2_HUMAN URB2 170.43 3.1984 2 2 sp|Q02978|M2OM_HUMAN SLC25A11 34.04 3.1901 2 2 sp|Q9H6R0|DHX33_HUMAN DHX33 78.82 3.1845 2 2 sp|Q9HCD5|NCOA5_HUMAN NCOA5 65.5 3.1842 2 2 sp|Q86WX3|AROS_HUMAN RPS19BP1 15.42 3.1782 2 2 sp|Q9H4L4|SENP3_HUMAN SENP3 64.97 3.1604 2 2 sp|P61313|RL15_HUMAN RPL15 24.13 3.1313 2 2 sp|Q9BQF6|SENP7_HUMAN SENP7 119.58 3.1128 2 2 sp|P38646|GRP75_HUMAN HSPA9 73.63 3.1102 2 2 sp|Q5SY16|NOL9_HUMAN NOL9 79.27 3.107 2 2 sp|P07197|NFM_HUMAN NEFM 102.41 3.1027 2 2 sp|P31942|HNRH3_HUMAN HNRNPH3 36.9 3.0985 2 2 sp|Q9Y265|RUVB1_HUMAN RUVBL1 50.2 3.0511 2 2 sp|Q9Y3B9|RRP15_HUMAN RRP15 31.46 3.0423 2 2 sp|Q9NSI2|F207A_HUMAN FAM207A 25.44 3.0287 2 2 sp|Q6IQ32|ADNP2_HUMAN ADNP2 122.75 3.0046 2 2 sp|P55081|MFAP1_HUMAN MFAP1 51.93 2.9802 2 2 sp|P30050|RL12_HUMAN RPL12 17.81 2.9606 2 2 sp|O00257|CBX4_HUMAN CBX4 61.33 2.9486 2 2 sp|Q9BW27|NUP85_HUMAN NUP85 74.97 2.9405 2 2 sp|P83731|RL24_HUMAN RPL24 17.77 2.9294 2 2 sp|O95983|MBD3_HUMAN MBD3 32.82 2.9113 2 2 sp|P06576|ATPB_HUMAN ATP5B 56.52 2.9104 2 2 sp|O75530|EED_HUMAN EED 50.17 2.8773 2 2 sp|Q8NAV1|PR38A_HUMAN PRPF38A 37.45 2.8489 2 2 sp|O60318|GANP_HUMAN MCM3AP 218.27 2.8449 2 2 sp|Q02878|RL6_HUMAN RPL6 32.71 2.7827 2 2 sp|O94805|ACL6B_HUMAN ACTL6B 46.85 2.773 2 2 sp|P08621|RU17_HUMAN SNRNP70 51.53 2.7581 2 2 tr|A0A096LNW1|A0A096LNW1_HUMAN ATRX 106.9 2.7254 2 2 sp|P62318|SMD3_HUMAN SNRPD3 13.91 2.7076 2 2 sp|Q6P0N0|M18BP_HUMAN MIS18BP1 129.01 2.7046 2 2 sp|P13010|XRCC5_HUMAN XRCC5 82.65 2.6965 2 2 sp|P30414|NKTR_HUMAN NKTR 165.58 2.6914 2 2 sp|P08574|CY1_HUMAN CYC1 35.4 2.6877 2 2 sp|Q9P1Y6|PHRF1_HUMAN PHRF1 178.56 2.5945 2 2 sp|Q9NW13|RBM28_HUMAN RBM28 85.68 2.5606 2 2 sp|P18847|ATF3_HUMAN ATF3 20.56 2.5504 2 2 sp|Q9UQ88|CD11A_HUMAN CDK11A 91.31 2.4514 2 2 sp|Q9Y483|MTF2_HUMAN MTF2 67.06 2.3506 2 2 sp|Q9H6R4|NOL6_HUMAN NOL6 127.51 2.2599 2 2 sp|O00148|DX39A_HUMAN DDX39A 49.1 2.1977 2 2 sp|Q9P2K5|MYEF2_HUMAN MYEF2 64.08 2.1848 1 2 tr|Q68E03|Q68E03_HUMAN DKFZp686L22104 30.92 5.1845 1 2 sp|Q16576|RBBP7_HUMAN RBBP7 47.79 3.3349 1 2 sp|Q9BYX7|ACTBM_HUMAN POTEKP 41.99 3.1033 1 2 sp|P46777|RL5_HUMAN RPL5 34.34 2.8247 1 1 sp|O95299|NDUAA_HUMAN NDUFA10 40.72 5.8592 1 1 tr|A0A0M3HER2|A0A0M3HER2_HUMAN CENPV 18.8 5.626 1 1 sp|P04280|PRP1_HUMAN PRB1 38.52 5.4758 1 1 tr|Q53GL6|Q53GL6_HUMAN RALY 32.53 5.4233 1 1 tr|S4R341|S4R341_HUMAN NOLC1 8.05 5.3679 1 1 sp|Q9NW64|RBM22_HUMAN RBM22 46.87 5.1218 1 1 sp|Q9BYD2|RM09_HUMAN MRPL9 30.22 5.0825 1 1 sp|P46782|RS5_HUMAN RPS5 22.86 5.0071 1 1 sp|Q9NZM5|GSCR2_HUMAN GLTSCR2 54.36 4.9887 1 1 sp|Q92804|RBP56_HUMAN TAF15 61.79 4.88 1 1 sp|Q6NSZ9|ZSC25_HUMAN ZSCAN25 61.44 4.8752 1 1 sp|P16989|YBOX3_HUMAN YBX3 40.07 4.8301 1 1 tr|Q6IPH7|Q61PH7_HUMAN RPL14 23.77 4.7072 1 1 sp|Q9BQ04|RBM4B_HUMAN RBM4B 40.12 4.7065 1 1 sp|Q9HCK1|ZDBF2_HUMAN ZDBF2 265.45 4.7004 1 1 sp|Q5T653|RM02_HUMAN MRPL2 33.28 4.6443 1 1 sp|P62136|PP1A_HUMAN PPP1CA 37.49 4.5999 1 1 sp|Q92782|DPF1_HUMAN DPF1 42.47 4.5393 1 1 sp|Q13151|ROAO_HUMAN HNRNPA0 30.82 4.5336 1 1 sp|P35232|PHB_HUMAN PHB 29.79 4.5324 1 1 sp|Q9BTC0|DIDO1_HUMAN DIDO1 243.72 4.5276 1 1 sp|Q9Y230|RUVB2_HUMAN RUVBL2 51.12 4.517 1 1 sp|P82673|RT35_HUMAN MRPS35 36.82 4.5017 1 1 sp|Q99567|NUP88_HUMAN NUP88 83.49 4.4729 1 1 sp|P05412|JUN_HUMAN JUN 35.65 4.4638 1 1 sp|Q8NDX5|PHC3_HUMAN PHC3 106.1 4.4539 1 1 sp|P50914|RL14_HUMAN RPL14 23.42 4.3217 1 1 sp|O60506|HNRPQ_HUMAN SYNCRIP 69.56 4.2631 1 1 sp|Q14974|IMB1_HUMAN KPNB1 97.11 4.2381 1 1 sp|P62913|RL11_HUMAN RPL11 20.24 4.2253 1 1 sp|O95391|SLU7_HUMAN SLU7 68.34 4.215 1 1 sp|P09234|RU1C_HUMAN SNRPC 17.38 4.2105 1 1 tr|A8K7N0|A8K7N0_HUMAN 23.63 4.193 1 1 tr|B2RC06|B2RC06_HUMAN 39.25 4.162 1 1 sp|Q8TBK6|ZCH10_HUMAN ZCCHC10 20.95 4.1487 1 1 sp|Q9UBB5|MBD2_HUMAN MBD2 43.23 4.1327 1 1 IGKC_MOUSE 11.77 4.1063 1 1 sp|P37198|NUP62_HUMAN NUP62 53.22 4.0927 1 1 sp|P35579|MYH9_HUMAN MYH9 226.39 4.0738 1 1 sp|P22695|QCR2_HUMAN UQCRC2 48.41 4.0303 1 1 sp|P07355|ANXA2_HUMAN ANXA2 38.58 4.0038 1 1 sp|O15381|NVL_HUMAN NVL 94.99 3.9989 1 1 sp|Q07157|ZO1_HUMAN TJP1 195.34 3.9914 1 1 sp|Q9Y3C1|NOP16_HUMAN NOP16 21.18 3.9883 1 1 sp|Q9Y625|GPC6_HUMAN GPC6 62.69 3.9801 1 1 tr|B2RWN5|B2RWN5_HUMAN HEATR1 242.11 3.9638 1 1 sp|Q15910|EZH2_HUMAN EZH2 85.31 3.9454 1 1 sp|Q9NQV6|PRD10_HUMAN PRDM10 130.05 3.9287 1 1 sp|Q8TF76|HASP_HUMAN GSG2 88.44 3.8857 1 1 sp|Q9Y3B4|SF3B6_HUMAN SF3B6 14.58 3.8364 1 1 sp|P62314|SMD1_HUMAN SNRPD1 13.27 3.8351 1 1 tr|Q6FI97|Q6FI97_HUMAN BAF53A 47.35 3.8176 1 1 sp|Q9P013|CWC15_HUMAN CWC15 26.61 3.8025 1 1 sp|P62899|RL31_HUMAN RPL31 14.45 3.7783 1 1 sp|P82663|RT25_HUMAN MRPS25 20.1 3.7565 1 1 sp|P35249|RFC4_HUMAN RFC4 39.66 3.7202 1 1 sp|P49759|CLK1_HUMAN CLK1 57.25 3.7089 1 1 sp|Q9NVH1|DJC11_HUMAN DNAJC11 63.24 3.7074 1 1 sp|Q9UHR5|S30BP_HUMAN SAP30BP 33.85 3.691 1 1 tr|B0UZZ8|B0UZZ8_HUMAN C6orf11 68 3.6905 1 1 sp|Q8N9T8|KRI1_HUMAN KRI1 82.55 3.6636 1 1 sp|P51991|ROA3_HUMAN HNRNPA3 39.57 3.6618 1 1 sp|O14880|MGST3_HUMAN MGST3 16.51 3.6611 1 1 tr|Q05CW7|Q05CW7_HUMAN NAT10 62.35 3.6602 1 1 sp|O75934|SPF27_HUMAN BCAS2 26.11 3.6364 1 1 sp|Q8NHW5|RLA0L_HUMAN RPLP0P6 34.34 3.6321 1 1 sp|Q9NQ50|RM40_HUMAN MRPL40 24.48 3.629 1 1 sp|P12235|ADT1_HUMAN SLC25A4 33.04 3.6148 1 1 sp|O14519|CDKA1_HUMAN CDK2AP1 12.36 3.6134 1 1 sp|Q96EP5|DAZP1_HUMAN DAZAP1 43.36 3.6108 1 1 sp|Q9NYK5|RM39_HUMAN MRPL39 38.69 3.5747 1 1 sp|P05023|AT1A1_HUMAN ATP1A1 112.82 3.565 1 1 sp|P35250|RFC2_HUMAN RFC2 39.13 3.526 1 1 sp|Q9Y676|RT18B_HUMAN MRPS18B 29.38 3.5192 1 1 sp|O94832|MYO1D_HUMAN MYO1D 116.13 3.5141 1 1 sp|O60762|DPM1_HUMAN DPM1 29.62 3.5123 1 1 sp|Q9UBU9|NXF1_HUMAN NXF1 70.14 3.5123 1 1 sp|Q96JM3|CHAP1_HUMAN CHAMP1 89.04 3.5122 1 1 sp|Q9BYD3|RM04_HUMAN MRPL4 34.9 3.4916 1 1 sp|Q9H7H0|MET17_HUMAN METTL17 50.7 3.483 1 1 sp|Q9NX24|NHP2_HUMAN NHP2 17.19 3.472 1 1 tr|Q562V5|Q562V5_HUMAN ACT 11.42 3.4708 1 1 sp|Q9Y5Q9|TF3C3_HUMAN GTF3C3 101.21 3.4705 1 1 sp|Q9ULU4|PKCB1_HUMAN ZMYND8 131.61 3.4689 1 1 sp|Q9HBE1|PATZ1_HUMAN PATZ1 74.01 3.4686 1 1 sp|P51571|SSRD_HUMAN SSR4 18.99 3.4601 1 1 sp|Q06587|RING1_HUMAN RING1 42.4 3.4513 1 1 sp|O14795|UN13B_HUMAN UNC13B 180.56 3.4394 1 1 sp|P33992|MCM5_HUMAN MCM5 82.23 3.3963 1 1 sp|Q13148|TADBP_HUMAN TARDBP 44.71 3.3906 1 1 sp|Q8WY36|BBX_HUMAN BBX 105.06 3.3426 1 1 sp|Q86X18|CS068_HUMAN C19orf68 70.03 3.3403 1 1 tr|B7Z8Y3|B7Z8Y3_HUMAN 106.95 3.3372 1 1 sp|Q8NDF8|PAPD5_HUMAN PAPD5 63.23 3.3133 1 1 sp|P32119|PRDX2_HUMAN PRDX2 21.88 3.312 1 1 sp|Q96SK2|TM209_HUMAN TMEM209 62.88 3.3026 1 1 tr|D3DTH7|D3DTH7_HUMAN MYO1C 98.87 3.2974 1 1 tr|B4DR34|B4DR34_HUMAN 36.86 3.2621 1 1 sp|Q9NP66|HM20A_HUMAN HMG20A 40.12 3.2532 1 1 sp|O75955|FLOT1_HUMAN FLOT1 47.33 3.2455 1 1 sp|Q96GN5|CDA7L_HUMAN CDCA7L 52.17 3.2347 1 1 sp|Q7Z6E9|RBBP6_HUMAN RBBP6 201.44 3.2342 1 1 sp|A8CG34|P121C_HUMAN POM121C 124.98 3.2079 1 1 sp|P41208|CETN2_HUMAN CETN2 19.73 3.2075 1 1 sp|P62829|RL23_HUMAN RPL23 14.86 3.2074 1 1 sp|Q9NXE4|NSMA3_HUMAN SMPD4 93.29 3.2044 1 1 sp|Q86Y91|KI18B_HUMAN KIF18B 94.16 3.1821 1 1 sp|Q6DRA6|H2B2D_HUMAN HIST2H2BD 18.01 3.1771 1 1 sp|P62263|RS14_HUMAN RPS14 16.26 3.1597 1 1 sp|Q9UHX1|PUF60_HUMAN PUF60 59.84 3.1255 1 1 sp|E9PRG8|CK098_HUMAN C11orf98 13.79 3.1088 1 1 sp|Q75QN2|INT8_HUMAN INTS8 113.02 3.1011 1 1 sp|Q58FF8|H90B2_HUMAN HSP90AB2P 44.32 3.0769 1 1 sp|O75477|ERLN1_HUMAN ERLIN1 38.9 3.0624 1 1 sp|Q15843|NEDD8_HUMAN NEDD8 9.07 3.0468 1 1 sp|Q15365|PCBP1_HUMAN PCBP1 37.47 3.0189 1 1 sp|P0DMR1|HNRC4_HUMAN HNRNPCL4 32.01 3.0131 1 1 sp|Q9ULV3|CIZ1_HUMAN CIZ1 99.98 3.0112 1 1 sp|P00558|PGK1_HUMAN PGK1 44.59 3.0094 1 1 sp|Q8TE59|ATS19_HUMAN ADAMTS19 133.96 3.0037 1 1 sp|Q8NAP3|ZBT38_HUMAN ZBTB38 134.17 2.9639 1 1 KV2A7_MOUSE 12.27 2.9512 1 1 sp|Q8IXKO|PHC2_HUMAN PHC2 90.66 2.9338 1 1 sp|O60341|KDM1A_HUMAN KDM1A 92.84 2.9303 1 1 sp|Q9Y6F7|CDY2_HUMAN CDY2A 60.49 2.9091 1 1 sp|O95235|KI20A_HUMAN KIF20A 100.22 2.8771 1 1 sp|Q9UJZ1|STML2_HUMAN STOML2 38.51 2.8728 1 1 sp|P04843|RPN1_HUMAN RPN1 68.53 2.868 1 1 sp|Q9BVA1|TBB2B_HUMAN TUBB2B 49.92 2.8339 1 1 sp|P43246|MSH2_HUMAN MSH2 104.68 2.8127 1 1 sp|Q8WUB8|PHF10_HUMAN PHF10 56.02 2.7847 1 1 sp|Q5JVF3|PCID2_HUMAN PCID2 46 2.7464 1 1 sp|Q9Y3Y2|CHTOP_HUMAN CHTOP 26.38 2.714 1 1 sp|Q9Y2X9|ZN281_HUMAN ZNF281 96.85 2.6924 1 1 sp|Q53GQ0|DHB12_HUMAN HSD17B12 34.3 2.6858 1 1 tr|A0A024R5M9|A0A024R5M9_HUMAN NUMA1 236.37 2.6796 1 1 sp|P17010|ZFX_HUMAN ZFX 90.46 2.6696 1 1 sp|P54652|HSP72_HUMAN HSPA2 69.98 2.6579 1 1 sp|Q9UKD2|MRT4_HUMAN MRTO4 27.54 2.6442 1 1 sp|P12273|PIP_HUMAN PIP 16.56 2.6385 1 1 sp|Q14151|SAFB2_HUMAN SAFB2 107.41 2.6256 1 1 sp|Q9UHA3|RLP24_HUMAN RSL24D1 19.61 2.6188 1 1 sp|Q5H9F3|BCORL_HUMAN BCORL1 182.41 2.5997 1 1 sp|Q8NDD1|CA131_HUMAN C1orf131 32.75 2.5842 1 1 sp|Q9NWU5|RM22_HUMAN MRPL22 23.63 2.5746 1 1 sp|Q9NZ01|TECR_HUMAN TECR 36.01 2.5526 1 1 sp|Q16352|AINX_HUMAN INA 55.36 2.5377 1 1 sp|O15504|NUPL2_HUMAN NUPL2 44.84 2.5352 1 1 sp|Q9NR22|ANM8_HUMAN PRMT8 45.26 2.5291 1 1 sp|P68871|HBB_HUMAN HBB 15.99 2.5078 1 1 sp|Q5SNT2|TM201_HUMAN TMEM201 72.19 2.4788 1 1 sp|P46776|RL27A_HUMAN RPL27A 16.55 2.3924 1 1 sp|P13639|EF2_HUMAN EEF2 95.28 2.3879 1 1 sp|Q8NDX1|PSD4_HUMAN PSD4 116.18 2.347 1 1 sp|P62861|RS30_HUMAN FAU 6.64 2.3442 1 1 sp|P13804|ETFA_HUMAN ETFA 35.06 2.3145 1 1 sp|Q9NV06|DCA13_HUMAN DCAF13 51.37 2.3115 1 1 sp|O75190|DNJB6_HUMAN DNAJB6 36.06 2.3041 1 1 sp|P62857|RS28_HUMAN RPS28 7.84 2.2822 1 1 sp|Q9NY93|DDX56_HUMAN DDX56 61.55 2.2165 1 1 sp|P51784|UBP11_HUMAN USP11 109.75 2.2148 1 1 sp|Q9NVH2|INT7_HUMAN INTS7 106.77 2.2143 1 1 sp|O60673|REV3L_HUMAN REV3L 352.55 2.2069 1 1 sp|Q9P015|RM15_HUMAN MRPL15 33.4 2.2064 1 1 sp|P61353|RL27_HUMAN RPL27 15.79 2.1707 1 1 sp|O75323|NIPS2_HUMAN NIPSNAP2 33.72 2.1553 1 1 tr|Q9UMG4|Q9UMG4_HUMAN hANK1 3.65 2.1374 1 1 sp|Q8IXM6|NRM_HUMAN NRM 29.36 2.1313

TABLE-US-00015 TABLE 5B HA-SS18SSX1_NE_peptides Unique Total reference Gene Symbol MWT(kDa) AVG 48 308 sp|P51532|SMCA4_HUMAN SMARCA4 184.53 2.8323 71 221 sp|O14497|ARI1A_HUMAN ARID1A 241.89 3.2897 60 221 sp|P51531|SMCA2_HUMAN SMARCA2 181.17 2.9484 37 213 sp|Q92922|SMRC1_HUMAN SMARCC1 122.79 2.94 51 174 sp|Q8TAQ2|SMRC2_HUMAN SMARCC2 132.8 2.952 52 108 sp|Q8NFD5|ARI1B_HUMAN ARID1B 235.97 3.1352 27 94 sp|Q96GM5|SMRD1_HUMAN SMARCD1 58.2 3.4459 78 89 sp|P78527|PRKDC_HUMAN PRKDC 468.79 3.2311 34 89 sp|Q9NZM4|BICRA_HUMAN BICRA 158.39 3.2316 26 77 sp|Q969G3|SMCE1_HUMAN SMARCE1 46.62

3.4283 28 75 sp|Q6STE5|SMRD3_HUMAN SMARCD3 54.98 2.9487 19 74 sp|O96019|ACL6A_HUMAN ACTL6A 47.43 3.1679 21 46 sp|Q92925|SMRD2_HUMAN SMARCD2 58.88 3.1179 20 46 sp|P49411|EFTU_HUMAN TUFM 49.51 2.681 10 46 sp|P62736|ACTA_HUMAN ACTA2 41.98 2.4533 16 45 sp|Q12824|SNF5_HUMAN SMARCB1 44.11 2.9132 7 41 sp|P63261|ACTG_HUMAN ACTG1 41.77 3.2375 24 40 sp|Q9H8M2|BRD9_HUMAN BRD9 66.96 3.3845 24 39 sp|P25705|ATPA_HUMAN ATP5A1 59.71 3.347 14 35 sp|Q92785|REQU_HUMAN DPF2 44.13 3.6206 23 34 sp|Q9UJS0|CMC2_HUMAN SLC25A13 74.13 3.1068 14 31 sp|P12236|ADT3_HUMAN SLC25A6 32.85 2.5554 15 30 sp|P52272|HNRPM_HUMAN HNRNPM 77.46 3.2894 22 25 sp|P06576|ATPB_HUMAN ATP5B 56.52 3.6483 24 24 sp|O75643|U520_HUMAN SNRNP200 244.35 3.1225 23 23 sp|Q14980|NUMA1_HUMAN NUMA1 238.12 3.7937 23 23 sp|Q14204|DYHC1_HUMAN DYNC1H1 532.07 3.2127 18 23 sp|P05023|AT1A1_HUMAN ATP1A1 112.82 3.7257 15 23 sp|O95831|AIFM1_HUMAN AIFM1 66.86 3.2173 12 23 sp|P50402|EMD_HUMAN EMD 28.98 2.7585 12 22 sp|P68371|TBB4B_HUMAN TUBB4B 49.8 3.5227 10 22 sp|Q00325|MPCP_HUMAN SLC25A3 40.07 2.6968 16 21 sp|Q08211|DHX9_HUMAN DHX9 140.87 2.9913 14 21 sp|P11021|GRP78_HUMAN HSPA5 72.29 3.2487 2 21 sp|Q15532|SSXT_HUMAN SS18 45.9 2.4719 15 20 sp|P38646|GRP75_HUMAN HSPA9 73.63 3.5062 7 20 sp|Q4VC05|BCL7A_HUMAN BCL7A 22.8 3.3097 18 19 sp|P52701|MSH6_HUMAN MSH6 152.69 3.4726 11 19 sp|Q6AI39|BICRL_HUMAN BICRAL 115.01 3.2661 9 19 sp|P68104|EF1A1_HUMAN EEF1A1 50.11 3.1829 16 18 sp|P09874|PARP1_HUMAN PARP1 113.01 3.0283 16 18 sp|Q10570|CPSF1_HUMAN CPSF1 160.78 2.7977 16 17 sp|P20700|LMNB1_HUMAN LMNB1 66.37 3.3547 15 17 sp|Q16891|MIC60_HUMAN IMMT 83.63 3.5711 16 16 sp|Q8N1F7|NUP93_HUMAN NUP93 93.43 3.0683 13 16 sp|P40939|ECHA_HUMAN HADHA 82.95 3.3237 13 16 sp|O00567|NOP56_HUMAN NOP56 66.01 3.1798 9 16 sp|P62805|H4_HUMAN HIST1H4A 11.36 3.0197 14 15 sp|P42704|LPPRC_HUMAN LRPPRC 157.81 3.2864 13 15 sp|Q9UJV9|DDX41_HUMAN DDX41 69.79 2.6016 5 15 sp|Q71DI3|H32_HUMAN HIST2H3A 15.38 1.934 14 14 sp|Q6P2Q9|PRP8_HUMAN PRPF8 273.43 2.9384 13 14 sp|Q9NVI7|ATD3A_HUMAN ATAD3A 71.32 3.1974 13 14 sp|Q92621|NU205_HUMAN NUP205 227.78 3.1896 7 14 sp|Q8WUZ0|BCL7C_HUMAN BCL7C 23.45 3.5497 13 13 sp|Q9Y4W6|AFG32_HUMAN AFG3L2 88.53 3.0599 13 13 sp|Q9BUQ8|DDX23_HUMAN DDX23 95.52 3.0451 12 13 sp|Q9C0J8|WDR33_HUMAN WDR33 145.8 3.3842 11 13 sp|Q96A33|CCD47_HUMAN CCDC47 55.84 3.3429 9 13 sp|P43243|MATR3_HUMAN MATR3 94.56 2.9462 12 12 sp|P04844|RPN2_HUMAN RPN2 69.24 3.9026 12 12 sp|Q9Y5B6|PAXB1_HUMAN PAXBP1 104.74 3.5313 12 12 sp|A0FGR8|ESYT2_HUMAN ESYT2 102.29 3.287 10 12 sp|O75746|CMC1_HUMAN SLC25A12 74.71 3.0669 6 12 sp|P55795|HNRH2_HUMAN HNRNPH2 49.23 3.0612 5 12 sp|Q96KK5|H2A1H_HUMAN HIST1H2AH 13.9 2.5583 2 12 sp|P05141|ADT2_HUMAN SLC25A5 32.83 3.4462 11 11 sp|Q15029|U5S1_HUMAN EFTUD2 109.37 3.6326 11 11 sp|Q02978|M2OM_HUMAN SLC25A11 34.04 3.5222 10 11 sp|Q9Y2X3|NOP58_HUMAN NOP58 59.54 4.2232 10 11 sp|Q7L0Y3|MRRP1_HUMAN TRMT10C 47.32 2.9714 9 11 sp|P22695|QCR2_HUMAN UQCRC2 48.41 3.5992 9 11 sp|Q9BQE3|TBA1C_HUMAN TUBA1C 49.86 3.3342 9 11 sp|Q53GQ0|DHB12_HUMAN HSD17B12 34.3 3.2769 9 11 sp|Q6UN15|FIP1_HUMAN FIP1L1 66.49 3.1953 9 11 sp|Q9H9B4|SFXN1_HUMAN SFXN1 35.6 3.0828 8 11 sp|O75306|NDUS2_HUMAN NDUFS2 52.51 3.1642 5 11 sp|P33778|H2B1B_HUMAN HIST1H2BB 13.94 2.5844 10 10 sp|O14980|XPO1_HUMAN XPO1 123.31 3.5888 10 10 sp|Q6NUK1|SCMC1_HUMAN SLC25A24 53.32 3.4159 10 10 sp|P04181|OAT_HUMAN OAT 48.5 3.4156 10 10 sp|Q96I99|SUCB2_HUMAN SUCLG2 46.48 3.3701 10 10 sp|Q86WJ1|CHD1L_HUMAN CHD1L 100.92 3.2369 10 10 sp|P10809|CH60_HUMAN HSPD1 61.02 2.9893 10 10 sp|Q9UBB9|TFP11_HUMAN TFIP11 96.76 2.8601 10 10 sp|Q9BQG0|MBB1A_HUMAN MYBBP1A 148.76 2.733 4 10 sp|Q16384|SSX1_HUMAN SSX1 21.92 3.3189 4 10 IGH1M_MOUSE lghg1 43.36 2.7234 9 9 sp|Q96TA2|YMEL1_HUMAN YME1L1 86.4 4.2777 9 9 sp|P30837|AL1B1_HUMAN ALDH1B1 57.17 4.1396 9 9 sp|P35251|RFC1_HUMAN RFC1 128.18 3.6089 9 9 sp|Q16822|PCKGM_HUMAN PCK2 70.68 3.2754 9 9 sp|Q9NTI5|PDS5B_HUMAN PDS5B 164.56 2.6945 8 9 sp|P11142|HSP7C_HUMAN HSPA8 70.85 3.7908 8 9 sp|Q12769|NU160_HUMAN NUP160 162.02 3.4687 8 9 sp|Q15758|AAAT_HUMAN SLC1A5 56.56 3.2819 8 9 sp|P55084|ECHB_HUMAN HADHB 51.26 3.1472 7 9 sp|P21796|VDAC1_HUMAN VDAC1 30.75 3.0601 7 9 sp|Q9P035|HACD3_HUMAN HACD3 43.13 2.84 6 9 sp|Q5T280|CI114_HUMAN SPOUT1 41.98 3.0837 6 9 sp|P22087|FBRL_HUMAN FBL 33.76 2.914 6 9 sp|P36542|ATPG_HUMAN ATP5C1 32.98 2.6524 6 9 sp|Q9P2R7|SUCB1_HUMAN SUCLA2 50.29 2.518 8 8 sp|Q29RF7|PDS5A_HUMAN PDS5A 150.73 3.8078 8 8 sp|P11310|ACADM_HUMAN ACADM 46.56 3.2564 8 8 sp|Q9NUL7|DDX28_HUMAN DDX28 59.54 3.2067 8 8 sp|Q92841|DDX17_HUMAN DDX17 80.22 3.1678 8 8 sp|P50570|DYN2_HUMAN DNM2 98 3.0011 8 8 sp|Q5SRE5|NU188_HUMAN NUP188 195.92 2.437 7 8 sp|P16615|AT2A2_HUMAN ATP2A2 114.68 3.0714 7 8 sp|Q8N8A6|DDX51_HUMAN DDX51 72.41 3.0036 6 8 sp|Q15637|SF01_HUMAN SF1 68.29 3.1692 6 8 sp|P53985|MOT1_HUMAN SLC16A1 53.91 2.686 7 7 sp|Q53H12|AGK_HUMAN AGK 47.11 3.4069 7 7 sp|O43175|SERA_HUMAN PHGDH 56.61 3.249 7 7 sp|P45880|VDAC2_HUMAN VDAC2 31.55 3.0615 7 7 sp|P53621|COPA_HUMAN COPA 138.26 3.0594 7 7 sp|P48047|ATPO_HUMAN ATP5O 23.26 2.8331 7 7 sp|P46977|STT3A_HUMAN STT3A 80.48 2.6162 6 7 sp|P42167|LAP2B_HUMAN TMPO 50.64 3.4633 6 7 sp|O14983|AT2A1_HUMAN ATP2A1 110.18 3.3261 6 7 sp|Q96T37|RBM15_HUMAN RBM15 107.12 2.7759 6 6 sp|O75489|NDUS3_HUMAN NDUFS3 30.22 3.7176 6 6

sp|Q9NXE4|NSMA3_HUMAN SMPD4 93.29 3.5606 6 6 sp|Q92576|PHF3_HUMAN PHF3 229.34 3.519 6 6
sp|Q9Y2R4|DDX52_HUMAN DDX52 67.46 3.4811 6 6 sp|P38432|COIL_HUMAN COIL 62.57 3.3461 6 6
sp|Q12931|TRAP1_HUMAN TRAP1 80.06 3.332 6 6 sp|O75367|H2AY_HUMAN H2AFY 39.59 3.2768 6 6
sp|Q9P2I0|CPSF2_HUMAN CPSF2 88.43 3.2559 6 6 sp|Q96NB2|SFXN2_HUMAN SFXN2 36.21 3.2502 6 6
sp|P43246|MSH2_HUMAN MSH2 104.68 3.2395 6 6 sp|Q92616|GCN1_HUMAN GCN1 292.57 3.0917 6 6
sp|O60313|OPA1_HUMAN OPA1 111.56 3.0052 6 6 sp|Q6JQN1|ACD10_HUMAN ACAD10 118.76 2.8882 5 6
sp|P35250|RFC2_HUMAN RFC2 39.13 4.048 5 6 sp|P53618|COPB_HUMAN COPB1 107.07 3.5299 5 6
sp|Q969V3|NCLN_HUMAN NCLN 62.93 3.1625 5 6 sp|P50213|IDH3A_HUMAN IDH3A 39.57 3.0362 5 6
sp|P49590|SYHM_HUMAN HARS2 56.85 3.0295 5 6 sp|P53007|TXTP_HUMAN SLC25A1 33.99 2.6822 5 6
sp|Q96CS3|FAF2_HUMAN FAF2 52.59 2.5882 4 6 sp|P50416|CPT1A_HUMAN CPT1A 88.31 3.1041 5 5
sp|Q9HC07|TM165_HUMAN TMEM165 34.88 4.4695 5 5 sp|Q9NVH2|INT7_HUMAN INTS7 106.77 3.8125 5 5
sp|P52597|HNRPF_HUMAN HNRNPF 45.64 3.783 5 5 sp|Q9HCM4|E41L5_HUMAN EPB41L5 81.8 3.7077 5 5
sp|P45954|ACDSB_HUMAN ACADSB 47.46 3.5191 5 5 sp|P35613|BASI_HUMAN BSG 42.17 3.5115 5 5
sp|Q9NSE4|SYIM_HUMAN IARS2 113.72 3.5039 5 5 sp|Q9UKM7|MA1B1_HUMAN MAN1B1 79.53 3.3681 5 5
sp|P33993|MCM7_HUMAN MCM7 81.26 3.3539 5 5 sp|Q14683|SMC1A_HUMAN SMC1A 143.14 3.3185 5 5
sp|Q3SY69|AL1L2_HUMAN ALDH1L2 101.68 3.3065 5 5 sp|Q8IXI1|MIRO2_HUMAN RHOT2 68.07 3.2378 5 5
sp|Q08945|SSRP1_HUMAN SSRP1 81.02 3.2282 5 5 sp|P55786|PSA_HUMAN NPEPPS 103.21 3.17 5 5
sp|P18074|ERCC2_HUMAN ERCC2 86.85 3.1631 5 5 sp|Q96EY1|DNJA3_HUMAN DNAJA3 52.46 3.1378 5 5
sp|Q9UH62|ARMX3_HUMAN ARMCX3 42.47 2.9452 5 5 sp|P39656|OST48_HUMAN DDOST 50.77 2.9053 5 5
sp|Q8NDT2|RB15B_HUMAN RBM15B 97.15 2.8733 5 5 sp|Q8IXI2|MIRO1_HUMAN RHOT1 70.74 2.8581 5 5
sp|Q9Y305|ACOT9_HUMAN ACOT9 49.87 2.7301 5 5 sp|Q8IZL8|PELP1_HUMAN PELP1 119.62 2.691 5 5
sp|P49756|RBM25_HUMAN RBM25 100.12 2.6029 4 5 sp|O00411|RPOM_HUMAN POLRMT 138.53 3.9144 4 5
sp|Q92784|DPF3_HUMAN DPF3 43.06 3.5795 4 5 sp|P31943|HNRH1_HUMAN HNRNPH1 49.2 3.2832 4 5
sp|P24539|AT5F1_HUMAN ATP5F1 28.89 3.0786 4 5 sp|P34931|HS71L_HUMAN HSPA1L 70.33 3.0763 4 5
sp|Q14739|LBR_HUMAN LBR 70.66 3.0611 4 5 sp|P11177|ODPB_HUMAN PDHB 39.21 2.8626 3 5
sp|P40938|RFC3_HUMAN RFC3 40.53 2.7258 4 4 sp|O75600|KBL_HUMAN GCAT 45.26 4.4371 4 4
sp|Q9UH99|SUN2_HUMAN SUN2 80.26 4.1201 4 4 sp|Q5T9A4|ATD3B_HUMAN ATAD3B 72.53 4.0914 4 4
sp|Q49A26|GLYR1_HUMAN GLYR1 60.52 3.9554 4 4 sp|Q9BW27|NUP85_HUMAN NUP85 74.97 3.9325 4 4
sp|Q8TED0|UTP15_HUMAN UTP15 58.38 3.9212 4 4 sp|P28331|NDUS1_HUMAN NDUFS1 79.42 3.8974 4 4
sp|Q3ZCQ8|TIM50_HUMAN TIMM50 39.62 3.8468 4 4 sp|Q9BQE9|BCL7B_HUMAN BCL7B 22.18 3.7012 4 4
sp|O14828|SCAM3_HUMAN SCAMP3 38.26 3.6839 4 4 sp|P0DMV9|HS71B_HUMAN HSPA1B 70.01 3.6515 4 4
sp|Q14684|RRP1B_HUMAN RRP1B 84.38 3.6515 4 4 sp|P35249|RFC4_HUMAN RFC4 39.66 3.6418 4 4
sp|Q96SK2|TM209_HUMAN TMEM209 62.88 3.6406 4 4 sp|Q9NX63|MIC19_HUMAN CHCHD3 26.14 3.628 4 4
sp|P26368|U2AF2_HUMAN U2AF2 53.47 3.6016 4 4 sp|A6NJ78|MET15_HUMAN METTL15 46.09 3.5927 4 4
sp|O75400|PR40A_HUMAN PRPF40A 108.74 3.5794 4 4 sp|O15269|SPTC1_HUMAN SPTLC1 52.71 3.5413 4 4
sp|O43615|TIM44_HUMAN TIMM44 51.32 3.5271 4 4 sp|Q9Y5B9|SP16H_HUMAN SUPT16H 119.84 3.4159 4 4
sp|Q14974|IMB1_HUMAN KPNB1 97.11 3.3547 4 4 sp|O60762|DPM1_HUMAN DPM1 29.62 3.2916 4 4
sp|Q9NRK6|ABCBA_HUMAN ABCB10 79.1 3.2617 4 4 sp|Q01831|XPC_HUMAN XPC 105.89 3.2417 4 4
sp|Q9BSD7|NTPCR_HUMAN NTPCR 20.7 3.2165 4 4 sp|Q9P0J0|NDUAD_HUMAN NDUFA13 16.69 3.1941 4 4
sp|Q9UJZ1|STML2_HUMAN STOML2 38.51 3.13 4 4 sp|Q13601|KRR1_HUMAN KRR1 43.64 3.1253 4 4
sp|P04843|RPN1_HUMAN RPN1 68.53 3.0947 4 4 sp|Q9BPW8|NIPS1_HUMAN NIPSNAP1 33.29 3.0705 4 4
sp|Q9H9P8|L2HDH_HUMAN L2HGDH 50.28 3.0617 4 4 sp|P17844|DDX5_HUMAN DDX5 69.1 3.0394 4 4
sp|O43809|CPSF5_HUMAN NUDT21 26.21 3.0246 4 4 sp|P26641|EF1G_HUMAN EEF1G 50.09 3.0126 4 4
sp|Q96HW7|INT4_HUMAN INTS4 108.1 3.0126 4 4 sp|Q8WYP5|ELYS_HUMAN AHCTF1 252.34 2.9942 4 4
sp|O43837|IDH3B_HUMAN IDH3B 42.16 2.9905 4 4 sp|Q03701|CEBPZ_HUMAN CEBPZ 120.9 2.9709 4 4
sp|Q14966|ZN638_HUMAN ZNF638 220.49 2.9484 4 4 sp|P00367|DHE3_HUMAN GLUD1 61.36 2.9254 4 4
sp|O76031|CLPX_HUMAN CLPX 69.18 2.9124 4 4 sp|Q9UDR5|AASS_HUMAN AASS 102.07 2.8915 4 4
sp|Q9BVA1|TBB2B_HUMAN TUBB2B 49.92 2.8539 4 4 sp|Q9NXF1|TEX10_HUMAN TEX10 105.61 2.846 4 4
sp|Q9H936|GHC1_HUMAN SLC25A22 34.45 2.8371 4 4 sp|P08559|ODPA_HUMAN PDHA1 43.27 2.8323 4 4
sp|P46821|MAP1B_HUMAN MAP1B 270.47 2.6704 4 4 sp|Q96PK6|RBM14_HUMAN RBM14 69.45 2.6673 4 4
sp|Q9NR30|DDX21_HUMAN DDX21 87.29 2.5986 4 4 sp|Q9NRG9|AAAS_HUMAN AAAS 59.54 2.5796 4 4
sp|A3KMH1|VWA8_HUMAN VWA8 214.69 2.2576 4 4 sp|Q86UT6|NLRX1_HUMAN NLRX1 107.55 2.2278 4 4
sp|Q9ULK5|VANG2_HUMAN VANGL2 59.68 2.1911 3 4 sp|P48735|IDHP_HUMAN IDH2 50.88 3.7507 3 4
sp|P13804|ETFA_HUMAN ETFA 35.06 3.5786 3 4 sp|Q9H2S9|IKZF4_HUMAN IKZF4 64.07 3.5119 3 4
sp|Q9H7H0|MET17_HUMAN METTL17 50.7 3.5054 3 4 sp|Q12905|ILF2_HUMAN ILF2 43.04 3.3415 3 4
sp|Q96EP5|DAZP1_HUMAN DAZAP1 43.36 3.3015 2 4 sp|Q93079|H2B1H_HUMAN HIST1H2BH 13.88 2.1574
3 3 sp|Q00839|HNRPU_HUMAN HNRNPU 90.53 4.8236 3 3 sp|Q5UIP0|RIF1_HUMAN RIF1 274.29 4.3695 3 3
sp|Q9Y5M8|SRPRB_HUMAN SRPRB 29.68 4.3599 3 3 sp|Q96BW9|TAM41_HUMAN TAMM41 51.03 4.3564 3 3
sp|Q8NI60|COQ8A_HUMAN COQ8A 71.9 4.3561 3 3 sp|P08195|4F2_HUMAN SLC3A2 67.95 4.1193 3 3

sp|O94813|SLIT2_HUMAN SLIT2 169.76 4.0788 3 3 sp|Q9UQE7|SMC3_HUMAN SMC3 141.45 4.0404 3 3
sp|Q6NSZ9|ZSC25_HUMAN ZSCAN25 61.44 4.0337 3 3 sp|P51648|AL3A2_HUMAN ALDH3A2 54.81 4.0321 3 3 sp|Q6IAN0|DRS7B_HUMAN DHRS7B 35.1 4.0295 3 3 sp|Q14498|RBM39_HUMAN RBM39 59.34 3.9624 3 3
sp|P23634|AT2B4_HUMAN ATP2B4 137.83 3.9226 3 3 sp|Q13435|SF3B2_HUMAN SF3B2 100.16 3.8939 3 3
sp|Q07021|C1QBP_HUMAN C1QBP 31.34 3.806 3 3 sp|O75616|ERAL1_HUMAN ERAL1 48.32 3.7878 3 3
sp|A8CG34|P121C_HUMAN POM121C 124.98 3.7762 3 3 sp|O94906|PRP6_HUMAN PRPF6 106.86 3.7503 3 3
sp|P53597|SUCA_HUMAN SUCLG1 36.23 3.721 3 3 sp|Q86Y07|VRK2_HUMAN VRK2 58.1 3.697 3 3
sp|Q5JTV8|TOIP1_HUMAN TOR1AIP1 66.21 3.6383 3 3 sp|P32189|GLPK_HUMAN GK 61.21 3.5989 3 3
sp|P51571|SSRD_HUMAN SSR4 18.99 3.5681 3 3 sp|Q96D53|COQ8B_HUMAN COQ8B 60.03 3.559 3 3
sp|P02545|LMNA_HUMAN LMNA 74.09 3.5564 3 3 tr|F8VXC8|F8VXC8_HUMAN SMARCC2 136.1 3.4843 3 3
tr|Q3B7X4|Q3B7X4_HUMAN IMMT 40.47 3.4796 3 3 sp|Q9Y679|AUP1_HUMAN AUP1 52.99 3.3718 3 3
sp|P13010|XRCC5_HUMAN XRCC5 82.65 3.3491 3 3 sp|P32322|P5CR1_HUMAN PYCR1 33.34 3.3089 3 3
sp|Q9UMS4|PRP19_HUMAN PRPF19 55.15 3.2974 3 3 sp|P12956|XRCC6_HUMAN XRCC6 69.8 3.2967 3 3
sp|P49755|TMEDA_HUMAN TMED10 24.96 3.2947 3 3 sp|Q9Y678|COPG1_HUMAN COPG1 97.66 3.2391 3 3
sp|P43307|SSRA_HUMAN SSR1 32.22 3.212 3 3 sp|O75592|MYCB2_HUMAN MYCBP2 509.76 3.1986 3 3
sp|Q14103|HNRPD_HUMAN HNRNPD 38.41 3.1817 3 3 sp|Q9BX10|GTPB2_HUMAN GTPBP2 65.73 3.1504 3 3
sp|O95299|NDUAA_HUMAN NDUFA10 40.72 3.1004 3 3 sp|P49792|RBP2_HUMAN RANBP2 357.97 3.071 3 3
sp|P28288|ABCD3_HUMAN ABCD3 75.43 3.0169 3 3 sp|Q92947|GCDH_HUMAN GCDH 48.1 3.0023 3 3
sp|Q9BW92|SYTM_HUMAN TARS2 80.99 2.9891 3 3 sp|O43824|GTPB6_HUMAN GTPBP6 56.85 2.9889 3 3
sp|Q9H583|HEAT1_HUMAN HEATR1 242.22 2.9281 3 3 sp|Q9H0U3|MAGT1_HUMAN MAGT1 38.01 2.916 3 3
sp|Q8IXB1|DJC10_HUMAN DNAJC10 91.02 2.9101 3 3 sp|P57088|TMM33_HUMAN TMEM33 27.96 2.8935 3 3
sp|P20020|AT2B1_HUMAN ATP2B1 138.67 2.8139 3 3 sp|Q9H857|NT5D2_HUMAN NT5DC2 60.68 2.811 3 3
sp|P54136|SYRC_HUMAN RARS 75.33 2.791 3 3 sp|P40937|RFC5_HUMAN RFC5 38.47 2.7891 3 3
sp|Q9BTX1|NDC1_HUMAN NDC1 76.26 2.7416 3 3 sp|Q5T160|SYRM_HUMAN RARS2 65.46 2.7067 3 3
sp|Q9Y4W2|LAS1L_HUMAN LAS1L 83.01 2.6959 3 3 sp|Q9NZ01|TECR_HUMAN TECR 36.01 2.6888 3 3
sp|Q8WWC4|MAIP1_HUMAN MAIP1 32.52 2.6849 3 3 sp|Q9UL03|INT6_HUMAN INTS6 100.33 2.6767 3 3
sp|Q13123|RED_HUMAN IK 65.56 2.6747 3 3 sp|O60318|GANP_HUMAN MCM3AP 218.27 2.6195 3 3
sp|Q9HBE1|PATZ1_HUMAN PATZ1 74.01 2.6111 3 3 sp|O94874|UFL1_HUMAN UFL1 89.54 2.5668 3 3
sp|Q96HS1|PGAM5_HUMAN PGAM5 31.98 2.5394 3 3 sp|O43143|DHX15_HUMAN DHX15 90.88 2.4658 3 3
sp|P63092|GNAS2_HUMAN GNAS 45.64 2.3729 3 3 sp|Q9HC21|TPC_HUMAN SLC25A19 35.49 2.0792 2 3
sp|Q01650|LAT1_HUMAN SLC7A5 54.97 4.3582 2 3 sp|O95674|CDS2_HUMAN CDS2 51.38 4.0086 2 3
sp|O75251|NDUS7_HUMAN NDUFS7 23.55 3.655 2 3 sp|Q9UBM7|DHCR7_HUMAN DHCR7 54.45 3.0868 2 3
sp|Q9Y277|VDAC3_HUMAN VDAC3 30.64 3.0513 2 3 sp|P04792|HSPB1_HUMAN HSPB1 22.77 2.9885 2 3
sp|O14925|TIM23_HUMAN TIMM23 21.93 2.985 2 3 sp|P14618|KPYM_HUMAN PKM 57.9 2.9626 2 3
sp|Q9UM00|TMCO1_HUMAN TMCO1 21.16 2.6723 2 3 sp|Q92782|DPF1_HUMAN DPF1 42.47 1.8419 2 2
sp|Q9NS69|TOM22_HUMAN TOMM22 15.51 5.021 2 2 sp|Q99805|TM9S2_HUMAN TM9SF2 75.73 4.898 2 2
sp|A1L0T0|ILVBL_HUMAN ILVBL 67.82 4.8795 2 2 sp|Q9BSF4|TIM29_HUMAN TIMM29 29.22 4.7691 2 2
sp|Q9NNW5|WDR6_HUMAN WDR6 121.65 4.6122 2 2 sp|Q70CQ3|UBP30_HUMAN USP30 58.47 4.5749 2 2
sp|Q9GZR7|DDX24_HUMAN DDX24 96.27 4.5606 2 2 sp|P09622|DLDH_HUMAN DLD 54.14 4.506 2 2
sp|P00403|COX2_HUMAN MT-CO2 25.55 4.3826 2 2 sp|Q8TB37|NUBPL_HUMAN NUBPL 34.06 4.2762 2 2
sp|Q5JPH6|SYEM_HUMAN EARS2 58.65 4.2723 2 2 sp|P07910|HNRPC_HUMAN HNRNPC 33.65 4.2686 2 2
sp|P47985|UCRI_HUMAN UQCRFS1 29.65 4.2492 2 2 sp|Q9NPI1|BRD7_HUMAN BRD7 74.09 4.2027 2 2
sp|P41208|CETN2_HUMAN CETN2 19.73 4.1872 2 2 sp|O00165|HAX1_HUMAN HAX1 31.6 4.1829 2 2
sp|P55265|DSRAD_HUMAN ADAR 135.98 4.1552 2 2 sp|Q9NUQ2|PLCE_HUMAN AGPAT5 42.04 4.1535 2 2
sp|Q68CQ7|GL8D1_HUMAN GLT8D1 41.91 4.0458 2 2 sp|Q5HYI7|MTX3_HUMAN MTX3 35.07 4.0096 2 2
sp|P07437|TBB5_HUMAN TUBB 49.64 3.9765 2 2 sp|P13674|P4HA1_HUMAN P4HA1 61.01 3.9399 2 2
sp|Q8WY36|BBX_HUMAN BBX 105.06 3.912 2 2 sp|Q6DD88|ATLA3_HUMAN ATL3 60.5 3.8159 2 2
sp|O00116|ADAS_HUMAN AGPS 72.87 3.8089 2 2 sp|P03886|NU1M_HUMAN MT-ND1 35.64 3.785 2 2
sp|O15270|SPTC2_HUMAN SPTLC2 62.88 3.7695 2 2 sp|Q9UDX5|MTFP1_HUMAN MTFP1 18 3.7587 2 2
sp|P52292|IMA1_HUMAN KPNA2 57.83 3.7421 2 2 sp|O95639|CPSF4_HUMAN CPSF4 30.23 3.7332 2 2
sp|Q96C36|P5CR2_HUMAN PYCR2 33.62 3.6947 2 2 sp|P11498|PYC_HUMAN PC 129.55 3.6878 2 2
sp|O00410|IPO5_HUMAN IPO5 123.55 3.6863 2 2 sp|O14654|IRS4_HUMAN IRS4 133.68 3.685 2 2
sp|Q9NVH1|DJC11_HUMAN DNAJC11 63.24 3.6387 2 2 sp|Q86VP6|CAND1_HUMAN CAND1 136.29 3.6258 2 2 sp|P05412|JUN_HUMAN JUN 35.65 3.6102 2 2 sp|Q00587|BORG5_HUMAN CDC42EP1 40.27 3.6034 2 2
sp|P62987|RL40_HUMAN UBA52 14.72 3.5869 2 2 sp|P12004|PCNA_HUMAN PCNA 28.75 3.5839 2 2
sp|Q9NRZ9|HELLS_HUMAN HELLS 97.01 3.5446 2 2 sp|Q15393|SF3B3_HUMAN SF3B3 135.49 3.5349 2 2
sp|Q9NVI1|FANCI_HUMAN FANCI 149.23 3.532 2 2 sp|P33527|MRP1_HUMAN ABCC1 171.48 3.5224 2 2
sp|Q03252|LMNB2_HUMAN LMNB2 69.91 3.5134 2 2 sp|O00400|ACATN_HUMAN SLC33A1 60.87 3.5111 2 2
sp|Q92522|H1X_HUMAN H1FX 22.47 3.4867 2 2 sp|Q9UBU9|NXF1_HUMAN NXF1 70.14 3.471 2 2

sp|Q15365|PCBP1_HUMAN PCBP1 37.47 3.4358 2 2 sp|Q9H061|T126A_HUMAN TMEM126A 21.51 3.4112 2 2 sp|P52429|DGKE_HUMAN DGKE 63.88 3.3745 2 2 sp|O00257|CBX4_HUMAN CBX4 61.33 3.344 2 2 sp|P54886|P5CS_HUMAN ALDH18A1 87.25 3.3363 2 2 sp|O43491|E41L2_HUMAN EPB41L2 112.52 3.3288 2 2 sp|Q5JTZ9|SYAM_HUMAN AARS2 107.27 3.3224 2 2 sp|P08574|CY1_HUMAN CYC1 35.4 3.3097 2 2 sp|P48651|PTSS1_HUMAN PTDSS1 55.49 3.2672 2 2 sp|Q53R41|FAKD1_HUMAN FASTKD1 97.35 3.263 2 2 sp|Q92542|NICA_HUMAN NCSTN 78.36 3.2618 2 2 sp|Q8IY17|PLPL6_HUMAN PNPLA6 149.9 3.2524 2 2 sp|P30825|CTR1_HUMAN SLC7A1 67.59 3.2044 2 2 sp|O15321|TM9S1_HUMAN TM9SF1 68.82 3.1856 2 2 sp|P31689|DNJA1_HUMAN DNAJA1 44.84 3.1792 2 2 sp|Q5JVF3|PCID2_HUMAN PCID2 46 3.1732 2 2 sp|Q8WUK0|PTPM1_HUMAN PTPMT1 22.83 3.1618 2 2 sp|O75396|SC22B_HUMAN SEC22B 24.58 3.154 2 2 sp|Q9Y2J2|E41L3_HUMAN EPB41L3 120.6 3.1507 2 2 sp|O95573|ACSL3_HUMAN ACSL3 80.37 3.1429 2 2 sp|Q96N66|MBOA7_HUMAN MBOAT7 52.73 3.0857 2 2 sp|Q9UG63|ABCF2_HUMAN ABCF2 71.24 3.0777 2 2 sp|Q10469|MGAT2_HUMAN MGAT2 51.52 3.06 2 2 sp|Q68CP9|ARID2_HUMAN ARID2 197.27 3.0415 2 2 sp|Q8NC56|LEMD2_HUMAN LEMD2 56.94 3.0367 2 2 sp|Q9UBD5|ORC3_HUMAN ORC3 82.2 3.0212 2 2 sp|O00159|MYO1C_HUMAN MYO1C 121.61 3.0205 2 2 sp|P56134|ATPK_HUMAN ATP5J2 10.91 3.0031 2 2 sp|P38117|ETFB_HUMAN ETFB 27.83 3.001 2 2 sp|P12235|ADT1_HUMAN SLC25A4 33.04 2.929 2 2 sp|Q5C9Z4|NOM1_HUMAN NOM1 96.2 2.8956 2 2 sp|Q8TCT9|HM13_HUMAN HM13 41.46 2.8954 2 2 sp|Q9H2D1|MFTC_HUMAN SLC25A32 35.38 2.892 2 2 sp|P17987|TCPA_HUMAN TCP1 60.31 2.8893 2 2 sp|Q9BYN8|RT26_HUMAN MRPS26 24.2 2.8798 2 2 sp|Q13428|TCOF_HUMAN TCOF1 152.02 2.8687 2 2 sp|Q8N442|GUF1_HUMAN GUF1 74.28 2.8578 2 2 sp|Q9NSI2|F207A_HUMAN FAM207A 25.44 2.8529 2 2 sp|O75323|NIPS2_HUMAN NIPSNAP2 33.72 2.8489 2 2 sp|Q8NF37|PCAT1_HUMAN LPCAT1 59.11 2.8362 2 2 sp|P46087|NOP2_HUMAN NOP2 89.25 2.83 2 2 sp|Q15120|PDK3_HUMAN PDK3 46.91 2.8249 2 2 sp|Q9UGN5|PARP2_HUMAN PARP2 66.16 2.8221 2 2 sp|Q12788|TBL3_HUMAN TBL3 88.98 2.8072 2 2 sp|Q13505|MTX1_HUMAN MTX1 51.44 2.802 2 2 sp|Q9H0A0|NAT10_HUMAN NAT10 115.66 2.785 2 2 sp|Q5VV42|CDKAL_HUMAN CDKAL1 65.07 2.7816 2 2 sp|Q9H223|EHD4_HUMAN EHD4 61.14 2.7606 2 2 sp|P38435|VKGC_HUMAN GGCX 87.5 2.7259 2 2 sp|Q99459|CDC5L_HUMAN CDC5L 92.19 2.7222 2 2 sp|Q99653|CHP1_HUMAN CHP1 22.44 2.7104 2 2 sp|Q9HD45|TM9S3_HUMAN TM9SF3 67.84 2.6916 2 2 sp|Q9Y4A5|TRRAP_HUMAN TRRAP 437.32 2.6864 2 2 sp|Q9Y6J9|TAF6L_HUMAN TAF6L 67.77 2.6785 2 2 sp|O95347|SMC2_HUMAN SMC2 135.57 2.6624 2 2 sp|Q92544|TM9S4_HUMAN TM9SF4 74.47 2.6428 2 2 sp|P11387|TOP1_HUMAN TOP1 90.67 2.6258 2 2 sp|Q9P032|NDUF4_HUMAN NDUFAF4 20.25 2.619 2 2 sp|Q12906|ILF3_HUMAN ILF3 95.28 2.6162 2 2 sp|Q9BXW7|HDHD5_HUMAN HDHD5 46.29 2.6115 2 2 sp|P49821|NDUV1_HUMAN NDUFV1 50.78 2.5884 2 2 sp|O00192|ARVC_HUMAN ARVCF 104.58 2.521 2 2 sp|Q9BVK6|TMED9_HUMAN TMED9 27.26 2.5109 2 2 sp|Q6ZXV5|TMTC3_HUMAN TMTC3 103.94 2.5017 2 2 sp|O76062|ERG24_HUMAN TM7SF2 46.38 2.4483 2 2 sp|Q9Y512|SAM50_HUMAN SAMM50 51.94 2.4034 2 2 sp|Q5TA45|INT11_HUMAN INTS11 67.62 2.4003 2 2 sp|P28370|SMCA1_HUMAN SMARCA1 122.53 2.3685 2 2 sp|P17480|UBF1_HUMAN UBTF 89.35 2.3649 2 2 sp|P84103|SRSF3_HUMAN SRSF3 19.32 2.2759 2 2 sp|Q14527|HLTF_HUMAN HLTF 113.86 2.2533 2 2 sp|P61619|S61A1_HUMAN SEC61A1 52.23 2.2427 2 2 sp|Q9Y2R9|RT07_HUMAN MRPS7 28.12 2.1584 2 2 sp|Q9BSJ2|GCP2_HUMAN TUBGCP2 102.47 1.8316 1 2 sp|Q01130|SRSF2_HUMAN SRSF2 25.46 4.3513 1 2 sp|P16104|H2AX_HUMAN H2AFX 15.14 3.4679 1 2 sp|O94805|ACL6B_HUMAN ACTL6B 46.85 3.0937 1 2 sp|Q99959|PKP2_HUMAN PKP2 97.35 3.0573 1 2 sp|Q9BYX7|ACTBM_HUMAN POTEKP 41.99 3.0304 1 2 sp|Q9Y3E0|GOT1B_HUMAN GOLT1B 15.42 2.5183 1 2 sp|Q9Y2X0|MED16_HUMAN MED16 96.73 2.3369 1 2 sp|Q9P104|DOK5_HUMAN DOK5 35.44 2.0759 1 1 sp|Q9NPL8|TIDC1_HUMAN TIMMDC1 32.16 5.5872 1 1 sp|Q75QN2|INT8_HUMAN INTS8 113.02 5.1795 1 1 sp|Q7L8L6|FAKD5_HUMAN FASTKD5 86.52 5.1458 1 1 sp|Q15287|RNPS1_HUMAN RNPS1 34.19 4.9637 1 1 sp|Q8WUA4|TF3C2_HUMAN GTF3C2 100.62 4.8457 1 1 sp|Q9Y2Q3|GSTK1_HUMAN GSTK1 25.48 4.8083 1 1 sp|Q9H845|ACAD9_HUMAN ACAD9 68.72 4.7486 1 1 sp|P61978|HNRPK_HUMAN HNRNPK 50.94 4.7158 1 1 sp|O94766|B3GA3_HUMAN B3GAT3 37.1 4.7011 1 1 sp|O15554|KCNN4_HUMAN KCNN4 47.66 4.684 1 1 sp|O60884|DNJA2_HUMAN DNAJA2 45.72 4.6496 1 1 sp|P62995|TRA2B_HUMAN TRA2B 33.65 4.633 1 1 sp|Q8N6R0|MET13_HUMAN METTL13 78.72 4.5641 1 1 tr|B7ZAF6|B7ZAF6_HUMAN SUCLA2 43.83 4.5502 1 1 sp|Q04837|SSBP_HUMAN SSBP1 17.25 4.5487 1 1 sp|Q6B0I6|KDM4D_HUMAN KDM4D 58.57 4.5398 1 1 sp|P56962|STX17_HUMAN STX17 33.38 4.5351 1 1 sp|Q9Y3D7|TIM16_HUMAN PAM16 13.82 4.4422 1 1 sp|P08708|RS17_HUMAN RPS17 15.54 4.4124 1 1 sp|Q9BT22|ALG1_HUMAN ALG1 52.48 4.402 1 1 tr|Q2M1J6|Q2M1J6_HUMAN OXA1L 55.35 4.3966 1 1 sp|O75528|TADA3_HUMAN TADA3 48.87 4.3954 1 1 sp|Q9UNQ2|DIM1_HUMAN DIMT1 35.21 4.3536 1 1 sp|O95470|SGPL1_HUMAN SGPL1 63.48 4.3015 1 1 sp|Q86Y91|KI18B_HUMAN KIF18B 94.16 4.2375 1 1 sp|Q9NWW5|CLN6_HUMAN CLN6 35.9 4.2186 1 1 sp|Q969Y2|GTPB3_HUMAN GTPBP3 52.03 4.1992 1 1 sp|O43159|RRP8_HUMAN RRP8 50.68 4.1825 1 1 sp|Q8N6L1|KTAP2_HUMAN KRTCAP2 14.67 4.1782 1 1 sp|O75531|BAF_HUMAN BANF1 10.05 4.1148 1 1 sp|O94887|FARP2_HUMAN FARP2 119.81 4.1008 1 1 sp|Q96GC9|VMP1_HUMAN VMP1 46.21 4.0888 1 1 sp|Q9BPX6|MICU1_HUMAN MICU1 54.32 4.0864 1 1 sp|O43347|MSI1H_HUMAN MSI1 39.1 4.0779 1 1 tr|B4DR34|B4DR34_HUMAN 36.86 4.074 1 1 sp|Q8TAE8|G45IP_HUMAN GADD45GIP1 25.37 4.0691 1 1

sp|Q8N465|D2HDH_HUMAN D2HGDH 56.38 4.0528 1 1 sp|O95140|MFN2_HUMAN MFN2 86.35 4.0323 1 1
sp|P00387|NB5R3_HUMAN CYB5R3 34.21 4.0153 1 1 sp|Q15334|L2GL1_HUMAN LLGL1 115.35 4.0019 1 1
sp|Q9UHX1|PUF60_HUMAN PUF60 59.84 3.985 1 1 sp|Q96DA6|TIM14_HUMAN DNAJC19 12.49 3.9537 1 1
sp|O43390|HNRPR_HUMAN HNRNPR 70.9 3.9371 1 1 sp|P62820|RAB1A_HUMAN RAB1A 22.66 3.9346 1 1
sp|Q9HDC5|JPH1_HUMAN JPH1 71.64 3.9134 1 1 sp|Q8TBP6|S2540_HUMAN SLC25A40 38.1 3.8986 1 1
sp|O75352|MPU1_HUMAN MPDU1 26.62 3.8941 1 1 sp|O75533|SF3B1_HUMAN SF3B1 145.74 3.8767 1 1
sp|P35558|PCKGC_HUMAN PCK1 69.15 3.8646 1 1 sp|Q15459|SF3A1_HUMAN SF3A1 88.83 3.8594 1 1
sp|O14880|MGST3_HUMAN MGST3 16.51 3.8593 1 1 sp|Q6NTF9|RHBD2_HUMAN RHBDD2 39.18 3.819 1 1
sp|Q15233|NONO_HUMAN NONO 54.2 3.7878 1 1 sp|Q9H4L4|SENP3_HUMAN SENP3 64.97 3.785 1 1
sp|Q96QD8|S38A2_HUMAN SLC38A2 55.99 3.7721 1 1 sp|Q9H8H3|MET7A_HUMAN METTL7A 28.3 3.7715 1 1 sp|Q9NVH0|EXD2_HUMAN EXD2 70.31 3.7582 1 1 sp|Q9H5Q4|TFB2M_HUMAN TFB2M 45.32 3.7433 1 1
sp|Q8IWA4|MFN1_HUMAN MFN1 84.05 3.742 1 1 tr|J3KN66|J3KN66_HUMAN TOR1AIP1 67.78 3.734 1 1
sp|O00148|DX39A_HUMAN DDX39A 49.1 3.7271 1 1 sp|O75976|CBPD_HUMAN CPD 152.84 3.726 1 1
sp|Q3SXY8|AR13B_HUMAN ARL13B 48.61 3.7066 1 1 sp|P35232|PHB_HUMAN PHB 29.79 3.701 1 1
sp|Q5SY16|NOL9_HUMAN NOL9 79.27 3.6809 1 1 sp|Q9Y3A6|TMED5_HUMAN TMED5 25.99 3.655 1 1
sp|Q9BZE1|RM37_HUMAN MRPL37 48.09 3.6351 1 1 sp|Q8NBN7|RDH13_HUMAN RDH13 35.91 3.6132 1 1
sp|Q6NUN9|ZN746_HUMAN ZNF746 69.09 3.6064 1 1 sp|P41252|SYIC_HUMAN IARS 144.41 3.601 1 1
sp|Q96BN2|TADA1_HUMAN TADA1 37.36 3.5969 1 1 sp|P54652|HSP72_HUMAN HSPA2 69.98 3.5946 1 1
sp|Q02338|BDH_HUMAN BDH1 38.13 3.5915 1 1 tr|A0A1B0GTJ8|A0A1B0GTJ8_HUMAN ARID1B 163.28 3.5712 1 1 sp|Q9Y5Y0|FLVC1_HUMAN FLVCR1 59.82 3.5588 1 1 sp|Q9H0H0|INT2_HUMAN INTS2 134.24 3.534 1 1 tr|H7BXI1|H7BXI1_HUMAN ESYT2 97.95 3.5218 1 1 sp|O00571|DDX3X_HUMAN DDX3X 73.2 3.5084 1 1 sp|Q6P9B9|INT5_HUMAN INTS5 107.93 3.4997 1 1 sp|Q969X6|UTP4_HUMAN UTP4 76.84 3.4982 1 1 sp|Q7Z5K2|WAPL_HUMAN WAPL 132.86 3.4906 1 1 sp|Q8NI27|THOC2_HUMAN THOC2 182.66 3.4868 1 1
sp|Q6YN16|HSDL2_HUMAN HSDL2 45.37 3.4781 1 1 sp|P83731|RL24_HUMAN RPL24 17.77 3.4775 1 1
sp|Q5BKZ1|ZN326_HUMAN ZNF326 65.61 3.4679 1 1 IGKC_MOUSE 11.77 3.4619 1 1
sp|Q5T0B9|ZN362_HUMAN ZNF362 45.79 3.4575 1 1 sp|Q16698|DECR_HUMAN DECR1 36.04 3.4391 1 1
sp|Q6PIW4|FIGL1_HUMAN FIGNL1 74.03 3.4381 1 1 sp|Q14517|FAT1_HUMAN FAT1 505.96 3.4316 1 1
sp|Q8TAA9|VANG1_HUMAN VANGL1 59.94 3.4174 1 1 sp|Q8NBI6|XXLT1_HUMAN XXYLT1 43.78 3.3993 1 1
sp|P61225|RAP2B_HUMAN RAP2B 20.49 3.3924 1 1 sp|Q9P2E9|RRBP1_HUMAN RRBP1 152.38 3.3909 1 1
sp|O14776|TCRG1_HUMAN TCERG1 123.82 3.3829 1 1 sp|P08670|VIME_HUMAN VIM 53.62 3.3766 1 1
sp|P68363|TBA1B_HUMAN TUBA1B 50.12 3.3736 1 1 sp|P62316|SMD2_HUMAN SNRPD2 13.52 3.3706 1 1
sp|Q8WY07|CTR3_HUMAN SLC7A3 67.13 3.3446 1 1 sp|O95070|YIF1A_HUMAN YIF1A 31.99 3.3423 1 1
sp|Q9Y289|SC5A6_HUMAN SLC5A6 68.6 3.3106 1 1 sp|P62701|RS4X_HUMAN RPS4X 29.58 3.3014 1 1
sp|Q9BVJ6|UT14A_HUMAN UTP14A 87.92 3.2855 1 1 sp|Q9Y3T9|NOC2L_HUMAN NOC2L 84.87 3.2781 1 1
sp|P05067|A4_HUMAN APP 86.89 3.2681 1 1 sp|P98194|AT2C1_HUMAN ATP2C1 100.51 3.2635 1 1
sp|Q06587|RING1_HUMAN RING1 42.4 3.2538 1 1 sp|Q13148|TADBP_HUMAN TARDBP 44.71 3.2457 1 1
tr|Q59G16|Q59G16_HUMAN 127.3 3.2436 1 1 sp|Q9UKU7|ACAD8_HUMAN ACAD8 45.04 3.2308 1 1
sp|P06493|CDK1_HUMAN CDK1 34.07 3.2145 1 1 sp|Q86VI3|IQGA3_HUMAN IQGAP3 184.58 3.2033 1 1
sp|Q71UI9|H2AV_HUMAN H2AFV 13.5 3.2018 1 1 sp|Q9Y584|TIM22_HUMAN TIMM22 20.02 3.1815 1 1
sp|Q96AA3|RFT1_HUMAN RFT1 60.3 3.1606 1 1 sp|O60830|TI17B_HUMAN TIMM17B 18.26 3.1592 1 1
sp|Q8TEM1|PO210_HUMAN NUP210 204.98 3.1521 1 1 sp|P62241|RS8_HUMAN RPS8 24.19 3.1446 1 1
sp|P04406|G3P_HUMAN GAPDH 36.03 3.1344 1 1 sp|Q96DA2|RB39B_HUMAN RAB39B 24.61 3.1279 1 1
sp|Q9ULH0|KDIS_HUMAN KIDINS220 196.42 3.1272 1 1 sp|O60264|SMCA5_HUMAN SMARCA5 121.83 3.1176 1 1 sp|Q15572|TAF1C_HUMAN TAF1C 95.15 3.1073 1 1 sp|Q9Y6M5|ZNT1_HUMAN SLC30A1 55.26 3.1001 1 1 sp|P19404|NDUV2_HUMAN NDUFV2 27.37 3.0934 1 1 sp|Q9Y2W1|TR150_HUMAN THRAP3 108.6 3.092 1 1 sp|Q9H0U9|TSYL1_HUMAN TSPYL1 49.16 3.0855 1 1 sp|Q9Y256|FACE2_HUMAN RCE1 35.81 3.0848 1 1 sp|O43913|ORC5_HUMAN ORC5 50.25 3.0745 1 1 sp|O00483|NDUA4_HUMAN NDUFA4 9.36 3.0468 1 1 sp|Q15366|PCBP2_HUMAN PCBP2 38.56 3.0445 1 1 sp|P35659|DEK_HUMAN DEK 42.65 3.0425 1 1
sp|P24468|COT2_HUMAN NR2F2 45.54 3.0302 1 1 sp|Q16563|SYPL1_HUMAN SYPL1 28.55 3.0247 1 1
sp|Q9H300|PARL_HUMAN PARL 42.16 3.0049 1 1 sp|P05091|ALDH2_HUMAN ALDH2 56.35 2.975 1 1
sp|Q13415|ORC1_HUMAN ORC1 97.29 2.9662 1 1 sp|Q9Y5A9|YTHD2_HUMAN YTHDF2 62.3 2.954 1 1
sp|P68871|HBB_HUMAN HBB 15.99 2.9512 1 1 sp|O14656|TOR1A_HUMAN TOR1A 37.78 2.9413 1 1
sp|Q9UJ14|GGT7_HUMAN GGT7 70.42 2.9379 1 1 sp|O95168|NDUB4_HUMAN NDUFB4 15.2 2.9313 1 1
sp|Q15392|DHC24_HUMAN DHCR24 60.06 2.9241 1 1 sp|Q9ULK4|MED23_HUMAN MED23 156.37 2.9228 1 1
sp|Q09161|NCBP1_HUMAN NCBP1 91.78 2.9153 1 1 sp|Q9Y3Z3|SAMH1_HUMAN SAMHD1 72.15 2.9067 1 1
tr|B3KUE6|B3KUE6_HUMAN 30.19 2.8998 1 1 sp|Q92643|GPI8_HUMAN PIGK 45.22 2.8928 1 1
sp|Q9BUN8|DERL1_HUMAN DERL1 28.78 2.8902 1 1 sp|Q9BQ39|DDX50_HUMAN DDX50 82.51 2.889 1 1
sp|Q9HAV4|XPO5_HUMAN XPO5 136.22 2.8793 1 1 sp|O75964|ATP5L_HUMAN ATP5L 11.42 2.8756 1 1
sp|Q9NTJ3|SMC4_HUMAN SMC4 147.09 2.8579 1 1 sp|Q14008|CKAP5_HUMAN CKAP5 225.35 2.8551 1 1

sp|Q8NCH0|CHSTE_HUMAN CHST14 42.97 2.8501 1 1 sp|Q92604|LGAT1_HUMAN LPGAT1 43.06 2.8286 1 1 sp|P50454|SERPH_HUMAN SERPINH1 46.41 2.8283 1 1 sp|P56192|SYMC_HUMAN MARS 101.05 2.8127 1 1 sp|Q5JU69|TOR2A_HUMAN TOR2A 35.69 2.8096 1 1 sp|Q9Y232|CDYL1_HUMAN CDYL 66.44 2.7991 1 1 sp|P50897|PPT1_HUMAN PPT1 34.17 2.7912 1 1 sp|Q9NZ08|ERAP1_HUMAN ERAP1 107.17 2.7866 1 1 sp|Q9NQ50|RM40_HUMAN MRPL40 24.48 2.7723 1 1 sp|Q96GQ7|DDX27_HUMAN DDX27 89.78 2.766 1 1 sp|O14735|CDIPT_HUMAN CDIPT 23.52 2.7651 1 1 sp|O95409|ZIC2_HUMAN ZIC2 54.97 2.7592 1 1 sp|Q3SXM5|HSDL1_HUMAN HSDL1 36.98 2.7545 1 1 sp|Q86Y39|NDUAB_HUMAN NDUFA11 14.84 2.7501 1 1 sp|Q9UBF2|COPG2_HUMAN COPG2 97.56 2.7403 1 1 sp|Q8TCJ2|STT3B_HUMAN STT3B 93.61 2.736 1 1 sp|O14802|RPC1_HUMAN POLR3A 155.54 2.7169 1 1 sp|Q15269|PWP2_HUMAN PWP2 102.39 2.711 1 1 KV2A7_MOUSE 12.27 2.7025 1 1 sp|Q12830|BPTF_HUMAN BPTF 338.05 2.6977 1 1 sp|Q8N8L6|ARL10_HUMAN ARL10 27.44 2.6741 1 1 sp|Q9Y697|NFS1_HUMAN NFS1 50.16 2.6592 1 1 sp|Q92797|SYMPK_HUMAN SYMPK 141.06 2.6557 1 1 sp|Q9UI10|EI2BD_HUMAN EIF2B4 57.52 2.6485 1 1 sp|Q9BVQ7|SPA5L_HUMAN SPATA5L1 80.66 2.6331 1 1 sp|Q15388|TOM20_HUMAN TOMM20 16.29 2.6315 1 1 sp|P82933|RT09_HUMAN MRPS9 45.81 2.6192 1 1 sp|Q9BXW9|FACD2_HUMAN FANCD2 164.02 2.6144 1 1 sp|Q6PML9|ZNT9_HUMAN SLC30A9 63.47 2.6142 1 1 sp|Q8TBF5|PIGX_HUMAN PIGX 28.77 2.6142 1 1 sp|Q9Y5J1|UTP18_HUMAN UTP18 61.96 2.6063 1 1 sp|Q86TJ2|TAD2B_HUMAN TADA2B 48.44 2.5991 1 1 sp|P68366|TBA4A_HUMAN TUBA4A 49.89 2.5969 1 1 sp|Q6DRA6|H2B2D_HUMAN HIST2H2BD 18.01 2.5889 1 1 sp|Q9BWM7|SFXN3_HUMAN SFXN3 35.48 2.5811 1 1 sp|Q96H55|MYO19_HUMAN MYO19 109.07 2.5717 1 1 sp|Q9Y4F1|FARP1_HUMAN FARP1 118.56 2.5565 1 1 sp|Q8N684|CPSF7_HUMAN CPSF7 52.02 2.5537 1 1 sp|Q8NFQ8|TOIP2_HUMAN TOR1AIP2 51.23 2.5378 1 1 sp|O43823|AKAP8_HUMAN AKAP8 76.06 2.5116 1 1 sp|O15260|SURF4_HUMAN SURF4 30.37 2.5109 1 1 sp|P52948|NUP98_HUMAN NUP98 197.46 2.5109 1 1 sp|Q8WVM8|SCFD1_HUMAN SCFD1 72.33 2.4935 1 1 sp|P14678|RSMB_HUMAN SNRPB 24.59 2.483 1 1 sp|Q12962|TAF10_HUMAN TAF10 21.7 2.4811 1 1 sp|Q8NB90|SPAT5_HUMAN SPATA5 97.84 2.4769 1 1 tr|Q96DP0|Q96DP0_HUMAN 49.22 2.4718 1 1 sp|Q8NBU5|ATAD1_HUMAN ATAD1 40.72 2.4518 1 1 sp|P10515|ODP2_HUMAN DLAT 68.95 2.4454 1 1 sp|Q643R3|LPCT4_HUMAN LPCAT4 57.18 2.4415 1 1 sp|P23258|TBG1_HUMAN TUBG1 51.14 2.4387 1 1 sp|Q9Y6K0|CEPT1_HUMAN CEPT1 46.52 2.4294 1 1 sp|Q9Y230|RUVB2_HUMAN RUVBL2 51.12 2.4188 1 1 sp|O60725|ICMT_HUMAN ICMT 31.92 2.4021 1 1 sp|Q9BYW2|SETD2_HUMAN SETD2 287.42 2.3755 1 1 sp|Q8N4U5|T11L2_HUMAN TCP11L2 58.05 2.3706 1 1 sp|Q9H2V7|SPNS1_HUMAN SPNS1 56.59 2.3667 1 1 sp|Q9BSK2|S2533_HUMAN SLC25A33 35.35 2.3493 1 1 sp|P51530|DNA2_HUMAN DNA2 120.34 2.3388 1 1 sp|Q9Y265|RUVB1_HUMAN RUVBL1 50.2 2.3132 1 1 sp|Q15043|S39AE_HUMAN SLC39A14 54.18 2.3094 1 1 sp|Q6P4A7|SFXN4_HUMAN SFXN4 37.97 2.3065 1 1 sp|Q92878|RAD50_HUMAN RAD50 153.8 2.2886 1 1 sp|P31327|CPSM_HUMAN CPS1 164.83 2.2879 1 1 sp|P61353|RL27_HUMAN RPL27 15.79 2.2777 1 1 sp|Q8N6I1|EID2_HUMAN EID2 25.17 2.2718 1 1 sp|Q9NZJ7|MTCH1_HUMAN MTCH1 41.52 2.2539 1 1 sp|P33121|ACSL1_HUMAN ACSL1 77.89 2.2518 1 1 sp|Q8NHH9|ATLA2_HUMAN ATL2 66.19 2.2317 1 1 sp|Q7L2E3|DHX30_HUMAN DHX30 133.85 2.2277 1 1 sp|Q6VAB6|KSR2_HUMAN KSR2 107.56 2.2255 1 1 sp|Q9BVK2|ALG8_HUMAN ALG8 60.05 2.2052 1 1 sp|P35914|HMGCL_HUMAN HMGCL 34.34 2.1941 1 1 sp|Q15526|SURF1_HUMAN SURF1 33.31 2.1791 1 1 sp|O14681|EI24_HUMAN EI24 38.94 2.1725 1 1 sp|Q9ULD4|BRPF3_HUMAN BRPF3 135.66 2.1675 1 1 sp|Q02539|H11_HUMAN HIST1H1A 21.83 2.15

TABLE-US-00016 TABLE 5C HA-SS18WT_CHR_peptides Unique Total reference Gene Symbol MWT(kDa) AVG 43 156 sp|P51532|SMCA4_HUMAN SMARCA4 184.53 3.0834 46 128 sp|P51531|SMCA2_HUMAN SMARCA2 181.17 2.9702 8 108 sp|P33778|H2B1B_HUMAN HIST1H2BB 13.94 2.3822 45 69 sp|O14497|ARI1A_HUMAN ARID1A 241.89 3.2558 24 52 sp|Q8TAQ2|SMRC2_HUMAN SMARCC2 132.8 3.3159 34 50 sp|Q8NFD5|ARI1B_HUMAN ARID1B 235.97 3.1174 24 43 sp|Q92922|SMRC1_HUMAN SMARCC1 122.79 3.4232 5 40 sp|Q96KK5|H2A1H_HUMAN HIST1H2AH 13.9 2.7079 31 33 sp|Q14839|CHD4_HUMAN CHD4 217.87 3.3364 24 33 sp|Q08211|DHX9_HUMAN DHX9 140.87 3.1075 29 31 sp|Q14980|NUMA1_HUMAN NUMA1 238.12 3.7685 28 31 sp|Q6P2Q9|PRP8_HUMAN PRPF8 273.43 3.1678 21 31 sp|Q9NYF8|BCLF1_HUMAN BCLAF1 106.06 3.1061 29 30 sp|O75643|U520_HUMAN SNRNP200 244.35 3.4233 20 30 sp|Q6STE5|SMRD3_HUMAN SMARCD3 54.98 3.0867 24 29 sp|P11388|TOP2A_HUMAN TOP2A 174.28 3.2664 15 29 sp|Q96GM5|SMRD1_HUMAN SMARCD1 58.2 3.3365 17 27 sp|Q00839|HNRPU_HUMAN HNRNPU 90.53 3.1686 15 27 sp|P52272|HNRPM_HUMAN HNRNPM 77.46 3.1827 11 27 sp|P22087|FBRL_HUMAN FBL 33.76 3.0616 25 26 sp|O75691|UTP20_HUMAN UTP20 318.18 2.9961 25 25 sp|Q9H583|HEAT1_HUMAN HEATR1 242.22 3.3547 24 25 sp|Q8WYP5|ELYS_HUMAN AHCTF1 252.34 3.4786 9 25 sp|P62805|H4_HUMAN HIST1H4A 11.36 2.9256 14 24 sp|Q9Y2X3|NOP58_HUMAN NOP58 59.54 3.8372 15 23 sp|Q9Y2W1|TR150_HUMAN THRAP3 108.6 2.9034 22 22 sp|Q14690|RRP5_HUMAN PDCD11 208.57 3.3884 18 22 sp|O00567|NOP56_HUMAN NOP56 66.01 3.6951 20 21 sp|Q9Y5B9|SP16H_HUMAN SUPT16H 119.84 3.3847 13 21 sp|O76021|RL1D1_HUMAN RSL1D1 54.94 2.4994 12 21 sp|O96019|ACL6A_HUMAN ACTL6A 47.43 3.0129 18 20 sp|Q02880|TOP2B_HUMAN TOP2B 183.15 3.2243 19 19 sp|Q9UIG0|BAZ1B_HUMAN BAZ1B 170.8 3.4521 16 19 sp|P08670|VIME_HUMAN VIM 53.62 3.4552 10 19 sp|Q12824|SNF5_HUMAN SMARCB1 44.11 3.4839 18 18 sp|Q9NTI5|PDS5B_HUMAN PDS5B 164.56 2.7474 16 17 sp|Q9H0A0|NAT10_HUMAN NAT10 115.66 3.0877 14 17 sp|Q969G3|SMCE1_HUMAN SMARCE1 46.62 3.741 6 17 sp|P62736|ACTA_HUMAN ACTA2 41.98 2.916 11 16 sp|Q9H307|PININ_HUMAN PNN 81.56 3.2407 10 16 sp|P43243|MATR3_HUMAN MATR3 94.56 3.0893 6 16 sp|P63261|ACTG_HUMAN ACTG1 41.77 3.2621 15 15 sp|Q03164|KMT2A_HUMAN KMT2A 431.5 3.5526 15 15 sp|O75533|SF3B1_HUMAN SF3B1 145.74 3.4533 14 15 sp|O60264|SMCA5_HUMAN SMARCA5 121.83 2.9348 13 15 sp|Q08945|SSRP1_HUMAN SSRP1 81.02 3.2569 13 15 sp|Q9Y3T9|NOC2L_HUMAN NOC2L 84.87 3.1816 6 15 sp|P07910|HNRPC_HUMAN HNRNPC 33.65 2.9851 14 14 sp|Q6PL18|ATAD2_HUMAN ATAD2 158.46 3.699 13 14 sp|Q14683|SMC1A_HUMAN SMC1A 143.14 3.4429 13 14 sp|Q96GQ7|DDX27_HUMAN DDX27 89.78 3.3349 13 14 sp|Q9UQE7|SMC3_HUMAN SMC3 141.45 3.3166 11 14 sp|Q12905|ILF2_HUMAN ILF2 43.04 3.537 11 14 sp|Q13601|KRR1_HUMAN KRR1 43.64 3.053 11 14 sp|Q9BQG0|MBB1A_HUMAN MYBBP1A 148.76 2.9411 5 14 IGH1M_MOUSE Ighg1 43.36 2.7484 13 13 sp|P20700|LMNB1_HUMAN LMNB1 66.37 3.6005 13 13 sp|P78527|PRKDC_HUMAN PRKDC 468.79 3.1715 12 13 sp|Q9UKV3|ACINU_HUMAN ACIN1 151.77 3.2842 11 13 sp|Q92925|SMRD2_HUMAN SMARCD2 58.88 3.4743 11 13 sp|Q9ULI0|ATD2B_HUMAN ATAD2B 164.81 2.9966 10 13 sp|Q92785|REQU_HUMAN DPF2 44.13 3.5859 8 13 sp|Q14978|NOLC1_HUMAN NOLC1 73.56 2.752 12 12 sp|Q8IY81|SPB1_HUMAN FTSJ3 96.5 3.5724 12 12 sp|P33993|MCM7_HUMAN MCM7 81.26 3.0588 11 12 sp|Q8WWQ0|PHIP_HUMAN PHIP 206.56 3.385 10 12 sp|P46087|NOP2_HUMAN NOP2 89.25 3.5911 9 12 sp|Q9H2P0|ADNP_HUMAN ADNP 123.49 3.844 10 11 sp|Q9NZM4|BICRA_HUMAN BICRA 158.39 3.6731 10 11 sp|Q15029|U5S1_HUMAN EFTUD2 109.37 3.5515 10 11 sp|P18583|SON_HUMAN SON 263.66 3.4212 10 10 sp|Q7Z3K3|POGZ_HUMAN POGZ 155.24 3.1926 10 10 sp|Q6KC79|NIPBL_HUMAN NIPBL 315.85 3.1909 10 10 sp|P46013|KI67_HUMAN MKI67 358.47 2.9203 9 10 sp|Q12906|ILF3_HUMAN ILF3 95.28 3.5285 9 10 sp|Q86U86|PB1_HUMAN PBRM1 192.83 3.1412 9 10 sp|Q96T58|MINT_HUMAN SPEN 402 3.1272 9 10 sp|O60216|RAD21_HUMAN RAD21 71.64 3.0948 8 10 sp|Q9Y2R4|DDX52_HUMAN DDX52 67.46 3.9244 7 10 sp|Q07955|SRSF1_HUMAN SRSF1 27.73 2.7409 9 9 sp|O14646|CHD1_HUMAN CHD1 196.57 4.0207 9 9 sp|Q99459|CDC5L_HUMAN CDC5L 92.19 3.7929 9 9 sp|O43143|DHX15_HUMAN DHX15 90.88 3.4603 9 9 sp|O75367|H2AY_HUMAN H2AFY 39.59 3.3062 9 9 sp|Q15393|SF3B3_HUMAN SF3B3 135.49 3.3021 9 9 sp|P24928|RPB1_HUMAN POLR2A 217.04 3.2373 9 9 sp|Q9NVP1|DDX18_HUMAN DDX18 75.36 3.2151 9 9 sp|Q8IXT5|RB12B_HUMAN RBM12B 118.03 3.1281 9 9 sp|P17480|UBF1_HUMAN UBTF 89.35 3.0431 9 9 sp|Q9NR30|DDX21_HUMAN DDX21 87.29 2.897 9 9 sp|Q9UJV9|DDX41_HUMAN DDX41 69.79 2.7643 7 9 sp|Q96T23|RSF1_HUMAN RSF1 163.72 3.2299 7 9 sp|P55795|HNRH2_HUMAN HNRNPH2 49.23 2.8092 6 9 sp|Q86VM9|ZCH18_HUMAN ZC3H18 106.32 2.8814 5 9 sp|Q8WUZ0|BCL7C_HUMAN BCL7C 23.45 3.9306 4 9 sp|Q4VC05|BCL7A_HUMAN BCL7A 22.8 3.6922 8 8 sp|Q8WTT2|NOC3L_HUMAN NOC3L 92.49 3.9612 8 8 sp|P55265|DSRAD_HUMAN ADAR 135.98 3.7284 8 8 sp|Q8TDD1|DDX54_HUMAN DDX54 98.53 3.4328 8 8 sp|Q9UQ35|SRRM2_HUMAN SRRM2 299.44 3.0215 7 8 sp|Q9NY61|AATF_HUMAN AATF 63.09 3.6956 7 8 sp|Q1KMD3|HNRL2_HUMAN HNRNPUL2 85.05 3.1539 6 8 sp|Q49A26|GLYR1_HUMAN GLYR1 60.52 3.8419 6 8 sp|Q9P0M6|H2AW_HUMAN H2AFY2 40.03 3.6549 6 8 sp|B2RXH8|HNRC2_HUMAN HNRNPCL2 32.05 2.8517 7 7 sp|Q13435|SF3B2_HUMAN SF3B2 100.16 3.6675 7 7 sp|Q14676|MDC1_HUMAN MDC1 226.53 3.6627 7 7 sp|Q9H6F5|CCD86_HUMAN CCDC86 40.21 3.6182 7 7 sp|Q12788|TBL3_HUMAN TBL3 88.98 3.609 7 7 sp|Q9H8M2|BRD9_HUMAN BRD9 66.96 3.5025 7 7 sp|Q8TDL5|BPIB1_HUMAN BPIFB1 52.41 3.4522 7 7 sp|P50402|EMD_HUMAN EMD 28.98 3.4145 7 7 sp|P19338|NUCL_HUMAN NCL 76.57 3.3742 7 7 sp|Q12873|CHD3_HUMAN CHD3 226.45 3.1365 7 7 sp|Q9BZE4|NOG1_HUMAN GTPBP4 73.92 3.0581 7 7 sp|Q8IWA0|WDR75_HUMAN WDR75 94.44 3.0082 7 7 sp|P28370|SMCA1_HUMAN SMARCA1 122.53 2.9683 7 7 sp|Q92841|DDX17_HUMAN DDX17 80.22 2.9366 7 7 sp|Q8N1F7|NUP93_HUMAN NUP93 93.43 2.9068 7 7 sp|Q6ZRS2|SRCAP_HUMAN SRCAP 343.34 2.8432 7 7 sp|Q03188|CENPC_HUMAN CENPC 106.77 2.6862 6 7 sp|O00566|MPP10_HUMAN MPHOSPH10 78.82 3.9439 6 7 sp|Q9H9B4|SFXN1_HUMAN SFXN1 35.6 3.5359 6 7 sp|O60306|AQR_HUMAN AQR 171.19 3.4683 6 7 sp|P62995|TRA2B_HUMAN TRA2B 33.65 3.0926 6 7 sp|Q8N8A6|DDX51_HUMAN DDX51 72.41 3.0592 5 7 sp|Q5QJE6|TDIF2_HUMAN DNTTIP2 84.42 3.1584 4 7 sp|P16403|H12_HUMAN HIST1H1C 21.35 3.2204 6 6 sp|Q9HCS7|SYF1_HUMAN XAB2 99.95 3.9611 6 6 sp|Q9Y5B6|PAXB1_HUMAN PAXBP1 104.74 3.8492 6 6 sp|O60281|ZN292_HUMAN ZNF292 304.62 3.8342 6 6 sp|Q14692|BMS1_HUMAN BMS1 145.72 3.8277 6 6 sp|Q9BVP2|GNL3_HUMAN GNL3 61.95 3.8227 6 6 sp|Q14202|ZMYM3_HUMAN ZMYM3 152.28 3.5194 6 6 sp|Q8N3U4|STAG2_HUMAN STAG2 141.24 3.5054 6 6 sp|Q9NVH2|INT7_HUMAN INTS7 106.77 3.419 6 6 sp|P49792|RBP2_HUMAN RANBP2 357.97 3.3617 6 6 sp|Q9UK61|TASOR_HUMAN FAM208A 188.91 3.0003 6 6 sp|Q9UKM9|RALY_HUMAN RALY 32.44 2.8877 6 6 sp|Q96PK6|RBM14_HUMAN RBM14 69.45 2.8047 5 6 sp|O00159|MYO1C_HUMAN MYO1C 121.61 4.3268 5 6 sp|Q6NSZ9|ZSC25_HUMAN ZSCAN25 61.44 3.5224 5 6 sp|Q7Z7K6|CENPV_HUMAN CENPV 29.93 3.357 5 6 sp|P68371|TBB4B_HUMAN TUBB4B 49.8 3.1922 5 6 sp|Q9BVJ6|UT14A_HUMAN UTP14A 87.92 3.1504 5 6 sp|Q96KR1|ZFR_HUMAN ZFR 116.94 2.9326 5 6 sp|P62277|RS13_HUMAN RPS13 17.21 2.654 4 6 sp|P31943|HNRH1_HUMAN HNRNPH1 49.2 3.278 4 6 sp|Q16629|SRSF7_HUMAN SRSF7 27.35 2.6251 2 6 sp|Q5BKZ1|ZN326_HUMAN ZNF326 65.61 2.8796 2 6 sp|Q93079|H2B1H_HUMAN HIST1H2BH 13.88 2.0968 5

5 sp|Q99549|MPP8_HUMAN MPHOSPH8 97.12 4.0198 5 5 sp|P33991|MCM4_HUMAN MCM4 96.5 3.8766 5 5 sp|Q14137|BOP1_HUMAN BOP1 83.58 3.8261 5 5 sp|Q15269|PWP2_HUMAN PWP2 102.39 3.7907 5 5 sp|Q03701|CEBPZ_HUMAN CEBPZ 120.9 3.7691 5 5 sp|P57740|NU107_HUMAN NUP107 106.31 3.6565 5 5 sp|Q86YP4|P66A_HUMAN GATAD2A 68.02 3.5678 5 5 sp|Q9BSC4|NOL10_HUMAN NOL10 80.25 3.5655 5 5 sp|Q8TED0|UTP15_HUMAN UTP15 58.38 3.526 5 5 sp|P25440|BRD2_HUMAN BRD2 88.01 3.4882 5 5 sp|Q9NXF1|TEX10_HUMAN TEX10 105.61 3.4606 5 5 sp|Q15459|SF3A1_HUMAN SF3A1 88.83 3.4356 5 5 sp|P07199|CENPB_HUMAN CENPB 65.13 3.4356 5 5 sp|Q9BYG3|MK67I_HUMAN NIFK 34.2 3.296 5 5 sp|Q8N7H5|PAF1_HUMAN PAF1 59.94 3.2547 5 5 sp|Q13620|CUL4B_HUMAN CUL4B 103.92 3.2437 5 5 sp|P52701|MSH6_HUMAN MSH6 152.69 3.2215 5 5 sp|P62424|RL7A_HUMAN RPL7A 29.98 3.2058 5 5 sp|Q15397|PUM3_HUMAN PUM3 73.54 3.2016 5 5 sp|Q9NQS7|INCE_HUMAN INCENP 105.36 3.0804 5 5 sp|Q9H8H0|NOL11_HUMAN NOL11 81.07 2.9959 5 5 sp|O43390|HNRPR_HUMAN HNRNPR 70.9 2.9694 5 5 sp|Q96H22|CENPN_HUMAN CENPN 39.53 2.9578 5 5 sp|P45880|VDAC2_HUMAN VDAC2 31.55 2.9465 5 5 sp|Q6AI39|BICRL_HUMAN BICRAL 115.01 2.8687 5 5 sp|P46100|ATRX_HUMAN ATRX 282.41 2.8152 5 5 sp|Q9UIF9|BAZ2A_HUMAN BAZ2A 211.07 2.7088 4 5 sp|Q9Y5J1|UTP18_HUMAN UTP18 61.96 3.2081 4 5 sp|Q92794|KAT6A_HUMAN KAT6A 224.89 2.736 2 5 sp|Q15532|SSXT_HUMAN SS18 45.9 3.3233 1 5 tr|Q6PJV4|Q6PJV4_HUMAN THRAP3 41.83 3.8114 4 4 sp|P61978|HNRPK_HUMAN HNRNPK 50.94 4.6312 4 4 sp|O96028|NSD2_HUMAN NSD2 152.16 4.0798 4 4 sp|Q9NYH9|UTP6_HUMAN UTP6 70.15 3.9237 4 4 sp|Q12931|TRAP1_HUMAN TRAP1 80.06 3.661 4 4 sp|Q16531|DDB1_HUMAN DDB1 126.89 3.4707 4 4 sp|Q15287|RNPS1_HUMAN RNPS1 34.19 3.4291 4 4 sp|P51398|RT29_HUMAN DAP3 45.54 3.3906 4 4 sp|P05412|JUN_HUMAN JUN 35.65 3.3333 4 4 sp|P15880|RS2_HUMAN RPS2 31.3 3.2743 4 4 sp|P38159|RBMX_HUMAN RBMX 42.31 3.2614 4 4 sp|Q8WXH0|SYNE2_HUMAN SYNE2 795.94 3.2386 4 4 sp|Q9Y4W2|LAS1L_HUMAN LAS1L 83.01 3.218 4 4 sp|Q8WXI9|P66B_HUMAN GATAD2B 65.22 3.2163 4 4 sp|P21796|VDAC1_HUMAN VDAC1 30.75 3.1761 4 4 sp|O94776|MTA2_HUMAN MTA2 74.98 3.1723 4 4 sp|Q16891|MIC60_HUMAN IMMT 83.63 3.1454 4 4 sp|Q9Y2P8|RCL1_HUMAN RCL1 40.82 3.1052 4 4 sp|O75400|PR40A_HUMAN PRPF40A 108.74 3.1025 4 4 sp|Q9UJS0|CMC2_HUMAN SLC25A13 74.13 3.0994 4 4 sp|P06733|ENOA_HUMAN ENO1 47.14 3.0929 4 4 sp|O60287|NPA1P_HUMAN URB1 254.23 3.0688 4 4 sp|P11498|PYC_HUMAN PC 129.55 3.0253 4 4 sp|P11021|GRP78_HUMAN HSPA5 72.29 2.9973 4 4 sp|Q92621|NU205_HUMAN NUP205 227.78 2.9481 4 4 sp|P06576|ATPB_HUMAN ATP5B 56.52 2.8745 4 4 sp|Q68CP9|ARID2_HUMAN ARID2 197.27 2.8522 4 4 sp|P18124|RL7_HUMAN RPL7 29.21 2.7909 4 4 sp|Q9H582|ZN644_HUMAN ZNF644 149.47 2.7228 4 4 sp|P17844|DDX5_HUMAN DDX5 69.1 2.7058 4 4 sp|Q96ME7|ZN512_HUMAN ZNF512 64.64 2.6681 4 4 sp|Q5VWN6|F208B_HUMAN FAM208B 268.68 2.5746 4 4 sp|P36578|RL4_HUMAN RPL4 47.67 2.534 4 4 sp|P49756|RBM25_HUMAN RBM25 100.12 2.5337 4 4 sp|P49750|YLPM1_HUMAN YLPM1 219.85 2.2901 3 4 tr|B2R5W2|B2R5W2_HUMAN HNRNPC 31.93 4.877 3 4 sp|P62701|RS4X_HUMAN RPS4X 29.58 3.4819 3 4 sp|Q96KQ7|EHMT2_HUMAN EHMT2 132.29 3.1695 3 4 sp|P62987|RL40_HUMAN UBA52 14.72 3.1637 3 4 sp|Q9NU22|MDN1_HUMAN MDN1 632.42 3.0818 3 4 sp|Q9H501|ESF1_HUMAN ESF1 98.73 2.9547 3 4 IGKC_MOUSE 11.77 2.9149 3 4 sp|Q12769|NU160_HUMAN NUP160 162.02 2.8295 3 4 sp|Q5JTH9|RRP12_HUMAN RRP12 143.61 2.7939 3 4 sp|Q9Y277|VDAC3_HUMAN VDAC3 30.64 2.7441 3 4 sp|Q13247|SRSF6_HUMAN SRSF6 39.56 2.6268 2 4 sp|Q96PV6|LENG8_HUMAN LENG8 86.07 2.0279 3 3 sp|Q15061|WDR43_HUMAN WDR43 74.84 4.5687 3 3 sp|Q13185|CBX3_HUMAN CBX3 20.8 4.3351 3 3 sp|Q969X6|UTP4_HUMAN UTP4 76.84 4.238 3 3 sp|Q7L2E3|DHX30_HUMAN DHX30 133.85 4.2339 3 3 sp|Q9Y3C1|NOP16_HUMAN NOP16 21.18 4.2066 3 3 sp|A8CG34|P121C_HUMAN POM121C 124.98 4.0062 3 3 sp|P25705|ATPA_HUMAN ATP5A1 59.71 3.966 3 3 sp|Q9BQE3|TBA1C_HUMAN TUBA1C 49.86 3.8249 3 3 sp|Q5SSJ5|HP1B3_HUMAN HP1BP3 61.17 3.8158 3 3 sp|P62753|RS6_HUMAN RPS6 28.66 3.8152 3 3 sp|Q96QD9|UIF_HUMAN FYTTD1 35.8 3.7604 3 3 sp|P52597|HNRPF_HUMAN HNRNPF 45.64 3.7157 3 3 sp|Q9BUJ2|HNRL1_HUMAN HNRNPUL1 95.68 3.6878 3 3 sp|P31942|HNRH3_HUMAN HNRNPH3 36.9 3.6784 3 3 sp|Q9H9B1|EHMT1_HUMAN EHMT1 141.38 3.6655 3 3 sp|Q6PD62|CTR9_HUMAN CTR9 133.42 3.5951 3 3 sp|O15213|WDR46_HUMAN WDR46 68.03 3.5419 3 3 sp|Q9NQ55|SSF1_HUMAN PPAN 53.16 3.5364 3 3 sp|Q9GZR7|DDX24_HUMAN DDX24 96.27 3.5302 3 3 sp|Q9BVI4|NOC4L_HUMAN NOC4L 58.43 3.5196 3 3 sp|P49411|EFTU_HUMAN TUFM 49.51 3.4818 3 3 tr|F8VXC8|F8VXC8_HUMAN SMARCC2 136.1 3.4615 3 3 sp|Q13330|MTA1_HUMAN MTA1 80.74 3.4299 3 3 sp|Q8TDI0|CHD5_HUMAN CHD5 222.91 3.4258 3 3 sp|P02545|LMNA_HUMAN LMNA 74.09 3.4069 3 3 tr|A0A1P0AZG4|A0A1P0AZG4_HUMAN LCOR 137.14 3.3907 3 3 sp|Q8IZL8|PELP1_HUMAN PELP1 119.62 3.3676 3 3 sp|Q9BQE9|BCL7B_HUMAN BCL7B 22.18 3.2712 3 3 sp|P68104|EF1A1_HUMAN EEF1A1 50.11 3.2285 3 3 sp|Q9UMS4|PRP19_HUMAN PRPF19 55.15 3.2237 3 3 sp|P14618|KPYM_HUMAN PKM 57.9 3.2008 3 3 sp|P56537|IF6_HUMAN EIF6 26.58 3.1738 3 3 sp|P04406|G3P_HUMAN GAPDH 36.03 3.1734 3 3 sp|Q9NRL2|BAZ1A_HUMAN BAZ1A 178.59 3.1621 3 3 sp|Q96G21|IMP4_HUMAN IMP4 33.74 3.1542 3 3 sp|P78316|NOP14_HUMAN NOP14 97.61 3.0703 3 3 sp|Q9H0D6|XRN2_HUMAN XRN2 108.51 3.0665 3 3 sp|P56182|RRP1_HUMAN RRP1 52.81 3.0433 3 3 sp|Q15050|RRS1_HUMAN RRS1 41.17 2.9918 3 3 sp|Q9H8H2|DDX31_HUMAN DDX31 94.03 2.9841 3 3 sp|P42285|SK2L2_HUMAN SKIV2L2 117.73 2.9221 3 3 sp|P09874|PARP1_HUMAN PARP1 113.01 2.9162 3 3

sp|Q99848|EBP2_HUMAN EBNA1BP2 34.83 2.9109 3 3 sp|Q96T37|RBM15_HUMAN RBM15 107.12 2.894 3 3 sp|P32119|PRDX2_HUMAN PRDX2 21.88 2.8741 3 3 sp|O95793|STAU1_HUMAN STAU1 63.14 2.8703 3 3 sp|Q9UGU0|TCF20_HUMAN TCF20 211.64 2.8664 3 3 sp|P62750|RL23A_HUMAN RPL23A 17.68 2.8052 3 3 sp|P42167|LAP2B_HUMAN TMPO 50.64 2.7995 3 3 sp|P61313|RL15_HUMAN RPL15 24.13 2.793 3 3 sp|P23396|RS3_HUMAN RPS3 26.67 2.7615 3 3 sp|Q6P0N0|M18BP_HUMAN MIS18BP1 129.01 2.7492 3 3 sp|Q53HL2|BOREA_HUMAN CDCA8 31.3 2.7294 3 3 sp|Q8N9T8|KRI1_HUMAN KRI1 82.55 2.7263 3 3 sp|P38432|COIL_HUMAN COIL 62.57 2.6522 3 3 sp|Q5JTV8|TOIP1_HUMAN TOR1AIP1 66.21 2.5133 3 3 sp|P52292|IMA1_HUMAN KPNA2 57.83 2.5042 3 3 sp|Q9NZM5|GSCR2_HUMAN GLTSCR2 54.36 2.4664 3 3 sp|Q9HC52|CBX8_HUMAN CBX8 43.37 2.4195 3 3 sp|P62851|RS25_HUMAN RPS25 13.73 2.387 3 3 sp|Q8NI36|WDR36_HUMAN WDR36 105.26 2.3862 3 3 sp|Q02878|RL6_HUMAN RPL6 32.71 2.3594 3 3 sp|P46781|RS9_HUMAN RPS9 22.58 2.3229 3 3 sp|Q9NVI7|ATD3A_HUMAN ATAD3A 71.32 2.1869 2 3 sp|Q14103|HNRPD_HUMAN HNRNPD 38.41 3.6101 2 3 sp|Q02539|H11_HUMAN HIST1H1A 21.83 2.8548 2 3 sp|Q71UI9|H2AV_HUMAN H2AFV 13.5 2.7277 2 3 sp|Q13610|PWP1_HUMAN PWP1 55.79 2.6004 2 3 sp|O15226|NKRF_HUMAN NKRF 77.62 2.3415 2 3 sp|P84103|SRSF3_HUMAN SRSF3 19.32 2.2047 1 3 sp|P07305|H10_HUMAN H1F0 20.85 3.9925 1 3 sp|P16104|H2AX_HUMAN H2AFX 15.14 3.8965 2 2 sp|Q9NS69|TOM22_HUMAN TOMM22 15.51 5.0427 2 2 sp|P32455|GBP1_HUMAN GBP1 67.89 4.7845 2 2 sp|Q14498|RBM39_HUMAN RBM39 59.34 4.5672 2 2 sp|P78364|PHC1_HUMAN PHC1 105.47 4.4038 2 2 sp|P62241|RS8_HUMAN RPS8 24.19 4.3331 2 2 sp|Q96EU6|RRP36_HUMAN RRP36 29.8 4.2929 2 2 sp|P83916|CBX1_HUMAN CBX1 21.4 4.1311 2 2 sp|O14983|AT2A1_HUMAN ATP2A1 110.18 4.1184 2 2 sp|P35580|MYH10_HUMAN MYH10 228.86 4.1122 2 2 sp|P49711|CTCF_HUMAN CTCF 82.73 4.0754 2 2 sp|O95251|KAT7_HUMAN KAT7 70.6 4.0575 2 2 sp|P35453|HXD13_HUMAN HOXD13 36.08 4.0572 2 2 sp|Q07020|RL18_HUMAN RPL18 21.62 4.0568 2 2 sp|Q92769|HDAC2_HUMAN HDAC2 55.33 4.028 2 2 sp|Q9H7B2|RPF2_HUMAN RPF2 35.56 4.0267 2 2 sp|Q92784|DPF3_HUMAN DPF3 43.06 3.9575 2 2 sp|Q8WVM7|STAG1_HUMAN STAG1 144.34 3.8979 2 2 sp|Q9H4L4|SENP3_HUMAN SENP3 64.97 3.8924 2 2 sp|P39023|RL3_HUMAN RPL3 46.08 3.7618 2 2 sp|P67809|YBOX1_HUMAN YBX1 35.9 3.751 2 2 sp|Q8NEJ9|NGDN_HUMAN NGDN 35.87 3.7496 2 2 sp|Q13111|CAF1A_HUMAN CHAF1A 106.86 3.7075 2 2 sp|O00267|SPT5H_HUMAN SUPT5H 120.92 3.697 2 2 sp|Q6DKI1|RL7L_HUMAN RPL7L1 28.64 3.6794 2 2 sp|Q5VZL5|ZMYM4_HUMAN ZMYM4 172.68 3.6656 2 2 sp|Q9H7H0|MET17_HUMAN METTL17 50.7 3.6536 2 2 sp|O95983|MBD3_HUMAN MBD3 32.82 3.6402 2 2 sp|P32969|RL9_HUMAN RPL9 21.85 3.6356 2 2 tr|B2RC06|B2RC06_HUMAN 39.25 3.6326 2 2 sp|Q92522|H1X_HUMAN H1FX 22.47 3.6307 2 2 sp|P04792|HSPB1_HUMAN HSPB1 22.77 3.6296 2 2 sp|P05141|ADT2_HUMAN SLC25A5 32.83 3.6195 2 2 sp|P63244|RACK1_HUMAN RACK1 35.05 3.6189 2 2 sp|P04040|CATA_HUMAN CAT 59.72 3.5759 2 2 sp|Q8WXF0|SRS12_HUMAN SRSF12 30.49 3.5688 2 2 sp|Q9Y6K1|DNM3A_HUMAN DNMT3A 101.79 3.5118 2 2 sp|Q86WX3|AROS_HUMAN RPS19BP1 15.42 3.5097 2 2 sp|Q96GD4|AURKB_HUMAN AURKB 39.29 3.4963 2 2 sp|Q96HW7|INT4_HUMAN INTS4 108.1 3.3641 2 2 sp|Q9NSI2|F207A_HUMAN FAM207A 25.44 3.3636 2 2 sp|O75494|SRS10_HUMAN SRSF10 31.28 3.3457 2 2 sp|Q9NW13|RBM28_HUMAN RBM28 85.68 3.3224 2 2 sp|P62906|RL10A_HUMAN RPL10A 24.82 3.319 2 2 sp|Q13123|RED_HUMAN IK 65.56 3.3104 2 2 sp|Q8WUM0|NU133_HUMAN NUP133 128.9 3.3062 2 2 sp|P14678|RSMB_HUMAN SNRPB 24.59 3.3027 2 2 sp|Q15517|CDSN_HUMAN CDSN 51.49 3.2564 2 2 sp|P17661|DESM_HUMAN DES 53.5 3.1482 2 2 sp|Q8NC56|LEMD2_HUMAN LEMD2 56.94 3.1402 2 2 sp|Q9UNQ2|DIM1_HUMAN DIMT1 35.21 3.1391 2 2 sp|Q00325|MPCP_HUMAN SLC25A3 40.07 3.1288 2 2 sp|Q9NP55|BPIA1_HUMAN BPIFA1 26.7 3.1215 2 2 sp|Q9BZJ0|CRNL1_HUMAN CRNKL1 100.39 3.1064 2 2 sp|Q9NV31|IMP3_HUMAN IMP3 21.84 3.0688 2 2 sp|O15523|DDX3Y_HUMAN DDX3Y 73.11 3.0673 2 2 sp|Q53GS7|GLE1_HUMAN GLE1 79.79 3.0633 2 2 sp|Q15717|ELAV1_HUMAN ELAVL1 36.07 3.0494 2 2 sp|Q9BQF6|SENP7_HUMAN SENP7 119.58 3.0117 2 2 sp|P83731|RL24_HUMAN RPL24 17.77 3.0103 2 2 sp|P08574|CY1_HUMAN CYC1 35.4 2.999 2 2 sp|Q7KZ85|SPT6H_HUMAN SUPT6H 198.95 2.9924 2 2 sp|P30050|RL12_HUMAN RPL12 17.81 2.9874 2 2 sp|Q15059|BRD3_HUMAN BRD3 79.49 2.943 2 2 sp|P62263|RS14_HUMAN RPS14 16.26 2.9348 2 2 sp|Q9Y6C9|MTCH2_HUMAN MTCH2 33.31 2.925 2 2 sp|P38919|IF4A3_HUMAN EIF4A3 46.84 2.9081 2 2 sp|Q8TAE8|G45IP_HUMAN GADD45GIP1 25.37 2.8833 2 2 sp|Q03252|LMNB2_HUMAN LMNB2 69.91 2.8508 2 2 sp|P22695|QCR2_HUMAN UQCRC2 48.41 2.8347 2 2 sp|Q5TAP6|UT14C_HUMAN UTP14C 87.13 2.8136 2 2 sp|Q13595|TRA2A_HUMAN TRA2A 32.67 2.7892 2 2 sp|P10599|THIO_HUMAN TXN 11.73 2.7799 2 2 sp|Q9UHA3|RLP24_HUMAN RSL24D1 19.61 2.7718 2 2 sp|Q8N0S6|CENPL_HUMAN CENPL 38.97 2.7494 2 2 sp|Q14781|CBX2_HUMAN CBX2 56.05 2.7332 2 2 sp|Q15910|EZH2_HUMAN EZH2 85.31 2.7153 2 2 sp|P62304|RUXE_HUMAN SNRPE 10.8 2.7029 2 2 sp|Q99590|SCAFB_HUMAN SCAF11 164.55 2.6846 2 2 sp|P52746|ZN142_HUMAN ZNF142 187.76 2.636 2 2 sp|Q96L73|NSD1_HUMAN NSD1 296.46 2.6321 2 2 sp|Q15149|PLEC_HUMAN PLEC 531.47 2.6286 2 2 sp|P05023|AT1A1_HUMAN ATP1A1 112.82 2.5941 2 2 sp|P26373|RL13_HUMAN RPL13 24.25 2.5795 2 2 sp|P52948|NUP98_HUMAN NUP98 197.46 2.5777 2 2 sp|Q9NW64|RBM22_HUMAN RBM22 46.87 2.521 2 2 sp|O14979|HNRDL_HUMAN HNRNPDL 46.41 2.5206 2 2 sp|Q96HS1|PGAM5_HUMAN PGAM5 31.98 2.5018 2 2 sp|P62249|RS16_HUMAN RPS16 16.44 2.3822 2 2 sp|Q14966|ZN638_HUMAN ZNF638 220.49 2.3648 2 2 sp|Q9NRG9|AAAS_HUMAN AAAS 59.54 2.3486 2 2 sp|Q86U38|NOP9_HUMAN NOP9 69.39 2.3397 2 2 sp|P25311|ZA2G_HUMAN AZGP1 34.24 2.2809 2 2 sp|Q13895|BYST_HUMAN BYSL 49.57 2.2751 1 2 sp|Q01130|SRSF2_HUMAN SRSF2 25.46 5.126 1 2 tr|A0A0M3HER2|A0A0M3HER2_HUMAN CENPV 18.8 4.1321 1 2 sp|O75151|PHF2_HUMAN PHF2 120.7 3.6742 1 2 sp|O75323|NIPS2_HUMAN NIPSNAP2 33.72 2.0829 1 1 sp|P08708|RS17_HUMAN RPS17 15.54 5.521 1 1 sp|P08865|RSSA_HUMAN RPSA 32.83 5.1704 1 1 sp|P07355|ANXA2_HUMAN ANXA2 38.58 5.1093 1 1 sp|Q6PK04|CC137_HUMAN CCDC137 33.21 5.0221 1 1 sp|O94906|PRP6_HUMAN PRPF6 106.86 4.9042 1 1 sp|Q3ZCQ8|TIM50_HUMAN TIMM50 39.62 4.8852 1 1 tr|A8K7N0|A8K7N0_HUMAN 23.63 4.8498 1 1 tr|A0A0A0MQS2|A0A0A0MQS2_HUMAN CLASRP 77.12 4.7872 1 1 sp|Q9NQZ2|SAS10_HUMAN UTP3 54.53 4.7178 1 1 sp|Q13243|SRSF5_HUMAN SRSF5 31.25 4.7078 1 1 sp|Q96MU7|YTDC1_HUMAN YTHDC1 84.65 4.7045 1 1 tr|Q6IPH7|Q6IPH7_HUMAN RPL14 23.77 4.6788 1 1 tr|S4R341|S4R341_HUMAN NOLC1 8.05 4.663 1 1 sp|O43159|RRP8_HUMAN RRP8 50.68 4.6332 1 1 sp|P50914|RL14_HUMAN RPL14 23.42 4.618 1 1 sp|Q9GZL7|WDR12_HUMAN WDR12 47.68 4.5869 1 1 sp|Q9GZR2|REXO4_HUMAN REXO4 46.64 4.5713 1 1 sp|Q8NAF0|ZN579_HUMAN ZNF579 60.47 4.5123 1 1 sp|P62913|RL11_HUMAN RPL11 20.24 4.4569 1 1 sp|P16989|YBOX3_HUMAN YBX3 40.07 4.4487 1 1 sp|Q75QN2|INT8_HUMAN INTS8 113.02 4.428 1 1 sp|Q13263|TIF1B_HUMAN TRIM28 88.49 4.4118 1 1 sp|Q8WYB5|KAT6B_HUMAN KAT6B 231.23 4.3886 1 1 tr|Q9UL78|Q9UL78_HUMAN 11.64 4.3545 1 1 sp|O94880|PHF14_HUMAN PHF14 99.99 4.3129 1 1 sp|Q7Z4V5|HDGR2_HUMAN HDGFL2 74.27 4.3001 1 1 sp|Q13867|BLMH_HUMAN BLMH 52.53 4.268 1 1 sp|Q86Y91|KI18B_HUMAN KIF18B 94.16 4.2378 1 1 sp|Q32P51|RA1L2_HUMAN HNRNPA1L2 34.2 4.2125 1 1 sp|P55081|MFAP1_HUMAN MFAP1 51.93 4.1959 1 1 sp|Q96SK2|TM209_HUMAN TMEM209 62.88 4.194 1 1 sp|P62829|RL23_HUMAN RPL23 14.86 4.1813 1 1 sp|Q9NQ39|RS10L_HUMAN RPS10P5 20.11 4.1778 1 1 tr|B2RWN5|B2RWN5_HUMAN HEATR1 242.11 4.1764 1 1 sp|Q9HCM4|E41L5_HUMAN EPB41L5 81.8 4.1232 1 1 sp|P26599|PTBP1_HUMAN PTBP1 57.19 4.1098 1 1 sp|Q96EY7|PTCD3_HUMAN PTCD3 78.5 4.0773 1 1 sp|P20226|TBP_HUMAN TBP 37.67 4.068 1 1 tr|B7Z8Y3|B7Z8Y3_HUMAN 106.95 4.036 1 1 sp|P35658|NU214_HUMAN NUP214 213.49 4.017 1 1 sp|Q9UGM3|DMBT1_HUMAN DMBT1 260.57 4.0093 1 1 sp|Q9BW27|NUP85_HUMAN NUP85 74.97 4.0026 1 1 sp|O15381|NVL_HUMAN NVL 94.99 3.9636 1 1 sp|Q14974|IMB1_HUMAN KPNB1 97.11 3.9488 1 1 sp|P0DMV9|HS71B_HUMAN HSPA1B 70.01 3.9319 1 1 sp|Q76FK4|NOL8_HUMAN NOL8 131.54 3.9233 1 1 sp|O60508|PRP17_HUMAN CDC40 65.48 3.9111 1 1 sp|P09234|RU1C_HUMAN SNRPC 17.38 3.8968 1 1 sp|Q9NRZ9|HELLS_HUMAN HELLS 97.01 3.8903 1 1 sp|Q9Y3A2|UTP11_HUMAN UTP11 30.43 3.8848 1 1 sp|Q13415|ORC1_HUMAN ORC1 97.29 3.8764 1 1 sp|Q69YH5|CDCA2_HUMAN CDCA2 112.61 3.8703 1 1 sp|Q6IQ32|ADNP2_HUMAN ADNP2 122.75 3.8567 1 1 sp|O43795|MYO1B_HUMAN MYO1B 131.9 3.833 1 1 sp|Q9P035|HACD3_HUMAN HACD3 43.13 3.8266 1 1 sp|Q9BYN8|RT26_HUMAN MRPS26 24.2 3.8217 1 1 sp|P07437|TBB5_HUMAN TUBB 49.64 3.818 1 1 sp|O15042|SR140_HUMAN U2SURP 118.22 3.8052 1 1 sp|Q01831|XPC_HUMAN XPC 105.89 3.8043 1 1 tr|B0UZZ8|B0UZZ8_HUMAN C6orf11 68 3.7793 1 1 sp|P13929|ENOB_HUMAN ENO3 46.96 3.7735 1 1 sp|Q9BWN1|PRR14_HUMAN PRR14 64.29 3.7614 1 1 sp|Q9UIS9|MBD1_HUMAN MBD1 66.56 3.7254 1 1 sp|Q9BYD2|RM09_HUMAN MRPL9 30.22 3.7014 1 1 sp|Q9NPI1|BRD7_HUMAN BRD7 74.09 3.6873 1 1 sp|Q9BYD3|RM04_HUMAN MRPL4 34.9 3.6809 1 1 sp|P42696|RBM34_HUMAN RBM34 48.54 3.6619 1 1 sp|Q13206|DDX10_HUMAN DDX10 100.83 3.6542 1 1 sp|Q07021|C1QBP_HUMAN C1QBP 31.34 3.6263 1 1 sp|Q8WVC0|LEO1_HUMAN LEO1 75.36 3.6184 1 1 sp|Q69YN4|VIR_HUMAN KIAA1429 201.9 3.6178 1 1 sp|P12277|KCRB_HUMAN CKB 42.62 3.5894 1 1 sp|Q9Y3B9|RRP15_HUMAN RRP15 31.46 3.5847 1 1 sp|O43251|RFOX2_HUMAN RBFOX2 41.35 3.5765 1 1 sp|Q8N1G0|ZN687_HUMAN ZNF687 129.45 3.5753 1 1 sp|Q14669|TRIPC_HUMAN TRIP12 220.3 3.5724 1 1 sp|P10523|ARRS_HUMAN SAG 45.09 3.563 1 1 sp|Q9HAF1|EAF6_HUMAN MEAF6 21.62 3.5254 1 1 sp|P01834|IGKC_HUMAN IGKC 11.76 3.5209 1 1 sp|Q5T280|CI114_HUMAN SPOUT1 41.98 3.519 1 1 sp|P38646|GRP75_HUMAN HSPA9 73.63 3.5173 1 1 sp|Q9H6R0|DHX33_HUMAN DHX33 78.82 3.5069 1 1 sp|O00571|DDX3X_HUMAN DDX3X 73.2 3.5066 1 1 sp|Q06587|RING1_HUMAN RING1 42.4 3.4787 1 1 sp|Q13151|ROA0_HUMAN HNRNPA0 30.82 3.4486 1 1 sp|O75934|SPF27_HUMAN BCAS2 26.11 3.4476 1 1 sp|Q9UBB5|MBD2_HUMAN MBD2 43.23 3.4443 1 1 sp|Q9NX63|MIC19_HUMAN CHCHD3 26.14 3.4382 1 1 tr|Q05CW7|Q05CW7_HUMAN NAT10 62.35 3.3784 1 1 sp|P49759|CLK1_HUMAN CLK1 57.25 3.3643 1 1 sp|Q58FF8|H90B2_HUMAN HSP90AB2P 44.32 3.358 1 1 sp|P62258|1433E_HUMAN YWHAE 29.16 3.3441 1 1 sp|Q15365|PCBP1_HUMAN PCBP1 37.47 3.3429 1 1 sp|Q13148|TADBP_HUMAN TARDBP 44.71 3.3393 1 1 sp|P62847|RS24_HUMAN RPS24 15.41 3.2951 1 1 sp|P41219|PERI_HUMAN PRPH 53.62 3.2937 1 1 sp|O00148|DX39A_HUMAN DDX39A 49.1 3.2936 1 1 sp|P12236|ADT3_HUMAN SLC25A6 32.85 3.2759 1 1 sp|Q8NHW5|RLA0L_HUMAN RPLP0P6 34.34 3.2647 1 1 sp|O60832|DKC1_HUMAN DKC1 57.64 3.2545 1 1 sp|Q13129|RLF_HUMAN RLF 217.81 3.2327 1 1 sp|Q02241|KIF23_HUMAN KIF23 109.99 3.2283 1 1 sp|P26368|U2AF2_HUMAN U2AF2 53.47 3.2053 1 1 sp|Q9ULW3|ABT1_HUMAN ABT1 31.06 3.2034 1 1 KV2A7_MOUSE 12.27 3.1941 1 1 sp|P21333|FLNA_HUMAN FLNA 280.56 3.1928 1 1 sp|Q8TF76|HASP_HUMAN GSG2 88.44 3.1876 1 1 sp|Q92665|RT31_HUMAN MRPS31 45.29 3.1529 1 1 sp|P04843|RPN1_HUMAN RPN1 68.53 3.1371 1 1 sp|O75223|GGCT_HUMAN GGCT 20.99 3.1294 1 1 sp|P40939|ECHA_HUMAN HADHA 82.95 3.0859 1 1 sp|E9PRG8|CK098_HUMAN C11orf98 13.79 3.0795 1 1 sp|Q9BXF3|CECR2_HUMAN CECR2 164.11 3.075 1 1 sp|Q8IXM6|NRM_HUMAN NRM 29.36 3.0489 1 1 sp|O94901|SUN1_HUMAN SUN1 90.01 3.041 1 1 sp|P01876|IGHA1_HUMAN IGHA1 37.63 3.0351 1 1 tr|A0A024R383|A0A024R383_HUMAN hCG_21098 172.58 3.0327 1 1 sp|O94805|ACL6B_HUMAN ACTL6B 46.85 3.0307 1 1 sp|Q96EP5|DAZP1_HUMAN DAZAP1 43.36 3.0232 1 1 tr|Q53F64|Q53F64_HUMAN 35.97 3.005 1 1 sp|P0DOX7|IGK_HUMAN 23.36 2.9832 1 1 sp|P34931|HS71L_HUMAN HSPA1L 70.33 2.971 1 1 sp|Q7Z5J4|RAI1_HUMAN RAI1 203.23 2.9686 1 1 sp|Q9Y6A4|CFA20_HUMAN CFAP20 22.76 2.9642 1 1 sp|P43246|MSH2_HUMAN MSH2 104.68 2.9358 1 1 sp|Q9NQ50|RM40_HUMAN MRPL40 24.48 2.9286 1 1 sp|Q8TEM1|PO210_HUMAN NUP210 204.98 2.9195 1 1 sp|Q09161|NCBP1_HUMAN NCBP1 91.78 2.9158 1 1 sp|Q13242|SRSF9_HUMAN SRSF9 25.53 2.9054 1 1 sp|Q9Y2R9|RT07_HUMAN MRPS7 28.12 2.8919 1 1 sp|Q9NY12|GAR1_HUMAN GAR1 22.33 2.8874 1 1 sp|Q9UQ88|CD11A_HUMAN CDK11A 91.31 2.8526 1 1 sp|Q8N6I1|EID2_HUMAN EID2 25.17 2.8514 1 1 sp|Q9UNL2|SSRG_HUMAN SSR3 21.07 2.8211 1 1 sp|O75530|EED_HUMAN EED 50.17 2.8165 1 1 sp|P47914|RL29_HUMAN RPL29 17.74 2.801 1 1 sp|Q12830|BPTF_HUMAN BPTF 338.05 2.7991 1 1 sp|Q8IWT3|CUL9_HUMAN CUL9 281.05 2.7852 1 1 sp|O95232|LC7L3_HUMAN LUC7L3 51.44 2.7789 1 1 sp|P0DMR1|HNRC4_HUMAN HNRNPCL4 32.01 2.7683 1 1 sp|Q5T3J3|LRIF1_HUMAN LRIF1 84.52 2.7663 1 1 sp|P01857|IGHG1_HUMAN IGHG1 36.08 2.7607 1 1 sp|P51571|SSRD_HUMAN SSR4 18.99 2.7559 1 1 sp|P46783|RS10_HUMAN RPS10 18.89 2.7434 1 1 sp|Q01469|FABP5_HUMAN FABP5 15.15 2.6709 1 1 sp|O60506|HNRPQ_HUMAN SYNCRIP 69.56 2.6267 1 1 sp|P68871|HBB_HUMAN HBB 15.99 2.6157 1 1 sp|Q5H9F3|BCORL_HUMAN BCORL1 182.41 2.6122 1 1 sp|Q9Y232|CDYL1_HUMAN CDYL 66.44 2.6118 1 1 sp|Q09028|RBBP4_HUMAN RBBP4 47.63 2.6044 1 1 sp|Q01844|EWS_HUMAN EWSR1 68.44 2.603 1 1 sp|Q9NWU5|RM22_HUMAN MRPL22 23.63 2.5935 1 1 sp|A8MTJ3|GNAT3_HUMAN GNAT3 40.33 2.5839 1 1 sp|O14647|CHD2_HUMAN CHD2 211.21 2.5688 1 1 sp|P56134|ATPK_HUMAN ATP5J2 10.91 2.5572 1 1 sp|Q9HBE1|PATZ1_HUMAN PATZ1 74.01 2.5483 1 1 sp|Q6DRA6|H2B2D_HUMAN HIST2H2BD 18.01 2.5415 1 1 sp|P54652|HSP72_HUMAN HSPA2 69.98 2.5309 1 1 sp|Q9UKD2|MRT4_HUMAN MRTO4 27.54 2.5209 1 1 sp|Q8N201|INT1_HUMAN INTS1 244.14 2.5195 1 1 sp|Q5SRE5|NU188_HUMAN NUP188 195.92 2.4771 1 1 sp|Q9Y4F1|FARP1_HUMAN FARP1 118.56 2.4621 1 1 sp|P17010|ZFX_HUMAN ZFX 90.46 2.439 1 1 sp|Q7Z2K6|ERMP1_HUMAN ERMP1 100.17 2.4223 1 1 sp|Q9UJZ1|STML2_HUMAN STOML2 38.51 2.4195 1 1 sp|P62269|RS18_HUMAN RPS18 17.71 2.417 1 1 sp|Q16352|AINX_HUMAN INA 55.36 2.4143 1 1 sp|Q9BRL6|SRSF8_HUMAN SRSF8 32.27 2.3771 1 1 sp|Q9HC84|MUC5B_HUMAN MUC5B 595.96 2.3769 1 1 sp|P11488|GNAT1_HUMAN GNAT1 40.02 2.3735 1 1 sp|Q9HCD5|NCOA5_HUMAN NCOA5 65.5 2.3666 1 1 sp|P43304|GPDM_HUMAN GPD2 80.8 2.3647 1 1 sp|Q15388|TOM20_HUMAN TOMM20 16.29 2.3634 1 1 sp|P61353|RL27_HUMAN RPL27 15.79 2.3498 1 1 sp|Q9BXY5|CAYP2_HUMAN CAPS2 63.8 2.3229 1 1 sp|Q9BZE1|RM37_HUMAN MRPL37 48.09 2.288 1 1 sp|Q9P2K5|MYEF2_HUMAN MYEF2 64.08 2.2862 1 1 tr|A0A024R5M9|A0A024R5M9_HUMAN NUMA1 236.37 2.2646 1 1 sp|Q06830|PRDX1_HUMAN PRDX1 22.1 2.2423 1 1 sp|Q9UI42|CBPA4_HUMAN CPA4 47.32 2.2271 1 1 sp|O75152|ZC11A_HUMAN ZC3H11A 89.08 2.2039 1 1 sp|Q9NVX2|NLE1_HUMAN NLE1 53.29 2.1771 1 1 sp|Q14156|EFR3A_HUMAN EFR3A 92.86 2.1643 1 1 sp|P22314|UBA1_HUMAN UBA1 117.77 2.1283

TABLE-US-00017 TABLE 5D HA-SS18WT_NE_peptides Unique Total reference Gene Symbol MWT(kDa) AVG 55 595 sp|P51532|SMCA4_HUMAN SMARCA4 184.53 3.203 86 536 sp|P51531|SMCA2_HUMAN SMARCA2 181.17 3.0037 105 448 sp|O14497|ARI1A_HUMAN ARID1A 241.89 3.1848 56 414 sp|Q92922|SMRC1_HUMAN SMARCC1 122.79 2.9463 69 368 sp|Q8TAQ2|SMRC2_HUMAN SMARCC2 132.8 2.9867 95 307 sp|Q8NFD5|ARI1B_HUMAN ARID1B 235.97 2.9764 45 158 sp|Q9NZM4|BICRA_HUMAN BICRA 158.39 3.144 29 157 sp|O96019|ACL6A_HUMAN ACTL6A 47.43 3.1706 32 144 sp|Q96GM5|SMRD1_HUMAN SMARCD1 58.2 3.2163 31 137 sp|Q6STE5|SMRD3_HUMAN SMARCD3 54.98 3.1622 34 120 sp|Q969G3|SMCE1_HUMAN SMARCE1 46.62 3.2044 13 115 sp|P62736|ACTA_HUMAN ACTA2 41.98 2.6075 23 109 sp|Q12824|SNF5_HUMAN SMARCB1 44.11 3.0533 83 98 sp|P78527|PRKDC_HUMAN PRKDC 468.79 3.3759 9 93 sp|P63261|ACTG_HUMAN ACTG1 41.77 3.302 29 89 sp|Q92925|SMRD2_HUMAN SMARCD2 58.88 3.204 13 70 sp|Q4VC05|BCL7A_HUMAN BCL7A 22.8 3.2931 33 69 sp|Q9H8M2|BRD9_HUMAN BRD9 66.96 3.3574 21 48 sp|P49411|EFTU_HUMAN TUFM 49.51 2.863 17 46 sp|Q92785|REQU_HUMAN DPF2 44.13 3.6127 26 42 sp|P25705|ATPA_HUMAN ATP5A1 59.71 3.5053 13 41 sp|P12236|ADT3_HUMAN SLC25A6 32.85 2.6018 35 36 sp|P35580|MYH10_HUMAN MYH10 228.86 3.6421 34 34 sp|O75643|U520_HUMAN SNRNP200 244.35 3.2978 28 33 sp|P06576|ATPB_HUMAN ATP5B 56.52 3.6358 11 33 sp|Q8WUZ0|BCL7C_HUMAN BCL7C 23.45 3.2939 20 29 sp|P52272|HNRPM_HUMAN HNRNPM 77.46 3.2563 16 29 sp|Q6AI39|BICRL_HUMAN BICRAL 115.01 3.0716 25 27 sp|P35579|MYH9_HUMAN MYH9 226.39 4.1382 23 25 sp|Q08211|DHX9_HUMAN DHX9 140.87 3.6402 20 25 sp|P05023|AT1A1_HUMAN ATP1A1 112.82 3.7054 22 24 sp|Q9UJS0|CMC2_HUMAN SLC25A13 74.13 3.3988 13 24 sp|Q00325|MPCP_HUMAN SLC25A3 40.07 2.7509 18 23 sp|P68371|TBB4B_HUMAN TUBB4B 49.8 3.6902 17 23 sp|O95831|AIFM1_HUMAN AIFM1 66.86 3.2073 21 21 sp|Q92621|NU205_HUMAN NUP205 227.78 3.3918 21 21 sp|Q6P2Q9|PRP8_HUMAN PRPF8 273.43 2.9541 15 20 sp|Q9NVI7|ATD3A_HUMAN ATAD3A 71.32 3.2389 9 20 sp|P68104|EF1A1_HUMAN EEF1A1 50.11 3.3548 4 20 tr|B9EGQ8|B9EGQ8_HUMAN SMARCA4 189.33 3.7789 14 19 sp|P11021|GRP78_HUMAN HSPA5 72.29 3.312 18 18 sp|P52701|MSH6_HUMAN MSH6 152.69 3.2055 18 18 sp|Q8N1F7|NUP93_HUMAN NUP93 93.43 2.826 17 17 sp|Q10570|CPSF1_HUMAN CPSF1 160.78 3.3065 12 17 sp|P20700|LMNB1_HUMAN LMNB1 66.37 3.3674 14 16 sp|P38646|GRP75_HUMAN HSPA9 73.63 3.5339 15 15 sp|Q15029|U5S1_HUMAN EFTUD2 109.37 3.4816 15 15 sp|Q14204|DYHC1_HUMAN DYNC1H1 532.07 3.4659 15 15 sp|Q7L0Y3|MRRP1_HUMAN TRMT10C 47.32 3.2489 15 15 sp|Q9BQG0|MBB1A_HUMAN MYBBP1A 148.76 2.8622 13 15 sp|Q16891|MIC60_HUMAN IMMT 83.63 3.7338 12 15 sp|P16615|AT2A2_HUMAN ATP2A2 114.68 2.973 14 14 sp|P04844|RPN2_HUMAN RPN2 69.24 3.8313 14 14 sp|Q14980|NUMA1_HUMAN NUMA1 238.12 3.6168 13 14 sp|Q9BQE3|TBA1C_HUMAN TUBA1C 49.86 3.2879 2 14 sp|Q15532|SSXT_HUMAN SS18 45.9 2.6617 13 13 sp|Q02978|M2OM_HUMAN SLC25A11 34.04 3.6267 12 13 sp|P30837|AL1B1_HUMAN ALDH1B1 57.17 3.9563 12 13 sp|P42704|LPPRC_HUMAN LRPPRC 157.81 3.2168 11 13 sp|P22087|FBRL_HUMAN FBL 33.76 3.2519 12 12 sp|Q96A33|CCD47_HUMAN CCDC47 55.84 3.6465 12 12 sp|P10809|CH60_HUMAN HSPD1 61.02 3.3748 11 12 sp|P40939|ECHA_HUMAN HADHA 82.95 3.7455 11 12 sp|P43243|MATR3_HUMAN MATR3 94.56 3.2881 8 12 sp|P36542|ATPG_HUMAN ATP5C1 32.98 2.4526 5 12 sp|Q9BQE9|BCL7B_HUMAN BCL7B 22.18 2.6066 11 11 sp|O75746|CMC1_HUMAN SLC25A12 74.71 3.7205 11 11 sp|P04181|OAT_HUMAN OAT 48.5 3.6399 11 11 sp|Q9Y4W6|AFG32_HUMAN AFG3L2 88.53 3.6226 11 11 sp|O75306|NDUS2_HUMAN NDUFS2 52.51 3.5563 11 11 sp|O43795|MYO1B_HUMAN MYO1B 131.9 3.3594 11 11 sp|Q9BUQ8|DDX23_HUMAN DDX23 95.52 3.0862 10 11 sp|P21796|VDAC1_HUMAN VDAC1 30.75 3.361 10 11 sp|Q53H12|AGK_HUMAN AGK 47.11 3.1775 9 11 sp|Q6UN15|FIP1_HUMAN FIP1L1 66.49 3.5179 9 11 sp|Q9H9B4|SFXN1_HUMAN SFXN1 35.6 3.4442 10 10 sp|O00567|NOP56_HUMAN NOP56 66.01 3.7799 10 10 sp|Q6NUK1|SCMC1_HUMAN SLC25A24 53.32 3.4896 10 10 sp|O94832|MYO1D_HUMAN MYO1D 116.13 3.1152 10 10 sp|Q12769|NU160_HUMAN NUP160 162.02 3.0307 9 10 sp|Q53GQ0|DHB12_HUMAN HSD17B12 34.3 3.7669 9 10 sp|Q9Y5B6|PAXB1_HUMAN PAXBP1 104.74 3.6482 9 10 sp|P11310|ACADM_HUMAN ACADM 46.56 3.5587 9 10 sp|Q9C0J8|WDR33_HUMAN WDR33 145.8 3.513 9 10 sp|Q96I99|SUCB2_HUMAN SUCLG2 46.48 3.1081 8 10 sp|P11177|ODPB_HUMAN PDHB 39.21 3.3188 5 10 IGH1M_MOUSE Ighg1 43.36 2.8688 5 10 sp|P53985|MOT1_HUMAN SLC16A1 53.91 2.6482 9 9 sp|Q9H857|NT5D2_HUMAN NT5DC2 60.68 3.0761 9 9 sp|Q5SRE5|NU188_HUMAN NUP188 195.92 2.5925 8 9 sp|Q92616|GCN1_HUMAN GCN1 292.57 3.5758 8 9 sp|Q9UBB9|TFP11_HUMAN TFIP11 96.76 2.5113 7 9 sp|Q9P035|HACD3_HUMAN HACD3 43.13 3.1073 5 9 sp|Q5T280|CI114_HUMAN SPOUT1 41.98 3.3565 8 8 sp|O00159|MYO1C_HUMAN MYO1C 121.61 4.128 8 8 sp|O75489|NDUS3_HUMAN NDUFS3 30.22 3.6044 8 8 sp|Q8NI60|COQ8A_HUMAN COQ8A 71.9 3.597 8 8 sp|Q969V3|NCLN_HUMAN NCLN 62.93 3.2065 8 8 sp|P33993|MCM7_HUMAN MCM7 81.26 3.0851 8 8 sp|Q9UJV9|DDX41_HUMAN DDX41 69.79 2.6071 7 8 sp|P11142|HSP7C_HUMAN HSPA8 70.85 3.9598 7 8 sp|O14980|XPO1_HUMAN XPO1 123.31 3.6204 7 8 sp|P39656|OST48_HUMAN DDOST 50.77 3.2075 7 8 sp|O60313|OPA1_HUMAN OPA1 111.56 2.9217 7 8 sp|Q8N8A6|DDX51_HUMAN DDX51 72.41 2.8323 6 8 sp|P22695|QCR2_HUMAN UQCRC2 48.41 3.6726 3 8 sp|P05141|ADT2_HUMAN SLC25A5 32.83 3.2983 7 7 sp|Q9Y2X3|NOP58_HUMAN NOP58 59.54 3.9978 7 7 sp|P43246|MSH2_HUMAN MSH2 104.68 3.7982 7 7 sp|Q9Y2R4|DDX52_HUMAN DDX52 67.46 3.695 7 7 sp|Q16822|PCKGM_HUMAN PCK2 70.68 3.5129 7 7 sp|A0FGR8|ESYT2_HUMAN ESYT2 102.29 3.375 7 7 sp|O14983|AT2A1_HUMAN ATP2A1 110.18 3.3628 7 7 sp|Q29RF7|PDS5A_HUMAN PDS5A 150.73 3.3405 7 7 sp|Q96T37|RBM15_HUMAN RBM15 107.12 3.1355 7 7 sp|Q9P2I0|CPSF2_HUMAN CPSF2 88.43 2.6885 7 7 sp|P53007|TXTP_HUMAN SLC25A1 33.99 2.3894 6 7 sp|O00411|RPOM_HUMAN POLRMT 138.53 4.0187 6 7 sp|Q3ZCQ8|TIM50_HUMAN TIMM50 39.62 3.4954 6 7 sp|P50213|IDH3A_HUMAN IDH3A 39.57 2.9987 6 7 sp|Q92841|DDX17_HUMAN DDX17 80.22 2.8887 6 7 sp|Q14739|LBR_HUMAN LBR 70.66 2.7688 6 6 sp|Q96TA2|YMEL1_HUMAN YME1L1 86.4 4.3425 6 6 sp|Q8TED0|UTP15_HUMAN UTP15 58.38 3.7408 6 6 sp|P13674|P4HA1_HUMAN P4HA1 61.01 3.6475 6 6 sp|Q3SY69|AL1L2_HUMAN ALDH1L2 101.68 3.5282 6 6 sp|P26368|U2AF2_HUMAN U2AF2 53.47 3.4804 6 6 tr|F8VXC8|F8VXC8_HUMAN SMARCC2 136.1 3.4048 6 6 sp|P45954|ACDSB_HUMAN ACADSB 47.46 3.1749 6 6 sp|P34931|HS71L_HUMAN HSPA1L 70.33 3.111 6 6 sp|P26641|EF1G_HUMAN EEF1G 50.09 3.109 6 6 sp|Q9NUL7|DDX28_HUMAN DDX28 59.54 3.0823 6 6 sp|P55084|ECHB_HUMAN HADHB 51.26 3.0592 6 6 sp|P35251|RFC1_HUMAN RFC1 128.18 3.0236 6 6 sp|Q9BPW8|NIPS1_HUMAN NIPSNAP1 33.29 2.9544 6 6 sp|Q9NXE4|NSMA3_HUMAN SMPD4 93.29 2.9319 6 6 sp|O43615|TIM44_HUMAN TIMM44 51.32 2.8937 6 6 sp|P50416|CPT1A_HUMAN CPT1A 88.31 2.7685 5 6 sp|O43175|SERA_HUMAN PHGDH 56.61 3.9221 5 6 sp|Q8NDT2|RB15B_HUMAN RBM15B 97.15 3.6373 5 6 sp|Q15758|AAAT_HUMAN SLC1A5 56.56 3.3323 5 6 sp|P53597|SUCA_HUMAN SUCLG1 36.23 3.2971 5 6 sp|P50570|DYN2_HUMAN DNM2 98 3.1087 4 6 sp|Q92784|DPF3_HUMAN DPF3 43.06 4.3143 4 6 sp|P51571|SSRD_HUMAN SSR4 18.99 2.9063 5 5 sp|Q68CP9|ARID2_HUMAN ARID2 197.27 3.8774 5 5 sp|Q5UIP0|RIF1_HUMAN RIF1 274.29 3.8562 5 5 sp|O14828|SCAM3_HUMAN SCAMP3 38.26 3.8314 5 5 sp|A1L0T0|ILVBL_HUMAN ILVBL 67.82 3.7405 5 5 sp|P47985|UCRI_HUMAN UQCRFS1 29.65 3.6437 5 5 sp|Q03701|CEBPZ_HUMAN CEBPZ 120.9 3.564 5 5 sp|P45880|VDAC2_HUMAN VDAC2 31.55 3.5043 5 5 sp|Q9BW27|NUP85_HUMAN NUP85 74.97 3.4294 5 5

sp|P53621|COPA_HUMAN COPA 138.26 3.3088 5 5 sp|Q9UDR5|AASS_HUMAN AASS 102.07 3.296 5 5
sp|P04843|RPN1_HUMAN RPN1 68.53 3.2012 5 5 sp|Q9BSD7|NTPCR_HUMAN NTPCR 20.7 3.1848 5 5
sp|Q6JQN1|ACD10_HUMAN ACAD10 118.76 3.1462 5 5 sp|P08670|VIME_HUMAN VIM 53.62 3.0791 5 5
sp|Q6P4A7|SFXN4_HUMAN SFXN4 37.97 2.9877 5 5 sp|Q9NTI5|PDS5B_HUMAN PDS5B 164.56 2.9774 5 5
sp|P46977|STT3A_HUMAN STT3A 80.48 2.9334 5 5 sp|Q96EY1|DNJA3_HUMAN DNAJA3 52.46 2.8714 5 5
sp|P00367|DHE3_HUMAN GLUD1 61.36 2.6644 5 5 sp|Q9NRK6|ABCBA_HUMAN ABCB10 79.1 2.6502 5 5
sp|Q9P2R7|SUCB1_HUMAN SUCLA2 50.29 2.0854 4 5 sp|Q9H9P8|L2HDH_HUMAN L2HGDH 50.28 3.836 4 5
sp|Q15637|SF01_HUMAN SF1 68.29 3.8277 4 5 sp|P28331|NDUS1_HUMAN NDUFS1 79.42 3.6792 4 5
sp|Q9H7H0|MET17_HUMAN METTL17 50.7 3.5298 4 5 sp|Q9NPI1|BRD7_HUMAN BRD7 74.09 3.3321 4 5
sp|Q14974|IMB1_HUMAN KPNB1 97.11 3.287 4 5 sp|P53618|COPB_HUMAN COPB1 107.07 3.1059 4 5
sp|P07437|TBB5_HUMAN TUBB 49.64 2.9809 4 5 sp|Q96CS3|FAF2_HUMAN FAF2 52.59 2.8438 4 4
sp|P13804|ETFA_HUMAN ETFA 35.06 4.3735 4 4 sp|P52597|HNRPF_HUMAN HNRNPF 45.64 4.2399 4 4
sp|Q96BW9|TAM41_HUMAN TAMM41 51.03 4.2029 4 4 sp|Q5T9A4|ATD3B_HUMAN ATAD3B 72.53 3.9984 4 4 sp|Q9HC07|TM165_HUMAN TMEM165 34.88 3.9971 4 4 sp|Q9GZR7|DDX24_HUMAN DDX24 96.27 3.86 4 4
sp|Q92947|GCDH_HUMAN GCDH 48.1 3.8323 4 4 sp|P55795|HNRH2_HUMAN HNRNPH2 49.23 3.6838 4 4
sp|Q92576|PHF3_HUMAN PHF3 229.34 3.6333 4 4 sp|Q9Y512|SAM50_HUMAN SAMM50 51.94 3.6212 4 4
sp|O43837|IDH3B_HUMAN IDH3B 42.16 3.5936 4 4 sp|P0DMV9|HS71B_HUMAN HSPA1B 70.01 3.5795 4 4
sp|O15269|SPTC1_HUMAN SPTLC1 52.71 3.5667 4 4 sp|P48047|ATPO_HUMAN ATP5O 23.26 3.3792 4 4
sp|Q96NB2|SFXN2_HUMAN SFXN2 36.21 3.3362 4 4 sp|Q96EP5|DAZP1_HUMAN DAZAP1 43.36 3.3271 4 4
sp|P49590|SYHM_HUMAN HARS2 56.85 3.2496 4 4 sp|Q96D53|COQ8B_HUMAN COQ8B 60.03 3.2252 4 4
sp|Q14978|NOLC1_HUMAN NOLC1 73.56 3.1412 4 4 sp|Q9NSE4|SYIM_HUMAN IARS2 113.72 3.0973 4 4
sp|P49756|RBM25_HUMAN RBM25 100.12 3.0237 4 4 sp|O75616|ERAL1_HUMAN ERAL1 48.32 2.8867 4 4
sp|P17844|DDX5_HUMAN DDX5 69.1 2.8479 4 4 sp|P55786|PSA_HUMAN NPEPPS 103.21 2.8037 4 4
sp|P24539|AT5F1_HUMAN ATP5F1 28.89 2.7676 4 4 sp|Q12905|ILF2_HUMAN ILF2 43.04 2.7356 4 4
sp|O60762|DPM1_HUMAN DPM1 29.62 2.7256 4 4 sp|Q9Y305|ACOT9_HUMAN ACOT9 49.87 2.7236 4 4
sp|O60318|GANP_HUMAN MCM3AP 218.27 2.6029 3 4 sp|P35250|RFC2_HUMAN RFC2 39.13 3.773 3 4
sp|O75027|ABCB7_HUMAN ABCB7 82.59 3.4147 3 4 sp|Q92782|DPF1_HUMAN DPF1 42.47 2.7282 3 4
sp|Q9UM00|TMCO1_HUMAN TMCO1 21.16 2.6108 3 4 sp|P46459|NSF_HUMAN NSF 82.54 2.133 3 3
sp|O75400|PR40A_HUMAN PRPF40A 108.74 4.5658 3 3 sp|Q14498|RBM39_HUMAN RBM39 59.34 4.3685 3 3
sp|O95299|NDUAA_HUMAN NDUFA10 40.72 4.3305 3 3 sp|Q9NX63|MIC19_HUMAN CHCHD3 26.14 4.2124 3 3 sp|O94906|PRP6_HUMAN PRPF6 106.86 4.1852 3 3 sp|A6NJ78|MET15_HUMAN METTL15 46.09 4.1633 3 3
sp|Q9H583|HEAT1_HUMAN HEATR1 242.22 4.1273 3 3 sp|O00165|HAX1_HUMAN HAX1 31.6 4.0022 3 3
sp|Q9H845|ACAD9_HUMAN ACAD9 68.72 3.9765 3 3 sp|P48735|IDHP_HUMAN IDH2 50.88 3.946 3 3
sp|Q00839|HNRPU_HUMAN HNRNPU 90.53 3.9089 3 3 sp|Q9BW92|SYTM_HUMAN TARS2 80.99 3.8276 3 3
sp|Q96SK2|TM209_HUMAN TMEM209 62.88 3.826 3 3 sp|Q9HCM4|E41L5_HUMAN EPB41L5 81.8 3.7752 3 3
sp|Q8IY17|PLPL6_HUMAN PNPLA6 149.9 3.7737 3 3 sp|P35249|RFC4_HUMAN RFC4 39.66 3.7582 3 3
sp|Q9UQ90|SPG7_HUMAN SPG7 88.18 3.7499 3 3 sp|Q9Y5M8|SRPRB_HUMAN SRPRB 29.68 3.7264 3 3
sp|Q14103|HNRPD_HUMAN HNRNPD 38.41 3.6717 3 3 sp|P38432|COIL_HUMAN COIL 62.57 3.6706 3 3
sp|Q15120|PDK3_HUMAN PDK3 46.91 3.629 3 3 sp|P19105|ML12A_HUMAN MYL12A 19.78 3.6243 3 3
sp|Q6IAN0|DRS7B_HUMAN DHRS7B 35.1 3.6008 3 3 sp|P28288|ABCD3_HUMAN ABCD3 75.43 3.5716 3 3
sp|P31689|DNJA1_HUMAN DNAJA1 44.84 3.5604 3 3 sp|Q49A26|GLYR1_HUMAN GLYR1 60.52 3.5595 3 3
sp|Q9H0A0|NAT10_HUMAN NAT10 115.66 3.5176 3 3 sp|Q9UKM7|MA1B1_HUMAN MAN1B1 79.53 3.4936 3 3 sp|P24468|COT2_HUMAN NR2F2 45.54 3.4642 3 3 sp|O75600|KBL_HUMAN GCAT 45.26 3.4574 3 3
sp|P18074|ERCC2_HUMAN ERCC2 86.85 3.446 3 3 sp|O75533|SF3B1_HUMAN SF3B1 145.74 3.3905 3 3
sp|Q68CQ7|GL8D1_HUMAN GLT8D1 41.91 3.3887 3 3 sp|P50402|EMD_HUMAN EMD 28.98 3.3781 3 3
sp|O94813|SLIT2_HUMAN SLIT2 169.76 3.3702 3 3 sp|Q12931|TRAP1_HUMAN TRAP1 80.06 3.3555 3 3
sp|Q8N201|INT1_HUMAN INTS1 244.14 3.3539 3 3 sp|Q9NVI1|FANCI_HUMAN FANCI 149.23 3.3187 3 3
sp|Q9UM54|MYO6_HUMAN MYO6 149.6 3.2785 3 3 sp|Q5JTZ9|SYAM_HUMAN AARS2 107.27 3.2518 3 3
sp|O15270|SPTC2_HUMAN SPTLC2 62.88 3.2179 3 3 sp|P43307|SSRA_HUMAN SSR1 32.22 3.2065 3 3
sp|Q5T160|SYRM_HUMAN RARS2 65.46 3.2025 3 3 sp|Q9H936|GHC1_HUMAN SLC25A22 34.45 3.1966 3 3
sp|P33527|MRP1_HUMAN ABCC1 171.48 3.1945 3 3 sp|P31943|HNRH1_HUMAN HNRNPH1 49.2 3.1369 3 3
sp|P32189|GLPK_HUMAN GK 61.21 3.1141 3 3 sp|Q92542|NICA_HUMAN NCSTN 78.36 3.0897 3 3
sp|O94874|UFL1_HUMAN UFL1 89.54 3.0748 3 3 sp|O00400|ACATN_HUMAN SLC33A1 60.87 3.047 3 3
sp|Q07021|C1QBP_HUMAN C1QBP 31.34 3.0334 3 3 sp|O76031|CLPX_HUMAN CLPX 69.18 3.0094 3 3
sp|Q9NZ01|TECR_HUMAN TECR 36.01 2.9946 3 3 sp|Q9BWM7|SFXN3_HUMAN SFXN3 35.48 2.993 3 3
sp|O43913|ORC5_HUMAN ORC5 50.25 2.9825 3 3 sp|Q92945|FUBP2_HUMAN KHSRP 73.07 2.977 3 3
sp|P12235|ADT1_HUMAN SLC25A4 33.04 2.958 3 3 sp|Q9H2S9|IKZF4_HUMAN IKZF4 64.07 2.9251 3 3
sp|O43824|GTPB6_HUMAN GTPBP6 56.85 2.9127 3 3 sp|O43809|CPSF5_HUMAN NUDT21 26.21 2.8367 3 3
sp|Q5SY16|NOL9_HUMAN NOL9 79.27 2.8134 3 3 sp|Q9NXF1|TEX10_HUMAN TEX10 105.61 2.7769 3 3

sp|Q9BX10|GTPB2_HUMAN GTPBP2 65.73 2.7706 3 3 sp|P60660|MYL6_HUMAN MYL6 16.92 2.7542 3 3
sp|P09874|PARP1_HUMAN PARP1 113.01 2.7471 3 3 sp|Q16795|NDUA9_HUMAN NDUFA9 42.48 2.7254 3 3
sp|Q86UT6|NLRX1_HUMAN NLRX1 107.55 2.6517 3 3 sp|Q9Y678|COPG1_HUMAN COPG1 97.66 2.637 3 3
sp|Q15393|SF3B3_HUMAN SF3B3 135.49 2.5952 3 3 sp|Q14684|RRP1B_HUMAN RRP1B 84.38 2.5659 3 3
sp|P23258|TBG1_HUMAN TUBG1 51.14 2.4401 3 3 sp|Q96PK6|RBM14_HUMAN RBM14 69.45 2.1696 2 3
sp|P56192|SYMC_HUMAN MARS 101.05 3.3287 2 3 sp|Q8TCT9|HM13_HUMAN HM13 41.46 3.0227 2 3
sp|P40938|RFC3_HUMAN RFC3 40.53 2.7393 2 2 sp|Q9NS69|TOM22_HUMAN TOMM22 15.51 4.8805 2 2
sp|Q9NVH1|DJC11_HUMAN DNAJC11 63.24 4.8216 2 2 sp|Q8IY37|DHX37_HUMAN DHX37 129.46 4.5684 2 2
sp|P07910|HNRPC_HUMAN HNRNPC 33.65 4.5425 2 2 sp|Q8TB37|NUBPL_HUMAN NUBPL 34.06 4.5419 2 2
sp|P23634|AT2B4_HUMAN ATP2B4 137.83 4.4613 2 2 sp|Q99805|TM9S2_HUMAN TM9SF2 75.73 4.4071 2 2
sp|O95674|CDS2_HUMAN CDS2 51.38 4.4009 2 2 sp|Q6DD88|ATLA3_HUMAN ATL3 60.5 4.3202 2 2
sp|Q8N465|D2HDH_HUMAN D2HGDH 56.38 4.2164 2 2 sp|Q8NF37|PCAT1_HUMAN LPCAT1 59.11 4.138 2 2
sp|O95470|SGPL1_HUMAN SGPL1 63.48 4.1063 2 2 sp|Q96C36|P5CR2_HUMAN PYCR2 33.62 4.0071 2 2
sp|P31040|SDHA_HUMAN SDHA 72.65 3.9895 2 2 sp|Q5JTV8|TOIP1_HUMAN TOR1AIP1 66.21 3.8342 2 2
sp|P52292|IMA1_HUMAN KPNA2 57.83 3.7956 2 2 sp|Q5JU69|TOR2A_HUMAN TOR2A 35.69 3.7897 2 2
sp|P61247|RS3A_HUMAN RPS3A 29.93 3.7885 2 2 sp|Q9BVA1|TBB2B_HUMAN TUBB2B 49.92 3.7692 2 2
sp|Q01650|LAT1_HUMAN SLC7A5 54.97 3.7618 2 2 sp|Q99623|PHB2_HUMAN PHB2 33.28 3.7184 2 2
sp|P61225|RAP2B_HUMAN RAP2B 20.49 3.709 2 2 sp|Q9UDX5|MTFP1_HUMAN MTFP1 18 3.6947 2 2
sp|Q13435|SF3B2_HUMAN SF3B2 100.16 3.6764 2 2 sp|O14654|IRS4_HUMAN IRS4 133.68 3.6741 2 2
sp|O43143|DHX15_HUMAN DHX15 90.88 3.6482 2 2 sp|P51648|AL3A2_HUMAN ALDH3A2 54.81 3.6364 2 2
sp|Q8NHH9|ATLA2_HUMAN ATL2 66.19 3.6352 2 2 sp|P49821|NDUV1_HUMAN NDUFV1 50.78 3.6288 2 2
sp|A3KMH1|VWA8_HUMAN VWA8 214.69 3.5989 2 2 sp|P35613|BASI_HUMAN BSG 42.17 3.5353 2 2
sp|O60264|SMCA5_HUMAN SMARCA5 121.83 3.5339 2 2 sp|Q9UBX3|DIC_HUMAN SLC25A10 31.26 3.5314 2
2 sp|Q9UBM7|DHCR7_HUMAN DHCR7 54.45 3.5297 2 2 sp|Q96GC9|VMP1_HUMAN VMP1 46.21 3.5212 2 2
sp|Q9UJZ1|STML2_HUMAN STOML2 38.51 3.4846 2 2 sp|P08195|4F2_HUMAN SLC3A2 67.95 3.4622 2 2
sp|Q8IXI1|MIRO2_HUMAN RHOT2 68.07 3.4496 2 2 sp|Q92544|TM9S4_HUMAN TM9SF4 74.47 3.4436 2 2
sp|Q9H061|T126A_HUMAN TMEM126A 21.51 3.4371 2 2 sp|P62987|RL40_HUMAN UBA52 14.72 3.3766 2 2
sp|P56134|ATPK_HUMAN ATP5J2 10.91 3.3667 2 2 sp|Q14966|ZN638_HUMAN ZNF638 220.49 3.3667 2 2
sp|Q96KK5|H2A1H_HUMAN HIST1H2AH 13.9 3.3625 2 2 sp|P57088|TMM33_HUMAN TMEM33 27.96 3.3465
2 2 sp|P42167|LAP2B_HUMAN TMPO 50.64 3.3447 2 2 sp|P49755|TMEDA_HUMAN TMED10 24.96 3.3296 2 2
sp|O00257|CBX4_HUMAN CBX4 61.33 3.3272 2 2 sp|P49792|RBP2_HUMAN RANBP2 357.97 3.3175 2 2
sp|P42285|SK2L2_HUMAN SKIV2L2 117.73 3.3157 2 2 sp|Q9NRZ9|HELLS_HUMAN HELLS 97.01 3.3123 2 2
sp|O75251|NDUS7_HUMAN NDUFS7 23.55 3.3089 2 2 sp|Q86U86|PB1_HUMAN PBRM1 192.83 3.3065 2 2
sp|Q8WYP5|ELYS_HUMAN AHCTF1 252.34 3.2824 2 2 sp|Q9Y5J1|UTP18_HUMAN UTP18 61.96 3.2445 2 2
sp|Q5TA45|INT11_HUMAN INTS11 67.62 3.2252 2 2 sp|Q8IZL8|PELP1_HUMAN PELP1 119.62 3.2022 2 2
sp|P19404|NDUV2_HUMAN NDUFV2 27.37 3.2008 2 2 sp|Q9H0U3|MAGT1_HUMAN MAGT1 38.01 3.1984 2 2
sp|P33778|H2B1B_HUMAN HIST1H2BB 13.94 3.1968 2 2 sp|Q9P0J0|NDUAD_HUMAN NDUFA13 16.69 3.1856
2 2 sp|Q9UNQ2|DIM1_HUMAN DIMT1 35.21 3.1738 2 2 sp|Q9BTV4|TMM43_HUMAN TMEM43 44.85 3.1711
2 2 sp|Q9NUQ2|PLCE_HUMAN AGPAT5 42.04 3.1429 2 2 sp|Q00587|BORG5_HUMAN CDC42EP1 40.27 3.1195
2 2 sp|P61619|S61A1_HUMAN SEC61A1 52.23 3.0941 2 2 sp|Q9BVP2|GNL3_HUMAN GNL3 61.95 3.0916 2 2
sp|Q9ULK4|MED23_HUMAN MED23 156.37 3.0903 2 2 sp|Q9NVH0|EXD2_HUMAN EXD2 70.31 3.0877 2 2
sp|P51553|IDH3G_HUMAN IDH3G 42.77 3.0735 2 2 sp|Q8IXI2|MIRO1_HUMAN RHOT1 70.74 3.0472 2 2
sp|P54652|HSP72_HUMAN HSPA2 69.98 2.9928 2 2 sp|Q969X6|UTP4_HUMAN UTP4 76.84 2.9848 2 2
sp|O95573|ACSL3_HUMAN ACSL3 80.37 2.972 2 2 sp|P05412|JUN_HUMAN JUN 35.65 2.9631 2 2
sp|Q9Y2X9|ZN281_HUMAN ZNF281 96.85 2.8979 2 2 sp|Q9NNW5|WDR6_HUMAN WDR6 121.65 2.8824 2 2
sp|Q13505|MTX1_HUMAN MTX1 51.44 2.8361 2 2 sp|Q13123|RED_HUMAN IK 65.56 2.8316 2 2
sp|Q9NR30|DDX21_HUMAN DDX21 87.29 2.794 2 2 sp|Q9H8H2|DDX31_HUMAN DDX31 94.03 2.7931 2 2
sp|Q5JVF3|PCID2_HUMAN PCID2 46 2.7734 2 2 sp|Q9H300|PARL_HUMAN PARL 42.16 2.7484 2 2
sp|Q8N684|CPSF7_HUMAN CPSF7 52.02 2.7454 2 2 sp|Q9NRG9|AAAS_HUMAN AAAS 59.54 2.738 2 2
sp|O43347|MSI1H_HUMAN MSI1 39.1 2.7273 2 2 sp|O94805|ACL6B_HUMAN ACTL6B 46.85 2.7223 2 2
sp|P30825|CTR1_HUMAN SLC7A1 67.59 2.6943 2 2 sp|Q9P032|NDUF4_HUMAN NDUFAF4 20.25 2.6784 2 2
sp|Q10469|MGAT2_HUMAN MGAT2 51.52 2.6709 2 2 sp|P38117|ETFB_HUMAN ETFB 27.83 2.6625 2 2
sp|Q9H2V7|SPNS1_HUMAN SPNS1 56.59 2.6569 2 2 sp|Q9HD45|TM9S3_HUMAN TM9SF3 67.84 2.6083 2 2
sp|Q96HS1|PGAM5_HUMAN PGAM5 31.98 2.5617 2 2 sp|P11166|GTR1_HUMAN SLC2A1 54.05 2.5583 2 2
sp|Q9NVP1|DDX18_HUMAN DDX18 75.36 2.4152 2 2 sp|P40937|RFC5_HUMAN RFC5 38.47 2.3758 2 2
sp|Q92643|GPI8_HUMAN PIGK 45.22 2.3695 2 2 sp|Q7L3T8|SYPM_HUMAN PARS2 53.23 2.3537 2 2
sp|Q96AA3|RFT1_HUMAN RFT1 60.3 2.3164 2 2 sp|P12081|SYHC_HUMAN HARS 57.37 2.3032 2 2
sp|Q8TDD1|DDX54_HUMAN DDX54 98.53 2.3024 2 2 sp|Q86TJ2|TAD2B_HUMAN TADA2B 48.44 2.2102 2 2
sp|Q6PI48|SYDM_HUMAN DARS2 73.52 2.2029 2 2 sp|P62851|RS25_HUMAN RPS25 13.73 2.0777 2 2

sp|Q9HC21|TPC_HUMAN SLC25A19 35.49 1.7524 1 2 sp|A8CG34|P121C_HUMAN POM121C 124.98 5.0358 1 2 sp|Q15388|TOM20_HUMAN TOMM20 16.29 4.3863 1 2 sp|Q01844|EWS_HUMAN EWSR1 68.44 4.2847 1 2 tr|C8C3P2|C8C3P2_HUMAN DPF1 45.07 4.2698 1 2 sp|Q01130|SRSF2_HUMAN SRSF2 25.46 3.8434 1 2 sp|Q9BYX7|ACTBM_HUMAN POTEKP 41.99 3.5392 1 2 sp|P41252|SYIC_HUMAN IARS 144.41 2.8477 1 2 sp|Q569K6|CC157_HUMAN CCDC157 83.89 2.5838 1 2 sp|Q7Z2K6|ERMP1_HUMAN ERMP1 100.17 2.5131 1 2 sp|Q7RTS9|DYM_HUMAN DYM 75.89 2.1052 1 1 sp|Q92804|RBP56_HUMAN TAF15 61.79 6.3231 1 1 tr|B7ZAF6|B7ZAF6_HUMAN SUCLA2 43.83 5.7056 1 1 sp|P39210|MPV17_HUMAN MPV17 19.72 5.6581 1 1 sp|Q9NXW2|DJB12_HUMAN DNAJB12 41.79 5.557 1 1 sp|Q8IXB1|DJC10_HUMAN DNAJC10 91.02 5.3514 1 1 sp|Q9Y3D7|TIM16_HUMAN PAM16 13.82 5.3043 1 1 sp|O75431|MTX2_HUMAN MTX2 29.74 5.2556 1 1 tr|H7BXI1|H7BXI1_HUMAN ESYT2 97.95 5.0949 1 1 sp|Q7L2E3|DHX30_HUMAN DHX30 133.85 5.0651 1 1 sp|O60306|AQR_HUMAN AQR 171.19 5.0401 1 1 sp|Q9NU22|MDN1_HUMAN MDN1 632.42 5.0035 1 1 tr|S4R341|S4R341_HUMAN NOLC1 8.05 4.9157 1 1 sp|Q8NE86|MCU_HUMAN MCU 39.84 4.8613 1 1 sp|P62314|SMD1_HUMAN SNRPD1 13.27 4.8513 1 1 sp|Q7L8L6|FAKD5_HUMAN FASTKD5 86.52 4.8453 1 1 sp|Q9UH99|SUN2_HUMAN SUN2 80.26 4.8348 1 1 tr|B2R5W2|B2R5W2_HUMAN HNRNPC 31.93 4.8315 1 1 sp|Q7LGA3|HS2ST_HUMAN HS2ST1 41.85 4.7987 1 1 sp|Q9H0H0|INT2_HUMAN INTS2 134.24 4.7963 1 1 sp|Q9Y3T9|NOC2L_HUMAN NOC2L 84.87 4.7054 1 1 tr|F8W7T1|F8W7T1_HUMAN DPF3 46.42 4.6927 1 1 sp|Q9Y679|AUP1_HUMAN AUP1 52.99 4.6866 1 1 sp|P12004|PCNA_HUMAN PCNA 28.75 4.6666 1 1 sp|P38435|VKGC_HUMAN GGCX 87.5 4.6623 1 1 sp|O75529|TAF5L_HUMAN TAF5L 66.11 4.6056 1 1 sp|Q8WY36|BBX_HUMAN BBX 105.06 4.5974 1 1 sp|O14925|TIM23_HUMAN TIMM23 21.93 4.5899 1 1 sp|Q6NSZ9|ZSC25_HUMAN ZSCAN25 61.44 4.5825 1 1 sp|Q13601|KRR1_HUMAN KRR1 43.64 4.5489 1 1 sp|Q9BXW9|FACD2_HUMAN FANCD2 164.02 4.517 1 1 sp|Q9Y2Q3|GSTK1_HUMAN GSTK1 25.48 4.509 1 1 sp|P09622|DLDH_HUMAN DLD 54.14 4.4997 1 1 sp|P62136|PP1A_HUMAN PPP1CA 37.49 4.4917 1 1 sp|Q5C9Z4|NOM1_HUMAN NOM1 96.2 4.4584 1 1 sp|Q5HYI7|MTX3_HUMAN MTX3 35.07 4.4263 1 1 sp|P55265|DSRAD_HUMAN ADAR 135.98 4.4253 1 1 sp|P17812|PYRG1_HUMAN CTPS1 66.65 4.4181 1 1 sp|Q8NBN7|RDH13_HUMAN RDH13 35.91 4.4096 1 1 sp|P35232|PHB_HUMAN PHB 29.79 4.3597 1 1 sp|Q9BT22|ALG1_HUMAN ALG1 52.48 4.3474 1 1 sp|Q9BQ67|GRWD1_HUMAN GRWD1 49.39 4.3332 1 1 sp|P35558|PCKGC_HUMAN PCK1 69.15 4.2123 1 1 sp|Q9UNL2|SSRG_HUMAN SSR3 21.07 4.1982 1 1 sp|O60884|DNJA2_HUMAN DNAJA2 45.72 4.1793 1 1 sp|Q9Y3A6|TMED5_HUMAN TMED5 25.99 4.1645 1 1 sp|P15880|RS2_HUMAN RPS2 31.3 4.1633 1 1 sp|Q9NVV4|PAPD1_HUMAN MTPAP 66.13 4.1628 1 1 sp|Q96QD8|S38A2_HUMAN SLC38A2 55.99 4.1444 1 1 sp|Q92797|SYMPK_HUMAN SYMPK 141.06 4.142 1 1 sp|Q75QN2|INT8_HUMAN INTS8 113.02 4.124 1 1 sp|O75528|TADA3_HUMAN TADA3 48.87 4.0685 1 1 sp|Q6NTF9|RHBD2_HUMAN RHBDD2 39.18 4.0674 1 1 sp|Q5JPH6|SYEM_HUMAN EARS2 58.65 4.0427 1 1 sp|Q8N0V3|RBFA_HUMAN RBFA 38.34 4.0312 1 1 sp|Q9H5Q4|TFB2M_HUMAN TFB2M 45.32 4.0306 1 1 sp|Q7Z7K6|CENPV_HUMAN CENPV 29.93 4.0181 1 1 sp|P62995|TRA2B_HUMAN TRA2B 33.65 4.0136 1 1 sp|P0CG08|GPHRB_HUMAN GPR89B 52.88 3.9848 1 1 sp|Q9BYD2|RM09_HUMAN MRPL9 30.22 3.9633 1 1 sp|Q12788|TBL3_HUMAN TBL3 88.98 3.963 1 1 sp|Q8NAN2|MIGA1_HUMAN MIGA1 70.96 3.9529 1 1 sp|P57740|NU107_HUMAN NUP107 106.31 3.932 1 1 sp|O75439|MPPB_HUMAN PMPCB 54.33 3.9211 1 1 sp|Q8IWA4|MFN1_HUMAN MFN1 84.05 3.906 1 1 sp|Q9UH62|ARMX3_HUMAN ARMCX3 42.47 3.8838 1 1 sp|O60830|TI17B_HUMAN TIMM17B 18.26 3.8619 1 1 sp|Q8TBP6|S2540_HUMAN SLC25A40 38.1 3.8378 1 1 sp|Q86Y07|VRK2_HUMAN VRK2 58.1 3.8287 1 1 sp|Q96BN2|TADA1_HUMAN TADA1 37.36 3.7447 1 1 sp|Q9HDC5|JPH1_HUMAN JPH1 71.64 3.7388 1 1 sp|Q6NUN9|ZN746_HUMAN ZNF746 69.09 3.7249 1 1 sp|O14579|COPE_HUMAN COPE 34.46 3.7075 1 1 sp|Q9Y2J2|E41L3_HUMAN EPB41L3 120.6 3.7065 1 1 sp|Q8N442|GUF1_HUMAN GUF1 74.28 3.7034 1 1 sp|P50454|SERPH_HUMAN SERPINH1 46.41 3.7024 1 1 sp|P52429|DGKE_HUMAN DGKE 63.88 3.6861 1 1 sp|Q96H55|MYO19_HUMAN MYO19 109.07 3.6814 1 1 sp|Q9Y2G8|DJC16_HUMAN DNAJC16 90.53 3.6556 1 1 sp|Q9Y6J9|TAF6L_HUMAN TAF6L 67.77 3.6543 1 1 sp|P48651|PTSS1_HUMAN PTDSS1 55.49 3.649 1 1 sp|Q08945|SSRP1_HUMAN SSRP1 81.02 3.6213 1 1 sp|Q9UBU9|NXF1_HUMAN NXF1 70.14 3.6188 1 1 sp|O43159|RRP8_HUMAN RRP8 50.68 3.6157 1 1 sp|Q9UMS4|PRP19_HUMAN PRPF19 55.15 3.6146 1 1 sp|Q8N6R0|MET13_HUMAN METTL13 78.72 3.5755 1 1 sp|Q8TAA9|VANG1_HUMAN VANGL1 59.94 3.5563 1 1 sp|O14776|TCRG1_HUMAN TCERG1 123.82 3.556 1 1 sp|Q14146|URB2_HUMAN URB2 170.43 3.5345 1 1 sp|Q9BVI4|NOC4L_HUMAN NOC4L 58.43 3.5218 1 1 sp|P35637|FUS_HUMAN FUS 53.39 3.4774 1 1 sp|P32322|P5CR1_HUMAN PYCR1 33.34 3.4646 1 1 sp|Q8N6L1|KTAP2_HUMAN KRTCAP2 14.67 3.4492 1 1 IGKC_MOUSE 11.77 3.4442 1 1 sp|P37198|NUP62_HUMAN NUP62 53.22 3.4196 1 1 sp|Q96RQ1|ERGI2_HUMAN ERGIC2 42.52 3.4166 1 1 sp|P49711|CTCF_HUMAN CTCF 82.73 3.3935 1 1 sp|Q02338|BDH_HUMAN BDH1 38.13 3.3731 1 1 sp|Q86VP6|CAND1_HUMAN CAND1 136.29 3.3721 1 1 sp|Q86U38|NOP9_HUMAN NOP9 69.39 3.3659 1 1 sp|O75352|MPU1_HUMAN MPDU1 26.62 3.3542 1 1 sp|Q8WWC4|MAIP1_HUMAN MAIP1 32.52 3.3373 1 1 sp|A4D1E9|GTPBA_HUMAN GTPBP10 42.91 3.3326 1 1 sp|P17858|PFKAL_HUMAN PFKL 84.96 3.3144 1 1 sp|Q8WUY9|DEP1B_HUMAN DEPDC1B 61.73 3.3118 1 1 sp|Q14318|FKBP8_HUMAN FKBP8 44.53 3.275 1 1 sp|P17987|TCPA_HUMAN TCP1 60.31 3.2565 1 1 sp|P35658|NU214_HUMAN NUP214 213.49 3.2536 1 1 sp|Q9NZB8|MOCS1_HUMAN MOCS1 70.06 3.2531 1 1 sp|Q96CU9|FXRD1_HUMAN FOXRED1 53.78 3.2527 1 1 sp|Q99653|CHP1_HUMAN CHP1 22.44 3.2414 1 1 sp|O43306|ADCY6_HUMAN ADCY6 130.53 3.2381 1 1 sp|Q8NB90|SPAT5_HUMAN SPATA5 97.84 3.2371 1 1 sp|Q6PML9|ZNT9_HUMAN SLC30A9 63.47 3.2286 1 1 sp|Q8TCJ2|STT3B_HUMAN STT3B 93.61 3.213 1 1 sp|Q9UI10|EI2BD_HUMAN EIF2B4 57.52 3.2073 1 1 sp|O76021|RL1D1_HUMAN RSL1D1 54.94 3.2001 1 1 sp|O75494|SRS10_HUMAN SRSF10 31.28 3.1959 1 1 sp|Q8WUK0|PTPM1_HUMAN PTPMT1 22.83 3.1906 1 1 sp|P62805|H4_HUMAN HIST1H4A 11.36 3.1841 1 1 sp|Q15366|PCBP2_HUMAN PCBP2 38.56 3.1618 1 1 sp|Q96HW7|INT4_HUMAN INTS4 108.1 3.1575 1 1 sp|Q8TE59|ATS19_HUMAN ADAMTS19 133.96 3.1453 1 1 sp|Q9NQ50|RM40_HUMAN MRPL40 24.48 3.1291 1 1 sp|Q66K74|MAP1S_HUMAN MAP1S 112.14 3.1258 1 1 sp|Q96JP5|ZFP91_HUMAN ZFP91 63.41 3.1186 1 1 sp|P82933|RT09_HUMAN MRPS9 45.81 3.1013 1 1 sp|O14981|BTAF1_HUMAN BTAF1 206.76 3.0838 1 1 sp|Q14008|CKAP5_HUMAN CKAP5 225.35 3.0668 1 1 sp|Q8IZ69|TRM2A_HUMAN TRMT2A 68.68 3.0458 1 1 sp|Q9H1X3|DJC25_HUMAN DNAJC25 42.38 3.0282 1 1 sp|Q7Z406|MYH14_HUMAN MYH14 227.73 3.0134 1 1 tr|Q3B7X4|Q3B7X4_HUMAN IMMT 40.47 3.0118 1 1 sp|Q8WY07|CTR3_HUMAN SLC7A3 67.13 2.9945 1 1 sp|Q6UX07|DHR13_HUMAN DHRS13 40.82 2.9688 1 1 sp|O00217|NDUS8_HUMAN NDUFS8 23.69 2.948 1 1 sp|Q9Y3D6|FIS1_HUMAN FIS1 16.93 2.9299 1 1 sp|Q92522|H1X_HUMAN H1FX 22.47 2.9236 1 1 sp|Q9H490|PIGU_HUMAN PIGU 50.02 2.9229 1 1 sp|P49368|TCPG_HUMAN CCT3 60.5 2.917 1 1 sp|Q9H974|QTRT2_HUMAN QTRT2 46.68 2.9127 1 1 sp|Q8WUB8|PHF10_HUMAN PHF10 56.02 2.9119 1 1 KV2A7_MOUSE 12.27 2.8807 1 1 sp|Q9NXS2|QPCTL_HUMAN QPCTL 42.9 2.8763 1 1 sp|Q9Y584|TIM22_HUMAN TIMM22 20.02 2.8643 1 1 sp|P42356|PI4KA_HUMAN PI4KA 236.68 2.8615 1 1 sp|Q86TW2|ADCK1_HUMAN ADCK1 60.54 2.8425 1 1 sp|Q8TAE8|G45IP_HUMAN GADD45GIP1 25.37 2.8196 1 1 sp|O15523|DDX3Y_HUMAN DDX3Y 73.11 2.8124 1 1 sp|O75323|NIPS2_HUMAN NIPSNAP2 33.72 2.7929 1 1 sp|Q9UBD5|ORC3_HUMAN ORC3 82.2 2.7077 1 1 sp|O95070|YIF1A_HUMAN YIF1A 31.99 2.6979 1 1 sp|P42695|CNDD3_HUMAN NCAPD3 168.78 2.6917 1 1 sp|P06493|CDK1_HUMAN CDK1 34.07 2.6561 1 1 sp|Q09161|NCBP1_HUMAN NCBP1 91.78 2.6544 1 1 sp|Q8IUX1|T126B_HUMAN TMEM126B 25.93 2.6456 1 1 tr|B3KUE6|B3KUE6_HUMAN 30.19 2.6448 1 1 sp|Q9BVK8|TM147_HUMAN TMEM147 25.24 2.642 1 1 sp|Q53R41|FAKD1_HUMAN FASTKD1 97.35 2.6172 1 1 sp|O00483|NDUA4_HUMAN NDUFA4 9.36 2.6166 1 1 sp|P61026|RAB10_HUMAN RAB10 22.53 2.596 1 1 sp|Q96A65|EXOC4_HUMAN EXOC4 110.43 2.5958 1 1 tr|Q6FI97|Q6FI97_HUMAN BAF53A 47.35 2.5795 1 1 sp|Q15392|DHC24_HUMAN DHCR24 60.06 2.5722 1 1 sp|P83731|RL24_HUMAN RPL24 17.77 2.5706 1 1 sp|Q9NZJ7|MTCH1_HUMAN MTCH1 41.52 2.5516 1 1 sp|Q9BUN8|DERL1_HUMAN DERL1 28.78 2.5224 1 1 sp|P46783|RS10_HUMAN RPS10 18.89 2.5189 1 1 sp|O43390|HNRPR_HUMAN HNRNPR 70.9 2.5031 1 1 sp|O60673|REV3L_HUMAN REV3L 352.55 2.4985 1 1 sp|Q8IVH4|MMAA_HUMAN MMAA 46.51 2.4963 1 1 sp|Q16563|SYPL1_HUMAN SYPL1 28.55 2.4909 1 1 sp|O14735|CDIPT_HUMAN CDIPT 23.52 2.475 1 1 sp|Q9BSJ2|GCP2_HUMAN TUBGCP2 102.47 2.4439 1 1 sp|P62701|RS4X_HUMAN RPS4X 29.58 2.4418 1 1 sp|Q14683|SMC1A_HUMAN SMC1A 143.14 2.4391 1 1 sp|P08559|ODPA_HUMAN PDHA1 43.27 2.4288 1 1 sp|Q9Y651|SOX21_HUMAN SOX21 28.56 2.4124 1 1 sp|Q96JX3|SRAC1_HUMAN SERAC1 74.1 2.3968 1 1 sp|Q8NDZ4|DIA1_HUMAN C3orf58 49.45 2.3757 1 1 sp|P46013|KI67_HUMAN MKI67 358.47 2.3711 1 1 sp|P68871|HBB_HUMAN HBB 15.99 2.3665 1 1 sp|P12036|NFH_HUMAN NEFH 112.41 2.3587 1 1 sp|Q5T0B9|ZN362_HUMAN ZNF362 45.79 2.3304 1 1 sp|Q9H0D6|XRN2_HUMAN XRN2 108.51 2.3224 1 1 tr|Q96DP0|Q96DP0_HUMAN 49.22 2.2814 1 1 sp|O60725|ICMT_HUMAN ICMT 31.92 2.2483 1 1 sp|Q8N4U5|T11L2_HUMAN TCP11L2 58.05 2.2363 1 1 sp|Q9UL03|INT6_HUMAN INTS6 100.33 2.1871 1 1 sp|P43007|SATT_HUMAN SLC1A4 55.69 2.1855 1 1 sp|Q6VAB6|KSR2_HUMAN KSR2 107.56 2.1832 1 1 sp|Q9NWU5|RM22_HUMAN MRPL22 23.63 2.1812 1 1 tr|B4DL14|B4DL14_HUMAN 27.5 2.1811 1 1 sp|Q6YN16|HSDL2_HUMAN HSDL2 45.37 2.1314 1 1 sp|Q9NVH2|INT7_HUMAN INTS7 106.77 2.1225 1 1 sp|P41219|PERI_HUMAN PRPH 53.62 2.117 1 1 sp|Q9P0U1|TOM7_HUMAN TOMM7 6.24 2.094 1 1 sp|P01857|IGHG1_HUMAN IGHG1 36.08 2.092 1 1 sp|Q9Y4F1|FARP1_HUMAN FARP1 118.56 2.0479 1 1 sp|O75964|ATP5L_HUMAN ATP5L 11.42 2.0435 1 1 tr|A0A1W2PP06|A0A1W2PP06_HUMAN KCNMA1 136.96 2.0205 1 1 sp|Q92604|LGAT1_HUMAN LPGAT1 43.06 2.0192 1 1 sp|Q8WZA0|LZIC_HUMAN LZIC 21.48 1.9913

TABLE-US-00018 TABLE 5E Ubiquitination report HA-SS18SSX_CHR ↓ Gene ScanF Z XCorr ΔCorr # Ions Reference Redun Peptide Symbol 18257 3 4.935 0.033 30/76 sp|Q96KK5|H2A1H_HUMAN 19 K.VTIAQGGVLPNIQAVLLPK#K.T HIST1H2AH (SEQ ID NO: 229)

[0536] Purification of SS18-SSX-bound complexes followed by density sedimentation using 10-30% glycerol gradients revealed larger-sized fusion-containing BAF complexes migrating in fractions 15-19, compared to WT SS18-bound complexes in fractions 13-14, as expected (Mashtalir et al. (2018) *Cell* 175:1272-1288), indicating high-affinity, stable binding of SS18-SSX-bound BAF complexes to the full histone octamer (FIG. **1**C and FIG. **2**H). In addition, histones bound to the ATPase module components were observed in isolation as well as to free SS18-SSX in fractions 9-13 and 2-4, respectively. These results indicate that fusion-containing BAF complexes exhibit exceptionally strong binding to nucleosomes, able to withstand separation even in a high centrifugal force environment, in contrast to WT BAF complexes which exhibit weaker interactions with nucleosomes, as seen consistently in BAF complex purifications performed to date (Mashtalir et al. (2018) Cell 175:1272-1288). Finally, to determine the relative chromatin affinities of WT BAF complexes versus SS18-SSX-containing BAF complexes, differential salt extraction in both SS cell lines and HEK-293T cells expressing SS18-SSX was performed (FIG. **1**D, FIGS. **2**C and **2**D). Normal extraction profiles for WT complexes were observed (elution at 300-500 mM NaCl), consistent with previous findings (Nakayama et al. (2017) *Nat. Genet.* 49:1613-1623; Pan et al. (2019) *Nat. Genet.* 51:618-626). However, fusion-containing complexes remained insoluble in up to 1M NaCl. In support of these findings, fluorescence recovery after photobleaching (FRAP) experiments in HEK-293T cells infected with either GFP-SS18 or GFP-SS18-SSX revealed substantially increased chromatin residency times for SS18-SSX-bound BAF complexes (FIG. **1**E and FIG. **2**E). Taken together, these findings indicate an unexpected, uniquely high-affinity conjugation of SS18-SSX-bound BAF complexes to nucleosomes, a property specific to this disease-associated BAF complex perturbation, indicating this as a feature that can underlie the site-specific targeting of SS18-SSX complexes on chromatin.

Example 3: A Minimal 34-Aa Region of SSX is Necessary and Sufficient for Direct Binding to Repressive Nucleosomes and SS18-SSX-Mediated Oncogenic Functions

[0537] Given these results, it was next determined whether the 78 residues of SSX in isolation (not fused to the SS18 subunit and hence not part of BAF complexes) could directly bind nucleosomes and could be responsible for conferring the unique affinity and nucleosome binding properties of the SS18-SSX fusion protein. Indeed, pull-down experiments revealed that the C-terminal 78 residues of SSX (aa 111-188) were sufficient for its nucleosomal interactions (FIG. **3**A and FIGS. **4**A-**4**B). In addition, it was found that binding to mammalian nucleosomes (purified via MNase digestion of HEK-293T cell chromatin and hence representing the diverse array of histone variants and modifications) was stronger than binding to recombinant, unmodified nucleosomes (FIGS. **4**C and **4**G), indicating that a mammalian histone modification can provide added affinity and site specificity. In agreement with this, targeted quantitative mass-spectrometry (MS) analysis of SSX-bound mammalian nucleosomes (pooled, purified by MNase digestion from HEK-293T cells, containing the full diversity of histone marks) revealed strong enrichment of nucleosomes decorated with known repressive histone marks and depletion of nucleosomes marked with known activation marks (FIG. **3**B, FIGS. **4**D-**4**F, and FIG. **4**H, Tables 6A-6C). For example, SSX-mediated enrichment of nucleosomes decorated with repressive marks such as H3K27me3 and H3K9me3, and SSX-mediated depletion of nucleosomes decorated with activating marks such as H4 lysine acetylation and H3K4me2/3 were detected (while nucleosomes containing unmodified H4 and H3 were enriched). Further, immunofluorescence (IF) analyses revealed strong colocalization of SS18-SSX as well as SSX in isolation (SSX aa 1-188, as expressed in testes) to Barr bodies marked with repressive PRC1 and PRC2 complexes and their marks (FIG. **3**C and FIGS. **5**A-**5**B).

TABLE-US-00019 TABLE 6A log2norm: (Light/Heavy) intensity ratios are normalized to the (Light/Heavy) intensity ratio of respective histone “norm” peptide, and brought into log2 space. Cells with #N/A were below l.o.d Histone Mark MM01_A01 MM02_B01 MM03_C01 MM04_D01 MM05_E01 Cell Line HEK293T HEK293T HEK293T HEK293T HEK293T Perturbation pull down pull down pull down SSX1 SSX1 Parent Peptide input input input pulldown pulldo Experiment 1 1 1 1 1 H4(4to17)ac0me0 H4K5K8K12K16 −0.999 −0.768 −0.681 −0.613 −0.573 H4(4to17)K5ac1me0 H4K5K8K12K16 0.048 0.332 0.333 −0.053 0.075 H4(4to17)K12ac1me0 H4K5K8K12K16 −0.327 0.036 0.140 −0.207 −0.275 H4(4to17)K16ac1me0 H4K5K8K12K16 −0.265 −0.168 −0.258 −0.391 −0.337 H4(4to17)K8ac1K12ac1me0 H4K5K8K12K16 −0.117 0.026 0.240 −0.198 −0.093 H4(4to17)K5ac1K8ac1me0 H4K5K8K12K16 0.502 0.475 0.651 0.107 0.094 H4(4to17)K5ac1K16ac1me0 H4K5K8K12K16 0.618 0.676 0.621 0.226 0.367 H4(4to17)K12ac1K16ac1me0 H4K5K8K12K16 0.411 0.391 0.424 0.204 0.304 H4(4to17)K5ac1K8ac1K12ac1me0 H4K5K8K12K16 −0.054 0.091 0.312 −0.223 −0.270 H4(4to17)K8ac1K12ac1K16ac1me0 H4K5K8K12K16 0.816 0.897 1.090 0.677 0.625 H4(4to17)K5ac1K8ac1K16ac1me0 H4K5K8K12K16 0.819 0.755 0.972 0.474 0.311 H4(4to17)K5ac1K8ac1K12ac1K16ac1me0 H4K5K8K12K16 0.718 0.864 1.040 0.276 0.245 H4(20to23)K20me0 H4K20 −0.045 −0.175 0.077 0.017 0.092 H4(20to23)K20me1 H4K20 −0.658 −0.526 −0.731 −0.427 −0.443 H4(20to23)K20me2 H4K20 −0.424 −0.453 −0.437 −0.414 −0.391 H4(20to23)K20me3 H4K20 0.161 0.261 0.136 −0.109 0.055 H2AZ(1to19)ac0 H2A.Z −0.160 −0.062 −0.047 −0.784 −0.697 H2AZ(1to19)K4ac1 H2A.Z 0.890 1.031 0.906 −0.217 0.048 H2B(1to29)ac0 H2B 0.112 −0.187 −0.042 0.152 0.160 H2B(1to29)K5ac1 H2B 0.234 0.001 0.124 0.273 0.323 H2A(4to11)ac0 H2AK5K9 −0.187 −0.105 0.115 −0.026 0.061 H2A(4to11)K5ac1 H2AK5K9 0.275 0.713 0.614 0.569 0.624 H2A(4to11)K9ac1 H2AK5K9 −0.837 −0.552 −0.586 −0.254 −0.375 H2A(4to11)K5ac1K9ac1 H2AK5K9 −0.224 −0.043 0.025 −0.031 −0.049 H2A(12to17)ac0 H2AK13K15 −0.521 −0.203 0.070 0.201 −0.120 H2A(12to17)K13ac1 H2AK13K15 −0.459 −0.129 −0.116 0.032 −0.047 H2A(12to17)K15ac1 H2AK13K15 −0.430 −0.070 0.035 0.042 −0.086 H3K4me0 H3K4 −0.049 0.108 0.047 0.549 0.339 H3K4me1 H3K4 0.393 0.542 0.415 0.740 0.353 H3K4me2 H3K4 0.848 0.992 0.801 0.778 0.532 H3K4me3 H3K4 0.953 1.142 0.953 0.766 0.505 H3K4ac1 H3K4 0.485 0.651 0.588 0.263 0.320 H3K9me0K14ac0 H3K9K14 −0.262 −0.323 −0.280 0.274 −0.036 H3K9me1K14ac0 H3K9K14 −0.501 −0.392 −0.314 0.206 0.046 H3K9me2K14ac0 H3K9K14 −0.585 −0.359 −0.568 0.035 −0.284 H3K9me3K14ac0 H3K9K14 −0.364 −0.401 −0.153 0.348 0.123 H3K9ac1K14ac0 H3K9K14 1.020 1.120 0.965 1.203 1.075 H3K9me0K14ac1 H3K9K14 0.271

0.259 −0.018 0.637 0.469 H3K9me1K14ac1 H3K9K14 0.565 0.691 0.585 0.769 0.508 H3K9me2K14ac1 H3K9K14 0.390 0.458 0.421 0.724 0.530 H3K9me3K14ac1 H3K9K14 0.657 0.663 0.268 0.828 0.602 H3K9ac1K14ac1 H3K9K14 1.842 1.990 1.786 2.038 1.804 H3K9me0S10ph1K14ac0 H3K9K14 −2.614 −3.006 −2.823 #N/A #N/A H3K9me1S10ph1K14ac0 H3K9K14 −2.261 −2.090 −2.268 #N/A #N/A H3K9me2S10ph1K14ac0 H3K9K14 −2.004 −1.899 −2.048 −0.683 −1.092 H3K9me3S10ph1K14ac0 H3K9K14 −0.920 −0.606 −0.834 −0.566 −1.026 H3K9me0S10ph1K14ac1 H3K9K14 −2.471 −2.280 −1.882 #N/A #N/A H3K9me1S10ph1K14ac1 H3K9K14 −2.299 −3.153 −3.422 −2.668 −2.507 H3K9me2S10ph1K14ac1 H3K9K14 −1.957 −2.023 −2.085 −0.870 −1.009 H3K18ac0K23ac0 H3K18K23 −0.505 −0.140 −0.317 0.268 0.069 H3K18ac1K23ac0 H3K18K23 1.168 1.411 1.229 1.341 1.185 H3K18ac0K23ac1 H3K18K23 0.768 1.061 0.574 1.141 1.021 H3K18ac1K23ac1 H3K18K23 2.134 2.522 2.107 2.137 1.988 H3K27me0K36me0 H3K27K36 −0.169 0.187 −0.678 0.552 0.233 H3K27me0K36me1 H3K27K36 −0.407 −0.152 −0.659 0.384 −0.066 H3K27me0K36me2 H3K27K36 −0.476 −0.061 −0.419 0.674 0.270 H3K27me0K36me3 H3K27K36 −0.057 −0.001 0.070 0.678 0.669 H3K27me1K36me0 H3K27K36 −0.086 0.026 −0.148 0.396 0.104 H3K27me1K36me1 H3K27K36 −0.194 0.078 −0.086 0.535 0.196 H3K27me1K36me2 H3K27K36 0.248 0.397 0.323 0.741 0.461 H3K27me1K36me3 H3K27K36 0.229 0.344 0.296 1.283 0.859 H3K27me2K36me0 H3K27K36 −0.337 −0.004 −0.152 0.382 0.209 H3K27me2K36me1 H3K27K36 −0.092 −0.102 0.017 0.363 0.186 H3K27me2K36me2 H3K27K36 0.249 0.342 0.242 0.553 0.417 H3K27me2K36me3 H3K27K36 1.050 1.018 0.643 #N/A #N/A H3K27me3K36me0 H3K27K36 0.209 0.291 0.081 0.689 0.037 H3K27me3K36me1 H3K27K36 −0.214 0.212 −0.178 0.391 0.170 H3K27me3K36me2 H3K27K36 0.236 0.259 0.066 0.085 −0.151 H3K27me3K36me3 H3K27K36 1.439 1.614 1.393 #N/A #N/A H3K27ac1K36me0 H3K27K36 1.564 #N/A 1.201 1.481 #N/A H3K27ac1K36me1 H3K27K36 1.131 1.759 1.594 #N/A #N/A H3K27ac1K36me3 H3K27K36 1.652 1.675 1.562 2.134 2.004 H3K56me0 H3K56 0.111 0.303 −0.071 0.856 0.297 H3K79me1 H3K79 0.268 0.441 0.301 0.954 0.540 H3K79me2 H3K79 0.498 0.469 0.455 0.826 0.557

TABLE-US-00020 TABLE 6B replace na: values from Tables 6A are copy/pasted and “#N/A” values are removed. The third to seventh columns separate values by experiment. Columns to the right of the matrix calculate required averages and medians for subsequent analyses. Histone Mark MM01_A01 MM02_B01 MM03_C01 MM04_D01 MM05_E01 Cell Line HEK293T HEK293T HEK293T HEK293T HEK293T Perturbation pull down pull down pull down SSX1 SSX1 pulldown Median across Parent Peptide input input input pulldown pulldown ctrl avg all experiments H4(4to17)ac0me0 H4K5K8K12K1 −0.999 −0.768 −0.681 −0.613 −0.573 −0.816 −0.681 H4(4to17)K5ac1me0 H4K5K8K12K1 0.048 0.332 0.333 −0.053 0.075 0.237 0.075 H4(4to17)K12ac1me0 H4K5K8K12K1 −0.327 0.036 0.140 −0.207 −0.275 −0.050 −0.207 H4(4to17)K16ac1me0 H4K5K8K12K1 −0.265 −0.168 −0.258 −0.391 −0.337 −0.230 −0.265 H4(4to17)K8ac1K12ac1me0 H4K5K8K12K1 −0.117 0.026 0.240 −0.198 −0.093 0.050 −0.093 H4(4to17)K5ac1K8ac1me0 H4K5K8K12K1 0.502 0.475 0.651 0.107 0.094 0.543 0.475 H4(4to17)K5ac1K16ac1me0 H4K5K8K12K1 0.618 0.676 0.621 0.226 0.367 0.638 0.618 H4(4to17)K12ac1K16ac1me H4K5K8K12K1 0.411 0.391 0.424 0.204 0.304 0.409 0.391 H4(4to17)K5ac1K8ac1K12ac H4K5K8K12K1 −0.054 0.091 0.312 −0.223 −0.270 0.116 −0.054 H4(4to17)K8ac1K12ac1K16a H4K5K8K12K1 0.816 0.897 1.090 0.677 0.625 0.935 0.816 H4(4to17)K5ac1K8ac1K16ac H4K5K8K12K1 0.819 0.755 0.972 0.474 0.311 0.849 0.755 H4(4to17)K5ac1K8ac1K12ac H4K5K8K12K1 0.718 0.864 1.040 0.276 0.245 0.874 0.718 H4(20to23)K20me0 H4K20 −0.045 −0.175 0.077 0.017 0.092 −0.048 0.017 H4(20to23)K20me1 H4K20 −0.658 −0.526 −0.731 −0.427 −0.443 −0.638 −0.526 H4(20to23)K20me2 H4K20 −0.424 −0.453 −0.437 −0.414 −0.391 −0.438 −0.424 H4(20to23)K20me3 H4K20 0.161 0.261 0.136 −0.109 0.055 0.186 0.136 H2AZ(1to19)ac0 H2A.Z −0.160 −0.062 −0.047 −0.784 −0.697 −0.090 −0.160 H2AZ(1to19)K4ac1 H2A.Z 0.890 1.031 0.906 −0.217 0.048 0.943 0.890 H2B(1to29)ac0 H2B 0.112 −0.187 −0.042 0.152 0.160 −0.039 0.112 H2B(1to29)K5ac1 H2B 0.234 0.001 0.124 0.273 0.323 0.120 0.234 H2A(4to11)ac0 H2AK5K9 −0.187 −0.105 0.115 −0.026 0.061 −0.059 −0.026 H2A(4to11)K5ac1 H2AK5K9 0.275 0.713 0.614 0.569 0.624 0.534 0.614 H2A(4to11)K9ac1 H2AK5K9 −0.837 −0.552 −0.586 −0.254 −0.375 −0.658 −0.552 H2A(4to11)K5ac1K9ac1 H2AK5K9 −0.224 −0.043 0.025 −0.031 −0.049 −0.081 −0.043 H2A(12to17)ac0 H2AK13K15 −0.521 −0.203 0.070 0.201 −0.120 −0.218 −0.120 H2A(12to17)K13ac1 H2AK13K15 −0.459 −0.129 −0.116 0.032 −0.047 −0.234 −0.116 H2A(12to17)K15ac1 H2AK13K15 −0.430 −0.070 0.035 0.042 −0.086 −0.155 −0.070 H3K4me0 H3K4 −0.049 0.108 0.047 0.549 0.339 0.035 0.108 H3K4me1 H3K4 0.393 0.542 0.415 0.740 0.353 0.450 0.415 H3K4me2 H3K4 0.848 0.992 0.801 0.778 0.532 0.881 0.801 H3K4me3 H3K4 0.953 1.142 0.953 0.766 0.505 1.016 0.953 H3K4ac1 H3K4 0.485 0.651 0.588 0.263 0.320 0.575 0.485 H3K9me0K14ac0 H3K9K14 −0.262 −0.323 −0.280 0.274 −0.036 −0.288 −0.262 H3K9me1K14ac0 H3K9K14 −0.501 −0.392 −0.314 0.206 0.046 −0.402 −0.314 H3K9me2K14ac0 H3K9K14 −0.585 −0.359 −0.568 0.035 −0.284 −0.504 −0.359 H3K9me3K14ac0 H3K9K14 −0.364 −0.401 −0.153 0.348 0.123 −0.306 −0.153 H3K9ac1K14ac0 H3K9K14 1.020 1.120 0.965 1.203 1.075 1.035 1.075 H3K9me0K14ac1 H3K9K14 0.271 0.259 −0.018 0.637 0.469 0.170 0.271 H3K9me1K14ac1 H3K9K14 0.565 0.691 0.585 0.769 0.508 0.614 0.585 H3K9me2K14ac1 H3K9K14 0.390 0.458 0.421 0.724 0.530 0.423 0.458 H3K9me3K14ac1 H3K9K14 0.657 0.663 0.268 0.828 0.602 0.529 0.657 H3K9ac1K14ac1 H3K9K14 1.842 1.990 1.786 2.038 1.804 1.872 1.842 H3K9me0S10ph1K14ac0 H3K9K14 −2.614 −3.006 −2.823 −2.814 −2.823 H3K9me1S10ph1K14ac0 H3K9K14

−2.261 −2.090 −2.268 −2.206 −2.261 H3K9me2S10ph1K14ac0 H3K9K14 −2.004 −1.899 −2.048 −0.683 −1.092 −1.984 −1.899 H3K9me3S10ph1K14ac0 H3K9K14 −0.920 −0.606 −0.834 −0.566 −1.026 −0.787 −0.834 H3K9me0S10ph1K14ac1 H3K9K14 −2.471 −2.280 −1.882 −2.211 −2.280 H3K9me1S10ph1K14ac1 H3K9K14 −2.299 −3.153 −3.422 −2.668 −2.507 −2.958 −2.668 H3K9me2S10ph1K14ac1 H3K9K14 −1.957 −2.023 −2.085 −0.870 −1.009 −2.022 −1.957 H3K18ac0K23ac0 H3K18K23 −0.505 −0.140 −0.317 0.268 0.069 −0.321 −0.140 H3K18ac1K23ac0 H3K18K23 1.168 1.411 1.229 1.341 1.185 1.270 1.229 H3K18ac0K23ac1 H3K18K23 0.768 1.061 0.574 1.141 1.021 0.801 1.021 H3K18ac1K23ac1 H3K18K23 2.134 2.522 2.107 2.137 1.988 2.254 2.134 H3K27me0K36me0 H3K27K36 −0.169 0.187 −0.678 0.552 0.233 −0.220 0.187 H3K27me0K36me1 H3K27K36 −0.407 −0.152 −0.659 0.384 −0.066 −0.406 −0.152 H3K27me0K36me2 H3K27K36 −0.476 −0.061 −0.419 0.674 0.270 −0.319 −0.061 H3K27me0K36me3 H3K27K36 −0.057 −0.001 0.070 0.678 0.669 0.004 0.070 H3K27me1K36me0 H3K27K36 −0.086 0.026 −0.148 0.396 0.104 −0.069 0.026 H3K27me1K36me1 H3K27K36 −0.194 0.078 −0.086 0.535 0.196 −0.067 0.078 H3K27me1K36me2 H3K27K36 0.248 0.397 0.323 0.741 0.461 0.322 0.397 H3K27me1K36me3 H3K27K36 0.229 0.344 0.296 1.283 0.859 0.289 0.344 H3K27me2K36me0 H3K27K36 −0.337 −0.004 −0.152 0.382 0.209 −0.164 −0.004 H3K27me2K36me1 H3K27K36 −0.092 −0.102 0.017 0.363 0.186 −0.059 0.017 H3K27me2K36me2 H3K27K36 0.249 0.342 0.242 0.553 0.417 0.278 0.342 H3K27me2K36me3 H3K27K36 1.050 1.018 0.643 0.904 1.018 H3K27me3K36me0 H3K27K36 0.209 0.291 0.081 0.689 0.037 0.194 0.209 H3K27me3K36me1 H3K27K36 −0.214 0.212 −0.178 0.391 0.170 −0.060 0.170 H3K27me3K36me2 H3K27K36 0.236 0.259 0.066 0.085 −0.151 0.187 0.085 H3K27me3K36me3 H3K27K36 1.439 1.614 1.393 1.482 1.439 H3K27ac1K36me0 H3K27K36 1.564 1.201 1.481 1.382 1.481 H3K27ac1K36me1 H3K27K36 1.131 1.759 1.594 1.494 1.594 H3K27ac1K36me3 H3K27K36 1.652 1.675 1.562 2.134 2.004 1.630 1.675 H3K56me0 H3K56 0.111 0.303 −0.071 0.856 0.297 0.114 0.297 H3K79me1 H3K79 0.268 0.441 0.301 0.954 0.540 0.337 0.441 H3K79me2 H3K79 0.498 0.469 0.455 0.826 0.557 0.474 0.498

TABLE-US-00021 TABLE 6C norm to ctrl avg: normalizing experiment values to average value for control samples. Histone Mark MM01_A01 MM02_B01 MM03_C01 MM04_D01 MM05_E01 Cell Line HEK293T HEK293T HEK293T HEK293T HEK293T Perturbation pull down pull down pull down SSX1 SSX1 Parent Peptide input input input pulldown pulldown H4(4to17)ac0me0 H4K5K8K12K16 −0.183 0.048 0.135 0.203 0.243 H4(4to17)K5ac1me0 H4K5K8K12K16 −0.190 0.094 0.095 −0.291 −0.162 H4(4to17)K12ac1me0 H4K5K8K12K16 −0.277 0.086 0.191 −0.156 −0.225 H4(4to17)K16ac1me0 H4K5K8K12K16 −0.034 0.062 −0.028 −0.161 −0.106 H4(4to17)K8ac1K12ac1me0 H4K5K8K12K16 −0.166 −0.024 0.190 −0.247 −0.143 H4(4to17)K5ac1K8ac1me0 H4K5K8K12K16 −0.041 −0.067 0.108 −0.436 −0.448 H4(4to17)K5ac1K16ac1me0 H4K5K8K12K16 −0.020 0.038 −0.018 −0.412 −0.272 H4(4to17)K12ac1K16ac1me0 H4K5K8K12K16 0.002 −0.018 0.016 −0.204 −0.105 H4(4to17)K5ac1K8ac1K12ac1me0 H4K5K8K12K16 −0.170 −0.026 0.195 −0.340 −0.387 H4(4to17)K8ac1K12ac1K16ac1me0 H4K5K8K12K16 −0.118 −0.037 0.155 −0.258 −0.309 H4(4to17)K5ac1K8ac1K16ac1me0 H4K5K8K12K16 −0.029 −0.094 0.123 −0.375 −0.537 H4(4to17)K5ac1K8ac1K12ac1K16ac1me0 H4K5K8K12K16 −0.156 −0.010 0.166 −0.599 −0.629 H4(20to23)K20me0 H4K20 0.003 −0.127 0.125 0.065 0.140 H4(20to23)K20me1 H4K20 −0.019 0.112 −0.093 0.211 0.195 H4(20to23)K20me2 H4K20 0.014 −0.015 0.001 0.024 0.047 H4(20to23)K20me3 H4K20 −0.025 0.075 −0.050 −0.295 −0.131 H2AZ(1to19)ac0 H2A.Z −0.071 0.028 0.043 −0.694 −0.608 H2AZ(1to19)K4ac1 H2A.Z −0.052 0.089 −0.036 −1.159 −0.894 H2B(1to29)ac0 H2B 0.151 −0.148 −0.003 0.191 0.199 H2B(1to29)K5ac1 H2B 0.114 −0.118 0.004 0.154 0.203 H2A(4to11)ac0 H2AK5K9 −0.128 −0.046 0.174 0.033 0.120 H2A(4to11)K5ac1 H2AK5K9 −0.259 0.179 0.080 0.036 0.090 H2A(4to11)K9ac1 H2AK5K9 −0.179 0.106 0.072 0.404 0.283 H2A(4to11)K5ac1K9ac1 H2AK5K9 −0.144 0.038 0.105 0.050 0.032 H2A(12to17)ac0 H2AK13K15 −0.303 0.015 0.288 0.419 0.098 H2A(12to17)K13ac1 H2AK13K15 −0.224 0.106 0.118 0.266 0.188 H2A(12to17)K15ac1 H2AK13K15 −0.275 0.085 0.190 0.197 0.069 H3K4me0 H3K4 −0.084 0.072 0.011 0.514 0.304 H3K4me1 H3K4 −0.057 0.092 −0.035 0.290 −0.097 H3K4me2 H3K4 −0.032 0.111 −0.079 −0.102 −0.348 H3K4me3 H3K4 −0.063 0.126 −0.063 −0.250 −0.511 H3K4ac1 H3K4 −0.090 0.076 0.013 −0.312 −0.255 H3K9me0K14ac0 H3K9K14 0.026 −0.034 0.008 0.562 0.252 H3K9me1K14ac0 H3K9K14 −0.099 0.010 0.088 0.608 0.449 H3K9me2K14ac0 H3K9K14 −0.081 0.146 −0.064 0.539 0.221 H3K9me3K14ac0 H3K9K14 −0.058 −0.095 0.153 0.654 0.429 H3K9ac1K14ac0 H3K9K14 −0.015 0.085 −0.070 0.168 0.040 H3K9me0K14ac1 H3K9K14 0.101 0.088 −0.189 0.466 0.299 H3K9me1K14ac1 H3K9K14 −0.049 0.077 −0.029 0.155 −0.105 H3K9me2K14ac1 H3K9K14 −0.033 0.035 −0.002 0.302 0.107 H3K9me3K14ac1 H3K9K14 0.128 0.134 −0.261 0.299 0.072 H3K9ac1K14ac1 H3K9K14 −0.031 0.117 −0.086 0.165 −0.068 H3K9me0S10ph1K14ac0 H3K9K14 0.200 −0.191 −0.009 H3K9me1S10ph1K14ac0 H3K9K14 −0.055 0.117 −0.062 H3K9me2S10ph1K14ac0 H3K9K14 −0.020 0.085 −0.064 1.301 0.892 H3K9me3S10ph1K14ac0 H3K9K14 −0.133 0.181 −0.048 0.221 −0.240 H3K9me0S10ph1K14ac1 H3K9K14 −0.260 −0.069 0.329 H3K9me1S10ph1K14ac1 H3K9K14 0.659 −0.195 −0.464 0.290 0.451 H3K9me2S10ph1K14ac1 H3K9K14 0.064 −0.001 −0.063 1.152 1.012 H3K18ac0K23ac0 H3K18K23 −0.184 0.181 0.004 0.588 0.390 H3K18ac1K23ac0 H3K18K23 −0.101 0.141 −0.040 0.072 −0.084 H3K18ac0K23ac1 H3K18K23 −0.033 0.260 −0.227 0.340 0.220 H3K18ac1K23ac1 H3K18K23 −0.120 0.267 −0.147 −0.117 −0.266 H3K27me0K36me0 H3K27K36 0.051 0.407 −0.458 0.773 0.453 H3K27me0K36me1 H3K27K36 −0.001 0.254

−0.253 0.790 0.340 H3K27me0K36me2 H3K27K36 −0.157 0.257 −0.100 0.993 0.589 H3K27me0K36me3 H3K27K36 −0.060 −0.005 0.066 0.674 0.666 H3K27me1K36me0 H3K27K36 −0.017 0.095 −0.079 0.465 0.173 H3K27me1K36me1 H3K27K36 −0.127 0.145 −0.019 0.602 0.263 H3K27me1K36me2 H3K27K36 −0.075 0.074 0.000 0.419 0.139 H3K27me1K36me3 H3K27K36 −0.060 0.054 0.006 0.993 0.569 H3K27me2K36me0 H3K27K36 −0.173 0.161 0.012 0.546 0.373 H3K27me2K36me1 H3K27K36 −0.033 −0.043 0.076 0.421 0.245 H3K27me2K36me2 H3K27K36 −0.028 0.064 −0.036 0.276 0.140 H3K27me2K36me3 H3K27K36 0.146 0.115 −0.261 H3K27me3K36me0 H3K27K36 0.015 0.097 −0.112 0.496 −0.157 H3K27me3K36me1 H3K27K36 −0.154 0.272 −0.118 0.452 0.230 H3K27me3K36me2 H3K27K36 0.049 0.072 −0.121 −0.102 −0.338 H3K27me3K36me3 H3K27K36 −0.043 0.132 −0.089 H3K27ac1K36me0 H3K27K36 0.181 −0.181 0.099 H3K27ac1K36me1 H3K27K36 −0.364 0.264 0.100 H3K27ac1K36me3 H3K27K36 0.023 0.045 −0.068 0.504 0.374 H3K56me0 H3K56 −0.004 0.189 −0.185 0.742 0.182 H3K79me1 H3K79 −0.069 0.104 −0.035 0.617 0.203 H3K79me2 H3K79 0.024 −0.005 −0.018 0.352 0.083

[0538] SSX-like protein sequences are only found in mammalian SSX family proteins (e.g., human SSX1-9) and members of the vertebrate-specific PRDM7/9 methyltransferases. A 34aa region of SSX (SSX aa155-188) that is highly conserved across vertebrate species of SSX (putative PFAM SSXRD domain) and is similar to that of PRDM7/9 proteins was identified (FIG. **3**D). Pull-down experiments using biotinylated peptides corresponding to this region indicated it was sufficient for SSX nucleosome binding while shorter 23- (SSX aa166-188) and 24- (SSX aa165-188) residue peptides (lacking the W164 residue) failed to do so (FIG. **3**E). This SSX-nucleosome interaction was specific as biotin pulldowns were outcompeted by addition of unlabeled SSX 34-residue peptide and could not be competed by a scrambled control peptide corresponding to the same SSX 34aa region (FIGS. **5**C and **5**G).

[0539] To define whether SSX 34-residue peptide can be used as a probe for repressive Barr bodies/polycomb bodies in cells, a peptide hybridization approach performed on methanol-fixed (non-crosslinked) IMR90 fibroblasts incubated with biotinylated SSX peptides and subsequently co-stained with the Barr body marker H2A K119Ub was implemented. Clear labeling of Barr bodies was observed, which indicated an innate ability of the SSX 34 residue region to selectively localize to repressed chromatin regions (FIG. **5**D). 34aa regions corresponding to most human SSX proteins exhibited interactions with nucleosomes, while shorter SSX-like sequences found in PRDM7/9 proteins lacking the W164 and first R residues of the basic region (R167) failed to do so, indicating a newly evolved, mammalian-specific function of this full protein region (FIGS. **5**E-**5**F). Finally, to identify residues important for nucleosome binding, a library of 34-residue SSX peptides containing alanine substitutions in either single conserved residues or alanine substitutions across the full basic and acidic regions was designed. Importantly, these experiments revealed that the core residues of the 6-aa basic region of the SSX (RLRERK (SEQ ID NO: 219)) were required as single residue and full region alanine substitutions in this region completely abrogated SSX-nucleosome binding (FIG. **3**F).

[0540] To determine whether these minimal regions were sufficient for the genome-wide targeting of fully-formed, endogenous SS18-SSX-containing BAF complexes in cells, either WT SS18, SS18-SSX, or SS18 fused to a range of mutant SSX variants for lentiviral infection in to CRL7250 human fibroblasts was expressed. ChIP-seq experiments revealed that the 34aa SSX tail fused to SS18 was sufficient to achieve SS18-SSX targeting, while the 24aa fusion was unable to do so (FIGS. **3**G and **6**A). Notably, deletion of either the basic or the acidic conserved regions of SSX resulted in complete loss of oncogenic fusion complex targeting, indicating that both of these regions are required for SS18-SSX-specific properties. These findings were consistent with biochemical results indicating that the full 34aa tail is needed to confer tight affinity of SS18-SSX to chromatin in cells (FIG. **6**B). Importantly, these changes in chromatin targeting resulted in corresponding changes in gene expression by RNA-seq, as evidenced by clustering of the transcriptional profiles of the 34-residue tail fusion with the full SS18-SSX fusion (78-aa fusion tail), while deletion of either basic or acidic conserved regions or 24aa SSX tail variants clustered with SS18 WT gene expression profiles (FIG. **3**H). These findings were further corroborated using IF for SS18-SSX Barr body localization (FIG. **6**C) as well as beta-galactosidase senescence assays in IMR90 fibroblasts performed across SS18-SSX and SSX (alone) variants (FIG. **6**D). Finally, both SS18-SSX-78aa and -34aa minimal fusions rescued proliferation in synovial sarcoma cell lines that are well-established to be dependent on the function of SS18-SSX and bearing shRNA-mediated KD of the endogenous SS18-SSX fusion. Taken together, these data indicate that the 34aa minimal region of SSX that contains the conserved basic and acidic regions, is responsible for the maintenance of oncogenic gene expression and proliferation in SS cell lines driven by the SS18-SSX fusion oncoprotein (FIG. **3**I and FIG. **6**E).

Example 4: An RLR Motif within the SSX Basic Region Competes with SMARCB1 for Nucleosome Acidic Patch Binding, Facilitating SS18-SSX-Bound BAF Complex-Mediated Chromatin Remodeling of Polycomb-Repressed Regions

[0541] Using systematic mutagenesis on the SSX 34-residue region, it was found that single residue perturbations to the basic region, which includes a Kaposi's sarcoma-associated herpesvirus (KSHV) LANA-like RLR motif, resulted in complete loss of nucleosome binding (FIG. **7**A). These data indicated that this highly basic region binds directly to the H2A/H2B acidic patch of the nucleosome. To identify the specific sites involved in acidic patch engagement, reactive diazirine probes were introduced at various residues within the nucleosome acidic patch and performed photocrosslinking studies (Dao et al. (2019) *Nat. Chem. Biol*. doi:10.1038/s41589-019-0413-4) with SSX 34-residue peptides (FIG. **7**B and FIGS. **8**A-**8**B). Histone-SSX crosslinks were identified at several positions across the extended acidic patch region, most prominently at positions H2A E56 and H2B E113, which importantly, were substantially reduced when key RLR basic residues in SSX were mutated (FIGS. **7**B-**7**C). To probe this further, nucleosomes containing H2A mutant variants D90N, E92K, and E113K were assembled which disrupt the integrity of the acidic patch for GST-SSX pull down experiments. Both H2A E113 and H2B E113 are important (crosslinks were made at H2B E113) for histone-SSX interaction and mutant variants disrupt the integrity of the acidic patch demonstrating that reciprocally disrupting the integrity of the acidic patch brakes SSX binding interaction. These experiments showed near complete loss of SSX binding to acidic patch-mutant nucleosomes, indicating the importance of this highly conserved and important docking site for the SSX-chromatin interaction (FIG. **7**D (homotypic) and FIG. **8**C (heterotypic)). These results were further corroborated by the fact that direct nucleosome binding competition between LANA peptide and SSX was observed (FIG. **7**E and FIG. **8**G), as the LANA peptide is well-established to bind the nucleosome acidic patch (Barbera et al. (2006) *Science* 311:856-861). Notably, solution NMR studies coupled with TALOS secondary structure prediction indicated that the SSX 78aa tail protein has a disordered N-terminal region (aa 110-154) followed by a predicted alpha helical region spanning the conserved stretch of basic amino acids, WTHRLRERKQ (SEQ ID NO: 230) (FIG. **7**F and FIGS. **8**D-**8**F). In cells, single-residue mutations within the nucleosome acidic patch binding region of SSX (SSX R169A as well as W164A) resulted in attenuation of SS18-SSX-specific BAF complex chromatin occupancy, recruitment to Barr bodies, gene expression activation, and proliferative maintenance in SS cell lines (FIGS. **7**G and **9**A-**9**G). Taken together, these data establish the role for the basic region, specifically the RLR motif, in mediating SS18-SSX-nucleosome binding, in conferring SS18-SSX-containing BAF complex chromatin binding properties, as well as function in gene expression and proliferative maintenance.

[0542] It was previously demonstrated that upon SS18-SSX expression and incorporation in to BAF complexes, the SMARCB1 (BAF47) subunit of BAF complexes, part of the core module (Mashtalir et al. (2018) Cell 175:1272-1288), is destabilized and proteasomally degraded (Kadoch and Crabtree (2013) *Cell* 153:71-85) (see also FIGS. **10**A-**10**B, Tables 7A-7C). Intriguingly, using pull down competition assays, it was found that SSX competed with the recently-identified SMARCB1 C-terminal alpha helix (aa351-385) region (Valencia et al. (2019) *Cell* 179:1342-1356; Ye et al. (2019) *Science* doi:10.1126/science.aay0033) for nucleosome acidic patch binding (FIG. **7**H) (Valencia et al. (2019) *Cell* 179:1342-1356; Ye et al. (2019) *Science* doi:10.1126/science.aay0033). However, the reverse was not true as the SMARCB1 C-term alpha helix was unable to outcompete SSX from binding the nucleosomes, implicating stronger affinity of SSX compared to SMARCB1 C-term for nucleosomes. This result, coupled with the positioning of SS18 at the very N-terminus SMARCA4 subunit within the core module of BAF complexes (defined by CX-MS, FIG. **10**C and recent structural insights (Valencia et al. (2019) *Cell* 179:1342-1356; Ye et al. (2019) *Science* doi:10.1126/science.aay0033) and assessment of SMARCB1 levels across SS18-SSX mutant conditions (FIG. **10**D) indicates the mechanism of degradation of SMARCB1 observed in SS cell lines (Kadoch and Crabtree (2013) *Cell* 153:71-85; McBride et al. (2018) *Cancer Cell* 33:1128-1141; Kohashi et al. (2010) *Mod. Path.* 23:981-990) can be explained by the dominant, higher affinity SSX binding to the nucleosome acidic patch and the resulting configurational changes within the BAF core module.

TABLE-US-00022 TABLE 7A BAF complex components Razor Protein Peptides Gene Description SeqLenn BrgIP_Aska BrgIP_hFib BrgIP_293t NSAF_BrgIP_Aska NSAF_BrgIP_hFib NSAF_BrgIP_293t sp|P51532|SMCA4_HUMAN 189 SMARCA4 SMCA4_HUMAN 1647 50 39 100 0.00294971 0.00150089 0.00377919 Transcription activator BRG1 sp|Q92922|SMRC1_HUMAN 11 SMARCC1 SMRC1_HUMAN 1105 44 19 86 0.00386895 0.00108986 0.00484427 SWI/SNF complex subunit SMARCC1 sp|Q969G3|SMCE1_HUMAN 96 SMARCE1 SMCE1_HUMAN 411 39 21 36 0.00921988 0.00323859 0.00545196 SWI/SNF-related matrix-associated actin-dependent regulator of chromatin subfamily E member 1 sp|Q8TAQ2|SMRC2_HUMAN 136 SMARCC2 SMRC2_HUMAN 1214 36 51 69 0.00288129 0.00266275 0.00353771 SWI/SNF complex subunit SMARCC2 sp|O14497|ARI1A_HUMAN 192 ARID1A ARI1A_HUMAN AT- 2285 35 44 121 0.00148828 0.00122052 0.00329603 rich interactive domain-containing protein 1A sp|Q8TAQ2-3|SMRC2_HUMAN 2 SMARCC2 SMRC2_HUMAN 1152 33 50 55 0.00278333 0.00275103 0.00297168 Isoform 3 of SWI/SNF complex subunit SMARCC2 sp|Q8NFD5-3|ARI1B_HUMAN 117 ARID1B ARI 1B_HUMAN 2289 28 28 61 0.00118854 0.00077534 0.00165873 Isoform 3 of AT-rich interactive domain- containing protein 1B sp|P60709|ACTB_HUMAN 19 ACTB ACTB_HUMAN Actin, 375 19 18 18 0.00492294 0.00304242 0.00298767 cytoplasmic 1 sp|O96019|ACL6A_HUMAN 55 ACTL6A ACL6A_HUMAN Actin- 429 14 16 25 0.00317083 0.00236397 0.00362723 like protein 6A sp|Q96GM5|SMRD1_HUMAN 30 SMARCD1 SMRD1_HUMAN 515 14 6 18 0.00264133 0.00073845 0.00217549 SWI/SNF-related matrix-associated actin-dependent regulator of chromatin subfamily D member 1 sp|Q92925|SMRD2_HUMAN 42 SMARCD2 SMRD2_HUMAN 531 12 9 21 0.00219578 0.0010743 0.0024616 SWI/SNF-related matrix-associated actin-dependent regulator of chromatin subfamily D member 2 sp|Q9NZM4|GSCR1_HUMAN 28 GLTSCR1 GSCR1_HUMAN 1560 10 4 14 0.00062284 0.00016252

0.00055859 Glioma tumor suppressor candidate region gene 1 protein sp|Q6AI39|GSC1L_HUMAN 0 GLTSCR1L GSC1L_HUMAN 1079 7 2 4 0.00063035 0.00011749 0.00023074 GLTSCR 1-like protein sp|Q6STE5|SMRD3_HUMAN 7 SMARCD3 SMRD3_HUMAN 483 6 6 12 0.001207 0.00078738 0.00154641 SWI/SNF-related matrix-associated actin-dependent regulator of chromatin subfamily D member 3 sp|Q9H8M2|BRD9_HUMAN 10 BRD9 BRD9_HUMAN 597 5 1 4 0.00081376 0.00010617 0.00041704 Bromodomain- containing protein 9 sp|Q92785|REQU_HUMAN 34 DPF2 REQU_HUMAN Zinc 391 5 5 24 0.0012425 0.00081054 0.00382055 finger protein ubi-d4 sp|Q12824-2|SNF5_HUMAN 31 SMARCB1 SNF5_HUMAN 376 5 10 18 0.00129207 0.00134859 0.00297973 Isoform B of SWI/SNF-related matrix-associated actin-dependent regulator of chromatin subfamily B member 1 sp|Q15532|SSXT_HUMAN 6 SS18 SSXT_HUMAN Protein 418 2 2 2 0.0004649 0.00030327 0.00029781 SSXT sp|Q4VC05-2|BCL7A_HUMAN 3 BCL7A BCL7A_HUMAN 231 0 0 4 0 0 0.0010778 Isoform 2 of B-cell CLL/lymphoma 7 protein family member A TABLE-US-00023 TABLE 7B All Spectral Counts Razor Protein Peptides Gene Description SeqLen BrgIP_Aska BrgIP_hFib BrgIP_293t NSAF_BrgIP_Aska NSAF_BrgIP_hFib NSAF_BrgIP_293t sp|P51532|SMCA4_HUMAN 189 SMARCA4 SMCA4_HUMAN 1647 50 39 100 0.00294971 0.00150089 0.00377919 Transcription activator BRG1 sp|Q92922|SMRC1_HUMAN 11 SMARCC1 SMRC1_HUMAN 1105 44 19 86 0.00386895 0.00108986 0.00484427 SWI/SNF complex subunit SMARCC1 sp|Q969G3|SMCE1_HUMAN 96 SMARCE1 SMCE1_HUMAN 411 39 21 36 0.00921988 0.00323859 0.00545196 SWI/SNF-related matrix-associated actin-dependent regulator of chromatin subfamily E member 1 sp|P22626|ROA2_HUMAN 104 HNRNPA2B1 ROA2_HUMAN 353 37 32 53 0.0101843 0.00574584 0.0093453 Heterogeneous nuclear ribonucleoproteins A2/B1 sp|Q8TAQ2|SMRC2_HUMAN 136 SMARCC2 SMRC2_HUMAN 1214 36 51 69 0.00288129 0.00266275 0.00353771 SWI/SNF complex subunit SMARCC2 sp|O14497|ARI1A_HUMAN 192 ARID1A ARI1A_HUMAN AT-rich 2285 35 44 121 0.00148828 0.00122052 0.00329603 interactive domain- containing protein 1A sp|Q8TAQ2-3|SMRC2_HUMAN 2 SMARCC2 SMRC2_HUMAN 1152 33 50 55 0.00278333 0.00275103 0.00297168 Isoform 3 of SWI/SNF complex subunit SMARCC2 sp|Q8NFD5-3|ARI1B_HUMAN 117 ARID1B ARI1B_HUMAN 2289 28 28 61 0.00118854 0.000775337 0.00165873 Isoform 3 of AT-rich interactive domain- containing protein 1B sp|P35637|FUS_HUMAN 53 FUS FUS_HUMAN RNA- 526 23 15 33 0.00424859 0.00180752 0.00390499 binding protein FUS sp|P78527|PRKDC_HUMAN 53 PRKDC PRKDC_HUMAN DNA- 4128 23 13 22 0.000541366 0.00019961 0.000331722 dependent protein kinase catalytic subunit sp|P60709|ACTB_HUMAN 19 ACTB ACTB_HUMAN Actin, 375 19 18 18 0.00492294 0.00304242 0.00298767 cytoplasmic 1 sp|P62736|ACTA_HUMAN 49 ACTA2 ACTA_HUMAN Actin, 377 14 22 13 0.00360819 0.00369879 0.00214632 aortic smooth muscle sp|O96019|ACL6A_HUMAN 55 ACTL6A ACL6A_HUMAN Actin- 429 14 16 25 0.00317083 0.00236397 0.00362723 like protein 6A sp|Q96GM5|SMRD1_HUMAN 30 SMARCD1 SMRD1_HUMAN 515 14 6 18 0.00264133 0.000738452 0.00217549 SWI/SNF-related matrix-associated actin-dependent regulator of chromatin subfamily D member 1 sp|P17844|DDX5_HUMAN 19 DDX5 DDX5_HUMAN 614 12 4 7 0.00189896 0.000412924 0.000709613 Probable ATP- dependent RNA helicase DDX5 sp|P02751-15|FINC_HUMAN 18 FN1 FINC_HUMAN Isoform 2477 12 6 0 0.000470715 0.000153534 0 15 of Fibronectin sp|Q92925|SMRD2_HUMAN 42 SMARCD2 SMRD2_HUMAN 531 12 9 21 0.00219578 0.0010743 0.0024616 SWI/SNF-related matrix-associated actin-dependent regulator of chromatin subfamily D member 2 sp|P51991|ROA3_HUMAN 30 HNRNPA3 ROA3_HUMAN 378 11 8 11 0.0028275 0.00134146 0.00181131 Heterogeneous nuclear ribonucleoprotein A3 sp|Q9NZM4|GSCR1_HUMAN 28 GLTSCR1 GSCR1_HUMAN 1560 10 4 14 0.000622842 0.000162523 0.000558593 Glioma tumor suppressor candidate region gene 1 protein sp|P31942|HNRH3_HUMAN 54 HNRNPH3 HNRH3_HUMAN 346 8 22 24 0.00224655 0.00403019 0.00431745 Heterogeneous nuclear ribonucleoprotein H3 sp|Q9Y3I0|RTCB_HUMAN 9 RTCB RTCB_HUMAN tRNA- 505 8 12 9 0.00153922 0.00150615 0.00110928 splicing ligase RtcB homolog sp|Q92841|DDX17_HUMAN 16 DDX17 DDX17_HUMAN 729 7 4 5 0.000932981 0.000347785 0.000426908 Probable ATP- dependent RNA helicase DDX17 sp|O00571|DDX3X_HUMAN 20 DDX3X DDX3X_HUMAN ATP- 662 7 6 8 0.00102741 0.000574476 0.000752184 dependent RNA helicase DDX3X sp|Q6AI39|GSC1L_HUMAN 0 GLTSCR1L GSC1L_HUMAN 1079 7 2 4 0.000630346 0.000117486 0.000230744 GLTSCR 1-like protein sp|Q00839|HNRPU_HUMAN 33 HNRNPU HNRPU_HUMAN 825 7 6 21 0.000824416 0.000460973 0.00158437 Heterogeneous nuclear ribonucleoprotein U sp|Q99729-2|ROAA_HUMAN 17 HNRNPAB ROAA_HUMAN 332 6 4 7 0.00175596 0.000763661 0.00131236 Isoform 2 of Heterogeneous nuclear ribonucleoprotein A/B sp|Q96PK6|RBM14_HUMAN 6 RBM14 RBM14_HUMAN RNA- 669 6 11 14 0.00087142 0.00104219 0.00130255 binding protein 14 sp|P38159|RBMX_HUMAN 22 RBMX RBMX_HUMAN RNA- 391 6 3 13 0.001491 0.000486321 0.00206947 binding motif protein, X chromosome sp|Q6STE5|SMRD3_HUMAN 7 SMARCD3 SMRD3_HUMAN 483 6 6 12 0.001207 0.000787377 0.00154641 SWI/SNF-related matrix-associated actin-dependent regulator of chromatin subfamily D member 3 sp|Q92804|RBP56_HUMAN 24 TAF15 RBP56_HUMAN TATA- 592 6 7 11 0.000984764 0.000749471 0.00115655 binding protein- associated factor 2N sp|Q9H8M2|BRD9_HUMAN 10 BRD9 BRD9_HUMAN 597 5 1 4 0.000813763 0.000106171 0.00041704 Bromodomain- containing protein 9 sp|Q92785|REQU_HUMAN 34 DPF2 REQU_HUMAN Zinc 391 5 5 24 0.0012425 0.000810535 0.00382055 finger protein ubi-d4 sp|P09651|ROA1_HUMAN 9 HNRNPA1 ROA1_HUMAN 372 5 4 9 0.00130596 0.000681546 0.00150588 Heterogeneous nuclear ribonucleoprotein A1 sp|P11142|HSP7C_HUMAN 19 HSPA8 HSP7C_HUMAN Heat 646 5 5 10 0.000752038 0.000490587 0.000963517 shock cognate 71 kDa protein sp|P38646|GRP75_HUMAN 12 HSPA9 GRP75_HUMAN 679 5 10 8 0.000715489 0.000933488 0.000733351 Stress-70 protein, mitochondrial sp|Q86U86|PB1_HUMAN 43 PBRM1 PB1_HUMAN Protein 1689 5 5 33 0.000287636 0.000187637 0.00121612 polybromo-1 sp|Q15149|PLEC_HUMAN 28 PLEC PLEC_HUMAN Plectin 4684 5 23 0 0.000103718 0.000311236 0 sp|Q12824-2|SNF5_HUMAN 31 SMARCB1 SNF5_HUMAN 376 5 10 18 0.00129207 0.00134859 0.00297973 Isoform B of SWI/SNF- related matrix- associated actin- dependent regulator of chromatin subfamily B member 1 sp|P68371|TBB4B_HUMAN 5 TUBB4B TBB4B_HUMAN 445 5 4 8 0.00109172 0.000569742 0.00111898 Tubulin beta-4B chain sp|P07900-2|HS90A_HUMAN 10 HSP90AA1 HS90A_HUMAN 854 4 4 2 0.000455098 0.00029688 0.000145769 Isoform 2 of Heat shock protein HSP 90- alpha sp|P07437|TBB5_HUMAN 4 TUBB TBB5_HUMAN Tubulin 444 4 5 11 0.000875345 0.000713782 0.00154206 beta chain sp|Q9NWB6-2|ARGL1_HUMAN 4 ARGLU1 ARGL1_HUMAN 273 3 0 1 0.00106773 0 0.000227997 Isoform 2 of Arginine and glutamate-rich protein 1 sp|P02452|CO1A1_HUMAN 4 COL1A1 CO1A1_HUMAN 1464 3 9 7 0.000199105 0.000389655 0.000297611 Collagen alpha-1(I) chain sp|P55084|ECHB_HUMAN 8 HADHB ECHB_HUMAN 474 3 5 0 0.000614958 0.000668606 0 Trifunctional enzyme subunit beta, mitochondrial sp|Q13151|ROA0_HUMAN 0 HNRNPA0 ROA0_HUMAN 305 3 0 4 0.000955705 0 0.000816304 Heterogeneous nuclear ribonucleoprotein A0 sp|Q9BUJ2|HNRL1_HUMAN 9 HNRNPUL1 HNRL1_HUMAN 856 3 4 2 0.000340526 0.000296186 0.000145428 Heterogeneous nuclear ribonucleoprotein U- like protein 1 sp|Q12906-7|ILF3_HUMAN 18 ILF3 ILF3_HUMAN Isoform 898 3 9 6 0.000324599 0.00063525 0.000415879 7 of Interleukin enhancer-binding factor 3 sp|Q9Y2W1|TR150_HUMAN 0 THRAP3 TR150_HUMAN 955 3 1 4 0.000305225 6.63705E−05 0.000260704 Thyroid hormone receptor-associated protein 3 sp|Q13509|TBB3_HUMAN 15 TUBB3 TBB3_HUMAN Tubulin 450 3 5 7 0.000647756 0.000704265 0.000968228 beta-3 chain sp|Q09666|AHNK_HUMAN 0 AHNAK AHNK_HUMAN 5890 2 36 0 3.29926E−05 0.000387405 0 Neuroblast differentiation- associated protein AHNAK sp|P25705|ATPA_HUMAN 3 ATP5A1 ATPA_HUMAN ATP 553 2 1 0 0.000351405 0.000114618 0 synthase subunit alpha, mitochondrial sp|Q9NPI1-2|BRD7_HUMAN 4 BRD7 BRD7_HUMAN Isoform 652 2 0 2 0.000298047 0 0.00019093 2 of Bromodomain- containing protein 7 sp|Q9Y224|CN166_HUMAN 3 C14orf166 CN166_HUMAN 244 2 3 1 0.000796421 0.000779309 0.000255095 UPF0568 protein C14orf166 sp|Q92499|DDX1_HUMAN 0 DDX1 DDX1_HUMAN ATP- 740 2 6 7 0.000262604 0.000513923 0.000588787 dependent RNA helicase DDX1 sp|Q9UJV9|DDX41_HUMAN 6 DDX41 DDX41_HUMAN 622 2 3 1 0.000312422 0.00030571 0.000100069 Probable ATP- dependent RNA helicase DDX41 sp|Q14103|HNRPD_HUMAN 8 HNRNPD HNRPD_HUMAN 355 2 3 3 0.000547399 0.000535638 0.000525999 Heterogeneous nuclear ribonucleoprotein D0 sp|O14979|HNRDL_HUMAN 8 HNRNPDL HNRDL_HUMAN 420 2 3 3 0.000462683 0.000452742 0.000444594 Heterogeneous nuclear ribonucleoprotein D- like sp|P31943|HNRH1_HUMAN 11 HNRNPH1 HNRH1_HUMAN 449 2 6 7 0.000432799 0.000847 0.000970384 Heterogeneous nuclear ribonucleoprotein H sp|Q1KMD3|HNRL2_HUMAN 6 HNRNPUL2 HNRL2_HUMAN 747 2 4 1 0.000260143 0.000339405 8.33242E−05 Heterogeneous nuclear ribonucleoprotein U- like protein 2 sp|P11021|GRP78_HUMAN 0 HSPA5 GRP78_HUMAN 78 654 2 5 1 0.000297136 0.000484586 9.51731E−05 kDa glucose-regulated protein sp|B9A064|IGLL5_HUMAN 2 IGLL5 IGLL5_HUMAN 214 2 4 8 0.000908069 0.00118474 0.00232685 Immunoglobulin lambda-like polypeptide 5 sp|Q13523|PRP4B_HUMAN 3 PRPF4B PRP4B_HUMAN 1007 2 0 1 0.000192976 0 6.18105E−05 Serine/threonine- protein kinase PRP4 homolog sp|P62913|RL11_HUMAN 5 RPL11 RL11_HUMAN 60S 178 2 1 2 0.00109172 0.000356089 0.000699362 ribosomal protein L11 sp|P40429|RL13A_HUMAN 5 RPL13A RL13A_HUMAN 60S 203 2 1 2 0.000957274 0.000312236 0.000613233 ribosomal protein L13a sp|P15880|RS2_HUMAN 6 RPS2 RS2_HUMAN 40S 293 2 3 1 0.000663231 0.000648981 0.000212434 ribosomal protein S2 sp|P05141|ADT2_HUMAN 0 SLC25A5 ADT2_HUMAN 298 2 2 2 0.000652103 0.000425395 0.00041774 ADP/ATP translocase 2 sp|Q15532|SSXT_HUMAN 6 SS18 SSXT_HUMAN Protein 418 2 2 2 0.000464896 0.000303272 0.000297814 SSXT sp|Q13263|TIF1B_HUMAN 5 TRIM28 TIF1B_HUMAN 835 2 1 2 0.000232727 7.59088E−05 0.000149086 Transcription intermediary factor 1- beta sp|Q9UHD8|SEPT9_HUMAN 2 9-Sep SEPT9_HUMAN Septin- 586 1 1 0 0.000165808 0.000108164 0 9 sp|Q68CP9|ARID2_HUMAN 15 ARID2 ARID2_HUMAN AT- 1835 1 1 13 5.29501E−05 3.45416E−05 0.00044096 rich interactive domain-containing protein 2 sp|Q8WUZ0-2|BCL7C_HUMAN 8 BCL7C BCL7C_HUMAN 242 1 0 7 0.000401501 0 0.00180042 Isoform 2 of B-cell CLL/lymphoma 7 protein family member C sp|P02461|CO3A1_HUMAN 1 COL3A1 CO3A1_HUMAN 1466 1 1 1 6.62779E−05 4.32359E−05 4.24578E−05 Collagen alpha-1(III) chain sp|P27658|CO8A1_HUMAN 0 COL8A1 CO8A1_HUMAN 744 1 0 0 0.000130596 0 0 Collagen alpha-1(VIII) chain sp|Q9Y678|COPG1_HUMAN 2 COPG1 COPG1_HUMAN 874 1 3 0 0.000111171 0.000217565 0 Coatomer subunit gamma-1 sp|Q8IXB1|DJC10_HUMAN 3 DNAJC10 DJC10_HUMAN DnaJ 793 1 2 0 0.000122526 0.000159858 0 homolog subfamily C member 10 sp|Q5VTE0|EF1A3_HUMAN 6 EEF1A1P5 EF1A3_HUMAN 462 1 4 1 0.00021031 0.000548778 0.000134726 Putative elongation factor 1-alpha-like 3 sp|P26641|EF1G_HUMAN 2 EEF1G EF1G_HUMAN 437 1 1 0 0.000222342 0.000145043 0 Elongation factor 1-

gamma sp|Q01844-5|EWS_HUMAN 8 EWSR1 EWS_HUMAN Isoform 661 1 6 1 0.000146994 0.000575345 9.41652E−05 5 of RNA-binding protein EWS sp|Q8NCA5|FA98A_HUMAN 6 FAM98A FA98A_HUMAN 519 1 4 1 0.000187213 0.000488507 0.000119929 Protein FAM98A sp|P21333|FLNA_HUMAN 40 FLNA FLNA_HUMAN Filamin- 2647 1 39 0 0.000036707 0.000933876 0 A sp|Q8IY81|SPB1_HUMAN 0 FTSJ3 SPB1_HUMAN pre- 847 1 0 1 0.000114715 0 7.34867E−05 rRNA processing protein FTSJ3 sp|P28799|GRN_HUMAN 4 GRN GRN_HUMAN 593 1 2 1 0.00016385 0.000213773 0.000104963 Granulins sp|P52597|HNRPF_HUMAN 0 HNRNPF HNRPF_HUMAN 415 1 3 2 0.000234129 0.000458196 0.000299967 Heterogeneous nuclear ribonucleoprotein F sp|P08107|HSP71_HUMAN 5 HSPA1A HSP71_HUMAN Heat 641 1 1 5 0.000151581 9.88827E−05 0.000485516 shock 70 kDa protein 1A/1B sp|Q07666|KHDR1_HUMAN 7 KHDRBS1 KHDR1_HUMAN KH 443 1 1 5 0.00021933 0.000143079 0.000702519 domain-containing, RNA-binding, signal transduction- associated protein 1 sp|P56192|SYMC_HUMAN 8 MARS SYMC_HUMAN 900 1 4 3 0.000107959 0.000281706 0.000207477 Methionine--tRNA ligase, cytoplasmic sp|Q9BU76|MMTA2_HUMAN 2 MMTAG2 MMTA2_HUMAN 263 1 0 1 0.000369442 0 0.000236666 Multiple myeloma tumor-associated protein 2 sp|P09874|PARP1_HUMAN 0 PARP1 PARP1_HUMAN Poly 1014 1 2 1 9.58218E−05 0.000125017 6.13838E−05 [ADP-ribose] polymerase 1 sp|Q9NR12|PDLI7_HUMAN 16 PDLIM7 PDLI7_HUMAN PDZ 457 1 15 0 0.000212611 0.00208043 0 and LIM domain protein 7 sp|Q8WUB8|PHF10_HUMAN 2 PHF10 PHF10_HUMAN PHD 498 1 0 1 0.000195107 0 0.000124986 finger protein 10 sp|Q14498|RBM39_HUMAN 2 RBM39 RBM39_HUMAN RNA- 530 1 0 1 0.000183327 0 0.00011744 binding protein 39 sp|P27635|RL10_HUMAN 2 RPL10 RL10_HUMAN 60S 214 1 0 1 0.000454034 0 0.000290856 ribosomal protein L10 sp|P18621-3|RL17_HUMAN 1 RPL17 RL17_HUMAN Isoform 228 1 0 0 0.000426155 0 0 3 of 60S ribosomal protein L17 sp|P83731|RL24_HUMAN 4 RPL24 RL24_HUMAN 60S 157 1 1 2 0.000618875 0.000403719 0.000792907 ribosomal protein L24 sp|P46776|RL27A_HUMAN 3 RPL27A RL27A_HUMAN 60S 148 1 1 1 0.000656509 0.000428269 0.000420569 ribosomal protein L27a sp|P62424|RL7A_HUMAN 4 RPL7A RL7A_HUMAN 60S 266 1 1 2 0.000365276 0.000238285 0.000467994 ribosomal protein L7a sp|P62269|RS18_HUMAN 2 RPS18 RS18_HUMAN 40S 152 1 0 2 0.000639233 0 0.000818989 ribosomal protein S18 sp|P62753|RS6_HUMAN 3 RPS6 RS6_HUMAN 40S 249 1 1 1 0.000390214 0.000254554 0.000249973 ribosomal protein S6 sp|Q9Y265|RUVB1_HUMAN 2 RUVBL1 RUVB1_HUMAN RuvB- 456 1 0 1 0.000213078 0 0.000136498 like 1 sp|Q9Y230|RUVB2_HUMAN 4 RUVBL2 RUVB2_HUMAN RuvB- 463 1 0 3 0.000209856 0 0.000403304 like 2 sp|P12236|ADT3_HUMAN 4 SLC25A6 ADT3_HUMAN 298 1 2 1 0.000326051 0.000425395 0.00020887 ADP/ATP translocase 3 sp|Q13242|SRSF9_HUMAN 6 SRSF9 SRSF9_HUMAN 221 1 2 3 0.000439653 0.000573609 0.00084493 Serine/arginine-rich splicing factor 9 sp|O75177|CREST_HUMAN 0 SS18L1 CREST_HUMAN 396 1 0 1 0.000245362 0 0.00015718 Calcium-responsive transactivator sp|P51571|SSRD_HUMAN 3 SSR4 SSRD_HUMAN 173 1 3 0 0.000561638 0.00109914 0 Translocon-associated protein subunit delta sp|P68363|TBA1B_HUMAN 15 TUBA1B TBA1B_HUMAN 451 1 5 9 0.00021544 0.000702703 0.0012421 Tubulin alpha-1B chain sp|P10599|THIO_HUMAN 3 TXN THIO_HUMAN 105 1 1 1 0.000925365 0.000603655 0.000592792 Thioredoxin sp|Q6NZY4|ZCHC8_HUMAN 2 ZCCHC8 ZCHC8_HUMAN Zinc 707 1 1 0 0.00013743 8.96518E−05 0 finger CCHC domain- containing protein 8 ##sp|Q6L8G9|KRA56_HUMAN 0 ##KRTAP5-6 ##KRA56_HUMAN 129 0 0 1 0 0 0.000482505 ##Keratin-associated protein 5-6 sp|A8K2U0|A2ML1_HUMAN 0 A2ML1 A2ML1_HUMAN 1454 0 2 0 0 8.71854E−05 0 Alpha-2- macroglobulin-like protein 1 sp|P12814-4|ACTN1_HUMAN 2 ACTN1 ACTN1_HUMAN 930 0 2 0 0 0.000136309 0 Isoform 4 of Alpha- actinin-1 sp|O43707|ACTN4_HUMAN 2 ACTN4 ACTN4_HUMAN 911 0 2 0 0 0.000139152 0 Alpha-actinin-4 sp|O95831|AIFM1_HUMAN 2 AIFM1 AIFM1_HUMAN 613 0 1 1 0 0.000103399 0.000101539 Apoptosis-inducing factor 1, mitochondrial sp|Q86V81|THOC4_HUMAN 1 ALYREF THOC4_HUMAN THO 257 0 0 1 0 0 0.000242191 complex subunit 4 sp|P15144|AMPN_HUMAN 2 ANPEP AMPN_HUMAN 967 0 6 0 0 0.000393281 0 Aminopeptidase N sp|P06576|ATPB_HUMAN 1 ATP5B ATPB_HUMAN ATP 529 0 1 0 0 0.000119818 0 synthase subunit beta, mitochondrial sp|Q4VC05-2|BCL7A_HUMAN 3 BCL7A BCL7A_HUMAN 231 0 0 4 0 0 0.0010778 Isoform 2 of B-cell CLL/lymphoma 7 protein family member A sp|Q9NYF8|BCLF1_HUMAN 3 BCLAF1 BCLF1_HUMAN Bcl-2- 920 0 0 3 0 0 0.000202967 associated transcription factor 1 sp|Q96MY1|CT112_HUMAN 1 C20orf112 CT112_HUMAN 436 0 1 0 0 0.000145376 0 Uncharacterized protein C20orf112 sp|Q05682|CALD1_HUMAN 4 CALD1 CALD1_HUMAN 793 0 4 0 0 0.000319717 0 Caldesmon sp|O43852-3|CALU_HUMAN 1 CALU CALU_HUMAN Isoform 323 0 1 0 0 0.000196235 0 3 of Calumenin sp|Q14444|CAPR1_HUMAN 1 CAPRIN1 CAPR1_HUMAN 709 0 0 1 0 0 8.77901E−05 Caprin-1 sp|Q76M96-2|CCD80_HUMAN 3 CCDC80 CCD80_HUMAN 961 0 3 0 0 0.000197868 0 Isoform 2 of Coiled-coil domain-containing protein 80 sp|Q15517|CDSN_HUMAN 3 CDSN CDSN_HUMAN 529 0 3 0 0 0.000359455 0 Corneodesmosin sp|Q07065|CKAP4_HUMAN 0 CKAP4 CKAP4_HUMAN 602 0 5 0 0 0.000526444 0 Cytoskeleton- associated protein 4 sp|Q00610|CLH1_HUMAN 2 CLTC CLH1_HUMAN Clathrin 1675 0 2 0 0 7.56822E−05 0 heavy chain 1 sp|P51911|CNN1_HUMAN 2 CNN1 CNN1_HUMAN 297 0 2 0 0 0.000426827 0 Calponin-1 sp|Q15417|CNN3_HUMAN 3 CNN3 CNN3_HUMAN 329 0 3 0 0 0.000577968 0 Calponin-3 sp|Q99715|COCA1_HUMAN 8 COL12A1 COCA1_HUMAN 3063 0 8 0 0 0.000165547 0 Collagen alpha-1(XII) chain sp|P08123|CO1A2_HUMAN 0 COL1A2 CO1A2_HUMAN 1366 0 1 4 0 0.000046401 0.000182264 Collagen alpha-2(I) chain sp|P53618|COPB_HUMAN 4 COPB1 COPB_HUMAN 953 0 7 0 0 0.000465568 0 Coatomer subunit beta sp|Q9BZJ0|CRNL1_HUMAN 1 CRNKL1 CRNL1_HUMAN 848 0 0 1 0 0 0.0000734 Crooked neck-like protein 1 sp|O00622|CYR61_HUMAN 0 CYR61 CYR61_HUMAN 381 0 1 0 0 0.000166362 0 Protein CYR61 sp|Q16531|DDB1_HUMAN 2 DDB1 DDB1_HUMAN DNA 1140 0 2 0 0 0.0001112 0 damage-binding protein 1 sp|O43143|DHX15_HUMAN 0 DHX15 DHX15_HUMAN 795 0 0 2 0 0 0.000156587 Putative pre-mRNA- splicing factor ATP- dependent RNA helicase DHX15 sp|Q08211|DHX9_HUMAN 1 DHX9 DHX9_HUMAN ATP- 1270 0 0 7 0 0 0.000343073 dependent RNA helicase A sp|Q14204|DYHC1_HUMAN 0 DYNC1H1 DYHC1_HUMAN 4646 0 3 0 0 0.000040928 0 Cytoplasmic dynein 1 heavy chain 1 sp|Q99848|EBP2_HUMAN 1 EBNA1BP2 EBP2_HUMAN 306 0 0 1 0 0 0.000203409 Probable rRNA- processing protein EBP2 sp|P00533|EGFR_HUMAN 1 EGFR EGFR_HUMAN 1210 0 1 0 0 5.23833E−05 0 Epidermal growth factor receptor sp|P50402|EMD_HUMAN 1 EMD EMD_HUMAN Emerin 254 0 1 0 0 0.000249543 0 sp|PO7814|SYEP_HUMAN 0 EPRS SYEP_HUMAN 1512 0 2 1 0 0.000083841 4.11661E−05 Bifunctional glutamate/proline-- tRNA ligase sp|Q9NZB2-6|F120A_HUMAN 4 FAM120A F120A_HUMAN 1146 0 4 0 0 0.000221235 0 Isoform F of Constitutive coactivator of PPAR- gamma-like protein 1 sp|Q8WVX9|FACR1_HUMAN 1 FAR1 FACR1_HUMAN Fatty 515 0 1 0 0 0.000123075 0 acyl-CoA reductase 1 sp|Q9Y4F1-2|FARP1_HUMAN 3 FARP1 FARP1_HUMAN 1076 0 3 0 0 0.000176721 0 Isoform 2 of FERM, RhoGEF and pleckstrin domain-containing protein 1 sp|P22087|FBRL_HUMAN 2 FBL FBRL_HUMAN rRNA 2′- 321 0 0 2 0 0 0.000387808 O-methyltransferase fibrillarin sp|Q14315|FLNC_HUMAN 9 FLNC FLNC_HUMAN Filamin- 2725 0 9 0 0 0.000209341 0 C sp|P04899-4|GNAI2_HUMAN 9 GNAI2 GNAI2_HUMAN 366 0 9 0 0 0.00155862 0 Isoform sGi2 of Guanine nucleotide- binding protein G(i) subunit alpha-2 sp|Q9BVP2|GNL3_HUMAN 3 GNL3 GNL3_HUMAN 549 0 0 3 0 0 0.000340127 Guanine nucleotide- binding protein-like 3 sp|P62805|H4_HUMAN 0 HIST1H4A H4_HUMAN Histone 103 0 1 0 0 0.000615377 0 H4 sp|P55795|HNRH2_HUMAN 4 HNRNPH2 HNRH2_HUMAN 449 0 3 3 0 0.0004235 0.000415879 Heterogeneous nuclear ribonucleoprotein H2 sp|P61978-2|HNRPK_HUMAN 1 HNRNPK HNRPK_HUMAN 464 0 0 1 0 0 0.000134145 Isoform 2 of Heterogeneous nuclear ribonucleoprotein K sp|P14866|HNRPL_HUMAN 1 HNRNPL HNRPL_HUMAN 589 0 0 1 0 0 0.000105676 Heterogeneous nuclear ribonucleoprotein L sp|P08238|HS90B_HUMAN 2 HSP90AB1 HS90B_HUMAN Heat 724 0 1 1 0 8.75467E−05 8.59713E−05 shock protein HSP 90- beta sp|P01857|IGHG1_HUMAN 0 IGHG1 IGHG1_HUMAN Ig 330 0 1 1 0 0.000192072 0.000188616 gamma-1 chain C region sp|Q12905|ILF2_HUMAN 0 ILF2 ILF2_HUMAN 390 0 0 1 0 0 0.000159598 Interleukin enhancer- binding factor 2 sp|Q9H0H0|INT2_HUMAN 1 INTS2 INT2_HUMAN 1204 0 0 1 0 0 0.000051697 Integrator complex subunit 2 sp|Q9NVH2|INT7_HUMAN 2 INTS7 INT7_HUMAN 962 0 0 2 0 0 0.000129404 Integrator complex subunit 7 sp|Q9H1B7|I2BPL_HUMAN 0 IRF2BPL I2BPL_HUMAN 796 0 1 0 0 7.96279E−05 0 Interferon regulatory factor 2-binding protein-like sp|Q63ZY3-2|KANK2_HUMAN 2 KANK2 KANK2_HUMAN 859 0 2 0 0 0.000147576 0 Isoform 2 of KN motif and ankyrin repeat domain-containing protein 2 sp|P52294|IMA5_HUMAN 1 KPNA1 IMA5_HUMAN 538 0 1 0 0 0.000117814 0 Importin subunit alpha-5 sp|P52292|IMA1_HUMAN 1 KPNA2 IMA1_HUMAN 529 0 0 2 0 0 0.000235324 Importin subunit alpha-1 sp|Q32P28-3|P3H1_HUMAN 3 LEPRE1 P3H1_HUMAN Isoform 804 0 3 0 0 0.000236507 0 3 of Prolyl 3-hydroxylase 1 sp|P09382|LEG1_HUMAN 0 LGALS1 LEG1_HUMAN 135 0 2 0 0 0.00093902 0 Galectin-1 sp|PO2545|LMNA_HUMAN 6 LMNA LMNA_HUMAN 664 0 6 0 0 0.000572745 0 Prelamin-A/C sp|Q8WWI1|LMO7_HUMAN 3 LMO7 LMO7_HUMAN LIM 1683 0 3 0 0 0.000112984 0 domain only protein 7 sp|P46821|MAP1B_HUMAN 0 MAP1B MAP1B_HUMAN 2468 0 4 0 0 0.000102729 0 Microtubule- associated protein 1B sp|P27816|MAP4_HUMAN 1 MAP4 MAP4_HUMAN 1152 0 1 0 0 5.50207E−05 0 Microtubule- associated protein 4 sp|P33993|MCM7_HUMAN 3 MCM7 MCM7_HUMAN DNA 719 0 0 3 0 0 0.000259707 replication licensing factor MCM7 sp|P52701|MSH6_HUMAN 2 MSH6 MSH6_HUMAN DNA 1360 0 0 2 0 0 9.15341E−05 mismatch repair protein Msh6 sp|P35579|MYH9_HUMAN 4 MYH9 MYH9_HUMAN 1960 0 6 0 0 0.000194032 0 Myosin-9 sp|Q969V3|NCLN_HUMAN 1 NCLN NCLN_HUMAN Nicalin 563 0 1 0 0 0.000112582 0 sp|O15226-2|NKRF_HUMAN 3 NKRF NKRF_HUMAN Isoform 705 0 0 3 0 0 0.000264865 2 of NF-kappa-B-repressing factor sp|O95478|NSA2_HUMAN 0 NSA2 NSA2_HUMAN 260 0 0 1 0 0 0.000239397 Ribosome biogenesis protein NSA2 homolog sp|P21589|5NTD_HUMAN 7 NT5E 5NTD_HUMAN 5′- 574 0 7 0 0 0.000772973 0 nucleotidase sp|P13674|P4HA1_HUMAN 10 P4HA1 P4HA1_HUMAN Prolyl 534 0 11 0 0 0.00130566 0 4-hydroxylase subunit alpha-1 sp|O15460|P4HA2_HUMAN 4 P4HA2 P4HA2_HUMAN Prolyl 535 0 4 0 0 0.000473898 0 4-hydroxylase subunit alpha-2 sp|P07237|PDIA1_HUMAN 4 P4HB PDIA1_HUMAN 508 0 4 0 0 0.000499085 0 Protein disulfide- isomerase sp|Q13310-3|PABP4_HUMAN 6 PABPC4 PABP4_HUMAN 660 0 3 3 0 0.000288108 0.000282924 Isoform 3 of Polyadenylate-binding protein 4 sp|Q8WX93|PALLD_HUMAN 1 PALLD PALLD_HUMAN 1383 0 1 0 0 4.58307E−05 0 Palladin sp|Q9NVD7|PARVA_HUMAN 1 PARVA PARVA_HUMAN 372 0 1 0 0 0.000170387 0 Alpha-parvin sp|Q96HS1|PGAM5_HUMAN 1 PGAM5 PGAM5_HUMAN 289 0 0 1 0 0 0.000215374 Serine/threonine- protein phosphatase PGAM5, mitochondrial sp|Q9UHX1|PUF60_HUMAN 4 PUF60 PUF60_HUMAN 559 0 4 0 0 0.000453552 0 Poly(U)-binding- splicing factor PUF60 sp|P11234|RALB_HUMAN 1 RALB RALB_HUMAN Ras- 206 0 1 0 0 0.000307688 0 related protein Ral-B sp|P62826|RAN_HUMAN 2 RAN RAN_HUMAN GTP- 216 0 0 2 0 0 0.000576326 binding nuclear protein Ran sp|P54136|SYRC_HUMAN 1 RARS SYRC_HUMAN 660 0 1 0 0 9.60361E−05 0 Arginine--tRNA ligase, cytoplasmic sp|P78332|RBM6_HUMAN 5 RBM6 RBM6_HUMAN RNA- 1123 0 0 5 0 0 0.000277129 binding protein 6 sp|Q6XE24|RBMS3_HUMAN 1 RBMS3 RBMS3_HUMAN RNA- 437 0 1 0 0 0.000145043 0 binding motif, single- stranded-interacting protein 3 sp|P26373|RL13_HUMAN 6 RPL13 RL13_HUMAN 60S 211 0 2 4 0 0.000600795 0.00117997 ribosomal protein L13 sp|P61353|RL27_HUMAN 1 RPL27 RL27_HUMAN 60S 136 0 1 0 0 0.000466058 0 ribosomal protein L27 sp|P42766|RL35_HUMAN 1 RPL35 RL35_HUMAN 60S 123 0 0 1 0 0 0.000506042 ribosomal protein L35 sp|P18124|RL7_HUMAN 2 RPL7 RL7_HUMAN 60S 248 0 2 2 0 0.00051116 0.000501961 ribosomal protein L7 sp|P05388|RLA0_HUMAN 2 RPLP0 RLA0_HUMAN 60S 317 0 1 1 0 0.000199949 0.000196351 acidic ribosomal protein P0 sp|P04843|RPN1_HUMAN 1 RPN1 RPN1_HUMAN 607 0 2 0 0 0.000208843 0 Dolichyl- diphosphooligosaccharide-- protein glycosyltransferase subunit 1 sp|P04844|RPN2_HUMAN 2 RPN2 RPN2_HUMAN 631 0 2 0 0 0.0002009 0 Dolichyl- diphosphooligosaccharide--protein glycosyltransferase subunit 2 sp|P62280|RS11_HUMAN 4 RPS11 RS11_HUMAN 40S 158 0 2 2 0 0.000802327 0.000787889 ribosomal protein S11 sp|P62263|RS14_HUMAN 1 RPS14 RS14_HUMAN 40S 151 0 2 0 0 0.000839521 0 ribosomal protein S14 sp|P62244|RS15A_HUMAN 2 RPS15A RS15A_HUMAN 40S 130 0 1 1 0 0.000487568 0.000478794 ribosomal protein S15a sp|P62249|RS16_HUMAN 1 RPS16 RS16_HUMAN 40S 146 0 1 0 0 0.000434136 0 ribosomal protein S16 sp|P42677|RS27_HUMAN 2 RPS27 RS27_HUMAN 40S 84 0 1 1 0 0.000754569 0.00074099 ribosomal protein S27 sp|P23396-2|RS3_HUMAN 6 RPS3 RS3_HUMAN Isoform 259 0 3 3 0 0.000734176 0.000720964 2 of 40S ribosomal protein S3 sp|P62241|RS8_HUMAN 5 RPS8 RS8_HUMAN 40S 208 0 2 3 0 0.00060946 0.000897738 ribosomal protein S8 sp|P46781|RS9_HUMAN 2 RPS9 RS9_HUMAN 40S 194 0 0 2 0 0 0.000641682 ribosomal protein S9 sp|Q9P2E9|RRBP1_HUMAN 6 RRBP1 RRBP1_HUMAN 1410 0 6 0 0 0.000269718 0 Ribosome-binding protein 1 sp|Q15424-3|SAFB1_HUMAN 3 SAFB SAFB1_HUMAN 917 0 0 3 0 0 0.000203631 Isoform 3 of Scaffold attachment factor B1 sp|P50454|SERPH_HUMAN 10 SERPINH1 SERPH_HUMAN Serpin 418 0 10 0 0 0.00151636 0 H1 sp|Q7Z333-4|SETX_HUMAN 1 SETX SETX_HUMAN Isoform 2706 0 0 1 0 0 2.30019E−05 4 of Probable helicase senataxin sp|Q92629-2|SGCD_HUMAN 1 SGCD SGCD_HUMAN 290 0 1 0 0 0.000218565 0 Isoform 2 of Delta- sarcoglycan sp|O60264|SMCA5_HUMAN 0 SMARCA5 SMCA5_HUMAN 1052 0 0 4 0 0 0.000236666 SWI/SNF-related matrix-associated actin-dependent regulator of chromatin subfamily A member 5 sp|Q9Y651|SOX21_HUMAN 0 SOX21 SOX21_HUMAN 276 0 0 1 0 0 0.000225519 Transcription factor SOX-21 sp|Q13813-2|SPTN1_HUMAN 4 SPTAN1 SPTN1_HUMAN 7.67668E−05 2.51285E−05 Isoform 2 of Spectrin alpha chain, non- 2477 0 3 1 0 erythrocytic 1 sp|Q01082|SPTB2_HUMAN 4 SPTBN1 SPTB2_HUMAN 2364 0 5 0 0 0.000134061 0 Spectrin beta chain, non-erythrocytic 1 sp|Q9UQ35|SRRM2_HUMAN 0 SRRM2 SRRM2_HUMAN 2752 0 0 3 0 0 6.78523E−05 Serine/arginine repetitive matrix protein 2 sp|O75494|SRS10_HUMAN 1 SRSF10 SRS10_HUMAN 262 0 1 0 0 0.000241923 0 Serine/arginine-rich splicing factor 10 sp|Q04837|SSBP_HUMAN 1 SSBP1 SSBP_HUMAN Single- 148 0 0 1 0 0 0.000420562 stranded DNA-binding protein, mitochondrial sp|O60506|HNRPQ_HUMAN 1 SYNCRIP HNRPQ_HUMAN 623 0 1 0 0 0.00010174 0 Heterogeneous nuclear ribonucleoprotein Q sp|P21980|TGM2_HUMAN 2 TGM2 TGM2_HUMAN 687 0 2 0 0 0.000184523 0 Protein-glutamine gamma-glutamyltransferase 2 sp|P07996|TSP1_HUMAN 1 THBS1 TSP1_HUMAN 1170 0 4 0 0 0.000216697 0 Thrombospondin-1 sp|Q9UPQ9|TNR6B_HUMAN 3 TNRC6B TNR6B_HUMAN 1833 0 2 1 0 6.91586E−05 0.000033957 Trinucleotide repeat- containing gene 6B protein sp|P62995|TRA2B_HUMAN 1 TRA2B TRA2B_HUMAN 288 0 1 0 0 0.000220083 0 Transformer-2 protein homolog beta sp|Q14157-5|UBP2L_HUMAN 2 UBAP2L UBP2L_HUMAN 1104 0 1 1 0 5.74129E−05 5.63797E−05 Isoform 5 of Ubiquitin- associated protein 2-like sp|P0CG48|UBC_HUMAN 2 UBC UBC_HUMAN 685 0 1 1 0 9.25311E−05 0.000090866 Polyubiquitin-C sp|P22695|QCR2_UMAN 1 UQCRC2 QCR2_HUMAN 453 0 1 0 0 0.00013992 0 Cytochrome b-c1 complex subunit 2, mitochondrial sp|P0C7P4|UCRIL_HUMAN 1 UQCRFS1P1 UCRIL HUMAN 283 0 1 0 0 0.000223971 0 Putative cytochrome b-c1 complex subunit Rieske-like protein 1 sp|Q14146|URB2_HUMAN 0 URB2 URB2_HUMAN 1524 0 0 1 0 0 0.000040842 Unhealthy ribosome biogenesis protein 2 homolog sp|P08670|VIME_HUMAN 1 VIM VIME_HUMAN 466 0 1 0 0 0.000136017 0 Vimentin sp|Q9H0D6|XRN2_HUMAN 1 XRN2 XRN2_HUMAN 5′-3′ 950 0 0 1 0 0 6.55192E−05 exoribonuclease 2 sp|P49750-4|YLPM1_HUMAN 13 YLPM1 YLPM1_HUMAN 2146 0 9 4 0 0.000265822 0.000116017 Isoform 4 of YLP motif- containing protein 1 sp|O75152|ZC11A_HUMAN 0 ZC3H11A ZC11A_HUMAN Zinc 810 0 0 1 0 0 7.68435E−05 finger CCCH domain- containing protein 11A sp|Q5BKZ1|ZN326_HUMAN 3 ZNF326 ZN326_HUMAN DBIRD 582 0 0 7 0 0 0.00074863 complex subunit ZNF326

TABLE-US-00024 TABLE 7C Aska-SS −+ shSSX Aska Aska (shControl)/Aska (shControl)/Aska (shSS18-SSX) (shSS18-SSX) LFQ Intensity LFQ Intensity (Norm to (Norm to SMARCA4 SMARCA4 % Gene names Rep 1) Rep 2) Coverage SSX1; SSX8 8.69195075 66.78165769 12.2 SMARCA4 1 1 43 SS18 0.959463613 1.263985576 7.7 BCL7C 1.744253828 1.537120215 57.1 BCL7B 1.339078001 1.126236646 21.8 ACTL6A 0.783942638 1.104979803 56.6 SMARCE1 0.893791358 0.854705908 61.8 SMARCD1 0.770993178 0.949816024 79.4

SMARCC1 0.505600572 0.622079125 47 ARID1B 0.721229344 0.649778539 41 SMARCB1 0.022092646 0.063656869 60.9

[0543] Finally, to evaluate whether SS18-SSX-containing BAF complexes that are tethered to the nucleosome acidic patch via SSX in place of the BAF core module SMARCB1 C-terminal acidic patch binding region (Valencia et al. (2019) *Cell* 179:1342-1356) are competent in remodeling, chromatin remodeling assays were performed using restriction enzyme accessibility assays (REAA) on endogenous BAF complexes containing either SS18 WT or SS18-SSX as well as assay for transposase-accessible chromatin using sequencing (ATAC-seq) in both CRL7250 fibroblasts and SS cell lines. Remodeling efficiency and ATPase activity of SS18-SSX-bound BAF complexes was slightly lower than that of WT SS18-bound complexes (FIGS. **7**I, **10**E and **10**G), however, this reduced activity was sufficient to enable DNA accessibility over SS18-SSX target sites genome-wide (FIGS. **7**J and **10**F).

[0544] Taken together, these data resolve SSX as a nucleosome acidic patch binding ligand fused to SS18, a subunit bound to the BAF complex ATPase subunit, SMARCA4 at its N-terminal region within the core structural module (Valencia et al. (2019) *Cell* 179:1342-1356; Ye et al. (2019) *Science* doi:10.1126/science.aay0033), that dominantly competes for acidic patch binding with BAF core module subunit SMARCB1, resulting in its partial destabilization and degradation. These oncogenic SS18-SSX-containing complexes are still proficient in chromatin remodeling and catalytic activity, resulting in the aberrant activation of normally repressed chromatin regions.

Example 5: SSX Exhibits Preference for H2A K119Ub-Marked Nucleosomes, Mediated by its Conserved C-Terminal Acidic Region

[0545] Previously, it was found that SS18-SSX-bound BAF complexes localize to polycomb-repressed regions (McBride et al. (2018) *Cancer Cell* 33:1128-1141). The engagement between the conserved SSX basic region and the nucleosome acidic patch is not, in itself, sufficient to explain why SS18-SSX complexes are preferentially recruited to repressed chromatin. It was therefore reasoned that the SSX-nucleosome acidic patch interaction can be augmented in some manner by the presence of specific histone repressive marks. To explore this possibility, CRISPR-Cas9-based screening of genes encoding proteins that are responsible for decorating and maintaining repressive chromatin was performed. These studies were performed in the SS cell line, SYO-1, as well as in a cell line that is an SS histologic mimic lacking the SS18-SSX fusion, SW982 (FIG. **11**A). Notably, it was found that PRC1 subunits (specifically, Ring1A/B, as well as PCGF5 and PCGF3 components of PRC1.3 and PRC1.5 complexes) were selectively enriched as synthetic lethal dependencies in SS cell lines SYO1 as well as other SS cell lines including Yamato and SCS241 (FIGS. **11**A, **12**A and **12**D). Importantly, all SS cell lines profiled exhibited significant dependency on SS18 and SSX (and hence the SS18-SSX fusion), relative to all other cell lines profiled (FIGS. **12**E and **12**F).

[0546] Given that the key histone modification placed by PRC1 is the H2A K119Ub mark, it was determined whether SSX exhibited any preferential binding to nucleosomes decorated with this modification. Notably, it was found that in SS cell lines, H2AK119Ub directly co-localized with sites of SS18-SSX BAF complex occupancy (FIGS. **11**B-**11**C and FIG. **12**B). This was consistent with the IF observations indicating substantial co-localization at Barr bodies (FIGS. **3**C, **5**A, **5**B, **5**D, **6**C and **9**C). Indeed, pulldown experiments and AlphaLisa binding assays performed with GST-SSX 78aa protein revealed significantly higher affinity to H2A K119Ub-decorated nucleosomes relative to unmodified nucleosomes or H2B K120Ub nucleosomes (FIGS. **11**D-**11**E and **12**G). Incubation of SSX 78aa with mammalian mononucleosomes also captured the higher molecular weight H2AUb species (FIG. **12**C), as did SS18-SSX-bound BAF complexes (FIGS. **1**A-**1**B, Tables 5A-5E). Importantly, endogenously purified SS18-SSX-bound BAF complexes enriched for binding of recombinant H2A K119Ub-modified nucleosomes over unmodified nucleosomes (FIGS. **11**F, **12**H and **12**I), consistent with the finding that SS18-SSX fusion target sites directly overlay H2AK119Ub sites genome-wide in SS cell lines (FIG. **11**B). Finally, a screen for SSX binding to a range of differentially-marked recombinant mononucleosomes as well as mammalian (pooled) nucleosomes was performed, and again, it was identified that GST-SSX 78aa preferentially bound to H2A K119Ub and mammalian nucleosomes over unmodified nucleosomes or nucleosomes with other histone marks (FIGS. **12**J-K). Fluorescence polarization (FP) experiments performed using fluoro-labeled SSX (in place of GST tag) also confirmed higher binding affinity to H2A K119Ub-decorated nucleosomes compared to unmodified nucleosomes (FIG. **12**I). To understand the role of H2A K119Ub in SSX-BAF localization, the core, catalytic subunits of the PRC1 complex, RING1A and RING1B, were next double deleted using CRISPR-Cas9 in HEK-293T cells (RING1A/1B-dKO HEK-293T cells) and expressed SS18-SSX (FIG. **13**A). Following immunofluorescence, complete loss of SS18-SSX localization to Barr bodies was observed as compared to RING1A/B WT cells (FIGS. **11**G-**11**H). To address whether the catalytic activity of PRC1 rather than PRC1 complex formation is required for SS18-SSX Barr body recruitment, structure-guided mutagenesis was performed to selectively disrupt the ubiquitin ligase activity of PRC1 and hence block its placement of H2A K119Ub (FIG. **11**G). a series of point mutations in RING1B were designed to disrupt acidic patch recognition (R98A), zinc binding (H69Y/R70C) and the E2 binding interface (R91A and I53A/D56K (Blackledge et al. (2019) *BioRxiv* 667667, doi:10.1101/667667)) (FIGS. **11**G-**11**H and **13**B-**13**C). Rescue of WT RING1B in RING1A/1B-dKO cells was able to completely rescue SSX localization. However, restoration of RING1B mutant variants affected SS18-SSX localization in a manner directly proportional to the degree to which these RING1B mutations impacted H2A K119Ub deposition. Significantly for this study, RING1A ligase-deficient R91E and I53A/D56K were able to form polycomb foci but were unable to recruit SS18-SSX, further highlighting the importance of the H2A K119Ub mark placement. As controls, R98A, and combined H69Y/R70C mutants had similar loss-of-function effects on SS18-SSX localization (Wang et al. (2004) *Nature* 431:873-878; McGinty et al. (2014) *Nature* 514:591-596) (FIGS. **11**G-**11**H and **13**B-**13**C). The widely-used I53A mutant (Buchwald et al. (2006) *EMBO J.* 25:2465-2474; Eskeland et al. (2010) *Mol. Cell* 38:452-464; Illingworth et al. (2015) *Genes* & *Dev.* 29:1897-1902) only partially attenuated H2A ubiquitination, and therefore had little effect on SSX targeting. As further support for a role for H2A K119Ub in SSX recruitment, a peptide hybridization assay performed on IMR90 cells pretreated with the deubiquitinating enzyme, USP2 was used. USP2-mediated removal of H2A K119Ub disrupted SSX peptide hybridization to Barr bodies specifically and without affecting its overall nuclear staining pattern, consistent with the general ability of SSX to bind unmodified nucleosomes via its acidic patch binding region (FIGS. **11**I and **13**D). EZH2 inhibitor treatment performed in WT HEK-293T or RING1A/B dKO HEK-293T cells further highlighted the requirement for H2A Ub119 placement (and hence PRC1) rather than H3K27me3 and PRC2 activity (FIGS. **13**F-**13**G). Somewhat surprisingly, given the clear role for H2A K119Ub in recruiting SSX to chromatin, direct binding between SSX and free ubiquitin was not observed, as assessed by a Ub-agarose pull down assay (FIG. **13**E), however, it is conceivable that SSX only engages Ub in the context of H2AK119Ub nucleosomes, as seen with other readers such as Dot1L (Anderson et al. (2019) *Cell Rep.* 26:1681-1690; Worden et al. (2019) *Cell* 176:1490-1501; Valencia-Sanchez et al. (2019) *Mol. Cell* 74:1010-1019), or alternatively, that it can recognize specific features of the nucleosome core itself that are sterically or allosterically affected by the presence of the ubiquitylation mark.

[0547] Finally, given that the conserved C-terminal acidic region of SSX did not disrupt SSX-nucleosome binding (FIG. **3**F) but did affect SS18-SSX-specific BAF complex targeting and resultant gene expression and proliferation (FIGS. **3**G-**3**I) in a manner comparable to loss of the basic region (acidic patch binding region), it was determined whether this region mediates the preference of SSX for H2A K119Ub-decorated nucleosomes. Excitingly, it was found that mutation of the C-terminal acidic region of SSX to alanines (i.e., DPEEDDE (SEQ ID NO: 221) to AAAAAAA (SEQ ID NO: 222)) relieved the preference of SSX for H2AK119Ub nucleosomes, while not altering SSX binding to nucleosomes (FIGS. **11**J-**11**K). These data collectively indicate that the conserved C-terminal acidic amino acids are required to drive the preference of SSX for H2A K119Ub nucleosomes and hence SS18-SSX-bound BAF complex targeting to repressive regions genome-wide, as observed in cells.

[0548] Here an unexpected set of properties have been identified and their functional ramifications were found in the fusion oncoprotein, SS18-SSX, the oncogenic driver of human synovial sarcoma (FIGS. **14**A-**14**B, model). An unusual, and new reported case in which an additional nucleosome acidic patch binding domain is fused to a subunit of a major chromatin remodeling complex, the mammalian SWI/SNF (BAF) complex, causing a generally tumor suppressor complex to gain oncogenic properties was found. Although several SNF2 helicase-based chromatin remodeling complexes are increasingly recognized to require the H2A/H2B nucleosome binding hub, it was found that the minimal, conserved SSX 34 aa region dominantly binds the acidic patch of nucleosomes, with preference for H2A K119Ub histone modification, altering the interaction between the nucleosome-SMARCB1 C-terminal alpha helix- interaction found in WT BAF complexes (Valencia et al. (2019) *Cell* 179:1342-1356; Ye et al. (2019) *Science*, doi:10.1126/science.aay0033), and resulting in higher affinity nucleosome binding properties augmented by specific repressive histone mark preferences (Valencia et al. (2019) *Cell* 179:1342-1356). While these data coupled with recent structural insights in yeast RSC and SWI/SNF complexes provide strong support for SS18-SSX-mediated displacement of SMARCB1 from the acidic patch and its destabilization at the nucleosome-proximal region of the core (base) module of BAF complexes, a high resolution, 3D structure of human BAF complexes containing SS18, as well as those containing SS18-SSX will be needed to define the full repertoire of structural changes to nucleosome-bound BAF complexes upon incorporation of SS18-SSX. This is particularly true given that the SS18 subunit is metazoan-specific and hence is not found in yeast complexes.

[0549] The expression of full length SSX is normally restricted to testes where it likely plays a role in sperm development, potentially involving polycomb-driven XY-body repression through engagement of H2A K119Ub-decorated sex chromosomes (Baarends et al. (1999) *Dev. Biol.* 207:322-333). Remarkably, this normal function of SSX as a binder of the nucleosome acidic patch and “reader” of this repressive state is leveraged in synovial sarcoma to alter BAF chromatin remodeling complex localization and gene expression patterns. Normally in testes, full-length SSX can function as a ligand for nucleosomes in this H2A K119Ub repressive state to promote further transcriptional repression through use of its N-terminal KRAB domain (Huntley et al. (2006) *Genome Res.* 16:669-677). In the case of SS, the KRAB domain is lost and replaced with essentially the whole ATPase module of the BAF chromatin remodeling complex via its fusion to SS18. This unfortunate scenario leads to gain of altered repressive chromatin reading properties of BAF complexes, loss of normal BAF complex-nucleosome acidic patch engagement, tight affinity and longer residency times at normally polycomb- repressed regions, and the activation of genes found in these regions (FIGS. **14**A-**14**B, model).

[0550] The SSX 78 aa tail, particularly the conserved 34aa C-terminus was therefore characterized (FIG. **3**D) as a ligand of the nucleosome acidic patch and the H2A K119Ub histone mark. These data indicate two non-mutually exclusive explanations for this reading preference: H2A K119Ub modification influences nucleosome structure by further exposes the acidic patch binding site; or, SSX exhibits a direct physical engagement with ubiquitin in the nucleosomal context. While studies that indicate that SSX does not bind directly to free (bead-bound) uniquitin was performed (FIG. **13**E), this does not rule out the possibility of direct ubiquitin engagement by the acidic C-terminal region of SSX when SS18-SSX-bound complexes are docked on nucleosomes. In this manner, Dot1L, for example, does not bind free ubiquitin but is only poised to interact with H2B UbK120 during substrate engagement (Anderson et al. (2019) *Cell Rep.* 26:1681-1690; Worden et al. (2019) *Cell* 176:1490-1501; Valencia-Sanchez et al. (2019) *Mol. Cell* 74:1010-1019). Understanding this binding preference requires future 3D high resolution structural characterization of SS18-SSX-bound human BAF complexes. Nonetheless, these results indicate that SSX acts as a nucleosome-specific binding ligand for the acidic patch on H2A K119Ub-decorated nucleosomes and that this property underlies the chromatin localization, gene expression, and synthetic lethal dependency profiles of this tumor type.

[0551] The biochemical and structural properties of fusion partner SSX elucidated here underpin the dependency of SS on PRC1 complex activity that have been detected in fitness screening efforts and in the structure-guided mutagenesis studies. In contrast to other reports (Banito et al. (2018) *Cancer cell* 33:527-541), direct binding to PRC1 by the SS18-SSX fusion is not found (or by SSX specifically), as has been indicated, nor a selective dependency on KDM2B; rather, it was found that SS18-SSX-bound complexes bind preferentially to H2A K119Ub-marked nucleosomes, and hence require PRC1 complex activity to place the H2AUbK119 mark. In all MS experiments here, peptides corresponding to PRC1 or PRC2 were not detected, rather, highly abundant peptides corresponding to histones and ubiquitin itself were found, and enrichment of peptides corresponding to the ATPase module subunits of BAF complexes (SMARCA4, BCL7A, beta-actin, ACTL6A) to which SS18 was tethered. The increased abundance of BAF complexes bound to SS18-SSX over PRC1-decorated sites and hence the frequency of molecules co-localized on chromatin can help reconcile these previous indications.

[0552] Finally, these results indicate that strategies to directly and specifically inhibit the SSX- or SS18-SSX-bound BAF complex- H2A K119Ub nucleosome interactions can represent viable new strategies for small molecule or inhibitory peptide identification and therapeutic development for synovial sarcoma. In conclusion, this disclosure presents an unexpected nucleosome acidic patch binding function of SSX, a partner within a fusion oncoprotein that lacks a canonical TF DNA-binding domain or recognizable chromatin reader domain and hence has remained a longstanding challenge to understand and target, that drives the altered behavior of the BAF chromatin remodeling complex, activating oncogenic programs in a cancer-specific manner.

EQUIVALENTS

[0553] Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments encompassed by the present invention described herein. Such equivalents are intended to be encompassed by the following claims.

Claims

1. A method of treating a subject afflicted with synovial sarcoma comprising administering to the subject a therapeutically effective amount of an agent that inhibits binding of a SS18-SSX fusion protein to a nucleosome, optionally wherein the nucleosome is an H2A K119Ub-marked nucleosome, optionally wherein a) the SS18-SSX fusion protein comprises a C-terminal region containing a basic region, and an acidic region of a SSX protein, further optionally wherein the basic region comprises a minimal 34-amino acid region; b) the SS18-SSX fusion protein is selected from Table 2; c) the SS18-SSX fusion protein comprises SS18 protein fused with a c-terminal portion of a SSX protein; d) the SS18-SSX fusion protein comprises c-terminal 34 amino acids (aa155-188) of a SSX protein; e) the SS18-SSX fusion protein comprises c-terminal 78 amino acids (aa 111-188) of a SSX protein; f) the SSX protein is selected form the group consisting of human SSX1, SSX2, SSX3, SSX4, SSX6, SSX7, SSX8, and SSX9; g) the SS18-SSX fusion protein comprises W164, R167, L168, R169 and/or R171 of SEQ ID: 3, 7, 13, 17, 21, 25, or 31, or orthologs thereof; h) the SS18-SSX fusion protein is a part of a BAF complex; i) the nucleosome comprises H2A protein comprising E56, E64, D90, E91, E92 and/or E113 of human, mouse, rat, or *Xenopus* H2A, or orthologs thereof; and/or H2B protein comprising E105 and/or E113 of human, mouse, rat, or *Xenopus* H2B, or orthologs thereof; j) the subject is an animal model of the cancer, optionally wherein the animal model is a mouse model; and/or k) the subject is a mammal, optionally wherein the mammal is a mouse or a human.

2-3. (canceled)

4. The method of claim 1, wherein the agent a) inhibits binding of the basic region of the SS18-SSX fusion protein to an acidic patch of the nucleosome, optionally wherein the nucleosome is an H2A K119Ub-marked nucleosome; b) is a small molecule inhibitor, a small molecule degrader, CRISPR guide RNA (gRNA), RNA interfering agent, oligonucleotide, peptide or peptidomimetic inhibitor, aptamer, antibody, or intrabody, optionally wherein the RNA interfering agent is a small interfering RNA (siRNA), CRISPR RNA (crRNA), CRISPR guide RNA (gRNA), a small hairpin RNA (shRNA), a microRNA (miRNA), or a piwi-interacting RNA (piRNA); c) comprises an antibody and/or intrabody, or an antigen binding fragment thereof, which specifically binds to the SS18-SSX fusion protein, the SSX tail, and/or the H2AK119Ub-marked nucleosome, optionally wherein the SSX tail is SSX tail (34 amino acid) and/or SSX tail (78 amino acid); d) comprises an antibody and/or intrabody, or an antigen binding fragment thereof, which specifically binds to at least one of the following regions: (1) the basic region of the SS18-SSX fusion protein; (2) the acidic region of the SS18-SSX fusion protein; (3) the acidic patch of the H2AK119Ub-marked nucleosome; and/or (4) the H2AK119Ub mark, further optionally wherein the antibody and/or intrabody, or antigen binding fragment thereof, i) is chimeric, humanized, composite, or human, and/or ii) comprises an effector domain, comprises an Fc domain, and/or is selected from the group consisting of Fv, Fav, F(ab′)2, Fab′, dsFv, scFv, sc(Fv)2, and diabodies fragments; e) induces deletion or mutation of the basic region of the SS18-SSX fusion protein, the acidic region of the SS18-SSX fusion protein, the acidic patch of the H2AK119Ub-marked nucleosome, and/or a region within the SSX tail (34 amino acid); f) inhibits H2A ubiquitination; g) inhibits ubiquitin ligase activity of a PRC1 complex; h) reduces expression, copy number, and/or ubiquitin ligase activity of RING1A and/or RING1B; i) inhibits recruitment of a SS18-SSX fusion protein-bound BAF complex to an H2AK119Ub-marked nucleosome; j) inhibits activation of at least one oncogenic target gene of the SS18-SSX fusion protein, optionally wherein the oncogenic target gene of the SS18-SSX fusion protein is selected from the group consisting of WNT16 and oncogenic target genes listed in McBride et al. (2018) *Cancer Cell* 33:1128-1141; k) reduces the number of viable or proliferating cells in the cancer, and/or reduces the volume or size of a tumor comprising the cancer cells; and/or l) is administered in a pharmaceutically acceptable formulation.

5-18. (canceled)

19. The method of claim 1, further comprising administering to the subject il an immunotherapy and/or cancer therapy and/or 2) at least one additional therapeutic agent or regimen for treating the cancer, optionally wherein a) the immunotherapy and/or cancer therapy is administered before, after, or concurrently with the agent, b) the immunotherapy is cell-based; c) the immunotherapy comprises a cancer vaccine and/or virus; d) the immunotherapy inhibits an immune checkpoint, optionally wherein the immune checkpoint is selected from the group consisting of CTLA-4, PD-1, VISTA, B7-H2, B7-H3, PD-L1, B7-H4, B7-H6, ICOS, HVEM, PD-L2, CD160, gp49B, PIR-B, KIR family receptors, TIM-1, TIM-3, TIM-4, LAG-3, GITR, 4-IBB, OX-40, BTLA, SIRPalpha (CD47), CD48, 2B4 (CD244), B7.1, B7.2, ILT-2, ILT-4, TIGIT, HHLA2, butyrophilins, and A2aR; and/or e) the cancer therapy is selected from the group consisting of radiation, a radiosensitizer, and a chemotherapy.

20-25. (canceled)

26. A method of reducing viability or proliferation of synovial sarcoma cells comprising contacting the synovial sarcoma cells with an agent that inhibits binding of a SS18-SSX fusion protein according to claim 1 to a nucleosome, optionally wherein the nucleosome is an H2AK119Ub-marked nucleosome and/or the step of administering or contacting occurs in vivo, ex vivo, or in vitro.

27-28. (canceled)

29. The method of claim 26, wherein the agent is as according to claim 4.

30-42. (canceled)

43. The method of claim 26, further comprising contacting the cancer cells with an immunotherapy and/or cancer therapy, optionally wherein a) the immunotherapy and/or cancer therapy is administered before, after, or concurrently with the agent, b) the immunotherapy is cell-based; c) the immunotherapy comprises a cancer vaccine and/or virus; d) the immunotherapy inhibits an immune checkpoint, optionally wherein the immune checkpoint is selected from the group consisting of CTLA-4, PD-1, VISTA, B7-H2, B7-H3, PD-L1, B7-H4, B7-H6, ICOS, HVEM, PD-L2, CD160, gp49B, PIR-B, KIR family receptors, TIM-1, TIM-3, TIM-4, LAG-3, GITR, 4-IBB, OX-40, BTLA, SIRPalpha (CD47), CD48, 2B4 (CD244), B7.1, B7.2, ILT-2, ILT-4, TIGIT, HHLA2, butyrophilins, and A2aR; and/or e) the cancer therapy is selected from the group consisting of radiation, a radiosensitizer, and a chemotherapy.

44-48. (canceled)

49. A method of assessing the efficacy of the agent according to claim 26 for treating synovial sarcoma in a subject, comprising; a) detecting in a subject sample at a first point in time the number of viable and/or proliferating cancer cells; b) repeating step a) during at least one subsequent point in time after administration of the agent; and c) comparing number of viable and/or proliferating cancer cells detected in steps a) and b), wherein the absence of, or a significant decrease in number of viable and/or proliferating cancer cells in the subsequent sample as compared to the amount in the sample at the first point in time, indicates that the agent treats synovial sarcoma in the subject.

50. The method of claim 49, wherein a) between the first point in time and the subsequent point in time, the subject has undergone treatment, completed treatment, and/or is in remission for synovial sarcoma; b) the first and/or at least one subsequent sample is selected from the group consisting of ex vivo and in vivo samples; c) the first and/or at least one subsequent sample is obtained from an animal model of synovial sarcoma; d) the first and/or at least one subsequent sample is a portion of a single sample or pooled samples obtained from the subject; e) the sample comprises cells, serum, peritumoral tissue, and/or intratumoral tissue obtained from the subject; f) the method further comprises determining responsiveness to the agent by measuring at least one criteria selected from the group consisting of clinical benefit rate, survival until mortality, pathological complete response, semi-quantitative measures of pathologic response, clinical complete remission, clinical partial remission, clinical stable disease, recurrence-free survival, metastasis free survival, disease free survival, circulating tumor cell decrease, circulating marker response, and RECIST criteria.

51-57. (canceled)

58. A cell-based assay for screening for agents that reduce viability or proliferation of a synovial sarcoma cell comprising; a) contacting the synovial sarcoma cell with a test agent; and b) determining the ability of the test agent to inhibit binding of a SS18-SSX fusion protein according to claim 1, a SSX (78 amino acid) region, and/or a SSX (34 amino acid) minimal region to a nucleosome, optionally wherein the nucleosome is a H2AK119Ub-marked nucleosome and/or the step of contacting occurs in vivo, ex vivo, or in vitro.

59-61. (canceled)

62. The cell-based assay of claim 58, further comprising a) determining the ability of the test agent to inhibit recruitment of a SS18-SSX fusion protein-bound BAF complex to an H2AK119Ub-marked nucleosome and/or H2AK 119Ub-marked region of chromatin in cells, optionally wherein the cellular chromatin comprises a PRC1/H2A Ub domain; b) determining the ability of the test agent to inhibit activation of at least one oncogenic target gene of the SS18-SSX fusion protein, optionally wherein the oncogenic target gene of the SS18-SSX fusion protein is selected from the group consisting of WNT16 and oncogenic target genes listed in McBride et al. (2018) *Cancer Cell* 33:1128-1141; and/or c) determining a reduction in the viability or proliferation of the cancer cells.

63-65. (canceled)

66. An in vitro assay for screening for agents that reduce viability or proliferation of a synovial sarcoma cell comprising; a) mixing a protein comprising a c-terminal basic region and a c-terminal acidic region of a SSX protein and a nucleosome together, optionally wherein the nucleosome is a H2AK119Ub-marked nucleosome; b) adding a test agent to the mixture; and c) determining the ability of the test agent to decrease binding of the protein to the nucleosome.

67. The in vitro assay of claim 66, wherein the protein a) comprises c-terminal 34 amino acids (aa155-188) of a SSX protein, optionally wherein the SS18-SSX fusion protein is selected from Table 2; and/or b) comprises c-terminal 78 amino acids (aa 111-188) of a SSX protein, optionally wherein the SS18-SSX fusion protein is selected from Table 2.

68-81. (canceled)

82. An isolated modified protein complex selected from the group consisting of protein complexes listed in Table 3, wherein the isolated modified protein complex comprises at least one subunit that is modified, optionally wherein the at least one modified subunit is a fragment of the subunit.

83. (canceled)

84. The isolated modified protein complex of claim 82, wherein the fragment of the subunit a) binds to at least one binding partner of the subunit to form the isolated modified protein complex; b) comprises the basic region and/or the acidic region of a SSX protein; c) comprises c-terminal 34 amino acids (aa155-188) of a SSX protein; d) comprises c-terminal 78 amino acids (aa 111-188) of a SSX protein; and/or e) comprises the acidic patch of a nucleosome and/or the H2A K119 Ub mark.

85-87. (canceled)

88. The isolated modified protein complex of claim 82, wherein the SSX protein is selected form the group consisting of human SSX1, SSX2, SSX3, SSX4, SSX6, SSX7, SSX8, and SSX9.

89. (canceled)

90. The isolated modified protein complex of claim 82, wherein at least one subunit a) is linked to at least another subunit; b) is linked to at least another subunit through covalent cross-links; c) is linked to at least another subunit through a peptide linker; d) comprises a heterologous amino acid sequence; and/or e) is selected from the group consisting of HA-SS18-SSX1, V5-SS18-SSX1, V5-SS18-SSX1 34aa tail, V5-SS18-SSX1 78aa tail, H2A, and H2B.

91-93. (canceled)

94. The isolated modified protein complex of claim 90, wherein the heterologous amino acid sequence comprises an affinity tag or a label.

95. The isolated modified protein complex of claim 94, wherein the affinity tag is selected from the group consisting of Glutathione-S-Transferase (GST), calmodulin binding protein (CBP), protein C tag, Myc tag, HaloTag, HA tag, Flag tag, His tag, biotin tag, and V5 tag.

96. The isolated modified protein complex of claim 95, wherein the label is a fluorescent protein.

97. (canceled)

98. A pharmaceutical composition comprising the isolated modified protein complex according to claim 82 and a carrier.